Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Critical-Density Feedback Bridge

This chapter records the critical-density principle used to organize the current frontier. It is a bridge from heuristic insight to proof obligations, not a replacement for the remaining row/column, two-point, or RH exit proofs.

Definition 2.1 (Self-sieving density model)**.** Let a self-sieving set have local density

$$\rho(x) \sim \frac{C}{(\log x)^\alpha}.$$

Its previous elements impose a first-order one-residue sieve intensity

$$S(x) = \sum_{a \le x,\ a \in A} \frac{1}{a} \sim \int_2^x \frac{\rho(t)}{t}\, dt.$$

The associated model survivor density is $\exp(-S(x))$.

Lemma 2.2 (Critical fixed point)**.** *Inside the self-sieving density model, the only logarithmic fixed-point exponent is*

$$\rho(x) \asymp \frac{1}{\log x}.$$

More precisely, if $\rho(x) \sim C/(\log x)^\alpha$ and the model survivor density is required to have the same logarithmic order as $\rho(x)$, then $\alpha = 1$; at that layer, exponent matching gives $C = 1$.

Proof. If $\alpha > 1$, then

$$\int_2^x \frac{C\, dt}{t(\log t)^\alpha}$$

converges, so the model survivor density stays bounded away from zero while $C/(\log x)^\alpha$ tends to zero. This is not self-consistent.

If $\alpha < 1$, then the same integral is

$$\frac{C}{1-\alpha}(\log x)^{1-\alpha} + O(1),$$

so the survivor density is

$$\exp\{-c(\log x)^{1-\alpha} + O(1)\},$$

which is smaller than any fixed negative power of $\log x$. This is again not of order $C/(\log x)^\alpha$.

Thus $\alpha = 1$. Then $S(x) \sim C \log\log x$, and the survivor density is $(\log x)^{-C+o(1)}$. Matching the logarithmic exponent with $C/\log x$ forces $C = 1$ at the model level. $\square$

Remark 2.3 (Proof boundary)**.** The lemma is a fixed-point calculation for the self-sieving model. It does not by itself prove a short-interval prime, a prime pair, or RH. To become a theorem, the fixed-point pressure must be upgraded into local lower bounds, new-modulus anti-concentration estimates, or controlled-exit inequalities.

Reduction-closed Statement 2.4 (Critical feedback bridge for the three programs)**.** *The critical-density principle routes the three current frontiers as follows.*

1. *For the prime-matrix row/column program, a fully covered row or column is a local collapse below the* $1/\log x$ *fixed point. Such a collapse must be explained by fixed CRT concentration, by moving high-factor slots that pay Rankin/SAE mass, or by a failure of the local survivor lower bound. The active narrow gate is*

   $$\mathit{AffineTwinFillResidueArrivalBoundOrFillCatchUpMassPDECExclusion}.$$

2. *For the two-point secondary sieve, the corresponding fixed point is the two-residue product*

   $$\prod_{q\le y}\left(1-\frac{\nu_q(w)}{q}\right)\asymp\frac{\mathfrak{S}(w)}{(\log y)^2}.$$

   The remaining theorem-level input is not a one-point density estimate but the true-residual new-modulus correlation bound in the I3-Core/TRC package.

3. *For the RH contradiction field, an off-critical zero creates a smooth prime anomaly relative to the same* $1/\log x$ *fixed point. The anomaly must pass through the recorded sparse, dense, tail, internal, and global exits; the manuscript can be promoted only after those controlled exits are independently verified under one load convention.*

Status and reference. This is a routing statement. The detailed version is archived as

`docs/monograph/three-claims-critical-density-feedback-frontier.md`.

It identifies the next prime-matrix input as a critical fill-arrival anti-overfilling theorem: persistent AffineTwin threshold crossing must either reuse a residue and become a reset/ColumnCRT/PDEC certificate, or introduce new fill residues whose Rankin/SAE mass exceeds the critical-density carrying capacity. That final carrying-capacity inequality is the new proof obligation; it is not asserted here as already proved. □

Reduction-closed Statement 2.5 (Critical fill-arrival pressure packet frontier)**.** *In the AffineTwin branch, write* $g=q-2$*,* $f=q$*, and let* A_g, A_f *be the generator-side and fill-side residue counts. The safe integer gate is*

$$A_gA_f\le T_q:=\lfloor\sqrt{gf}\rfloor.$$

For a threshold-crossing route with final counts A'_g, A'_f*, crossing is equivalent to*

$$A'_gA'_f>T_q,\qquad \text{hence}\qquad \frac{(A'_g)^2}{g}\,\frac{(A'_f)^2}{f}>1.$$

Thus fill catch-up is not merely a raw residue-arrival event. It is a threshold-relevant pressure packet: for fixed A'_g *the minimum fill target is*

$$A'_{f,\min}(A'_g)=\left\lfloor\frac{T_q}{A'_g}\right\rfloor+1.$$

Consequently a persistent row/column counterexample in this branch must enter one of four named exits: repeated-residue reset/ColumnCRT PDEC, fixed pressure packet recurrence PDEC, moving pressure packet SAE/Rankin, or non-sparse AffineTwin pressure demand.

Status and reference. The product gate and the pressure form are algebraic equivalents. The minimum fill target follows by solving $A'_g A'_f > T_q$ in integers. The repeated-residue exit is the previous fill-catchup dichotomy; the fixed-packet exit is the AffineTwin slot phase lock, where the combined modulus $q(q-2)$ exceeds the common phase support; and the moving-packet exit is the existing moving-family SAE/ColumnCRT split. The remaining unproved input is the global pressure-packet carrying ceiling, recorded in

`docs/monograph/prime-matrix-critical-fill-arrival-anti-overfilling-router.md`.

□

Reduction-closed Statement 2.6 (Pressure-packet carrying ceiling)**.** *Let $M_q = A_g A_f$ be the pressure-packet multiplicity for an AffineTwin modulus q, so that q and $q-2$ are prime. The packet SAE mass is*

$$\mu_q = \frac{M_q}{q(q-2)}.$$

If the square-root packet gate

$$M_q^2 \le q(q-2)$$

holds, then

$$\mu_q \le \frac{1}{\sqrt{q(q-2)}} \le \frac{1}{q-2}.$$

Consequently, any standard Brun–Selberg upper-bound sieve of the form

$$\Pi_2(X) := \#\{q \le X : q,\ q-2 \text{ prime}\} \le C\frac{X}{(\log X)^2}$$

implies by partial summation that the moving pressure-packet tail is summable. If the square-root packet gate fails, then

$$M_q^2 > q(q-2),$$

which is exactly the named SuperSqrt/PressureProduct PDEC exit.

Status and reference. The first inequalities are immediate from the square-root gate. The Brun–Selberg bound gives

$$\sum_{\substack{q \ge Q \\ q,\ q-2 \text{ prime}}} \frac{1}{q} \le \frac{\Pi_2(Q)}{Q} + \int_Q^\infty \frac{\Pi_2(t)}{t^2}\,dt \ll \frac{1}{(\log Q)^2} + \frac{1}{\log Q},$$

and $1/(q-2) \ll 1/q$ for $q \ge 5$. Thus all packets satisfying the square-root gate are absorbed by the reciprocal ceiling. The complementary case is the previous pressure-product inequality with product > 1, hence the registered PDEC/ColumnCRT exit. The detailed route is recorded in the pressure-packet carrying-ceiling router under `docs/monograph/`. This still leaves two explicit tasks before final row/column promotion. First, internalize or cite the Brun–Selberg reciprocal ceiling under the monograph's convention. Second, exclude the SuperSqrt/PressureProduct PDEC family. □

Reduction-closed Statement 2.7 (Critical-error frontier). *After the critical-density fixed point has been fixed, the remaining frontier is best measured by a normalized critical error*

$$\mathcal{E} = \frac{\textit{actual load}}{\textit{critical capacity}} - 1.$$

A positive critical error is not allowed to remain an anonymous remainder in the contradiction field. In the prime-matrix AffineTwin branch it is the pressure product

$$\frac{A_g^2}{g}\frac{A_f^2}{f} - 1;$$

in the two-point sieve it is the excess of the large-incidence mean above the Buchstab semiprime transition constant $K(\alpha)$, together with the terminal gap to the threshold 1; in the RH branch it is the off-critical zero load after zero-frequency CRT subtraction, normalized by the capacity of each controlled exit. Thus every positive critical error must become a fixed CRT/PDEC defect, a moving SAE/Rankin or analytic-average packet, a denominator-floor defect, or a named controlled exit.

Status and reference. This is a frontier reduction and audit rule, not a final theorem. The detailed ledger is recorded in

`docs/monograph/three-claims-critical-error-frontier-audit.md`.

It sharpens the current next tasks to three interfaces: a non-PDEC square-root phase-support bound for the prime-matrix pressure packets, a critical denominator-floor check for the two-point BMD-to-TLI transfer, and a common baseline-subtracted load convention for every RH controlled exit. □

Reduction-closed Statement 2.8 (Actual-packet square-root phase support). *In the prime-matrix AffineTwin branch, distinguish the formal product*

$$M_q^{\mathrm{form}} = A_g A_f$$

from the actual non-PDEC packet count N_q. The latter counts only generator–fill packets whose two slot conditions are compatible, whose common phase support is nonempty, and which have not already become a reset or ColumnCRT/PDEC event. For a primitive AffineTwin slot pair with $q \geq 13$, the recorded depth identities give common support width

$$W_q = \frac{q+9}{2}.$$

Since

$$W_q \leq \sqrt{q(q-2)}$$

for all $q \geq 13$, and since a repeated actual packet projection inside this support is by definition a PDEC/ColumnCRT event, every non-PDEC primitive packet family satisfies

$$N_q \leq W_q \leq \sqrt{q(q-2)}.$$

Thus the square-root gate is closed for actual primitive packets. A remaining formal SuperSqrt failure must be either product-accounting looseness (M_q^{form} overcounts N_q), a projection-collision PDEC, or a primitive support escape.

Status and reference. The inequality follows by squaring:

$$4q(q-2)-(q+9)^2 = 3q^2-26q-81 \geq 0$$

for $q \geq 13$. The injection of actual packets into the primitive support is the non-PDEC projection rule: two actual packets with the same primitive projection would be a repeated fixed CRT phase and therefore a reset/ColumnCRT/PDEC event. The detailed audit is archived in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`.

This does not yet prove the row/column theorem. It reduces the remaining Prime-Matrix obligation to product-accounting tightening and to excluding or routing primitive-support escape packets. □

Reduction-closed Statement 2.9 (Formal envelope versus actual load). *Across the three programs, a critical contradiction can only be triggered by actual load, not by a formal envelope. If F is a formal upper envelope, A is the actual load, and C is the critical capacity, then*

$$0 \leq A \leq F.$$

The case $F \leq C$ is absorbed. The case $F > C$ but $A \leq C$ is only an accounting issue and requires tightening the envelope to the actual load. Only the case $A > C$ can force a structural exit. In the prime-matrix AffineTwin branch this is the distinction between $(A_g A_f)^2$ and N_q^2; in the two-point sieve it is the distinction between model numerator/denominator envelopes and the actual ratio

$$|U_Y(I)|^{-1} \sum_{x \in U_Y(I)} D_Y^P(x);$$

in the RH branch it is the distinction between routed atom envelopes and source-deleted final loads in the GEE ledger.

Status and reference. This is a status-discipline and reduction rule. It prevents any of the three frontiers from closing by a coarse upper envelope alone. The detailed cross-program audit is recorded in

`docs/monograph/three-claims-formal-to-actual-critical-load-frontier.md`.

The resulting active interfaces are actual packet enumeration for the prime-matrix branch, actual denominator-floor control for the two-point BMD-to-TLI transfer, and actual final-load normalization for the RH controlled exits. □

Reduction-closed Statement 2.10 (Actual-load closure contracts). *The current frontier is organized into three actual-load contracts. In the prime-matrix branch, PM–ALC requires that the actual AffineTwin packet count N_q replace the formal product $A_g A_f$ in the critical load; the current machine contract records candidate $q = 31, 43, 103$, total formal product upper bound 40, but only one actual packet in the current sweep. In the two-point branch, TP–ALC requires both a BMD/KLS numerator bound and a denominator floor for $|U_Y(I)|$ in the same Buchstab convention, with*

$$K(\alpha)+\varepsilon(\alpha) < 1-\delta(\alpha).$$

In the RH branch, RH–ALC requires every controlled exit to be written with baseline-subtracted source-deleted final load. Failure of any contract is not an anonymous error; it is a named PDEC/SAE/parity/contro exit obligation.

Status and reference. The PM current-sweep contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py`.

The unified contract ledger is archived in

`docs/monograph/three-claims-actual-load-closure-contracts.md`.

This is a reduction and evidence-organizing step. It does not close the row/column theorem, the two-point theorem, or RH. □

Reduction-closed Statement 2.11 (AffineTwin formal-pair pruning). *In the current prime-matrix AffineTwin sweep, the formal product universe has total size* 40*, while the actual packet count is* 1*. The difference is now fully explained at the current level:* 11 *formal residue pairs are removed by an exact CRT-window emptiness check, and* 28 *formal residue pairs have no matching gap-fill source materialization in the required orientation. Thus the current formal-to-actual gap is not an unaccounted critical load.*

Status and reference. The pruning audit is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py`.

For $q = 31$, the audit enumerates the $3 \cdot 4$ formal residue pairs and finds only the pair $(19, 8)$ with CRT representative 2687 in the support interval $[2669, 2688]$. The remaining current candidates $q = 43, 103$ have no matching gap-fill source in the required direction. The global obligation remains: future formal pairs must be routed to CRT-window emptiness, source materialization failure as PDEC/SAE, or primitive-support escape. This does not close the row/column theorem. □

Reduction-closed Statement 2.12 (AffineTwin source materialization gate). *The current source-materialization failures in the AffineTwin audit are not unclassified missing mass. An actual AffineTwin packet must have a matching gap-fill source with gap q, generator modulus $q - 2$, fill modulus q, the prescribed orientation, and affine delay*

$$p_{\text{fill}} - p_{\text{gen}} = \frac{11q - 21}{4}.$$

In the current sweep, $q = 31$ is the only candidate passing this gate. The candidate $q = 43$ has a same-gap source, but it uses generator 47*, orientation plus-to-minus, and delay* 74 *instead of generator* 41*, orientation minus-to-plus, and delay* 113*. The candidate $q = 103$ has no gap-fill source. Thus the* 28 *source-unmaterialized formal pairs from the pruning audit are fully classified at the current level.*

Status and reference. The source gate audit is generated by

`experiments/prime_matrix_affine_twin_source_materialization_gate_audit.py`.

It records 16 blocked formal pairs for the wrong-source $q = 43$ row and 12 blocked formal pairs for the no-source $q = 103$ row. The remaining global obligation is to prove that every future source-gate failure routes to SameGapWrongSource-PDEC/SAE or NoGapSource-PDEC/SAE, rather than contributing to actual load. This does not close the row/column theorem. □

Reduction-closed Statement 2.13 (AffineTwin CRT window gap). *After the source gate, the remaining current AffineTwin formal pairs are controlled by a CRT window test. For $q = 31$, the exact source has generator modulus* 29*, fill modulus* 31*, combined modulus* 899*, and common phase support* $[2669, 2688]$ *of width* 20*. Among the* 12 *formal residue pairs, exactly one pair,* $(19, 8)$*, has a CRT representative in the support, namely* 2687*. The other* 11 *pairs have nearest CRT representatives at positive distance from the support; the minimum such distance is* 40*.*

Status and reference. The CRT-window distance audit is generated by

`experiments/prime_matrix_affine_twin_crt_window_gap_audit.py`.

It upgrades current CRTWindowEmpty rows from a Boolean non-hit to positive phase-gap data. The global obligation remains to prove that future source-materialized non-hits have a uniform CRT window gap, or otherwise route the failure to the named PDEC/SAE exits recorded in the audit. This does not close the row/column theorem. □

Reduction-closed Statement 2.14 (Window edge collision). *The current CRT-window non-hits can be expressed as residue-grid displacement requirements. For $q = 31$, the phase window* $[2669, 2688]$ *induces* 20 *target residue pairs. Each of the* 11 *non-hit formal pairs has a nearest target pair; the least L^1 displacement is* 1*, realized by*

$$(19, 9) \longrightarrow (19, 8).$$

Thus the narrowest current edge-collision atom is a one-step fill-residue collision with the already realized actual pair $(19, 8)$*.*

Status and reference. The edge-collision displacement audit is generated by

`experiments/prime_matrix_affine_twin_window_edge_collision_audit.py`.

It records two non-hit pairs whose nearest target is the existing actual pair, and nine non-hit pairs whose nearest target requires unused target-residue arrival. The global obligation remains to prove that such residue movements cannot be supplied persistently by a counterexample chain, or else to route the failure to residue-collision, target-arrival, or support-motion exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.15 (Existing actual CRT jump). *In the current AffineTwin edge-collision audit, the existing-actual branch is not a small boundary slip. The two non-hit formal pairs whose nearest target is the already realized pair* $(19, 8)$ *have CRT phase jumps* 120 *and* 435*, whereas the common phase support has width* 20*. In particular the narrowest residue atom*

$$(19, 9) \longrightarrow (19, 8)$$

has L^1 residue displacement 1*, but the fill-residue unit step in the combined CRT modulo* $29 \cdot 31$ *is* 435*.*

Status and reference. The CRT jump audit is generated by

`experiments/prime_matrix_affine_twin_existing_actual_collision_jump_audit.py`.

It records generator and fill CRT unit steps 465 and 435, respectively, and verifies the exact signed jumps from the empty-pair nearest representatives to the existing actual point. This closes the current-sweep existing-actual collision jump ledger. The global obligation remains to prove that such jumps cannot be reset persistently by moving support, or to route the failure to repeated-residue, ColumnCRT, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.16 (Unused target arrival). *In the current AffineTwin edge-collision audit, the unused-target branch also has no hidden actual load. The* 9 *non-hit formal pairs whose nearest targets are unused collapse to* 5 *target residue pairs:*

$$(10, 30), \ (16, 5), \ (17, 6), \ (18, 7), \ (20, 9).$$

None of these pairs belongs to the current formal product. To make them available, one must introduce new generator residues

$$10, 16, 17, 18, 20$$

and new fill residues

$$5, 6, 7, 30.$$

Moreover every corresponding CRT jump has absolute size at least 59*, while the common phase support has width* 20*.*

Status and reference. The unused-target arrival audit is generated by

`experiments/prime_matrix_affine_twin_unused_target_arrival_audit.py`.

It records that all unused targets are outside the current formal product, all need a new side residue, all need a new generator residue, and all CRT jump identities exceed the support width in the current sweep. The global obligation remains to control new generator/fill residue arrival or route the failure to support-motion, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.17 (Support motion depth). *In the current AffineTwin CRT-window audit, support motion is not a free escape. The common support is the intersection of*

$$[2669, 2693] \quad \textit{and} \quad [2659, 2688],$$

namely $[2669, 2688]$*. Therefore an empty CRT representative can enter by support motion only if the corresponding generator endpoint and shifted-fill endpoint are both released. In the current sweep all* 11 *empty pairs require such a two-endpoint release, and the common side depth required ranges from* 58 *to* 435*.*

Status and reference. The support-motion depth audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_depth_audit.py`.

For the narrowest row $(19, 12)$, the nearest CRT representative is 2629. To include it, the generator left depth must increase by 40 and the fill left depth by 30, so the total endpoint release is 70. The audit records the analogous depth-inflation vector for every empty row. This closes the current-sweep support-motion depth ledger. The global obligation remains to prove non-persistence of such synchronized depth inflation, or to route it to moving-support, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.18 (Support motion primitive defect). *The current support-motion branch cannot be absorbed by the fixed* $q = 31$ *AffineTwin primitive key. The slot-lock identities give*

$$d_g^- = 18, \quad d_g^+ = 6, \quad d_f^- = 28, \quad d_f^+ = 1.$$

Every support-motion candidate requires a common side depth different from the corresponding generator and fill depths. The smallest defect is the row $(19, 12)$*: it requires left common depth* 58*, producing defects*

$$58 - 18 = 40, \qquad 58 - 28 = 30.$$

Thus even the narrowest support-motion atom breaks both fixed primitive depth identities.

Status and reference. The primitive-depth defect audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_primitive_defect_audit.py`.

It records 11 support-motion candidates, all of which break both the generator and fill affine-depth identities for the fixed $q = 31$ primitive key. This closes the current-sweep fixed-key absorption route. The global obligation remains to exclude moving primitive keys, or to route them to moving-key, depth-inflation, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.19 (Moving-key depth formula)**.** *Even if the AffineTwin key is allowed to move, same-orientation depth absorption is blocked in the current support-motion rows. On the lower side, a common depth* D *would have to satisfy*

$$(q + 5)/2 = D, \qquad q - 3 = D,$$

so $q = 2D - 5$ *and* $q = D + 3$ *must agree. This occurs only at* $D = 8$*. The narrowest lower support-motion row has* $D = 58$*, giving* $q = 111$ *from the generator formula and* $q = 61$ *from the fill formula. On the upper side, the fill depth is fixed at* 1*, while the current required depths are at least* 342*.*

Status and reference. The moving-key depth formula audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_depth_formula_audit.py`.

It records 11 support-motion candidates. None admits a common moving q with the same orientation and both depth formulas. This closes the current-sweep same-orientation moving-key absorption route. The global obligation remains to exclude orientation changes, source rematerialization, or other primitive-key migration, or to route them to PDEC/SAE/endpoint-coupling exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.20 (Moving-key source rematerialization)**.** *The one-sided* q *values produced by the moving-key depth formulas also fail to become AffineTwin sources with the same orientation in the current sweep. Across the* 11 *support-motion rows the formulas produce* 19 *distinct one-sided candidate keys. Thirteen are composite. The six prime candidates*

$$61,\ 181,\ 293,\ 761,\ 1499,\ 1747$$

all fail the AffineTwin prime/source gate: none satisfies the full same-orientation requirement that q *and* $q - 2$ *are prime,* $q \equiv 3 \pmod 4$*, the delay* $(11q - 21)/4$ *is integral, and a matching gap-fill source with generator* $q - 2$*, fill* q*, and sides minus-to-plus exists.*

Status and reference. The moving-key source-rematerialization audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_source_rematerialization_audit.py`.

It records no exact rematerialized candidate. The narrowest prior depth atom 19:12 gives $q = 111$ and $q = 61$; 111 is composite, while 61 has $61 \equiv 1 \pmod 4$ and hence a nonintegral AffineTwin delay. The nearest available gap source has gap 59, with generator 61, fill 59, and the wrong orientation for a $q = 61$ AffineTwin source. Thus the current-sweep same-orientation moving-key source-rematerialization route is closed. The global obligation remains to exclude orientation changes or persistent source rematerialization, or to route them to the named PDEC/SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.21 (Endpoint-release critical error)**.** *In the current AffineTwin sweep, support motion has no anonymous critical-error channel left. If an empty CRT representative is pulled into the primitive support by endpoint motion, the actual load is the total generator and shifted fill endpoint release. The critical capacity is the current common support width, which is* 20*. All* 11 *support-motion candidates are supercritical: the total endpoint-release load is* 4929 *against total capacity* 220*, and the narrowest atom* 19:12 *already has load* 70*, hence critical error*

$$\frac{70}{20} - 1 = \frac{50}{20} = 2.5.$$

Status and reference. The endpoint-release critical-error audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_critical_error_audit.py`.

It records that every support-motion row requires both endpoints to release, that every endpoint-release load exceeds the support width, and that the fixed primitive-depth defect total matches the endpoint-release load row by row. Together with the moving-key source-rematerialization audit, this eliminates same-orientation source rematerialization as a hidden carrier for the positive local error. Thus, in the current sweep, the positive endpoint-release critical error is routed to `EndpointReleaseCoupling-PDEC`. The global obligation remains to exclude persistent endpoint-release coupling, orientation changes, and source rematerialization, or to route them to named exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.22 (Endpoint-release feedback loss)**.** *Even after the current support-motion rows are given full geometric credit for the larger endpoint movement, a supercritical feedback loss remains. For each row, write*

$$L = \textit{generator release} + \textit{fill release}, \qquad G = \max(\textit{generator release}, \textit{fill release}).$$

The credited support gain is G, so the unrecovered feedback loss is

$$L - G = \min(\textit{generator release}, \textit{fill release}).$$

In the current audit, the total release load is 4929*, the total geometric credit is* 2512*, and the remaining feedback loss is* 2417 *against support-width budget* 220*. The narrowest atom* 19:12 *has load* 70*, credit* 40*, and feedback loss* 30*, still larger than the support width* 20*.*

Status and reference. The endpoint-release feedback-loss audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_loss_audit.py`.

It records that all 11 feedback losses are positive, equal to the smaller synchronized endpoint release, and still exceed the original support width. Thus the current sweep has no self-feedback absorption channel after geometric support gain is fully credited. The remaining positive residual is routed to `EndpointReleaseFeedbackLoss-PDEC`. The global obligation is to exclude that persistent feedback-loss family, or route it to the named moving-key, orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.23 (Endpoint-release feedback horizon). *The endpoint feedback loss has an equivalent phase-horizon form in the current sweep. A support-motion representative can be absorbed by endpoint self-feedback only if its distance from the current support window is at most*

$$\textit{support width} + |\textit{generator side depth} - \textit{fill side depth}|.$$

For all 11 *support-motion rows the representative lies outside this horizon. The narrowest row is* 19:12*: its representative is at distance* 40*, while the support width is* 20 *and the side-depth skew is* 10*, so the feedback horizon is* 30 *and the phase-horizon surplus is* 10*.*

Status and reference. The endpoint-release feedback-horizon audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_horizon_audit.py`.

It proves, for the current sweep, the identities

$$\text{feedback loss} = \text{window distance} - \text{side-depth skew},$$

and

$$\text{post-credit overcritical units} = \text{window distance} - (\text{support width} + \text{side-depth skew}).$$

Thus the residual critical error is exactly the distance by which the CRT representative crosses the feedback horizon. All rows have positive horizon surplus, so the current self-feedback absorption route is closed and the residual is routed to `EndpointReleaseFeedbackHorizon-PDEC`. The global obligation remains to exclude persistent horizon surplus, or to route it to orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.24 (Endpoint-release feedback-horizon slack). *In the same current sweep, the feedback-horizon surplus is exactly the missing side-depth skew needed for absorption. If a representative has distance d from the current support window and the support width is W, then absorption requires total side-depth skew at least*

$$\max(0, d - W).$$

Hence the required extra skew is

$$\max(0, d - W) - |\textit{generator side depth} - \textit{fill side depth}|,$$

which equals the phase-horizon surplus row by row. The total required skew is 2292*, while the current skew supplies only* 95*; the missing extra skew is* 2197*. The narrowest row is again* 19:12*, where distance* 40 *and support width* 20 *require total skew* 20*, but the current skew is* 10*.*

Status and reference. The endpoint-release feedback-horizon slack audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
feedback_horizon_slack_audit.py
```

It verifies that all 11 rows have positive required extra skew, that this extra skew equals the phase-horizon surplus, and that a pure orientation flip does not change the absolute skew. It also inherits from the moving-key source-rematerialization ledger that same-orientation exact rematerialization is absent. Thus the current anonymous slack absorption route is closed and the remaining obstruction is routed to `EndpointSkewGrowth-PDEC/OrientationChangingPrimitiveKey-SAE`. The global row/column theorem remains open until those named exits are excluded or absorbed. □

Reduction-closed Statement 2.25 (Endpoint-release bidirectional skew hull). *The current slack rows also have a simultaneous CRT-hull form. Each formal generator/fill residue pair gives a local alignment condition in the variable P,*

$$P \equiv r_g \pmod{q-2}, \qquad P \equiv r_f - \Delta \pmod{q},$$

where Δ is the affine delay. For the current $q = 31$ atom, the combined modulus is 899 and the primitive support interval is [2669, 2688], of width 20. The nearest representatives of all 12 formal pairs lie in the hull [2304, 3122], of width 819. Thus a single support hull would need left and right extensions 365 and 434; after the feedback horizons are credited, positive two-sided deficits 335 and 409 remain. Moreover

$$899 - 819 = 80,$$

which equals the affine delay.

Status and reference. The bidirectional skew-hull audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
bidirectional_skew_hull_audit.py
```

It imports the inverse-alignment equivalence only as a local CRT-alignment discipline: the global statement that the minimal zero-row alignment x exceeds P is not used as a theorem. The audit proves that the current formal-pair system cannot be absorbed by one-sided skew growth, because below and above representatives both remain outside the credited feedback horizons. The remaining obstruction is routed to `BidirectionalSkewHull-PDEC/ColumnCRT` or to the moving-family multiplicity exits. This is a current-sweep closure, not a final unconditional row/column theorem. □

Reduction-closed Statement 2.26 (Endpoint-release circular aperture). *The bidirectional hull also has to be read on the circle modulo $q(q-2)$, not only in the current linear cut. For the current $q = 31$ atom, the previous interval [2304, 3122] has width 819 in one chosen cut of the modulus 899. Removing the largest circular open gap instead gives the minimal circular alignment arc*

$$[3029, 3586],$$

of width 558. The largest open gap runs from the actual pair 19:8 to the pair 13:9 and has open width 341. The affine delay gap of width 80 is present, but it is only the third largest circular gap, so the previous equality $899 - 819 = 80$ is a linear-cut rigidity, not the complement of the circular minimal arc.

Status and reference. The circular-aperture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
circular_aperture_audit.py
```

It verifies that even after the optimal circular cut, the minimal arc has width 558, whereas the primitive support has width 20. After optimally shifting the support interval to [3568, 3587], the total extension still required is 539; after crediting the most favorable one-sided feedback horizon, a conservative deficit 509 remains. Thus the current single-sided circular-aperture absorption route is closed for this sweep. The remaining obstruction is routed to `CircularAperture-PDEC/ColumnCRT`, `SAE`, or the already named moving-family multiplicity exits. This refinement corrects the linear-hull wording but does not close the global row/column theorem. □

Reduction-closed Statement 2.27 (Endpoint-release anchored circular depth). *The circular aperture can be sharpened by keeping the unique actual packet as an anchor. In the current sweep the minimal circular arc is* $[3029, 3586]$*, and its right endpoint* 3586 *is the actual pair* $19{:}8$ *after one period of the modulus* 899*. Hence an actual-anchored absorption of the whole arc requires the generator phase and the shifted-fill phase to both reach the left endpoint* 3029*. The required common left depth is*

$$D = 3586 - 3029 = 557.$$

The current generator and fill left depths are 18 *and* 28*, so the required endpoint releases are* 539 *and* 529*, for a total of* 1068*. This is* 53.4 *times the primitive support width* 20*.*

Status and reference. The anchored circular-depth audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_circular_depth_audit.py
```

It shows that the previous one-sided circular support extension 539 hides a second endpoint release 529: the intersection support moves only when both underlying phase intervals move. The same-orientation moving-key explanation also fails at this anchored depth. The lower-side AffineTwin depth formulas would require

$$q = 2D - 5 = 1109, \qquad q = D + 3 = 560,$$

so there is no common odd-prime key. Thus the current same-orientation anchored circular-depth absorption route is closed for this sweep. The remaining obstruction is routed to anchored circular-depth PDEC, ColumnCRT, SAE, orientation-changing keys, or the moving-family multiplicity exits; the global row/column theorem remains open. □

Reduction-closed Statement 2.28 (Anchored parity no-go). *For a same-orientation lower-side AffineTwin key to absorb an anchored common left depth* D*, the generator and fill depth formulas must give the same prime parameter:*

$$q_g = 2D - 5, \qquad q_f = D + 3.$$

Thus a common key forces $2D - 5 = D + 3$*, hence* $D = 8$ *and* $q = 11$*. The actual-anchored circular arc in the current sweep instead forces* $D = 557$*. It gives*

$$q_g = 1109, \qquad q_f = 560.$$

The fill-side candidate is an even integer larger than 2*, so it cannot be an odd-prime AffineTwin key.*

Status and reference. The anchored parity no-go audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_parity_nogo_audit.py
```

It records both obstructions: the forced depth is 549 away from the only common-depth solution $D = 8$, and its odd parity makes the fill-side candidate $D + 3$ even. Therefore the current same-orientation anchored parity absorption route is closed for this sweep. The remaining obstruction is a global family version of this no-go, or one of the already named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. The global row/column theorem remains open. □

Reduction-closed Statement 2.29 (Endpoint-release cut-anchor sweep)**.** *The anchored parity obstruction can be checked against every circular cut, not only the optimal one. In the current* $q = 31$ *atom there are* 12 *adjacent CRT cuts on the circle modulo* 899*. The unique actual pair* 19:8 *is retained in every lifted arc: it is the left endpoint for one cut, the right endpoint for one cut, and an interior point for the remaining ten cuts. For each cut we anchor the actual representative in the lifted arc and charge both the generator interval and the shifted-fill interval for the left and right depths needed to cover that arc.*

The smallest arc is still the previous cut $19{:}8 \to 13{:}9$*, with width* 558 *and actual pair at the right endpoint. Its total endpoint release is* $1068 = 53.4 \cdot 20$*, where* 20 *is the primitive support width. Across all cuts, the endpoint release ranges from* 1068 *to* 1737*, so no cut fits inside the support budget.*

Status and reference. The cut-anchor sweep audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_sweep_audit.py
```

It also checks the same-orientation moving-key formulas for every cut. On the left lower side a common key can occur only at $D = 8$; the closest positive left depth in the sweep is $D = 58$. On the right above side the shifted-fill right depth is fixed at $D = 1$; the closest positive right depth in the sweep is $D = 342$. Hence changing the circular cut or moving the actual endpoint does not reopen the same-orientation absorption channel. Endpoint cuts fall into the left anchored parity no-go or the right fixed-fill no-go; interior cuts require both side releases simultaneously. The current sweep is closed, but the global theorem still requires a family version of this cut-anchor no-go, or one of the named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. □

Reduction-closed Statement 2.30 (Cut-anchor ColumnCRT compression)**.** *The circular cut is a gauge choice, not a new actual residue. In the current cut-anchor sweep all* 12 *cuts retain the same actual pair* 19:8*. The actual representative is either* 2687 *or* $3586 = 2687 + 899$*, and in both cases*

$$P \equiv 889 \pmod{899}.$$

Thus the fixed $q = 31$ *actual slot contributes one ColumnCRT atom of mass* $1/899$*, not twelve independent atoms of total mass* $12/899$*. The apparent extra mass*

$$\frac{11}{899}$$

is only cut multiplicity.

Status and reference. The cut-anchor ColumnCRT compression audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_columncrt_compression_audit.py
```

It registers the persistent fixed-residue escape as a `CutAnchorColumnCRT-PDEC` object with phase $P \equiv 889 \pmod{899}$, equivalently $P \equiv 19 \pmod{29}$ and $P \equiv 21 \pmod{31}$ after the fill shift. Therefore, after same-orientation cut-anchor absorption is closed, changing the cut cannot supply additional capacity. If q or the residues move, the case leaves this fixed ledger and returns to the existing AffineTwin moving-family SAE/ColumnCRT ledger. The current sweep closes cut multiplicity as an independent capacity source; it does not yet exclude the global ColumnCRT/PDEC or orientation-changing exits. □

Reduction-closed Statement 2.31 (Actual-anchor replacement). *After cut-anchor compression, keeping the actual pair* 19:8 *gives the single fixed residue*

$$P \equiv 889 \pmod{899}.$$

If the counterexample chain tries to abandon this anchor, then some other formal pair must become actual, or else an unused target residue must arrive. Both alternatives require a CRT phase jump larger than the primitive support width 20*. Among existing formal pairs, the closest replacement is* 19:12*, at CRT distance* 58*, and it requires total endpoint release* 70*. Among unused targets, the closest replacement is* 20:9*, at CRT jump* 59*, and it already needs a new side residue.*

Status and reference. The actual-anchor replacement audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
actual_anchor_replacement_audit.py
```

It imports the support-motion and unused-target ledgers only as audited local interfaces. The first interface covers the 11 unsupported formal pairs: every replacement CRT jump exceeds the support width, every replacement requires both endpoint releases, and the minimum release is $70 = 3.5 \cdot 20$. The second interface covers the unused target residues: every CRT jump again exceeds the support width, and every target requires a new side residue. Thus actual-anchor replacement is closed for the current sweep. The global theorem still requires a family version of this no-go, or exclusion of the named support-motion, unused-target, ColumnCRT/PDEC, and moving-family exits. □

Reduction-closed Statement 2.32 (Near-miss 61/59 phase fracture). *The closest apparent bridge between actual-anchor replacement and moving-key rematerialization is not a bridge in the current sweep. The closest existing formal replacement is still* 19:12*: its CRT jump is* 58*, its required common side depth is* 58*, and its endpoint release is* 70*. The corresponding one-sided moving-key formulas produce* $q = 61$ *and* $q = 111$*. Meanwhile the closest unused-target replacement is* $19{:}12 \to 20{:}9$*, with CRT jump* 59*, and the nearest available gap source to the candidate* $q = 61$ *also has gap* 59*.*

Status and reference. The near-miss phase-fracture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
near_miss_61_59_phase_fracture_audit.py
```

It records that $q = 61$ is prime and $q-2 = 59$ is prime, but $61 \equiv 1 \pmod 4$, so the AffineTwin delay $(11q - 21)/4$ is not integral, and there is no matching gap-61 source. The nearest actual source has signature

$$\text{gap} = 59, \quad \text{generator} = 61, \quad \text{fill} = 59, \quad \text{sides} = \text{minus} \to \text{minus}, \quad p \text{ delay} = 70.$$

This differs from the $q = 61$ AffineTwin source signature

$$\text{gap} = 61, \quad \text{generator} = 59, \quad \text{fill} = 61, \quad \text{sides} = \text{minus} \to \text{plus}.$$

The other moving-key candidate, $q = 111$, is composite. Thus the integer 59 appears in two nearby roles but not in the same phase role: as the nearest available gap source to $q = 61$, and as the unused-target CRT jump. The latter still exceeds the support width 20 and introduces a new side residue. Hence the current-sweep 61/59 near miss cannot carry hidden actual load. The global theorem still requires a family version of this no-go, or routing to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.33 (Phase-scale bridge exhaustion). *In the current sweep there is no other exact phase-scale bridge among the moving-key candidates, the available gap sources, and the unused-target CRT jumps. Across all one-sided moving-key candidates, no candidate q is equal to an available gap-source scale, and no candidate q is equal to an unused-target jump scale. The only exact offset bridge of the form $q - 2 =$ gap source $=$ unused-target jump is the already fractured case $q = 61$, $q - 2 = 59$.*

Status and reference. The phase-scale bridge exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
phase_scale_bridge_exhaustion_audit.py
```

It compares the 19 moving-key candidates

$$61, 65, 96, 111, 119, 123, 154, 181, 235, 293, 297, 355, 386, 575, 699, 761, 1375, 1499, 1747$$

with the available source scales

$$31,\ 43,\ 59$$

and the unused-target jump scales

$$59,\ 60,\ 90,\ 150,\ 281,\ 343,\ 345,\ 374,\ 375.$$

The audit records

$$\{q : q = \text{gap source}\} = \varnothing, \qquad \{q : q = \text{unused jump}\} = \varnothing,$$

and

$$\{q : q - 2 = \text{gap source} = \text{unused jump}\} = \{61\}.$$

The previous near-miss audit already closes this unique offset bridge: the gap-59 source has the wrong role and orientation for a $q = 61$ AffineTwin source, and the unused-target jump 59 still introduces a new side residue. Therefore the current sweep has no anonymous exact or offset scale bridge. The global theorem still requires a family version of this exhaustion, or a route to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.34 (Support-width near-scale fracture). *Even after exact phase-scale bridges are exhausted, the current sweep has no support-width near-scale bridge. With support width 20, the only moving-key candidates that are simultaneously within distance 20 of an available source scale and an unused-target jump scale are $q = 61$ and $q = 65$. The first is the already fractured prime-but-not-AffineTwin case; the second is composite.*

Status and reference. The support-width near-scale audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_width_nearscale_fracture_audit.py
```

It enumerates all events satisfying

$$|s - q| \le 20, \quad |s - (q - 2)| \le 20, \quad |j - q| \le 20, \quad |j - (q - 2)| \le 20,$$

where s is a current gap-source scale and j is an unused-target CRT jump. It records

$$\{q : \text{near source}\} = \{61, 65\},$$

$$\{q : \text{near unused jump}\} = \{61, 65, 96, 111, 154, 293, 297, 355, 386\},$$

and

$$\{q : \text{near both source and unused jump}\} = \{61, 65\}.$$

There is no viable near-scale bridge. For $q = 61$, the AffineTwin source gate fails by the same delay, congruence, and missing-source invariants as in the near-miss audit. For $q = 65$, q is composite. The remaining near-jump candidates have no near source and cannot form a three-way phase bridge. Thus support-width proximity cannot replace exact CRT/source signatures in the current sweep. A global proof must still show the corresponding family near-scale no-go, or route failures to source-rematerialization, unused-target, ColumnCRT/PDEC, or moving-family exits. □

Reduction-closed Statement 2.35 (Orphan near-jump source deficit). *The remaining support-width near-scale candidates that are near an unused-target jump but not near a source scale are source-deficient in the current sweep. The orphan near-jump candidates are*

$$96,\ 111,\ 154,\ 293,\ 297,\ 355,\ 386.$$

For all of them, the nearest source scale is farther than the support width 20*. The smallest source distance is* 35*, attained at* $q = 96$*, leaving a positive source deficit* $35 - 20 = 15$*.*

Status and reference. The orphan near-jump source-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
orphan_nearjump_source_deficit_audit.py
```

It records that the seven orphan near-jump candidates split as six composite candidates and one prime-but-not-AffineTwin candidate. Every corresponding unused-target near-jump event introduces a new side residue, and no moving source is materialized. Thus target-side proximity does not create a three-way bridge: the source side remains outside the support-width tolerance. Any attempt to repair the orphan near-jump would have to move or create a source family, which is precisely the source-rematerialization or moving-family exit, not an anonymous primitive-support absorption. A global proof still requires the corresponding family no-go or routing to named PDEC/SAE exits. □

Reduction-closed Statement 2.36 (Near-jump carrier exhaustion). *All near-jump carriers are exhausted in the current sweep. The carrier set is*

$$61,\ 65,\ 96,\ 111,\ 154,\ 293,\ 297,\ 355,\ 386.$$

It splits into two disjoint parts:

$$\{61, 65\}$$

near both a source scale and a target jump, and

$$\{96, 111, 154, 293, 297, 355, 386\}$$

near only a target jump. The first part fails the source gate; the second part has source distance greater than the support width. In every case the target-side near-jump event requires a new side residue.

Status and reference. The near-jump carrier exhaustion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
nearjump_carrier_exhaustion_audit.py
```

It records the source-status histogram

$$\{\text{near-source gate fracture} : 2, \quad \text{orphan source deficit} : 7\}$$

and the moving-route histogram

$$\{\text{CompositeQ} : 7, \quad \text{PrimeButNotTwinAffine} : 2\}.$$

The nearest orphan source distance is 35, still larger than the support width 20, and all carrier jump events need new side residues. Hence a near-target phase cannot by itself carry actual load. Repairing it forces one of the named exits: source rematerialization, unused-target residue arrival, ColumnCRT/PDEC, or moving-family migration. This closes the current-sweep near-jump carrier but not the global row/column theorem. □

Reduction-closed Statement 2.37 (Carrier-arrival routing)**.** *Every target-side near-jump event from the exhausted carrier set routes to an already registered unused-target arrival atom. There are* 27 *such events, using all* 9 *arrival atoms and* 5 *target pairs. The missing-match count is* 0*. The event-level new-side-residue demand is* 50*, compressed to generator residues*

$$10, \ 16, \ 17, \ 18, \ 20$$

and fill residues

$$5, \ 6, \ 7, \ 30.$$

The only exact zero-phase event is $q = 61$*,* $q - 2 = 59$*, jump* 59*; it still requires the new generator residue* 20*.*

Status and reference. The carrier-arrival routing audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_routing_audit.py
```

It matches each carrier event against the unused-target arrival ledger by source pair, target pair, and CRT jump. All matches are present, and every matched target is outside both the current formal product and the current actual support. Therefore near-target proximity is only a repeated projection of the unused-target arrival packet, not a new actual-load carrier. A persistent version must prove a global new-residue arrival bound or route to new-generator, new-fill, ColumnCRT/PDEC, or moving-family exits. This closes the current-sweep anonymous target-side carrier but not the global row/column theorem. □

Reduction-closed Statement 2.38 (Arrival pressure product)**.** *If the carrier-arrival packet is materialized as side residues, then the current sweep crosses the square-root pressure gate. The base side counts are*

$$(A_g, A_f) = (3, 4), \qquad A_g A_f = 12 < \sqrt{29 \cdot 31}.$$

The carrier packet adds generator residues

$$10, \ 16, \ 17, \ 18, \ 20$$

and fill residues

$$5, \ 6, \ 7, \ 30.$$

Thus full materialization gives side counts $(8, 8)$ *and*

$$(8 \cdot 8)^2 - 29 \cdot 31 = 3197 > 0.$$

Moreover the minimal crossing is exact to one unit: any two two-sided target atoms give counts $(5,6)$ *and*

$$(5 \cdot 6)^2 - 29 \cdot 31 = 1.$$

Status and reference. The carrier-arrival pressure-product audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_pressure_product_audit.py
```

It starts from the already routed carrier-arrival packet and updates the formal side-residue counts exactly. Since the updated product is above the square-root gate, the packet cannot remain an anonymous unused-target arrival. It is a named `SuperSqrt/PressureProduct-PDEC` atom. Therefore, after excluding anonymous target-side carriers, the remaining global obligation is to exclude this PDEC family or keep it as the explicit named exit; the row/column theorem is not yet unconditionally closed. □

Reduction-closed Statement 2.39 (Arrival projection deficit)**.** *The preceding square-root crossing is not an actual overload in the current sweep. Project the formal side product back to the common support window. For each minimal crossing row the formal product is* 30*, while the actual projection consists of exactly three points:*

$$\{\textit{actual anchor } 19{:}8\} \cup \{\textit{the two selected target atoms}\}.$$

Thus the projection deficit is 27*, and the actual square-root slack is* $29-3=26$*. For the full carrier packet the formal product is* 64*, but the actual projection has only* 6 *hits, with deficit* 58 *and slack* 23*.*

Status and reference. The carrier-arrival projection-deficit audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
carrier_arrival_projection_deficit_audit.py
```

It enumerates the side-residue Cartesian product for each minimal crossing and intersects it with the 20 residue pairs actually present in the common support window. The excess $30 > 29$ is therefore a formal-product excess, not an actual-load excess. Any global continuation must either tighten product accounting to actual projection, or produce a named projection-collision or support-escape PDEC. The current sweep has no anonymous actual overload. □

Reduction-closed Statement 2.40 (Support-graph cap)**.** *In the current fixed AffineTwin slot the common support is a graph, not a two-dimensional product set. The target window consists of the* 20 *pairs*

$$(1,21),(2,22),\ldots,(10,30),(11,0),\ldots,(20,9),$$

so each generator residue has at most one fill partner and conversely. Hence any actual projection from side sets is bounded by the graph size 20*. Since*

$$20 \le \lfloor \sqrt{29 \cdot 31} \rfloor = 29,$$

the fixed slot cannot produce an actual square-root overload.

Status and reference. The support-graph cap audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
support_graph_cap_audit.py
```

It verifies that the common window is functional in both directions and has constant fill offset modulo 31. The general fixed-slot inequality is also recorded: for an AffineTwin modulus $q \geq 13$, the support width $W = (q+9)/2$ satisfies $W^2 \leq q(q-2)$, equivalently $3q^2 - 26q - 81 \geq 0$. Therefore the current ProductAccounting problem is closed for fixed slots. Any remaining escape must move or change the support graph, which is the named primitive-slot support-escape exit, not an anonymous load surplus. □

Reduction-closed Statement 2.41 (Moving-slot graph-cap route)**.** *In the current endpoint-release sweep there is no anonymous moving-slot actual overload. A fixed or same-orientation AffineTwin slot remains bounded by the graph cap; the candidate fixed-family moduli are* $q = 31, 43, 103$*, and each satisfies* $W^2 \leq q(q-2)$*. If the support is truly moved while keeping the current primitive identity, the narrowest atom already requires endpoint release* 70 *against support width* 20*.*

Status and reference. The moving-slot route audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_slot_graph_cap_route_audit.py
```

It records that all eleven support-motion candidates require both endpoint releases and that the fixed primitive depth identities break on both sides. Allowing a same-orientation moving key does not repair this: at the narrowest atom the depth formulas give $q_g = 111$ and $q_f = 61$. Taking all one-sided candidate q values through the source-rematerialization gate gives no exact rematerialized source. Thus the present sweep routes every moving-slot escape to a named exit: orientation-changing primitive key, source rematerialization, ColumnCRT/PDEC, SAE, or unused-target arrival control. This is a reduction of the remaining interface, not a final unconditional closure of the row/column theorem. □

Reduction-closed Statement 2.42 (Moving-family persistence pressure)**.** *The current moving-family escape is not an anonymous capacity reservoir. Among the candidate moduli* $q = 31, 43, 103$*, only* $q = 31$ *is source-materialized. The candidate* $q = 43$ *has a same-gap source but with the wrong orientation and delay, and* $q = 103$ *has no gap source. The non-realized candidates have one-sided generator pressure, but their paired pressure products remain below the square-root gate.*

Status and reference. The moving-family persistence pressure audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
moving_family_persistence_pressure_audit.py
```

It combines the source-materialization gate, epoch-pair sparse ledger, paired pressure ledger, threshold route classifier, and fill catchup budget. The source gate blocks 28 formal pairs: 16 from $q = 43$ and 12 from $q = 103$. All minimal threshold-crossing routes require fill-side residue arrival; if all three candidates crossed, the minimum fill-arrival demand would be 11 new fill residues, with Rankin mass 23339/137299. Repeated fill arrival is routed to reset or ColumnCRT-PDEC. Hence the present sweep routes moving-family persistence to named exits: fill-arrival control, high-density epoch-pair PDEC/ColumnCRT, pressure-product PDEC, or source rematerialization. This is still a reduction boundary, not a final global proof. □

Reduction-closed Statement 2.43 (Fill-arrival projection gate)**.** *In the current sweep, fill-side residue arrival alone cannot create an actual overload. The only source-materialized candidate is* $q = 31$*, and its minimal threshold crossing is not fill-only: it requires at least two new generator residues. When such co-arrivals are counted formally, the minimal side product is* 30*, but its actual projection has only* 3 *hits and remains below* $\lfloor\sqrt{29 \cdot 31}\rfloor$*.*

Status and reference. The fill-arrival projection gate audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
fill_arrival_projection_gate_audit.py
```

For $q = 31$, every fill-arrival threshold crossing requires generator co-arrival and then returns to the support-graph projection ledger. The unused-target ledger shows that the needed target residues also require new generator residues and have CRT jumps beyond the current support width. For $q = 43$ and $q = 103$, formal fill-only routes exist, but they are blocked before projection: $q = 43$ has the wrong source signature and $q = 103$ has no gap source. Thus the present sweep routes fill-arrival persistence to generator co-arrival, ProductAccounting tightening, source rematerialization, or ColumnCRT/PDEC. This remains a reduction boundary rather than a global exclusion theorem. □

Reduction-closed Statement 2.44 (Generator-coarrival projection accounting)**.** *In the current sweep, forcing fill arrival to co-arrive with generator residues still does not create an actual overload. For the realized candidate $q = 31$, all 31 nonempty subsets of the five unused-target atoms have been enumerated. There are 22 formally super-square-root subsets, but none projects above $\lfloor\sqrt{29 \cdot 31}\rfloor$.*

Status and reference. The generator-coarrival projection-accounting audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_projection_accounting_audit.py
```

The base packet has formal side product $3 \cdot 4 = 12$ and one actual projection hit, the anchor $19 : 8$. The minimal coarrival crossing has formal product 30, but its actual projection has only 3 hits and slack 26 below $\lfloor\sqrt{29 \cdot 31}\rfloor = 29$. The full carrier packet has formal product 64, but only 6 actual hits and slack 23. Moreover every projection hit is exactly the anchor plus the selected target atoms; no hidden projection collision is being counted as free capacity. Thus the present sweep routes generator coarrival to ProductAccounting tightening, source rematerialization, ColumnCRT/PDEC, or moving-family persistence. This remains a local closed audit and not a global unconditional row/column proof. □

Reduction-closed Statement 2.45 (Coarrival schema)**.** *For every fixed AffineTwin support graph with $q \geq 13$, generator coarrival cannot by itself create an actual overload. The graph width is $W = (q + 9)/2$, and*

$$W^2 \leq q(q - 2) \quad\Longleftrightarrow\quad 3q^2 - 26q - 81 \geq 0.$$

The right hand side is already positive at $q = 13$ and is increasing for $q \geq 13$.

Status and reference. The family-schema audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
generator_coarrival_family_schema_audit.py
```

The current candidate moduli $q = 31, 43, 103$ all pass this fixed-graph inequality. Hence the only way for coarrival to become an actual overload is to leave the fixed functional support graph. The present sweep routes those escapes to named ledgers: support motion requires endpoint release 70 against support width 20, source materialization blocks 28 formal pairs, the fixed actual anchor compresses to the single ColumnCRT atom $P \equiv 889 \pmod{899}$, and moving-family persistence remains a named pressure/source/fill exit. This closes the current-sweep schema, but the global row/column theorem still needs the schema promoted to all persistent AffineTwin families and the named exits excluded or absorbed. □

Reduction-closed Statement 2.46 (Persistent-family promotion)**.** *The current sweep promotes the coarrival schema through the active persistent AffineTwin family ledger. Fixed support graphs are bounded by the graph-width inequality; fixed q and fixed residue recurrence is routed to ColumnCRT/PDEC; moving q or moving residues are routed to epoch-pair SAE unless they become high-density.*

Status and reference. The promotion audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
persistent_family_promotion_audit.py
```

In the current ledger the candidate moduli are $q = 31, 43, 103$, with realized modulus 31. Their epoch-pair occupancy upper sum is 0.023577117628562343, below the threshold $\eta = 0.025$, and there is no high-density epoch pair. The fixed-slot recurrence count is 0; the 12 fixed-residue recurrences all drift in slot and are therefore the next ColumnCRT/PDEC input. Source materialization blocks 28 formal pairs, paired pressure does not cross, and fill arrival is routed to new-fill or reset PDEC. Thus the current sweep has no anonymous persistent-family capacity source. The remaining global obligations are an epoch-pair multiplicity bound and exclusion of fixed-residue slot drift by ColumnCRT/PDEC or SAE/Rankin. □

Reduction-closed Statement 2.47 (Transport-frontier integration)**.** *In the current AffineTwin endpoint-release sweep, fixed-residue slot drift is not an anonymous ColumnCRT capacity source. The* 12 *slot-drift pairs match* 12 *distinct transport cells, with no transport-cell recurrence and no exact phase translate. Chained recurrence has finite forward lifetime; non-chained recurrence must repeat the full transport-cell key, and the current reset-PDEC atom count is* 0*.*

Status and reference. The transport-frontier integration audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
transport_frontier_integration_audit.py
```

It imports the persistent-family promotion ledger and the fixed-residue transport-cell, iteration-drift, reset-PDEC, and persistence-frontier partition ledgers. The aggregate values are: 12 fixed-residue slot-drift pairs, 12 transport cells, 12 unique transport cells, 0 transport-cell recurrences, 0 exact phase translates, maximum forward transition count 8, and reset-PDEC atom count 0. The frontier partition has 324 physical records, split as 24 transport-cell records and 300 singleton-residue records, with no unclassified record. Thus the present sweep routes the fixed-residue drift branch to TransportResetPDECExclusion or SingletonResidue SAE/Rankin, while the global row/column theorem still requires those exits and the epoch-pair multiplicity bound to be proved globally. □

Reduction-closed Statement 2.48 (Remaining-frontier bridge)**.** *After transport-frontier integration, the computable current-sweep remainder bridges to three narrower ledgers. Transport reset has no current atom; singleton residue SAE is localized to the active prime-band endpoint problem; and AffineTwin epoch-pair multiplicity is currently below the $\eta = 1/40$ sparse gate.*

Status and reference. The bridge audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
remaining_frontier_bridge_audit.py
```

It reads seven ledgers: transport integration, singleton profile, Rankin budget, one slot capacity, active source, active band, and epoch-pair multiplicity. The current singleton profile has 300 singleton packets and Rankin mass about 4.328474446, with one slot mass about 4.291749691 and two slot share about 0.008484457. The one slot route has 40 active epochs, maximum occupancy ratio about 0.309859155, and minimum spare ratio about 0.690140845.

The active ℓ source is exactly the prime band $23, \ldots, 109$, with core $29, \ldots, 107$ and minus-only endpoints $23, 109$. The AffineTwin candidate epoch-pair product-mass upper sum is $0.023577118 < 0.025$, with no high-density row.

Thus the current remainder is sharpened to endpoint growth or reset PDEC, transport reset PDEC exclusion, global epoch-pair multiplicity, and moving residue SAE/Rankin. These remain global obligations, not a completed unconditional proof. □

Reduction-closed Statement 2.49 (Endpoint motion stencil). *In the current H-lower singleton route, endpoint motion of the active prime band has only named exits. The outward neighbor primes* 19 *and* 113 *have no current record; the endpoints* 23 *and* 109 *are minus-side singleton atoms; and the inward core edges* 29 *and* 107 *already have both-side support.*

Status and reference. The endpoint-motion stencil audit is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_stencil_audit.py
```

The six-point stencil is

$$19 \to 23 \to 29, \qquad 107 \to 109 \to 113.$$

The outer entries $19, 113$ are empty in the current sweep. The endpoint entries $23, 109$ each have one minus-side record, one residue, and one slot. The inward entries $29, 107$ have both-side core support, with 4 and 6 records respectively. Hence current endpoint motion cannot be counted as anonymous capacity: outward motion requires a new endpoint arrival, while inward motion is absorbed into the core. The global obligations are endpoint outward arrival control, endpoint atom PDEC/SAE exclusion, and core-edge multiplicity control. □

Reduction-closed Statement 2.50 (Endpoint motion gap). *In the current H-lower singleton route, endpoint expansion of the active prime band creates only three temporary internal prime gaps. The missing prime moduli are* $31, 43, 59$, *and all are filled within the current sweep; the largest observed fill delay is* 80 *in the* P *parameter.*

Status and reference. The endpoint-motion gap router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_endpoint_motion_gap_router.py
```

It orders one-slot ℓ values by their first activation. The three non-interval snapshots are: activation of 47 at $P = 2063$ leaves 43 missing until delay 74; activation of 29 at $P = 2687$ leaves 31 missing until delay 80; activation of 61 at $P = 3187$ leaves 59 missing until delay 70. Every 1000-level prefix snapshot from 3000 to 10000 is already a complete prime interval, and the final active band is $23, \ldots, 109$. Thus the current sweep routes endpoint motion to a gap-fill bound or to a Gap-PDEC/SAE if such a gap persists globally. □

Reduction-closed Statement 2.51 (Gap-fill pair and corridor). *In the current H-lower singleton route, every endpoint-motion gap is repaired by the next first activation in* ℓ*-activation order, and every repair lies inside the short corridor*

$$P_{\text{fill}} - P_{\text{gen}} \le 3\ell_{\text{gap}}.$$

The three gap-fill keys are distinct in the current sweep.

Status and reference. The gap-fill pair and repair-corridor routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_fill_pair_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_gap_repair_corridor_router.py.
```

The three repairs are $47 \to 43$ with delay 74, $29 \to 31$ with delay 80, and $61 \to 59$ with delay 70. Each has activation-rank delay 1, and each restores the exact prime interval after the fill. The corresponding full GapFillPair keys are pairwise distinct. The corridor ratios are bounded by 3: the tight row is $\ell_{\text{gap}} = 31$, where $80 \leq 3 \cdot 31$ with margin 13; a $2\ell_{\text{gap}}$ corridor is already false there. Thus the global remaining obligation is a short gap-repair corridor bound, or exclusion of a Corridor-PDEC/SAE when that bound fails. □

Reduction-closed Statement 2.52 (Corridor phase budget)**.** *In the current H-lower singleton route, the short repair corridor decomposes into the exact three-term identity*

$$P_{\text{fill}} - P_{\text{gen}} = d_{\text{gen,right}} + g_{\text{bridge}} + d_{\text{fill,left}}.$$

All three current gap repairs satisfy this identity and keep positive $3\ell_{\text{gap}}$ slack. The unique component excess is the phase bridge for $\ell_{\text{gap}} = 31$, but that excess is absorbed by the neighboring depth spares.

Status and reference. The corridor phase-budget router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_corridor_phase_budget_router.py.
```

The three identities are

ℓ_{gap}	$d_{\text{gen,right}}$	g_{bridge}	$d_{\text{fill,left}}$	$P_{\text{fill}} - P_{\text{gen}}$
43	12	40	22	74
31	6	46	28	80
59	31	31	8	70.

The minimum remaining $3\ell_{\text{gap}}$ slack is 13. The only component excess is $46 - 31 = 15$ in the $\ell_{\text{gap}} = 31$ phase bridge; the other two components provide total spare 28, leaving the same positive corridor slack 13. Hence the current short-corridor obstruction has been sharpened to a phase-budget obstruction: a global proof must bound the bridge component by neighboring depth spare, or route any persistent failure to a PhaseBridge-PDEC or SAE exit. □

Reduction-closed Statement 2.53 (Phase-bridge excess return to AffineTwin)**.** *The unique current phase-bridge excess is not a new anonymous capacity source. It collapses to the already registered AffineTwin fixed slot atom with*

$$q = 31, \qquad q - 2 = 29.$$

The excess is 15, the absorbing spare is 28, and the post-absorption slack is 13. The same atom satisfies the primitive affine identities and the coprime CRT lock

$$P \equiv 19 \pmod{29}, \qquad P \equiv 21 \pmod{31}.$$

Hence the combined modulus $29 \cdot 31 = 899$ exceeds the common support width 20 and isolates the fixed representative.

Status and reference. This return chain is generated by the phase-bridge excess atom/source routers, the margin-slot normal-form and primitive-identity routers, the primitive affine-collapse router, and the AffineTwin slot-phase-lock router:

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_atom_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_phase_bridge_excess_source_router.py

experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_margin_slot_primitive_identity_router.py
```

and the downstream AffineTwin phase-lock and moving-family routers. They record

$$\text{excess} = 15 = \text{generator margin} = 3\rho_{\text{jump}}, \qquad \text{slack}_{\text{after}} = |\Delta b| = 13,$$

and the affine primitive identities

$$\begin{gathered} \ell_{\text{fill}} = \ell_{\text{gap}} = 31, \quad \ell_{\text{gen}} = \ell_{\text{gap}} - 2 = 29, \\ m_{\text{gen}} = (\ell_{\text{gap}} - 1)/2 = 15, \quad |\Delta b| = (\ell_{\text{gap}} - 5)/2 = 13, \quad |\Delta u| = (\ell_{\text{gap}} - 3)/4 = 7. \end{gathered}$$

Thus the bridge-excess atom is exactly the $q = 31$ AffineTwin atom. Since 29 and 31 are coprime, the two slot congruences have modulus 899; the pair-support width is only 20, so the fixed slot contributes a single representative, already recorded as the fixed actual-anchor ColumnCRT atom. If the same q and residues recur, recurrence is a fixed-modulus ColumnCRT/PDEC object. If q or the residues move, the packet returns to the AffineTwin epoch-pair SAE/Rankin and moving-family exits. This closes the new current-sweep anonymous phase-bridge outlet, but the global row/column theorem still requires the moving-family multiplicity, endpoint-growth/reset, transport-reset, and moving-residue exits to be bounded or excluded. □

Reduction-closed Statement 2.54 (AffineTwin sparse SAE envelope)**.** *After the phase-bridge outlet returns to the AffineTwin moving-family ledger, the current candidate epoch-pairs pass the* $\eta = 1/40$ *sparse gate. The candidate moduli are*

$$q = 31, 43, 103,$$

the current product-mass upper sum is approximately

$$0.023577117629 < 0.025,$$

and there is no high-density epoch-pair row in the current sweep. Moreover, the single fixed-slot SAE mass has the telescoping envelope

$$\frac{1}{q(q-2)} = \frac{1}{2}\left(\frac{1}{q-2} - \frac{1}{q}\right),$$

so the odd tail from $q \geq Q$ *is at most* $1/(2(Q-2))$.

Status and reference. The epoch-pair multiplicity and sparse-SAE envelope routers are generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_epoch_pair_
multiplicity_router.py
```

and

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sparse_sae_
global_envelope_router.py.
```

The current realized atom is the $q = 31$ fixed slot with mass $1/899$, while the single-atom tail from $q \geq 31$ is bounded by

$$\frac{1}{2(31-2)} = \frac{1}{58}.$$

This is a genuine global summability statement for one fixed AffineTwin atom per modulus. It does not, however, close the moving-family problem by itself: the η sparse gate alone would still permit $O(\eta q(q-2))$ atoms at a fixed q, and each such atom has mass $1/(q(q-2))$. A saturated sparse row could therefore contribute about η for each q, which is not summable over infinitely many moduli. Hence the latest narrow obstruction is not the single-atom SAE mass; it is the per-q multiplicity bound. A global proof must show $O(1)$, decay, or another summable bound for realized AffineTwin atoms per q, or route excess multiplicity to a HighDensityEpochPair PDEC/ColumnCRT exit. □

Reduction-closed Statement 2.55 (AffineTwin square-root product gate)**.** *The per-q multiplicity target can be weakened to a square-root product gate. Let M_q be the candidate two-side residue product upper bound for one AffineTwin modulus q. If*

$$M_q^2 \leq q(q-2),$$

then the corresponding SAE contribution is bounded by

$$\frac{M_q}{q(q-2)} \leq \frac{1}{\sqrt{q(q-2)}} \leq \frac{1}{q-2}.$$

In the current sweep the candidate moduli $31, 43, 103$ *all pass this gate. The largest product-to-capacity ratio is about* 0.400222408*.*

Status and reference. The square-root product and Brun-conditional router is generated by

```
experiments/prime_matrix_square_phase_offband_prefix_
gap_shadow_selector_h_lower_affine_twin_sqrt_product_
brun_router.py.
```

It records

q	M_q	$q(q-2)$	$M_q^2 \leq q(q-2)$
31	12	899	yes
43	16	1763	yes
103	12	10403	yes.

The minimum squared slack is 755. Thus the current finite sweep has no SuperSqrtEpochPair row. If the square-root product gate is proved globally, then accepting the classical Brun convergence of reciprocal twin-prime sums would make the AffineTwin SAE tail summable, since AffineTwin moduli satisfy that both q and $q-2$ are prime. This is recorded only as an external-input lane: the author-side self-contained route has not accepted Brun as an inline proof and has not proved the global square-root product bound. Therefore the latest self-contained obstruction is sharper: prove $M_q^2 \leq q(q-2)$ for every persistent AffineTwin epoch-pair, or route every violation to a SuperSqrtEpochPair PDEC/ColumnCRT exit. □

Reduction-closed Statement 2.56 (Return to actual-packet support)**.** *The square-root product hard point is an actual-packet question, not merely a formal product question. Let $M_q^{\mathrm{form}} = A_g A_f$ be the side-residue Cartesian product upper bound, and let $N_q = |\Pi_q|$ be the number of actual non-PDEC packets passing the two slots, CRT compatibility, common phase support, and non-reuse gates. If every such packet lies in the primitive AffineTwin two-slot support, then*

$$N_q \le \frac{q+9}{2} \le \sqrt{q(q-2)} \qquad (q \ge 13).$$

Status and reference. The reduction is recorded in

```
docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md
```

and is compatible with the pressure-packet carrying ceiling file. The primitive support width is $W_q = (q+9)/2$. For $q \ge 13$,

$$W_q \le \sqrt{q(q-2)} \quad \Longleftrightarrow \quad 3q^2 - 26q - 81 \ge 0,$$

and the right side is positive at $q = 13$ and then increasing. A non-PDEC packet has a primitive phase projection into the common support set Ω_q. If two distinct actual packets had the same primitive projection, the recurrence would be a repeated-residue/reset event or a fixed projection ColumnCRT/PDEC event. After those named exits are removed, the projection is injective:

$$\Pi_q \hookrightarrow \Omega_q, \qquad N_q \le |\Omega_q| \le W_q.$$

Thus a formal SuperSqrt row has only three interpretations: product accounting used M_q^{form} where N_q is required; an actual projection collision gives a PDEC/ColumnCRT certificate; or an actual packet escaped the primitive two-slot support. The latest narrow task is therefore PrimitiveTwinSlotSupportExhaustion, together with replacing remaining SAE and Rankin ledgers by the actual-packet count N_q where they still use the formal product upper bound. □

Reduction-closed Statement 2.57 (Current actual-packet contract)**.** *In the current H-lower AffineTwin sweep, the formal product upper count is*

$$\sum_q M_q^{\mathrm{form}} = 40,$$

but the actual non-PDEC packet count is only

$$\sum_q N_q = 1.$$

The single actual packet occurs at $q = 31$; the candidate rows $q = 43$ and $q = 103$ have no current actual packet.

Status and reference. The actual-packet contract is generated by

```
experiments/prime_matrix_affine_twin_actual_packet_contract.py
```

and archived at

```
docs/monograph/prime_matrix_affine_twin_actual_packet_contract.md.
```

The current table is

q	M_q^{form}	N_q	W_q	$q(q-2)$
31	12	1	20	899
43	16	0	26	1763
103	12	0	56	10403.

Every current actual packet is in primitive support and passes the actual square-root gate $N_q \leq W_q \leq \sqrt{q(q-2)}$. The formal-to-actual gap is 39, and there is no current projection-collision PDEC row. Therefore the current SuperSqrt pressure is product-accounting looseness, not actual load. The global obligations remain: prove the product-accounting tightening for all persistent AffineTwin families, prove primitive twin-slot support exhaustion, or route support escapes to their named PDEC/SAE exits. □

Reduction-closed Statement 2.58 (Current formal-pair pruning)**.** *In the current sweep, the whole formal-to-actual gap*

$$40 - 1 = 39$$

is accounted for by named gates:

$$39 = 11 + 28,$$

where 11 *formal pairs are CRT-window empty and* 28 *formal pairs fail source materialization. There is no unresolved current formal pair.*

Status and reference. The pruning certificate is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py`

and archived at

`docs/monograph/prime_matrix_affine_twin_formal_pair_pruning_audit.md.`

The current rows are

q	M_q^{form}	N_q	CRT empty	source unmaterialized
31	12	1	11	0
43	16	0	0	16
103	12	0	0	12.

For $q = 31$, the only supported pair is the residue pair $(19, 8)$; after the fill shift it gives the CRT class 889 mod 899, represented in the support window $[2669, 2688]$ by 2687. The other eleven formal pairs have no support-window representative. For $q = 43$, the same gap exists only with the wrong generator/orientation/delay data; for $q = 103$, there is no current gap-fill source. Hence the present product-accounting gap is not hidden actual load. To globalize this step one must prove that every persistent AffineTwin formal pair either materializes as an actual primitive-supported packet, is CRT-window empty, fails source materialization as a named PDEC/SAE, or is explicitly routed as a primitive support escape. □

Reduction-closed Statement 2.59 (Current formal-to-actual cutset)**.** *In the current sweep, the formal-to-actual pipeline has no remaining anonymous capacity source. The formal product universe has* 40 *pairs, the actual packet count is* 1*, and the gap*

$$39 = 11 + 28$$

is exactly the sum of CRT-window-empty pairs and source-unmaterialized pairs. The active CRT modulus is 899*, the support width is* 20*, and the minimum empty-window distance is* 40*.*

Status and reference. The cutset router is generated by

```
experiments/prime_matrix_formal_to_actual_global_cutset_router.py
```

and archived at

```
docs/monograph/prime_matrix_formal_to_actual_global_cutset_router.md.
```

It imports the actual-packet, formal-pruning, source-gate, CRT-window, moving-slot, remaining-frontier, and endpoint-atom ledgers. The current support escape branch is already routed: the fixed support width is 20, the square-root floor is 29, exact rematerialized moving q-values are empty, transport-reset atoms are zero, the active singleton band is $23, \dots, 109$, and the endpoint band has two minus-only atoms. Thus a current counterexample cannot keep the formal product as hidden actual load. The global obligation is to promote this cutset to all persistent families and to exclude or summably absorb the named exits: source failure, CRT-window failure, primitive-support escape, endpoint/reset, transport reset, epoch-pair multiplicity, and moving-residue SAE/Rankin. This is still a reduction boundary, not a global unconditional row/column proof. □

Reduction-closed Statement 2.60 (Cutset completeness)**.** *The formal-to-actual cutset has a deterministic completeness law. For a fixed AffineTwin modulus $q \geq 13$, let*

$$F_q = G_q \times H_q$$

be the formal generator/fill residue product. Then F_q is the disjoint union of source failures, CRT-window failures, primitive actual packets, and named primitive-support or projection escapes:

$$F_q = S_q \sqcup C_q \sqcup P_q \sqcup E_q.$$

Consequently

$$M_q^{\text{form}} = |S_q| + |C_q| + |P_q| + |E_q|, \qquad N_q = |P_q|.$$

Status and reference. The deterministic lemma is archived at

```
docs/monograph/prime_matrix_formal_to_actual_cutset_
completeness_lemma.md.
```

For a formal pair $a \in F_q$, first test whether the matching gap-fill source exists with the required gap, generator, fill, orientation, delay, and slot-lock key. If not, $a \in S_q$. If the source exists, combine the generator and shifted-fill congruences modulo $q(q-2)$. If no representative lies in the primitive pair support, $a \in C_q$. If the representative exists and the primitive depth identities, non-reuse, and non-ColumnCRT conditions all hold, then $a \in P_q$. The remaining cases are precisely source-and-CRT hits that fail the primitive actual conditions; they form E_q, routed to ProjectionCollision/ColumnCRT/PDEC or PrimitiveTwinSlotSupportEscape-PDEC/SAE. The four alternatives are selected by the first failed gate, hence are disjoint and exhaustive. Thus formal overcount is never hidden actual load. What remains open is not classification, but global exclusion or summable absorption of the named exits. □

Reduction-closed Statement 2.61 (After-cutset named-exit frontier)**.** *After the deterministic cutset split, the current sweep has no anonymous post-cutset capacity or phase escape. The* 40 *formal pairs split as*

$$40 = 1 + 28 + 11,$$

where 1 *pair is an actual primitive packet,* 28 *pairs are source failures, and* 11 *pairs are CRT-window empty. The latter* 11 *pairs are routed to the named frontier consisting of window-edge or unused-target arrival, support motion and endpoint release, primitive identity shift, moving-slot families, transport reset, epoch-pair multiplicity, and moving-residue or singleton SAE.*

Status and reference. The frontier router is generated by

```
experiments/prime_matrix_after_cutset_named_exit_frontier_router.py
```

and archived at

```
docs/monograph/prime_matrix_after_cutset_named_exit_frontier_router.md.
```

The current numerical gates are: edge-collision candidates 11, support motion candidates 11, minimum endpoint release $70 = 3.5 \cdot 20$, minimum affine depth defect 70, no fixed highfactor slot isolation failure at the replay base, no transport reset atom, epoch-pair mass $0.023577117628562343 < 1/40$, 37 moving residue shapes, and 300 singleton residue packets. Hence the current after-cutset frontier is closed as a classification and routing statement. The next global obligation is still to exclude or summably absorb these named exits: SourceMaterializationFailure, CRTWindowEmpty, WindowEdge/UnusedTargetArrival, SupportMotion, PrimitiveIdentityShift, MovingSlotFamily, TransportReset, EpochPairMultiplicity, MovingResidueShape, and SingletonResidue. This is a sharper reduction boundary, not an unconditional row/column theorem. □

Reduction-closed Statement 2.62 (Terminal named-exit choke set)**.** *The ten after-cutset named exits in the current frontier compress to five terminal chokes: Source/CRT materialization, support-motion primitive identity, transport–singleton active-ℓ, epoch-pair paired pressure, and moving-residue SAE. All five are closed in the current sweep as routing and finite-capacity statements, but none is yet globally excluded.*

Status and reference. The terminal choke router is generated by

```
experiments/prime_matrix_global_named_exit_terminal_choke_router.py
```

and archived at

```
docs/monograph/prime_matrix_global_named_exit_terminal_choke_router.md.
```

Its current explicit margins are:

quantity	current value
unresolved formal pairs	0
CRT phase margin	879
minimum extra endpoint release over support width	50
transport reset atoms	0
one-slot total spare ratio	0.9038893044128646
one-slot minimum epoch spare ratio	0.6901408450704225
paired-pressure slack	0.8398220244716351.

Thus the latest obstruction is not a local capacity deficit. A persistent counterexample family must repeatedly realize one of the five terminal chokes; the task is to exclude those recurrences or summably absorb them through the registered PDEC/SAE/ColumnCRT ledgers. This is still a global reduction frontier, not an unconditional row/column proof. □

Reduction-closed Statement 2.63 (Terminal choke amplification barrier)**.** *In the current terminal choke set, the narrowest free phase jump is the CRT-window jump of size $40 = 2 \cdot 20$, but that jump is not a terminal escape: it is routed to unused-target arrival or window-edge PDEC/SAE. Among the terminal capacity barriers, the narrowest one-slot transport epoch is the minus $\ell = 71$ epoch, which uses 22 of 71 residues and therefore needs 50 new residues to overflow. The support-motion endpoint release barrier has the same extra size $70 - 20 = 50$.*

Status and reference. The amplification-barrier router is generated by

`experiments/prime_matrix_terminal_choke_amplification_barrier_router.py`

and archived at

`docs/monograph/prime_matrix_terminal_choke_amplification_barrier_router.md.`

The current ordered barriers are

barrier	factor	additive units
CRTWindowPhaseJump	2	40
OneSlotTransportOverflow	71/22	50
SupportEndpointRelease	70/20	50
EpochPairPairedPressure	899/144	1 to 8 fill units
FixedMovingSlotCRT	731/104	703.

Thus the current narrow terminal obstruction is a fifty-unit cross-lock: persistent recurrence must pay a common release/residue package of size 50, or else fall into reset, PDEC, SAE, or ColumnCRT. This identifies the next global interface, but does not yet exclude that interface for all persistent families. □

Reduction-closed Statement 2.64 (Fifty-unit cross-lock carrier separation)**.** *The fifty-unit cross-lock is not an immediate same-carrier contradiction in the current sweep. The support side is the* $q = 31$ *atom with source pair* $(19, 12)$*, generator prime* $P = 2687$*, and endpoint release* $70-20 = 50$*. The one-slot side is the minus* $\ell = 71$ *epoch, which uses* 22 *of* 71 *residues and needs* 50 *new residues to overflow. These two carriers have distinct moduli, distinct phase taxonomy, and disjoint current* P*-bands; therefore any persistent recurrence must pass through the cross-carrier modulus* $31 \cdot 71 = 2201$.

Status and reference. The carrier-separation router is generated by

`experiments/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.py`

and archived at

`docs/monograph/prime_matrix_fifty_unit_cross_lock_carrier_separation_router.md.`

It records

`same_q_or_ell=false`, `same_side_taxonomy=false`, `same_p_band=false`,
`direct_same_carrier_contradiction=false`, `combined_carrier_modulus=2201`.

The fixed delay of the support atom is 80. Modulo 71 this is 9, and $\gcd(80, 71) = 1$, so the first 50 transported residues are distinct. The first such block fitting the current minus-71 epoch runs from $P = 4207$ to $P = 8127$, inside the recorded epoch range $4177 \le P \le 9257$. Thus a short repeated-residue PDEC is not automatic. If, however, these 50 transported arrivals are all new residues in the same minus-71 epoch, then $22 + 50 > 71$, so one-slot capacity fails. If they are not new, or if they cannot stay synchronized, the failure is routed to TransportReset-PDEC, SingletonResidue-SAE, support-motion/unused-target exits, or moving-carrier ColumnCRT. The next global interface is therefore

`CrossCarrierFiftyUnitSynchronizationPDECOrSAE.`

This closes the same-carrier shortcut and sharpens the remaining obstruction; it is not yet an unconditional row/column proof. □

Reduction-closed Statement 2.65 (Cross-carrier fifty-unit residue saturation). *After reconstructing the complete singleton physical records, the current cross-carrier synchronization does not overflow the minus-*71 *epoch inside the recorded* P*-band. The already-used minus-*71 *residue set has* 22 *elements. The support lattice points admitted by the current epoch range contribute* 64 *distinct residues;* 17 *of them are already used, so the combined set has size* 69*, leaving exactly two missing residues,* 0 *and* 62*.*

Status and reference. The saturation router is generated by

`experiments/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.py`

and archived at

`docs/monograph/prime_matrix_cross_carrier_fifty_unit_residue_saturation_router.md.`

It records

`used_residue_count_reconstructed=22`, `exact_used_matches_capacity_ledger=true`,
`admitted_step_range=[19,82]`, `admitted_distinct_residue_count=64`,
`admitted_intersection_existing_count=17`, `admitted_new_residue_count=47`,
`union_size_after_admitted_band=69`, `spare_after_admitted_band=2`.

Every contiguous 50-step block inside the current admitted band remains below capacity: the largest block union has size 62. Thus the branch "fifty arrivals are automatically fifty new residues" is false on the actual frontier. The precise remaining phase data are:

$$P = 9647 \mapsto 62, \qquad P = 9727 \mapsto 0, \qquad P = 9807 \mapsto 9.$$

The first two arrivals fill the two missing residues; the third is an old residue and hence must be routed to transport reset-PDEC if the same epoch persists. Therefore the next interface is

`TwoResidueSpareEndpointExtensionOrTransportResetPDEC`.

This is a sharper current-frontier obstruction, not a proof of the global row/column theorem. □

Reduction-closed Statement 2.66 (Two-residue spare prime-anchor filter). *The two-residue near-fill route is not a valid actual-chain closure after the prime-anchor condition on* P *is imposed. In the same admitted lattice* $19 \le t \le 82$*, only* 17 *of the* 64 *values* $P = 2687 + 80t$ *are prime anchors. They contribute only* 13 *new residues to the minus-*71 *epoch; together with the* 22 *existing residues this covers* 35 *of the* 70 *nonzero residue classes. The two near endpoint arrivals from the previous router are composite or structurally impossible as prime anchors.*

Status and reference. The prime-anchor filter router is generated by

`experiments/prime_matrix_two_residue_spare_prime_anchor_filter_router.py`

and archived at

`docs/monograph/prime_matrix_two_residue_spare_prime_anchor_filter_router.md.`

It records

`admitted_lattice_step_count=64`, `admitted_prime_anchor_count=17`,
`admitted_prime_anchor_new_residue_count=13`, `prime_filtered_union_size=35`,
`prime_filtered_nonzero_spare=35`, `residue_zero_prime_anchor_impossible=true`.

The previous near endpoints satisfy

$$9647 \mapsto 62, \qquad 9727 \mapsto 0, \qquad 9807 \mapsto 9,$$

but all three P-values fail the prime-anchor test. Moreover the residue 0 arithmetic progression has $P \equiv 4047 \pmod{5680}$ and $\gcd(4047, 5680) = 71$, so for $P > 71$ it cannot contain a prime anchor. The first prime anchor for residue 62 is delayed to $P = 26687$. Full nonzero-residue coverage in the scanned progression is delayed until $P = 98047$, and the first prime-anchor repeat after that is $P = 98207$. Thus the current interface is

`PrimeAnchorFilteredNonzeroResidueCoverageOrTransportResetPDEC`.

This closes the two-residue near-fill shortcut. It still leaves a global obligation: exclude persistent long arithmetic-progression prime-anchor coverage, or route the first repeat after coverage to transport reset-PDEC. □

Reduction-closed Statement 2.67 (Prime-anchor post-band immediate repeat)**.** *The long prime-anchor nonzero-residue coverage branch is preempted in the current primitive epoch. After the admitted band* $19 \le t \le 82$*, the first prime anchor in the progression* $P = 2687 + 80t$ *is*

$$t = 90, \qquad P = 9887, \qquad \mathrm{residue}_{71}(P) = 18.$$

The residue 18 *is already in the original minus-*71 *used-residue set. Thus no new missing nonzero residue is added before the first repeat.*

Status and reference. The immediate-repeat router is generated by

`experiments/prime_matrix_prime_anchor_postband_immediate_repeat_router.py`

and archived at

`docs/monograph/prime_matrix_prime_anchor_postband_immediate_repeat_router.md.`

It records

`first_postband_prime_anchor=(90,9887,18)`, `first_postband_prime_is_repeat=true`,
`first_postband_prime_is_original_used_repeat=true`,
`new_prime_anchor_count_before_first_repeat=0`,
`missing_nonzero_remaining_at_first_repeat=35`,
`coverage_before_reset_possible_in_same_epoch=false`.

The intervening candidates $t = 83, \ldots, 89$, namely

$$9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

are composite. Therefore, if the same reset-free epoch persists past the admitted band, it hits a used residue before it hits any new missing nonzero class and must enter transport reset-PDEC. If the epoch does not persist to that anchor, the branch has already moved to endpoint motion/SAE. The latest interface is

`ImmediatePrimeAnchorRepeatTransportResetPDECOrEndpointMotionSAE`.

This closes the current long-coverage shortcut. It does not globally exclude transport reset or endpoint motion. □

Reduction-closed Statement 2.68 (Prime-anchor repeat reset atom)**.** *The first post-band repeat is an exact one-slot reset atom. The original minus-*71 *singleton record with residue* 18 *occurs at* $P = 7757$*. The first post-band prime anchor has* $P = 9887$ *and the same residue. Their difference is*

$$9887 - 7757 = 2130 = 30 \cdot 71.$$

Thus the first prime anchor after the admitted band is not a new coverage packet, but a lift by 30 *full* ℓ*-periods of an existing residue packet.*

Status and reference. The repeat-reset atom router is generated by

```
experiments/prime_matrix_prime_anchor_repeat_reset_atom_router.py
```

and archived at

```
docs/monograph/prime_matrix_prime_anchor_repeat_reset_atom_router.md.
```

It records

`original_record_p=7757`, `repeat_prime_anchor_p=9887`, `repeat_residue=18`, `p_delta=2130`, `p_delta_over_ell=30`, `exact_same_residue_ell_translate=true`, `reset_atom_instantiated=true`, `missing_nonzero_remaining_at_reset_atom=35`.

The associated original slot key is `644:128:71`, and the formal unit is

```
one-slot-repeat-reset|side=minus|ell=71|residue=18|
lift_delta=30|p=7757->9887 .
```

Consequently, if the same reset-free epoch extends to $P = 9887$, the reset-PDEC atom is already present. To avoid it, the endpoint must cut before step 90, leaving only the composite buffer $t = 83, \ldots, 89$. The next interface is

`OneSlotPrimeAnchorRepeatResetPDECExclusionOrEndpointMotionSAE`.

This instantiates the local reset atom. It does not yet globally exclude all one-slot reset atoms or endpoint-motion escapes. □

Reduction-closed Statement 2.69 (Endpoint cut zero-gain)**.** *If the endpoint is cut before the first post-band prime anchor in order to avoid the instantiated reset atom, the cut has no prime-anchor gain in the current primitive epoch. The only buffer before the reset is*

$$t = 83, \ldots, 89, \qquad P = 9327, 9407, 9487, 9567, 9647, 9727, 9807,$$

and none of these P*-values is prime.*

Status and reference. The zero-gain endpoint router is generated by

```
experiments/prime_matrix_endpoint_cut_zero_gain_router.py
```

and archived at

```
docs/monograph/prime_matrix_endpoint_cut_zero_gain_router.md.
```

It records

$$\texttt{cut\_buffer\_step\_count=7},\quad \texttt{formal\_gap\_composite\_row\_count=2},$$
$$\texttt{formal\_gap\_composite\_residues=[62,0]},$$
$$\texttt{actual\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_new\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{endpoint\_cut\_actual\_gain\_zero=true},$$
$$\texttt{reset\_atom\_instantiated\_at\_first\_prime\_anchor=true}.$$

The apparent gap residues 62 and 0 occur at $P = 9647$ and $P = 9727$, respectively, but 9647 has factor 11 and 9727 is divisible by 71. Thus they are formal gaps only, not actual prime-anchor arrivals. If the endpoint is not cut, the next value $P = 9887$ is the one-slot repeat-reset atom. Hence the local dichotomy is

$$\texttt{reset-PDEC} \quad \text{or} \quad \texttt{zero-gain endpoint cut SAE}.$$

The latest interface is

$$\texttt{ZeroGainEndpointCutSAEOrOneSlotResetPDECExclusion}.$$

This closes the current endpoint-gain explanation, not the global SAE or reset-PDEC exclusion. □

Reduction-closed Statement 2.70 (Endpoint cut no-payload SAE)**.** *In the current primitive epoch, the endpoint-cut branch has empty actual payload. The cut buffer* $t = 83, \ldots, 89$ *contains no prime-anchor row, no new prime-anchor row, and no filtered repeat prime-anchor row. The apparent gap residues* 62 *and* 0 *do not survive prime-anchor filtering.*

Status and reference. The no-payload endpoint router is generated by

```
experiments/prime_matrix_endpoint_cut_no_payload_sae_router.py
```

and archived at

```
docs/monograph/prime_matrix_endpoint_cut_no_payload_sae_router.md.
```

It records

$$\texttt{actual\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_new\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{filtered\_repeat\_prime\_anchor\_count\_before\_reset=0},$$
$$\texttt{actual\_payload\_empty=true},\quad \texttt{actual\_payload\_mass=0},$$
$$\texttt{missing\_nonzero\_before\_cut=35},\quad \texttt{missing\_nonzero\_after\_cut=35},$$
$$\texttt{missing\_nonzero\_set\_preserved\_by\_cut=true}.$$

Thus the endpoint cut cannot supply the capacity required by the counterexample chain; it is a no-payload SAE branch. If that branch is not taken, the first prime anchor is already the one-slot reset-PDEC atom. The current local dichotomy is

$$\texttt{one-slot reset-PDEC} \quad \text{or} \quad \texttt{no-payload endpoint SAE}.$$

The latest interface is

$$\texttt{OneSlotResetPDECExclusionOrNoPayloadEndpointSAESummability}.$$

This closes the current endpoint-cut actual-load exit, not the global reset or SAE summability problem. □

Reduction-closed Statement 2.71 (One-slot reset prefix no-relief). *If the endpoint branch is not cut, the line first accepts the one-slot reset at* $t = 90$*,* $P = 9887$*, residue* 18*. After that accepted reset, there is still no new missing nonzero residue until* $t = 115$*,* $P = 11887$*, residue* 30*.*

Status and reference. The reset-prefix no-relief router is generated by

`experiments/prime_matrix_one_slot_reset_prefix_no_relief_router.py`

and archived at

`docs/monograph/prime_matrix_one_slot_reset_prefix_no_relief_router.md.`

It records

```
reset_step=90,   reset_p=9887,   reset_residue=18,
prefix_prime_anchor_count_before_first_relief=7,
prefix_new_missing_nonzero_count_before_first_relief=0,
prefix_repeat_prime_anchor_count_before_first_relief=7,
first_relief_step_gap_after_reset=25,   first_relief_p_gap_after_reset=2000.
```

Thus a new actual-capacity path must first cross the accepted reset-PDEC atom; before the first relief all actual prime anchors remain old residues. If the reset is rejected, the endpoint branch has already been reduced to a no-payload SAE. The latest interface is

`AcceptedResetPDECExclusionOrDelayedReliefSupportMotionSAE.`

This closes the explanation that reset immediately supplies fresh capacity, not the global exclusion of accepted reset-PDEC atoms. □

Reduction-closed Statement 2.72 (Accepted reset full relief horizon). *After accepting the one-slot reset at* $P = 9887$*, complete relief of the remaining* 35 *missing nonzero residues is not local. It first completes at* $t = 1192$*,* $P = 98047$*, after* 1102 *synchronized steps.*

Status and reference. The accepted-reset full-relief horizon router is generated by

`experiments/prime_matrix_accepted_reset_full_relief_horizon_router.py`

and archived at

`docs/monograph/prime_matrix_accepted_reset_full_relief_horizon_router.md.`

It records

```
missing_nonzero_count_at_reset=35,
first_relief=(step=115,P=11887,residue=30),
full_relief=(step=1192,P=98047,residue=67),
full_relief_step_gap_after_reset=1102,
full_relief_p_gap_after_reset=88160,
new_relief_prime_anchor_count_until_full_relief=35,
repeat_prime_anchor_count_until_full_relief=225.
```

In addition, 101 formal missing-residue hits before full relief are composite and cannot be counted as actual load. Hence full relief is a long-horizon support-motion obligation, not a local repair inside the current primitive epoch. The latest interface is

`AcceptedResetPDECExclusionOrLongReliefHorizonSupportMotionSAE.`

This quantifies the delayed relief branch; it does not yet exclude the global accepted-reset family. □

Reduction-closed Statement 2.73 (Long relief cycle debt)**.** *The long-relief branch has an explicit* 71*-cycle phase debt. Among the remaining* 35 *nonzero residues, only* 8 *receive prime relief on the first post-reset phase hit; the other* 27 *require later* 71*-period returns.*

Status and reference. The long-relief cycle-debt router is generated by

`experiments/prime_matrix_long_relief_cycle_debt_router.py`

and archived at

`docs/monograph/prime_matrix_long_relief_cycle_debt_router.md.`

It records

$$\texttt{ell=71},\quad \texttt{period\_p=5680},$$
$$\texttt{zero\_cycle\_relief\_count=8},\quad \texttt{positive\_cycle\_debt\_residue\_count=27},$$
$$\texttt{total\_cycle\_debt=101},\quad \texttt{total\_composite\_wait\_count=101},$$
$$\texttt{matches\_full\_horizon\_composite\_missing\_count=true},$$
$$\texttt{max\_cycle\_debt=15},\quad \texttt{max\_cycle\_debt\_residue=67}.$$

Thus the 101 composite formal missing-residue hits in the full-horizon ledger are exactly the sum of the per-residue cycle debts. The worst residue waits from $P = 12847$ to $P = 98047$, a 15-period delay. The latest interface is

$$\texttt{LongReliefCycleDebtPDECExclusionOrSupportMotionSAESummability}.$$

This records the phase cost of long relief; it does not yet prove global summability or global PDEC exclusion. □

Reduction-closed Statement 2.74 (One-period relief deficit)**.** *In the first complete* 71*-residue cycle after the accepted reset, every missing nonzero residue appears formally once, but only* 8 *of the* 35 *formal hits are actual prime relief.*

Status and reference. The one-period relief deficit router is generated by

`experiments/prime_matrix_one_period_relief_deficit_router.py`

and archived at

`docs/monograph/prime_matrix_one_period_relief_deficit_router.md.`

It records

$$\texttt{period\_is\_complete\_residue\_cycle=true},$$
$$\texttt{missing\_nonzero\_required=35},\quad \texttt{formal\_missing\_hit\_count=35},$$
$$\texttt{actual\_relief\_count\_in\_one\_period=8},$$
$$\texttt{composite\_missing\_count\_in\_one\_period=27},$$
$$\texttt{repeat\_prime\_anchor\_count\_in\_one\_period=10},$$
$$\texttt{relief\_deficit\_after\_one\_period=27}.$$

The 27 composite missing hits match exactly the positive cycle-debt residues. Thus one reset-local period is not enough actual load; it forces cycle debt or a support-motion SAE branch. The latest interface is

$$\texttt{OnePeriodReliefDeficitForcesCycleDebtPDECOrSupportMotionSAE}.$$

This is still a reduced hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.75 (Cycle-debt CRT cover pressure)**.** *The* 27 *positive cycle-debt residues are not free local drift. Their composite waits form a CRT cover in the cycle coordinate, using* 24 *distinct blocking prime factors.*

Status and reference. The CRT cover pressure router is generated by

`experiments/prime_matrix_cycle_debt_crt_cover_pressure_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_crt_cover_pressure_router.md.`

It records

$$\texttt{positive\_cycle\_debt\_residue\_count=27},$$
$$\texttt{total\_composite\_waits=101},$$
$$\texttt{global\_unique\_blocker\_factor\_count=24},$$
$$\texttt{global\_blocker\_lcm=337212073559813724487421695331234639247},$$
$$\texttt{max\_row\_residue=67}, \quad \texttt{max\_row\_cycle\_debt=15},$$
$$\texttt{max\_row\_blocker\_lcm=55140500775337593}.$$

For a fixed missing residue the same-residue column is

$$P(k) = P_0 + 5680k.$$

Each composite wait imposes one congruence class in the cycle coordinate modulo its least prime blocker. Hence persistent reproduction of the debt pattern requires a large CRT phase package, not merely a local endpoint translation. The latest interface is

$$\texttt{CycleDebtCRTCoverPressurePDECOrGlobalSupportMotionSAE}.$$

This records the global-family pressure, while leaving the PDEC exclusion or SAE summability proof open. □

Reduction-closed Statement 2.76 (Cycle-debt transverse CRT independence)**.** *The cycle-debt CRT cover is transverse to the local period* 5680*. It cannot be absorbed by a local period translation or by a same-slot drift.*

Status and reference. The transverse CRT independence router is generated by

`experiments/prime_matrix_cycle_debt_transverse_crt_independence_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_transverse_crt_independence_router.md.`

It records

$$\texttt{period\_p=5680=2\^{}4*5*71},$$
$$\texttt{global\_unique\_blocker\_factor\_count=24}, \quad \texttt{transverse\_blocker\_factor\_count=24},$$
$$\texttt{all\_blocker\_factors\_coprime\_to\_period\_p=true},$$
$$\texttt{gcd\_global\_blocker\_lcm\_with\_period\_p=1},$$
$$\texttt{combined\_period\_equals\_product=true},$$
$$\texttt{all\_row\_shift\_replay\_classes\_zero=true},$$
$$\texttt{all\_row\_lcm\_exceeds\_phase\_support\_width=true}.$$

For a fixed residue column $P(k) = P_0 + 5680k$, every composite wait has a blocking prime q with $\gcd(q, 5680) = 1$. Replaying the same wait after a cycle shift K therefore forces $K \equiv 0 \pmod{q}$. Rowwise this gives $K \equiv 0 \pmod{\mathrm{lcm}_{\mathrm{row}}}$, and across all rows it gives

$$K \equiv 0 \pmod{3372120735598137244874216953312346392 47}.$$

The modulus is coprime to 5680, so this is an independent transverse CRT phase package, not a factor of the local period. The latest interface is

`TransverseCRTCoverPDECExclusionOrGlobalSupportMotionSAE`.

Thus local period absorption is closed in this certificate, while global PDEC exclusion or SAE summability remains open. □

Reduction-closed Statement 2.77 (Cycle-debt sparse replay barrier)**.** *If the transverse CRT cover keeps the same blocking graph, exact replay is extremely sparse. A non-sparse persistent branch must move the blocker graph and hence enters a moving-cover PDEC interface.*

Status and reference. The sparse replay barrier router is generated by

`experiments/prime_matrix_cycle_debt_sparse_replay_barrier_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_sparse_replay_barrier_router.md.`

It records

`global_exact_replay_cycle_modulus= 337212073559813724487421695331234639247`,

`global_exact_replay_p_gap= 1915364577819741955088555229481412750922960`,

`max_cycle_debt_support_width=15`,

`full_relief_cycle_span_ceiling=16`,

`single_full_debt_copy_per_full_relief_horizon=true`,

`exact_replay_branch_is_sae_sparse_current_certificate=true`,

`non_sparse_persistence_forces_moving_blocker_map=true`.

The full debt word can replay exactly only after a shift by the global transverse lcm. The observed full-relief horizon has length $\lceil 1102/71 \rceil = 16$ residue cycles, whereas the replay modulus is the large integer displayed above. Thus any local or medium support-motion window shorter than that modulus contains at most one exact copy of the full word. The exact branch is therefore a sparse SAE branch. If a counterexample branch requires positive-density or short-window persistence, it must alter the blocking primes, phases, or row combination; this is the moving transverse cover PDEC branch. The latest interface is

`SparseReplaySAEOrMovingTransverseCoverPDEC`.

This is a sharper reduced hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.78 (Cycle-debt near-shift exit boundary)**.** *The moving-cover branch cannot be a small shift of the old debt prefix. Any near shift either hits an exit prime inside the shifted word, or else has no old prefix left to reuse and must build a fresh cover.*

Status and reference. The near-shift exit-boundary router is generated by

`experiments/prime_matrix_cycle_debt_near_shift_exit_boundary_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_near_shift_exit_boundary_router.md.`

It records

$$\texttt{near\_shift\_limit\_cycles=16}, \quad \texttt{positive\_cycle\_debt\_residue\_count=27},$$
$$\texttt{total\_cycle\_debt\_mass=101}, \quad \texttt{max\_cycle\_debt=15},$$
$$\texttt{all\_near\_shifts\_close\_old\_prefix\_reuse=true},$$
$$\texttt{shift\_1\_exit\_prime\_collision\_row\_count=27},$$
$$\texttt{shift\_1\_exit\_prime\_collision\_debt\_mass=101},$$
$$\texttt{shift\_max\_cycle\_debt\_exit\_prime\_collision\_row\_count=1},$$
$$\texttt{shift\_near\_limit\_fresh\_cover\_required\_row\_count=27}.$$

For every positive-debt residue, the old composite prefix is immediately followed by the first relief prime in that same residue column. A shift $1 \le K \le d$ for a row of debt d pulls that exit prime into the shifted debt word. Hence $K = 1$ breaks all 27 rows, while every $1 \le K \le 15$ breaks all rows with $d \ge K$. At $K = 16$ no row has old prefix support left, so all 27 rows require fresh cover material. The latest interface is

$$\texttt{FreshMovingCoverPDECOrGlobalSupportMotionSAE}.$$

Thus near-shift reuse of the old CRT cover is closed in this certificate; the remaining branch is fresh moving-cover PDEC or genuine global support-motion SAE. □

Reduction-closed Statement 2.79 (Cycle-debt fresh-cover prime obstacle)**.** *On the same* 27 *positive-debt support rows, every near shifted fresh window contains actual prime obstacles. Same-support fresh cover is therefore closed unless those actual primes are deleted by a PDEC.*

Status and reference. The fresh-cover prime-obstacle router is generated by

`experiments/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_fresh_cover_prime_obstacle_router.md.`

It records

$$\texttt{total\_cycle\_debt\_mass\_per\_shift=101}, \quad \texttt{tested\_shift\_count=16},$$
$$\texttt{total\_tested\_same\_support\_slots=1616},$$
$$\texttt{total\_prime\_obstacles\_all\_near\_shifts=364},$$
$$\texttt{total\_new\_prime\_obstacles\_all\_near\_shifts=263},$$
$$\texttt{all\_near\_shift\_same\_support\_windows\_have\_prime\_obstacles=true},$$
$$\texttt{min\_prime\_obstacle\_count\_per\_shift=17},$$
$$\texttt{shift\_near\_limit\_prime\_obstacle\_count=27},$$
$$\texttt{same\_support\_fresh\_cover\_closed\_current\_certificate=true}.$$

For each $K = 1, \dots, 16$, the shifted window on the same support rows has the same total length 101 as the original debt word. Direct primality checking of those actual P-values shows at least 17 prime obstacles in every such shifted word; the fully fresh $K = 16$ window still has 27, all new. Hence a same-support fresh moving cover cannot be an all-composite word. The latest interface is

`FreshCoverPrimeObstaclePDECOrSupportRowReplacementSAE`.

The remaining branch must either delete actual prime obstacles, forming a prime-obstacle PDEC, or replace support rows and enter a support-row replacement SAE/PDEC interface. □

Reduction-closed Statement 2.80 (Cycle-debt support-row replacement Hall barrier)**.** *Support-row replacement is a Hall capacity problem over the* 35 *missing residue candidates. Fourteen of the sixteen near phases fail the Hall threshold test; only* $K = 13, 14$ *survive.*

Status and reference. The support-row replacement Hall router is generated by

`experiments/prime_matrix_cycle_debt_support_row_replacement_hall_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_support_row_replacement_hall_router.md`.

It records

`candidate_missing_residue_count=35`, `needed_positive_debt_row_count=27`,
`total_demand_width=101`, `max_required_width=15`,
`hall_fail_shift_count=14`,
`hall_survivor_shifts=[13,14]`,
`k13_assigned_immediate_relief_rows=6`,
`k14_assigned_immediate_relief_rows=8`.

For a phase K, a candidate row has capacity equal to the longest composite run starting at K, capped at the maximum required width 15. For every threshold t, Hall requires

$$\#\{\text{candidate rows with capacity} \geq t\} \geq \#\{\text{demand rows with debt} \geq t\}.$$

This condition fails for

$$K = 1, 2, \dots, 12, 15, 16.$$

Only $K = 13$ and $K = 14$ pass, and both are tight survivor phases requiring immediate-relief rows and large-scale row replacement. The latest interface is

`K13K14SupportReplacementSurvivorPDECOrGlobalSAE`.

This is a two-phase survivor hardpoint, not a global theorem closure. □

Reduction-closed Statement 2.81 (K13/K14 tight-Hall survivor island)**.** *The two support-row replacement survivors* $K = 13, 14$ *are not free moving replacement phases. They form a tight Hall island whose zero-slack layers force row replacement, immediate-relief support, and transverse CRT pressure.*

Status and reference. The K13/K14 survivor rigidity router is generated by

```
experiments/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_k13_k14_survivor_rigidity_router.md.
```

It records

$$\begin{gathered}\texttt{k13\_zero\_slack\_thresholds=[1,4,7,8]},\\ \texttt{k14\_zero\_slack\_thresholds=[7,8,13,15]},\\ \texttt{k13\_minimum\_replacement\_count=20},\quad \texttt{k14\_minimum\_replacement\_count=19},\\ \texttt{k13\_minimum\_immediate\_relief\_rows=6},\quad \texttt{k14\_minimum\_immediate\_relief\_rows=6},\\ \texttt{k13\_minimum\_overstretch\_units=61},\quad \texttt{k14\_minimum\_overstretch\_units=65},\\ \texttt{all\_tight\_layers\_have\_transverse\_lcm\_exceeding\_period=true}.\end{gathered}$$

Here a zero-slack threshold t means that the number of candidate rows with capacity at least t is exactly the number of demand rows with debt at least t. Consequently the whole supply set of that layer is forced: deleting one such row, or moving one of its first t composite slots to a prime obstacle, breaks the Hall inequality at that threshold. For $K = 13$, the forced shells include the disjoint high shell

$$\{1, 13, 19, 30\} \to \{17, 23, 58, 67\}$$

at threshold 8, plus the singleton shell $31 \to 15$ at threshold 7. For $K = 14$, the top shells force $13 \to 67$, $19 \to 23$, and $70 \to 15$ at the corresponding thresholds.

The router then solves exact minimum-cost matching problems rather than relying on the previous greedy witness. Even after optimization, $K = 13$ requires at least 20 row replacements and 6 immediate-relief rows, while $K = 14$ requires at least 19 row replacements and 6 immediate-relief rows. The forced composite slots in every tight layer have transverse CRT least common multiple larger than the local period 5680. The neighboring phases are not survivors: $K = 12$ fails Hall at threshold 1, and $K = 15$ fails at threshold 6. Thus the remaining support-row branch is compressed to

$$\texttt{K13K14TightHallCRTIslandPDECOrMovingSupportSAE}.$$

This still is not a global theorem closure; it replaces the previous free replacement interpretation with a named tight-Hall CRT island. □

Reduction-closed Statement 2.82 (Shell phase graph obstruction). *The $K = 13, 14$ tight-Hall island cannot be absorbed by a single residue translation of the support. Its forced shells split into multiple incompatible phase deltas.*

Status and reference. The shell phase graph router is generated by

```
experiments/prime_matrix_cycle_debt_shell_phase_graph_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_shell_phase_graph_router.md.
```

It records

$$\texttt{k13\_forced\_shell\_matching\_product=252829237248000},$$
$$\texttt{k14\_forced\_shell\_matching\_product=2},$$
$$\texttt{k13\_forced\_edge\_count=1}, \quad \texttt{k14\_forced\_edge\_count=3},$$
$$\texttt{k13\_single\_translation\_support\_possible=false},$$
$$\texttt{k14\_single\_translation\_support\_possible=false},$$
$$\texttt{k13\_core\_min\_distinct\_delta\_lower\_bound=7},$$
$$\texttt{k14\_core\_min\_distinct\_delta\_lower\_bound=4}.$$

For each forced Hall shell the router builds the bipartite graph from supply residues to demand residues, keeping only edges whose capacity can cover the demand width. A single translation support motion would require one common $\delta \pmod{71}$ across all forced shells. For $K = 14$ three forced edges already have incompatible deltas:

$$13 \mapsto 67 \quad (\delta = 54), \qquad 19 \mapsto 23 \quad (\delta = 4), \qquad 70 \mapsto 15 \quad (\delta = 16).$$

The remaining 8..12 shell has only two possible matchings; the whole forced core still needs at least four distinct deltas. For $K = 13$, the singleton forced edge is $31 \mapsto 15$ with $\delta = 55$, and the enumerable core shells $>= 8$, 7..7, and 4..6 already require at least seven distinct deltas. The large 1..3 shell is not used to lower this count; it can only preserve or increase the number of deltas.

Thus the remaining branch is not an ordinary single-phase support translation. It is compressed to

$$\texttt{NonAffineShellPhaseFragmentPDECOrMultiDeltaSupportSAE}.$$

This still is not a global theorem closure; it closes the single-translation support-motion explanation and leaves a non-affine, multi-delta support branch. □

Reduction-closed Statement 2.83 (Multi-delta core CRT load)**.** *The non-affine multi-delta support branch carries a large transverse CRT load on its forced core. It is not a lightweight phase-noise explanation.*

Status and reference. The multi-delta core CRT-load router is generated by

```
experiments/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_multi_delta_core_crt_load_router.md.
```

It records

$$\texttt{k13\_enumerated\_core\_matching\_count=96},$$
$$\texttt{k13\_minimum\_delta\_count=7}, \quad \texttt{k13\_minimum\_delta\_matching\_count=2},$$
$$\texttt{k13\_best\_min\_delta\_global\_lcm\_log10=36.165},$$
$$\texttt{k14\_enumerated\_core\_matching\_count=2},$$
$$\texttt{k14\_minimum\_delta\_count=4}, \quad \texttt{k14\_minimum\_delta\_matching\_count=1},$$
$$\texttt{k14\_best\_min\_delta\_global\_lcm\_log10=31.716}.$$

For each core edge $a \mapsto b$, the router expands the source row from the given shift K for the full demand width of b, records the actual composite slots, and multiplies the distinct smallest blocking factors. The resulting lanes are grouped by $\delta = b - a \pmod{71}$. For $K = 14$, the minimum multi-delta core is unique; its four lanes $\delta = 4, 16, 28, 54$ carry total demand width 52 and have global least common multiple about $10^{31.716}$. For $K = 13$, the enumerable rigid core has 96 matchings, only two of which realize the minimum of seven deltas; the best such core has total demand width 71 and global least common multiple about $10^{36.165}$.

Both least common multiples are far larger than the local period 5680. Thus the latest branch is compressed to

$$\texttt{MultiDeltaCoreCRTLoadPDECOrLowShellFullResidueSAE}.$$

This still is not a global theorem closure: it closes the light multi-delta noise interpretation, while leaving a heavy CRT-load PDEC or the $K = 13$ low-shell full-residue SAE as the next explicit interface. □

Reduction-closed Statement 2.84 (K13 low-shell delta skeleton)**.** *The $K = 13$ low shell does not require a full-residue escape. Relative to the rigid core, it adds only two delta lanes and turns $K = 13$ into a full-debt nine-lane CRT-load branch.*

Status and reference. The low-shell delta skeleton router is generated by

```
experiments/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.py
```

and archived at

```
docs/monograph/prime_matrix_cycle_debt_low_shell_delta_skeleton_router.md.
```

It records

$$\texttt{low\_shell\_possible\_delta\_count=71},$$
$$\texttt{low\_shell\_minimum\_delta\_count=6},$$
$$\texttt{low\_shell\_minimum\_delta\_witness=[23,35,38,58,68,70]},$$
$$\texttt{core\_delta\_count\_before\_low\_shell=7},$$
$$\texttt{low\_shell\_new\_delta\_count\_over\_core=2},$$
$$\texttt{low\_shell\_new\_deltas\_over\_core=[0,66]},$$
$$\texttt{full\_k13\_delta\_count\_after\_low\_shell=9},$$
$$\texttt{full\_k13\_total\_required\_width=101},$$
$$\texttt{full\_k13\_global\_lcm\_log10=42.095}.$$

The router formulates the low shell as a binary matching problem. Edge variables choose the supply-to-demand matching, while delta variables record which residue lanes are used. First, with unit cost on every delta, the low shell alone needs only six deltas despite having all 71 deltas available. Second, after the seven core deltas from the $K = 13$ rigid core are given zero cost, the low shell requires only two new lanes, $\delta = 0$ and $\delta = 66$. Expanding the core and low-shell edges into their actual composite slots gives a full 101-width $K = 13$ load with nine delta lanes and global CRT least common multiple about $10^{42.095}$.

Thus the latest branch is compressed to

$$\texttt{K13FullDebtNineLaneCRTLoadPDECOrResidualSlackTailSAE}.$$

This still is not a global theorem closure; it closes the low-shell full-residue free-escape interpretation and leaves a full-debt CRT-load PDEC or a residual slack-tail SAE. □

Reduction-closed Statement 2.85 (K13 full-debt nine-lane residual tail)**.** *The residual slack-tail in the $K = 13$ full-debt nine-lane branch is not an artifact of the chosen matching. It is forced by the layer Hall ledger and already carries a transverse CRT load larger than the local period.*

Status and reference. The K13 full-debt nine-lane tail router is generated by

`experiments/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_full_debt_nine_lane_tail_router.md.`

It records

`capacity_row_count=27`, `demand_row_count=27`,
`all_capacity_rows_mandatory=true`,
`total_capacity=119`, `total_demand_width=101`,
`unavoidable_tail_slot_count=18`,
`zero_slack_layers=[1,4,7,8]`,
`allowed_deltas=[0,8,22,39,54,55,58,59,66]`,
`min_tail_factor_count=8`, `min_tail_lcm_log10=11.488`.

At threshold 1, the Hall count is 27 supply rows against 27 demand rows. Hence every positive-capacity row is mandatory in any full-debt matching. The total capacity is 119, while the full demand is 101; equivalently the sum of the layer slacks is 18. Thus the residual tail contains exactly 18 slots independently of the particular nine-lane matching witness.

The zero-slack layers $1, 4, 7, 8$ are the local rigidity gates. Moving a slot through one of these layers without a compensating arrival would immediately break Hall capacity. Within the fixed nine-lane family, a MILP over all allowed 71 supply-demand edges optimizes the active tail blockers. Even under the minimizing objective, the tail activates at least eight distinct blocking factors,

$$3, 7, 11, 13, 19, 127, 151, 281,$$

whose least common multiple has logarithm about 11.488, already exceeding the local period 5680.

Thus the latest branch is compressed to

`K13LayerSlackTailCRTInvariantPDECOrMovingFamilySAE`.

This still is not a global theorem closure. It closes the interpretation that the residual slack-tail is a matching artifact or a cost-free tail; the remaining branch is a layer-invariant tail-CRT PDEC or a genuinely moving family SAE. □

Reduction-closed Statement 2.86 (K13 tail gate drift)**.** *The $K = 13$ layer-tail profile has no near-shift replay. The only other Hall-surviving shift is $K = 14$, and it changes the zero-slack gates and increases the tail balance.*

Status and reference. The K13 tail gate-drift router is generated by

`experiments/prime_matrix_cycle_debt_k13_tail_gate_drift_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k13_tail_gate_drift_router.md.`

It records

`hall_pass_shifts=[13,14]`,

`same_slack_vector_shifts=[13]`, `same_zero_gate_set_shifts=[13]`,

`k13_tail_balance=18`, `k13_zero_slack_layers=[1,4,7,8]`,

`k14_tail_balance=31`, `k14_tail_increase_over_k13=13`,

`k14_zero_slack_layers=[7,8,13,15]`,

`k14_slack_l1_distance_from_k13=19`.

The router compares every near shift $1 \le K \le 16$ with the $K = 13$ slack vector. The complete vector is repeated only at the base shift itself. The same is true for the zero-gate set. Since the Hall-surviving shifts are only 13 and 14, any near moving branch that leaves $K = 13$ must enter $K = 14$.

But $K = 14$ is not a same-profile replay. It preserves only the common gates $7, 8$, loses the low gates $1, 4$, gains the high gates $13, 15$, and raises the tail balance from 18 to 31. Hence a moving family cannot carry the $K = 13$ layer-tail invariant through the near-shift window without changing the gate structure.

Thus the latest branch is compressed to

`K13GateProfileNoNearReplayPDECOrK14HighGateDriftSAE`.

This still is not a global theorem closure. It closes the near same-profile moving replay interpretation; the remaining branch is either a fixed $K = 13$ gate-profile PDEC or a $K = 14$ high-gate drift SAE/PDEC. □

Reduction-closed Statement 2.87 (K14 high-gate low-shell skeleton)**.** *The $K = 14$ high-gate drift branch is not an unregistered moving family. Its tight shells leave only two high-core matchings, and the remaining low shell compresses to a finite delta skeleton.*

Status and reference. The K14 high-gate low-shell skeleton router is generated by

`experiments/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_k14_high_gate_low_shell_skeleton_router.md`.

It records

`k14_zero_slack_layers=[7,8,13,15]`,

`k14_high_gate_core_alternative_count=2`,

`k14_best_core_deltas=[4,16,28,54]`,

`k14_low_shell_minimum_delta_count=6`,

`k14_low_shell_new_deltas_over_core=[2,48,58,70]`,

`k14_full_delta_count_after_low_shell=8`,

`k14_full_total_required_width=101`,

`k14_full_global_lcm_log10=48.508`.

The high gates force the singleton shells $13 \mapsto 67$, $19 \mapsto 23$, and $70 \mapsto 15$, together with one of two matchings on the 8..12 shell. For each high-core candidate, the router removes the already-used supply and demand residues and solves the remaining low-shell matching by a binary delta-skeleton

MILP. The best full branch has core deltas $4, 16, 28, 54$, and the low shell adds only the four lanes $2, 48, 58, 70$.

Expanding the resulting twenty-seven edges into actual composite slots covers all 101 demand-width units. The eight-lane load has global CRT least common multiple about $10^{48.508}$, far beyond the local period 5680. Thus the $K = 14$ moving branch returns to a full-debt CRT-load PDEC; it is no longer an unnamed high-gate drift SAE.

Thus the latest branch is compressed to

`K13FixedGateProfilePDECOrK14FullDebtEightLaneCRTLoadPDEC`.

This still is not a global theorem closure. It closes the unregistered $K = 14$ high-gate moving-family explanation and leaves a fixed $K = 13$ gate-profile PDEC or the registered $K = 14$ eight-lane CRT-load PDEC. □

Reduction-closed Statement 2.88 (Two-survivor terminal CRT bifurcation). *The $K = 13$ and $K = 14$ terminal branches are not two views of the same support motion. They form a named terminal CRT bifurcation.*

Status and reference. The two-survivor terminal CRT bifurcation router is generated by

`experiments/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.py`

and archived at

`docs/monograph/prime_matrix_cycle_debt_two_survivor_terminal_crt_bifurcation_router.md`.

It records

```
k13_delta_count=9,  k14_delta_count=8,
common_deltas=[54,58],
delta_union_count=15,  delta_symmetric_difference_count=13,
source_intersection_count=19,  source_symmetric_difference_count=16,
target_intersection_count=27,  pair_intersection_count=1,
union_lcm_log10=57.156,  tail_union_lcm_log10=24.632.
```

Both terminal branches cover the same 27 target demand residues. However, their source supports overlap in only 19 rows, and the actual source-to-target matching overlaps in only one edge, $13 \mapsto 67$. At the lane level, only the two deltas 54 and 58 are common; the symmetric difference contains 13 lanes. Thus $K = 13$ and $K = 14$ cannot be treated as a single support pattern with a small phase perturbation.

The combined blocker set has least common multiple about $10^{57.156}$, and the combined tail blocker set has least common multiple about $10^{24.632}$. Both are far larger than the local period 5680. Therefore the remaining branch is no longer an unnamed support-motion SAE. It is a terminal two-survivor CRT bifurcation PDEC that must be excluded directly.

Thus the latest branch is compressed to

`TwoSurvivorTerminalCRTBifurcationPDECExclusion`.

This still is not a global theorem closure. It closes the interpretation that the two survivor branches can absorb each other as one unnamed moving family. □

Reduction-closed Statement 2.89 (Two-survivor switch-arrival pressure)**.** *The terminal two-survivor bifurcation cannot be absorbed by the common anchor or by the two common delta lanes. Avoiding a fixed-branch PDEC forces a fresh-arrival ColumnCRT load or a branch-exclusive CRT-load exclusion.*

Status and reference. The two-survivor PDEC exclusion router is generated by

```
experiments/prime_matrix_cycle_debt_two_survivor_pdec_exclusion_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-two-survivor-pdec-exclusion-router.md.
```

It records

```
common_pair_count=1,  common_pair_width=15
forced_rematched_target_count=26,  forced_rematched_target_width=86
minimum_branch_exclusive_delta_width=70
minimum_branch_exclusive_delta_lcm_log10=32.582
k14_only_assigned_width=29,  k14_only_capacity_at_k13=0
k14_only_capacity_at_k14=39,  k14_arrival_lcm_log10=20.205
k13_same_support_deficit_at_k14=8,  common_source_changed_count=18
```

The only common actual source-target edge is $13 \mapsto 67$, and it accounts for only 15 units of target width. Hence the remaining 26 targets, of total width 86, must be genuinely rematched. The common delta lanes $54, 58$ do not carry this rematching load: each terminal branch still has at least 70 units of width on branch-exclusive deltas, with least common multiple at least $10^{32.582}$ in the current certificate.

On the switching side, the $K = 14$ branch uses eight source rows that have zero capacity at $K = 13$. These rows acquire 39 units of capacity at $K = 14$ and carry 29 units of assigned demand, with arrival blocker lcm about $10^{20.205}$. Conversely, dragging the $K = 13$ source set to $K = 14$ leaves only 93 units of capacity against the 101-unit demand, a deficit of 8.

Thus the latest branch is compressed to

$$\texttt{TerminalSwitchArrivalColumnCRTPDECOrBranchExclusiveCRTLoadExclusion}.$$

This is still not a global theorem closure. It closes the interpretation that the terminal bifurcation can be repaired by the common anchor or the common delta lanes; the next obligation is to exclude the switch-arrival ColumnCRT load or the persistent branch-exclusive CRT loads. □

Reduction-closed Statement 2.90 (Terminal switch arrival wall)**.** *The $K = 14$ fresh-arrival source is not a smoothly opening capacity source. It is an eight-prime entry wall followed by a post-wall CRT load.*

Status and reference. The terminal switch arrival wall router is generated by

```
experiments/prime_matrix_cycle_debt_terminal_switch_arrival_wall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-terminal-switch-arrival-wall-router.md.
```

It records

```
arrival_source_count=8,  all_arrival_sources_zero_capacity_at_k13=true
all_entry_wall_cycles_equal_13=true,  all_entry_wall_positions_zero_at_k13=true
entry_wall_lcm_log10=39.482
postwall_capacity_total=39,  postwall_assigned_width_total=29
postwall_assigned_lcm_log10=20.205
entry_plus_assigned_lcm_log10=59.687
entry_plus_postwall_lcm_log10=68.361
```

The eight $K = 14$-only source rows all have zero capacity at $K = 13$. More precisely, each of them is cut at $K = 13$ by a prime obstacle at window-position 0. These eight entry primes have least common multiple about $10^{39.482}$, coprime to the local period 5680.

After crossing this entry wall, the $K = 14$ branch obtains 39 post-wall capacity slots. It uses 29 of them for the actual assignment, with post-wall assigned blocker lcm about $10^{20.205}$. Combining the entry wall with the assigned post-wall load gives lcm about $10^{59.687}$, and combining the entry wall with all post-wall slots gives lcm about $10^{68.361}$, still coprime to 5680.

Thus the latest branch is compressed to

`EightPrimeEntryWallPostWallCRTExclusionOrBranchExclusiveCRTLoadExclusion`.

This is still not a global theorem closure. It closes the interpretation that the $K = 14$ fresh-arrival capacity is an unnamed smooth deformation; the next obligation is to exclude the eight-prime entry-wall/post-wall CRT load, or the parallel branch-exclusive CRT loads. □

Reduction-closed Statement 2.91 (Coupled branch and entry-wall load). *The entry-wall/post-wall load and the branch-exclusive load are not independent terminal exits. They couple on the $K = 14$ branch; the $K = 13$ branch leaves its own branch-exclusive CRT load.*

Status and reference. The coupled branch entry-wall router is generated by

`experiments/prime_matrix_cycle_debt_coupled_branch_entry_wall_router.py`

and archived at

`docs/monograph/prime-matrix-cycle-debt-coupled-branch-entry-wall-router.md.`

It records

```
k13_branch_exclusive_width=73,  k13_branch_exclusive_lcm_log10=36.678
k14_branch_exclusive_width=70,  k14_arrival_assigned_width=29
k14_arrival_branch_overlap_width=21
k14_branch_arrival_union_width=78
k14_branch_plus_entry_plus_postwall_lcm_log10=85.024
both_branches_plus_k14_entry_postwall_lcm_log10=99.349
```

Thus the $K = 14$ terminal branch does not choose between entry-wall load and branch-exclusive load. It carries both. The 29-unit arrival load overlaps the 70-unit branch-exclusive load in 21 units, but their union still forces 78 units of actual load. After adding the entry wall and all post-wall slots, the $K = 14$ coupled lcm is about $10^{85.024}$, coprime to the local period 5680.

The other terminal possibility is $K = 13$. There the remaining branch-exclusive load has width 73 and lcm about $10^{36.678}$. If both branch-exclusive loads are combined with the $K = 14$ entry/post-wall load, the audit lcm is about $10^{99.349}$.

Thus the latest branch is compressed to

`K13BranchExclusiveCRTLoadExclusionOrK14CoupledEntryBranchCRTWallExclusion`.

This is still not a global theorem closure. It replaces the loose entry-wall/branch-exclusive pair by two branch-specific CRT-load exclusions. □

Reduction-closed Statement 2.92 (Branch replay support gap)**.** *The registered terminal branch loads cannot be reproduced by a local moving-slot replay. Any replay of the same blocker package has period at least its CRT lcm, which is already far larger than the phase support width.*

Status and reference. The branch replay support-gap router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_support_gap_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-support-gap-router.md.
```

It records

```
period_p=5680,  near_shift_limit_cycles=16
k13_branch_exclusive_width=73
k14_branch_arrival_union_width=78
k14_coupled_audit_slot_count_all_postwall=117
smallest_log10_margin_over_support_width=30.737
all_blocks_coprime_to_period=true
all_nonzero_replay_moduli_exceed_support_width=true
local_moving_slot_replay_excluded_for_registered_blocks=true.
```

The proof is a direct CRT replay calculation. Let $M = 5680$. If a registered blocker package B reappears after T local periods with the same blockers, then for every $q \in B$

$$n_q + TM \equiv 0 \pmod q, \qquad n_q \equiv 0 \pmod q.$$

All registered q's are coprime to M. Hence $T \equiv 0 \pmod q$ for every $q \in B$, and the least nonzero replay period is $\mathrm{lcm}(B)$.

For the $K = 13$ branch-exclusive block this replay modulus is about $10^{36.678}$, while the phase support width is 73. For the $K = 14$ coupled branch-plus-entry-plus-postwall block the replay modulus is about $10^{85.024}$, while the actual support width is 78 and the full audit slot count is 117. Therefore the registered high-factor absorption slots cannot be carried by local moving-slot support motion.

Thus the latest branch is compressed to

$$\texttt{BranchReplayColumnCRTPDECExclusionOrIsolatedTerminalAtomAbsorption}.$$

This is still not a global theorem closure. It closes the local replay interpretation; any far replay is a named ColumnCRT/PDEC package, and any non-replaying survivor must be treated as an isolated finite atom. □

Reduction-closed Statement 2.93 (Global dichotomy for branch replay)**.** *The isolated-atom alternative is not a global structural escape. A persistent registered replay block is a Column-CRT/PDEC family; a nonpersistent registered block is only a finite atom requiring finite base checking.*

Status and reference. The branch replay global dichotomy router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_global_dichotomy_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-global-dichotomy-router.md.
```

It records

```
registered_replay_block_count=6
minimum_cycle_replay_modulus_log10=32.582
minimum_p_space_columncrt_modulus_log10=36.337
maximum_p_space_columncrt_modulus_log10=103.103
persistent_registered_replay_routes_to_columncrt_pdec=true
isolated_atoms_cannot_form_infinite_registered_family=true
finite_atom_base_check_required=true.
```

There are only finitely many registered replay blocks in the current terminal certificate. If a global counterexample chain used one of them infinitely often with the same blocker package, the previous replay lemma gives a fixed cycle congruence $T \equiv 0 \pmod{\operatorname{lcm}(B)}$. In the original P-coordinate this is a fixed ColumnCRT class modulo

$$5680 \cdot \operatorname{lcm}(B).$$

The smallest such P-space modulus in the registered list is already about $10^{36.337}$.

If no registered replay block occurs infinitely often, then the registered atoms are finite and cannot be a global structural escape. They still require finite base checking; this router does not erase that obligation. If the blocker package changes, then the object is no longer the same replay block and must return to a PDEC/SAE route or to a new named router.

Thus the latest branch is compressed to

$$\texttt{BranchReplayColumnCRTPDECExclusionOrFiniteAtomBaseCheck}.$$

This remains below an unconditional theorem. The genuine structural hard point is now the exclusion of persistent branch-replay ColumnCRT/PDEC, plus the finite base check for any isolated atoms. □

Reduction-closed Statement 2.94 (Finite atom boundary bridge)**.** *The finite-atom check splits into a covered finite prefix and a post-*100000 *tail packet. The older direct grid verification up to* $P \leq 5000$ *must not be used to absorb the present cycle-debt atoms.*

Status and reference. The finite atom boundary bridge router is generated by

```
experiments/prime_matrix_cycle_debt_finite_atom_boundary_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-finite-atom-boundary-bridge-router.md.
```

It records

```
direct_grid_verified_max_p=5000
dynamic_finite_bridge_range=[3001,99991]
dynamic_finite_bridge_closed=true
cover_pressure_atom_count=155
cover_pressure_all_atoms_in_dynamic_finite_bridge=true
arrival_wall_atom_count=55
arrival_wall_tail_atom_count=31
tail_atom_p_range=[101087,134047]
finite_atom_base_check_fully_closed=false.
```

The cycle-debt cover-pressure packet has 155 recorded P-coordinates, all lying in the already archived dynamic finite bridge $3001 \leq P < 100000$. These finite-prefix atoms are therefore absorbed by the existing bridge, not by the older direct grid verification up to $P \leq 5000$.

The terminal arrival-wall packet is different. It has 55 recorded P-coordinates. Only 24 lie below 100000; the remaining 31 are post-100000 tail atoms, with P-range 101087 to 134047. They consist of assigned slots, post-wall first primes, and tail slots. Hence the finite-atom alternative is not completely closed by the finite bridge.

Thus the latest branch is compressed to

`BranchReplayColumnCRTPDECExclusionOrPost100000TailAtomRunner`.

This is still not a global theorem closure. It closes the finite-prefix part of the finite-atom check and leaves the post-100000 tail packet to a tail-runner, tail lower-sieve, or ColumnCRT/PDEC route. □

Reduction-closed Statement 2.95 (Post-100000 tail atom exact runner)**.** *The post-*100000 *finite tail atoms in the registered terminal packet are exactly verified. The finite-atom branch is closed for the current registered objects, leaving the persistent branch-replay ColumnCRT/PDEC as the active structural obstruction.*

Status and reference. The post-100000 tail atom exact runner is generated by

```
experiments/prime_matrix_cycle_debt_post100000_tail_atom_exact_runner.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-post100000-tail-atom-exact-runner.md.
```

It records

```
tail_atom_count=31,  tail_atom_p_range=[101087,134047]
kind_counts={assigned_slot:15,postwall_first_prime:8,tail_slot:8}
all_composite_atoms_divisible_by_recorded_factor=true
all_composite_recorded_factors_are_smallest=true
all_postwall_first_primes_verified=true
all_preprime_slots_composite_in_arrival_rows=true
finite_atom_branch_closed_for_registered_atoms=true.
```

The 31 tail atoms split into 23 composite slots and 8 post-wall first-prime boundaries. For every composite slot the recorded factor divides the slot and equals its smallest factor. For every post-wall first-prime boundary, direct trial division confirms primality in the recorded range. Moreover, in each arrival row all slots before the first-prime boundary are verified composite. Thus the finite post-100000 tail packet is no longer a parallel exit for the current registered terminal configuration.

Thus the latest branch is compressed to

`BranchReplayColumnCRTPDECExclusion`.

This still does not prove the row/column theorem. It removes the finite atom alternative and leaves the global persistent replay family as a named ColumnCRT/PDEC obstruction. □

Reduction-closed Statement 2.96 (Fresh-modulus escalation)**.** *A fixed finite ColumnCRT replay class is not a terminal structure for an infinite counterexample chain. New prime layers are coprime to the registered finite modulus and force unbounded modulus escalation, a PDEC/ColumnCRT defect, or a tail-sieve stability contradiction.*

Status and reference. The branch replay fresh-modulus escalation router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_fresh_modulus_escalation_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-fresh-modulus-escalation-router.md.
```

It records

```
finite_atom_branch_closed_for_registered_atoms=true
registered_replay_block_count=6
fresh_prime_sample_count_per_block=8
all_registered_blocks_have_coprime_fresh_layers=true
minimum_first_fresh_log10_gain=2.400
minimum_sample_log10_gain=19.435
finite_crt_terminal_description_excluded=true.
```

Let a registered replay class have finite modulus Q. Every prime $\ell \nmid Q$ is a new CRT coordinate independent of the old replay class. When such an ℓ enters the active wheel, the old congruence modulo Q cannot decide the ℓ-layer. A persistent counterexample chain must therefore absorb ℓ into a larger modulus, fail at that layer as a named PDEC/ColumnCRT defect, or prove that the tail sieve absorbs the new coordinate.

The current six registered replay blocks make this explicit. Each has coprime fresh-prime layers. Adding only the first fresh prime increases the modulus by at least $10^{2.400}$, and adding the first eight fresh primes in the recorded sample increases it by at least $10^{19.435}$. Thus a finite registered ColumnCRT class cannot be a final stable object for an infinite counterexample chain.

Thus the latest branch is compressed to

$$\texttt{UnboundedFreshModulusEscalationPDECOrTailSieveStabilityContradiction}.$$

This is still not an unconditional closure. It identifies the remaining global obstruction as unbounded fresh-modulus escalation or a tail-sieve stability contradiction, rather than a fixed finite CRT atom. □

Reduction-closed Statement 2.97 (Fresh-modulus tail-sieve bridge)**.** *After finite ColumnCRT terminal classes have been excluded, a persistent branch replay family has only two remaining branch-replay exits. Either a fresh prime layer creates a PDEC/ColumnCRT defect, or the non-defective unbounded fresh-prime layers form the same one-residue-per-prime tail rough object used in the $B = 3$ lower-sieve ledger.*

Status and reference. The fresh-modulus tail-sieve bridge router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail_sieve_bridge_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail-sieve-bridge-router.md.
```

It records

```
fresh_modulus_escalation_registered=true
finite_columncrt_terminal_excluded=true
non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
tail_object_interface_closed=true
conditional_external_tail_sieve_closed=true
strict_self_contained_tail_sieve_closed=false
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

Let Q be the finite modulus of a registered replay class. Once the fixed finite terminal explanation is removed, every subsequent fresh prime $\ell \nmid Q$ is an independent CRT coordinate. If the ℓ-layer cannot be registered without a projection collision, residue repetition, moving-support escape, or ColumnCRT defect, the branch has entered the named fresh-layer PDEC exit. Otherwise the only information carried by that layer is one forbidden residue class modulo ℓ. Repeating this over the unbounded fresh-prime layers gives the tail object

$$S(P) = \#\{1 \le k < P : \ k \text{ avoids one prescribed residue class modulo each } q \le P^{0.43}\}.$$

For each squarefree $d < P$, CRT counting gives

$$A_d(P) = \frac{P-1}{d} + r_d, \qquad |r_d| \le 1,$$

so this is exactly the same interface as the existing $B = 3$ lower-sieve rough-object ledger. The imported tail ledger has

$$\alpha = 0.43, \qquad \text{model main at } P = 100000 = 4896.256004,$$

and its ten-percent main term still exceeds 401 by 88.625600. Thus, if external explicit Mertens/Dusart estimates or the standard beta-sieve input are accepted, the tail-sieve stability branch is conditionally closed.

The strict self-contained branch is not closed by this router. It still requires the internal Mertens/PNT/Dusart reciprocal-prime tail proof and a separate exclusion of fresh-layer PDEC/ColumnCRT. The current branch-replay frontier is therefore

$$\texttt{FreshLayerPDECColumnCRTExclusionOrSelfContainedTailSieveStabilityClosure}.$$

This is a compression of the latest branch, not an unconditional proof of the row/column theorem.

□

Reduction-closed Statement 2.98 (Self-contained tail-sieve synchronization)**.** *For the branch replay fresh-modulus branch, the old self-contained tail-sieve opening is no longer an active branch-replay obstruction after the later strict Mertens/PNT and $B = 3$ total-variation synchronization certificates are imported. The remaining branch-replay obstruction is the fresh-layer PDEC/ColumnCRT exit.*

Status and reference. The synchronization router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_modulus_tail
_self_contained_sync_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-modulus-tail
-self-contained-sync-router.md.
```

It records

```
previous_tail_object_interface_closed=true
previous_non_pdec_unbounded_fresh_layers_force_tail_rough_object=true
strict_self_contained_mertens_tail_proved_latest=true
b3_tv_strict_self_contained_synchronized_latest=true
strict_self_contained_tail_sieve_closed_for_branch_replay=true
fresh_layer_pdec_excluded=false
row_column_unconditional_closed=false.
```

The previous bridge had reduced the non-PDEC unbounded fresh layers to the one-residue-per-prime $B = 3$ tail rough object, but it kept the strict self-contained branch open under the coarse name

$$\texttt{SelfContainedDusartReciprocalPrimeProofAppendixXGe10372}.$$

Later strict synchronization certificates refine this coarse atom: the theta/PNT envelope and the Meissel–Mertens B_1 interval are imported as self-contained inputs, and the $B = 3$ total-variation ledger is synchronized with that Mertens tail closure. Hence the branch-replay strict remaining basis changes from

```
FreshLayerPDECColumnCRTExclusion
AND SelfContainedDusartReciprocalPrimeProofAppendixXGe10372
```

to

```
FreshLayerPDECColumnCRTExclusion.
```

This still does not close the row/column theorem. It only removes the old tail-sieve analytic opening from this branch; the fresh-layer PDEC/ColumnCRT obstruction remains active. □

Reduction-closed Statement 2.99 (Local fresh-layer collision exclusion)**.** *For the registered branch-replay blocks, a fresh-prime layer cannot create a local projection collision inside the recorded primitive support windows. Any remaining fresh-layer PDEC must therefore be a support-motion escape, a remote P-space ColumnCRT/PDEC recurrence, or a new moving-family block.*

Status and reference. The local collision router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_layer_local_collision_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-layer-local-collision-router.md.
```

It records

```
period_p=5680
registered_block_count=6
all_sample_fresh_primes_coprime_to_period_p=true
all_first_fresh_primes_exceed_support_width=true
all_first_fresh_primes_exceed_audit_slots=true
all_registered_samples_injective_on_local_windows=true
minimum_first_fresh_minus_support_width=178
minimum_first_fresh_minus_audit_slots=178
local_fresh_layer_projection_collision_excluded=true
row_column_unconditional_closed=false.
```

Let the local period step be $M = 5680$, and let ℓ be a registered fresh prime. In every recorded sample, $\gcd(M, \ell) = 1$. If two slots j_1, j_2 in a local window of length W collide modulo ℓ, then

$$a + j_1 M \equiv a + j_2 M \pmod{\ell},$$

so $\ell \mid (j_1 - j_2)M$. Since M is invertible modulo ℓ, this forces $\ell \mid j_1 - j_2$. But the recorded windows satisfy $W < \ell$, with minimum first-fresh surplus 178 over both support width and audit-slot count. Hence $|j_1 - j_2| < \ell$, and the only possible collision is $j_1 = j_2$.

Thus the local projection-collision subcase of fresh-layer PDEC is excluded for the registered blocks. The remaining branch-replay obstruction is

$$\texttt{FreshLayerSupportMotionEscapeOrRemoteColumnCRTPDECExclusion}.$$

This is still a named open exit, not a global unconditional proof. □

Reduction-closed Statement 2.100 (Registered support-motion escape removal)**.** *For the registered branch-replay blocks, the support-motion escape is not a remaining local mechanism. After local fresh-layer projection collisions and same-package local support drift are excluded, a persistent registered replay must be a remote P-space ColumnCRT/PDEC class; if the blocker package changes, it is an unregistered moving-family branch.*

Status and reference. The fresh support-motion global router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_support_motion_global_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-support-motion-global-router.md.
```

It records

```
period_p=5680
registered_block_count=6
local_fresh_layer_projection_collision_excluded=true
registered_local_support_motion_excluded=true
persistent_registered_replay_routes_to_columncrt_pdec=true
registered_support_motion_escape_closed=true
minimum_cycle_log10_margin_over_support_width=30.737
minimum_p_space_columncrt_modulus_log10=36.337
remote_pspace_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The preceding local-collision certificate removes collisions inside the recorded primitive support windows. The earlier support-gap replay lemma also removes same-package local support drift: a nonzero replay by the same blocker package must move by a multiple of $\mathrm{lcm}(B)$, and the least recorded margin over support width is $10^{30.737}$. Hence a registered block cannot persist through local support motion.

The global dichotomy then leaves only two possibilities. If a registered block recurs infinitely often, it is promoted to a fixed P-space ColumnCRT/PDEC class, whose smallest recorded modulus is about $10^{36.337}$. If the blocker package changes, the branch is not registered support motion but an unregistered moving-family branch requiring a new named router. The current branch-replay frontier is therefore

$$\texttt{RemotePspaceColumnCRTExclusionOrUnregisteredMovingFamilyRouter}.$$

This is a further compression of the branch-replay interface, not a proof of the row/column theorem.

□

Reduction-closed Statement 2.101 (Remote ColumnCRT feedback router)**.** *Within the registered branch-replay chain, a remote P-space ColumnCRT class is no longer allowed to remain as a bare fixed-period terminal. Such a bare class is absorbed by the already recorded finite-atom runner, the fresh-modulus escalation router, and the strict tail-sieve synchronization. Therefore the current branch-replay remainder is narrowed to a materialized fresh-layer PDEC/ColumnCRT event or to an unregistered moving-family event.*

Status and reference. The feedback router is generated by

```
experiments/prime_matrix_cycle_debt_remote_columncrt_feedback_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-remote-columncrt-feedback-router.md.
```

It records

```
registered_remote_block_count=6
minimum_remote_pspace_columncrt_modulus_log10=36.337
maximum_remote_pspace_columncrt_modulus_log10=103.103
minimum_first_fresh_log10_gain=2.400
minimum_sample_fresh_log10_gain=19.435
registered_finite_atoms_absorbed=true
fixed_finite_remote_crt_not_terminal=true
non_pdec_fresh_tail_sieve_closed=true
bare_remote_pspace_columncrt_terminal_closed=true
materialized_fresh_layer_pdec_columncrt_excluded=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

The point is only a feedback reduction. If the remote class is an isolated or finite registered atom, it has already been handled by the exact finite runner. If it is a fixed finite CRT replay class, the fresh-modulus escalation router shows that it cannot be a terminal infinite counterexample description. If fresh layers do not materialize a PDEC/ColumnCRT event, the non-PDEC unbounded fresh-layer branch has already been routed to and closed by the strict tail-sieve synchronization. Hence the broad remote ColumnCRT label can be removed as a bare terminal explanation.

The remaining branch-replay interface is consequently

`MaterializedFreshLayerPDECColumnCRTExclusionOrUnregisteredMovingFamilyRouter`.

This still leaves a real open obligation: one must exclude materialized fresh-layer PDEC/ColumnCRT events or unregistered moving-family recurrences. It is not a proof of the row/column theorem. □

Reduction-closed Statement 2.102 (Fresh-layer PDEC admission firewall)**.** *Within the current cycle-debt branch-replay corpus, the materialized fresh-layer PDEC/ColumnCRT label has no unnamed admitted instance. A future fresh-layer PDEC obstruction must be submitted as an explicit primitive same-formal-unit schema; otherwise the branch is an unregistered moving family.*

Status and reference. The firewall router is generated by

```
experiments/prime_matrix_cycle_debt_fresh_layer_pdec_admission_firewall_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-fresh-layer-pdec-admission-firewall-router.md.
```

It records

```
registered_block_count=6
minimum_first_fresh_minus_max_window=178
minimum_remote_plus_first_fresh_log10=38.803
local_registered_materialized_pdec_closed=true
current_materialized_pdec_frontier_closed=true
pdec_family_explicit_input_boundary_closed=true
newlayer_schema_admission_closed=true
newlayer_ranktwo_budget_independent_gate_removed=true
current_corpus_materialized_fresh_layer_pdec_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_excluded=false
row_column_unconditional_closed=false.
```

For the registered blocks, every sampled first fresh prime is larger than the support and audit windows, and the local affine projection is injective. Thus local fresh-layer phase reuse cannot materialize a PDEC event inside the registered support. If a remote event is to be admitted as PDEC, it must pass the already registered PDEC boundary: the same formal unit, one fixed phase map, at least three physical primitive atoms after quotienting, non-tautology, rank at least two, and cap stability. ColumnCRT or displacement cases are absorbed before admission; sparse fallbacks require their own packet extractor.

Consequently the current branch-replay interface is narrowed to

`FutureExplicitPrimitiveFreshLayerPDECSchemaIfNewOrUnregisteredMovingFamilyRouter`.

This does not prove that no future primitive schema can exist, and it does not exclude unregistered moving families. It only removes the current unnamed materialized fresh-layer PDEC/ColumnCRT label. □

Reduction-closed Statement 2.103 (Cycle-debt branch-replay current frontier zero)**.** *In the present materialized cycle-debt branch-replay corpus, the remaining fresh-layer PDEC/ColumnCRT and unregistered moving-family labels are reduced to future admission firewalls rather than active terminal instances. The branch-replay subfrontier is therefore empty in the current corpus, while the global row/column theorem still depends on the separate final-input gates.*

Status and reference. The current-frontier-zero router is generated by

```
experiments/prime_matrix_cycle_debt_branch_replay_current_frontier_zero_router.py
```

and archived at

```
docs/monograph/prime-matrix-cycle-debt-branch-replay-current-frontier-zero-router.md.
```

It records

```
current_fresh_layer_pdec_frontier_closed=true
future_explicit_primitive_fresh_layer_pdec_schema_submitted=false
unregistered_moving_family_schema_submitted=false
future_pdec_schema_admission_discipline_closed=true
future_sparse_schema_admission_discipline_closed=true
final_input_firewall_boundary_closed=true
no_hidden_terminal_remaining=true
cycle_debt_branch_replay_current_materialized_frontier_zero=true
future_explicit_schema_global_nonexistence_proved=false
row_column_unconditional_closed=false.
```

The proof is a current-instance compression. A future primitive fresh-layer PDEC schema has not been submitted, so it is not an active mathematical object in the current branch-replay corpus; it is only an admission discipline. The same applies to an unregistered moving family unless it supplies source, shape, phase, persistence, and no-loss ledger fields. The PDEC and sparse schema firewalls specify how such a future object would have to re-enter.

Thus the current materialized cycle-debt branch-replay frontier is

`CycleDebtBranchReplayCurrentMaterializedFrontierZeroWithFutureSchemaFirewall`.

This is still not a proof of the row/column theorem. The global remainder is

`GlobalFinalInputsStillOpen`,

including the noncanonical legal-closure mode and the DStructure/Rankin promotion gate, unless future explicit schemas are submitted and separately excluded. □

Chapter 3

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r, c) = (r-1)P + c, \qquad 1 \leq r, c \leq P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non-P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Reduction-closed Statement 3.1 (A/B reduction, recorded form)**.** *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem. □

3.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer $w > 0$ and ask whether there is an entry x in the back half of the $P \times P$ matrix such that both x and $x - w$ are nonzero modulo every prime $p \leq P$, or equivalently both x and $w - x$ are nonzero modulo every prime $p \leq P$.

Proposition 3.2 (Equivalence and strength of the two-point rough-pair problem)**.** *For fixed even $w > 0$, the two conditions*

$$p \nmid x(x-w) \quad (p \leq P)$$

and

$$p \nmid x(w-x) \quad (p \leq P)$$

are identical. In the region $1 < x \leq P^2$, any integer nonzero modulo every prime $p \leq P$ is prime. Consequently, a proof of this two-point assertion for every sufficiently large P would imply infinitely many prime pairs of gap w; in particular $w = 2$ would imply the twin-prime conjecture.

Proof. For each $p \le P$, the condition $p \nmid x - w$ is $x \not\equiv w \pmod p$, and the condition $p \nmid w - x$ is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and w modulo each p, with possible collision when $p \mid w$.

If $1 < x \le P^2$ is nonzero modulo every prime $p \le P$ and is composite, then it has a prime factor $q \le \sqrt{x} \le P$, contradicting the nonzero-residue condition. Thus such an x is prime. If the two-point assertion holds for all sufficiently large $P > w$, then x and $x - w$ are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap w. □

Remark 3.3 (Status)**.** This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

3.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md`.

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod p \qquad (p \le P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

Definition 3.4 (Two-point secondary-sieve field)**.** Fix an even integer $w > 0$. A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate x fails at least one of the two congruence conditions $p \nmid x$ or $p \nmid x - w$ for some prime $p \le P$. Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

Lemma 3.5 (Fixed-order local crossing complexity)**.** *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \le R, \qquad R \le (\log P)^{C_R},$$

with coefficient mass $\sum_{d \le R} |\lambda_d| \le (\log P)^{C_\lambda}$ and short-block shifts $|h| < L \asymp (\log\log P)^2$. For every fixed order k, the effective historical complexity satisfies

$$Q_{\mathrm{eff},k} \ll_k \left(\sum_{d \le R} |\lambda_d| \right)^k R^k L^{O(k)} \le (\log P)^{C_k}.$$

In particular the required $k = 2, 3$ local crossing tests have polylogarithmic effective complexity.

Proof. After expanding the k truncated weights, only moduli $d_1, \ldots, d_k \le R$ occur. For fixed new-line primes p_i, phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\operatorname{lcm}(d_1, \ldots, d_k) \operatorname{lcm}(p_1, \ldots, p_k).$$

The primes p_i are the explicit variables being summed in the crossing estimate; they contribute the usual Y^k scale, not hidden historical complexity. The historical part is bounded by $\operatorname{lcm}(d_1, \ldots, d_k) \le R^k$, the block shifts contribute $L^{O(k)}$ patterns, and the coefficient expansion contributes $(\sum |\lambda_d|)^k$. This proves the displayed bound. The true residual set U_Y is not expanded into a full primorial period. □

Lemma 3.6 (Zero-mass and smoothing discharge). *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \le 0.45, \qquad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \ge 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \le 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \le C_{\text{end}}\Delta + o(1),$$

then choosing $\Delta \le 0.01/C_{\text{end}}$ gives a smooth-to-sharp loss < 0.03 for all sufficiently large P.

Proof. For every integer $R \ge 0$,

$$1_{R=0} \ge 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by $X_B \ge 0$ and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 1 - M_1 + M_2 - M_3 \ge 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs $W_- \le 1_{|t|<1} \le W_+$ whose difference is supported in $1 - \Delta \le |t| \le 1 + \Delta$. The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most $C_{\text{end}}\Delta + o(1)$, which is < 0.03 after the stated choice of Δ and then taking P large. □

Lemma 3.7 (Adaptive true-hit layering dichotomy). *Let U_Y be the current true residual set and define, for each new prime $p > Y$,*

$$a_p = \frac{1}{|U_Y|}\#\{x \in U_Y : x \equiv 0 \text{ or } w \pmod{p}\}.$$

Fix $\eta = 0.05$. Either some p has $a_p > \eta$, in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers $\mathcal{P}_j$ satisfying

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \le 0.4,$$

and, except for the final layer, $\nu_j^{\text{real}} > 0.4 - \eta \ge 0.35$.

Proof. If $a_p > \eta$ for some p, then a fixed positive proportion of U_Y lies in the two residue classes $0, w$ modulo p. This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all $a_p \le \eta$. Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most η, every non-final completed layer has sum greater than $0.4 - \eta$. This gives the claimed adaptive true-hit decomposition. □

Proposition 3.8 (Recorded conditional closure chain for the two-point secondary sieve)**.** *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime* $45°$ *main-synchronization peak, Single-Prime CRTDefect exits, the normalized* $q = 2$ *parity skeleton, the moment bounds* M_1, M_2, M_3*, and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice

$$\alpha_0 = 0.75, \qquad c_0 = 1/2, \qquad \nu_j^{\mathrm{real}} \le 0.4$$

for adaptive true-hit layers, subject to the same fixed-w *convention used in the research manuscript.*

Proof sketch and status. The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The $45°$ main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when $w = 2$, is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof. □

Proposition 3.9 (Window-compressed conditional variant)**.** *Keep the same fixed even integer* $w > 0$ *and the same remaining I3-Core input as in the preceding proposition. Let* I *be any consecutive back-half window of* $H(P)$ *rows in the* $P \times P$ *matrix, so that* $|I| = H(P)P + O(P)$*. By the fixed-order complexity lemma, write*

$$Q_{\mathrm{eff}} \le C_Q (\log P)^C.$$

For a prescribed middle-scale error budget $\varepsilon_{\mathrm{SC2}} > 0$*, the conditional row threshold supplied by the current ledger is*

$$H_{\min}^{\mathrm{cond}}(P; \varepsilon_{\mathrm{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\mathrm{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H_{\min}^{\mathrm{asym}}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \qquad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \qquad p \nmid x(x-w) \quad (p \leq P).$$

When $x - w > 1$, both x and $x - w$ are primes.

Recorded scale check. The only scale-sensitive term in the recorded chain is the middle/high-scale $K = 2$ near-crossing error. With top layer $Y = P^{3/4}$ and $Q_{\mathrm{eff}} \leq (\log P)^C$, replacing the original back-half length $\asymp P^2$ by $|I| \asymp H(P)P$ changes the normalized error to

$$\frac{Q_{\mathrm{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on $H(P)$ makes this at most $\varepsilon_{\mathrm{SC2}}$ in the explicit form, or $o(1)$ in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted. □

Remark 3.10 (Referee boundary)**.** The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

Remark 3.11 (Current obstruction)**.** The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

Remark 3.12 (Latest TRC-1G reduction)**.** The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes $0, w \pmod p$ for some new prime $p > Y$, then Parseval gives a non-zero Fourier coefficient modulo p of size $> 1/5$ after normalization. Equivalently, the three long-direction difference families $0, \pm w \pmod p$ carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

Remark 3.13 (Total large-incidence reduction)**.** The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to P, and in the P^2 matrix range this forces both x and $x - w$ to be prime. The CRT model predicts the normalized left side to be $2\log(1/\alpha) + o(1)$; for $\alpha = 3/4$ this is $2\log(4/3) = 0.57536\ldots$, leaving margin $0.42463\ldots$. The unproved part is the true-residual total-incidence estimate with an error below that margin.

Remark 3.14 (RB–TLI audit boundary)**.** The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing $x = pm$ or $x = w + pm$, the variable m again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \le 2C_+|I|V_2(P^\alpha)/p + R_p.$$

The denominator requires a model-level lower bound $|U_{P^\alpha}(I)| \ge c_-|I|V_2(P^\alpha)$. For $\alpha = 3/4$, this route would need $C_+/c_- < 1/(2\log(4/3)) = 1.738\ldots$. This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

Remark 3.15 (Local collapse when $q \mid w$)**.** If an old odd prime q divides the fixed even difference w, then the two forbidden classes $0, w \pmod q$ coalesce into a single forbidden class. The local density is therefore $1-1/q$ instead of $1-2/q$, giving the singular local gain $(q-1)/(q-2)$. This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both $|U_Y|$ and the model numerator $A_p \sim 2|I|V_w(Y)/p$ for all new primes $p > Y > |w|$. Thus the main relative constant remains $2\log(1/\alpha)$; the collapse improves finite constants but does not by itself close the hardest case $w = 2$.

Remark 3.16 (Buchstab semiprime transition for $w = 2$)**.** The latest numerical audit of the hardest case $w = 2$ shows that the naive constant $2\log(1/\alpha)$ is not the correct conditional main term after the old primes have been sieved. For $\alpha > 2/3$, any new-prime hit $p \mid x$ or $p \mid x - 2$ with $Y = P^\alpha < p \le P$ forces the complementary quotient to be prime, because it is Y-rough and smaller than Y^2. Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2\log((2-\alpha)/\alpha)}{1 + \log((2-\alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example, $E_U D = 0.691696$ at $P = 10007, \alpha = 3/4$, compared with $K(3/4) = 0.676220$. The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

Remark 3.17 (BST error decomposition)**.** The refined scan decomposes the remaining error. At $P = 10007, \alpha = 3/4$, the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the x-side and $(x-2)$-side large-incidence counts is approximately $-9.3\cdot 10^{-5}$. The zero-incidence set is independent of the intermediate cutoff α, as it is exactly the set unsieved by all primes up to P. Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I,\ pm - 2 \in U_Y\} \le (B(\alpha) + o(1))|U_Y|.$$

This is the current BST–2 obstruction.

Remark 3.18 (BMD reduction)**.** The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting $x = pm$, the condition $pm - 2$ is unsieved by old primes is, for each $q \le Y$,

$$pm \not\equiv 2 \pmod q.$$

The prime $q = 2$ is peeled off deterministically: for odd primes p, m, the integer $pm-2$ is odd. Thus the multiplicative sieve is taken over odd q. For each odd $q \le Y$, since $p, m > Y \ge q$, the forbidden curve has $(q-1)$ points inside $(\mathbb{F}_q^\times)^2$, out of $(q-1)^2$ possible non-zero pairs. The local forbidden density is therefore $1/(q-1)$. The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences $pm \equiv 2 \pmod d$, with odd squarefree d, must be $o(|U_Y|)$ for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

Remark 3.19 (Row-column input for BMD)**.** If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd d, expanding

$$1_{pm\equiv 2 \pmod d} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi\neq\chi_0} \overline{\chi}(2)\chi(p)\chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d\le D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi\neq\chi_0} \left| \sum_{Y<p\le P} \chi(p) \sum_{\substack{m\in J_p \\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

Remark 3.20 (Direct BMD attack via an E_2 Bombieri–Vinogradov input)**.** Set $N \asymp P^2$ and

$$a_n = \#\{(p,m) : Y < p \le P,\ m \text{ prime},\ n = pm,\ n \in I\}.$$

For $\alpha > 2/3$ the multiplicity of a_n is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree d is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n\in I \\ n\equiv 2 \pmod d}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n\in I \\ (n,d)=1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted E_2 sequences at level

$$d \le P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d\le P/\log^B P} \max_{(a,d)=1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is $o(|U_Y|)$ since $|U_Y| \asymp N/(\log P)^2$. The condition $\alpha < 1$ ensures $Y < P/\log^B P$ for large P, so all one-prime forbidden curves $q \le Y$ are included in the available distribution range. Therefore BMD is closed if this standard E_2 BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

Remark 3.21 (Self-contained BV–E_2 obligation)**.** The local BV–E_2 appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block $p \sim P_1$, $m \sim M_1$, with $P_1 \asymp M_1 \asymp P$ and $Q \asymp P/\log^B P$, Cauchy plus the large sieve gives size roughly $P^3/\log^{2B} P$, while the required Bombieri–Vinogradov error is $P^2/\log^A P$. Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

Remark 3.22 (Weighted BE2–3K core)**.** BMD does not require the full $\max_a$ Bombieri–Vinogradov statement. It only needs the fixed class $2 \bmod d$ against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \le Q} \lambda_d \left(\sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening $\sum_r |T_r|^2$, the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \qquad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\left(-\frac{2h\overline{s_1}\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\overline{d_1}}{d_2} \right).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2–3K in the notes. Proving BE2–3K gives BE2–3, then weighted BV–E_2, then BMD.

Remark 3.23 (Referee status of the no-black-box claim)**.** At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2–3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2–3K remains the unique missing line-by-line proof obligation.

Remark 3.24 (Further reduction of BE2–3K)**.** The BE2–3K core can be narrowed further. Pointwise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the d_1, d_2, h families. The common-divisor strata $(d_1, d_2) > 1$ are not the main obstruction: compatibility forces $s_1 \equiv s_2$ modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

Remark 3.25 (External-theorem closure of KLS-window)**.** The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

Chapter 4

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 4.1 (RH contradiction-field synthesis, recorded form)**.** *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. □

Chapter 5

Unified Dependency Map

The two programs share the same logic:

counterexample $\Rightarrow$ structured covering field $\Rightarrow$ named exits $\Rightarrow$ capacity/no-cycle contradiction.

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 6

Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

6.1 Proposed Book Structure

Part	Contents	Status
I	Unified contradiction-field method: entrance, local rigidity, exits, capacity, no-cycle potential	Methodological framework
II	Prime-matrix row/column program: CRT skeleton, A/B reduction, Structured-EHPD interface	Reduction package; not final theorem
III	Two-point secondary sieve: TLI, BST, BMD, WBE2, BE2–3K, KLS-window	External-deep-theorem closure for BMD
IV	Zeta-zero contradiction field: explicit-formula entrance and controlled exits	Verification package; not final RH theorem
V	Dependency map, referee checklist, optimization routes	Audit and future work

6.2 Latest Two-Point Chain

For the $w = 2$ hardest two-point branch, the current analytic chain is

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST–2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

6.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

$$\begin{aligned} 13 \le P < 61 &: \text{ low-range final certificate,} \\ 61 \le P \le 103 &: \text{ narrow-band collision-energy certificate,} \\ 107 \le P \le 229 &: \text{ exact } P/5 \text{ splitting recursion,} \\ 233 \le P \le 13207 &: \text{ integer cross-multiplied product certificate,} \\ P \ge 13208 &: \text{ Rosser–Schoenfeld explicit constants.} \end{aligned}$$

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at $P = 41$ close by exact finite set-cover dynamic programming. The narrow band $61 \le P \le 103$ covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since $P \ge 13208$ implies $\lfloor P/5 \rfloor \ge 2641$, the factor $1 + 1/(2\log^2\lfloor P/5 \rfloor)$ is below 1.009, hence the manuscript's conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

6.4 Prime-Matrix WSH-Hall Phase Certificate

The latest row-program compression keeps the endpoint status honest. The terminal RCI branch has been reduced to the comparison between no-tail primes and balanced two-tail semiprimes. The next local interface is

$$\text{RCI/PDEC} \Rightarrow \text{Distributed-RCI} \Rightarrow \text{WSH-Hall/PDEC}.$$

Here WSH-Hall means that every balanced two-tail semiprime in a row can be injectively matched to a same-row prime through an offset allowed by the small-prime wheel. If such a Hall matching fails, the failing interval must be routed to a named exit: persistent endpoint CRT defect, tail-anchor concentration, or endpoint prime deficit.

A finite phase certificate is archived in

`docs/monograph/prime-matrix-wsh-hall-phase-certificate.md`.

For $17 \le p \le 2000$, $y = \lfloor p/e \rfloor$, and wheel primes $2, 3, 5, 7, 11, 13$, it scans 215074 rows containing balanced two-tail semiprimes. It finds no Hall matching failure, no failure under the candidate radius $3\log^2 q$, maximum minimal matching radius 132, maximum ratio $R/\log^2 q = 2.3771082468335174$, and minimum margin #primes − #semiprimes = 1. The certificate rows list semiprime labels, matched primes, offsets, fixed-offset loads, and wheel-phase loads; each matched offset is checked against the congruence condition $d \not\equiv -b \pmod{r}$.

This is a computational-certificate update, not a global theorem. The remaining proof obligation is either a uniform WSH-Hall theorem or a proof that every Hall defect necessarily triggers PDEC, Tail-anchor, or Endpoint deficit.

6.5 WSH-Hall Defect Trichotomy

The corresponding defect audit is archived in

`docs/monograph/prime-matrix-wsh-hall-defect-trichotomy.md`.

It converts a possible Hall failure into a consecutive block B_0 of balanced two-tail semiprimes with

$$|N_R(B_0)| < |B_0|.$$

The same block is then examined through five projections: endpoint prime deficit, tail-factor load, small-wheel residue load, fixed-offset capacity, and the terminal mirror $n \mapsto q^2 - n$. The resulting target is the named interface

WSH-Expansion-or-Defect.

If Tail-anchor and fixed-offset/PDEC thresholds do not fire, one must prove the expansion inequality

$$|N_R(B_0)| \geq |B_0|$$

for $R = C \log^2 q$; otherwise the failing block must be absorbed by Endpoint/PDEC. A forced-radius audit on the 44 tight certificate rows produces 124 explicit Hall defects and records their tail, phase, offset, and mirror data. This audit fixes the certificate fields and exit geometry; it is not a substitute for the uniform expansion theorem.

6.6 WSH Expansion Margin Ledger

The expansion term itself is audited in

`docs/monograph/prime-matrix-wsh-expansion-margin-ledger.md`.

For $17 \leq p \leq 2000$, $R = \lceil 3 \log^2 q \rceil$, and the same $y = \lfloor p/e \rfloor$, the audit scans every consecutive balanced-semiprime block in every relevant row. It checks 8440419 blocks across 215074 rows and finds

$$\min_B \bigl(|N_R(B)| - |B|\bigr) = 1,$$

with no zero-surplus block. Only seven blocks have surplus at most 1.

This strengthens the finite certificate from "a matching exists" to "every interval-Hall block has positive margin" in the audited range. It still does not prove the uniform theorem. The remaining target is:

WSH Positive Expansion or Named Defect.

After Tail-anchor, fixed-offset/PDEC, and endpoint-mirror deficit exits are removed, one must prove the positive expansion inequality for all formal blocks.

6.7 Short Critical Block Reduction

The short-critical-block audit is archived in

`docs/monograph/prime-matrix-wsh-short-critical-block-reduction.md`.

It reruns the expansion-margin audit with tight surplus threshold 2. Among the same 8440419 consecutive semiprime blocks, only 45 have surplus at most 2, and every such block has at most three semiprimes. The distribution is:

surplus	$\|B\|$	count
1	1	4
1	2	3
2	1	24
2	2	10
2	3	4

This suggests the next reduction:

$$\text{SCB–1: } |B| \geq 4 \Rightarrow \text{positive expansion or named defect,}$$

$$\text{SCB–2: } |B| \leq 3 \Rightarrow \text{local exclusion or named defect.}$$

This is still finite evidence and a proof target, not a theorem. It is useful because SCB–2 is a small local problem involving only one, two, or three two-tail semiprimes, where tail labels, fixed offsets, wheel residues, and endpoint mirrors can be written explicitly.

6.8 SCB–2 Local Routing Certificate

The SCB–2 local certificate is archived in

`docs/monograph/prime-matrix-wsh-scb2-local-exclusion-template.md`.

It enumerates all consecutive blocks with $|B| \leq 3$ for $17 \leq p \leq 2000$. The block counts are

$\|B\|$	count	$\min(\|N_R(B)\| - \|B\|)$
1	1496400	1
2	1281326	1
3	1088870	2

There are no negative-surplus or zero-surplus short blocks. The local routing statement is:

$$|B| \leq 3, \quad |N_R(B)| < |B| \Rightarrow \text{Endpoint/PDEC deficit,}$$

with additional Tail-repeat and fixed-offset-full-load checks on tight two- and three-point blocks. Thus SCB–2 is closed as a routing lemma modulo the still-open Endpoint/PDEC exclusion.

6.9 SCB–1 Long-Block Certificate

The complementary long-block certificate is archived in

`docs/monograph/prime-matrix-wsh-scb1-long-block-expansion-template.md`.

It enumerates all consecutive balanced two-tail semiprime blocks with $|B| \geq 4$ for $17 \leq p \leq 2000$ and the same radius $R = \lceil 3 \log^2 q \rceil$. The audit checks 4573823 long blocks and finds

$$\min_{|B| \geq 4} \big(|N_R(B)| - |B|\big) = 3,$$

with no negative-surplus or zero-surplus long block. The four tight long blocks have sizes 4 or 5, and every tight block carries both Endpoint-margin and fixed-offset-full-load labels.

This is a finite certificate, not the uniform SCB–1 theorem. Its proof-theoretic role is to isolate the next target:

$$\text{long-block positive expansion} \quad \text{or} \quad \text{fixed-offset/PDEC absorption.}$$

Together with SCB–2, the present WSH chain is therefore:

$$\text{SCB–2 routing modulo Endpoint/PDEC}, \qquad \text{SCB–1 finite long-block margin},$$

The remaining global obstructions are fixed-offset/PDEC absorption and Endpoint/PDEC exclusion.

6.10 Fixed-Offset/PDEC Absorption

The fixed-offset absorption interface is archived in

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-absorption.md`.

It proves the no-third-escape lemma behind the fixed-offset-full-load label. If $p < q$ are consecutive primes and $1 < n < q^2$ is composite, then n has a prime factor at most p. Hence, if a fixed offset d is allowed by the wheel $2, 3, 5, 7, 11, 13$ for every $b \in B$, every missing candidate $b + d$ is explained by some factor in $(13, p]$.

The finite ledger

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-ledger.md`

checks the tight SCB–1 long blocks. It finds 11 full-load offset rows and 47 candidates, among which 15 are prime and 32 are missing. Every missing candidate has an explaining factor at most p; there is no unexplained candidate.

Thus fixed-offset-full-load is no longer an unnamed escape. It routes to expansion, Tail-anchor, PDEC/low-mod CRTDefect, or SAE/Endpoint. The still-open global target is

$$\text{FO-PDEC: distributed explaining factors} \Rightarrow \text{persistent PDEC or sparse SAE/Endpoint.}$$

6.11 FO-PDEC Low-Mod Equations

The low-mod equation audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-hard-attack.md`.

For every missing candidate $n = b + d = (r - 1)q + c$ and every explaining factor ℓ, the exact CRT equation is

$$r \equiv 1 - cq^{-1} \pmod{\ell}.$$

If $b = uv$ is the corresponding two-tail semiprime, the same missing candidate also satisfies

$$uv + d \equiv 0 \pmod{\ell}.$$

The finite audit checks 43 such low-mod equations in the tight fixed-offset ledger. There are no CRT-equation failures and no bilinear-equation failures.

This closes the equation layer of FO-PDEC. It does not close the global theorem. The remaining target is the quantitative inequality

$$\text{low-mod defect energy} \geq \text{PDEC threshold},$$

unless the non-persistent part is absorbed by SAE/Endpoint.

6.12 FO-PDEC Energy Lower Bound

The energy ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-energy-lemma.md`.

For a multiset E of explaining equations, let t_ℓ be the number of equations using ℓ and let s_ℓ be the number of occupied row residues modulo ℓ. Define

$$\mathcal{E}_\ell(E) = t_\ell - \frac{s_\ell t_\ell}{\ell}, \qquad \mathcal{E}(E) = \sum_\ell \mathcal{E}_\ell(E).$$

For every $0 < \theta < 1$, either some explaining factor has high load $t_\ell > \theta\ell$, which is a Tail/PDEC exit, or

$$\mathcal{E}(E) \geq (1-\theta)|E|.$$

Thus FO-PDEC now produces low-mod energy unconditionally, modulo the named high-load exit.

The finite ledger with $\theta = 0.25$ gives global energy per equation

$$0.9547817296210457$$

and no high-load factor. The remaining obstruction is not energy production; it is the final threshold comparison on the same bad-window set:

$$U_{\mathrm{CRT}}(E) < \mathcal{E}(E) \quad \text{or} \quad L_{\mathrm{PDEC}}(E) \leq (1-\theta)|E|.$$

6.13 FO-PDEC Explicit Threshold

The explicit threshold ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-threshold-comparison.md`.

For each explaining factor ℓ, it computes the nonzero Fourier threshold

$$M_\ell(E) = \max_{1 \leq h < \ell} \left| \sum_{\rho \bmod \ell} g_\ell(\rho) e^{2\pi i h \rho/\ell} \right|.$$

The current best projection is

$$\ell = 199, \qquad h = 95, \qquad M_\ell(E) = 3.959247567099438.$$

Thus the remaining task is now concrete:

$$U_{\mathrm{CRT},199} < 3.959247567099438$$

for the same vector g_{199}. Otherwise this projection must be absorbed by SAE/Endpoint.

6.14 FO-PDEC Dual-Cluster Obstruction

The dual-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-dual-cluster-route.md`.

For the best projection, the mass is 4 and the Fourier threshold is 3.959247567099438, so the trivial mass bound misses closure by only

$$0.04075243290056196.$$

The best frequency $h = 95$ sends the support to a short dual arc of length 11 modulo 199:

$$40, 126, 61 \mapsto 19, 30, 24.$$

Thus the last obstruction is a concrete dual-cluster exclusion problem: this short dual arc must be ruled out by a PDEC dual certificate, or else absorbed as a non-persistent SAE/Endpoint event.

6.15 FO-PDEC Multiplicity Legitimacy

The primitive-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-multiplicity-legitimacy.md`.

It checks whether the best projection counts independent bad-window equations or repeated physical candidates. Under the equation or block-local convention, the projection has mass 4 and Fourier value

$$3.959247567099438.$$

Under physical deduplication, it has mass 2 and Fourier value

$$1.9699193446802263.$$

Thus the final PDEC comparison must first fix the multiplicity convention. The lower bound and the upper bound must be computed on the same multiset. Otherwise the argument must switch to the primitive threshold or absorb the repeated events by SAE/Endpoint.

6.16 FO-PDEC Formal Unit Consistency

The formal-unit audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-formal-unit-route.md`.

It checks a stricter requirement: the PDEC vector $g(t)$ must come from one formal bad-window set or multiset with one phase map. The current strong value

$$3.959247567099438$$

is obtained by pooling the finite certificate library. After splitting by formal unit, the audit gives

convention	best Fourier value
global library	3.959247567099438
layer deduplication	2.9698366905785227
physical deduplication	1.9997507790353146
single coordinate-deduplicated formal unit	1.0.

Thus the strong threshold cannot yet be used as a single-branch PDEC lower bound. One must prove FormalUnit-Stitching for the cross-q events and NestedBlock-Independence for exact nested-coordinate duplicates, or route the repeated events to SAE/Endpoint. This is now the narrowest FO-PDEC obstruction before any global unconditional conclusion can be claimed.

6.17 FO-PDEC Branch Separation

The branch-separation theorem and stitching audit are archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-branch-separation-theorem.md`.

They separate two issues. First, events from different matrix widths q cannot be pooled into one PDEC vector without a cross-q persistence theorem. Second, two occurrences with the same

$$(q, row, candidate\_row, column, offset, n, \ell, \rho)$$

give the same CRT equation and must be quotiented unless a weighted Hall-dual independence row is supplied. In the present finite ledger there are 7 exact nested-coordinate duplicates and 9 cross-level physical reuses. After coordinate deduplication inside a single formal unit, the best Fourier value recorded by the audit is only 1.0. Therefore the next narrow obstruction is Weighted Hall Dual Independence, or else SAE/Endpoint absorption of the exact duplicates.

The subsequent dominance audits discharge these finite FO–PDEC subgates without promoting the full theorem. The nested duplicates are laminar repeated coordinates; the fractional weighted Hall differences add one semiprime and one neighboring prime, with zero surplus, so they cannot pay for a second independent unit of mass. The cross-q reuses are coordinate-chart overlaps of the same physical candidate, not independent persistent events. Thus the raw library value 3.959247567099438 and the coordinate-cap value 2.9698366905785227 cannot be used as single-formal-unit lower bounds in the audited branch.

6.18 FO-PDEC Physical Primitive Tautology and SAE Absorption

The physical primitive audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-physical-primitive-tautology.md`.

After physical deduplication the current 199-factor frontier has only two physical atoms. For a prime modulus ℓ and two distinct residues a, b, choosing

$$h \equiv (b-a)^{-1} \pmod{\ell}$$

makes the dual residues adjacent. Hence every two-point support satisfies

$$\max_{h\neq 0} |e(ha/\ell) + e(hb/\ell)| = 2\cos(\pi/\ell).$$

For $\ell = 199$ this is exactly

$$2\cos(\pi/199) = 1.9997507790353146.$$

Therefore the remaining physical primitive value is a two-point Fourier tautology, not a usable PDEC defect threshold.

The two atoms are then routed to the SAE/Endpoint interface and audited in

`docs/monograph/prime-matrix-wsh-fo-pdec-sae-endpoint-absorption.md`.

Each associated fixed-offset fiber has a local prime survivor: the 250541 atom has witness 250543, and the 1664237 atom has witness 1664227. In all four source rows the 199-factor load in the same fiber is 1. Consequently the audited two-atom primitive branch is absorbed by LocalSurvivor witnesses.

This closes only the current finite $\ell = 199$ FO–PDEC sample chain:

$$\begin{gathered}\text{raw library signal} \to \text{nested/weighted rejection,}\\ \text{coordinate-cap signal} \to \text{cross-}q \text{ chart-overlap rejection,}\\ \text{physical two-point signal} \to \text{Fourier tautology,}\\ \text{two physical atoms} \to \text{LocalSurvivor witness absorption.}\end{gathered}$$

It does not close the global prime-matrix theorem. The next honest frontier is the global LocalSurvivor certificate family, or a future non-tautological primitive PDEC family with at least three physical atoms or an extra fixed-frequency constraint. The non-tautological PDEC admission audit records that the current materialized frontier has no such candidate: the raw three-point signal is not admissible as one primitive formal unit, nested and weighted duplicate recovery are blocked, cross-q reuse is only chart overlap, and the physical primitive remnant has two atoms. Those two atoms are already absorbed by LocalSurvivor/SAE witnesses. Thus any future PDEC certificate must introduce a genuinely new same-formal-unit primitive family with at least three physical atoms, or an additional constraint defeating the two-point tautology.

6.19 Materialized LocalSurvivor Packet Ledger

The current LocalSurvivor packet ledger is archived in

`docs/monograph/prime-matrix-local-survivor-materialized-packet-ledger.md`.

It combines the Triad-A1 SparseCap ledger, the FO–PDEC two-atom SAE/Endpoint absorption ledger, and the RPZ Endpoint-SAE finite ledger. The combined current ledger records

quantity	value
materialized packets	9
materialized phase atoms	32
local survivor witnesses	5
finite PDEC atoms	25
open materialized obligations	0.

Thus every LocalSurvivor/SAE packet currently materialized by the proof frontier is either witnessed, vacuous in the current ledger, or routed to a finite PDEC packet beyond the early $P \times P$ rows.

The packet-generation contract is archived in

`docs/monograph/prime-matrix-local-survivor-packet-generation-contract.md`.

It records the structural dichotomy for any future sparse escape. If the window data can be extracted, it is a finite LocalSurvivor packet and must be checked by a witness or blocker-deficit certificate. If the same finite signature persists along an infinite counterexample family, it is no longer a sparse terminal: according to the repeated signature it returns to PDEC, ColumnCRT, endpoint PDEC, TailAnchor, or CofactorAnchor. If the finite signatures do not persist and the layers keep escaping, the branch enters CleanKLS/DLS admission.

Consequently the next narrow frontier is

$$\text{LocalSurvivorPacketGenerationOrNonTautologicalPDEC}.$$

This is still not a final unconditional theorem. It says that the current materialized LocalSurvivor obligations are exhausted and that no unnamed LocalSurvivor terminal remains; the remaining work is packet-extractor completeness, a genuinely non-tautological primitive PDEC family, or the CleanKLS/DLS and referee inputs.

The known-entry extractor audit is archived in

`docs/monograph/prime-matrix-local-survivor-packet-extractor-coverage.md`.

It checks eight currently named LocalSurvivor entry types. Four are covered by machine extractors: Triad-A1 SparseCap, FO–PDEC two-atom SAE/Endpoint, RPZ Endpoint-SAE, and the materialized packet aggregator. Four are covered by contract routes: generic SAE, persistent finite signatures, layer escape to CleanKLS/DLS, and descent/seam return. The audit has no missing or open known entry.

The new-entry admission audit is archived in

`docs/monograph/prime-matrix-new-sparse-entry-admission-audit.md`.

It checks the possible sources of a future sparse entrance: short-window or endpoint sparse packets, PDEC dual sparse caps, endpoint seams, Bohr caps, column-tail/cofactor sparse packets, descent seams, persistent signatures, and layer escape. Each is already admitted to a named route: SAE/LocalSurvivor, PDEC, ColumnCRT, TailAnchor/CofactorAnchor, or CleanKLS/DLS. Hence there is no additional unnamed LocalSurvivor entry route in the current contract system.

After these two audits the LocalSurvivor branch is conditionally exhausted: any future explicitly introduced sparse route must bring its own extractor schema and finite ledger. The current narrow frontier is therefore

$$\text{NonTautologicalPDECOrCleanKLS}.$$

This remains a reduction boundary, not the final prime-matrix theorem.

The clean KLS side is no longer an unnamed black-box exit in the current routing table. The A1 clean-admission router checks that K1–K9 failures return to PDEC/SAE/Multiplicity/Promotion, while a fully admitted clean residual enters the Kuznetsov–LS atom SC-9. The SC-9 frontier router expands this atom into KZ-A–KZ-E: KZ-A–KZ-D are routed by smoothing, trace specialization, Bessel decay, and the spectral-large-sieve/pretrace chain; KZ-E remains the actual WFD block non-concentration input NC-BLK, or else a precisely matched external DI/BFI original dispersion theorem. Thus the updated narrow frontier is

$$\text{NonTautologicalPDECOrNCBLK}.$$

This also remains a reduction boundary, not a final unconditional row/column theorem. The NC-BLK boundary reconciliation audit records the exact reading of this label. On the canonical RIW/Buchstab source branch, the NC-BLK clean-KLS chain is already absorbed by the existing same-set capacity boundary. On the broader full-S non-AP generic branch, NC-BLK remains an exact-source-entropy or external DI/BFI/Kuznetsov route and is not imported into the self-contained theorem. Thus the label no longer denotes an unnamed CleanKLS exit, but it also does not promote the whole row/column theorem.

The subsequent global-terminal-family boundary router updates the frontier one more time. After the NC-BLK reconciliation and the non-tautological PDEC admission audit, the current materialized frontier has no local terminal instance left to eliminate: the non-tautological primitive PDEC candidate count is zero, the materialized LocalSurvivor ledger has no open packet, all known sparse entry routes are admitted, and NC-BLK is reconciled with the canonical boundary. The router records the new narrow label

`CurrentMaterializedFrontierExhausted_GlobalTerminalFamiliesOpen`.

Equivalently, the next required theorem is no longer a fixed numerical improvement for the current $\ell = 199$ sample or a new reading of NC-BLK. It is the global terminal-family exclusion package: every terminal object emitted by a minimal counterexample must be certified by a PDEC-family exclusion, a LocalSurvivor witness/deficit certificate, a CleanKLS/DLS internal large-sieve certificate, or an explicitly accepted external/referee input. This is again a boundary closure, not a final unconditional row/column theorem.

The split router for this global terminal family then removes the remaining ambiguity in the word “family”. The non-final reductions are closed: there is no fourth terminal route, the current LocalSurvivor and NC-BLK ledgers are not independent blockers, and the finite-projection terminal dichotomy sends positive-limsup mass to PDEC-CAP and diffuse mass to CleanKLS/DLS. Thus the self-contained mathematical remainder is

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`,

while final promotion also keeps the D-structure/Rankin referee gate. In particular, the frontier is not an unnamed local sample problem; it is the pair of named analytic terminal estimates plus the referee interface.

The self-contained terminal bottleneck router then removes one more apparent branch. Inside the current canonical-source boundary, InternalCleanKLS is not an independent minimal blocker: a clean large-sieve failure returns a dual concentration object to PDEC/SAE, the canonical NC-BLK/CleanKLS branch is already absorbed by the same-set capacity boundary, and the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed. Hence the independent self-contained mathematical bottleneck is

`PDEC_CAP_SameSetGlobalDualCertificate`,

namely a same-bad-window global dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for the PDEC families. The D-structure/Rankin referee gate remains a separate final-promotion condition, so this is still not a final unconditional row/column theorem.

The next PDEC-CAP router then opens this named bottleneck rather than treating it as a black box. All currently materialized PDEC intermediate gates are routed: the DualCap families have same-M_Q mass sources and closed early P-row exits; the current PersistentCaps route through promotion deletion or NoDeletion–KL/PDEC/CleanKLS; the current ForcedCaps route into multi-bucket ActualPaymentStitching; and LHB projection-stitching is removed by the common full-period completion set. The profinite ActualPaymentStitching router then closes the remaining APS logic by the finite-projection pigeonhole dichotomy. The PDEC-CAP theorem is still not proved. Its current narrow internal frontier is now the single terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

Persistent real payment Γ and fixed-shell low-mod persistence are now treated as the same finite-signature formal-unit problem: the former is a phase-bucket/tail-column signature, and the latter is a shell/displacement signature. ColumnCRT displacement is absorbed as displacement PDEC, PDEC dual failure is absorbed as cap refinement, and formal-unit mismatch is absorbed by the multiplicity-stitching normalization. Thus the single persistent terminal is PersistentFiniteSignaturePDECColumnCRT. The flat-clean SC–9 branch has been reconciled with the canonical-source boundary: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not imported into the self-contained claim. Finally, the persistent terminal admission router removes raw ColumnCRT, dual-failure, multiplicity, and two-point-tautology pseudo-terminals. Thus the remaining admitted object must be a primitive multi-atom same-formal-unit PDEC certificate. The intermediate diffuse split is now routed: persistent KL or phase-residue mutual information returns to refined/new-layer PDEC, while only the KL-flat branch may enter CleanKLS/DLS; divergent deletion already implies support exhaustion or a named sparse/PDEC return, and HRO reduces deletion non-divergence to occupancy saturation or TailIndependence. The latter is already NoDeletion/CleanKLS. The occupancy-kernel router then applies the HRO injection bound $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$: sparse old-hole or residue-deficient phases cannot saturate, and any remaining saturation must contain a near-full old-hole selector kernel. The common-variable router writes $c_b = \rho_b + rk_b$, turning every old-prime ban into one forbidden residue for the same shell variable k_b. Hence no legal shell is a capacity/Hall deletion, fixed-shell mass is PDEC/ColumnCRT persistence, and shell-dispersed flat mass is the SC–9 atom. External KLS input closes the flat branch only as an external-theorem version.

6.20 Triad-A1 Self-Contained Theorem-Boundary Closure

The final Triad-A1 review separates the exact theorem that is now closed from a stronger statement that is not true under the recorded hypotheses. In plain terms, the closed result is the following. Once the source of the A1 payment measure is fixed to the canonical RIW/Buchstab decision-tree source, every remaining same-set capacity gate in the full-S terminal chain is internally accounted for. The proof no longer needs to borrow the external FullS–KLS theorem for this canonical-source branch.

Proved-in-text Statement 6.1 (Canonical-source self-contained boundary)**.** *The Triad-A1 same-set capacity / full-S terminal is self-contained on the canonical RIW/Buchstab source branch.*

Recorded proof certificate. The certificate is recorded in

`docs/monograph/prime-matrix-triad-a1-self-contained-theorem-boundary-review.md`.

It checks seven gates: exact theorem statement boundary, final closure certificate, absorption by the same-set frontier, canonical-source provenance, branch coverage without silent generic upgrade, no false generic claim, and separation of the external FullS–KLS contract. The review verdict is

`APPROVE_CANONICAL_SOURCE_SELF_CONTAINED_BOUNDARY`,

with no open review gate. □

Not-claimed Statement 6.2 (Unrestricted generic WFD self-contained theorem)**.** The manuscript does not claim an unrestricted generic full-S well-factorable WFD self-contained theorem.

Reason for non-claim. The generic anti-atom input needed for that broader statement is refuted by the moving-delta model recorded in the A1 no-go audit. Hence the generic WFD version is not an open missing lemma of the canonical-source proof; it is a different and stronger statement that is false under the current formal hypotheses. The external FullS–KLS contract remains available as an external-theorem route, but it is kept separate from the self-contained canonical-source claim. □

Remark 6.3 (Boundary in everyday language)**.** The row-program terminal proof has reached a closed theorem-boundary for the canonical source: the bookkeeping, provenance, and branch split now agree on what is proved. What remains outside this boundary is not a hidden gap in that theorem. It is either a broader generic WFD theorem that the current model refutes, or one of the separate global prime-matrix terminal certificate obligations listed below for promoting the whole row/column program to a final unconditional theorem.

6.21 Claim Status Legend

The monograph uses the following status labels.

Label	Meaning
Proved-in-text	The statement is proved inside the manuscript under the stated hypotheses.
Reduction-closed	The statement has been reduced to a smaller named interface.
External-theorem closed	The statement is closed after citing explicitly named external theorems.
Computational-certificate	The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs.
Referee-block	A proof package exists, but it still requires line-by-line referee verification.
Not claimed	The statement is not asserted as a theorem of the manuscript.

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md`.

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

6.22 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required $\pi(x)$ bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the $P \geq 13208$ tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter $B(A)$. The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

6.23 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

6.24 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.
2. Replace the overly strong $\max_a$ BV–E_2 formulation by the precise WBE2 well-factorable weighted statement in the main line.
3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.
4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.
5. Preserve the RH warning language until every controlled exit has independent referee verification.

6.25 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

Program	Hard interface	Required closure before theorem promotion

Prime matrix	Structured-EHPD / D-structure exclusion	Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.
Prime matrix	PDEC / SAE / Rankin exits	Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.
Prime matrix	RRD / OSPC / Selberg-uniform constants	Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.
Two-point sieve	I3-Core true-residual package	Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.
Two-point sieve	DI/BFI to KLS-window adaptation	State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption.
Two-point sieve	BMD to TLI transfer	Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.
RH branch	controlled exits	Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.

For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md`.

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md`.

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md`.

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md`.

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md`.

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md`.

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json`.

It gives finite $Q = 2310$ LHB column rows for $13 \le p \le 47$. Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md`.

They make explicit that support size is not a multiplicity bound for the persistent count vector $g(t)$; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md`.

It states the precise acceptance contract: for the same (p, Q, S, τ) one must prove $g(t) \le M(t)$, and only then may a diagnostic block C be promoted to the capacity row $\sum_{t\in C} g(t) \le \sum_{t\in C} M(t)$. The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json`.

It takes $M(t)$ to be the high-layer CRT completion count $C_P(t; Q)$ and verifies, for $Q = 2310$ and $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$, that the WHOLEDEF and BRIDGED support blocks have $\sum_{t\in C} M(t) = 0$. This is an allowed-set projection certificate: it can be used in the LHB branch after proving $S \subset Z_{\mathrm{LHB}}(p, Q)$, but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md`.

It proves that an LHB-typed full-row bad window is indeed an element of $Z_{\mathrm{LHB}}(p, Q)$. Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md`.

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md`.

It closes the bookkeeping case in which one attempted certificate mixes different (Q, τ, F, κ, p) types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md`.

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold D_0, a column-witness selector, and phase-compatible weights $W_D(t)$, while displacement rows use a label ℓ, a non-zero displacement residue a, a threshold L_D, and phase-compatible weights $W_{\ell,a}(t)$. Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json`.

It uses the refined finite phase $\tau_{\mathrm{fin}} = (p, q, \mathrm{row})$ for $p \leq 1000$ and the five tightest RCI rows for each p. In this finite domain, the blocks $D_{\mathrm{col}} > 81$ and displacement-residue load > 2 are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global D_0, L_D theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md`.

It proves the exact layer-peeling identity: after a p-sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The adjacent-shell refinement is recorded in

`docs/monograph/prime-matrix-adjacent-shell-recursive-descent-route.md`.

For adjacent primes $p < q$, it proves that the only composite old p-sieve survivor in $[1, q^2]$ is q^2. Hence a non-first q-zero row first descends to an old p-sieve q-window, with only the terminal q^2 puncture in the last row. That window either contains a full p-aligned zero row or is a seam window whose two guard intervals have lengths a and $p - (q - p) - a$. The resulting narrow target is SeamGuard-Elimination: guards cannot absorb all forced survivors without entering SAE, PDEC, or ColumnCRT. The multi-level seam descent audit is recorded in

`docs/monograph/prime-matrix-seam-multilevel-descent-route.md`.

In this conditional model, if a seam interval is already an old p-sieve zero window, then after descending to a lower prime $h < p$ the only resurrected points are those with $P^-(n) \in (h, p]$, together with the terminal q^2 puncture in the last q-row. Any full h-aligned row avoiding these resurrected points is therefore a forced h-zero row. The finite ledger checks all 21339 seam windows for $p \le 500$ and a deterministic $p \le 2000$ sample of 11488 seam windows, with no persistent blocking cases. This is not a global theorem; it sharpens the remaining gate to the SMD-Global Inequality, or else a proof that persistent revival blocking enters SAE, PDEC, or ColumnCRT. The tail-mirror refinement is recorded in

`docs/monograph/prime-matrix-seam-tail-mirror-descent-route.md`.

For a forced lower h-zero row with row number R, set $N_h = M_h/h$ and $\rho = ((R - 1) \bmod N_h) + 1$. If $\rho \le h$ or $N_h - \rho + 1 \le h$, then periodicity or CRT reflection places a conditional zero row inside the lower $h \times h$ matrix. The finite ledgers above again show full success in the same domains. This motivates the sharper TailMirror-SMD gate: every genuine seam counterexample must hit such a lower head/tail phase, or its persistent non-hit phases must be routed to SAE, PDEC, or ColumnCRT. The all-branch version is recorded in

`docs/monograph/prime-matrix-total-zero-row-recursive-descent-route.md`.

It treats aligned containment, seam windows, and the terminal q^2 puncture by the same descent rule. In the $p \le 500$ ledger all 21936 non-first q-rows, including 597 aligned rows and 21339 seam rows, force a lower matrix head/tail phase. In the deterministic $p \le 2000$ sample all 11583 checked rows do the same. The remaining theorem-level statement is TotalDescent-TM: every genuine higher zero row must force a lower $h \times h$ zero row by periodicity or tail reflection, or else its non-hit phase set must enter SAE, PDEC, or ColumnCRT. The fixed $h = 3$ refinement is recorded in

`docs/monograph/prime-matrix-total-descent-h3-margin-route.md`.

Each complete 3-row contains a unique candidate c_R coprime to 6. If $5 \le P^-(c_R) \le p$, that row is blocked by a resurrected point when descending from the p-sieve to the 3-sieve; if $P^-(c_R) > p$, then c_R is already a survivor of the old p-sieve and contradicts the assumed p-sieve zero window. Thus the narrowest remaining form of TotalDescent is the six-wheel inequality

$$\#\{R : J_R \subset I_s\} > \#\{R : J_R \subset I_s,\ 5 \le P^-(c_R) \le p\}.$$

The audit

`docs/monograph/prime-matrix-total-descent-h3-margin-audit.md` checks this for all 1,552,462 non-first q-rows with $p \le 5000$, with minimum margin 1. This is a finite certificate and a sharpened interface, not a global theorem: a final proof must establish the six-wheel inequality uniformly, or route full blocking to SAE/PDEC/ColumnCRT. The hard-attack reduction

`docs/monograph/prime-matrix-h3-sixwheel-hard-attack.md`

clarifies the remaining obstruction. The direct six-wheel inequality sits at the Buchstab/linear-sieve boundary $u = 2$, i.e. at square-root interval length near $x = q^2$. Therefore a proof cannot be obtained by a generic short-interval theorem or a finite template. If the six-wheel candidates are all blocked, they admit the first-factor partition $n = \ell m$ with $5 \le \ell \le p$, m in an interval of length q/ℓ, and $P^-(m) \ge \ell$. The next theorem-level gate is consequently an H3 full-blocking defect theorem: high first-factor load enters Tail/PDEC, distributed low-load coverage produces low-mod PDEC energy, and endpoint residue concentration is SAE or ColumnCRT. The near-miss audit

`docs/monograph/prime-matrix-h3-full-blocking-defect-audit.md`

checks the same ledger for all $p \le 5000$. Among 1,552,462 non-first q-rows, only 4015 have margin at most 20, and these occur only up to $p = 317$ and $q = 331$. Their aggregate first-factor load is led by the small-prime skeleton $5, 7, 11, 13, \ldots$. This data does not prove the theorem, but it identifies the correct next split: small-first-factor overload should be absorbed by Tail/PDEC, while low-load many-label blocking should produce low-mod PDEC/ColumnCRT energy. The small-factor envelope route

`docs/monograph/prime-matrix-h3-small-factor-envelope-route.md`

makes this split deterministic. For a cutoff y, let C_y be the load carried by first factors at most y, let $R_y = A - C_y$, and let $\ell_+(y)$ be the next prime. If full blocking holds, then either C_y is already a small-skeleton overload, or at least

$$K_y = \left\lceil \frac{R_y}{\lfloor q/\ell_+(y) \rfloor + 1} \right\rceil$$

middle-tail labels are active. The finite ledger for the near-miss rows gives maximum forced values $K_y = 7$ for $y = 31$ and $K_y = 9$ for $y = 43$. The remaining proof obligation is to convert small-skeleton overload into Tail/PDEC, or many active labels into H3-PDEC energy, with endpoint losses routed to SAE/ColumnCRT. The global scaling audit

`docs/monograph/prime-matrix-h3-global-scaling-law-route.md`

records the underlying scale. The natural six-wheel rough count is

$$\#A_s \prod_{5 \le \ell \le p} \left(1 - \frac{1}{\ell}\right) \asymp \frac{q}{\log q}.$$

In the finite ledger, from $p = 331$ onward every prime layer has minimum H3 margin at least 21, and bucket averages track this Mertens scale. Thus a formal counterexample must annihilate a growing $q/\log q$ rough-residue scale. The current proof task is to show that such annihilation forces small-skeleton Tail/PDEC overload, many-label H3-PDEC energy, or endpoint SAE/ColumnCRT. The constant-level formula is recorded in

`docs/monograph/prime-matrix-h3-universal-scaling-inequality.md`.

In the finite ledger, $M_{H3}(p,s) \geq 0.30\, q/\log q$ holds from $p = 113$ onward, and $0.40\, q/\log q$ holds from $p = 2011$ onward. These are not asserted as unconditional short-interval prime theorems. They are target constants for a defect inequality: if the H3 margin falls below $cq/\log q$, then SmallSkeletonOverload, ManyLabel-PDEC, or Endpoint-SAE/ColumnCRT must fire. The global tail-energy lemma

`docs/monograph/prime-matrix-h3-global-tail-energy-lemma.md`

proves the deterministic core of this defect inequality. For any cutoff y, target B, threshold L, and modulus

$$d \geq 2\left(\left\lfloor \frac{q}{\ell_+(y)} \right\rfloor + 1\right),$$

where $\ell_+(y)$ is the next prime after y, the implication

$$M_{H3}(p,s) < B \quad \Longrightarrow \quad C_y > \#A_s - B - 2L \quad \text{or} \quad E_y(d) > L$$

holds for every adjacent pair $p < q$ and every row window $2 \leq s \leq q$. Substituting $B = cq/\log q$ and $L = \lambda q/\log q$ gives a uniform infinite-scale defect formula. This is still not the final H3 lower bound: the remaining theorem-level exits are SmallSkeletonOverload $\Rightarrow$ Tail/PDEC and TailEnergy $\Rightarrow$ H3-PDEC/ColumnCRT. The first exit is bridged in

`docs/monograph/prime-matrix-h3-small-skeleton-pdec-bridge.md`.

Let

$$V_y = \#A_s \prod_{5 \leq r \leq y} \left(1 - \frac{1}{r}\right), \qquad D_y = V_y - R_y,$$

where $R_y = \#\{n \in A_s : P^-(n) > y\}$. The overload inequality $C_y > \#A_s - B - 2L$ gives $R_y < B + 2L$, hence

$$D_y > V_y - B - 2L.$$

Thus, whenever $V_y \geq (c + 2\lambda + \eta)q/\log q$, small-skeleton overload forces a PDEC defect of size $D_y > \eta q/\log q$. The remaining issue is not this bridge, but the exclusion of such persistent PDEC defects and of the tail-energy alternative. The tail alternative is put in Fourier form in

`docs/monograph/prime-matrix-h3-tail-energy-fourier-bridge.md`,

where Parseval gives $\mathcal{F}_y(d) = dE_y(d)$ for the non-zero residue frequencies of the tail first-factor labels. The synthesis

`docs/monograph/prime-matrix-h3-unified-defect-criterion.md`

therefore proves the conditional closure criterion

$$D_y \leq V_y - B - 2L \quad \text{and} \quad \mathcal{F}_y(d) \leq dL \quad \Longrightarrow \quad M_{H3}(p,s) \geq B.$$

With $B = cq/\log q$ and $L = \lambda q/\log q$, this is the current global scale theorem: the empirical $q/\log q$ law is explained by two explicit low-mod defect exclusions. The unconditional row theorem still requires proving those exclusions uniformly. The source of the $q/\log q$ scale is clarified in

`docs/monograph/prime-matrix-h3-first-row-scale-bridge.md`.

Let $\Pi_1(q) = \pi(q)$ be the number of primes in the first row up to q. By the prime number theorem, $\Pi_1(q) \sim q/\log q$. On the other hand, adjacent-shell singleton rigidity says that below q^2 every old p-sieve survivor, apart from the endpoint q^2, is prime. Hence the average H3 margin over the non-first q-rows satisfies

$$\frac{1}{q-1}\sum_{s=2}^{q} M_{H3}(p,s) \sim \frac{q}{2\log q} \sim \frac{1}{2}\Pi_1(q).$$

The factor $1/2$ is simply $\log q/\log(q^2)$. Thus H3 is the square-shell projection of the first-row prime scale; a zero H3 row would have to remove a full first-row-scale mass from one row and therefore must be accounted for by the low-mod defects above. The pointwise boundary is recorded in

`docs/monograph/prime-matrix-h3-pointwise-closure-boundary.md`.

The average identity above is rigorous, but it does not imply pointwise positivity. The missing statement is a pointwise scale-transfer theorem, for example

$$M_{H3}(p,s) \geq \kappa\,\pi(q) \qquad (2 \leq s \leq q)$$

for some fixed $\kappa > 0$, or equivalently the uniform exclusion of the D_y and $\mathcal{F}_y(d)$ defects in the H3 unified criterion. The analytic boundary is recorded in

`docs/monograph/prime-matrix-h3-square-root-short-interval-barrier.md`.

It identifies this pointwise transfer with a prime lower bound in aligned intervals of length $q = \sqrt{q^2}$ near $x = q^2$. This is exactly the square-root short-interval scale, where ordinary average PNT input and the linear sieve at $u = 2$ do not supply a pointwise positive lower bound. The remaining theorem is therefore a genuine square-root defect exclusion, not a bookkeeping consequence of the first-row average. The direct hard attack is archived in

`docs/monograph/prime-matrix-h3-square-root-defect-exclusion-hard-attack.md`.

It shows that the D_y branch can be controlled at a small sieve level by the standard one-dimensional interval-sieve fundamental lemma. The obstruction is the tail branch: for a natural low modulus d, the tail Fourier energy satisfies

$$\frac{1}{2}T_y \leq E_y(d) \leq T_y,$$

so it is essentially the tail-label count. The final hard core is therefore to exclude tail labels and double-rough points as an exact full-row filler, or to force them into ColumnCRT, endpoint phase, or cofactor-structure defects. The local rigidity of such a filler is recorded in

`docs/monograph/prime-matrix-h3-tail-filler-rigidity-hardcore.md`.

Adjacent H3 candidates differ by 2 or 4, and candidates two steps apart differ by 6. If both endpoints are tail-filled, all their large factor families are disjoint. A repeated tail label must be separated by at least $y/4$ candidate steps, and any block of diameter less than y consumes at least two fresh large prime factors per candidate. Thus the remaining obstruction is the global concatenation of these local 2/4/6 rough short-difference units into a whole row. The global concatenation interface is sharpened in

`docs/monograph/prime-matrix-h3-tail-filler-global-chain-capacity.md`.

For any continuous tail-filled block B, full concatenation forces

$$\mathrm{EdgeCap}(B; y) = |B| - 1, \qquad \mathrm{TriCap}(B; y) = |B| - 2.$$

Here EdgeCap counts compatible adjacent 2/4 tail edges and TriCap counts compatible 2/4/6 triples. If $y > \sqrt{q+6}$, a fixed ordered tail-label pair contributes at most once in the row; if $y > q^{2/3}$, every tail-filled point is a double-rough semiprime. Hence the final H3 hard input can now be stated as an explicit capacity exclusion: prove that these edge or triple capacities are strictly below the full-chain requirements, or that full capacity forces PDEC, ColumnCRT, endpoint phase, or cofactor-structure defects. This is a necessary-capacity reduction, not yet an unconditional proof of H3. The macroscopic full-tail branch is then attacked in

`docs/monograph/prime-matrix-h3-tail-edge-selberg-exclusion.md`.

For a fixed even shift $\delta \in \{2, 4, 6\}$ and an interval I of length H, the classical two-dimensional Selberg upper-bound sieve gives

$$\#\{n \in I : (n(n+\delta), P(y)) = 1\} \le A_\delta(\theta) \frac{H}{(\log y)^2} \qquad (3 \le y \le H^\theta).$$

Every filled adjacent edge gives a y-rough pair with shift 2 or 4, and every filled triple gives a y-rough endpoint pair with shift 6. Hence a continuous tail-filled block B must satisfy

$$|B| \le 2 + A_*(\theta) \frac{q+6}{(\log y)^2}.$$

Thus a macroscopic block of length comparable with q, in particular a whole H3 row of about $q/3$ candidates, cannot be filled only by tail labels at sufficiently large scale. The remaining H3 obstruction is consequently narrower: any surviving bad row must split tail points into many short blocks, and the small-factor cuts or short-block endpoint phases must be routed to PDEC, ColumnCRT, endpoint, or cofactor defects. The short-block routing is made exact in

`docs/monograph/prime-matrix-h3-shortblock-singleton-barrier.md`.

If the tail points are decomposed into maximal consecutive tail blocks, with total tail mass T, number of blocks J, and internal adjacent-tail edges E, then

$$T = J + E.$$

The two-dimensional sieve controls E, so a bad row with T still of order $q/\log y$ would have almost all tail mass in singleton tail blocks. The small-factor cuts themselves are not yet a PDEC contradiction, because the small skeleton has natural size comparable with q. The final local obstruction is therefore the singleton-tail system: a point $a_j = \ell_j m_j$ with $\ell_j > y$ is isolated between two small-skeleton neighbours, so $a_j \equiv 0 \pmod{\ell_j}$ and simultaneously satisfies two non-zero congruences modulo small labels $r_-, r_+ \le y$. One must prove that many such squeezed singleton phases force PDEC, ColumnCRT, endpoint, or cofactor defects. The squeezed singleton phase is converted into a CRT capacity criterion in

`docs/monograph/prime-matrix-h3-singleton-clamp-defect-criterion.md`.

For a fixed clamp datum $c = (\delta_-, \delta_+, r_-, r_+)$, the two small-skeleton congruences are either incompatible or define one class $\rho(c) \pmod{\mathrm{lcm}(r_-, r_+)}$. Adding the tail label $\ell > y$ gives one class modulo $\ell\,\mathrm{lcm}(r_-, r_+)$, hence one row contains at most

$$1 + \left\lfloor \frac{q + O(1)}{\ell\,\mathrm{lcm}(r_-, r_+)} \right\rfloor$$

points from that unit. Consequently many singleton tails must enter one of three branches: tail-label concentration, clamp low-mod concentration, or distributed singleton capacity. The first two are named defect entrances. The final unresolved branch is the distributed signed excess of double-rough semiprimes over the prime survivor channel. This distributed branch is made bilinear in

`docs/monograph/prime-matrix-h3-distributed-singleton-bilinear-obstruction.md`.

For $y > q^{2/3}$, every singleton tail is a product ℓm with $y < \ell \le p$ and m prime. The clamp class $\rho(c) \pmod{R(c)}$ is equivalent to the moving cofactor congruence

$$m \equiv \ell^{-1}\rho(c) \pmod{R(c)}.$$

Thus the final distributed obstruction is a signed bilinear prime-counting defect over the short cofactor intervals I/ℓ. Without tail-label concentration, clamp concentration, endpoint accumulation, or ColumnCRT/cofactor bias, one must prove that this signed bilinear error is only $O(q/\log^2 y)$; this is the current last H3 input. The non-zero-frequency bridge is recorded in

`docs/monograph/prime-matrix-h3-bilinear-large-sieve-defect-bridge.md`.

Expanding the cofactor residue condition modulo $R(c)$ into non-principal characters gives

$$\chi(\ell^{-1}\rho(c)) = \chi(\rho(c))\overline{\chi(\ell)}.$$

Hence a large distributed singleton error forces a non-principal bilinear character sum over the short cofactor intervals. A plain large-sieve bound is still too weak at this short window scale; the final input is a dispersion/Kloosterman-type estimate for this family, or a proof that the corresponding non-zero frequency is already a PDEC, ColumnCRT, or cofactor defect. The additive Kloosterman reduction is

`docs/monograph/prime-matrix-h3-dsb-kloosterman-window-reduction.md`.

The congruence $m \equiv \rho(c)\bar{\ell} \pmod{R(c)}$ gives, after finite additive Fourier expansion, the inverse phase

$$e\left(-\frac{h\rho(c)\bar{\ell}}{R(c)}\right).$$

Thus the final H3 frequency input is a short-window Kloosterman sum with variable ℓ, cofactor window I/ℓ, modulus $R(c)$, and frequency h. The remaining audit splits into four branches: KLS-window coverage of the active parameters, high-lcm clamp escape, high-frequency endpoint tail, and coefficient concentration. The high-lcm branch is routed further in

`docs/monograph/prime-matrix-h3-dsb-high-lcm-clamp-routing.md`.

If $R(c) > R_0$, then one row contributes at most

$$1 + \left\lfloor \frac{q + O(1)}{R_0} \right\rfloor$$

points from a fixed clamp unit. Hence any high-lcm mass of order $q/\log y$ must activate many almost-singleton high-lcm units. Across a persistent family of bad rows this gives a non-zero CRT/Fourier defect, hence a PDEC/ColumnCRT exit; if it is not persistent, it is a sparse single-window escape. This does not yet exclude the branch, but it removes high-lcm mass from the low-modulus KLS-window estimate and routes it to the existing final exits. The common energy source of the two high-lcm exits is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-fourier-energy-clamp.md.`

For a homogeneous high-lcm modulus block, with residue-counting measure μ and total mass U, finite Plancherel gives

$$\sum_{1\le h<R} |\widehat{\mu}(h)|^2 = R \sum_{a \bmod R} \mu(a)^2 - U^2.$$

In particular, when $U \le R/2$, this non-zero energy is at least $RU/2$. Hence high-lcm escape necessarily injects Fourier energy: persistent energy must be excluded by PDEC/ColumnCRT, and isolated energy by SAE/endpoint analysis. The persistent branch is converted to a PDEC threshold in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-threshold-bridge.md.`

For one homogeneous formal unit B with phase-counting vector g_B,

$$L_{\mathrm{HLC}}(B) = \left(\frac{R\sum_a g_B(a)^2 - U_B^2}{R-1}\right)^{1/2}$$

is a forced non-zero-frequency lower bound. Thus a persistent high-lcm unit is excluded by a same-unit certificate $U_{\mathrm{CRT}}(B) < L_{\mathrm{HLC}}(B)$. The case $U_B > R/2$ is not a new escape; it is dense effective-modulus concentration and routes back to KLS-window, PDEC/ColumnCRT, or SAE/endpoint. If this PDEC upper certificate fails, the failure is localized in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-failure-localization.md.`

For a failing non-zero frequency h and direction ζ, every cap

$$\mathcal{C}_\alpha = \{a : \Re(\zeta e(ha/R)) \ge \alpha\}, \qquad 0 \le \alpha < L/U,$$

has mass at least $(L-\alpha U)/(1-\alpha)$. Hence failure is a concrete Bohr-cap concentration, which must become a same-unit PDEC constraint or a sparse SAE/endpoint certificate. The cap is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-bohr-cap-component-route.md.`

With $d = (h, R)$ and $R' = R/d$, the cap is the d-fold lift of one arc in $\mathbb{Z}/R'\mathbb{Z}$. Each lifted component has length at most $d+\omega(\alpha)R$, where $\omega(\alpha) = \arccos(\alpha)/\pi$, and one component carries mass at least $(L-\alpha U)/(d(1-\alpha))$. Thus the remaining cap obstruction is either a large-gcd low-effective-modulus concentration or a short-arc component concentration. The large-gcd branch is descended in

`docs/monograph/prime-matrix-h3-dsb-hlc-high-gcd-descent.md.`

If $d = (h, R) > 1$ and $R' = R/d$, the projected vector $g'(b) = \sum_{a\equiv b \pmod{R'}} g(a)$ satisfies

$$\widehat{g}_R(h) = \widehat{g}_{R'}(h/d),$$

and the Bohr-cap mass is preserved. Hence each high-gcd step strictly lowers the effective modulus and must terminate in a low-effective-modulus PDEC/KLS exit or in the short-arc case. The short-arc case is quantified in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-density-pressure.md.`

If a short arc has length at most $D_0 + \omega(\alpha)R$ and mass forced by the Bohr-cap lower bound, then its density relative to the global density U/R is at least

$$\frac{R(L-\alpha U)}{D_0(1-\alpha)U(D_0+\omega(\alpha)R)}.$$

If this pressure exceeds $1+\epsilon$, a local-density PDEC row or SAE/endpoint certificate is forced; otherwise only the explicit parameter residual remains. The low-pressure residual is optimized in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-pressure-optimizer.md`.

Writing $t = L/U$ and

$$\Psi_{D_0,R}(t) = \sup_{0\le\alpha<t} \frac{t-\alpha}{D_0(1-\alpha)(D_0/R+\omega(\alpha))},$$

the residual is the one-dimensional condition $\Psi_{D_0,R}(t) \le 1+\epsilon$. This implies the same-unit flatness estimate

$$\sum_a g(a)^2 \le \frac{1+(R-1)\tau^2}{R} U^2$$

and the corresponding support lower bound. Thus low-pressure short-arc mass is routed back to the KLS-window coefficient-norm check or to coefficient concentration. The KLS admission step is separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-l2-flat-kls-admission.md`.

It lists six required checks for an HLC L2-flat residual: effective modulus, effective frequency, endpoint smoothing, coefficient L^2 norm, gcd/unit strata, and dyadic or tail-label splitting. The L2-flat estimate supplies the coefficient-norm check; any other failed item is a named exit, and only if all six pass may the KLS-window input be invoked. Clean admission is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-clean-kls-reduction.md`.

A clean HLC formal unit is one in which the six named exits do not occur; then the K1–K6 admission checks pass. The remaining object is the clean HLC Kloosterman window

$$\mathcal{K}_{\mathrm{HLC}}(B),$$

and the remaining external/self-contained input is an estimate $\mathcal{K}_{\mathrm{HLC}}(B) = O(q/\log^2 y)$, denoted HLC–KLS-ext in the notes. The external-theorem adaptation is now separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-external-adaptation.md`.

It matches the clean HLC object to a DI/BFI/Kuznetsov-type windowed Kloosterman input: the inverse phase is $e(-h\rho(c)\bar{\ell}/R(c))$, the modulus and frequency checks are supplied by K1–K2, endpoint smoothing by K3, coefficient L^2 control by K4, gcd/unit strata by K5, and dyadic or tail-label losses by K6. Thus the clean HLC branch is closed in the external-deep-theorem version. A fully self-contained version would still require reproving the corresponding spectral/dispersion Kloosterman theorem inside this manuscript. The self-contained obligation is further compressed in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-reduction.md`.

There the clean HLC window is decomposed into dyadic Type-I/II blocks with standard kernel $\mathfrak{S}(\mathcal{D})$, and the proof of HLC–KLS-ext is reduced to one core estimate $|\mathfrak{S}(\mathcal{D})| \le \mathcal{N}(\mathcal{D})/\log^A y$. Thus the remaining no-black-box line is this single windowed Kloosterman spectral mean estimate. The no-black-box spine is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-self-contained-spine.md`.

It completes the smooth completion, Kloosterman quadratic-form reduction, coefficient L^2 bookkeeping, and the implication from the Kuznetsov large-sieve atom to the core estimate. It also records why pointwise Weil bounds alone do not yield the required arbitrary logarithmic saving. Hence a fully self-contained version now has one precise remaining analytic line: prove the specialized Kuznetsov large-sieve atom for this window family. That line is expanded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kuznetsov-ls-atom-expansion.md`.

The atom is decomposed into smoothing, the specialized Kuznetsov trace formula, Bessel transform decay, the spectral large sieve including oldforms and Eisenstein spectrum, and the well-factorable dispersion step that supplies the logarithmic saving. The manuscript therefore records the exact no-black-box obligations rather than hiding them under the single phrase "spectral theory". The Bessel-transform part is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-c-bessel-transform-decay.md`.

It uses the Bessel differential equation, the self-adjoint Bessel operator, and repeated integration by parts to obtain the required rapid transform decay. The remaining no-black-box analytic atoms at this point are the specialized Kuznetsov formula, the spectral large sieve, and the well-factorable dispersion logarithmic saving. The spectral-large-sieve atom is further reduced in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-d-spectral-large-sieve-spine.md`.

This dualizes the spectral large sieve to a spectral projection kernel, accounts for oldforms, Eisenstein spectrum and holomorphic forms as polylogarithmic losses, and reduces KZ–D to a single pre-trace kernel bound with diagonal size T^2 and off-diagonal Schur size N_0. The pre-trace kernel is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-ptk-d-pretrace-kernel-bound.md`.

There the diagonal term is recorded as the local Weyl volume, and the remaining off-diagonal part is reduced to a spatial lattice-point/correlation row-column bound, denoted LPC–D. The spatial kernel is split in

`docs/monograph/prime-matrix-h3-dsb-hlc-lpc-d-spatial-correlation-split.md`.

The far tail is absorbed by kernel decay, the identity-near sector is empty or parabolic, and the parabolic cusp sector is controlled by divisor bookkeeping. The remaining spatial line is the generic hyperbolic local-correlation bound GHLC–D. That last spatial line is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-ghlc-d-local-schur-closure.md`.

The key correction is to estimate the pre-trace object as an integral kernel, not by pointwise hyperbolic neighbour counting. On the local radius $r_* = T^{-1}\log^B y$ one has

$$T^2 \int_0^{r_*} (1+Tr)^{-A} \sinh r \, dr = O_A(1),$$

so the local ball area cancels the T^2 height of the kernel. Combining this local L^1 kernel mass with the Poincare-packet Schur normalization gives GHLC–D, hence LPC–D, PTK–D, and KZ–D. At that point the remaining analytic atoms were KZ–B, the specialized Kuznetsov trace formula, and KZ–E, the well-factorable dispersion logarithmic saving. The KZ–B specialization is treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-b-kuznetsov-trace-specialization.md`.

It derives the needed trace formula from the automorphic kernel, Poincare-packet unfolding, the double-coset Kloosterman geometric side, spectral Plancherel, and Bessel-transform normalization. This closes the formula-conversion part; it does not supply the final logarithmic saving. Hence the remaining no-black-box analytic atom in the HLC branch is KZ–E, the well-factorable dispersion logarithmic-saving step. The current KZ–E attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-e-well-factorable-dispersion-spine.md`.

It internalizes the well-factorable convolution algebra, the dispersion variance identity, the CRT reduction to a standard Kloosterman phase, and the gcd/endpoint polylogarithmic ledger. This proves that KZ–E would follow from one remaining estimate, denoted WFD-core: a windowed well-factorable Kloosterman dispersion mean estimate. That core is further balanced in

`docs/monograph/prime-matrix-h3-dsb-hlc-wfd-core-balanced-factor-reduction.md`.

Using the square-root well-factorable decomposition $\lambda_c = \sum_{c=uv} \alpha_u \delta_v$, the condition $c \asymp C$ forces $u, v \asymp C^{1/2}$ up to polylogarithmic boundary blocks. The gcd strata are polylogarithmic, and CRT factors the phase modulo uv into coupled phases modulo u and v. At this intermediate stage the remaining atom is BWFD-core, a balanced two-modulus well-factorable Kloosterman dispersion mean estimate. The subsequent completion attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bwfd-core-spectral-completion-attack.md`.

It performs exact finite Fourier completion in the s variable and uses complete Kloosterman multiplicativity modulo uv. This proves that BWFD-core follows from one still narrower atom, BSC-core: balanced complete Kloosterman bilinear correlation with arbitrary logarithmic saving. Ordinary KZ–D spectral large sieve recovers only the raw quadratic norm scale, and pointwise Weil bounds give only single-sum square-root cancellation; neither supplies the required $\log^{-A}$ saving. The current direct attack on BSC-core is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bsc-core-kloosterman-fraction-attack.md`.

It expands the complete Kloosterman sums term by term and exposes the reciprocal-fraction phase $e(\bar{v}R/u+\bar{u}T/v)$. One-sided degeneracy is localized to quadratic congruences such as $(a_h+\ell)x^2+b_h \equiv 0 \pmod{u}$; the remaining no-black-box atom is KFLS-core, a balanced Kloosterman-fraction large-sieve estimate with arbitrary logarithmic saving. The next square-kernel attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kfls-core-square-kernel-attack.md`.

It expands the square of the reciprocal-fraction sum into a four-modulus phase kernel. This also rules out an absolute Schur proof of the full kernel: the exact diagonal recovers only the raw quadratic norm scale. The remaining atom is therefore CFQK-core, a centered four-modulus Kloosterman-fraction correlation estimate covering the semi-diagonal and true off-diagonal layers. The block-centering correction is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-cfqk-core-block-centering-attack.md`.

It shows that the entire same-(u, v) block diagonal is local variance and cannot be forced to be logarithmically small for arbitrary admissible coefficients. The remaining obligations are therefore BD-CEN, the block-centering identity, OSQK-core for one shared modulus, and TFQK-core for the true four-modulus layer. The BD-CEN audit is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-dispersion-centering-audit.md`.

It checks the current KZ–E dispersion spine and finds only the $h = 0$ main-term subtraction, not the same-(u, v) block projection required by BD-CEN. Thus the first blocking obligation is the identity (BDC-5); OSQK-core and TFQK-core are conditional downstream obligations until that identity is proved. The subsequent no-go note is

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-no-go-route-fork.md`.

It shows that, for the current unblocked WFD/KZ–E object, BD-CEN is false rather than merely unproved: a single nonzero block with nonzero frequency survives the $h = 0$ subtraction but has no cross-block counterpart. The remaining choices are therefore SOURCE-CEN, BLK-energy-core, or an explicit external DI/BFI input. The SOURCE-CEN no-go note

`docs/monograph/prime-matrix-h3-dsb-hlc-source-cen-no-go.md`

checks the first of these choices. It shows that source block-centering would replace the unblocked linear WFD target in (KE-13) by a different centered object and would leave a block mean term which still has to be estimated. Thus SOURCE-CEN is not an identity for the present target. The subsequent BLK-energy audit

`docs/monograph/prime-matrix-h3-dsb-hlc-blk-energy-core-obstruction.md`

checks the second choice. As an arbitrary coefficient-array theorem at the square-kernel level, a one-block one-atom test makes $\sum_b |S_b|^2$ comparable with the raw norm, so a bare arbitrary logarithmic saving for BLK-energy-core is false. The remaining self-contained obligation is therefore the sharper NC-BLK input: prove block non-concentration for the actual WFD coefficients produced by the Type-I/II, Fourier, and well-factorable structure. Without that new input, this branch is only closed in an external-theorem version using a precisely matched DI/BFI dispersion theorem. The CRT lift-parity variant is audited in

`docs/monograph/prime-matrix-crt-lift-parity-audit.md`.

For adjacent primes $p < q$, the non-trivial q-columns are not covered by the new prime q; therefore a hypothetical q-row zero does descend to an old p-sieve zero window of length q. However, the old p-sieve row period already has an even number of zero rows by reflection symmetry, so multiplying the period by the odd factor q preserves evenness and does not create a parity contradiction. The retained conclusion is only the early-window dichotomy: an aligned lower zero row or a seam window routed to PDEC/SAE. The attempted recursive continuation is audited in

`docs/monograph/prime-matrix-recursive-mirror-descent-audit.md`.

It separates three objects: zero rows in a full CRT period at fixed width q, zero rows after peeling the largest base prime, and zero rows in an actual smaller-width matrix. These objects do not imply one another. In the finite audit, full-period zero rows occur for several adjacent pairs, but no early q^2 zero row and no actual smaller matrix zero row is forced. The next usable gate remains seam-window exclusion, not recursive parity. The annulus-mirror localization audit

`docs/monograph/prime-matrix-annulus-mirror-localization-audit.md`

separates two reflections. The CRT reflection preserving zero residue classes is $n \mapsto -n \pmod{M_p}$ and sends a small row to the tail of the CRT period. The terminal reflection $n \mapsto q^2 - n$ does land in an early interval, but it changes the forbidden classes from 0 to $q^2 \bmod \ell$. It is therefore a terminal non-zero-class block, not a smaller prime-matrix zero row. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md`.

It shows that the scaled height nP/p is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md`.

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md`.

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at $T_0 = 4$ in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md`.

It proves that, if TailAnchor is excluded, the center interval $J_{T_0} = [L + A + T_0, R + B - T_0]$ is an h-sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md` finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT. The exact grid criterion is recorded in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-criterion.md`.

For $J = [u, v]$, $N = v - u + 1$, and $\delta_h(u) \equiv 1 - u \pmod h$, the interval contains an aligned h-row if and only if $N \geq \delta_h(u) + h$. Failure is therefore a finite endpoint seam phase. The audit in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-audit.md`

matches this criterion in all five samples and finds no endpoint-defect sample. The remaining gate is endpoint persistence exclusion: a failed grid phase must be isolated as SAE or persistent as PDEC/ColumnCRT. This endpoint gate is routed in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-persistence-route.md`

and audited in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-phase-ledger.md`.

For fixed (h, N), endpoint failure is a finite residue ledger in u mod h. In the known samples there are no actual endpoint failures and only two possible failure phases. Sparse loads are SAE obligations; persistent same-phase loads are PDEC/ColumnCRT objects. Thus the RPZ endpoint branch is now named, not closed: the remaining work is either certificate closure for SAE/PDEC/ColumnCRT or recursive descent from the lower h-zero row. The two remaining RPZ attacks are now advanced together in

`docs/monograph/prime-matrix-rpz-dual-track-closure-route.md`.

Track A fixes the certificate interfaces for RPZ-SAE, RPZ-PDEC, and RPZ-ColumnCRT. Track B uses the lower-zero descent audit

`docs/monograph/prime-matrix-rpz-lower-zero-descent-audit.md`

which shows that, in the known BCB samples, all six conditional lower zero rows descend to a $p = 2$ impossible zero row with no endpoint/grid obstruction. This is still a sample-level descent audit; the global task is to prove descent-grid persistence or route every obstruction to the named certificate interfaces. The obstruction side is now made explicit in

`docs/monograph/prime-matrix-rpz-lower-descent-obstruction-ledger.md`

and

`docs/monograph/prime-matrix-rpz-certificate-materialization-interface.md`.

For each adjacent descent $p \to r$, obstruction is a finite phase condition modulo $\prod_{\ell \leq r} \ell$ and is classified as either grid failure or endpoint-puncture blocking. The current descent tree has twenty actual transition nodes, all successful, with no actual blocked node. This does not close the global theorem; it reduces any future obstruction to the SAE finite package, PDEC endpoint phase row, or ColumnCRT displacement row. The certificate skeleton builder

`docs/monograph/prime-matrix-rpz-certificate-skeleton-package.md`

materializes these rows: two endpoint SAE candidates, two endpoint PDEC/ColumnCRT rows, and three lower-descent grid-failure PDEC/ColumnCRT rows. These are certificate obligations, not completed exclusions. The endpoint SAE finite certificate

`docs/monograph/prime-matrix-rpz-endpoint-sae-finite-certificate.md`

checks that the two endpoint candidates have possible load two but actual load zero in the current BCB ledger. Hence the current finite endpoint SAE line is vacuously closed; global SAE exclusion is still not proved. The lower grid-failure certificate

`docs/monograph/prime-matrix-rpz-lower-grid-fail-avoidance-certificate.md`

reduces every adjacent descent $p \to r$ to the exact test $\delta = -(a-1)(p-r) \bmod r$ and grid success iff $\delta \le p-r$. The current descent tree has twenty actual transition nodes and none hits grid failure. The remaining global task is to prove this phase inequality for formal descent paths, or route its failure to PDEC/ColumnCRT. The formal path note

`docs/monograph/prime-matrix-rpz-formal-descent-phase-inequality.md`

sharpens this statement: once a formal descent path exists, the phase inequality is automatic from interval containment. The true remaining task is path existence for every formal counterexample branch, or exclusion of the first obstruction phase through SAE/PDEC/ColumnCRT. The first-obstruction dichotomy

`docs/monograph/prime-matrix-rpz-first-obstruction-dichotomy.md`

removes endpoint puncture as a genuine obstruction: a contained lower row containing the endpoint ap would force $r \mid a$, contradicting the roughness condition needed for the endpoint to revive. Hence every non-descending branch has a first grid-failure seam phase, which must be discharged by SAE/PDEC/ColumnCRT certificates. The first-grid-failure seam certificate

`docs/monograph/prime-matrix-rpz-first-grid-fail-seam-certificate.md`

puts this obstruction in normal form. If $g = p - r$ and $g < \delta < r$, the bad row is the union of two adjacent lower-row caps of lengths δ and $r + g - \delta$, with conserved missing mass $r - g$ and at most the right endpoint ap as an r-rough survivor. The certificate produces twelve seam phase rows covering all 1752 grid-failure phases in the current ledger. It now also records the exact PDEC support row for each seam:

$$S_\tau = \{a \bmod Q : \ a \equiv \rho \pmod r\}, \qquad F_\tau = \mathbf{1}_{a\equiv\rho \pmod r} - 1/r,$$

whose non-zero Fourier support is contained in the frequencies jQ/r, $1 \le j < r$. The same certificate splits the endpoint ap into 1348 phases already killed by lower labels and 404 Q-unit endpoint phases. Thus the next narrow gate is no longer support extraction, but either an endpoint-PDEC upper bound $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for these same bad-window families or a ColumnCRT displacement certificate with explicit selectors Π, λ and threshold L_D. The unit-endpoint gate certificate

`docs/monograph/prime-matrix-rpz-unit-endpoint-columncrt-gate.md` pushes this one step further. For a unit endpoint seam,

$$c \equiv ap \equiv p\rho \equiv r + (p - r) - \delta \pmod r,$$

so the endpoint lies in a fixed non-trivial lower column. If $\pi = hr + c$ is a same-column prime witness, then the endpoint lower row satisfies $H \equiv -cr^{-1} \pmod p$, and the displacement $d = h - H \pmod p$ is independent of the row phase and non-zero. In the current ledger all twelve gate rows have explicit witnesses and non-zero displacements, covering all 404 unit endpoint phases. This routes persistent unit endpoint seams to fixed non-zero ColumnCRT candidates; it still does not exclude the ColumnCRT threshold L_D or prove that formal counterexample families avoid these gates. The threshold obstruction audit

`docs/monograph/prime-matrix-rpz-columncrt-threshold-obstruction.md`

then shows why this cannot be closed by merely lowering L_D. Inside a fixed unit gate, all unit endpoint residues already have the same label p and the same non-zero displacement class $d \bmod p$, so the intrinsic load of that class is

$$R_{p,d} \geq \prod_{\ell < r} (\ell - 1).$$

In the present ledger the maximum intrinsic load is 48, and the test value $L_D = 2$ would send ten gate rows, covering 400 phases, into ColumnCRTDefect rather than exclude them. Thus the remaining legitimate closures are a formal-family avoidance theorem, an endpoint-PDEC upper bound, or an independent exclusion theorem for the resulting ColumnCRTDefect. The three-route audit

`docs/monograph/prime-matrix-rpz-three-route-closure-audit.md`

compares these options under the same ledger. It finds no closed route yet, but ranks formal-family avoidance first: the present lower-descent ledger has twenty actual transition nodes and no grid-failure node, while the endpoint-PDEC route still lacks an admissible U_{CRT} upper bound and the ColumnCRT route has been reduced to an independent defect-exclusion theorem. The formal phase automaton

`docs/monograph/prime-matrix-rpz-formal-phase-automaton.md`

turns the first route into an explicit start-phase admission problem. For each prime level p it defines an accepted set $A_p \bmod P(p)$: phases in A_p descend canonically through successive prime levels to $p = 2$, while phases outside A_p hit a first-grid-fail seam and return to the already materialized exits. In the current ledger all six BCB-Core start rows belong to their accepted sets. The global missing theorem is that every formal counterexample-family start phase lies in A_p, or else is captured by the seam exits. The rejected-phase absorption certificate

`docs/monograph/prime-matrix-rpz-rejected-phase-absorption.md`

removes the second ambiguity within the present automaton range. It checks all 27924 rejected phases, finds twelve distinct first-fail seam rows, and verifies that every one of them is already materialized by the first-grid-fail seam certificate. Hence the RPZ formal route is now a strict two-way routing statement: an accepted start phase descends to $p = 2$, while a rejected start phase enters the named seam/PDEC/ColumnCRT certificate chain. This is an absorption theorem, not an exclusion theorem; the remaining hard point is the actual discharge of those seam/PDEC/ColumnCRT exits, or a proof that formal counterexample starts always lie in A_p. The seam exit pressure ledger

`docs/monograph/prime-matrix-rpz-seam-exit-pressure-ledger.md` then compresses this remaining exit. The 27924 rejected phases reduce to twelve seam rows; within those rows 1348 endpoint phases are already killed by lower labels, while the remaining 404 unit endpoint phases fall into ten fixed non-zero ColumnCRT displacement classes, with maximum aggregated load 96. Thus the next gate is no longer a phase search: one must either prove formal-family avoidance of the twelve seam rows, provide $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for their single-residue PDEC supports, or prove an independent exclusion theorem for the ten ColumnCRT displacement classes. The symbolic ladder certificate

`docs/monograph/prime-matrix-rpz-symbolic-ladder-certificate.md`

rewrites the preferred avoidance route as local adjacent-prime digits. For each adjacent pair $p > r$, with $g = p - r$, it uses

$$\delta_p(a) = -(a-1)g \pmod r, \qquad \delta_p(a) \le g$$

as the exact success digit. Failure of this inequality is precisely the first-grid-fail seam already materialized above. The certificate verifies the resulting counting formula by full enumeration through modulus $P(19) = 9699690$ and symbolically records the same ladder through $p = 97$. The remaining theorem is now sharply stated: derive these digit inequalities from the formal counterexample construction itself, not from a finite phase enumeration. The BCB start-digit ledger

`docs/monograph/prime-matrix-rpz-bcb-start-digit-ledger.md`

extracts these digits for the current BCB-Core starts. All twenty digit nodes are safe, but ten of them have zero margin $\delta_p(a) = p - r$. This is a useful negative result for the proof strategy: the desired avoidance theorem cannot be closed by a coarse positive margin estimate. It must use exact congruence information for the start row modulo each lower prime r. The accepted lower-row selector ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-row-selector.md`

weakens the required positive statement to the exact selector condition

$$C_h(J) \cap A_h \neq \varnothing, \qquad C_h(J) = \{m : \ J \supset [(m-1)h+1, mh]\}.$$

For the current BCB-Core samples all five core intervals have such a selector, and all six candidate lower rows are accepted. This is still only a current-ledger certificate; the global theorem must prove selector existence for every formal BCB core interval, or else route the all-rejected selector failure back to the seam/PDEC/ColumnCRT exits. Finally, the selector gap ledger

`docs/monograph/prime-matrix-rpz-selector-gap-threshold.md`

splits selector existence into two verifiable mechanisms. If the number of complete candidate rows in $C_h(J)$ exceeds the maximum cyclic rejected gap of A_h, length alone forces a selector. In the current samples this explains two of the five BCB cores; the remaining three are short-candidate cases and require exact endpoint phase information. Thus the next proof obligation is not merely to enlarge J, but to control the endpoint phase of short cores relative to the rejected gaps of A_h. The short-candidate endpoint phase ledger

`docs/monograph/prime-matrix-rpz-short-candidate-phase-ledger.md`

performs this refinement at modulus $hP(h)$ rather than just modulo h. In the current ledger the three short-candidate BCB records all hit an accepted selector. However, the full (h, ℓ) phase families still contain all-rejected endpoint phases: 588 for $(h, \ell) = (7, 13)$, 11880 for $(11, 19)$, and 344760 for $(13, 25)$. These rejected phases are not unnamed escapes; each carries a first-failure seam key and therefore returns to the seam/PDEC/ColumnCRT exits. The remaining theorem-level gate is consequently exact: prove that the formal BCB construction avoids these all-rejected $hP(h)$ endpoint phases, or close the corresponding seam/PDEC/ColumnCRT certificates. The follow-up forbidden-block ledger

`docs/monograph/prime-matrix-rpz-short-phase-block-formula.md`

removes another layer of enumeration. Since the relevant short cores have length $\ell < 2h$, a starting endpoint u contains at most one complete h-row. For a lower row m, the endpoint interval that contains it is exactly

$$mh - \ell + 1 \le u \le (m - 1)h + 1,$$

of width $\ell - h + 1$. Hence the all-rejected endpoint set is the disjoint union of such blocks over rejected row phases $m \bmod P(h)$. The formula matches enumeration in all three short families, and the current endpoints have positive distance from the rejected endpoint union. The remaining non-enumerative input is therefore narrower still: derive from the formal BCB construction that the induced candidate row phase belongs to A_h, or else close its named first-failure exit. The BCB candidate phase identity ledger

`docs/monograph/prime-matrix-rpz-bcb-candidate-phase-identity.md`

then makes the induced row phase explicit. If the top zero row is R, the top prime is P, the half-prime is h, the plateau extremes are s_-, s_+, and the tail threshold is T, the forced core is

$$L = (R - 1)P + 1 + s_- + T, \qquad U = RP + s_+ - T.$$

The complete lower rows inside this core are exactly

$$m_{\min} = \left\lfloor \frac{L + h - 2}{h} \right\rfloor + 1, \qquad m_{\max} = \left\lfloor \frac{U}{h} \right\rfloor.$$

Equivalently, after writing $P = Qh + d$, these phases depend only on $R \bmod hP(h)$ and the plateau parameters. In the current ledger the formula matches all five BCB samples and all six induced lower row phases lie in A_h. This is the sharp remaining input: prove the same residue implication for every formal BCB counterexample, or send any rejected induced phase to its first-failure exit. The accepted-preimage ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-preimage-ledger.md`

enumerates this final residue condition for the current BCB parameter families. The $P = 13, h = 5$ family has no bad residue. The other four families still have bad residues, split between no-candidate endpoint failures and all-rejected candidate phases. The actual top-row residues in the ledger all lie in the selector preimage, with positive distance from the bad set where the bad set is nonempty. Thus the residue preimage computation is now explicit, but the proof is still not global: the formal counterexample construction must be shown to land in the selector preimage, or the bad residue must be discharged through its named BCB endpoint or first-failure seam/PDEC/ColumnCRT exit. The core-run obstruction ledger

`docs/monograph/prime-matrix-rpz-bcb-core-run-obstruction.md` adds a stronger upstream test for the no-TailAnchor branch. In that branch BCB-Core forces J_T itself to be an h-sieve zero interval, so its length cannot exceed the maximum cyclic run of integers divisible by primes $\leq h$ modulo $P(h)$. For the current layers these exact maxima are

$$h = 5 : 5, \qquad h = 7 : 9, \qquad h = 11 : 13, \qquad h = 13 : 21.$$

The five current BCB cores have lengths $9, 13, 15, 19, 25$, respectively, so every one of them exceeds the corresponding low-sieve run bound, with minimum margin 4. Thus the current finite BCB no-TailAnchor samples are obstructed before the accepted-preimage question is reached. The global theorem-level input is now a Jacobsthal-type low-sieve run bound for the formal layers; without such a bound, the remaining bad cases still route to the named endpoint and first-failure exits. This input is isolated in

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-closure-interface.md`.

Writing $G(h)$ for the maximum length of a consecutive interval all of whose points are divisible by some prime $\leq h$, the exact BCB closure condition is

$$G(h) < P + m - 1 - 2T,$$

where m is the sliding platform length. Under this inequality the no-TailAnchor BCB branch closes immediately; if it fails, the proof must keep the TailAnchor, endpoint, first-failure, PDEC, and ColumnCRT exits active. The risk scan

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-risk-scan.md`

checks this linear closure route against the Ziller–Morack primorial Jacobsthal data. It shows that the current finite samples are safe, but the naive global specialization $m = 5$, $T = 4$, and only $P > 2h$ fails already at $h = 43$: there $G(h) = 89$, the minimal top prime is $P = 89$, and the core length is only 85. Thus the finite core-run obstruction cannot be promoted to a global theorem by a blanket linear bound. A global proof must either force longer platforms m, prove stronger restrictions on the formal BCB parameter set, or close the TailAnchor/endpoint/first-failure exits. The platform-threshold ledger

`docs/monograph/prime-matrix-rpz-bcb-platform-length-threshold.md`

rewrites this requirement as the exact integer condition

$$m \geq G(h) - P + 2 + 2T.$$

For $T = 4$, the first substantive unsafe layer with $h \geq 5$ is again $h = 43$, where the required platform length is 10 rather than 5. The next attack must therefore prove platform-length growth or prove that short platforms trigger one of the named exits. The short-platform embedding route

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-route.md`

handles the second alternative. If $N = P + m - 1 - 2T \leq G(h)$ and no TailAnchor is present, then the forced core J_T is an h-sieve zero interval of length N and hence must be embedded inside an extremal low-sieve covered block modulo $P(h)$. Therefore its left endpoint lies in a finite Jacobsthal endpoint set $E_{h,N}$. Persistent occurrence of such an endpoint phase is no longer an unnamed escape: it is routed to SAE if sparse, or to PDEC/ColumnCRT if persistent. The accompanying certificate

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-certificate.md` materializes the max-block subfamily $E_{h,N}^{\max}$ by reconstructing CRT left endpoints from the Ziller–Morack modulus representation. For the current $m = 5, T = 4$ interface, the first non-trivial short platform is $h = 43$, $P = 89$, $N = 85$, with 240 distinct max-block endpoint phases; across the scanned table, 1197 reconstructed longest blocks pass the coverage and boundary checks. This remains a certificate for the longest-block subfamily, not a proof that the full endpoint set $E_{h,N}$ has been exhausted.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

Chapter 7

Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

Definition 7.1 (Closure certificate)**.** A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

Theorem 7.2 (Current monograph closure)**.** *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

Proof. For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. Inside Triad-A1, the same-set capacity / full-S terminal is now closed as a canonical-source self-contained theorem-boundary: the source provenance is canonical RIW/Buchstab, the final closure certificate has no open gates, and the unrestricted generic WFD self-contained statement is explicitly refuted rather than claimed. These are genuine submodule and theorem-boundary closures. The D-structure exclusion and the remaining global PDEC/SAE/Rankin or LocalSurvivor/CleanKLS certificates remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction, certificate, and boundary package but not as a final unconditional prime-matrix theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph's logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification. □

Theorem 7.3 (Prime-matrix frontier). *The current prime-matrix row/column unconditional theorem is routed to named terminal gates but is not closed. The canonical-source Triad-A1 self-contained boundary is closed, and the unrestricted generic WFD self-contained theorem is refuted and not claimed. After the global terminal-family split and the self-contained bottleneck router, the independent self-contained mathematical bottleneck is the same-set PDEC capacity certificate*

PDEC_CAP_SameSetGlobalDualCertificate.

Inside that bottleneck, the transverse formal-unit embedding and the downstream canonical layer admission/nonzero-transfer/thin-return gate are now closed on the canonical-source self-contained branch. The remaining materialized open gate is the generic/external original DI/BFI route

DIBFIQuantifiedNoProjection
WindowCertificate ,

after the APS profinite dichotomy, the persistent finite-signature unification, and the SC–9 boundary reconciliation are closed as routing laws, and after the persistent terminal admission and finite-arc routers remove raw, low-rank, continuous-cap, and unnamed finite-arc pseudo-terminals. The previous transverse clean large-sieve atom is now routed to the A1 CleanKLS/SC–9 frontier, and its self-contained source-support branch is further embedded into the canonical A1/KZ-E source; the canonical layer closure then returns to the already closed A1 selector/decision-tree/source-provenance boundary. The PDEC-CAP theorem itself remains open in the broader global/external sense. Final theorem promotion still separately requires the D-structure/Tail-log4/finite Rankin referee interface. The boundary-lift router records the canonical-source branch as

NoFurtherCanonicalSourceSelfContainedPDECCapGap,

while keeping the generic/external DI/BFI route and the final D-structure/Rankin referee interface outside that self-contained closure. The subsequent terminal-promotion closure router sharpens the canonical-source self-contained boundary to

NoFurtherCanonicalSourceTerminalPromotionGap.

Proof. The frontier router

`docs/monograph/prime-matrix-row-column-unconditional-frontier-router.md`

checks the merged boundary audit, the Triad-A1 terminal confluence router, the terminal triad contract, the formal-unit audit, the stitching feasibility audit, the nested-duplicate dominance audit, the line-by-line referee matrix, and the claim-status table. It records

`row_column_unconditional_closed=false`,

while preserving the closed canonical-source boundary and the refuted generic branch. The terminal triad has no fourth exit, so any final counterexample must be absorbed by PDEC, LocalSurvivor, or CleanKLS/DLS certificates, together with the still guarded D-structure/Tail-log4/Rankin interfaces.

The global terminal-family boundary router and its split router then show that the current materialized frontier is exhausted and that all non-final reductions are closed. The open terminal family gates reduce to PDEC-CAP, KLS/external-or-internal large sieve, and the D-structure/Rankin referee interface. For the fully self-contained route, the remaining choice was

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`.

The self-contained bottleneck router closes the non-final CleanKLS ambiguity. CleanKLS failure returns to PDEC/SAE, the canonical clean branch is absorbed by the closed same-set boundary, the generic full-S self-contained strengthening is refuted and not claimed, and the SC-9 route has already been accounted for. Therefore InternalCleanKLS is no longer an independent self-contained blocker. What remains is exactly the global same-set PDEC dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$, plus the independent referee-promotion gate. This proves the stated frontier and also explains why it is not yet a final unconditional theorem.

The PDEC-CAP same-set global-dual router refines that frontier once more. It records

`closed_current_materialized_pdec_gates=true`

and

`pdec_cap_same_set_global_dual_closed=false`.

Thus the current local PDEC materialization is exhausted, but the global proof is not. The profinite APS router closes the missing dichotomy: along the inverse tower of signatures, the real payment graph Γ_n either has a positive-limsup persistent finite signature, which becomes a multi-bucket PDEC formal unit, or avoids every finite signature, which is a distributed CleanKLS/DLS input unless FiberDeletion or NoDeletion–KL has already routed it back to PDEC. The first persistent narrow hardpoint is therefore the terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

The diffuse terminal split router first sharpens the nonpersistent side. No unnamed FiberDeletion, NoDeletion–KL, or CleanKLS exit remains; the side is reduced to either SC–9 or a finite-shell low-mod persistence object. The latter is a PDEC/ColumnCRT input. The persistent-signature unification router then identifies persistent MFU and fixed-shell persistence as the same finite-signature formal-unit grammar; ColumnCRT displacement, PDEC cap failure, and formal-unit mismatch all route back into the PDEC/ColumnCRT family rather than creating a parallel terminal. The SC–9 boundary reconciliation then removes the flat-clean SC–9 branch as an independent canonical-source blocker: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not claimed as self-contained. The persistent terminal admission router then forces any remaining PDEC object to be primitive, multi-atom, same-formal-unit, non-tautological, and not already absorbed by SAE/Endpoint. The deletion support-exhaustion bridge closes the implication from divergent deletion to exhausted support or named sparse/PDEC return; the HRO router then shows that if deletion does not diverge, either TailIndependence returns to NoDeletion/CleanKLS or old-hole residue occupancy saturates. The occupancy-kernel router uses $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$ to remove the sparse-old-hole subcase. Therefore the deletion-side hardpoint first becomes the dense old-hole selector kernel. The common-variable router then rewrites each selected column as $c_b = \rho_b + rk_b$; every old-prime condition forbids exactly one residue of the same shell variable k_b. The resulting alternatives are named: no legal shell gives capacity/Hall deletion, positive mass on a fixed shell or finite shell packet gives PDEC/ColumnCRT persistence, and genuine multishell flatness enters the already reconciled SC–9/NC-BLK boundary. An external KLS/DI/BFI/Kuznetsov citation would close only the external-input version.

The primitive terminal is then reduced further. The primitive multi-atom rank router removes rank-zero and rank-one pseudo-terminals: after physical deduplication, such remnants are reuse defects, two-point Fourier tautologies, fixed-shell PDEC/ColumnCRT objects, or SAE objects. What remains is a rank-at-least-two cap-stable primitive kernel. The cap-stable kernel router rewrites its inequality by the contrapositive of cap localization: if a direction reaches L_{PDEC}, then a direction cap has mass at least $(L_{\mathrm{PDEC}} - \alpha M)/(1-\alpha)$ and must return to SAE, refined PDEC, ColumnCRT, or multiplicity normalization. Hence a genuinely cap-stable kernel only has to certify that all legal direction caps stay below their thresholds. The uniform-cap finite-basis router then removes the continuum in (ζ, α): for a finite formal unit, every such cap is the preimage of a finite cyclic arc under a nontrivial character. Finally, the finite-arc transverse router splits a high arc into low transverse support, persistent transverse bias, or transverse-flat dispersion. These route respectively to SAE/ColumnCRT/fixed-shell PDEC, refined PDEC, or CleanKLS/DLS or an external large-sieve input. The current narrow PDEC-CAP hardpoint is thereby reduced to transverse clean analysis. The transverse clean reduction router observes that a finite arc fixes only one character direction; a rank-at-least-two primitive kernel still leaves a transverse quotient. Nonflat transverse defects are named returns to SAE, refined PDEC, or ColumnCRT. If none occurs, the residual is exactly a transverse L^2-flat clean large-sieve atom. The current narrow PDEC-CAP hardpoint is therefore first reduced to

`TransverseQuotientCleanLargeSieveAtom`.

The transverse clean-atom frontier router then imports the already registered A1 CleanKLS/SC–9 grammar. K1–K9 admission failure returns to PDEC, SAE, ColumnCRT, or multiplicity normalization. If admission succeeds, the object enters the SC–9 frontier: self-contained actual-coefficient NC-BLK or external DI/BFI. The canonical NC-BLK absorption cannot be used for a generic transverse quotient until the quotient coefficients are shown to have the canonical RIW/Buchstab source support, or until actual transverse block non-concentration is proved directly. The naive factor-residue incidence bridge is blocked by the internal-fiber obstruction. Thus the current self-contained hardpoint is

`TransverseSourceSupportNonconcentrationCertificate`.

This certificate is then split once more. The A1/KZ-E actual-source provenance ledger is closed for the canonical RIW/Buchstab pre-Cauchy source, the canonical RIW support route is reduced to Buchstab-layer support, and the raw thick squarefree support count is closed. The transverse embedding router then proves that the transverse quotient formal unit is a legal restriction, quotient, or conditioning of that canonical A1/KZ-E source without changing its coefficients: actual payment is a deterministic pushforward, finite signatures are finite projections, direction arcs are preimage restrictions, and the transverse quotient is a finite factor. The canonical layer closure router then observes that the downstream layer-admission/nonzero-transfer/thin-return gate is exactly the already routed A1 chain: selector retention, finite signatures, RIW/Buchstab decision tree, actual source provenance, and canonical branch boundary. Hence the canonical-source self-contained PDEC-CAP branch has no remaining layer-transfer gate. The original external DI/BFI route is still separate and requires

`DIBFIQuantifiedNoProjectionWindowCertificate`,

unless one explicitly accepts a windowed KLS external theorem as an external input. This does not promote the full row/column statement to an unconditional theorem.

Finally, the boundary-lift router

`prime-matrix-self-contained-pdec-cap-boundary-lift-router`

propagates this conclusion back to the upper self-contained bottleneck. It records

`canonical_source_self_contained_pdec_bottleneck_closed=true`

and

`narrowest_self_contained_boundary=NoFurtherCanonicalSourceSelfContainedPDECCapGap`.

The same audit keeps

`pdec_cap_same_set_global_dual_closed=false`

and

`row_column_unconditional_closed=false`.

Thus the PDEC-CAP gap has been removed only from the canonical-source self-contained branch. The broader generic/external branch still needs the quantified no-projection DI/BFI certificate, and the whole prime-matrix theorem still needs the global terminal family promotion review and the D-structure/Rankin referee interface.

The canonical terminal-promotion closure router then reconciles the older global split with the boundary lift. It observes that the current materialized terminal frontier is exhausted, the terminal triad has no fourth exit, PDEC-CAP promotion is closed on the canonical-source branch, canonical CleanKLS/NC-BLK is absorbed or returns to PDEC/SAE, and all known LocalSurvivor entrances are covered by witness, extractor, or admission contracts. It records

`latest_self_contained_hardpoint_closed=true`

and

`open_self_contained_gates=[]`.

Thus the latest self-contained boundary is

`NoFurtherCanonicalSourceTerminalPromotionGap`.

This is still a canonical-source theorem-boundary closure, not an unrestricted global terminal-family exclusion theorem. The external DI/BFI route and the D-structure/Rankin referee gate remain outside the closure.

The final self-contained theorem router then packages the exact statement boundary. The approved self-contained statement is the Prime Matrix canonical-source terminal theorem boundary: the Triad-A1 same-set/full-S terminal together with canonical terminal promotion on the canonical RIW/Buchstab source branch. Under that exact boundary it records

`canonical_source_self_contained_theorem_closed=true`

and

`open_self_contained_gates=[]`.

Its terminal is

`NoFurtherCanonicalSourceSelfContainedTheoremBoundaryGap`.

The router also records the non-claims: the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed, and the unrestricted/global Prime Matrix row-column unconditional theorem is not claimed. The external DI/BFI route and the D-structure/Rankin referee promotion gate remain outside this self-contained theorem boundary.

The global-unconditional obstruction router makes the last point explicit. It records

`current_corpus_global_unconditional_self_contained_closure_possible=false`

and

`row_column_unconditional_closed=false`.

The named blockers are the refuted unrestricted generic WFD route, the still open actual full-S source bridge or strengthened source anti-atom theorem, the DI/BFI no-projection window certificate, and the D-structure/Tail-log4/finite Rankin promotion gate. Thus the current manuscript closes the exact canonical-source self-contained theorem boundary, but does not close the unrestricted global row-column theorem without new inputs.

The actual-source bridge reconciliation router sharpens this last obstruction list. It checks the canonical provenance ledger, the source-lock branch split, and the final canonical-source theorem certificate together. On the canonical RIW/Buchstab branch it records

`actual_source_bridge_closed_for_canonical_branch=true`.

Hence the former broad blocker

`ActualFullSSourceBridge`

is superseded inside the canonical self-contained theorem boundary. The global complement is still open, however:

`actual_source_bridge_closes_global_unrestricted=false`.

The updated global blockers are

`UnrestrictedGenericWFD`,
`NoncanonicalFullSComplementAntiAtomOrExternalDIBFI`,
`FullSNonAPStrengthenedSourceAntiAtom`,
`DIBFIQuantifiedNoProjectionWindowCertificate`,
`DStructureTailLog4FiniteRankinPromotion`.

Thus the actual-source bridge is closed only for the canonical branch. Full global unrestricted closure still requires a strengthened anti-atom for the noncanonical full-S complement, or an external quantified DI/BFI no-projection certificate, together with the final D-structure/Rankin promotion gate.

The noncanonical-complement input-contract router then fixes the remaining input boundary. It records

`contract_boundary_closed=true`, `canonical_branch_removed_from_remainder=true`,

but also

`generic_wfd_template_available=false`,
`noncanonical_complement_closed=false` for the current corpus.

Thus the remaining full-S complement is not an unrestricted generic WFD lemma. The moving-delta separation blocks that template, while the canonical branch has already been subtracted. Under the current terminal map there is no hidden fourth route: one must prove actual source identity with the canonical RIW/Buchstab source, prove a strengthened anti-atom for the actual noncanonical source, or provide the quantified no-projection DI/BFI route. Final unconditional row-column promotion still additionally requires the D-structure/Tail-log4/Rankin acceptance gate.

The noncanonical-complement trilemma router sharpens this boundary. It records

$$\texttt{trilemma\_boundary\_closed=true},$$

but also

$$\texttt{self\_contained\_noncanonical\_package\_closed=false},$$
$$\texttt{external\_contract\_package\_closed\_if\_fulls\_kls\_ext\_accepted=true}.$$

The router separates the only legal closure modes after the canonical branch has been subtracted:

actual source identity,
strengthened actual-source anti-atom,
or acceptance/proof of FullS-KLS-ext.

Its negative content is important: the generic full-S self-contained anti-atom route is refuted by the moving-delta capacity model under the recorded formal WFD, Type/Fourier, K4/K6, and naive-incidence hypotheses. Hence the noncanonical package cannot be closed by returning to the generic WFD template. Without the external FullS-KLS-ext input, one must prove actual source identity or a strengthened actual-source anti-atom; with the external input, the remaining primary-source issue is the line-by-line `DIBFIPrimarySourceSpecializationProof`.

The closure-input atlas finally reviews the explored models together. The CRT/MinRep equivalence, cylindrical line covering, the P-column anchor wheel, the layered $P^2 \pm k$ wheel, far-tail cofactor inversion, SN recursive peeling, named-exit absorption, and PDEC cap refinement no longer point to independent unnamed branches. They all feed the same terminal certificate package:

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert}, \quad \texttt{ColumnCRT-Cert},$$
$$\texttt{CleanMultishellKLS}, \quad \texttt{TotalDescent}.$$

Together with the noncanonical full-S complement package and the D-structure/Rankin promotion package, these are the minimal remaining input basis. Thus future work should prove these packages directly rather than search for another generic WFD template or a fixed numerical constant.

The D-structure/Rankin promotion package is now fixed as an acceptance boundary rather than a hidden terminal. The corresponding router records

$$\texttt{promotion\_package\_boundary\_closed=true},$$
$$\texttt{promotion\_package\_independently\_accepted=false},$$
$$\texttt{rankin\_sample\_pass=true}.$$

It checks that the package is listed in the minimal input basis, that it is separated from the canonical-source self-contained boundary, that PM-16 remains a BLOCK-REFEREE item in the line-by-line matrix, and that the Rankin ledger has a checkable certificate format. The current sample has exact smooth-core count 39, allowed budget 40.0, and both the Rankin and exact budget tests passing.

This is a format and sample verification, not a global acceptance of the whole promotion package. Final promotion still requires independent acceptance of the D-structure/Structured-EHPD entrance reductions, Tail-log4 BG/RKS theorem-number and adaptation checks, reproducible finite-verification hashes, and Rankin certificates for all formal colored corridors, with failures routed to PDEC/SAE.

The terminal-certificate package has a further compression. The no-unnamed-escape machine and the named-exit absorption contract are already closed at the routing level. The canonical-source clean KLS branch is absorbed by the final canonical boundary, and the noncanonical clean branch belongs to the noncanonical complement or to an external KLS route. TotalDescent is also not a separate terminal: a formal descent path reaches $p = 2$, while a first failure is a first-grid-fail seam already routed to PDEC, ColumnCRT, or SAE. Therefore the independent terminal certificate work is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert}, \quad \texttt{ColumnCRT-Cert}.$$

This is a structural compression of the first package, not yet a proof of those three certificate families.

The ColumnCRT branch is then absorbed once more. A persistent nonzero column displacement overload is a displacement PDEC after adjoining the displacement signature to the finite phase space; a sparse displacement overload is an SAE/endpoint event; and balanced displacement loads become admissible constraints in the PDEC dual problem. RPZ unit endpoint gates show that tuning the displacement threshold alone cannot prove exclusion, but they do not create a third terminal family. Hence the independent terminal remainder is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, and primitive variants.}$$

This is still not a proof of the terminal certificate package; it is the removal of ColumnCRT as an independent terminal.

The SAE branch is also removed as an independent terminal by the SAE absorption router. The router checks the SAE local-reduction lemma, the materialized LocalSurvivor packet ledger, the known extractor coverage audit, the new-sparse-entry admission audit, the packet-generation dichotomy, and the persistent-terminal admission boundary. It records

$$\texttt{sae\_independent\_terminal\_removed=true}.$$

The absorption law is the following. A sparse or single-window escape must either emit a finite LocalSurvivor packet with a witness or blocker-deficit certificate, or else a finite signature persists and returns to PDEC, ColumnCRT, TailAnchor, or CofactorAnchor; if the signature layers keep escaping, the branch enters CleanKLS/DLS admission or an external input package. The currently materialized LocalSurvivor/SAE packets are exhausted, the known sparse entries have extractors or contract fallbacks, and no additional unnamed sparse entry is admitted by the current contract system. Hence SAE is not a third terminal family. The first terminal package is now reduced to

$$\texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, primitive, and} \\ \text{non-tautological variants,}$$

together with the hygiene rule that any genuinely new sparse route must bring an explicit packet-extractor schema. This still does not prove the terminal package or the unrestricted row-column theorem; it deletes SAE as an independent terminal and pushes the remaining structural burden back to PDEC or to a future explicit sparse schema.

The broad PDEC family is then turned into an explicit input boundary. The PDEC-family router records

```
pdec_family_explicit_input_boundary_closed=true,
current_materialized_pdec_frontier_closed=true,
canonical_source_pdec_cap_closed=true,
global_pdec_family_unconditional_closed=false.
```

It checks the persistent-terminal admission gate, the non-tautological PDEC admission audit, the primitive-rank boundary, the rank-two cap-stable inverse step, the finite cyclic-arc reduction, the transverse clean reduction, the clean-frontier route, and the canonical-layer closure. Thus the currently materialized non-tautological primitive PDEC candidate count is zero, and the canonical-source PDEC-CAP route has returned to the final self-contained boundary. A future PDEC obstruction is not admitted as a label: it must come with a same-formal-unit phase map, at least three physical primitive atoms after deduplication, no two-point tautology, rank at least two after shell and column quotients, cap stability under finite cyclic-arc localization, and no sparse, persistent-biased, or clean-externalized transverse return. Hence the first package is now reduced to future explicit primitive PDEC schemas and future explicit sparse packet schemas, while generic branches remain in the noncanonical trilemma and final theorem promotion remains in the D-structure/Rankin acceptance package.

The future sparse branch is also turned from a hygiene rule into an explicit admission boundary. The sparse-schema router records

```
future_sparse_packet_schema_boundary_closed=true,
current_materialized_sparse_frontier_closed=true,
global_sparse_family_unconditional_closed=false.
```

It uses the materialized LocalSurvivor ledger, the known packet-extractor coverage audit, the new-sparse-entry admission audit, the SAE absorption router, the packet-generation dichotomy, and the PDEC-family boundary. Thus a future sparse obstruction is not admitted as a bare terminal label. It must specify the source class, a finite window or fixed-offset fiber, the candidate set, the blocker families and their incidence projection, a witness or a strict blocker-deficit inequality, phase and formal-unit normalization with deduplication, a finite-signature persistence test, a layer-escape test, and a reproducible ledger with no open local obligation. If the finite signature persists, the branch returns to explicit PDEC/ColumnCRT/Tail/Cofactor schemas; if the packet layers escape, it enters CleanKLS/DLS or an explicit external large-sieve input. Hence the current materialized sparse frontier is zero, but the global sparse family is not unconditionally excluded.

Combining these items gives the final input firewall. The firewall router records

```
final_input_firewall_boundary_closed=true,
current_materialized_terminal_frontier_closed=true,
no_hidden_terminal_remaining=true,
all_final_inputs_independently_accepted=false,
row_column_unconditional_closed=false.
```

The remaining named inputs are exactly future explicit primitive PDEC schemas, future explicit sparse packet-extractor schemas, one legal branch of the noncanonical full-S trilemma, and independent D-structure/Rankin promotion acceptance. Thus the author-side boundary and naming problem is closed: there is no hidden terminal left in the current material. The unrestricted row-column theorem is still not promoted, because the named inputs have not all been proved or independently accepted.

The four named inputs are then attacked one by one. The attack router records

current_materialized_frontier_zero=true, no_hidden_terminal_remaining=true, conditional_closure_chain_complete=true, all_current_obligations_closed=false, unconditional_closure_possible_from_current_corpus=false.

The future primitive PDEC and future sparse packet-extractor inputs have no present materialized obligation: they are admission firewalls for future routes. The real current obligations are the noncanonical full-S complement and D-structure/Rankin promotion. The canonical actual-source branch is closed, but the unrestricted noncanonical complement is not; the recorded primary-source audit rejects deriving the required full-S non-AP WFD Kloosterman large-sieve estimate from the existing DI/BFI sources. The D-structure/Rankin boundary is named and the sample Rankin certificate passes, but the full ledger and independent acceptance are not complete. Therefore the logical chain is conditionally complete: if future PDEC/sparse routes are discharged by their schemas, one legal noncanonical branch is proved or accepted, and the D-structure/Rankin package is independently accepted, then the no-hidden-terminal chain promotes the theorem. Without those new inputs, the present corpus cannot honestly claim the unrestricted unconditional row-column theorem.

The noncanonical input is finally narrowed further. The noncanonical-narrowing router records

noncanonical_narrowing_boundary_closed=true, ap_source_lift_rejected=true, generic_self_contained_antiatom_refuted=true, exact_source_entropy_closed=false, external_full_s_contract_closed_if_accepted=true, self_contained_noncanonical_closed=false.

The AP-source lift is rejected by the AP/non-AP branch definition and by the object ledger; the generic self-contained anti-atom is refuted by the moving-delta capacity model; the canonical branch has already been removed. Therefore the noncanonical complement has only two real remaining inputs: an internal theorem proving exact source entropy, equivalently a strengthened source anti-atom for the actual full-S non-AP source, or an external/new proof of the full-S non-AP WFD Kloosterman large-sieve input.

The last-remaining-atoms router packages this boundary into the final open atoms. It records

last_remaining_atom_boundaries_closed=true, conditional_logic_chain_complete=true, all_last_atoms_proved_or_accepted=false, row_column_unconditional_closed=false.

The final atoms are

ActualFullSNonAPExactSupportAtom, ModulusDependentCompletedFullSKLSInput, DStructureRankinIndependentAcceptance.

The first two are alternative closures for the noncanonical full-S complement: either an internal exact support/source-entropy theorem for the actual full-S non-AP source, or an external/new modulus-dependent completed full-S Kloosterman large-sieve input. The third is the independent promotion acceptance for the D-structure/Tail-log4/finite Rankin package. Thus the no-hidden-terminal logic chain is conditionally complete: one of the first two atoms, together with the D-structure/Rankin acceptance atom, promotes the row-column theorem. Since the present corpus has not proved or independently accepted these atoms, the unrestricted unconditional row-column theorem remains open.

The three-final-atoms hard-attack router then pushes this boundary one step further. It records

$$\texttt{three\_atom\_attack\_boundary\_closed=true},$$
$$\texttt{conditional\_logic\_chain\_complete=true},$$
$$\texttt{all\_three\_atoms\_proved\_or\_accepted=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The final mathematical choice is:

```
ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or
CDependentResidueWeightSpectralCancellationInput
```

The first input is a theorem that the actual full-S non-AP source capacity measure has no moving same-(u, v) atom. The generic version is refuted by the moving-delta model, and it is not forced by K4/K6 or naive incidence. The second input is a spectral/dispersion cancellation theorem for the completed c-dependent residue weights $B_{c,x} = \sum_k \beta_{x+kc}$; pointwise Weil, L^2 budgeting, the ordinary large sieve, and flat-residue shortcuts do not supply it. The final promotion input is:

```
DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

Thus the minimal unconditional input basis is

```
(ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus has not proved this mathematical alternative and has not completed the independent promotion acceptance, so the unrestricted unconditional row-column theorem is not yet a proved theorem of the manuscript.

The final unconditional-closure attempt router then checks whether the mathematical alternative can be closed from the current corpus. It records

$$\texttt{final\_attempt\_boundary\_closed=true},$$
$$\texttt{math\_lanes\_collapsed\_to\_common\_core=true},$$
$$\texttt{internal\_math\_proof\_found\_in\_current\_corpus=false},$$
$$\texttt{external\_math\_match\_found\_in\_current\_corpus=false},$$
$$\texttt{independent\_promotion\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The c-dependent completed-residue route is not an independent terminal: finite Fourier inversion connects it to the existing BWFD–BSC–KFLS chain, whose self-contained remaining duty is actual same-(u, v) block non-concentration, unless a precisely matched external DI/BFI/Kuznetsov dispersion theorem is supplied. The actual-source anti-atom is the same moving-block spread/source-entropy duty. Therefore the irreducible final input basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR PreciselyMatchedExternalDIBFIKuznetsovDispersionTheorem)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus does not prove moving-block spread, does not contain a line-by-line external theorem match for the current full-S, non-AP, uncentered, no-projection object, and has not completed independent D-structure/Rankin promotion acceptance. Thus the final input boundary is closed, but the unrestricted unconditional theorem is still not proved.

The irreducible-input refinement router then sharpens the external label. The phrase "precisely matched external DI/BFI/Kuznetsov dispersion theorem" cannot be read as a claim that the existing primary DI/BFI sources have already been matched to this object: the BFI AP theorem, the DI/Maynard J-scale, and the AP-source lift route are all blocked for the present full-S, non-AP, uncentered, no-projection WFD window. The refined basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here FullSNonAPWFDKLSTheoremInput means a new proof or an independently cited theorem which directly estimates the current full-S non-AP WFD Kloosterman/dispersion object. The router records

`refinement_boundary_closed=true`,

`row_column_unconditional_closed=false`.

The subsequent joint attack compares the two remaining mathematical lanes directly. The canonical RIW/Buchstab source branch has already been closed and subtracted from the noncanonical full-S complement, so its support theorem cannot be imported as a proof of the noncanonical moving-block anti-atom. The internal lane is therefore narrowed to an exact support package for the actual noncanonical source:

```
FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource.
```

The package has four subclauses:

exact factor-support lower bound,
balanced-range thresholds,
Type/Fourier capacity compatibility,
factor-residue incidence bridge or direct support proof.

The external lane remains

```
FullSNonAPWFDKLSTheoremInput.
```

The BFI, DI, and Maynard sources provide the AP/dispersion/Kuznetsov technology class, but the current audit finds no ready theorem covering the present c-dependent completed residue weights, uncentered no-projection full-S non-AP object, with arbitrary logarithmic saving. Thus the latest basis is

```
(FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The two lanes can be narrowed once more. On the internal side, the full-S balanced-range threshold is closed, since $C \asymp P/\log^{O(1)} P$ and the balanced factors of $c = uv$ dominate every fixed logarithmic threshold. The remaining exact-support and Type/Fourier-capacity clauses are therefore a single source anti-atom contract:

```
FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource.
```

On the external side, the full-S window may be completed modulo c; the real residue is the c-dependent completed weight

$$B_{c,x} = \sum_k \beta_{x+kc}.$$

Thus the external theorem input is narrowed to

```
CDependentResidueWeightSpectralCancellationInput.
```

The refined terminal basis is

```
(FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The dual-lane common-core reconciliation router then checks whether the internal and external lanes are still independent self-contained obstacles. It records

```
common_core_reconciliation_closed=true,
self_contained_common_core_proved=false,
external_contract_accepted_as_final_input=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The reconciliation law is this: if `CDependentResidueWeightSpectralCancellationInput` is accepted as a FullS–KLS or c-dependent spectral theorem, it is an external theorem input. If it is proved internally from the present corpus instead, finite Fourier completion sends it back to the BWFD–BSC–KFLS chain, and then to actual same-(u, v) block non-concentration. That is the same moving-block source-entropy core as the internal source anti-atom lane. Consequently the reconciled basis is

```
((ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput)
OR AcceptedFullSKLSExtOrCDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis, without an external black box, is

```
ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the independence gap between the two mathematical lanes is closed. What remains is not two separate self-contained obstacles, but one actual-source anti-atom/source-lock input plus the independent D-structure/Rankin promotion acceptance. The unrestricted unconditional row-column theorem is therefore still not proved.

The self-contained narrowest-core router then removes one more ambiguity from the no-external-black-box basis. It records

narrowest_core_reduction_closed=true,
canonical_source_lock_absorbed_for_canonical_branch=true,
source_lock_option_removed_from_global_remainder=true,
external_black_box_used=false,
generic_self_contained_antiatom_available=false,
noncanonical_actual_source_core_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.

The source-lock option has already been absorbed by the canonical RIW/Buchstab branch and the actual-source provenance ledger. It closes the canonical-source self-contained theorem, but it does not close the unrestricted global noncanonical full-S complement. The generic self-contained anti-atom is also unavailable, since the moving-delta model refutes it under the recorded formal WFD/Type/Fourier hypotheses. Therefore the latest fully self-contained global basis is

```
ActualNoncanonicalFullSSourceEntropyOrStrengthenedAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input means exact source entropy, equivalently no moving same-(u, v) atom, for the actual noncanonical full-S non-AP WFD coefficients. It is not an unrestricted generic WFD template. This is a sharper boundary, not a proof of the remaining source core or of the independent D-structure/Rankin promotion gate.

The noncanonical source-core atomization router then removes the remaining naming duality between "source entropy" and "strengthened anti-atom." It records

source_core_atomization_closed=true,
entropy_antiatom_duality_removed=true,
balanced_range_threshold_closed=true,
k4_k6_or_naive_incidence_suffices=false,
canonical_import_allowed=false,
actual_support_capacity_core_proved=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.

Exact source entropy and the strengthened anti-atom are two formulations of the same actual noncanonical source-capacity core. The balanced range threshold is already closed in the full-S regime. K4/K6 residue and dyadic controls do not imply moving factor-pair support, naive factor-residue incidence is blocked by the large internal fiber inside one (u, v) block, and canonical RIW/Buchstab support cannot be imported into the noncanonical complement. Hence the latest self-contained basis is

```
ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input asks for log-power lower absolute support of the exact u- and v-factors in every surviving actual noncanonical full-S non-AP balanced block, together with Type/Fourier capacity compatibility so that no moving (u, v) pair receives an unrecorded capacity multiplier. This closes a naming and shortcut boundary, but it is not yet a proof of that support-capacity theorem.

The actual capacity-ledger microatom router then checks one last possible shortcut inside that support-capacity statement. It records

$$\texttt{microatom\_boundary\_closed=true},$$
$$\texttt{support\_only\_suffices=false},$$
$$\texttt{registered\_multiplier\_discipline\_would\_suffice\_with\_support=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{registered\_multiplier\_discipline\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The correction is that raw support width is not itself a source anti-atom. A model with many raw (u, v) pairs but one unrecorded Type/Fourier/fiber multiplier can still concentrate the final capacity measure on a single moving pair. Conversely, if all such multipliers are registered in the same formal unit and bounded by a fixed logarithmic power, then exact support does imply the final anti-atom: with $L = \log y$, $|\alpha_u|, |\delta_v| \leq L^C$, registered multipliers $W_{u,v} \leq L^E$, and $S_u S_v \geq L^{2A+4C+E}$, one obtains

$$\max_{u,v} M_{u,v} / \sum M_{u,v} \leq L^{-2A}.$$

Thus the sharp self-contained input is best stated as

```
ActualFinalCapacityAntiAtomLedgerForNoncanonicalFullS
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

A proof may supply the package

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline.
```

This closes the support-only shortcut and fixes the conditional implication, but it still does not prove the final actual capacity anti-atom ledger or the independent D-structure/Rankin acceptance.

The registered capacity-multiplier discipline router then attacks the second micro-input in that package. It records

$$\texttt{registered\_capacity\_multiplier\_discipline\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

This is a bookkeeping closure, not a new cancellation theorem. The Type decomposition, dyadic summation, CRT phase normalization, the Fourier h-window and tail, the coefficient, gcd, endpoint and smoothing losses, the full-S completion fiber, and tail-label bookkeeping are all registered inside the same actual formal unit and cost only fixed logarithmic powers. These costs are absorbed into the final capacity measure $M_{u,v}$, or equivalently into a registered multiplier $W_{u,v} \leq L^E$. The router does not use the external DI/BFI no-projection theorem-match branch and does not prove factor support.

Consequently the fully self-contained source-side basis is reduced from

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline
```

to the single source micro-input

```
ActualNoncanonicalExactUVSupportLowerBound.
```

Together with the independent promotion gate, the latest basis is

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the multiplier-discipline item, but the exact u, v-support theorem and the independent D-structure/Rankin acceptance remain open.

The ExactUVSupport terminal-attack router then audits the remaining source micro-input. It records

$$\texttt{exact\_uv\_support\_terminal\_boundary\_closed=true},$$
$$\texttt{registered\_capacity\_multiplier\_discipline\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The audit removes the remaining shortcut routes. Formal WFD hypotheses allow point-supported factors, so they do not force actual u, v-support. K4/K6 and naive factor-residue incidence still live on the wrong projections. Raw Buchstab/Mertens counting gives enough squarefree products only after the canonical layer is admitted, and the canonical RIW/Buchstab support chain closes only the canonical-source branch. It cannot be imported into the noncanonical full-S complement.

Thus the current fully self-contained basis is still

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the exact-support theorem is supplied, the already closed multiplier discipline gives the final capacity anti-atom by the recorded inequality. Without that theorem, and without independent D-structure/Rankin acceptance, the unrestricted row-column theorem cannot yet be stated as unconditional.

The ExactUVSupport failure-packetization router then rewrites the remaining source input in its contrapositive finite form. It records

$$\texttt{failure\_packetization\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_support\_failure\_packet\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

After the actual noncanonical clean block, exact u, v-support notion, and registered multiplier threshold are fixed, the positive lower-bound input

```
ActualNoncanonicalExactUVSupportLowerBound
```

is equivalent to the absence of a positive-mass support-failure packet:

```
ActualNoncanonicalSupportFailurePacketExclusion.
```

Such a packet must carry the source class, formal-unit id, block key, exact u- and v-support sets, the $L^{2A+4C+E}$ threshold, the registered capacity profile, return tests to the named exits, and a reproducible certificate. Hence support failure can no longer remain an unnamed obstruction. This packetization is an equivalent boundary reformulation, not a proof of packet exclusion. The latest self-contained input basis can therefore be written as

```
ActualNoncanonicalSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The manuscript still lacks a proof that all such packets are absent or must return to the PDEC, SAE, ColumnCRT, CleanKLS, or external KLS exits, and it still lacks independent D-structure/Rankin acceptance.

The support-failure packet return-dichotomy router then closes the return alphabet for these packets. It records

$$\texttt{support\_failure\_packet\_return\_dichotomy\_closed=true},$$
$$\texttt{clean\_core\_packet\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

An actual noncanonical support-failure packet cannot be a fifth terminal family. If it is not a clean-core packet, then it must return through one of the existing named interfaces: an isolated finite packet gives a LocalSurvivor/SAE certificate; a persistent finite signature gives an explicit primitive PDEC schema; a column, displacement, endpoint or cofactor load returns through the ColumnCRT-to-PDEC/SAE absorption; and a drifting moving block enters CleanKLS/DLS, exact source entropy, or an explicit external KLS input. The canonical and generic-template escapes have already been blocked by the ExactUVSupport audit.

Thus the latest source-side micro-input is

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion.
```

A clean-core packet is a positive-mass actual noncanonical full-S non-AP balanced block, in the same formal unit, below the exact u, v-support threshold, with no canonical import, no finite sparse witness, no persistent PDEC signature, no column or displacement defect, and no external or generic CleanKLS exit. The latest self-contained input basis is therefore

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes return completeness, not clean-core exclusion. The manuscript still lacks a proof that such clean-core packets are impossible, or a direct proof of the final capacity anti-atom.

The clean-core moving-atom sharp-input router then corrects one final over-strong formulation. It records

$$\texttt{clean\_core\_moving\_atom\_sharp\_boundary\_closed=true},$$
$$\texttt{clean\_core\_moving\_atom\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Excluding every clean-core low-support packet is sufficient, but it is stronger than the final capacity anti-atom requires. Once the registered multiplier discipline is closed, the contrapositive of the recorded capacity inequality says that any final $M_{u,v}$-atom produces a low-support packet. The converse is not needed: a low-support packet that does not concentrate final capacity is not a terminal obstruction. Therefore the sharp source-side input is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion.
```

Here a clean-core moving atom is a positive-mass actual noncanonical full-S non-AP balanced block, after all return tests, with some pair (u, v) satisfying

$$M_{u,v}/\sum_{u,v} M_{u,v} > \log^{-2A}.$$

The latest self-contained input basis is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is only a sharp reformulation. It is not implied by formal WFD, Type decomposition, Fourier smoothing, or fixed-projection diffuse; the moving-delta obstruction remains.

The clean-core terminal normal-form router then removes the last naming ambiguity around that sharp input. It records

`clean_core_terminal_normal_form_closed=true`,
`internal_exact_entropy_proved=false`,
`external_completed_kls_accepted=false`,
`dstructure_rankin_independent_acceptance_completed=false`,
`row_column_unconditional_closed=false`.

The internal normal form of `ActualNoncanonicalCleanCoreMovingAtomExclusion` is exact clean-core source entropy: for every post-return clean-core moving block $b = (u, v)$,

$$\max_b M_b/\sum_b M_b \le \log^{-2A}.$$

The external alternative is not a generic DI/BFI or Kuznetsov citation; it is the completed, modulus-dependent full-S input `ModulusDependentCompletedFullSKLSInput`. Thus the latest conditional terminal input basis is

```
(ExactCleanCoreFullSNonAPWFDSourceEntropy
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ExactCleanCoreFullSNonAPWFDSourceEntropy
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the terminal input normal form only. It does not prove exact entropy, accept the completed KLS input, or complete the independent D-structure/Rankin gate.

Finally, the clean-core exact-entropy atom router rewrites the failure mode of that self-contained input. It records

$$\texttt{clean\_core\_exact\_entropy\_atom\_boundary\_closed=true},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{exact\_clean\_core\_entropy\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

If exact clean-core entropy fails, there is a clean-core moving atom. The registered multiplier discipline removes unrecorded Type/Fourier/fiber capacity as an explanation; packetization forces any support failure to be reproducible; and the return dichotomy routes every non-clean-core packet back to the named exits. The only remaining self-contained failure mode is a clean-core terminal support atom. A sufficient actionable proof package is therefore

```
CleanCoreTerminalSupportIncidenceTheorem
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here `CleanCoreTerminalSupportIncidenceTheorem` means that every positive-mass post-return actual clean-core full-S non-AP WFD block has enough exact u/v support, inside the same formal unit, to absorb the divisor bound and all registered capacity multipliers. This theorem is not proved in the current corpus.

The clean-core support-incidence attack router then audits whether this theorem follows from the already closed ingredients. It records

$$\texttt{clean\_core\_support\_incidence\_attack\_boundary\_closed=true},$$
$$\texttt{clean\_core\_exact\_layer\_transfer\_proved=false},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{external\_completed\_kls\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The range threshold, registered capacity budget, and raw squarefree/Buchstab counting in thick blocks are no longer the obstruction. The naive factor-residue incidence bridge is still blocked by the internal h, ℓ, x, z fiber, and canonical RIW/Buchstab support cannot be imported into the noncanonical clean-core branch. The remaining internal input is

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

The latest conditional basis is therefore

```
(CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis replaces the parenthesized alternative by

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

This input requires exact clean-core layer admission, nonzero transfer to the actual α/δ coefficients, and return of thin or rejected blocks to the named exits. It is still open.

The clean-core layer-transfer path router then separates the available canonical template from what is needed for the noncanonical clean-core residual. It records

```
clean_core_layer_transfer_path_boundary_closed=true,
clean_core_path_partition_proved=false,
clean_core_exact_layer_transfer_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The canonical RIW/Buchstab decision tree and source-provenance ledger are closed only on the canonical-source branch. The noncanonical clean-core residual must supply its own actual coefficient path partition, or leave through the completed KLS or named-return interfaces. The newest internal input is

```
CleanCoreExactCoefficientPathPartitionNoCancellationAndThinReturn.
```

It asks for a polylogarithmic partition of the actual clean-core α/δ coefficients into exact path signatures, nonzero same-path contribution without cancellation, and return of thin, over-budget, cancelling, or unidentified-source blocks to the named exits.

The clean-core path-source firewall router then moves this input to the coefficient source level. It records

```
clean_core_path_source_firewall_boundary_closed=true,
clean_core_precauchy_source_law_proved=false,
clean_core_path_partition_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The path partition must be built on the pre-Cauchy actual clean-core α/δ coefficient formula. The canonical RIW/Buchstab decision tree is only a canonical-source template, and generic WFD well-factorability is blocked by the moving-delta model. The latest conditional basis is

```
(CleanCorePreCauchyCoefficientSourceLawAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCorePreCauchyCoefficientSourceLawAndReturn.
```

It requires an exact source formula before Cauchy, Type/Fourier and completion, plus polylog path signatures, same-path nonzero contribution, and named return for source-law failure, over-budget paths, or thin/rejected blocks.

The clean-core pre-Cauchy source-law atom router then compresses this source-law input to an origin-generation ledger. It records

$$\texttt{clean\_core\_precauchy\_source\_law\_atom\_boundary\_closed=true},$$
$$\texttt{origin\_generation\_ledger\_implication\_closed=true},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{clean\_core\_precauchy\_source\_law\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The compression theorem is that a pre-Cauchy source law is exactly an origin ledger for the actual clean-core α/δ coefficients in one formal unit before Cauchy, dispersion, Type/Fourier and completion. If that ledger lists every summand, branch key, u/v map, sign and local factor, then the branch keys give the exact path signatures, the registered K6/tail-label costs give the polylog path budget, and same-key nonzero or sign-split refinement gives no cancellation. Missing source entries, over-budget paths, thin/rejected blocks or unresolved cancellation must return to the named exits or to the external spectral input. The latest conditional basis is

```
(CleanCoreOriginalCoefficientGenerationLedgerAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCoreOriginalCoefficientGenerationLedgerAndReturn.
```

This input is not proved here. The router only shows that the former source-law input has no smaller honest self-contained evidence than the actual origin ledger; the canonical RIW/Buchstab ledger remains scoped to the canonical-source branch, and generic WFD well-factorability cannot replace it.

The clean-core origin-source admission router then compresses the origin ledger to a primitive source-constructor admission gate. It records

$$\texttt{clean\_core\_origin\_source\_admission\_boundary\_closed=true},$$
$$\texttt{constructor\_admission\_implies\_origin\_ledger=true},$$
$$\texttt{unregistered\_source\_return\_absorbed=false},$$
$$\texttt{clean\_core\_primitive\_source\_constructor\_admission\_proved=false},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

An origin ledger is not an estimate; it first needs a source constructor. If the actual clean-core α/δ coefficients are emitted by a pre-Cauchy primitive constructor inside one formal unit, then the emitted-summand schema expands into the origin ledger. If no such constructor is registered, the object must return as an unregistered-source failure rather than remain a clean-core terminal. The latest conditional basis is

```
(CleanCorePrimitiveSourceConstructorAdmissionAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

`CleanCorePrimitiveSourceConstructorAdmissionAndReturn.`

This input is still open: the current material proves the canonical constructor only on the canonical-source branch and blocks generic WFD as a constructor, but it does not prove noncanonical clean-core constructor admission or absorption of every unregistered source.

The clean-core constructor source-class firewall then separates this admission gate by source class. It records

```
constructor_source_class_firewall_boundary_closed=true,
source_class_partition_closed=true,
unregistered_source_return_absorbed=true,
actual_noncanonical_primitive_constructor_formula_proved=false,
clean_core_primitive_source_constructor_admission_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The firewall has five classes. The canonical constructor is already closed but scoped to the canonical-source branch. A generic well-factorable template is not a constructor. Unregistered or mixed formal-unit sources return through Multiplicity/Stitching or the K7 formal-unit failure. The external spectral class remains the completed c-dependent residue-weight input. Thus the only self-contained source-side atom is

`ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn.`

Equivalently, the latest conditional basis is

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The source-side input asks for a pre-Cauchy definition of the actual noncanonical clean-core coefficients, a deterministic summand emitter for α/δ, branch keys, u/v maps, signs and local factors, and compatibility with the same formal unit used by the support, capacity and return ledgers. This formula is not proved here.

Finally, the external-lemma parameter-match router compares this remaining self-contained atom with the available DI/BFI/Kuznetsov tools. It records

```
external_lemma_parameter_match_boundary_closed=true,
external_lemmas_match_constructor_formula=false,
external_lemmas_close_self_contained_remainder=false,
external_spectral_atom_accepted=false,
actual_noncanonical_primitive_constructor_formula_proved=false,
row_column_unconditional_closed=false.
```

The comparison is structural. DI/Kuznetsov and BFI act after coefficients, residue weights, or Kloosterman phases have already been produced: they give spectral or AP-distribution control. The current self-contained atom is earlier in the chain: it asks for the pre-Cauchy actual noncanonical source constructor and summand emitter. Therefore those external lemmas cannot replace the constructor formula. They remain relevant only to the external spectral branch, whose narrow target is

```
CDependentResidueWeightSpectralCancellationInput.
```

The conditional basis is unchanged:

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis remains

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core reverse-provenance functor router then audits whether the older geometric and payment models can be used in the opposite direction. It records

```
reverse_provenance_functor_boundary_closed=true,
payment_pushforward_functoriality_available=true,
pushforward_reverse_uniqueness_rejected=true,
finite_projection_recovers_gamma_not_source=true,
constructor_formula_equivalent_to_registered_fiber_emitter=true,
registered_primitive_prepushforward_fiber_emitter_proved=false,
actual_noncanonical_primitive_constructor_formula_proved=false,
row_column_unconditional_closed=false.
```

The point is functorial rather than numerical. The actual payment graph Γ, its finite projections, cyclic arc restrictions, and transverse quotients are forward pushforwards or finite factors of a pre-Cauchy source. They preserve the registered formal unit and prevent source replacement, but they do not invert the pushforward: several primitive source disintegrations may have the same Γ. Thus one cannot recover a primitive constructor merely from the downstream payment graph.

Modulo the already closed return discipline, however, the constructor formula is equivalent to a more auditable input:

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn.
```

Such an emitter must list, before Cauchy/dispersion, finite or polylogarithmically many primitive preimage summands over each positive-mass clean-core payment atom, with α/δ, branch keys, u/v maps, signs, local factors, formal-unit registration, and coefficient identities. Empty fibers, incompatible registrations, over-budget fibers, thin blocks, or unregistered sources must return through named exits. The latest conditional basis is therefore

```
(RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis becomes

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core fiber-emitter field audit then separates the already available payment skeleton from the still missing signed coefficient lift. It records

`fiber_emitter_field_audit_boundary_closed=true`,
`payment_fiber_skeleton_closed=true`,
`first_cover_payment_map_closed=true`,
`payment_count_identity_closed=true`,
`alpha_delta_coefficient_lift_proved=false`,
`polylog_branch_schema_proved=false`,
`registered_primitive_prepushforward_fiber_emitter_proved=false`,
`row_column_unconditional_closed=false`.

The payment-level skeleton is now explicit: a completion y and a low hole c are sent by the first-cover rule $\mathrm{pay}(c, y)$ to a payment atom in Γ, and the count identity

$$\text{payment\_count} = \sum_{\text{phase}} M(\text{phase})\,|H_{\text{low}}(\text{phase})|$$

has been checked. ActualPaymentStitching also leaves no fourth payment-layer exit. What is not supplied by this skeleton is the signed clean-core primitive coefficient table. Thus the newest self-contained input is

```
ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The alpha/delta disintegration router then rewrites this lift in its intrinsic form. It records

`alpha_delta_disintegration_boundary_closed=true`,
`payment_base_map_closed=true`,
`lift_equivalent_to_signed_disintegration_dictionary=true`,
`canonical_decision_tree_template_scoped=true`,
`generic_or_unregistered_dictionary_blocked=true`,
`registered_alpha_delta_disintegration_dictionary_proved=false`,
`exact_alpha_delta_lift_proved=false`,
`row_column_unconditional_closed=false`.

Let Φ denote the first-cover payment map from primitive source summands to the completion-hole payment skeleton. An exact alpha/delta lift is equivalent to a registered signed disintegration dictionary for the actual noncanonical source measure over Φ: one must list the signed preimage summands, branch keys, u/v maps, signs, local factors, variation/support budgets, and the push-forward identity. The canonical RIW/Buchstab decision tree is a model for such a dictionary only on the canonical branch; generic well-factorability or unregistered sources do not provide it. The latest self-contained basis is therefore

```
RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Finally, the disintegration automaticity router removes a false hard point. It records

$$\begin{gathered}\texttt{disintegration\_automaticity\_boundary\_closed=true},\\ \texttt{discrete\_payment\_base\_map\_closed=true},\\ \texttt{signed\_fiber\_disintegration\_formal=true},\\ \texttt{pushforward\_identity\_is\_real\_gate=true},\\ \texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},\\ \texttt{row\_column\_unconditional\_closed=false}.\end{gathered}$$

Once a signed source measure ν and the first-cover map Φ are given, the fiber decomposition

$$\nu = \sum_a \nu|_{\Phi^{-1}(a)}$$

is formal on the discrete payment skeleton. The real compatibility gate is not disintegration itself, but the proof that ν is the actual noncanonical alpha/delta source measure in the same formal unit, that $\Phi_*\nu$ equals the target payment-side coefficient, and that variation, absolute support, and branch-key budgets are registered. The latest self-contained basis is

```
ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is a boundary refinement, not a proof of the signed source or budget identity.

The geometric Phi/budget bridge router then returns the line-covering, cylindrical, P-column anchor, layered wheel, and dynamic-capacity models to this same interface. It records

$$\begin{gathered}\texttt{geometric\_phi\_budget\_bridge\_boundary\_closed=true},\\ \texttt{geometric\_payment\_base\_available=true},\\ \texttt{geometry\_defines\_signed\_source\_measure=false},\\ \texttt{geometry\_proves\_phi\_pushforward\_identity=false},\\ \texttt{geometry\_supplies\_budget\_return\_shape=true},\\ \texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},\\ \texttt{row\_column\_unconditional\_closed=false}.\end{gathered}$$

The geometric models provide the unsigned payment base map, candidate variation/support budgets, and named returns such as PDEC, SAE, ColumnCRT, and CleanKLS. They do not define the pre-Cauchy signed α/δ source measure, nor do they prove the coefficient identity $\Phi_*\nu =$ target payment-side coefficient. Hence the previous single input splits into two atoms:

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn.
```

The latest conditional basis is therefore

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This again refines the boundary rather than proving the two remaining atoms.

The geometric variation/branch budget router then attacks the second atom. It records

geometric_budget_attack_boundary_closed=true,
geometry_ledger_alphabet_closed=true,
dprc_analytic_capacity_bound_proved=false,
signed_variation_branch_lift_proved=false,
geometric_variation_branch_budget_certificate_proved=false,
row_column_unconditional_closed=false.

The line-cylinder, bottom-deficit, P-column anchor, dynamic promoted wheel, layered wheel, and input-atlas ledgers close the unsigned support and named-return alphabet. They do not control the total variation of the actual signed source, since mass may cancel after pushforward along Φ. The budget atom therefore splits as

```
GeometricVariationBranchBudgetCertificateOrNamedReturn
=> DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the budget-side interface, not the two new inputs.

The DPRC centered-discrepancy router then attacks the DPRC atom itself. It records

dprc_centered_discrepancy_boundary_closed=true,
rsm_identity_closed=true,
bes_compression_closed=true,
dprc_centered_discrepancy_input_proved=false,
row_column_unconditional_closed=false.

The relative-sieve margin identity reduces $T_Y < S_Y$ to an explicit model-gap/finite ledger and a centered positive-discrepancy bound. The BES compression reduces that bound to the exclusion of simultaneous high L^1 and high L^2 positive beta-bucket synchronization. If such a danger intersection is not directly excluded, the DLS route requires a named return:

```
PointLoad/ColumnCRT,  ShortWindow/SAE,  LowPhase/PDEC.
```

Thus

```
DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
=> ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the DPRC interface, not the finite/model-gap or BES-DLS proofs.

The BES-DLS named-return router then attacks the BES-DLS atom. It records

`bes_dls_named_return_boundary_closed=true`,
`dls_kernel_and_danger_algebra_closed=true`,
`bes_dls_named_return_input_proved=false`,
`row_column_unconditional_closed=false`.

The centered kernel, danger intersection, spike-bucket pigeonhole, and quadratic energy expansion are now registered. If the BES danger intersection occurs, it may no longer remain an unnamed synchronization failure. It must enter one of three exits:

```
PointLoad -> ColumnCRT/tail-anchor/displacement PDEC,
ShortWindow -> SAE/LocalSurvivor/persistent sparse PDEC,
LowPhase -> W-unit PDEC/new-layer PDEC/flat DLS.
```

Thus

```
BESDangerIntersectionExclusionOrDLSNamedReturn
=> DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the named-return boundary, not the three microinputs.

The DLS LowPhase PDEC/flat-DLS router then attacks the LowPhase atom. It records

$$\texttt{dls\_lowphase\_boundary\_closed=true},$$
$$\texttt{lowphase\_input\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

LowPhase is not a single fixed-modulus law. If a fixed-wheel unit-class peak persists, it must return a W-unit PDEC. If a new Fourier layer concentrates in low dimension, it must return a new-layer PDEC. If both fixed-wheel and new-layer peaks dilute, the remaining mass is genuinely high-modulus and flat, so it must be absorbed by flat DLS/KLS. Thus

```
DLSLowPhasePDECNewLayerOrFlatDLSBound
=> DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the LowPhase boundary, not the three new microinputs.

The new-layer external-lemma matching router then compares the most structural LowPhase microinput with the already registered external FullS–KLS and CleanKLS/DLS interfaces. It records

$$\texttt{direct\_external\_lemma\_closes\_newlayer=false},$$
$$\texttt{self\_contained\_newlayer\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The comparison is not a new numerical experiment. It is a parameter-level boundary check: external KLS/FullS lemmas estimate a prepared, completed, coefficient-given flat spectral block. The current new-layer atom sits one layer earlier. It must first prove that low-dimensional concentration on the added wheel fibre is an actual PDEC certificate, or that after deleting all such fibres the remaining object satisfies the clean flat-admission hypotheses. Hence

```
DLSNewLayerFourierConcentrationPDECReturn
=> ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission.
```

The latest self-contained basis becomes

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the external theorem can attach only after a new-layer projection/admission morphism; it cannot be used directly to erase the self-contained new-layer obligation.

The new-layer PDEC projection router then attacks the first of these two atoms. It records

$$\texttt{newlayer\_projection\_slicer\_closed\_schema\_admission\_open},$$
$$\texttt{exact\_newlayer\_projection\_morphism\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Two deterministic sublemmas are closed. First, on the same full-period completion set C_P, lifting Q to rQ cannot create support outside the old projection. Second, a large $r \nmid h$ Fourier coefficient on the new fibre can be rotated and sliced by a layer-cake argument to produce a cyclic-arc or half-plane cap with comparable signed mass discrepancy. Hence

```
ExactNewLayerFiberPDECProjectionMorphism
=> RegisteredNewLayerPDECFormalUnitAndCapStableSchema.
```

The latest self-contained basis is therefore

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND RegisteredNewLayerPDECFormalUnitAndCapStableSchema
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The remaining obstruction is formal PDEC admission for the sliced cap: the same Ω, τ, w and phase map must be fixed, the cap must be non-tautological and cap-stable, and its lower and upper budgets must be computed in that same formal unit. $\square$

Theorem 7.4 (Early-zero gap carrier and CRT-asymmetry router)**.** *The early-zero gap-carrier observation is a valid entrance reduction but not by itself a final contradiction. If the k-th P-row*

$$I_{P,k} = [(k-1)P + 1, kP]$$

contains no prime, and if a and b are the nearest primes respectively to the left and to the right of $I_{P,k}$, then a, b are adjacent primes and $b - a > P$. The CRT period M_P repeats the small-factor covering pattern, but it does not repeat the primality of the carrier endpoints. Consequently a persistent endpoint phase is a PDEC/ColumnCRT object, an isolated endpoint phase is an SAE object, and the non-periodic prime endpoint case returns to the H3-DSB/KLS branch, whose current self-contained hard core is NC-BLK unless an external DI/BFI dispersion input is accepted.

Proof. The finite certificate is recorded in

`docs/monograph/prime-matrix-early-zero-gap-crt-asymmetry-router.md.`

If $I_{P,k}$ contains no prime, then every prime between a and b would have to lie inside $I_{P,k}$, contradicting the hypothesis. Hence a, b are adjacent. Since $a \le (k-1)P$ and $b \ge kP+1$,

$$b - a \ge (kP+1) - (k-1)P = P + 1 > P.$$

This proves the carrier-gap lemma.

The second assertion is a boundary statement. Small-factor coverage of a row is periodic modulo M_P; primality of the carrier endpoints is not periodic modulo M_P. Therefore the carrier cannot be used as a naked CRT-period contradiction. If its endpoint phase recurs as a fixed formal unit, it is a PDEC/ColumnCRT candidate. If it is isolated, it is SAE. If the endpoint primality cannot be made periodic while the covering pressure persists, the already established H3 singleton-tail reduction sends the obstruction to the H3-DSB/KLS chain. The present document has not closed NC-BLK internally and has not replaced it by an accepted external DI/BFI theorem, so the global row/column theorem remains open. □

Theorem 7.5 (Period-lift carrier drift router). *Let*

$$L_P = \prod_{\substack{q<P \\ q \text{ prime}}} q.$$

If x is a P-zero-row multiplier, then $x + tL_P$ is again a P-zero-row multiplier for every $t \ge 0$. The corresponding intervals are shifted by PL_P. However, the adjacent prime carriers around these intervals are not forced to translate by PL_P. Thus the period-lifted counterexample cover and the true adjacent-prime endpoint chain have an intrinsic asymmetry: persistent endpoint drift is a PDEC/ColumnCRT object, sparse drift is SAE, and non-periodic drift with persistent covering pressure returns to the H3-DSB/KLS NC-BLK or external DI/BFI branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-early-zero-period-lift-carrier-drift-router.md.`

For each non-trivial column $1 \le c < P$, the zero-row hypothesis gives a prime $q < P$ with $q \mid Px+c$. Since $q \mid L_P$,

$$P(x + tL_P) + c \equiv Px + c \pmod{q},$$

so the same small prime q covers the lifted entry. This proves exact period lift of the covering chain.

Let a_t, b_t be the adjacent primes immediately outside the lifted interval $[P(x+tL_P)+1, P(x+tL_P)+P]$. The congruence calculation above contains no condition forcing $a_t = a_0 + tPL_P$ or $b_t = b_0 + tPL_P$. It only proves absence of primes inside the lifted interval. Hence endpoint translation is not a CRT consequence. Any infinite structured use of the endpoint drift must therefore be named: fixed recurring drift gives a PDEC/ColumnCRT formal unit; isolated drift gives SAE; and non-periodic endpoint drift leaves the tail filler/short-window dispersion problem already registered as H3-DSB/KLS with NC-BLK or external DI/BFI as the remaining input. This proves the router but not the final row/column theorem. □

Theorem 7.6 (Q2-stage endpoint inversion router). *Let an early zero row be straddled by adjacent primes $Q_1 < Q_2$. If $Q_2 < P^2$, then every composite in the open carrier gap (Q_1, Q_2) already has* *a prime factor below P; otherwise the branch enters the square-anchor endpoint problem. At the Q_2-stage, the full wheel*

$$M_{\le Q_2} = \prod_{\substack{\ell \le Q_2 \\ \ell \text{ prime}}} \ell$$

does not reproduce the two prime endpoints. For every $t \ge 1$,

$$Q_1 + tM_{\le Q_2} \equiv 0 \pmod{Q_1}, \qquad Q_2 + tM_{\le Q_2} \equiv 0 \pmod{Q_2},$$

so the endpoint-stable full-wheel replay is impossible. The surviving branches are: a persistent closed carrier block, which is a ColumnCRT/PDEC object; a sparse carrier block, which is SAE; or moving endpoints/support, which returns to the moving-family H3-DSB/KLS branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-carrier-stage-crt-asymmetry-router.md.`

The below-square assertion follows because a composite $n < Q_2 < P^2$ has a least prime factor at most $\sqrt{n} < P$, while adjacency of Q_1, Q_2 excludes primes in the open gap. The endpoint inversion is the displayed congruence: both endpoints divide the full Q_2-wheel modulus, so any positive full-wheel copy of either endpoint is composite. If one uses only $M_{<Q_2}$, the left endpoint Q_1 is still killed by its own zero class, and the right endpoint Q_2 is not forced to remain prime by CRT data. Thus the full Q_2-stage gives a genuine local contradiction only to endpoint-stable replay. Without endpoint-stable replay, the obstruction is precisely one of the named ColumnCRT/PDEC, SAE, or moving-family exits. Hence the router is closed, but the global row/column theorem is not. ☐

Theorem 7.7 (Q2 endpoint replacement and aperture growth)**.** *In the Q_2-stage of the preceding theorem, full-wheel replay creates a replacement debt for the true adjacent-prime endpoints. More precisely, in the below-square branch the full Q_2-wheel copy of $[Q_1, Q_2]$ is a closed composite block. Hence if $A_t < B_t$ are the adjacent primes straddling that copied block, then*

$$B_t - A_t \ge Q_2 - Q_1 + 2.$$

Consequently a fixed bounded carrier aperture cannot support infinitely many endpoint-stable replays. A fixed aperture of width W, starting from closed carrier width $W_0 = Q_2 - Q_1 + 1$, permits at most

$$\left\lfloor \frac{W - W_0}{2} \right\rfloor$$

such replacements. Any infinite continuation must move or enlarge the aperture and is therefore routed to moving-aperture ColumnCRT/PDEC, SAE, or the H3-DSB/KLS moving-family branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-replacement-aperture-growth-router.md.`

For an early zero row $[(k-1)P + 1, kP]$, the left endpoint satisfies $Q_1 \le (k-1)P = kP - P$, with equality possible only in the $k = 2$ boundary case $Q_1 = P$; the right endpoint satisfies $Q_2 > kP$. In the $Q_2 < P^2$ branch, every composite in the open gap has least prime factor below P, and the full Q_2-wheel kills Q_1 and Q_2 themselves. Thus every point of the copied closed interval

$$[Q_1 + tM_{\le Q_2}, Q_2 + tM_{\le Q_2}]$$

is composite for $t \ge 1$. The nearest true primes around this copied block must lie outside it, giving the displayed $+2$ lower bound. Iterating this lower bound inside a fixed aperture gives the finite bound $\lfloor (W - W_0)/2 \rfloor$. Therefore the bounded-aperture branch is closed; the remaining branches are exactly the named moving, PDEC, SAE, and H3 exits. ☐

Theorem 7.8 (Fresh endpoint layer cascade). *After the bounded-aperture branch is removed, any continued full-wheel endpoint replay must introduce an unbounded sequence of fresh endpoint prime layers. Let M_j be the full-wheel modulus at stage j, and let B_{j+1} be the new right adjacent-prime endpoint forced after the copied closed composite block. Then*

$$B_{j+1} > M_j.$$

If $L_j = \log M_j$, the next full wheel satisfies

$$L_{j+1} \geq L_j + \log B_{j+1} > 2L_j.$$

Thus the endpoint cascade cannot terminate in a fixed finite CRT period. A persistent fresh endpoint phase is a ColumnCRT/PDEC object; a sparse cascade is SAE; and a non-PDEC unbounded fresh-layer cascade is routed to the one-residue per fresh prime tail-sieve/H3-DSB/KLS branch.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-endpoint-fresh-layer-cascade-router.md.`

At stage j, full-wheel replay moves the current closed carrier by at least M_j and turns it into a closed composite block. The true right adjacent prime around that copied block must therefore lie to the right of the copied block, hence beyond M_j. This prime is not contained in any fixed finite endpoint layer below the old bound. When the next wheel includes it, its next copy is divisible by itself, so the same replacement mechanism repeats. The displayed logarithmic inequality follows immediately. Since a finite CRT period contains only finitely many prime layers, it cannot support this infinite cascade. The only remaining possibilities are the named PDEC, SAE, or tail-sieve/H3 exits. □

Theorem 7.9 (Fresh-layer tail-mass dichotomy). *In the non-PDEC branch of the fresh endpoint layer cascade, assume the local aperture at stage j is W_j and the fresh endpoint prime is B_j. If the fresh layer does not create a persistent ColumnCRT/PDEC correlation, then its local one-residue contribution is bounded by*

$$\frac{W_j}{B_j}.$$

Under the controlled-aperture condition $\log W_j = o(\log B_j)$, and in particular under the endpoint-replacement model $W_j = W_0 + 2j$, the total fresh-tail mass is summable:

$$\sum_j \frac{W_j}{B_j} < \infty.$$

Thus the controlled non-PDEC fresh-tail branch is SAE. If instead the aperture repeatedly grows fast enough to track the fresh modulus, the branch is no longer local endpoint replacement; it is aperture explosion or global support motion and must be routed to H3-DSB or moving-support PDEC.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-fresh-layer-tail-mass-dichotomy-router.md.`

The previous cascade theorem gives $L_{j+1} > 2L_j$, where $L_j = \log M_j$ is the logarithm of the full-wheel modulus, and also gives $B_{j+1} > M_j$. Hence the denominators B_{j+1} have lower bounds $\exp(L_j)$

whose logarithms grow at least double-exponentially in the stage index. If $\log W_j = o(\log B_j)$, then for all sufficiently large j one has $W_j \le B_j^{1/2}$, so

$$\frac{W_j}{B_j} \le B_j^{-1/2},$$

and the right-hand series converges by the same double-exponential lower growth. The linear endpoint-replacement aperture $W_j = W_0 + 2j$ is a special case.

The only ways to avoid this SAE absorption are exactly the two named failures of the hypotheses. A persistent concentration of fresh-layer hits is, by definition, a ColumnCRT/PDEC formal unit. A failure of controlled aperture is an aperture explosion or support-motion event, not the local endpoint replay branch. Therefore the router closes the controlled fresh-tail outlet but not the global row/column theorem. □

Theorem 7.10 (Q2 aperture-explosion schema firewall)**.** *After the controlled fresh-tail branch has been absorbed into SAE, any remaining aperture-explosion branch must present an explicit persistent structure: support motion, a blocker-package change, a fresh-layer ColumnCRT/PDEC unit, or a moving-family schema. In the current materialized corpus, unnamed aperture explosion is not an admissible terminal. Registered support motion has already been routed away, current materialized fresh-layer PDEC/ColumnCRT has already been blocked by the admission firewall, and the branch-replay current frontier is zero. Hence the only surviving form is a future explicit schema, which must reopen under the stated firewall; the global row/column theorem is still not proved.*

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-aperture-explosion-schema-firewall-router.md.`

The previous theorem leaves only two ways out of SAE: persistent fresh-layer correlation, which is PDEC/ColumnCRT, or failure of controlled aperture, which is support motion or aperture explosion. The current registered support-motion certificate eliminates local support drift for the registered blocks and routes persistent registered replay to named ColumnCRT/PDEC objects. The fresh-layer admission firewall eliminates unnamed materialized PDEC/ColumnCRT in the current corpus. The current-frontier-zero certificate then states that no active branch-replay terminal instance remains in the materialized ledger.

Therefore an aperture-explosion claim cannot be kept as an unnamed terminal. It must specify its source family, blocker package, shape, phase map and persistence test, with a reproducible ledger. Such a future schema may reopen the branch, but it is not a present closed counterexample chain. This proves the schema-firewall synchronization only, not the final theorem. □

Theorem 7.11 (Q2 CRT ladder to final exact-source alignment)**.** *The Q2-stage CRT ladder has no remaining current unnamed local terminal: the controlled fresh tail is SAE, and aperture explosion must enter an explicit schema firewall. However, this does not by itself prove the row/column theorem. The CRT ladder controls positions and phases of allowed and forbidden classes; it does not produce a mass-dispersion estimate for the actual pre-Cauchy source over exact (u, v)-fibers. Therefore the Q2 branch can promote only through one of the final source inputs:*

$$\begin{gathered} \textit{NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource} \\ \textit{or ExternalDIBFIKuznetsovDispersionTheoremMatch,} \end{gathered}$$

together with independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-q2-to-final-exact-source-alignment-router.md.`

The preceding Q2 routers prove that a full Q_2-wheel repeats the composite cover but destroys endpoint-prime replay, that continued endpoint replacement forces fresh endpoint layers, that controlled fresh tails are summable, and that current unnamed aperture explosion is barred by the schema firewall. Thus there is no current Q2-local terminal left to carry an infinite counterexample chain.

The remaining obstruction is of a different type. The final source anti-atom concerns the distribution of the actual signed pre-Cauchy source among exact (u, v)-fibers before Cauchy, pair energy and terminal extraction. CRT wheel rigidity gives congruence positions, not a bound on that source mass. Hence the strict self-contained branch must prove the nonterminal exact-UV fiber aperiodicity estimate, equivalently the already named source-domain rank/no-collapse package. The alternative is an accepted external DI/BFI/Kuznetsov dispersion theorem. In either case the separate DStructure/Rankin promotion gate remains open. This proves alignment of the remaining Q2 branch, not unconditional closure. □

Theorem 7.12 (Global CRT homogeneity frontier)**.** *The Q1/Q2 adjacent-prime carrier gives a genuine endpoint obstruction: at the full Q_2-wheel the two endpoint primes cannot replay as primes. Nevertheless this obstruction is not, by itself, a global CRT phase contradiction. For any squarefree wheel modulus M_Y and any fresh prime $r \nmid M_Y$, the lifts $a + tM_Y$, $0 \le t < r$, of a fixed old residue a run through all residue classes modulo r. Hence the fresh layer deletes exactly one class in each old fiber. Finite CRT growth is therefore a homogeneous one-class deletion mechanism, not a deterministic short-interval occupancy theorem, unless the additional phase equations form a registered PDEC/ColumnCRT collision.*

Consequently the pure global-CRT phase-contradiction outlet is closed as a final proof route. The remaining strict self-contained branch must pass through actual-source exact-(u, v) fiber dispersion and then the pointwise primitive alpha/delta kernel table. Its first active atom is

$$AlphaRowAnchorPhaseEmissionFormulaLedger,$$

together with the independent pre-Cauchy arithmetic identity, the same-unit rank/multiplicity certificate, and independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-homogeneity-frontier-router.md.`

The previous Q2 routers already prove endpoint inversion, fixed-period removal, controlled fresh-tail SAE absorption, and the aperture-explosion schema firewall. These remove the local endpoint-replay and unnamed moving outlets. The remaining question is whether the infinite CRT wheel itself forces a global phase contradiction.

The lift calculation above is exact because M_Y is invertible modulo the fresh prime r. As t varies modulo r, the class $a + tM_Y$ also varies through all classes modulo r; precisely one value is 0. Thus each finite wheel extension applies the same relative deletion to every old fiber. The Euler product explains the critical density scale, but it does not supply a lower bound for the actual mass in a moving interval of length P near P^2.

Therefore CRT position and phase rigidity must be supplemented by actual source information. The already archived strict descent identifies that source information with exact-(u, v) fiber aperiodicity and then with the pointwise primitive alpha/delta kernel table. Since the alpha-row emission formula, the signed pre-Cauchy identity, the rank/multiplicity certificate, and DStructure/Rankin promotion acceptance remain open, this theorem is a frontier synchronization, not a proof of the row/column theorem. □

Theorem 7.13 (Global CRT route to terminal saturation)**.** *The global CRT/Q1-Q2 route does not stop at the coarse alpha-row atom. After the pure CRT homogeneity firewall is imported, the local alpha-row anchor/phase formula has already been synchronized to the terminal PDEC/CleanKLS gate. The PDEC/CleanKLS gate has then been attacked to the current internal saturation frontier: the CleanKLS/Kuznetsov-DLS arm returns to the strict acyclic terminal family, the nonrecursive constructor/signed-lift breaker returns to the signed coordinate-source cycle, and the seed-cycle-cut branch is saturated. Therefore the current non-cyclic strict inputs are*

$$\begin{gathered} \mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\ \text{or } \mathit{NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact}, \end{gathered}$$

together with the model-gap ledger, rate preservation for the moving-atom packet, and independent DStructure/Rankin promotion acceptance.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-terminal-saturation-sync-router.md.`

The previous theorem leaves the strict route at the pointwise alpha/delta kernel table. The archived alpha-row terminal synchronization shows that the unsigned carry-shell, source-tuple and phase-wheel parts of the alpha formula are local geometric data, while the signed-lift and anchor-overload parts return to the global PDEC/CleanKLS terminal gate and the model-gap ledger.

The archived terminal split then separates the terminal gate into a direct same-set PDEC scope arm and a clean Kloosterman large-sieve arm. Continuing the clean arm through the Kuznetsov/DLS files returns to the strict acyclic terminal family rather than producing a self-contained log-saving theorem. Continuing the nonrecursive breaker returns to the signed source-coordinate cycle; continuing seed-cycle-cut returns to the same row-level fixed point unless a new joint emitter formula is supplied.

Thus the CRT route, the alpha-row route and the current terminal-family route all meet at the same saturated frontier. A proof must either supply the same-set PDEC scope match or a new explicit joint alpha/delta constructor formula, while also paying the model, rate and DStructure/Rankin gates. Since none of these has been proved here, this is a sharpening of the frontier, not unconditional closure. □

Theorem 7.14 (Global CRT route to branch-trace frontier)**.** *The non-PDEC side of the current global CRT frontier can be sharpened further. The alternative "new explicit joint alpha/delta formula" is not a single undifferentiated atom. If it is split through the existing alpha-side, same-row, row-level and signed-source routes, it returns to the already registered source-coordinate fixed point. Therefore a genuinely new formula must be anti-split: it must produce atomic joint rows before alpha-side projection. The ordinary anti-split declaration still degrades to the old split cycle unless it contains built-in word/coefficient pairing, and that built-in pairing further requires an exact atomic joint branch trace. Thus the current strict frontier is*

$$\begin{gathered} \mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\ \text{or } \mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}, \end{gathered}$$

with the ExactUV, model-gap, rate-preservation and DStructure/Rankin gates still present.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-branch-trace-frontier-router.md.`

The same-set PDEC scope arm remains an independent possible new certificate, but it is not proved in the present strict corpus. On the other arm, the archived joint-formula obligation says that the old joint declaration, alpha-side primitive rule and same-row origin identity return to the signed-source fixed point. The anti-split router therefore requires a non-split joint primitive word/coefficient formula. The declaration firewall then sharpens this to atomic pre-Cauchy joint rows with built-in pairing, and the atomic pairing router sharpens it to a built-in signed coefficient pairing closed form.

The signed coefficient is odd branch data: it depends on orientation and local factors, not only on the unsigned CRT skeleton. Hence the closed-form frontier is an exact atomic branch trace that, before Cauchy, Phi and payment pushforward, gives the basis word, signed coefficient, alpha/delta payload, orientation/local factor, exact (u, v), and named returns in the same formal unit. The current corpus has not supplied that trace formula. This proves the frontier sharpening only, not the row/column theorem. ☐

Theorem 7.15 (Global CRT route to signed-payload frontier)**.** *After importing the strict atomic-payload audit, the branch-trace side of the current global CRT frontier sharpens from*

$$\mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}$$

to

$$\mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn}.$$

Moreover, in the present self-contained internal corpus the PDEC same-set arm is saturated: unless a new acyclic same-set scope certificate, or an external no-projection DI/BFI input, is supplied, that arm returns to the new-joint formula line, and the new-joint line has already been routed through branch trace to signed payload. Thus the internal self-contained frontier is

$$\begin{aligned}&\mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn}\\&\quad \textit{and } \mathit{ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem}\\&\quad \textit{and } \mathit{ExplicitModelGapAndFiniteDPRCLedger}\\&\quad \textit{and } \mathit{RatePreservationLedger\_FOR\_moving\_atom\_packet}\\&\quad \textit{and } \mathit{DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance}.\end{aligned}$$

If a new PDEC scope input is retained, the global activity basis is instead

$$\begin{aligned}&\mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate}\\&\quad \textit{or } \mathit{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn},\end{aligned}$$

with the same ExactUV, model, rate and DStructure/Rankin gates.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-global-crt-signed-payload-sync-router.md.`

The preceding theorem left the non-PDEC side at an exact atomic branch trace. The archived signed-payload audit separates such a trace into its visible coordinate part and its signed payload part. The coordinate part follows the CRT anchor, phase and word-coordinate data. The payload part consists of the orientation bit, local-factor product, signed coefficient, same-row word/coefficient identity, alpha/delta prepushforward identity and exact (u, v) key. These are not consequences of the unsigned CRT skeleton.

The PDEC arm is treated separately. The archived scope-saturation audit says that current internal PDEC arguments either require a new same-set scope certificate, require an external no-projection DI/BFI input, or return to the new-joint formula line. The branch-trace synchronization has already routed that new-joint line to exact atomic branch trace, and the payload audit routes exact atomic branch trace to signed payload. Therefore the self-contained internal route has the stated payload frontier, while the new PDEC scope route is retained as an independent unproved input. No signed payload constructor, PDEC scope certificate, ExactUV incidence theorem, model-gap ledger, rate preservation ledger, or DStructure/Rankin promotion is proved here, so this is a frontier sharpening rather than unconditional closure. □

Theorem 7.16 (Predecessor-gap P-CRT uniformity route)**.** *Let $p^- < P$ be consecutive primes. A large gap $P - p^-$ creates a local first-row actual-prime deficit in the columns $p^- + 1, \ldots, P - 1$, but it does not create an imbalance in the complete P-level CRT wheel. In the full period $M_{\le P} = PM_{<P}$, before imposing the P-zero class, each residue class modulo P contains exactly $\varphi(M_{<P})$ residues coprime to $M_{<P}$. After imposing the P-zero class, the P-column is deleted and every nonzero column $c \in \mathbb{F}_P^*$ still contains exactly $\varphi(M_{<P})$ surviving CRT classes. The reflection $n \mapsto -n \pmod{M_{\le P}}$ also pairs c with $P - c$ exactly. Therefore a predecessor-gap argument can close the row/column theorem only if it supplies a localization transfer from this complete-wheel identity to the initial $P \times P$ square, or else a new PDEC scope or signed-payload input.*

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-predecessor-gap-pcrt-uniformity-router.md.`

The complete-wheel count is a direct CRT bijection. For every unit $a \bmod M_{<P}$ and every $c \bmod P$, there is a unique residue $n \bmod PM_{<P}$ satisfying $n \equiv a \bmod M_{<P}$ and $n \equiv c \bmod P$. Hence each P-column has $\varphi(M_{<P})$ such classes before the P-sieve. The P-sieve removes exactly the class $c = 0$, leaving all nonzero columns equal. Negation gives the stated reflection symmetry.

The gap $P - p^-$ is different in kind. It says that several first-row columns contain no prime atom, not that the full wheel has unequal nonzero columns. Passing from the full-period identity to the initial $P \times P$ arc is precisely a short localization problem. The archived scan for $P \le 5000$ records the largest predecessor gaps and observes that the first-row missing columns are repaired later in those finite squares, but this empirical repair is not used as proof. Without a localization transfer, the route returns to the current frontier: same-set PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Thus this theorem closes a false shortcut, not the row/column theorem. □

Theorem 7.17 (Localized P-CRT transfer and the Linnik–2 barrier)**.** *The column side of*

$$\mathit{LocalizedPCRTColumnUniformityTransferToInitialPxPSquare}$$

is equivalent to the following pointwise least-prime-in-progressions assertion for prime moduli:

$$\forall c \in \mathbb{F}_P^* \quad \exists\, \ell \le P^2 \quad \textit{prime with} \quad \ell \equiv c \pmod{P}.$$

Thus any proof that localizes complete P-wheel nonzero-column uniformity to the initial $P \times P$ square must either prove this Linnik–2 type pointwise AP assertion, or else introduce a genuinely new structural input such as a same-set PDEC scope certificate or a signed-payload constructor.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-localized-pcrt-transfer-linnik2-barrier-router.md.`

The c-th non-P column of the initial square is exactly the finite progression

$$c,\ c+P,\ c+2P,\dots,c+(P-1)P,$$

intersected with the prime numbers. Since $1 \le c < P$, every term is congruent to $c \bmod P$ and is at most P^2. Therefore the column has a prime if and only if the displayed pointwise AP assertion holds for that residue c. Requiring this for every nonzero column gives the asserted equivalence.

The complete CRT wheel does not contain this pointwise first-prime information. It says that the nonzero columns have equal total mass over a full period $M_{\le P}$, while the column theorem asks for a hit in the first P terms of each residue class. Mean AP estimates or complete-period symmetry can at best control averaged exceptions; they do not remove a single bad residue class. The archived finite scan up to $P \le 3000$ records no missing class and a largest first-prime row 108, but this is only diagnostic. In the absence of a new pointwise AP theorem, the route returns to the active frontier consisting of PDEC scope, signed payload, ExactUV, model, rate and DStructure/Rankin gates. Hence this is a barrier identification, not an unconditional proof. □

Theorem 7.18 (Terminal-row CRT atoms and the square-root gate)**.** *For $P = 5$, the terminal-row and next-row atoms 23 and 29 are units modulo $M_{\le 5} = 2\cdot 3\cdot 5$. For $P = 7$, the atoms $43, 47$ and 53 are units modulo $M_{\le 7} = 2\cdot 3\cdot 5\cdot 7$. Hence none of these atoms can be absorbed by a prime factor $\le P$. More generally, if p_+ is the next prime after P and*

$$1 < n < p_+^2, \qquad (n, M_{\le P}) = 1,$$

then n is prime.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-crt-atom-router.md.`

For $P = 5$,

$$23 \equiv (1,2,3) \pmod{(2,3,5)}, \qquad 29 \equiv (1,2,4) \pmod{(2,3,5)}.$$

For $P = 7$,

$$\begin{aligned} 43 &\equiv (1,1,3,1) \pmod{(2,3,5,7)},\\ 47 &\equiv (1,2,2,5) \pmod{(2,3,5,7)},\\ 53 &\equiv (1,2,3,4) \pmod{(2,3,5,7)}. \end{aligned}$$

Every displayed coordinate is nonzero, so the hypothesis that any of these numbers is divisible by a prime $\le P$ contradicts its CRT vector.

For the general square-root gate, suppose n is composite and $(n, M_{\le P}) = 1$. Then the least prime factor of n is strictly larger than P, hence at least p_+. A composite integer all of whose prime factors are at least p_+ is at least p_+^2, contradicting $n < p_+^2$. Thus n is prime.

This proves an atom-level principle: a reduced CRT atom in the terminal row or the immediate post-square row is automatically a true prime. It does not prove that every prime P has such an atom in a prescribed short row. Complete CRT-period symmetry supplies full-period uniformity and reflection, but not the local pointwise existence needed in the initial square. The remaining global input is therefore a terminal-row reduced-residue existence theorem, the localized P-CRT/Linnik–2 route, or the already registered AP zero-packet, Page-sparsity, signed-payload, or PDEC frontiers. □

Theorem 7.19 (Terminal rows as square-phase Jacobsthal special phases)**.** *For $1 \le r < P$, the next-row and last-row reduced atoms are exactly*

$$(P^2 + r, M_{<P}) = 1, \qquad (P^2 - r, M_{<P}) = 1.$$

Equivalently, the next-row full-cover failure is the covering of $r = 1, \dots, P-1$ by the forbidden classes

$$r \equiv -P^2 \pmod q, \qquad q < P,$$

and the last-row full-cover failure is the covering by

$$r \equiv P^2 \pmod q, \qquad q < P.$$

Thus either full cover forces the square phase P^2 to start a covered block of length $P-1$ in the primorial period $M_{<P}$.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-terminal-row-square-phase-bridge-router.md.`

For every prime $q < P$, since P is a unit modulo q,

$$q \mid P^2 + r \iff r \equiv -P^2 \pmod q,$$

and

$$q \mid P^2 - r \iff r \equiv P^2 \pmod q.$$

Therefore the absence of a reduced atom on one side is exactly the statement that these one-residue-per-q forbidden classes cover the whole interval $[1, P-1]$. In the primorial period this is precisely a consecutive covered block of length $P-1$ beginning at the corresponding square phase.

The full-period Jacobsthal shortcut is not available in the present corpus: known and archived finite data already show primorial periods with covered blocks of length at least $P-1$. The useful remaining target is not a uniform period-wide maximum-run bound, but the special-phase assertion that the particular phase P^2 cannot align with such a long block, or that such alignment is a PDEC/SAE/ColumnCRT defect. Hence this is a route sharpening, not an unconditional row/column proof. □

Theorem 7.20 (Linnik–2 defects and nonprincipal character obstruction)**.** *For a prime modulus P, put*

$$\vartheta_a(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime} \\ \ell \equiv a \pmod P}} \log \ell, \qquad a \in \mathbb{F}_P^*,$$

and for each Dirichlet character χ mod P put

$$T_\chi(P) = \sum_{a \in \mathbb{F}_P^*} \chi(a) \vartheta_a(P).$$

Let $T_0(P)$ denote the principal-character mass and let

$$N_a(P) = \sum_{\chi \ne \chi_0} \overline{\chi(a)} T_\chi(P)$$

be the nonprincipal projection at the class a. Then

$$\vartheta_a(P) = \frac{T_0(P) + N_a(P)}{P-1}.$$

Consequently a Linnik–2 zero-column defect, namely $\vartheta_a(P) = 0$ for some $a \in \mathbb{F}_P^*$, forces the exact negative phase alignment

$$N_a(P) = -T_0(P)$$

and hence

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P-2}.$$

Thus the remaining AP route is equivalent to excluding this pointwise nonprincipal negative projection at $x = P^2$, or else returning to the same-set PDEC scope or signed-payload structural frontier.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-nonprincipal-character-obstruction-router.md.`

All identities are finite orthogonality identities on the group $\mathbb{F}_P^*$. Since

$$T_\chi(P) = \sum_{b \in \mathbb{F}_P^*} \chi(b)\vartheta_b(P),$$

character inversion gives the displayed formula for $\vartheta_a(P)$. If a nonzero column has no prime below P^2, then $\vartheta_a(P) = 0$, so the nonprincipal projection must cancel the entire principal mass. Applying Cauchy's inequality to

$$T_0(P) = \left| \sum_{\chi \neq \chi_0} \overline{\chi(a)} T_\chi(P) \right|$$

and using $\sum_{\chi \neq \chi_0} |\chi(a)|^2 = P - 2$ gives the stated energy lower bound.

The archived finite scan up to $P \leq 3000$ finds no theta-zero residue class, with largest normalized negative theta deficit about 0.686688 at $P = 73$, but this empirical fact is not used in the proof. The theorem only sharpens the hard point: complete CRT symmetry and averaged AP estimates do not by themselves exclude the required negative nonprincipal projection. No pointwise projection bound, PDEC scope certificate, signed-payload constructor, ExactUV incidence theorem, model-gap ledger, rate-preservation ledger, or DStructure/Rankin promotion is proved here. □

Theorem 7.21 (Energy capacity and rank-one phase for the Linnik–2 defect)**.** *In the notation of the preceding theorem, set*

$$\rho_a(P) = -\frac{N_a(P)}{T_0(P)}.$$

Then

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P-1)}.$$

Thus a zero-column defect is exactly the rank-one condition $\rho_a(P) = 1$. The energy condition

$$\sum_{\chi \neq \chi_0} |T_\chi(P)|^2 \geq \frac{T_0(P)^2}{P-2}$$

is only a necessary condition for such a defect. Therefore an L^2 capacity argument can close this route only if it proves the strict reverse inequality. Otherwise the remaining obligation is a pointwise phase statement: exclude a negative projection of size $T_0(P)$ in one evaluation direction.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-linnik2-rankone-phase-capacity-router.md.`

The first identity follows directly from $\vartheta_a(P) = (T_0(P) + N_a(P))/(P-1)$. Hence $\vartheta_a(P) = 0$ if and only if $\rho_a(P) = 1$. The displayed energy lower bound is exactly the Cauchy consequence proved in the preceding theorem. Its converse is false as a matter of logic: total nonprincipal energy above the floor does not force the energy to point in the single evaluation direction attached to a.

The evaluation vectors $v_a = (\overline{\chi(a)})_{\chi \neq \chi_0}$ satisfy

$$\langle v_a, v_b \rangle = \begin{cases} P-2, & a = b, \\ -1, & a \neq b, \end{cases}$$

so the residue tests are simplex directions in the nonprincipal character space. The archived scan up to $P \leq 3000$ records 423 prime moduli for which the total-energy floor is already exceeded while no zero residue class exists. This diagnostic is not a proof input; it identifies why the capacity route is too coarse. The remaining named input is the rank-one negative evaluation projection exclusion at $x = P^2$, or else a return to same-set PDEC scope or signed payload. □

Theorem 7.22 (Three-claim frontier and the sharp AP zero-packet barrier)**.** *The three main claims currently have distinct frontiers. The row/column chain has been sharpened to the rank-one condition*

$$\rho_a(P) < 1 \qquad (a \in \mathbb{F}_P^*),$$

which is equivalent to

$$\vartheta_a(P) > 0 \qquad (a \in \mathbb{F}_P^*).$$

The two-point secondary-sieve chain remains conditional on the I3-Core true residual total-incidence input, or on the external DI/BFI–Kloosterman large-sieve route. The RH contradiction-field chain remains a verification package awaiting independent acceptance of its controlled exits.

For the row/column AP branch, the rank-one condition is therefore precisely the sharp positivity statement

$$\theta(P^2; P, a) > 0 \quad \textit{for every } a \in \mathbb{F}_P^*.$$

In explicit-formula language this requires the signed nonprincipal zero packet in each residue class to be strictly smaller than the main term, which is of order P. Complete CRT averages, Bombieri–Vinogradov type mean estimates, standard Linnik bounds with exponent > 2, and GRH-shape errors $O(P \log^2 P)$ do not by themselves imply this $x = P^2$ pointwise positivity.

Proof. The certificate is recorded in

`docs/monograph/three-claims-frontier-rankone-explicit-formula-router.md.`

The equivalence $\rho_a(P) < 1 \iff \vartheta_a(P) > 0$ follows from the identity

$$\rho_a(P) = 1 - \frac{\vartheta_a(P)}{T_0(P)/(P-1)}$$

proved in the preceding theorem. Since $\vartheta_a(P)$ is exactly the Chebyshev prime weight in the class $a \bmod P$ up to P^2, positivity is equivalent to a prime in that nonzero class below the square barrier.

The explicit formula for $\theta(P^2; P, a)$ has a principal term of size about P, plus signed nonprincipal zero packets and controlled trivial terms. Therefore the sharp remaining analytic input is a pointwise bound on those signed packets for every a. Mean estimates and full-period CRT identities only control averages, not a single exceptional simplex direction. GRH-shape errors at $x = P^2$ are of size $P \log^2 P$, too large to beat a main term of size P without additional sharp constants or cancellation. Thus this theorem is a frontier separation and barrier identification, not a promotion of any of the three claims to final unconditional theorem status. □

Theorem 7.23 (Explicit AP zero-packet Siegel split). *The remaining row/column AP input*

`ExplicitAPZeroPacketBoundBeatingMainTermAtXEqualsP2`

splits into two genuine analytic branches. First, any exceptional real zero β of a real nonprincipal character modulo P contributes at $x = P^2$ on the scale

$$P^{2\beta-1}/\beta.$$

If $\beta = 1 - \lambda/\log P$, this is $P\exp(-2\lambda)/\beta$, hence still of the same order as the principal term for bounded λ. This is the square-scale Siegel-bias branch. Second, after extracting any real exceptional packet, the nonreal zeros leave a signed residue packet; closing it requires pointwise phase cancellation in every evaluation direction $a \bmod P$, not just average energy or complete CRT uniformity.

Consequently the current AP branch is not a single unnamed zero-packet error. It is the conjunction

```
SiegelExceptionalBiasExclusionAtSquareScale
AND NonrealZeroPacketResiduePhaseCancellationAtXEqualsP2,
```

with the trivial and finite explicit-formula terms kept as a downstream constant ledger once a positive main-term margin is available.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-explicit-ap-zero-packet-siegel-split-router.md.`

The character explicit formula for $\theta(P^2; P, a)$ has a principal term of size P and a signed sum of nonprincipal zero terms. A real zero close to 1 cannot be dismissed by a mean theorem, because its square-scale contribution remains comparable to the main term on one of the two real character half-classes. Deuring–Heilbronn repulsion organizes the remaining zeros but does not itself prove that this biased half-class is positive before P^2.

After this real branch is separated, the nonreal packet is still pointwise in the residue a. Bombieri–Vinogradov, complete CRT identities, total nonprincipal energy bounds, and GRH-shape estimates do not exclude a single rank-one evaluation direction from carrying too much signed mass at the square barrier. Therefore the latest analytic AP route remains open and must either prove the two displayed branches or return to the already named structural inputs: same-set PDEC scope, atomic signed payload, exact-(u, v) multiplicity, model/rate preservation, and the D-structure Rankin promotion gate. No unconditional row/column theorem is claimed here. □

Theorem 7.24 (Siegel branch as a quadratic half-class margin). *For prime modulus P, let χ_P be the quadratic character and define*

$$T_\chi(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime},\ \ell \ne P}} \chi_P(\ell) \log \ell, \qquad T_0(P) = \sum_{\substack{\ell \le P^2 \\ \ell \text{ prime},\ \ell \ne P}} \log \ell.$$

Then the real-character part of the Siegel branch is measured by

$$r_{\text{quad}}(P) = \frac{|T_\chi(P)|}{T_0(P)}.$$

The dangerous quadratic half-class retains an average main-term margin proportional to $1-r_{\mathrm{quad}}(P)$. *Thus the square-scale Siegel branch is reduced to either proving the margin theorem*

$\texttt{QuadraticHalfClassSquareScaleBiasMarginTheorem},$

or accepting an independently certified effective no-Siegel-zero or β*-gap input strong enough at* $x = P^2$.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-siegel-quadratic-halfclass-margin-router.md.`

The identity is the specialization of the nonprincipal character expansion to the unique real quadratic character modulo P. The sum T_χ is the difference between the Chebyshev weights falling in the quadratic-residue and quadratic-nonresidue half-classes. If the real projection almost equals the principal mass in magnitude, one half-class may have only a small average margin left before the nonreal packet and finite terms are charged. Conversely, any explicit bound $r_{\mathrm{quad}}(P) \le 1 - \eta(P)$ supplies exactly that half-class margin.

The archived finite scan $7 \le P \le 1000$ records a maximum diagnostic ratio about 0.13027, attained at $P = 7$. This scan is not used as an infinite proof. The theorem only identifies the exact remaining self-contained target for the real branch. The nonreal zero-packet phase-cancellation branch and the structural PDEC/signed-payload/ExactUV/D-structure inputs remain open. ☐

Theorem 7.25 (Nonreal packet as two half-class simplexes)**.** *After removing the principal character and the quadratic character, define*

$$R_a = (P-1)\theta(P^2; P, a) - T_0(P) - \chi_P(a)T_\chi(P).$$

Let w_a *be the evaluation vector over the remaining nonreal characters. Then*

$$\langle w_a, w_b\rangle = \begin{cases} P-3, & a = b, \\ -2, & a \ne b \text{ and } \chi_P(a) = \chi_P(b), \\ 0, & \chi_P(a) \ne \chi_P(b). \end{cases}$$

Thus the nonreal packet splits into two orthogonal regular simplexes indexed by the quadratic-residue and quadratic-nonresidue half-classes. A zero column in class a *would force*

$$R_a = -T_0(P) - \chi_P(a)T_\chi(P),$$

so the nonreal packet must supply a rank-one negative projection of size at least

$$(1 - |T_\chi(P)|/T_0(P))T_0(P).$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-simplex-phase-router.md.`

The character orthogonality relation over $\mathbb{F}_P^*$ gives $\sum_\chi \overline{\chi(a)}\chi(b) = P-1$ if $a = b$, and 0 otherwise. Removing the principal character subtracts 1, and removing the quadratic character subtracts $\chi_P(a)\chi_P(b)$. This gives the displayed inner products. The identity for R_a is the inverse character expansion with those two components removed. Setting $\theta(P^2; P, a) = 0$ gives the displayed zero-column requirement.

Cauchy's inequality gives a necessary energy floor

$$E_{\mathrm{nonreal}} \geq \frac{T_0(P)^2(1-|T_\chi(P)|/T_0(P))^2}{P-3},$$

but this is still only a capacity condition. The archived scan $7 \leq P \leq 1000$ records many cases where the nonreal energy is above this floor while no zero column occurs. Hence the remaining analytic AP input is not an energy statement, but

`NonrealHalfClassSimplexRankOneProjectionExclusionAtP2.`

No unconditional row/column theorem is claimed. □

Theorem 7.26 (Half-class compensation variance for the nonreal packet)**.** *Keep the notation of the preceding theorem and fix $s \in \{+1,-1\}$. Let*

$$H_s = \{a \in \mathbb{F}_P^* : \chi_P(a) = s\}, \qquad h = |H_s| = \frac{P-1}{2},$$

and define the half-class mean

$$\mu_s = \frac{T_0(P) + sT_\chi(P)}{P-1}.$$

For $a \in H_s$ the nonreal residual is exactly

$$R_a = (P-1)\big(\theta(P^2;P,a) - \mu_s\big),$$

and therefore

$$\sum_{a\in H_s} R_a = 0.$$

If a zero column occurs at $a_0 \in H_s$, then

$$R_{a_0} = -(P-1)\mu_s, \qquad \sum_{\substack{a\in H_s\\ a\neq a_0}} R_a = (P-1)\mu_s.$$

Consequently the sharp one-hole variance floor is

$$\sum_{a\in H_s} R_a^2 \geq ((P-1)\mu_s)^2 \frac{h}{h-1}.$$

Equality is attained, at the level of the centered half-class geometry, by the flat compensator in which the non-hole residues all have

$$\theta(P^2;P,a) = \mu_s \frac{h}{h-1}.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-nonreal-halfclass-compensation-variance-router.md.`

Since $T_0 + sT_\chi = 2\sum_{a\in H_s} \theta(P^2;P,a)$, the mean over H_s is μ_s. Substituting this identity into the previous residual formula gives $R_a = (P-1)(\theta_a - \mu_s)$, and summing over H_s gives the centering identity.

If $\theta_{a_0} = 0$, then the displayed value of R_{a_0} follows immediately, and the compensation identity follows from centering. Cauchy's inequality applied to the remaining $h-1$ residues gives

$$\sum_{\substack{a \in H_s \\ a \neq a_0}} R_a^2 \geq \frac{((P-1)\mu_s)^2}{h-1},$$

so adding $R_{a_0}^2$ gives the stated floor. Equality requires all remaining residuals to be equal, i.e. the flat compensator above.

Thus the previous energy floor is not only necessary but convexly sharp. It cannot by itself exclude a zero column, because the flat compensator is a nonnegative centered half-class vector. The remaining self-contained obligation is therefore the actual prime-phase input

`HalfClassCompensationMassNonconcentrationOrColumnCRTPDEC,`

which must either rule out such flat or near-flat compensation for the true prime-induced vector at $x = P^2$, or route persistent compensation phases to a registered ColumnCRT/PDEC or moving-family return. No unconditional row/column theorem is claimed here. □

Theorem 7.27 (Ratio-Fourier lock for half-class compensation)**.** *Let $Q \subset \mathbb{F}_P^*$ be the subgroup of quadratic residues and fix a candidate puncture $a_0 \in H_s$. Put*

$$f_{a_0}(u) = \theta(P^2; P, a_0 u), \qquad u \in Q,$$

so that $u = 1$ is the punctured coordinate. Let

$$z = f_{a_0}(1), \qquad c_{a_0} = \frac{\sum_{u \in Q,\ u \neq 1} f_{a_0}(u)}{h-1}, \qquad \mu_s = \frac{z + (h-1)c_{a_0}}{h}.$$

Then the excess over the one-point variance floor is exactly

$$\sum_{a \in H_s} R_a^2 - (P-1)^2 (z - \mu_s)^2 \frac{h}{h-1} = (P-1)^2 \sum_{\substack{u \in Q \\ u \neq 1}} \big(f_{a_0}(u) - c_{a_0}\big)^2.$$

If $\widehat{f}(\psi) = \sum_{u \in Q} f_{a_0}(u)\overline{\psi(u)}$, where ψ ranges over the characters of Q, then the flat compensator

$$f_{a_0}(u) = c_{a_0} \qquad (u \neq 1)$$

is equivalent to the simultaneous lock

$$\widehat{f}(\psi) = z - c_{a_0} \qquad (\psi \neq 1).$$

Moreover,

$$\sum_{\psi \neq 1} \left| \widehat{f}(\psi) - (z - c_{a_0}) \right|^2 = h \sum_{\substack{u \in Q \\ u \neq 1}} \big(f_{a_0}(u) - c_{a_0}\big)^2.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-ratio-fourier-lock-router.md.`

The normalization by a_0^{-1} maps the half-class H_s bijectively to Q, because a_0 has the same quadratic character as every element of H_s. Write

$$f_{a_0}(u) = \begin{cases} z, & u = 1, \\ c_{a_0} + d_u, & u \neq 1, \end{cases} \qquad \sum_{u \neq 1} d_u = 0.$$

Substituting this decomposition into $\sum_{a \in H_s} R_a^2 = (P-1)^2 \sum_{u \in Q} (f_{a_0}(u) - \mu_s)^2$ gives the displayed one-point excess identity; the cross term vanishes because the d_u sum to zero.

For the Fourier statement, if $d_u = 0$ for all $u \neq 1$, then for every nontrivial ψ

$$\widehat{f}(\psi) = z + c_{a_0} \sum_{u \neq 1} \overline{\psi(u)} = z - c_{a_0}.$$

Conversely, if all nontrivial coefficients equal $z - c_{a_0}$, Fourier inversion on the finite abelian group Q gives $d_u = 0$ for every $u \neq 1$. Applying Parseval to the function $d_1 = 0$, $d_u = f_{a_0}(u) - c_{a_0}$ for $u \neq 1$, gives the final identity.

Thus a flat zero-column compensator is not merely a low-variance event: after normalization by the missing residue it is an all-frequency lock on the quadratic-residue subgroup. The remaining input is the exclusion of persistent locks of this kind, or their registration as ColumnCRT/PDEC or moving-family returns. No unconditional row/column theorem is claimed here. □

Theorem 7.28 (Quadratic-twist pair character orbit lock)**.** *Let $\psi \neq 1$ be a character of the quadratic-residue subgroup Q, and let χ be either extension of ψ to $\mathbb{F}_P^*$. The other extension is $\chi\chi_P$. For $a_0 \in H_s$, define*

$$T_\chi(P) = \sum_{a \in \mathbb{F}_P^*} \theta(P^2; P, a)\overline{\chi(a)}.$$

Then the Q-Fourier coefficient in the preceding theorem satisfies

$$\widehat{f}(\psi) = \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$

Consequently the flat compensator condition is equivalent to the simultaneous orbit lock

$$\chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} = \theta(P^2; P, a_0) - c_{a_0} \qquad (\psi \neq 1).$$

Equivalently,

$$\sum_{\psi \neq 1} \left| \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2} - (\theta(P^2; P, a_0) - c_{a_0}) \right|^2 = h \sum_{\substack{u \in Q \\ u \neq 1}} \left(f_{a_0}(u) - c_{a_0}\right)^2.$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-twist-pair-character-lock-router.md.`

Since χ restricts to ψ on Q, for $a = a_0 u$ with $u \in Q$ we have $\overline{\psi(u)} = \chi(a_0)\overline{\chi(a)}$. Also the indicator of H_s is $(1 + s\chi_P(a))/2$. Therefore

$$\widehat{f}(\psi) = \chi(a_0) \sum_{a \in H_s} \theta(P^2; P, a)\overline{\chi(a)} = \chi(a_0) \frac{T_\chi(P) + sT_{\chi\chi_P}(P)}{2}.$$

Substituting this identity into the ratio-Fourier lock theorem gives the orbit lock and Parseval identity.

This reformulation is important for the global route: the obstruction is not a single exceptional character and not a scalar energy excess. A persistent zero-column compensator must synchronize every nontrivial character of Q, or equivalently every quadratic-twist pair of Dirichlet characters modulo P, onto the same a_0-dependent phase orbit. The remaining input is

`QuadraticTwistPairCharacterOrbitLockExclusionOrColumnCRTPDEC.`

No unconditional row/column theorem is claimed here. □

Theorem 7.29 (Log-prime degeneration of exact half-class orbit locks)**.** *For $a \in \mathbb{F}_P^*$, put*

$$S_a(P) = \{\ell \le P^2 : \ell \text{ prime},\ \ell \ne P,\ \ell \equiv a \bmod P\}$$

and $\theta_a = \sum_{\ell \in S_a(P)} \log \ell$. If $a \ne b$ and $\theta_a = \theta_b$, then $S_a(P) = S_b(P) = \varnothing$.

Consequently, for $P \ge 7$, an exact punctured half-class flatness condition

$$\theta(P^2; P, a_0 u) = c \qquad (u \in Q,\ u \ne 1)$$

can hold only in the degenerate case $c = 0$ and

$$S_{a_0 u}(P) = \varnothing \qquad (u \in Q,\ u \ne 1).$$

Thus the exact quadratic-twist pair orbit lock from the preceding theorem is routed to

`PuncturedHalfClassZeroSupportDegeneracyExclusionOrColumnCRTPDEC.`

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-log-independence-degeneracy-router.md.`

If $\theta_a = \theta_b$, then

$$\sum_{\ell \in S_a(P)} \log \ell - \sum_{\ell \in S_b(P)} \log \ell = 0.$$

Equivalently,

$$\prod_{\ell \in S_a(P)} \ell = \prod_{\ell \in S_b(P)} \ell.$$

Unique factorization forces the two prime multisets to be identical. Distinct nonzero residues modulo P are disjoint prime supports, so both supports must be empty.

For $P \ge 7$, the punctured set $\{u \in Q : u \ne 1\}$ contains at least two elements. Exact flatness makes all corresponding θ-values equal. Applying the first paragraph to any pair and then to a fixed representative against every other punctured residue gives empty support throughout the punctured half-class and $c = 0$.

Therefore a positive exact flat compensator cannot be the persistent obstruction. The only remaining exact-lock exit is the much sharper zero-support degeneracy: after removing a_0, the same half-class has no prime arrivals up to P^2. This theorem does not exclude that degeneracy and does not prove the row/column theorem unconditionally. □

Theorem 7.30 (Half-class mass versus single-residue capacity). *Let*

$$\Theta_s(P) = \sum_{\chi_P(a)=s} \theta(P^2; P, a) = \frac{T_0(P) + sT_\chi(P)}{2}, \qquad s \in \{\pm 1\}.$$

For every $a \in \mathbb{F}_P^*$,

$$\theta(P^2; P, a) \le P \log(P^2) = 2P \log P.$$

Consequently a punctured half-class zero-support degeneracy can occur only if

$$\min_{s=\pm 1} \Theta_s(P) \le 2P \log P.$$

Equivalently, the sufficient exclusion gate is

$$T_0(P)\left(1 - \frac{|T_\chi(P)|}{T_0(P)}\right) > 4P \log P.$$

Thus the latest exact-lock exit is routed to

> ***QuadraticHalfClassMassBeatsSingleResidueCapacityAtP2,***

or else to a named ColumnCRT/PDEC capacity-failure registration.

Proof. The certificate is recorded in

> `docs/monograph/prime-matrix-halfclass-single-residue-capacity-router.md.`

In a fixed nonzero residue $a \bmod P$, the integers not exceeding P^2 have the form $a + kP$, $0 \le k \le P - 1$. Hence at most P prime terms can occur in this residue before the square barrier, and each carries weight at most $\log(P^2)$. This gives the displayed single-residue capacity bound.

If a punctured half-class zero-support degeneracy occurs in H_s, then all prime weight in that half-class is concentrated in the single unpunctured residue a_0. Therefore

$$\Theta_s(P) = \theta(P^2; P, a_0) \le 2P \log P.$$

Taking the contrapositive gives the half-class capacity exclusion. Finally, since

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2},$$

the condition is exactly

$$T_0(P)\left(1 - \frac{|T_\chi(P)|}{T_0(P)}\right) > 4P \log P.$$

This is a strict narrowing of the previous support-degeneracy exit, but it is not yet a proof of the required global half-class mass lower bound. No unconditional row/column theorem is claimed here. □

Theorem 7.31 (Capacity margin as a logarithmic-over-linear projection gap). *The half-class single-residue capacity gate*

$$\min_{s=\pm 1} \Theta_s(P) > 2P \log P$$

is equivalent to

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4P \log P}{T_0(P)}.$$

In particular, if a principal Chebyshev lower bound

$$T_0(P) \geq c_0(P)P^2$$

is available, then it is enough to prove

$$1 - \frac{|T_\chi(P)|}{T_0(P)} > \frac{4\log P}{c_0(P)P}.$$

Thus any persistent failure of the capacity gate, in the presence of a P^2-scale principal mass lower bound, forces an ultra-near-one quadratic projection defect

$$\frac{|T_\chi(P)|}{T_0(P)} \geq 1 - O\left(\frac{\log P}{P}\right).$$

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-halfclass-capacity-margin-factorization-router.md.`

The identity

$$\min_s \Theta_s(P) = \frac{T_0(P) - |T_\chi(P)|}{2}$$

was proved in the preceding theorem. Therefore

$$\min_s \Theta_s(P) > 2P\log P$$

is equivalent to

$$\frac{T_0(P) - |T_\chi(P)|}{2} > 2P\log P,$$

and division by $T_0(P) > 0$ gives the stated projection-gap threshold. Substituting $T_0(P) \geq c_0(P)P^2$ gives the logarithmic-over-linear sufficient threshold.

This reformulation exposes the scale of the remaining obstruction. The capacity gate does not need a fixed-size half-class margin: a gap of order $\log P/P$, after the principal Chebyshev lower bound is in place, is already enough. Conversely, a counterexample surviving this gate must make the quadratic projection almost equal to the principal mass. The remaining input is

`QuadraticProjectionGapBeatsLogOverPThresholdAtSquareScale,`

or a named registration of the ultra-near-one projection defect as a ColumnCRT/PDEC/moving-family exit. No unconditional row/column theorem is claimed here. □

Theorem 7.32 (Low CRT insufficiency and the large splitting residual)**.** *Let*

$$\Theta_s(P) = \sum_{\substack{\ell \leq P^2,\ \ell \neq P \\ \chi_P(\ell) = s}} \log \ell, \qquad s \in \{+1, -1\},$$

where $\chi_P(\ell) = (\ell/P)$. Split

$$L_s(P) = \sum_{\substack{\ell < P \\ \chi_P(\ell) = s}} \log \ell, \qquad G_s(P) = \sum_{\substack{P < \ell \leq P^2 \\ \chi_P(\ell) = s}} \log \ell.$$

Then $\Theta_s(P) = L_s(P) + G_s(P)$, and the projection-gap capacity gate is equivalent to

$$\min_s \Theta_s(P) > 2P \log P.$$

Moreover the complete low CRT layer satisfies the elementary bound

$$L_+(P) + L_-(P) = \vartheta(P-1) \le P \log P < 2P \log P.$$

Consequently the low CRT layer alone cannot close the capacity gate. If the capacity gate fails, then for some sign s

$$G_s(P) \le 2P \log P.$$

Conversely, the large-splitting lower bound

$$\min_s G_s(P) > 2P \log P$$

is sufficient to close the capacity gate.

Proof. The certificate is recorded in

`docs/monograph/prime-matrix-quadratic-projection-large-splitting-router.md.`

The decomposition $\Theta_s = L_s + G_s$ is only the partition of the prime sum at the modulus P. The equivalence between the projection-gap gate and $\min_s \Theta_s(P) > 2P \log P$ is the preceding theorem.

For the low layer, every prime counted in $L_+ + L_-$ is $< P$, hence there are fewer than P summands and each summand is at most $\log P$. Thus $L_+ + L_- \le P \log P$. This is strictly below the minority-mass threshold $2P \log P$, so the low CRT period can be entirely absorbed by the failure allowance and cannot by itself force the required contradiction.

If the capacity gate fails, choose s with $\Theta_s(P) = \min_t \Theta_t(P) \le 2P \log P$. Since $L_s(P) \ge 0$, we have $G_s(P) \le 2P \log P$. Conversely, if both large layers satisfy $G_s(P) > 2P \log P$, then $\Theta_s(P) \ge G_s(P) > 2P \log P$ for both signs, which closes the capacity gate. Thus the remaining self-contained attack is

`LargePrimeQuadraticSplittingMinorityMassBeatsTwoPLogPAtSquareScale,`

or the registration of a persistent large-splitting desert as a Siegel/ColumnCRT/PDEC exit. No unconditional row/column theorem is claimed here. ☐

Theorem 7.33 (Large splitting as a beta-gap and zero-packet budget). *Let*

$$G_s(P) = \sum_{\substack{P<\ell\le P^2 \\ \chi_P(\ell)=s}} \log \ell, \qquad H_0(P) = G_+(P) + G_-(P), \qquad H_\chi(P) = G_+(P) - G_-(P).$$

Then the large-splitting gate

$$\min_s G_s(P) > 2P \log P$$

is equivalent to

$$1 - \frac{|H_\chi(P)|}{H_0(P)} > \frac{4P \log P}{H_0(P)}.$$

Consequently, any explicit-formula estimate of the form

$$\frac{|H_\chi(P)|}{H_0(P)} \le R_\beta(P) + E_{\text{zero}}(P)$$

closes the large-splitting gate whenever

$$R_\beta(P) + E_{\mathrm{zero}}(P) < 1 - \frac{4P\log P}{H_0(P)}.$$

For the real-zero shadow model

$$R_\beta(P) = \frac{P^{2\beta} - P^\beta}{\beta H_0(P)}, \qquad \beta = 1 - \delta,$$

the critical no-residual value of δ is asymptotic to $2/P$ when $H_0(P) \sim P^2$.

Proof. The certificate is recorded in

```
docs/monograph/prime-matrix-large-splitting-beta-gap-router.md.
```

The first equivalence is the identity

$$\min_s G_s(P) = \frac{H_0(P) - |H_\chi(P)|}{2}.$$

Thus $\min_s G_s(P) > 2P\log P$ is exactly

$$H_0(P) - |H_\chi(P)| > 4P\log P,$$

and division by $H_0(P) > 0$ gives the stated projection-gap form.

The explicit-formula budget statement is purely algebraic: if the normalized character projection is bounded by $R_\beta + E_{\mathrm{zero}}$, then it is enough for that upper bound to be below the allowed ratio $1 - 4P\log P/H_0(P)$.

For the scale calculation, put $x = P^2$, $\beta = 1 - \delta$, and $H_0(P) \sim P^2$. Near $\delta = 0$,

$$\frac{P^{2\beta} - P^\beta}{\beta H_0(P)} = 1 - \delta(2\log P - 1) + o(\delta\log P) + o(1/P).$$

The required missing ratio is

$$\frac{4P\log P}{H_0(P)} \sim \frac{4\log P}{P}.$$

Equating these first-order quantities gives

$$\delta \sim \frac{4\log P/P}{2\log P - 1} \sim \frac{2}{P}.$$

Hence a persistent failure must come either from a real zero within the $1/P$ scale, or from a non-real zero packet, endpoint, or prime-power residual consuming the same slack. The remaining self-contained target is

```
SquareScaleQuadraticBetaGapAndZeroPacketResidualBudgetAtP2,
```

or the named exit

```
UltraCloseRealZeroOrCoherentZeroPacketLargeSplittingPDEC.
```

No unconditional row/column theorem is claimed here. □

Theorem 7.34 (Page sparsity for ultra-close beta-gap carriers)**.** *Assume the Landau–Page uniqueness input in the following form: for primitive real characters of conductor $q \le Q$, at most one character has a real zero*

$$\beta > 1 - \frac{c_{\mathrm{Page}}}{\log Q}.$$

Let $E(Q)$ be the set of prime moduli $P \le Q$ for which the large-splitting beta-gap failure produces a real-zero carrier

$$\beta_P > 1 - \frac{C(P)}{P}, \qquad C(P) \le C_*.$$

Then

$$\left|E(Q) \cap ((C_*/c_{\mathrm{Page}}) \log Q, Q]\right| \le 1.$$

In particular, after this external Page input and its constant matching are accepted, a positive-density or fixed-period CRT family of such ultra-close real-zero carriers cannot be the persistent obstruction.

Proof. The certificate is recorded in

```
docs/monograph/prime-matrix-beta-gap-page-sparsity-router.md.
```

If $P > (C_*/c_{\mathrm{Page}}) \log Q$, then

$$\frac{C(P)}{P} \le \frac{C_*}{P} < \frac{c_{\mathrm{Page}}}{\log Q}.$$

Hence every zero counted by $E(Q)$ in this range satisfies

$$\beta_P > 1 - \frac{C(P)}{P} > 1 - \frac{c_{\mathrm{Page}}}{\log Q},$$

and so lies in the Page uniqueness region for conductors $q \le Q$. The Page input allows at most one such real-zero carrier. Therefore the high part of $E(Q)$ has cardinality at most one.

This eliminates only the multi-carrier CRT reproduction shape. It does not internalize the Page constant, exclude a moving singleton exceptional carrier, or control non-real zero packets, endpoint terms, or prime-power residuals. The remaining target is

```
PageExceptionalSingletonCarrierOrNonrealZeroPacketResidualBudget,
```

with fallback

```
MovingPageSingletonCarrierPDECOrCoherentZeroPacket.
```

No unconditional row/column theorem is claimed here. $\square$

Theorem 7.35 (Terminal-square downstream frontier synchronization)**.** *The terminal-row square-phase bridge does not leave the row/column program at the older square-phase long-block interface. In the current archive, that interface has already been synchronized with the half-grid and phase-band routers, with the localized P-CRT/AP-zero-packet route, and with the global CRT signed-payload route. Thus*

`SquarePhaseSpecialPhaseLongBlockPDECExclusion`

is an upstream alias rather than the deepest active terminal-row frontier.

Status and reference. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-terminal-square-phase-downstream-frontier-sync-router.md.
```

It reads the existing terminal CRT atom certificate, the terminal square-phase bridge, the half-grid boundary-word router, the no-slot phase-band router, the localized P-CRT/Linnik–2 barrier, the rank-one/AP-zero-packet routers, the Page-sparsity beta-gap router, and the global CRT signed-payload router.

The result is a bookkeeping theorem: the terminal-square interface is now aligned with the deeper consolidated basis consisting of AP zero-packet residuals, signed-payload construction, same-set PDEC input, and the recorded global incidence/rate/D-structure inputs. No new unconditional contradiction is asserted; the row/column theorem remains open until those downstream inputs are closed or independently accepted. □

Theorem 7.36 (Common source declaration packet for payload and ExactUV)**.** *In the strict internal source lane, the signed-payload obstruction and the ActualEmitterExactUV incidence obstruction have the same missing pre-Cauchy source root. More precisely, the current archive routes*

$$\mathtt{NoncircularAtomicBasisWordSignedCoefficientOriginIdentityBeforePushforward}$$

and

$$\mathtt{ActualEmitterSourceDomainEntropyLedger}$$
$$\wedge\ \mathtt{ExactUVMapFixedPairPolylogFiberBoundLedger}$$

to the single packet

$$\mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}.$$

Status and reference. The certificate is recorded in

```
docs/monograph/
prime-matrix-strict-source-declaration-payload-exactuv-
unification-router.md.
```

The packet must be declared before Cauchy, Φ, payment, or any zero-row recovery. Its fields are: an actual noncanonical source declaration line, primitive summand rows, basis-word/signed-coefficient origin identities, prepushforward α/δ equality, source-domain entropy, fixed exact (u, v) fiber bounds, and a named return partition for absent declarations, fiber collapse, entropy deficit, sign conflicts, local-factor zeros, over-budget rows, and canonical leakage.

This is a routing and consolidation statement. The packet has not been proved in the present corpus. The AP zero-packet branch and same-set PDEC branch remain open, as do the model/rate ledgers and D-structure/Rankin acceptance. Therefore no unconditional row/column theorem is claimed here. □

Theorem 7.37 (Downstream split of the common source packet)**.** *The packet*

$$\mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}$$

does not remain a single black box in the strict source lane. In the current archive it splits into a signed-payload lane and an ExactUV lane. The signed-payload lane is synchronized to

$$\mathtt{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows},$$

while the ExactUV lane remains

$$\mathtt{ActualEmitterSourceDomainEntropyLedger}$$
$$\wedge\ \mathtt{ExactUVMapFixedPairPolylogFiberBoundLedger}.$$

Status and reference. The certificate is recorded in

`docs/monograph/prime-matrix-strict-source-declaration-downstream-sync-router.md.`

The ordinary joint-constructor route is rejected as a proof route because its expansion returns through the alpha side, same-row identity, and row-level origin table to the signed-source fixed point. The anti-split route therefore requires an atomic pre-Cauchy rows declaration with built-in word/coefficient pairing; the current archive further reduces that target to the built-in signed coefficient pairing closed form above.

This theorem records a downstream synchronization only. The built-in pairing closed form and the ExactUV entropy/fiber bounds are not proved here, and the global rate and D-structure/Rankin gates remain open. □

Theorem 7.38 (Signed-lane cycle closure)**.** *In the strict source lane, the current signed-payload route is a closed dependency cycle:*

$$\begin{gathered}
\mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}\\
\to \mathtt{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows}\\
\to \mathtt{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}\\
\to \mathtt{AtomicSignedPayloadTraceConstructorBeforeAssignmentOrReturn}\\
\to \mathtt{NoncircularAtomicBasisWordSignedCoefficientOriginIdentityBeforePushforward}\\
\to \mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}.
\end{gathered}$$

Consequently no node in this lane is currently an independent self-contained proof of the signed payload.

Status and reference. The certificate is recorded in

`docs/monograph/prime-matrix-strict-signed-lane-cycle-closure-router.md.`

The first arrow is the downstream split of the common source packet. The second is the built-in pairing frontier. The third and fourth are the atomic branch-trace and atomic-payload routers: the visible branch trace supplies coordinates but not the signed payload. The final arrow is the source declaration unification, where the noncircular origin identity returns to the same common packet.

Thus the signed lane is diagnostic rather than terminal. A proof must add a new primitive signed payload or trace formula, prove a well-founded terminal descent, or use an independently proved same-set PDEC scope certificate. The ExactUV entropy/fiber, rate-preservation, and D-structure/Rankin gates remain open. □

Theorem 7.39 (New primitive payload alignment to the source atom)**.** *The exit*

$$\mathtt{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact}$$

is not an independent terminal name in the present archive. If it is to break the signed-lane cycle, it must supply a pre-Cauchy actual source object in the same formal unit and must include the three source-rank/no-collapse atoms

$$\begin{gathered}
\mathtt{ActualPreCauchySourceDomainAbsoluteEntropyLedger},\\
\mathtt{CompletePrimitiveEmitterKeyPartitionLedger},\\
\mathtt{FixedKeyExactUVLocalMultiplicity01Ledger}.
\end{gathered}$$

Otherwise it is only a renaming of an existing trace, payload, origin, or common-packet node, or else it leaves the lane through terminal descent, PDEC, or an external spectral input.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-new-primitive-payload-
source-atom-alignment-router.md.
```

The audit imports the signed-lane cycle closure, the atomic trace/payload frontier, the source declaration unification, and the final exact-source alignment. The visible trace supplies coordinates but no signed coefficient or local factor. The origin identity alone returns to the common source packet. Thus a genuinely new primitive artifact must name the actual pre-Cauchy source and control its exact (u, v) fibers before any Cauchy, Φ, payment, or terminal recovery step.

The deterministic target is therefore the source-rank/no-collapse package. Those three atoms are not proved in the present corpus, and the external spectral and D-structure/Rankin acceptance routes remain open. No unconditional row/column theorem is claimed here. $\square$

Theorem 7.40 (Source entropy downstream cycle guard)**.** *The source-domain atom*

$$\texttt{ActualPreCauchySourceDomainAbsoluteEntropyLedger}$$

is not a terminal self-contained proof point in the present archive. Its recorded downstream route is

$$\begin{gathered}\texttt{ActualPreCauchySourceDomainAbsoluteEntropyLedger}\\ \to \texttt{AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward}\\ \to \texttt{AcyclicSeedPreCauchyBasisWeightSourceFormulaForPrimitiveRows}\\ \to \texttt{AcyclicSeedInternalArithmeticBasisExpansionBeforeCauchy}\\ \to \texttt{AcyclicSeedNoncanonicalPreCauchyBasisAlphabetLedger},\end{gathered}$$

and continuing from the basis alphabet enters the signed coordinate-source cycle. That cycle cannot count as a proof of source entropy.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-source-entropy-
downstream-cycle-sync-router.md.
```

The imported cycle guard records that the word-coordinate formula, signed slot, coefficient assignment, coefficient value map, basis-word origin, row-level generation table, signed row emitter, primitive coefficient law, basis-weight source, internal basis expansion, basis alphabet, basis-word generation, source-tuple constructor, and basis-word formula form a closed dependency loop. This loop only shows mutual definition inside the present formalism. It does not produce an independent pre-Cauchy primitive basis and coefficient source.

Thus the current source-entropy branch needs a new acyclic primitive basis/coefficient source input, or else it must return through the terminal descent/PDEC/external spectral exits. The complete-key, fixed-key multiplicity, and D-structure/Rankin gates also remain open. $\square$

Theorem 7.41 (Unified cycle-cut and terminal-descent frontier)**.** *In the present archive the two immediate exits from the source-entropy cycle guard do not remain independent terminal proof points. The recorded reduction is*

```
ActualPreCauchySourceDomainAbsoluteEntropyLedger
-> AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
   OR AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
-> AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
   OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
```

```
AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
-> AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
   OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
   OR IndependentActualSourceBridgeNotFactoredThrough
      ExactUVPairEnergyOrJointConstructorLoop.
```

Moreover, the internal PDEC same-set branch is saturated in the current self-contained corpus, so the first productive internal atom of the remaining joint route is

```
PreCauchyJointWordCoefficientEmitterDeclarationLine
ForActualNoncanonicalSourceTuple.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-cyclecut-terminal-descent-
unified-frontier-router.md.
```

The seed-cycle-cut branch has already rejected sequential word-first or coefficient-first splitting and returns to the signed-source fixed point unless a new joint emitter is supplied. The terminal-descent branch forms the recorded macrocycle

$$\text{TERMINAL} \to \text{SOURCE} \to \text{PAIR} \to \text{JOINT} \to \text{TERMINAL},$$

so it is not a well-founded descent proof. The same-set PDEC arm remains a valid conditional or external input, but within the present strict internal corpus it no longer gives an independent non-cyclic exit.

Thus the current internal joint route must begin before Cauchy, payment, and Φ-pushforward with a declaration line for the same actual noncanonical source tuple. That line is still unproved; the row formula, word/coefficient identity, no-downstream-return ledger, complete/fixed-key controls, ExactUV, rate-preservation, and D-structure/Rankin gates remain open. Hence this is a frontier compression, not an unconditional row/column proof. □

Theorem 7.42 (Downstream anti-split synchronization of the unified frontier)**.** *The ordinary joint declaration line obtained in the unified frontier is not a new non-cyclic proof point. In the present archive it synchronizes to the ordinary explicit joint constructor, and that route returns to the signed-source fixed point unless it is replaced by an anti-split atomic declaration. The current downstream synchronization is*

```
PreCauchyJointWordCoefficientEmitterDeclarationLineForActualNoncanonicalSourceTuple
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple
-> signed-source fixed point unless replaced
AntiSplit route
-> AtomicPreCauchyJointRowsFormulaWithBuiltInWordCoefficientPairing
-> BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
ExactUV gate
-> ActualEmitterSourceDomainEntropyLedger
   AND ExactUVMapFixedPairPolylogFiberBoundLedger.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-cyclecut-unified-antisplit-
downstream-sync-router.md.
```

The joint declaration/constructor synchronization shows that a plain declaration line only selects the ordinary explicit joint alpha/delta rule. The direct constructor audit shows that this ordinary route expands through the alpha-side and same-row origin tables back to the signed-source fixed point. Consequently the declaration cannot be used as a proof merely by renaming it.

The anti-split firewall requires an atomic pre-Cauchy row declaration whose rows already contain the word/coefficient pairing and the pre-pushforward payload. The atomic-row audit then identifies the first remaining signed field as the built-in signed coefficient/pairing closed form for atomic joint rows. In parallel, the ExactUV incidence gate has been decomposed into actual source-domain entropy and fixed exact-pair polylog fiber bounds. None of these three inputs is proved in the present corpus, and the complete/fixed-key, model, rate-preservation, and D-structure/Rankin gates remain open. ☐

Theorem 7.43 (Anti-split trace cycle and ExactUV atomization)**.** *The built-in signed pairing frontier does not terminate at an ordinary branch trace. The next recorded synchronization is*

```
BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
-> ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn
-> signed-lane dependency cycle
-> NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact
   OR terminal descent OR same-set PDEC.
```

In parallel, the ExactUV side is atomized as

```
ActualEmitterSourceDomainEntropyLedger
-> ActualPreCauchySourceDomainEntropyFromSignedRowsAndRowMassLedger
-> AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward
ExactUVMapFixedPairPolylogFiberBoundLedger
-> RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
   AND FixedKeyExactUVLocalMultiplicity01Ledger.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-antisplit-trace-exactuv-
atomized-frontier-router.md.
```

The built-in pairing audit shows that signed coefficients require odd orientation and local-factor data, so unsigned row skeletons and parity shadows do not determine the pairing. The exact atomic branch trace would provide that data only conditionally. The signed-lane cycle audit then shows that the trace, payload constructor, origin identity, and common source declaration packet form a dependency cycle. Hence the trace cannot close the proof unless a genuinely new primitive payload/trace artifact, a well-founded terminal descent, or a same-set PDEC certificate is supplied.

The ExactUV incidence audit is independent of this signed pairing cycle. Its source-domain side is reduced to signed row-mass entropy and then to the primitive signed row coefficient law. Its fixed-pair fiber side is reduced to a registered complete key partition and fixed-key local multiplicity. These atoms, together with the model, rate-preservation, and D-structure/Rankin gates, remain unproved in the present corpus. ☐

Theorem 7.44 (Post-antisplit source-rank convergence)**.** *After the anti-split trace/ExactUV atomization, the recorded new-primitive and terminal-descent exits are not independent terminal proof points. The current synchronization is*

```
NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact
-> ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger,
AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
-> ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger,
ActualPreCauchySourceDomainRankAndExactUVNoCollapseLedger
-> PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate.
```

The first internal atom after this convergence is

```
AlphaRowAnchorPhaseEmissionFormulaLedger,
```

with parallel obligations

```
IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-antisplit-source-rank-
convergence-router.md.
```

The new-primitive payload audit already shows that a purported new signed payload only breaks the signed-lane cycle if it carries a pre-Cauchy actual source object, source entropy, a complete key partition, and fixed-key exact-UV local multiplicity. This is exactly the source-rank/no-collapse package. The terminal descent synchronization reaches the same package through its noncanonical leaf analysis. Thus neither exit supplies a third independent closure mechanism.

The source-rank package has three visible faces: source-domain entropy, registered complete keys, and fixed-key multiplicity. The entropy face returns to the primitive signed row coefficient law; the registered-key face requires an actual emitter source table; and the fixed-key face requires a same-unit rank/multiplicity certificate. These faces meet at the same formal-unit pointwise alpha/delta kernel table. The table is still unproved; in particular the alpha row anchor/phase emission formula, the independent pre-Cauchy arithmetic identity, and the same-unit exact-UV rank/multiplicity certificate remain open, as do same-set PDEC, external spectral input, model, rate-preservation, and D-structure/Rankin gates. □

Theorem 7.45 (Post-antisplit alpha return to terminal leaves)**.** *The apparent post-antisplit alpha-row target is not an independent final frontier. The current synchronization is*

```
PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate
-> AlphaRowAnchorPhaseEmissionFormulaLedger
   AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
   AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows,
alpha/weight legs -> PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve,
rank/multiplicity leg -> the same primitive table,
NonrecursivePointwisePrimitiveKernelTableConstructionWithoutTerminalReturn
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple
-> signed-source fixed point -> terminal leaf firewall.
```

Thus the active strict self-contained basis is

```
(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
 OR NoncanonicalFullSComplementLegalClosureMode)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-antisplit-alpha-terminal-
leaf-sync-router.md.
```

The alpha formula leg has already been synchronized to the terminal PDEC/CleanKLS gate: its unsigned carry-shell and phase parts give only a skeleton, while the signed lift and overload branches return to terminal capacity/model ledgers. The signed-weight identity leg returns to the same terminal gate through the moving-block taxonomy. The exact-UV rank/multiplicity leg cannot stand alone because it requires the same primitive source table. Hence a three-leg attack is a fixed point unless a nonrecursive pointwise table is supplied at once.

The nonrecursive table field contract then identifies the first productive field as an explicit joint alpha/delta constructor rule. Existing joint constructor expansions proceed through alpha-side, same-row origin, and row-level source tables back to the signed-source fixed point; without a new formula artifact this route enters the terminal leaf firewall. The currently active terminal leaf has been reduced to canonical-lock or noncanonical legal mode. Neither is proved in the present corpus, and ExactUV incidence, rate-preservation, and D-structure/Rankin remain independent open gates. ☐

Theorem 7.46 (Post-alpha terminal leaf latest noncycle synchronization)**.** *The post-alpha terminal leaf is not the deepest active frontier after importing the existing self-contained and noncycle audits. The synchronized chain is*

```
AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
 OR NoncanonicalFullSComplementLegalClosureMode
-> AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
   OR (AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
       AND IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed),
pair-energy old spine -> source-entropy fixed point
-> ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRuleForActualNoncanonicalSourceTuple
-> signed-source fixed point or terminal descent
-> TERMINAL-SOURCE-PAIR-JOINT macrocycle.
```

Thus the current strict internal noncycle basis is

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
 OR ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
 OR FullSNonAPStrengthenedSourceAntiAtomForActualSource
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND ExplicitModelGapAndFiniteDPRCLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-alpha-terminal-leaf-
latest-noncycle-sync-router.md.
```

The latest self-contained audit reduces the noncanonical legal mode to the actual-source entropy line and then to an acyclic seed plus independent pair energy input. The pair-energy noncycle audit shows that the present pair-energy/rate-packet/terminal route returns to the source-entropy fixed point and therefore cannot prove that target. The first productive field is the explicit joint alpha/delta constructor, but its direct expansion returns to the signed-source fixed point unless a new actual joint formula artifact is supplied.

The terminal descent alternative has also been audited: through the terminal leaf, source, pair-energy, and joint-constructor routes it returns to itself, forming the TERMINAL-SOURCE-PAIR-JOINT macrocycle. The only strict noncycle ways out currently recorded are a new joint formula, an actual-source bridge before the ExactUV/pair/joint loop, or a fresh same-set PDEC scope certificate. ExactUV bounded multiplicity, model/DPRC margin, rate preservation, and the D-structure/Rankin promotion gate remain independent open inputs. Hence this is a synchronization and cycle-elimination result, not an unconditional proof of the row/column theorem. ☐

Theorem 7.47 (Post-alpha noncycle synchronization to terminal three atoms)**.** *The post-alpha noncycle basis is not allowed to stop at* `NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact`*. After importing the branch-trace, signed-payload, source-declaration, signed-lane, and current-global-frontier audits, the present strict frontier is*

```
(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
 OR A1CleanBranchCanonicalSourceAdmission
 OR ActualNoncanonicalCleanCoreMovingAtomExclusion)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The next direct attack target is

```
IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-alpha-noncycle-to-
terminal-three-atoms-sync-router.md.
```

The earlier new-joint formula target is absorbed by the global CRT branch-trace audit into an exact atomic branch trace. The signed-payload audit then shows that visible CRT coordinates do not generate the needed orientation, local factor, or signed coefficient payload. The source-declaration and signed-lane audits close that payload line into a dependency cycle through the common pre-Cauchy source declaration packet. Finally, the current-global frontier audit removes the trace/source/terminal/pair-mass self-return routes, leaving only the three displayed terminal atoms. None of these atoms, and not the D-structure/Rankin promotion gate, is proved here; this is a sharper frontier synchronization, not the unconditional row/column proof. □

Theorem 7.48 (Terminal atoms to alpha-return bridge synchronization)**.** *The terminal three-atom frontier is still not the deepest active strict frontier. After importing the moving-atom entropy normal form, the post-Mertens nonrecursive synchronization, the same formal-unit kernel identity attack, the pointwise primitive table contract, the terminal leaf firewall, and the alpha-return firewall, the old moving-atom line is absorbed as*

```
ActualNoncanonicalCleanCoreMovingAtomExclusion
-> ExactCleanCoreFullSNonAPWFDSourceEntropy
-> SameFormalUnitPreCauchyAlphaDeltaKernelIdentityWithSignedPhiAndFiberDispersion
-> PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate
-> NonrecursivePointwisePrimitiveKernelTableConstructionWithoutTerminalReturn
-> terminal leaf firewall.
```

The current strict self-contained frontier is therefore

```
(AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
 OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn)
AND (SelfContainedExplicitZetaZeroFreeRegionThetaEnvelopeXGe20000
     AND SelfContainedMeisselMertensConstantIntervalLedgerAt20000)
AND SelfContainedDStructureTailLog4FiniteRankinReplacementPackage.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-terminal-atoms-to-alpha-return-
bridge-sync-router.md.
```

The A1 branch is only a scoped canonical statement and does not close the unrestricted noncanonical branch. The moving-atom atom is equivalent to an actual clean-core source entropy target; post-Mertens and kernel-identity audits then force the same formal-unit pointwise primitive table. That table cannot be assembled by separate alpha, weight, and rank legs: the field contract requires a single nonrecursive primitive table, while the old joint constructor expansion returns to the signed-source fixed point and then to the terminal leaf firewall.

The terminal leaf firewall has already reduced current materialized PDEC and sparse instances to future admission disciplines, leaving canonical-lock or the noncanonical legal mode. The non-canonical legal mode filters to an actual-source bridge, but the old pointwise/alpha descent route returns through PDEC/CleanKLS to the same terminal family. Hence that route is a backedge, not a well-founded descent. The remaining strict inputs are the displayed canonical-lock or independent actual-source bridge, together with the explicit Mertens/theta tail and D-structure/Rankin replacement gates. This is not an unconditional proof of the row/column theorem. □

Theorem 7.49 (Alpha-return bridge to concrete terminal split)**.** *The frontier*

```
AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
 OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn
```

can be sharpened one more level. The independent actual-source arm reduces to an A1 scoped source-admission statement or exact clean-core source entropy; the A1 statement is not a global contradiction, and the entropy arm requires a nonrecursive seed plus an independent pair-energy bound. The canonical-lock arm reduces to a mismatch/DLS/PDEC split. Thus the concrete terminal split is

```
((AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
  AND IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed)
 OR SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks
 OR (PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve
    AND ExplicitModelGapAndFiniteDPRCLedger)).
```

The next direct attack target is

```
IndependentExactPairL2EnergyOrMaxAtomBoundForAcyclicSeed.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-alpha-return-bridge-to-
concrete-terminal-split-sync-router.md.
```

On the independent actual-source arm, the clean A1 admission route is absorbed as a scoped canonical branch statement and cannot close the unrestricted noncanonical complement. The remaining entropy route is guarded by the nonrecursive pair-energy firewall: one may not prove the pair mass bound using the moving-atom or source-entropy conclusion that it is supposed to prove. It therefore requires both an acyclic pre-Cauchy noncanonical source seed and an independent exact pair L2 or max-atom estimate.

On the canonical-lock arm, direct attack gives only scoped absorption in the equality case. In the mismatch case the signed-source and alpha-weight routes return to the terminal family, while the clean analytic exit is a genuine self-contained Kuznetsov/DLS large-sieve inequality and the combinatorial exit is PDEC/CleanKLS together with the explicit model/DPRC margin. None of these three concrete alternatives is proved in the present corpus, and the rate preservation and D-structure/Rankin gates remain open. Hence this is a frontier synchronization, not an unconditional row/column proof. □

Theorem 7.50 (Pair-energy to seed-coordinate cycle synchronization)**.** *The pair-energy target singled out by the alpha-return split does not admit a new internal descent through the present signed-source machinery. Seed-only information and qualitative projection dissipation do not give a log-power rate; the old ExactUV/source-entropy pair-energy spine returns to the same source-entropy target; and the joint-constructor route enters the signed coordinate-source cycle unless a genuinely acyclic primitive basis/coefficient source input is supplied. Since the seed exists/does-not exist dichotomy has already been routed to the acyclic terminal family, the current pair-energy frontier is sharpened to*

```
SelfContainedExactPairEnergyLargeSieveInequalityForAcyclicSeed
 OR SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks
 OR (PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve
     AND ExplicitModelGapAndFiniteDPRCLedger).
```

The first term is the direct non-source-cycle pair-energy large-sieve obligation; the other two are the concrete terminal alternatives already exposed by the canonical-lock/direct-terminal audits.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-pair-energy-to-seed-
coordinate-cycle-sync-router.md.
```

The router imports the independent pair-energy attack, the pair-energy noncycle reconciliation, the rate-bearing large-pair packet split, the terminal recurrence firewall, the source-entropy downstream cycle guard, the acyclic seed coordinate-source cycle guard, the seed-terminal fusion ledger, and the terminal canonical-lock direct attack. These ledgers together show that every currently documented internal pair-energy descent either lacks a rate, reuses the theorem it is supposed to prove, or returns to the signed coordinate-source cycle and terminal family. Hence the remaining productive work is exactly a direct pair-energy large-sieve inequality, a self-contained Kuznetsov/DLS inequality, or a PDEC/CleanKLS bound with the explicit model margin. None of these three is proved here, and the rate-preservation and D-structure/Rankin gates remain open; this is therefore a sharper frontier, not an unconditional row/column proof. □

Theorem 7.51 (Pair-energy diagonal peeling terminal reduction)**.** *The direct exact-pair energy large-sieve atom is not an independent third frontier after diagonal peeling. Any such L2/max-atom estimate must first control the same formal-unit diagonal mass of a single exact pair; otherwise a delta packet supported on one exact pair violates every log-power no-heavy target. Failure of this diagonal ledger is precisely a rate-bearing large pair packet and therefore returns to the already materialized terminal trident. After the diagonal and low-dimensional returns are removed, the off-diagonal exact-pair bilinear form is the clean Kuznetsov/DLS object. With the existing KZ and PDEC/model ledgers imported, the strict frontier is therefore reduced to*

```
AcyclicNCBLKActualBlockNonconcentrationOrStrengthenedSourceAntiAtom
 OR (AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
     AND HighSegmentModelGapAlpha043C3AnalyticLedger).
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-pair-energy-diagonal-
peeling-terminal-reduction-router.md.
```

The router decomposes the direct pair-energy large-sieve statement into a diagonal no-heavy ledger and a clean off-diagonal large-sieve ledger. The single-pair delta model proves that the diagonal ledger is a necessary input, not a consequence of a black-box large-sieve bound. If that ledger fails, the existing pair-energy packetization sends the object to the rate-bearing packet terminal split. If the diagonal ledger is peeled away, the remaining off-diagonal bilinear form is exactly the clean Kloosterman/Kuznetsov-DLS problem, whose self-contained frontier has already been reduced to acyclic NC-BLK/source anti-atom. The PDEC/CleanKLS branch is simultaneously split into same-set PDEC scope matching, while the explicit model ledger has reduced its finite part to a high-segment analytic margin. This removes the direct pair-energy atom as an independent disjunct, but it does not prove either NC-BLK/source anti-atom or the PDEC/model branch, nor the rate-preservation and D-structure/Rankin gates. □

Theorem 7.52 (NCBLK/source anti-atom source-root frontier)**.** *After the pair-energy diagonal has been peeled, the NCBLK/source anti-atom label is not the sharpest remaining self-contained target. The generic well-factorable anti-atom route is blocked by the moving-delta model. The actual acyclic route must first produce a forward pre-Cauchy source-root packet: a declaration line before Cauchy/payment, primitive signed rows, built-in word/coefficient pairing, actual source-domain entropy, fixed exact (u, v) fiber control, and a same formal-unit exact-pair no-heavy or L2 energy ledger. If this packet is not produced forward, the existing taxonomy routes the failure to the moving-block/global PDEC-sparse terminal. Therefore the current strict frontier is*

```
ForwardAcyclicPreCauchySourceRootPacketOrNamedReturn
 OR (GlobalPDECorSparseTerminalExclusion
      AND HighSegmentModelGapAlpha043C3AnalyticLedger)
 OR (AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
      AND HighSegmentModelGapAlpha043C3AnalyticLedger).
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-ncblk-source-
antiatom-frontier-sync-router.md.
```

The router imports the pair-energy diagonal-peeling frontier, the strict acyclic NCBLK deduplication, the actual-source support seed split, the hypothetical zero-row seed no-go, the independent pre-Cauchy identity taxonomy, the moving-block return, and the source-declaration downstream split. These inputs show that NCBLK/source anti-atom remains open, but using the NCBLK name as the direct target now hides the first productive field: the forward source-root packet. The global PDEC/sparse terminal, the direct PDEC same-set scope, the high-segment model margin, rate preservation, and the D-structure/Rankin gate remain unproved. This is a frontier sharpening, not an unconditional row/column theorem. □

Theorem 7.53 (Forward source-root terminal-cycle synchronization)**.** *The forward source-root packet singled out by the NCBLK/source anti-atom frontier is not an independent non-circular proof route in the present corpus. When the common packet downstream, signed-lane cycle, new primitive source-rank convergence, cycle-cut/terminal descent, antisplit ExactUV atomization, pointwise primitive alpha/delta kernel table, alpha-row unsigned skeleton, signed-lift return, and anchor-collar overload return are imported, the documented source-root route returns to the terminal capacity gate*

```
PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve.
```

Hence the sharpened strict frontier is

```
(AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
 OR NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse)
AND HighSegmentModelGapAlpha043C3AnalyticLedger,
```

together with the still independent rate-preservation and D-structure/Rankin acceptance gates.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-forward-source-root-
terminal-cycle-sync-router.md.
```

The router records the following status flags:

```
forward_source_root_packet_proved=false
forward_source_root_independent_after_router=false
row_column_unconditional_closed=false.
```

Its positive content is only that the currently documented source-root subroutes are exhausted as non-circular descents: the signed lane is a self-cycle, the new primitive and terminal descent meet at the same source-rank/pointwise-kernel front, the unsigned alpha skeleton is already the geometric part, and the signed lift and overload lines return to the terminal gate. Therefore a CleanKLS/Kuznetsov-DLS closure that reuses NCBLK/source-root would be circular. The remaining productive alternatives are a direct same-set PDEC cap dual certificate or a Kuznetsov-DLS large-sieve proof that does not reuse this source-root loop, plus the high-segment model ledger and the rate and D-structure gates. This is a cycle obstruction and frontier synchronization, not a proof of the row/column theorem. □

Theorem 7.54 (Post-source-root PDEC scope saturation). *The direct PDEC arm left by the forward source-root synchronization is already saturated in the current internal corpus. Same-set PDEC protocol and the canonical-source capacity boundary may be used as tools, but the strict acyclic non-canonical branch still lacks a proof that its terminal certificate has the same formal unit, the same bad-window set, the same U_CRT/L_PDEC pushforward, and the same mass convention as the canonical same-set certificate. Following the branch through canonical-lock or ordinary Kuznetsov/DLS returns to scoped canonical absorption, mismatch/DLS, or the terminal-family cycle. Thus, unless a new PDEC scope certificate or an accepted external DIBFI no-projection input is supplied, the internal non-circular frontier becomes*

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
 OR NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-source-root-pdec-scope-
saturation-sync-router.md.
```

It records

```
post_source_root_pdec_scope_active=true
pdec_scope_branch_saturated_in_current_internal_corpus=true
pdec_scope_proved=false
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
new_explicit_joint_constructor_formula_artifact_present=false
row_column_unconditional_closed=false.
```

The positive information is a routing exclusion: direct PDEC is not a proved non-circular exit inside the present material. The conditional PDEC line is kept as a possible new scope certificate, and the external DIBFI line is kept as an external input; neither is supplied here. The non-circular Kuznetsov/DLS arm is likewise open because the documented KZ route would otherwise reuse the NCBLK/source-root loop. Hence this step narrows the active internal frontier to a new explicit joint alpha/delta constructor or a genuinely non-circular KZ/DLS proof, with model, rate, and D-structure gates still open. It is not an unconditional row/column proof. □

Theorem 7.55 (Post-PDEC new-joint noncycle synchronization). *The new explicit joint alpha/delta constructor line left by the post-PDEC scope audit is saturated in the current internal corpus. The documented joint/antisplit route only reaches exact atomic branch trace; branch trace only supplies visible coordinates; the missing signed coefficient and local factor require an atomic signed payload constructor; and the signed-payload line is already part of the signed-lane/source-rank/alpha terminal cycle. Thus the old internal new-joint route is not a non-circular exit. Unless a new joint/payload/PDEC/external input is supplied, the remaining internal non-circular frontier is*

```
NonCircularSelfContainedKuznetsovDLSLargeSieveWithoutNCBLKSourceRootReuse
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-pdec-new-joint-
noncycle-sync-router.md.
```

It records

```
new_joint_current_internal_route_saturated=true
new_explicit_joint_constructor_formula_artifact_present=false
exact_atomic_branch_trace_formula_proved=false
atomic_signed_payload_constructor_proved=false
new_primitive_artifact_independent_present=false
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
row_column_unconditional_closed=false.
```

The positive information is a non-circularity audit. Finite CRT phase replication does not create orientation, local factor, or signed coefficient data. A new primitive would have to carry a pre-Cauchy actual source-rank and no-collapse package from outside the signed-lane cycle; no such package is supplied here. Hence this step deletes the old new-joint self-proof route and moves the direct attack to a Kuznetsov/DLS large-sieve proof that does not reuse NCBLK/source-root, with the high-segment model, rate preservation, and D-structure gates still open. It is not an unconditional row/column proof. □

Theorem 7.56 (Post-new-joint KZ no-cycle gate)**.** *The non-circular KZ/DLS gate left by the post-new-joint synchronization is not closed by the currently documented KZ route. The windowed DLS formal layer and the KZ-A through KZ-D spectral spine may be imported, but the documented KZ-E log-saving route factors through NCBLK/source-antiatom projection and then returns to the source-root or terminal cycle. Since the current gate explicitly requires no NCBLK/source-root reuse, that route cannot count as a non-circular proof. The remaining internal frontier is therefore*

```
AcyclicKZEWellFactorableDispersionLogSavingWithoutNCBLKProjection
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-new-joint-kz-
nocycle-gate-sync-router.md.
```

It records

```
latest_kz_nocycle_gate_active=true
windowed_dls_formal_layer_imported=true
kz_abcd_spine_imported=true
existing_kz_e_route_factors_through_ncblk=true
ncblk_projection_forbidden_for_nocycle_gate=true
noncircular_kuznetsov_dls_without_source_root_reuse_proved=false
kz_e_direct_log_saving_without_ncblk_projection_proved=false
row_column_unconditional_closed=false.
```

The positive information is a gate refinement. KZ-A–KZ-D no longer hide the main obstruction; the obstruction is the direct KZ-E well-factorable dispersion saving. If the saving is obtained only by projecting to NCBLK/source anti-atoms, it reuses the cycle that the gate was designed to exclude. Thus the next direct attack is a KZ-E log-saving theorem that does not pass through NCBLK projection, or an explicitly external no-projection DI/BFI/Kuznetsov certificate. This is not an unconditional row/column proof. □

Theorem 7.57 (Post-KZ-E direct source bridge)**.** *The direct KZ-E no-projection gate is not a free generic WFD theorem. In the current corpus the generic self-contained WFD route is refuted by the moving-delta model, while the canonical-restricted route is closed only after the source is locked, before Cauchy and dispersion, to the canonical Rosser–Iwaniec/Buchstab decision-tree coefficient. Since noncanonical moving-atom or source-antiatom returns are terminal returns rather than a KZ no-cycle proof, the internal KZ-E direct gate is reduced to*

```
A1CleanBranchCanonicalSourceAdmission
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-kze-direct-
source-bridge-sync-router.md.
```

It records

```
kze_direct_no_projection_gate_active=true
self_contained_taxonomy_imported=true
canonical_provenance_closed_only_for_canonical_branch=true
source_lock_scoped_not_global=true
actual_source_bridge_obstruction_imported=true
noncanonical_moving_atom_not_kz_nocycle_proof=true
kze_direct_current_internal_route_reduced_to_source_admission=true
a1_clean_branch_canonical_source_admission_proved=false
row_column_unconditional_closed=false.
```

The positive information is a source-bridge refinement. The canonical branch may use the internal support chain only after the actual clean A1 branch is admitted to the canonical source branch. The generic complement remains external or terminal; it cannot be silently upgraded, and a return through moving atoms would reuse the terminal route rather than prove KZ-E directly. Thus the next direct attack is source admission, with an external no-projection DI/BFI/Kuznetsov certificate retained only as a conditional input. This is not an unconditional row/column proof. □

Theorem 7.58 (Post-source-admission macrocycle synchronization)**.** *The A1 source-admission target left by the KZ-E direct source bridge does not supply a new non-circular descent in the present corpus. The canonical equality case is absorbed only as a scoped canonical-source case. The non-canonical mismatch case is routed through T1 mismatch forcing, signed-lift failure registration, alpha signed-weight downstream synchronization, the PDEC/CleanKLS terminal gate, new-joint saturation, and the KZ no-cycle gate; that route returns to the KZ-E source bridge and hence to A1 source admission. Thus the currently documented internal route forms the macrocycle*

```
A1CleanBranchCanonicalSourceAdmission
  -> T1 classifier -> mismatch/signed lift
  -> PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve
  -> new-joint saturation -> non-circular KZ/DLS
  -> KZ-E direct source bridge
  -> A1CleanBranchCanonicalSourceAdmission.
```

Consequently the remaining non-circular break basis is

```
(AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
 OR NewPrimitiveJointPayloadArtifactOutsideSignedLaneCycle
 OR ExactExternalDIBFIKuznetsovNoProjectionCertificate_FOR_NONCIRCULAR_KZ_ONLY)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-strict-post-source-admission-
macrocycle-sync-router.md.
```

It records

```
post_kze_source_admission_active=true
a1_source_admission_scoped_not_global=true
t1_mismatch_no_silent_exit_imported=true
signed_lift_failure_return_only_registers=true
alpha_weight_downstream_returns_to_terminal=true
internal_pdec_clean_kls_cycle_imported=true
nonrecursive_breaker_hits_seed_cycle=true
new_joint_and_kz_return_to_a1_gate=true
a1_pdec_kz_macrocycle_detected=true
row_column_unconditional_closed=false.
```

The positive information is a non-circularity audit. It proves that the current A1/PDEC/KZ internal route returns to its starting gate and therefore cannot be counted as a proof of the row/column contradiction. It does not exclude the outside-cycle inputs listed in the theorem, nor does it close the high-segment model, rate-preservation, or D-structure gates. □

Theorem 7.59 (Row-gap supply and phase cycle-cut audit). *Fix an odd prime P and $1 \le k < P$, and consider the row interval*

$$I_k = \{kP + a : 1 \le a < P\}.$$

If this interval contains no prime, then each element of I_k has a prime divisor $q < P$. The correct uniform root split, however, is

```
low-root primes:  q <= floor(sqrt(kP)),
near-root primes:  floor(sqrt(kP)) < q <= floor(sqrt(kP+P-1)).
```

Thus it is too strong to assert that the primes below $\sqrt{kP}$ alone must cover the row. The zero-row obligation is instead the exact finite covering condition that the low-root residue classes together with the near-root slots cover every column $a = 1, \ldots, P-1$. Raw covering capacity with multiplicity is not a contradiction; the active non-circular inputs are therefore a uniform low-root sifted-residue deficit after near-root slots, or an adjacent-prime Q_1, Q_2 CRT transport defect forcing a stable short same-label return. The synchronized basis is

```
(UniformLowRootSiftedResidueDeficitAfterNearRootSlots
 OR AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
 OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
 OR ExactExternalPrimeGapSqrtBarrierCertificate_FOR_ROW_GAP_ONLY)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-row-gap-supply-phase-cycle-cut-router.md.
```

It records

```
row_gap_model_pinned=true
correct_divisor_supply_law=true
low_root_only_claim_rejected=true
capacity_only_contradiction_rejected=true
crt_period_mirror_not_contradiction=true
low_root_deficit_after_near_root_slots_proved=false
q1q2_transport_defect_or_stable_short_return_proved=false
row_column_unconditional_closed=false.
```

The positive information is the correction of the supply law and the separation of two possible non-circular inputs. If $Q_1 < kP$ and $Q_2 > (k+1)P$ are adjacent primes around the putative gap, CRT mirror symmetry by itself only gives a mirror covered block. It becomes a contradiction only after one proves a stable same-formal-unit same-label short return, or proves that all instabilities enter registered PDEC/SAE/ColumnCRT defects. Those assertions are not proved here. □

Theorem 7.60 (Low-root sifted deficit frontier). *The row-gap low-root deficit target is equivalent to a sharp short-interval rough-residue lower bound. The near-root contribution is confined to*

$$\sqrt{kP} < q \le \sqrt{kP + P - 1},$$

and each such prime contributes only one residue class in the row. Hence the near-root side has an elementary capacity envelope. The open point is not that envelope, but the following non-circular lower bound:

```
NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP.
```

Mertens density gives only a heuristic for this lower bound; it cannot be substituted for a pointwise row estimate. The current synchronized basis is

```
(NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP
 OR AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
 OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-lowroot-sifted-deficit-frontier-router.md.
```

It records

```
exact_deficit_identity_closed=true
near_root_slot_upper_bound_elementary=true
mertens_heuristic_not_proof=true
jacobsthal_barrier_identified=true
noncircular_short_interval_rough_residue_lower_bound_proved=false
row_column_unconditional_closed=false.
```

The exact identity is that, inside I_k, a slot not covered by any prime up to $\sqrt{kP + P - 1}$ must itself be prime. Thus a positive uncovered slot is precisely the desired row prime. This identity explains why the remaining rough-residue lower bound is not a harmless auxiliary estimate: it is the hard row-prime content in sieve form. This step therefore narrows the frontier but does not close it. □

Theorem 7.61 (Short-interval rough-residue barrier). *The direct target*

```
NonCircularShortIntervalRoughResidueLowerBoundLengthPLevelSqrtkP
```

is not an independent internal lower-sieve input in the self-contained row/column proof. For $I_k = \{kP + a : 1 \le a < P\}$, $1 \le k < P$, a slot uncovered by all primes $q \le \sqrt{kP + P - 1}$ is necessarily prime, since $kP + a < P^2$. Conversely, a prime slot is uncovered by every such q. After subtracting the near-root-only slots

$$\sqrt{kP} < q \le \sqrt{kP + P - 1},$$

the same equivalence remains: the positive deficit is exactly a row prime. Thus a proof of this direct rough-residue lower bound is already a sqrt-length prime-gap proof for the row. It may be registered as an external strong input, but it cannot be used as a non-circular internal black box. The current internal non-circular basis is therefore

```
(AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
 OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-short-interval-rough-residue-barrier-router.md.
```

It records

```
full_root_rough_equals_prime_in_row=true
near_root_subtraction_keeps_equivalence=true
mertens_average_insufficient=true
periodwide_jacobsthal_shortcut_rejected=true
noncircular_internal_use_rejected=true
noncircular_short_interval_rough_residue_lower_bound_proved=false
row_column_unconditional_closed=false.
```

Mertens density, finite samples, and period-wide Jacobsthal intuition do not give the required point-wise row estimate. The next internal attack is the adjacent-prime Q_1, Q_2 CRT transport defect, with seed cycle-cut and same-set PDEC scope matching kept as parallel non-circular routes. This is a barrier classification, not a final unconditional closure. $\square$

Theorem 7.62 (Phi-LPF $1 < k < P$ row interval difference)**.** *Fix an odd prime P and $1 < k < P$. The Phi-LPF endpoint identity applies to the closed interval $[kP, kP + P]$:*

$$\pi(kP + P) - \pi(kP - 1) = P + 1 - \sum_{p \le \sqrt{kP+P}} \left(\Phi\left(\left\lfloor \frac{kP + P}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{kP - 1}{p} \right\rfloor, p \right) \right).$$

Since both endpoints kP and $kP + P = (k+1)P$ are composite multiples of P, this closed interval count is exactly the internal row count

$$\pi(kP + P) - \pi(kP - 1) = \pi(kP + P - 1) - \pi(kP).$$

For the actual internal row

$$I_k = \{kP + a : 1 \le a < P\},$$

the corresponding exact formula is

$$\pi(kP + P - 1) - \pi(kP) = (P - 1) - \sum_{p \le \sqrt{kP+P-1}} \left(\Phi\left(\left\lfloor \frac{kP + P - 1}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{kP}{p} \right\rfloor, p \right) \right).$$

The Phi recurrence

$$\Phi(x, p_j) = \Phi(x, p_{j+1}) + \Phi\left(\left\lfloor \frac{x}{p_j} \right\rfloor, p_j \right)$$

computes each bucket endpoint exactly. Since $1 < k < P$ and $1 \le a < P$, every row slot satisfies $kP + a < P^2$. Hence a slot of I_k uncovered by all primes $q \le \sqrt{kP + P - 1}$ is a prime, and conversely every prime slot is uncovered. The endpoint-difference identity therefore gives an exact value for the number of row primes, but its positivity is precisely the inequality

$$\sum_{p \le \sqrt{kP+P-1}} \left(\Phi\left(\left\lfloor \frac{kP + P - 1}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{kP}{p} \right\rfloor, p \right) \right) < P - 1.$$

This inequality is not supplied by the identity itself; it is equivalent to the full-root uncovered-slot, hence row-prime, positive statement. The previous CRT aligned prime-free sample for $P = 5, k = 8166$ is not applicable to this $1 < k < P$ row problem. The remaining non-circular basis is still

```
(AdjacentPrimeQ1Q2CRTTransportDefectOrStableShortReturn
 OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-k-less-p-row-interval-
difference-router.md.
```

It records

```
k_less_p_endpoint_difference_identity_proved=true
strict_k_closed_interval_equals_internal_row_proved=true
internal_row_formula_proved=true
phi_recurrence_computes_bucket_deltas=true
inside_p_square_prime_uncovered_equivalence_proved=true
previous_crt_prime_free_block_not_applicable=true
interval_positivity_from_identity_alone_proved=false
full_root_uncovered_slot_positive_proved=false
row_column_unconditional_closed=false.
```

The positive information is the exact localization of what Phi-LPF does in the $1 < k < P$ row setting. It turns the closed interval count into the same mechanical endpoint difference as the internal row count, and it proves that full-root uncovered slots and row primes are the same objects below P^2. It also prevents importing the earlier $k > P$ CRT prime-free block as a counterexample to this row target. What it does not provide is the strict upper bound on the composite bucket increment sum. That bound remains the row-prime content itself, so the row/column theorem remains open. □

Theorem 7.63 (Phi-LPF strict-k endpoint bucket cancellation)**.** *In the setting of the preceding theorem, let Δ_p^{cl} be the Phi-LPF bucket increment for the closed interval $[kP, kP+P]$, and let Δ_p^{int} be the bucket increment for the internal interval $[kP+1, kP+P-1]$. Then*

$$\Delta_p^{\mathrm{cl}} - \Delta_p^{\mathrm{int}} = \mathbf{1}_{p=\mathrm{LPF}(k)} + \mathbf{1}_{p=\mathrm{LPF}(k+1)}.$$

Consequently

$$\sum_p \Delta_p^{\mathrm{cl}} - \sum_p \Delta_p^{\mathrm{int}} = 2.$$

Thus the two extra positions in the closed interval are exactly paid for by the two endpoint composite owner buckets. The closed interval length $P+1$ supplies no extra prime-positive slack over the internal row length $P-1$. The remaining positivity condition is still

$$\sum_p \Delta_p^{\mathrm{int}} < P - 1,$$

equivalently the existence of a full-root uncovered row slot.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-endpoint-
bucket-cancellation-router.md.
```

It records

```
strict_k_endpoint_composite_mass_two_proved=true
endpoint_lpf_owner_buckets_pinned=true
closed_minus_internal_bucket_delta_equals_endpoint_owners_proved=true
no_endpoint_slack_for_row_positivity_proved=true
internal_row_composite_delta_inequality_proved=false
full_root_uncovered_slot_positive_proved=false
row_column_unconditional_closed=false.
```

The left endpoint kP has least prime factor $\mathrm{LPF}(k)$, since $1 < k < P$. The right endpoint $(k+1)P$ has least prime factor $\mathrm{LPF}(k+1)$, with the boundary case $k+1 = P$ giving the owner P. The LPF ownership identity assigns each endpoint to exactly its owner bucket and to no other bucket. Removing the two endpoints therefore subtracts precisely these two bucket masses. This is a useful accounting closure, but it does not prove the internal row inequality; the row/column theorem remains open. □

Theorem 7.64 (Phi-LPF strict-k internal owner saturation)**.** *In the strict $1 < k < P$ row*

$$I_k = \{kP + a : 1 \le a < P\},$$

the Phi-LPF internal bucket increment equals the direct LPF-owner partition. If $M_p(k)$ denotes the number of slots a for which $\mathrm{LPF}(kP + a) = p$, then

$$\Delta_p^{\mathrm{int}} = M_p(k)$$

for every prime $p \le \sqrt{kP + P - 1}$. Hence

$$\#\{a : \ kP + a \text{ is prime}\} = (P-1) - \sum_p \Delta_p^{\mathrm{int}} = (P-1) - \sum_p M_p(k).$$

Moreover, a slot assigned to owner p satisfies the explicit owner equation

$$a \equiv -kP \pmod{p}, \qquad (kP + a)/p \text{ is } p\text{-rough}.$$

Therefore a zero row is equivalent to exact LPF-owner saturation:

$$\sum_p M_p(k) = P - 1.$$

The unresolved point is the strict positive saturation defect

$$P - 1 - \sum_p M_p(k) > 0.$$

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-internal-
owner-saturation-router.md.
```

It records

```
internal_phi_bucket_equals_lpf_owner_partition_proved=true
owner_residue_equation_pinned=true
saturation_defect_equals_prime_count_proved=true
zero_row_iff_owner_saturation_proved=true
positive_saturation_defect_proved=false
row_column_unconditional_closed=false.
```

The Phi-LPF identity counts a composite $kP + a$ once, in the bucket of its least prime factor. Conversely, a direct LPF-owner assignment has precisely the same bucket key and rough cofactor condition. Since $kP + a < P^2$, the unassigned slots are exactly the prime slots. Thus the certificate closes the owner accounting and the zero-row equivalence. It does not prove that the saturation defect is always positive; that remains the full-root uncovered slot problem, so the row/column theorem remains open. □

Theorem 7.65 (Phi-LPF strict-k raw rejection balance)**.** *In the same strict row $I_k = \{kP + a : 1 \le a < P\}$, let*

$$R(k) = \#\{(a, p) : 1 \le a < P,\ p \le \sqrt{kP + P - 1},\ p \mid kP + a\}$$

be the total raw root-prime incidence. Let $O(k) = \sum_p M_p(k)$ be the LPF-owner mass from the preceding theorem, and let $E(k)$ be the number of raw incidences whose prime p is not the least prime factor of the same slot. Then

$$R(k) = O(k) + E(k).$$

Consequently the row-prime count has the exact signed form

$$\#\{a :\ kP + a \text{ is prime}\} = E(k) - \big(R(k) - (P - 1)\big).$$

Thus a zero row is equivalent to the critical raw/rejection balance

$$E(k) = R(k) - (P - 1),$$

while a positive row is equivalent to the strict imbalance

$$E(k) > R(k) - (P - 1).$$

Raw incidence capacity alone therefore gives no contradiction: its excess over the row length can be absorbed exactly by non-owner LPF rejections. The remaining non-circular input is a proof of strict rejection excess, not merely a raw capacity estimate.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-raw-
rejection-balance-router.md.
```

It records

```
raw_incidence_owner_rejection_partition_proved=true
prime_count_equals_rejection_excess_proved=true
zero_row_iff_exact_raw_rejection_balance_proved=true
raw_capacity_only_contradiction_rejected=true
positive_rejection_excess_proved=false
row_column_unconditional_closed=false.
```

Every raw incidence is either the LPF-owner incidence of its slot or a hit by a larger prime factor after a smaller factor has already owned the slot. This gives $R(k) = O(k) + E(k)$. Substituting the preceding owner identity $\#\mathrm{PrimeSlots} = P - 1 - O(k)$ gives the displayed signed balance. The certificate narrows the zero-row obstruction to an exact raw/rejection critical point, but it does not prove the strict inequality $E(k) > R(k) - (P - 1)$. That inequality is still the full-root uncovered-slot problem in another exact coordinate system, so the row/column theorem remains open. □

Theorem 7.66 (Phi-LPF strict-k short-interval exponent barrier)**.** *The Phi-LPF endpoint difference gives an exact count for the interval $[kP, kP+P]$ when $1 < k < P$. If one tries to supply the remaining positivity from a general short-interval theorem of the form*

$$[x, x + Cx^{\theta}] \text{ contains a prime}$$

for all sufficiently large x, then applying it at $x = kP$ inside the row interval requires

$$P \geq C(kP)^{\theta}.$$

Equivalently,

$$k \leq C^{-1/\theta} P^{(1-\theta)/\theta}.$$

Thus every fixed exponent $\theta > 1/2$ covers only the lower part of the strict range $1 < k < P$. The top band

$$C^{-1/\theta} P^{(1-\theta)/\theta} < k < P$$

remains an infinite family as $P \to \infty$. Finite verification below a threshold cannot remove this asymptotic top band. Therefore the pure short-interval lane would need a square-root-scale input with effective constant at most 1*,*

$$[x, x + \sqrt{x}] \text{ contains a prime}$$

in the required aligned range, or else the proof must return to the structural strict-imbalance problem

$$\text{PositiveRejectionExcessForStrictKRawLPFIncidence}\,.$$

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-short-
interval-exponent-barrier-router.md.
```

It records

```
endpoint_difference_available=true
theta_greater_than_half_barrier_proved=true
finite_verification_plus_theta_gt_half_cannot_close_all_large_p=true
sqrt_scale_input_needed_for_pure_short_interval_lane=true
sqrt_scale_input_available_in_current_corpus=false
positive_rejection_excess_proved=false
row_column_unconditional_closed=false.
```

The calculation is only exponent arithmetic, but it is decisive for this route. The row interval has length P, while $x = kP$. A theorem with length Cx^{θ} fits inside the row only on the displayed k-range. For $\theta > 1/2$, the exponent $(1-\theta)/\theta$ is strictly less than one, so the uncovered top band has unbounded size. The certificate therefore does not prove row positivity; it rules out a common attempted closure by threshold plus finite verification unless a genuine square-root-scale interval input is supplied. The row/column theorem remains open. □

Theorem 7.67 (Phi-LPF strict-k square-root gap equivalence)**.** *For $1 < k < P$, the Phi-LPF endpoint difference gives the exact integer*

$$N_P(k) = \pi((k+1)P - 1) - \pi(kP).$$

Since the two endpoints kP and $(k+1)P$ are composite, the desired positivity is exactly

$$N_P(k) \geq 1 \quad \Longleftrightarrow \quad \text{nextprime}(kP) < (k+1)P.$$

Equivalently, no consecutive-prime gap may cover the whole P-grid row $(kP,(k+1)P)$. In the raw/rejection coordinates of the preceding strict Phi-LPF certificates, the same condition is

$$E(k) > R(k) - (P-1).$$

Moreover, if one had a square-root prime-gap input

$$\forall x \geq X_0 \quad \exists r \text{ prime} \quad x < r \leq x + \sqrt{x},$$

then all strict rows with $kP \geq X_0$ would close, because

$$\sqrt{kP} < P \qquad (1 < k < P).$$

The remaining rows would satisfy $kP < X_0$, hence $P < X_0/2$, and would be a finite verification problem. Thus the threshold-plus-finite-verification route is valid only after a genuine square-root-scale prime-gap theorem has been supplied.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-sqrt-
gap-equivalence-router.md.
```

It records

```
endpoint_difference_exact_integer_value_proved=true
ge_one_equivalent_to_positive_rejection_excess_proved=true
ge_one_equivalent_to_aligned_sqrt_gap_exclusion_proved=true
sqrt_gap_input_would_close_all_large_strict_rows=true
finite_verification_template_after_sqrt_gap_input_proved=true
sqrt_gap_input_proved_in_current_corpus=false
row_column_unconditional_closed=false.
```

The equivalence is immediate from the endpoint cancellation and the exact Phi-LPF count. The conditional closure follows from the strict inequality $\sqrt{kP} < P$. What is not present in the current corpus is the square-root prime-gap input itself. Therefore the theorem gives the correct closure template and the exact obstruction, but not an unconditional proof of $N_P(k) \geq 1$ for all strict rows. □

Theorem 7.68 (Phi-LPF strict-k top-row square collar)**.** *Among the strict rows $1 < k < P$, the top row $k = P-1$ is the prime-square upper collar*

$$I_{P-1} = (P^2 - P, P^2).$$

The Phi-LPF endpoint formula specializes to

$$N_{\text{top}}(P) = \pi(P^2 - 1) - \pi(P^2 - P)$$

and equivalently

$$N_{\text{top}}(P) = P - 1 - \sum_{q<P} \left(\Phi\left(\left\lfloor \frac{P^2-1}{q} \right\rfloor, q \right) - \Phi\left(\left\lfloor \frac{P^2-P}{q} \right\rfloor, q \right) \right).$$

Thus any proof of positivity for all strict rows must in particular prove

$$\pi(P^2 - 1) - \pi(P^2 - P) \geq 1$$

for every prime P. This top row is also the least-slack square-root row: the quantity $P - \sqrt{kP}$ decreases with k, and at $k = P - 1$

$$P - \sqrt{P(P-1)} = \frac{P}{P + \sqrt{P(P-1)}} \in (1/2, 1).$$

Hence the top row isolates the prime-square upper-collar hard core of the strict Phi-LPF positivity problem.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-top-row-
square-collar-router.md.
```

It records

```
top_row_necessary_hard_core_identified=true
top_row_minimal_sqrt_slack_proved=true
top_row_positive_proved_in_current_corpus=false
row_column_unconditional_closed=false.
```

The identities follow by setting $k = P - 1$ in the strict row endpoint formula. The square-root slack statement is the elementary monotonicity of $\sqrt{kP}$ in k. The finite audit finds no empty top row in the checked range, but the present corpus does not prove the global collar input $\pi(P^2 - 1) - \pi(P^2 - P) \geq 1$. Therefore this theorem extracts a necessary hard core; it does not close the row/column theorem. □

Theorem 7.69 (Phi-LPF top-row perfect-tiling reduction)**.** *Let*

$$T_P = \{P^2 - P + a : 1 \leq a < P\}$$

and, for primes $q < P$,

$$B_q(P) = \Phi\left(\left\lfloor \frac{P^2 - 1}{q} \right\rfloor, q\right) - \Phi\left(\left\lfloor \frac{P^2 - P}{q} \right\rfloor, q\right).$$

Then

$$N_{\text{top}}(P) = P - 1 - \sum_{q<P} B_q(P).$$

Consequently

$$N_{\text{top}}(P) = 0 \quad \Longleftrightarrow \quad \sum_{q<P} B_q(P) = P - 1,$$

and

$$N_{\text{top}}(P) \geq 1 \quad \Longleftrightarrow \quad \sum_{q<P} B_q(P) \leq P - 2.$$

Thus the exact Phi-LPF endpoint difference reduces top-row positivity to excluding a perfect LPF tiling of all $P - 1$ top-row slots by composite buckets.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-top-row-
perfect-tiling-router.md.
```

It records

```
perfect_tiling_excluded_globally=false
row_column_unconditional_closed=false.
```

The displayed equivalences are immediate from the top-row Phi-LPF formula. Writing $\Delta_j(A,B) = \Phi(B,p_j) - \Phi(A,p_j)$, the recursion gives

$$\Delta_j(A,B) = \Delta_{j+1}(A,B) + \Phi\left(\left\lfloor \frac{B}{p_j} \right\rfloor, p_j\right) - \Phi\left(\left\lfloor \frac{A}{p_j} \right\rfloor, p_j\right),$$

so the endpoint recursion only decomposes the same composite load into deeper rough-factor branches. It does not by itself produce an unfilled prime slot. The Sylvester–Schur consecutive-product input also stops at a named payment interface: the $P-1$ consecutive top-row slots have a prime factor $r > P-1$, and no top-row slot is divisible by P, hence $r > P$. If the top row has no prime slot, that large factor is carried by some cofactor $2 \le m < P$, which is a low-carrier high-prime payment rather than a contradiction. The current corpus has not globally excluded the perfect-tiling case. □

Theorem 7.70 (Phi-LPF top-row high-prime payment split)**.** *For the top row $P^2 - P < n < P^2$, define*

$$H_1(P) = \pi(P^2-1) - \pi(P^2-P),$$

and, for $2 \le m < P$,

$$H_m(P) = \pi\left(\left\lfloor \frac{P^2-1}{m} \right\rfloor\right) - \pi\left(\left\lfloor \frac{P^2-P}{m} \right\rfloor\right).$$

Let $S_P(P)$ be the number of top-row composite slots all of whose prime factors are at most P. Then

$$P - 1 = H_1(P) + \sum_{2 \le m < P} H_m(P) + S_P(P).$$

The target top-row positivity is exactly $H_1(P) \ge 1$. The terms $H_m(P)$ with $2 \le m < P$ are low-carrier high-prime payments, not prime slots.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-top-row-
high-prime-payment-split-router.md.
```

It records

```
low_carrier_payment_capacity_exceeded=false
row_column_unconditional_closed=false.
```

Every top-row slot either is prime, has a unique prime factor $r > P$ with cofactor $m = n/r < P$, or has all prime factors at most P. These cases are disjoint and give the displayed identity. The Sylvester–Schur input, under the contrary assumption $H_1(P) = 0$, only forces

$$\sum_{2 \le m < P} H_m(P) \ge 1.$$

It does not force $H_1(P) \ge 1$. A closure proof must therefore show a capacity deficit

$$\sum_{2 \le m < P} H_m(P) + S_P(P) \le P - 2,$$

or return to the previously named positive-rejection/prime-collar input. The current corpus has not proved this capacity deficit globally. □

Theorem 7.71 (Phi-LPF strict-row high-prime payment support). *Fix an odd prime P and $1 < k < P$. For the row*

$$I_{k,P} = \{kP + a : 1 \le a < P\},$$

if $n = mr \in I_{k,P}$ has a prime factor $r > P$, then

$$m = \frac{n}{r} < \frac{(k+1)P}{P} = k + 1.$$

Thus either $m = 1$, which is the prime-slot case, or $2 \le m \le k$. Hence

$$P - 1 = H_1(k,P) + \sum_{2 \le m \le k} H_m(k,P) + S_k(P),$$

where

$$H_1(k,P) = \pi((k+1)P - 1) - \pi(kP),$$
$$H_m(k,P) = \pi\left(\left\lfloor \frac{(k+1)P-1}{m} \right\rfloor\right) - \pi\left(\left\lfloor \frac{kP}{m} \right\rfloor\right) \qquad (2 \le m \le k),$$

and $S_k(P)$ counts row slots all of whose prime factors are at most P.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-row-
high-prime-payment-support-router.md.
```

It records

```
general_capacity_deficit_proved=false
row_column_unconditional_closed=false.
```

The support bound follows from $n < (k+1)P$. Since $n < P^2$, a row slot has at most one prime factor $> P$. Therefore the cases $m = 1$, $2 \le m \le k$, and no factor $> P$ are disjoint and exhaustive. The target positivity is $H_1(k,P) \ge 1$, so this route still requires

$$\sum_{2 \le m \le k} H_m(k,P) + S_k(P) \le P - 2.$$

The finite audit finds no failure through $P \le 1009$, but that audit only checks the implementation scale. It does not prove the global strict-row capacity deficit. $\square$

Theorem 7.72 (Phi-LPF strict-row load-phase tradeoff). *For $1 < k < P$, define normalized row loads*

$$\lambda(k,P) = \frac{\sum_{2 \le m \le k} H_m(k,P)}{P-1}, \qquad \sigma(k,P) = \frac{S_k(P)}{P-1}, \qquad \eta(k,P) = \frac{H_1(k,P)}{P-1}.$$

Then

$$\lambda(k,P) + \sigma(k,P) + \eta(k,P) = 1.$$

Moreover

$$H_1(k,P) \ge 1 \quad \Longleftrightarrow \quad \lambda(k,P) + \sigma(k,P) \le 1 - \frac{1}{P-1}.$$

Thus the strict-row Phi-LPF positivity problem is equivalent, in these variables, to excluding co-saturation of the low-carrier payment load λ and the P-smooth composite load σ.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-row-
load-phase-tradeoff-router.md.
```

It records

```
anti_cosaturation_inequality_proved=false
row_column_unconditional_closed=false.
```

The identity is the normalized form of the preceding high-prime payment support split. The equivalence follows because $H_1(k,P)$ is an integer. The finite audit through $P \le 1009$ shows a phase separation: the largest payment load in the sweep occurs near the top phase, while the largest smooth load occurs at very small k/P. It also finds no row with both loads at least 0.70. These are finite diagnostics only. The present corpus does not prove the global anti-co-saturation inequality. □

Theorem 7.73 (Phi-LPF strict-k Dusart interval bridge). *Assume the external Dusart 2010 interval input*

$$x \ge 396738 \quad \Longrightarrow \quad [x, x + x/(25\log^2 x)] \text{ contains a prime.}$$

For a strict row $1 < k < P$, put $x = kP$. If

$$kP \ge 396738 \qquad \text{and} \qquad k \le 25\log^2(kP),$$

then $H_1(k,P) = \pi((k+1)P-1) - \pi(kP) \ge 1$. The rows with $kP < 396738$ are separately covered by the finite bridge certificate. Consequently Dusart closes the finite-threshold and low-k logarithmic band, but it does not close all strict rows $1 < k < P$.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-dusart-
interval-bridge-router.md.
```

It records

```
dusart_covers_all_strict_rows=false
row_column_unconditional_closed=false.
```

The only external analytic input is Dusart's `arXiv:1002.0442` interval theorem, not reproved here. If the displayed inequality holds, then $x/(25\log^2 x) \le P$, so the Dusart prime lies in $[kP,(k+1)P]$. Both endpoints are composite, since $1 < k < P$; hence the prime lies in the open row interval and contributes to $H_1(k,P)$. The finite bridge verifies 257198 rows below $kP = 396738$. A sample audit through $P \le 10007$ closes about 0.968923 of strict rows by the finite bridge, the direct Dusart π-endpoint estimate, or the Dusart interval condition, but the first uncovered sampled row is $P = 8101, k = 8100$. Near $k \sim P$, the Dusart length has scale $P^2/\log^2(P^2)$, which can exceed the row length P. Thus the remaining global input is still a square-root-scale gap theorem or an internal anti-co-saturation/rejection-excess inequality. □

Theorem 7.74 (Phi-LPF strict-k low-carrier lower-half source cut). *In the strict row $1 < k < P$, every low-carrier high-prime payment slot is sourced from the lower half of the earlier P-rows. More precisely, if*

$$kP < n = mr < (k+1)P, \qquad 2 \le m \le k, \qquad r > P \text{ prime},$$

then

$$\frac{kP}{m} < r \le \frac{(k+1)P-1}{m} \quad \text{and hence} \quad r < \frac{(k+1)P}{2} < kP.$$

Consequently, writing $j = \lfloor r/P \rfloor$*, one has*

$$1 \le j \le \lfloor k/2 \rfloor.$$

For a fixed target row (P, k)*, the same source prime* $r > P$ *can pay at most one slot, because two different carriers would move by at least* $r > P$*, which is larger than the row width* $P - 1$*. Thus a zero row would have to be a perfect tiling by an acyclic lower-half prime-source injection together with the* P*-smooth slots.*

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-low-carrier-
lower-half-source-cut-router.md.
```

It records

```
all_payment_sources_in_lower_half=true
lower_half_payment_smooth_anti_tiling_proved=false
row_column_unconditional_closed=false.
```

The displayed source interval follows by dividing the row inequalities by m. Since $m \ge 2$ and $k > 1$, we have $((k+1)P)/2 < kP$. The source row bound follows immediately. If a single source prime r admitted two carriers $m_1 < m_2$ into the same target row, then $(m_2 - m_1)r \ge r > P$, while two numbers in the row differ by at most $P - 1$, a contradiction. The finite audit through $P \le 1009$ verifies the same bound on 76797 strict rows and finds maximum source ratio 0.5. This removes a payment source loop, but the remaining anti-tiling inequality between the lower-half injection image and the P-smooth slots is not yet proved. $\square$

Theorem 7.75 (Phi-LPF strict-k payment Beatty source map)**.** *In the strict row* $1 < k < P$*, the low-carrier high-prime payment relation is a unique near-multiple source map. A source prime* $r > P$ *can pay the target row only in the range*

$$P < r \le \left\lfloor \frac{(k+1)P - 1}{2} \right\rfloor.$$

For such an r*, the only possible carrier is*

$$m = \left\lfloor \frac{kP}{r} \right\rfloor + 1, \qquad a = mr - kP = r - (kP \bmod r).$$

It pays a row slot if and only if $1 \le a < P$*. Consequently*

$$\sum_{2 \le m \le k} H_m(k, P) = \#\{\, r > P : \ r \text{ prime}, \ r - (kP \bmod r) \in [1, P-1] \,\}.$$

Thus a zero strict row would have to be a perfect tiling by the image of this Beatty near-multiple prime-source map together with the P*-smooth slots.*

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-payment-
beatty-source-map-router.md.
```

It records

```
all_hm_counts_match_beatty_source_map=true
beatty_smooth_anti_tiling_proved=false
row_column_unconditional_closed=false.
```

The carrier uniqueness is the same row-width argument as in the lower-half source cut: for fixed $r > P$, two distinct carriers differ by at least $r > P$, while the open row has width $P-1$. Therefore any payment carrier must be the first integer above kP/r, namely $\lfloor kP/r \rfloor + 1$. The displayed residue criterion is just the condition $kP < mr < (k+1)P$ written as $1 \le mr - kP < P$. The finite audit through $P \le 257$ checks 6227 strict rows and verifies the equality between the H_m-payment count and the Beatty source count; larger sample rows $P = 571, k = 438$ and $P = 1009, k = 1008$ give matching counts 382 and 670. This removes payment source-table freedom, but it does not prove the remaining Beatty/smooth anti-tiling inequality. □

Theorem 7.76 (Phi-LPF strict-k endpoint Beatty–smooth exact value)**.** *For $1 < k < P$, let*

$$N_P(k) = \pi((k+1)P-1) - \pi(kP).$$

Then $N_P(k)$ is the number of primes in the closed interval $[kP, kP+P]$, because the two endpoints kP and $(k+1)P$ are composite. The Phi-LPF endpoint difference and the Beatty payment source map give the same exact value:

$$N_P(k) = (P-1) - \sum_{p \le \sqrt{(k+1)P-1}} \left[\Phi\left(\left\lfloor \frac{(k+1)P-1}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{kP}{p} \right\rfloor, p \right) \right]$$

and

$$N_P(k) = (P-1) - B_P(k) - S_P(k),$$

where $B_P(k)$ is the Beatty near-multiple high-prime payment count and $S_P(k)$ is the number of P-smooth composite slots in the row. Hence

$$N_P(k) \ge 1 \quad \Longleftrightarrow \quad B_P(k) + S_P(k) \le P-2.$$

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-endpoint-
beatty-smooth-value-router.md.
```

It records

```
all_endpoint_beatty_smooth_value_identities_hold=true
all_rows_positive_in_finite_sweep=true
minimum_prime_count=1
finite_evidence_not_used_as_global_proof=true.
```

The first displayed formula is the LPF bucket endpoint difference on the internal row $\{kP + a : 1 \le a < P\}$. Because all row entries are below P^2, every composite row entry either has all prime factors at most P, contributing to $S_P(k)$, or has a unique high-prime factor $r > P$. The preceding Beatty source-map theorem identifies the latter slots exactly with $B_P(k)$. Thus the composite bucket delta equals $B_P(k) + S_P(k)$, giving the exact value. The finite audit through $P \le 257$ verifies these identities on 6227 strict rows; its closest sampled fill is $P = 59, k = 42$, where $N_P(k) = 3$, $B_P(k) = 40$, and $S_P(k) = 15$. This is an exact coordinate reduction, not a proof of the remaining global Beatty/smooth anti-tiling inequality. □

Theorem 7.77 (Phi-LPF strict-k Beatty Euclidean source-window law)**.** *The Beatty payment term $B_P(k)$ admits an equivalent Euclidean quotient source-window expansion. Fix $1 < k < P$ and a carrier $2 \le m \le k$. Write*

$$j = \left\lfloor \frac{k}{m} \right\rfloor, \qquad k = mj + t, \qquad 0 \le t < m,$$

and write a source prime as $r = jP + b$. Then

$$kP < mr < (k+1)P \iff tP < mb < (t+1)P,$$

or equivalently

$$\left\lfloor \frac{tP}{m} \right\rfloor + 1 \le b \le \left\lfloor \frac{(t+1)P-1}{m} \right\rfloor.$$

Therefore

$$B_P(k) = \sum_{2 \le m \le k} \#\{\, r = jP + b : \ r \text{ prime}, \ b \text{ lies in the above window} \,\}.$$

In particular $j \le \lfloor k/2 \rfloor$, so the earlier lower-half source cut is a consequence of this quotient-source law.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-beatty-
euclidean-source-window-router.md.
```

It records

```
all_hm_counts_match_quotient_source_windows=true
all_source_rows_in_lower_half=true
finite_evidence_not_used_as_global_proof=true.
```

The equivalence follows by substituting $k = mj + t$ and $r = jP + b$:

$$mr = mjP + mb = (k-t)P + mb,$$

so $kP < mr < (k+1)P$ is exactly $tP < mb < (t+1)P$. Integer b then lies in the displayed window. Since $m \ge 2$, $j = \lfloor k/m \rfloor \le \lfloor k/2 \rfloor$. The finite audit through $P \le 1009$ verifies that summing these source-window prime counts reproduces the H_m-payment count on 76797 strict rows. This refines the payment source table; it does not prove the remaining source-window/P-smooth anti-tiling inequality. ☐

Theorem 7.78 (Phi-LPF strict-k smooth owner quotient-window law)**.** *Fix a prime P and $1 < k < P$. Let $S_P(k)$ denote the number of composite entries $n \in \{kP + a : 1 \le a < P\}$ whose prime factors are all at most P. Then*

$$S_P(k) = \sum_{p \le \sqrt{(k+1)P-1}} \# \left\{ q : \left\lfloor \frac{kP}{p} \right\rfloor < q \le \left\lfloor \frac{(k+1)P-1}{p} \right\rfloor, \mathrm{lpf}(q) \ge p, \mathrm{gpf}(q) \le P \right\},$$

where lpf *and* gpf *are respectively the least and greatest prime factors, with the condition $q \ge p$ implicit in $\mathrm{lpf}(q) \ge p$. Consequently the strict row value is the exact dual-window difference*

$$N_P(k) = (P-1) - B_P(k) - S_P(k), \qquad N_P(k) \ge 1 \iff B_P(k) + S_P(k) \le P - 2.$$

Status and reference. Every P-smooth composite row entry n has a unique least prime factor p, and then $n = pq$. The row condition is exactly the displayed quotient window for q. Minimality of p gives $\mathrm{lpf}(q) \ge p$, while P-smoothness gives $\mathrm{gpf}(q) \le P$. Conversely, any pair (p, q) counted by the displayed window produces a row entry pq whose least prime factor is p and whose greatest prime factor is at most P. The owner p is unique, so the sum has no double counting. Combining this with the preceding high-prime Beatty source-window theorem gives the dual-window value.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-smooth-owner-
quotient-window-router.md.
```

It records

```
all_dual_quotient_window_identities_hold=true
strict_row_count=6227
minimum_prime_count=1
finite_evidence_not_used_as_global_proof=true.
```

The finite audit through $P \le 257$ verifies the direct slot split, the high-prime Beatty windows, the smooth owner windows, and the dual formula on 6227 strict rows. This closes the smooth source-domain accounting; it does not prove the remaining dual-window anti-tiling inequality. $\square$

Theorem 7.79 (External short-interval bridge for strict k). *Let $x = kP$ with $1 < k < P$. Any external theorem of the form "there is a prime in $[x, x + H(x)]$ for $x \ge X_0$" proves positivity of the strict row $[kP, kP + P]$ precisely on the subrange where*

$$kP \ge X_0, \qquad H(kP) \le P = \frac{x}{k}.$$

For Dusart's explicit 2010 interval $H(x) = x/(25 \log^2 x)$, this gives only the logarithmic carrier band

$$k \le 25 \log^2(kP)$$

above the explicit threshold. For the Baker–Harman–Pintz exponent $H(x) = x^{0.525}$, this gives only

$$k \le x^{0.475}.$$

Since the strict condition $k < P$ allows k to approach $x^{1/2}$, these known unconditional gap inputs do not close all strict rows.

Status and reference. The bridge criterion is immediate: a prime in $[x, x + H(x)]$ lies in the row interval once $H(x) \le P$. Substituting $x = kP$ gives the displayed conditions. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-external-
gap-bridge-router.md.
```

It records that Dusart 2010 covers the explicit logarithmic k-band, while the Baker–Harman–Pintz 0.525 gap exponent covers only the $k \le x^{0.475}$ exponent band. Boundary samples with $k = P - 1$ already show that these bands do not reach the $k \approx P \approx \sqrt{x}$ edge. Thus the remaining non-circular input is either a genuine square-root-scale interval theorem, or an internal proof of the dual-window anti-tiling inequality

$$B_P(k) + S_P(k) \le P - 2.$$

$\square$

Theorem 7.80 (Unified positive core for strict k). *For a prime P and $1 < k < P$, the following four quantities are the same integer:*

$$\begin{aligned} N_P(k) &= \pi((k+1)P - 1) - \pi(kP) \\ &= (P-1) - B_P(k) - S_P(k) \\ &= (P-1) - \text{OwnerMass} \\ &= \text{RejectionMass} - (\text{RawTotal} - (P-1)). \end{aligned}$$

Moreover

$$N_P(k) \geq 1 \quad \Longleftrightarrow \quad \text{next\_prime}(kP) < (k+1)P.$$

Thus the dual-window anti-tiling inequality, the raw/rejection strict-excess statement, the owner-saturation defect, and the aligned square-root gap exclusion are not independent hard points; they are four coordinate systems for the same strict-row positive core.

Status and reference. The first equality is the definition of the strict row prime count. The second is the Phi-LPF endpoint difference after splitting composite slots into high-prime Beatty windows and P-smooth owner windows. The third is the LPF owner saturation identity: every composite row slot has exactly one least-prime owner and every prime row slot has none. The fourth follows by partitioning raw root-prime incidences into the owner incidence and rejected non-owner incidences. Finally, the row endpoints kP and $(k+1)P$ are composite because $1 < k < P$, so $N_P(k) \geq 1$ is exactly the assertion that the next prime after kP arrives before $(k+1)P$.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-unified-
positive-core-router.md.
```

It records

```
all_four_coordinate_identities_hold=true
strict_row_count=6228
minimum_positive_core_value=1
finite_evidence_not_used_as_global_proof=true.
```

This theorem does not prove the core positive for all P, k. It removes a false branching of the frontier: proving any of the named remaining statements amounts to proving the same integer $N_P(k)$ is positive. □

Theorem 7.81 (Top row and prime-indexed Oppermann left half)**.** *The least-slack strict row $k = P-1$ is exactly the left Oppermann half-window at the prime base $n = P$:*

$$N_{\text{top}}(P) = \pi(P^2 - 1) - \pi(P^2 - P).$$

Thus top-row positivity is the assertion that $(P^2 - P, P^2)$ contains a prime. This is stronger than merely applying Legendre's interval $((P-1)^2, P^2)$, because a Legendre prime may lie in the lower subinterval

$$((P-1)^2, P^2 - P].$$

Consequently neither Legendre-type finite verification nor the usual X^θ, $\theta > 1/2$, prime-gap inputs close the top row. The correct named subcore is

$$\pi(P^2 - 1) - \pi(P^2 - P) \geq 1 \qquad (P \text{ prime}),$$

or, equivalently, exclusion of a square-phase low-sieve full cover of all $r = 1, \ldots, P-1$ in $P^2 - r$.

Status and reference. Substituting $k = P - 1$ in the strict-row formula gives the displayed count. The interval $(P^2 - P, P^2)$ is exactly $(n^2 - n, n^2)$ at $n = P$. Since $(P-1)^2 = P^2 - 2P + 1 < P^2 - P$, the Legendre interval is strictly larger on the left and does not force a prime in the upper half. Known generic prime-gap exponents remain longer than P after setting $X = P^2$, so they also do not close this half-window.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-top-row-
oppermann-alignment-router.md.
```

It records

```
all_prime_bases_left_half_positive=true
all_prime_bases_right_half_positive=true
prime_base_count=669
minimum_left_oppermann_count=1
finite_evidence_not_used_as_global_proof=true.
```

This finite audit is only structural evidence. The theorem identifies the true top-row subcore; it does not prove Oppermann's half-window or the row/column theorem unconditionally. □

Theorem 7.82 (Finite sqrt-gap bridge and square-phase tail)**.** *Let P be prime and $1 < k < P$, and put $x = kP$. Since $\sqrt{x} < P$, any theorem guaranteeing a prime in $[x, x + \sqrt{x}]$ gives*

$$N_P(k) = \pi((k+1)P - 1) - \pi(kP) \geq 1.$$

Using the finite sqrt-gap input

$$117 \leq x \leq 10^{18} \quad \Longrightarrow \quad [x, x + \sqrt{x}] \text{ contains a prime},$$

the strict rows with $kP \leq 10^{18}$ are therefore closed, after a direct finite check of the rows with $kP < 117$. For the top row $x = P^2 - P$, this finite bridge reaches every prime base

$$P \leq 999999937.$$

Beyond this range the same top-row failure is exactly the square-phase low sieve full cover

$$N_{\text{top}}(P) = 0 \Longleftrightarrow \forall\, 1 \leq r < P, \quad \gcd\left(P^2 - r, \prod_{q<P} q\right) > 1,$$

equivalently the interval $1 \leq r < P$ is covered by the residue classes

$$r \equiv P^2 \pmod{q}, \qquad q < P,\ q \text{ prime}.$$

Status and reference. The bridge inequality $\sqrt{kP} < P$ is immediate from $k < P$. Hence a prime in $[x, x + \sqrt{x}]$ lies before $(k+1)P$; because the endpoints are composite in the strict range, this is exactly positivity of the Phi–LPF endpoint difference. The finite sqrt-gap input is Lemma 2.7 of Erdos–Harcos–Kharel–Maga–Mezei–Toroczkai, based on the computed prime-gap tables of Oliveira e Silva–Herzog–Pardi. The direct subthreshold audit checks 49 strict rows and finds minimum prime count 1. Solving $P^2 - P \leq 10^{18}$ gives integer bound $P \leq 10^9$, and the largest prime base not exceeding that bound is 999999937.

The last equivalence is the same least-prime-factor ownership identity applied to the top row $P^2 - P < n < P^2$. If no top-row entry is prime, then each $P^2 - r$ has a least prime factor $q < P$, because $P^2 - r < P^2$. Conversely, if every $P^2 - r$ is divisible by some $q < P$, no top-row entry is prime. For each fixed q, this divisibility is exactly $r \equiv P^2 \pmod{q}$. Thus the remaining infinite tail is not a finite verification problem; it is the named square-phase special long-cover problem.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-finite-
sqrt-square-phase-tail-router.md.
```

It records

```
external_range=[117,1000000000000000000]
small_checked_row_count=49
all_small_rows_positive=true
largest_prime_base_with_top_row_start_le_x_max=999999937.
```

This theorem closes only the declared finite segment. The row/column theorem still requires `SquarePhaseTailLongBlockPDECExclusion`, a uniform rough-survivor lower bound, or a genuine global square-root-scale short-interval theorem. □

Theorem 7.83 (Top-row half-rough semiprime shadow). *For a prime P, define*

$$R_{1/2}(P) = \#\left\{1 \le r < P : \gcd\left(P^2 - r, \prod_{q \le P/2} q\right) = 1\right\},$$

where q runs over primes, and define the reciprocal semiprime shadow

$$T_{1/2}(P) = \sum_{\substack{P/2<q<P \\ q \text{ prime}}} \#\left\{m \text{ prime} : \left\lfloor \frac{P^2 - P}{q} \right\rfloor < m \le \left\lfloor \frac{P^2 - 1}{q} \right\rfloor\right\}.$$

Then

$$\pi(P^2 - 1) - \pi(P^2 - P) = R_{1/2}(P) - T_{1/2}(P).$$

Consequently the prime-indexed Oppermann-left top row is positive exactly when the half-rough survivor count strictly exceeds this reciprocal semiprime shadow.

Status and reference. The half-rough survivor set consists of the top-row entries $P^2 - r$ not divisible by any prime $q \le P/2$. Such a survivor is either prime or composite. If it is composite, its least prime factor q must satisfy $P/2 < q < P$, since $P^2 - r < P^2$. Write $P^2 - r = qm$. The top-row inequality $P^2 - P < P^2 - r < P^2$ gives

$$\frac{P^2 - P}{q} < m < \frac{P^2}{q},$$

and because $q < P$ and $q > P/2$, this puts m in $(P, 2P)$. For $P \ge 11$, m cannot be composite: all prime factors of m would be at least $q > P/2$, forcing $m \ge q^2 > 2P$. The small prime bases are checked directly in the finite audit. Thus every composite half-rough survivor is counted exactly once by the displayed $T_{1/2}$, and every term of $T_{1/2}$ gives a composite top-row survivor. Subtracting the shadow from the half-rough survivor set leaves precisely the prime top-row entries.

For each fixed $q \in (P/2, P)$, the admissible m-window has length $P/q < 2$, so the shadow is a sparse reciprocal prime-pair graph, not a full LPF tree. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-top-row-half-
rough-semiprime-shadow-router.md.
```

It records

```
max_prime=5003
all_half_rough_identities_hold=true
minimum_half_rough_minus_shadow=1
P=5003: R_1/2=330, T_1/2=49, direct_top_primes=281.
```

This theorem is an exact reduction. It does not prove the uniform inequality $R_{1/2}(P) > T_{1/2}(P)$ in the infinite tail, but it makes the next hard point the named `HalfRoughSurvivorExcessOverReciprocalSemiprimeS`
□

Theorem 7.84 (Strict-k external bulk and square-band partition). *For $x = kP$ with $1 < k < P$, the Phi–LPF endpoint difference gives the exact count*

$$N_P(k) = \pi((k+1)P - 1) - \pi(kP).$$

The finite square-root input closes the initial segment $kP \le 10^{18}$. Dusart's explicit interval closes only the logarithmic subband

$$k \le 25 \log^2(kP),$$

and the Baker–Harman–Pintz $x^{21/40}$ input, when its asymptotic threshold is available, closes the bulk subband

$$P \ge (kP)^{21/40} \iff k \le P^{19/21}.$$

Therefore any strict-k counterexample not already handled by these external arms must lie in the high square-edge band

$$kP > 10^{18}, \qquad P^{19/21} < k < P.$$

The top row $k = P - 1$ has the sharper half-rough semiprime-shadow interface $R_{1/2}(P) > T_{1/2}(P)$.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-external-bulk-
square-band-partition-router.md.
```

It records

```
status=strict_k_external_bulk_covered_remaining_forced_to_high_k_square_band
bhp_strict_k_exponent=19/21
finite_sqrt_initial_segment=kP<=10^18
remaining_high_k_band=P^(19/21)<k<P.
```

This is a partition theorem, not a closure theorem. It shows that the present external-route residue is no longer all strict rows, but the named high-k square band; closing that band still requires a genuine square-root-scale short interval theorem, a structural Phi–LPF excess, or a semiprime-shadow/PDEC exclusion. □

Theorem 7.85 (Strict-k half-rough shadow band split). *Let P be prime and $1 < k < P$. Define*

$$R_{1/2}(P,k) = \#\left\{1 \le t < P : \gcd\left(kP + t, \prod_{q \le P/2} q\right) = 1\right\},$$

where q runs over primes, and

$$T_{1/2}(P,k) = \#\left\{(q,m) : q, m \text{ prime},\ P/2 < q \le m < 2P,\ kP < qm < (k+1)P\right\}.$$

Then

$$\pi((k+1)P - 1) - \pi(kP) = R_{1/2}(P,k) - T_{1/2}(P,k).$$

Moreover, if $4((k+1)P - 1) \le P^2$, then $T_{1/2}(P,k) = 0$.

Status and reference. Any composite survivor counted by $R_{1/2}(P,k)$ has least prime factor $q > P/2$. Since the number is below P^2, this q is also below P. Writing $kP + t = qm$, one has $m < 2P$. For $P \ge 11$, m cannot be composite, because all its prime factors would be at least $q > P/2$, forcing $m \ge q^2 > P^2/4 > 2P$. Taking $q \le m$ gives a unique two-prime shadow representative for every composite half-rough survivor, and every such shadow is composite and survives the $q \le P/2$ deletion. Subtraction leaves exactly the primes in the row. If $4((k+1)P - 1) \le P^2$, the whole row lies at or below $P^2/4$, while every product of two primes $> P/2$ is strictly larger than $P^2/4$, so the shadow is empty.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-strict-k-half-rough-
shadow-band-split-router.md.
```

It records

```
max_prime=1009
all_half_rough_shadow_identities_hold=true
identity_failure_count=0
next=HalfRoughSurvivorExistenceInShadowFreeHighBand
     OR HighKHalfRoughSurvivorExcessOverTwoPrimeShadow.
```

This is still a reduction, not a proof of the final positive core. $\square$

Theorem 7.86 (Shadow-free half-primorial special phase)**.** *Let*

$$M_{1/2}(P) = \prod_{\substack{q \le P/2 \\ q \text{ prime}}} q.$$

In the shadow-free subband $4((k+1)P-1) \le P^2$, the strict row is positive if and only if

$$\exists\, 1 \le t < P \quad \gcd(kP+t, M_{1/2}(P)) = 1.$$

Equivalently, a failure in this subband is exactly the assertion that the special phase $kP+1 \bmod M_{1/2}(P)$ starts a covered block of length $P-1$ in the half-primorial period.

Status and reference. The previous theorem gives $T_{1/2}(P,k) = 0$ in the stated subband. Thus a row prime is the same object as a half-rough survivor after deleting all primes $q \le P/2$, which is precisely a residue coprime to $M_{1/2}(P)$. Since the whole row lies below $P^2/4$, any such survivor is automatically prime: a composite with no prime factor $\le P/2$ would be a product of at least two factors $> P/2$, hence would exceed $P^2/4$. The residue formulation follows by reducing the interval $kP+1, \ldots, kP+P-1$ modulo $M_{1/2}(P)$.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-shadow-free-half-
primorial-phase-router.md.
```

It records

```
status=shadow_free_lane_reduced_to_half_primorial_special_phase_avoidance
all_sampled_rows_have_survivor=true
all_scanned_half_primorial_periods_have_max_run_less_than_P_minus_1=true
next=HalfPrimorialSpecialPhaseAvoidsLongCoveredBlockOrPDEC.
```

The finite scans are evidence only. Global closure still needs either a half-primorial Jacobsthal bound at this scale, or a special-phase avoidance/PDEC argument. $\square$

Theorem 7.87 (Upper-band reciprocal two-prime shadow)**.** *Let P be prime and $1 < k < P$. In the upper square band*

$$4((k+1)P-1) > P^2,$$

the two-prime shadow from the strict-k half-rough split has the exact reciprocal-window form

$$T_{1/2}(P,k) = \sum_{\substack{P/2<q<P \\ q \text{ prime}}} \#\{m \in I_q(P,k) : m \text{ prime}\},$$

where

$$I_q(P,k) = \left[\max\left(q, \left\lfloor \frac{kP}{q} \right\rfloor + 1\right), \left\lfloor \frac{(k+1)P-1}{q} \right\rfloor\right] \cap \mathbb{Z}.$$

Moreover $|I_q(P,k)| \le 2$ *for every such* q. *Hence an upper-band row failure is exactly a sparse reciprocal prime-pair saturation of the half-rough survivors, or a named half-rough-floor / reciprocal-shadow-ceiling defect.*

Status and reference. The previous half-rough split counts composite survivors by products qm with $P/2 < q \le m < 2P$ and

$$kP < qm < (k+1)P.$$

For a fixed q, this inequality is equivalent to

$$\left\lfloor \frac{kP}{q} \right\rfloor + 1 \le m \le \left\lfloor \frac{(k+1)P-1}{q} \right\rfloor.$$

The unique representative convention $q \le m$ supplies the additional lower cutoff $m \ge q$, giving the displayed $I_q(P,k)$. Its real length is strictly less than $P/q < 2$, so it contains at most two integers. Filtering those at-most-two candidates by primality gives the actual two-prime shadow.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-upper-band-two-
prime-shadow-excess-router.md.
```

It records

```
status=upper_band_two_prime_shadow_reduced_to_sparse_reciprocal_prime_pair_windows
max_prime=1009
all_reciprocal_window_identities_hold=true
identity_failure_count=0
all_sampled_upper_rows_have_positive_margin=true.
```

This theorem is still a reduction. It does not prove the required uniform inequality $R_{1/2}(P,k) > T_{1/2}(P,k)$; the remaining named interface is

```
UpperBandReciprocalPrimePairShadowSaturationOrPDEC
AND UpperBandHalfRoughFloorOrReciprocalPrimePairCeiling.
```

□

Theorem 7.88 (Upper-band reciprocal shadow graph forest)**.** *In the setting of the preceding theorem, form the bipartite graph whose left vertices are primes* $q \in (P/2, P)$*, whose right vertices are prime candidates* $m \in [q, 2P)$*, and whose edges are the pairs* (q,m) *with*

$$kP < qm < (k+1)P.$$

This reciprocal shadow graph has degree at most two on each side and is ordered non-crossing. In particular it is a forest. Therefore an upper-band counterexample cannot be produced by a cyclic shadow feedback; it must be a forest-shadow saturation of all half-rough survivors, or a named PDEC.

Status and reference. The degree bound is the reciprocal-window bound in both directions. For fixed q, the admissible m-interval has length $< P/q < 2$. For fixed m, the admissible q-interval has length $< P/m \le P/q < 2$ because $m \ge q$. Thus both sides have degree at most two.

If two edges crossed in the ordered sense, say $q_1 < q_2$ but $m_1 < m_2$, then

$$q_2 m_2 - q_1 m_1 = (q_2 - q_1)m_2 + q_1(m_2 - m_1) > P,$$

since all involved primes exceed $P/2$ and distinct odd primes differ by at least 2. Two such products cannot both lie in the same interval of length P. Hence the graph is non-crossing. If a cycle existed, take the smallest q_0 on the cycle and let its two cycle-neighbours be $m_a < m_b$. Any edge from another $q > q_0$ to a vertex $m > m_a$ would cross the edge (q_0, m_a), so m_b cannot have the second incident edge required by the cycle. This contradiction proves that the graph is a forest.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-upper-band-
reciprocal-graph-structure-router.md.
```

It records

```
status=upper_band_reciprocal_shadow_graph_reduced_to_ordered_degree_two_forest
max_prime=1009
all_upper_rows_have_forest_shadow_graph=true
graph_structure_failure_count=0
all_sampled_graphs_are_forests=true.
```

This still does not prove row positivity. The remaining upper-band interface is

```
UpperBandForestShadowSaturationExclusionOrPDEC.
```

□

Theorem 7.89 (Punctured half-primorial forest phase). *Let*

$$M_{1/2}(P) = \prod_{\substack{q \le P/2 \\ q \text{ prime}}} q$$

and let $F(P,k)$ be the forest-hole set

$$F(P,k) = \{\, t : t = qm - kP,\ q, m \text{ prime},\ P/2 < q \le m < 2P,\ kP < qm < (k+1)P \,\}.$$

Then the prime slots in the row $kP+1, \dots, kP+P-1$ are exactly

$$\{\, 1 \le t < P : \gcd(kP+t, M_{1/2}(P)) = 1,\ t \notin F(P,k) \,\}.$$

Consequently the shadow-free lane is the zero-hole case $F(P,k) = \varnothing$, and an upper-band row failure is exactly a punctured half-primorial covered block:

$$\{\, 1 \le t < P : \gcd(kP+t, M_{1/2}(P)) = 1 \,\} \subseteq F(P,k).$$

Status and reference. If t is coprime to $M_{1/2}(P)$, then $kP+t$ is not divisible by any prime $q \le P/2$. If this number is composite, its least prime factor satisfies $P/2 < q < P$. Writing $kP + t = qm$, one has $m < 2P$, and for $P \ge 11$ the same least-prime-factor argument used above forces m to be prime. With the convention $q \le m$, this composite survivor is counted by exactly one element of $F(P,k)$. Conversely every element of $F(P,k)$ is a composite half-rough survivor. Removing $F(P,k)$ from the half-primorial survivors leaves precisely the prime row slots.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-punctured-half-
primorial-forest-phase-router.md.
```

It records

```
status=shadow_free_and_upper_band_reduced_to_punctured_half_primorial_forest_phase
max_prime=1009
all_punctured_phase_identities_hold=true
saturation_row_count=0
sample_saturation_row_count=0.
```

This unifies the two current strict-k residues but does not prove the needed global phase avoidance. The remaining interface is

```
PuncturedHalfPrimorialForestPhaseAvoidanceOrPDEC.
```

□

Theorem 7.90 (Punctured Phi–LPF endpoint difference). *Let $p_{1/2}(P)$ be the least prime $> P/2$, and define*

$$\Delta\Phi_{1/2}(P,k) = \Phi((k+1)P-1, p_{1/2}(P)) - \Phi(kP, p_{1/2}(P)).$$

Then for every prime P and $1 < k < P$,

$$\pi((k+1)P-1) - \pi(kP) = \Delta\Phi_{1/2}(P,k) - |F(P,k)|,$$

where $F(P,k)$ is the reciprocal forest-hole set of the previous theorem. Thus the remaining strict-k counterexample condition is the endpoint inequality

$$\Delta\Phi_{1/2}(P,k) \le |F(P,k)|.$$

Status and reference. By definition, $\Delta\Phi_{1/2}(P,k)$ counts exactly the entries in the row whose least prime factor is at least $p_{1/2}(P)$, equivalently those not divisible by any prime $q \le P/2$. The preceding theorem says that the composite entries among these are exactly the forest holes $F(P,k)$. Subtracting those holes leaves precisely the primes in the row. The shadow-free case is the subcase $|F(P,k)| = 0$; the upper-band case is the same identity with a nonempty reciprocal forest.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-punctured-
endpoint-difference-router.md.
```

It records

```
status=strict_k_prime_count_equals_phi_half_endpoint_difference_minus_forest_holes
max_prime=1009
all_endpoint_difference_identities_hold=true
endpoint_difference_failure_count=0
minimum_sample_endpoint_margin=4385.
```

This theorem is an exact endpoint-difference reformulation, not a proof of global positivity. The remaining interface is

```
PuncturedPhiEndpointDifferencePositiveOrPDEC.
```

□

Theorem 7.91 (Punctured endpoint parity-capacity bound). *For a prime P, $1 < k < P$, and a high prime $q \in (P/2, P)$, put*

$$I_q(P,k) = \left[\max\left(q, \left\lfloor \frac{kP}{q} \right\rfloor + 1\right), \left\lfloor \frac{(k+1)P-1}{q} \right\rfloor\right] \cap \mathbf{Z}.$$

Let
$$W_{\mathrm{int}}(P,k) = \sum_{P/2<q<P} |I_q(P,k)|$$
and let $E_{\mathrm{even}}(P,k)$ be the number of pairs (q,m), with q prime in $(P/2,P)$ and $m \in I_q(P,k)$, for which $m>2$ is even. Set
$$C_{\mathrm{par}}(P,k) = W_{\mathrm{int}}(P,k) - E_{\mathrm{even}}(P,k).$$
Then
$$|F(P,k)| \le C_{\mathrm{par}}(P,k).$$
Consequently
$$\Delta\Phi_{1/2}(P,k) > C_{\mathrm{par}}(P,k) \implies \pi((k+1)P-1) - \pi(kP) > 0.$$

Status and reference. The reciprocal-window theorem shows that every forest hole is represented by a pair (q,m) with q prime in $(P/2,P)$, $m \in I_q(P,k)$, and m prime. The first bound $|F(P,k)| \le W_{\mathrm{int}}(P,k)$ is therefore pure containment. If $m>2$ is even, it cannot be prime, hence each such candidate may be deleted from the prime-candidate ceiling. This gives $|F(P,k)| \le C_{\mathrm{par}}(P,k)$. Combining with the preceding exact endpoint identity gives the displayed positivity implication.

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-punctured-
endpoint-parity-capacity-router.md.
```

It records

```
status=forest_holes_bounded_by_reciprocal_integer_windows_minus_forced_even_candidates
max_prime=1009
row_count=76789
closed_by_parity_ceiling_count=76788
parity_not_closed_count=1
parity_failure_count=0
minimum_sample_delta_minus_parity_ceiling=1769.
```

The lone finite equality case in this audit is $P=19, k=15$, where the direct row has one prime. This is a capacity narrowing, not a global proof of $\Delta\Phi_{1/2} > C_{\mathrm{par}}$. The remaining interface is

```
PuncturedParityEndpointCapacityInequalityOr
ReciprocalPrimePairSaturationPDEC.
```

□

Theorem 7.92 (Punctured endpoint 6-wheel capacity bound)**.** *For a prime P, $1<k<P$, and the reciprocal windows $I_q(P,k)$ above, let $E_{2,3}(P,k)$ be the number of pairs (q,m) with*
$$P/2 < q < P, \qquad q \text{ prime}, \qquad m \in I_q(P,k),$$
such that m is forced composite by the 6-wheel, namely
$$(m>2,\ 2 \mid m) \quad \textit{or} \quad (m>3,\ 3 \mid m).$$
Set
$$C_6(P,k) = W_{\mathrm{int}}(P,k) - E_{2,3}(P,k).$$
Then
$$|F(P,k)| \le C_6(P,k).$$
Consequently
$$\Delta\Phi_{1/2}(P,k) > C_6(P,k) \implies \pi((k+1)P-1) - \pi(kP) > 0.$$

Status and reference. The proof is the same reciprocal-window containment as in the parity-capacity bound, with one additional Euler 6-wheel deletion. If $m > 2$ is even, or if $m > 3$ is divisible by 3, then m is not prime and cannot be the cofactor of a prime-pair forest hole. Deleting all such candidates from the integer windows gives $|F(P,k)| \le C_6(P,k)$. The endpoint positivity implication follows from

$$\pi((k+1)P - 1) - \pi(kP) = \Delta\Phi_{1/2}(P,k) - |F(P,k)|.$$

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-punctured-endpoint-
wheel6-capacity-router.md.
```

It records

```
max_prime=1009
row_count=76789
closed_by_parity_ceiling_count=76788
closed_by_wheel6_ceiling_count=76789
parity_not_closed_count=1
wheel6_not_closed_count=0.
```

The preceding parity-capacity audit had a single equality case $P = 19, k = 15$. There $C_{\text{par}} = 3$, while the 6-wheel deletes the additional candidate $m = 27$, giving $C_6 = 2$ and $\Delta\Phi_{1/2} - C_6 = 1$. This finite strict closure is not used as a global proof. The remaining interface is

```
PuncturedWheel6EndpointCapacityInequalityOr
ReciprocalPrimePairWheel6SaturationPDEC.
```

□

Theorem 7.93 (Punctured endpoint 30-wheel capacity bound)**.** *For a prime P, $1 < k < P$, and the reciprocal windows $I_q(P,k)$ above, let $E_{2,3,5}(P,k)$ be the number of pairs (q,m) with*

$$P/2 < q < P, \qquad q \text{ prime}, \qquad m \in I_q(P,k),$$

such that m is forced composite by the 30-wheel, namely

$$(m > 2,\ 2 \mid m), \qquad (m > 3,\ 3 \mid m), \qquad (m > 5,\ 5 \mid m).$$

Set

$$C_{30}(P,k) = W_{\text{int}}(P,k) - E_{2,3,5}(P,k).$$

Then

$$|F(P,k)| \le C_{30}(P,k).$$

Consequently

$$\Delta\Phi_{1/2}(P,k) > C_{30}(P,k) \quad \Longrightarrow \quad \pi((k+1)P - 1) - \pi(kP) > 0.$$

Status and reference. This is the same reciprocal-window containment as in the 6-wheel bound, with the next Euler-wheel deletion included. A forest hole requires the cofactor m to be prime. Hence every candidate $m > 5$ divisible by 5 is impossible, just as the preceding layer deleted even $m > 2$ and 3-divisible $m > 3$. Deleting all candidates forced composite by $2, 3$, or 5 gives $|F(P,k)| \le C_{30}(P,k)$. The endpoint positivity implication again follows from

$$\pi((k+1)P - 1) - \pi(kP) = \Delta\Phi_{1/2}(P,k) - |F(P,k)|.$$

The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-punctured-endpoint-
wheel30-capacity-router.md.
```

It records

```
max_prime=1009
row_count=76789
closed_by_wheel6_ceiling_count=76789
closed_by_wheel30_ceiling_count=76789
wheel30_not_closed_count=0
minimum_sample_delta_minus_wheel30_ceiling=3215.
```

For the representative top row $P = 1009, k = 1008$, the audit gives

$$\Delta\Phi_{1/2} = 89, \qquad W_{\mathrm{int}} = 101, \qquad C_6 = 34, \qquad C_{30} = 28.$$

Thus this layer genuinely tightens the unsigned cofactor ceiling. It remains a capacity narrowing, not a global proof of $\Delta\Phi_{1/2} > C_{30}$. The new interface is

```
PuncturedWheel30EndpointCapacityInequalityOr
ReciprocalPrimePairWheel30SaturationPDEC.
```

$\square$

Theorem 7.94 (Primorial-wheel limit is exact but not an independent proof)**.** *The* 30*-wheel capacity bound extends monotonically to every primorial wheel. If S is a finite set of primes and $C_S(P, k)$ denotes the reciprocal-window capacity after deleting all $m \in I_q(P, k)$ with*

$$m > \ell, \qquad \ell \mid m, \qquad \ell \in S,$$

then $S_1 \subseteq S_2$ implies

$$C_{S_2}(P, k) \leq C_{S_1}(P, k).$$

In particular the chain

$$2, \quad 2 \cdot 3, \quad 2 \cdot 3 \cdot 5, \quad 2 \cdot 3 \cdot 5 \cdot 7, \quad \ldots$$

gives a genuine sequence of sharper forest-hole ceilings.

However, once S contains all primes $\ell \leq \sqrt{2P - 1}$, the capacity is exact:

$$C_S(P, k) = |F(P, k)|.$$

Consequently

$$\Delta\Phi_{1/2}(P, k) - C_S(P, k) = \pi((k + 1)P - 1) - \pi(kP).$$

Thus the primorial-wheel limit is an exact restatement of the target row positivity, not an independent proof of it.

Status and reference. Monotonicity is immediate: enlarging S only deletes more cofactor candidates that are forced composite. For exactness, every reciprocal cofactor satisfies $m < 2P$. If such an m is composite, it has a prime factor $\ell \leq \sqrt{m} \leq \sqrt{2P - 1}$. Hence the dynamic wheel containing all primes up to $\sqrt{2P - 1}$ deletes every composite m, while a prime m is not deleted by the rule $m > \ell$, $\ell \mid m$. The survivors are therefore exactly the prime cofactors, counted with the same q-window multiplicity as $F(P, k)$.

Combining this with the endpoint identity

$$\pi((k + 1)P - 1) - \pi(kP) = \Delta\Phi_{1/2}(P, k) - |F(P, k)|$$

gives the displayed equality. The machine certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-primorial-wheel-limit-audit.md.
```

It records, for $P \leq 1009$,

```
row_count=76789
all_exact_capacity_equals_holes=true
all_exact_margin_equals_direct_prime_count=true.
```

For example, at $P = 1009, k = 1008$,

$$W_{\text{int}} = 101, \qquad C_{30} = 28, \qquad C_{210} = 27, \qquad C_{2310} = 26, \qquad C_{\sqrt{2P}} = 19 = |F|.$$

This narrows the bookkeeping interface to

```
PuncturedSqrtWheelExactForestHolePositivityOr
SignedDispersionOrSpecialSquarePhaseLowerBound.
```

No unconditional row/column closure follows from the wheel limit alone. □

Theorem 7.95 (Fixed-wheel rough-composite residual decomposition)**.** *Let S be a finite set of wheel primes and let $C_S(P,k)$ be the reciprocal-window capacity obtained by deleting every candidate $m \in I_q(P,k)$ for which*

$$m > \ell, \qquad \ell \mid m, \qquad \ell \in S.$$

Define the fixed-wheel rough-composite residual

$$R_S(P,k) = \#\{(q,m) : P/2 < q < P,\ q \text{ prime},\ m \in I_q(P,k),\ m \text{ composite},\ m \text{ is not deleted by } S\}.$$

Then

$$C_S(P,k) = |F(P,k)| + R_S(P,k),$$

and therefore

$$\Delta\Phi_{1/2}(P,k) - C_S(P,k) = \big(\pi((k+1)P-1) - \pi(kP)\big) - R_S(P,k).$$

Consequently a fixed-wheel positivity proof

$$\Delta\Phi_{1/2}(P,k) > C_S(P,k)$$

is exactly the stronger dominance assertion

$$\pi((k+1)P-1) - \pi(kP) > R_S(P,k).$$

If S contains all primes $\ell \leq \sqrt{2P-1}$, then $R_S(P,k) = 0$.

Status and reference. The candidates counted by $C_S(P,k)$ are precisely the reciprocal-window pairs (q,m) that survive the S-wheel forced-composite deletion. Each surviving m is either prime or composite. The prime alternatives are exactly the forest holes $F(P,k)$, counted with the same q-window multiplicity; the composite alternatives are, by definition, $R_S(P,k)$. This proves $C_S = |F| + R_S$.

Subtracting this identity from the endpoint formula

$$\pi((k+1)P-1) - \pi(kP) = \Delta\Phi_{1/2}(P,k) - |F(P,k)|$$

gives the displayed residual equation. If S contains all primes up to $\sqrt{2P-1}$, then every composite $m < 2P$ has a prime factor in S and is deleted; hence $R_S = 0$.

The machine certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-fixed-wheel-
residual-rough-composite-audit.md.
```

It records

```
max_prime=1009
row_count=76789
all_sqrt_residual_zero=true
all_fixed_capacity_decomposition_holds=true
all_delta_minus_capacity_equals_prime_minus_residual=true.
```

For the representative row $P = 1009, k = 1008$, it gives

$$\Delta\Phi_{1/2} = 89, \qquad N = 70, \qquad |F| = 19,$$

and

$$R_{30} = 9, \qquad R_{210} = 8, \qquad R_{2310} = 7, \qquad R_{\sqrt{2P}} = 0.$$

Thus finite wheel promotion narrows the residual but does not remove the need for a genuine same-row prime-minus-rough-composite separation. The current hard input is

```
PrimeCountDominatesFixedWheelRoughCompositeResidual
OR same-object signed dispersion
OR special square-phase lower bound.
```

No unconditional row/column closure follows from this residual ledger alone. □

Theorem 7.96 (LPF shell decrement law for fixed-wheel residuals)**.** *For each composite reciprocal cofactor m, let $r = \mathrm{LPF}(m)$. Then*

$$m = ra, \qquad a \geq r,$$

and a has no prime factor below r. Let

$$\mathrm{Shell}_r(P, k)$$

denote the number of reciprocal-window pairs (q, m) with $\mathrm{LPF}(m) = r$, counted with the same multiplicity as in the forest-hole ledger. If $C_y(P, k)$ denotes the primorial-wheel capacity after deleting all cofactor candidates with least prime factor at most y, then

$$R_y(P, k) = \sum_{r>y} \mathrm{Shell}_r(P, k),$$

and for two cutoffs $y < y'$,

$$C_y(P, k) - C_{y'}(P, k) = \sum_{y<r\leq y'} \mathrm{Shell}_r(P, k).$$

Thus each primorial-wheel promotion strips exactly the next LPF shell, and the remaining fixed-wheel hard point is the tail dominance

$$\pi((k+1)P - 1) - \pi(kP) > \sum_{r>y} \mathrm{Shell}_r(P, k).$$

Status and reference. The factorization assertion is the defining property of the least prime factor: if $r = \mathrm{LPF}(m)$, then $m = ra$, $a \geq r$, and no prime below r divides a. Hence the composite cofactor candidates are partitioned disjointly by their least prime factor. A primorial wheel with cutoff y deletes exactly those composite candidates whose least prime factor is at most y, while it leaves the prime cofactors untouched. Therefore the residual after the cutoff is precisely the shell tail $\sum_{r>y} \mathrm{Shell}_r$, and the drop from cutoff y to cutoff y' is the intervening shell sum.

The certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-lpf-shell-
decrement-audit.md.
```

It records

```
max_prime=1009
row_count=76789
all_lpf_factorizations_ordered=true
all_capacity_reconstructed_from_lpf_shells=true
all_adjacent_decrements_equal_lpf_shells=true.
```

The aggregate finite shell ledger is

```
LPF=2:1269907, LPF=3:423339, LPF=5:169232,
LPF=7:96700, LPF=11:52080, LPF=13:44104,
tail_ge_17:107093.
```

For $P = 1009, k = 1008$, the shell profile is

$$2:47, \quad 3:20, \quad 5:6, \quad 7:1, \quad 11:1, \quad 13:2, \quad \text{tail}_{\geq 17}:5.$$

This proves the decrement bookkeeping law and identifies the current non-circular target as

```
PrimeCountDominatesLPFTailShellSum
OR signed shell cancellation
OR square-phase endpoint lower bound.
```

Existing rough-number short-interval or variance inputs provide useful density diagnostics, but they do not by themselves prove this reciprocal-window weighted, pointwise, same-row dominance statement. No unconditional row/column closure follows from the shell ledger alone. □

Theorem 7.97 (Adjacent-coprime parity trap after the 30-wheel). *Let $n = qm$ be a 30-wheel rough-composite residual candidate in the reciprocal-window ledger. Put $r = \mathrm{LPF}(m)$ and $m = ra$. Then*

$$q, \ m, \ r, \ a$$

are all odd, and $r \geq 7$. Hence the adjacent-coprime identities

$$(qm, qm \pm 1) = 1, \qquad (m, m \pm 1) = 1, \qquad (a, a \pm 1) = 1$$

do not create a prime channel: every integer $qm \pm 1$, $m \pm 1$, and $a \pm 1$ occurring here is even and larger than 2, hence composite. Moreover the quotient-adjacent lift

$$qr(a \pm 1) = qra \pm qr$$

cannot stay in the same row, because $q > P/2$ and $r \geq 7$, so $qr > P$.

Status and reference. The 30-wheel residual has already removed all cofactor candidates divisible by $2, 3$, or 5. Thus a composite residual cofactor m has $\mathrm{LPF}(m) = r \geq 7$, and m, r are odd. Since $a = m/r$, the quotient a is odd. Also $q \in (P/2, P)$ is an odd prime in the strict rows under discussion. Therefore qm, m, and a are odd. Their adjacent integers are even. The lower adjacent values are also larger than 2: $m - 1 \geq 6$, $a - 1 \geq 6$, and $qm - 1 > 2$. Hence all these adjacent integers are composite despite being coprime to their source.

For the quotient lift, changing a to $a \pm 1$ changes qra by $\pm qr$. Since $q > P/2$ and $r \geq 7$, this displacement is larger than the row length P. Thus the quotient-adjacent object cannot be a same-row prime payment.

The certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-adjacent-
coprime-parity-trap-audit.md.
```

It records, for $P \le 1009$,

```
active_residual_row_count=52697
total_R30=299977
total_same_row_adjacent_slots=595083
total_same_row_adjacent_prime_shadows=0
total_cofactor_adjacent_prime_shadows=0
total_quotient_adjacent_prime_shadows=0
total_quotient_lift_inside_row=0.
```

For the maximum residual row in the finite audit, $P = 971, k = 936$, one has

$$R_{30} = 23, \qquad \text{same-row adjacent slots} = 44, \qquad \text{adjacent prime shadows} = 0.$$

This closes the direct adjacent-coprime and quotient-adjacent-coprime payment routes. A Ford–Maynard style prime-producing sieve remains relevant only if one supplies a matching Type-I/Type-II or bilinear theorem for the present reciprocal-window object. The remaining hard input is still

```
PrimeCountDominatesLPFTailShellSum
OR same-object Type-II signed dispersion
OR square-phase endpoint lower bound.
```

No unconditional row/column closure follows from adjacent coprimality alone. □

Theorem 7.98 (LPF tail reciprocal-graph Type–II obligation)**.** *The 30-wheel LPF tail residual has the exact representation*

$$\begin{aligned} R_{30}(P,k) = \#\{(q,r,a) : P/2 < q < P,\ q \text{ prime},\ m = ra \in I_q(P,k), \\ r = \mathrm{LPF}(m) \ge 7,\ a \ge r,\ a \text{ is } r\text{−rough}\}, \end{aligned}$$

where

$$I_q(P,k) = \big[\max(q, \lfloor kP/q \rfloor + 1),\ \min(2P-1, \lfloor((k+1)P-1)/q\rfloor)\big].$$

This representation is a balanced q-m object in scale, but its support is not a rectangular Type–II box. It is a same-row reciprocal graph with thin fibres:

$$\#I_q(P,k) \le 2, \qquad \#\{q : m \in I_q(P,k)\} \le 2, \qquad \#\{a : kP < qra < (k+1)P\} \le 1.$$

Consequently a Ford–Maynard or Heath-Brown style Type–II continuation must prove dispersion for this exact reciprocal graph. It cannot be replaced by adjacent coprimality, ordinary rough-number density, or a rectangular bilinear box estimate.

Status and reference. The preceding LPF shell ledger partitions each 30-wheel residual cofactor m by $r = \mathrm{LPF}(m)$. Since the 30-wheel has removed factors $2, 3, 5$, every residual composite has $r \ge 7$ and $m = ra$ with $a \ge r$ and a r-rough. This gives the displayed triple representation.

The fibre bounds are deterministic. For fixed $q > P/2$, the row condition $kP < qm < (k+1)P$ gives an m-interval of length $P/q < 2$, hence at most two integers after intersecting with the reciprocal cofactor range. Conversely, for fixed $m \ge q > P/2$, the admissible q-interval has length $P/m < 2$. Finally, for fixed q and r, the admissible a-interval has length

$$\frac{P}{qr} < 1,$$

because $q > P/2$ and $r \ge 7$. Thus each (q, r)-fibre contains at most one a. In particular there is no quotient-fibre cancellation to harvest inside a single (q, r) line; any Type–II cancellation must run across the moving reciprocal graph itself.

The machine certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-lpf-tail-
typeii-obligation-audit.md.
```

For $P \leq 1009$ it records

```
row_count=76789
active_residual_row_count=52697
total_R30=299977
total_direct_prime_count=4172483
total_prime_count_minus_R30=3872506
all_q_m_windows_have_at_most_two_points=true
all_m_q_reverse_fibers_have_at_most_two_points=true
all_qr_a_fibers_have_at_most_one_point=true.
```

For the maximum finite residual row $P = 971, k = 936$, it gives

$$N = 80, \qquad W_{\text{int}} = 97, \qquad R_{30} = 23, \qquad N - R_{30} = 57,$$

with q-count 71, m-span 915, and reciprocal-graph support density 0.00149311 inside its rectangular hull. The sparsest finite graph recorded in the audit is $P = 1009, k = 965$, with density 0.0013077201.

The external-theorem boundary is unchanged: a Runbo Li $x^{0.52}$-level pointwise short-interval input still gives length $P^{1.04}$ at $x = P^2$, not the row length P; a Ford–Maynard prime-producing sieve framework is relevant only after Type–I/Type–II estimates are supplied for this exact sequence. The narrowed non-circular target is therefore

```
SameRowReciprocalWindowTypeIIDispersionForLPFTail
OR PrimeCountDominatesLPFTailShellSum
OR SquarePhaseEndpointLowerBound.
```

No unconditional row/column closure follows from this Type–II obligation ledger alone. □

Theorem 7.99 (CRT signed residue projection gate for the LPF tail)**.** *Fix a wheel S, put $W_S = \prod_{\ell \in S} \ell$, and let $R_S(P,k)$ be the corresponding rough-composite residual. For each residue class $a \bmod W_S$, define the same-row signed residue measure*

$$\mu_S(a; P, k) = \#\{n : kP < n < (k+1)P,\ n \text{ prime},\ n \equiv a \pmod{W_S}\}$$

$$-\#\{n = qm : kP < n < (k+1)P,\ n \equiv a \pmod{W_S},\ (q,m) \text{ counted by } R_S(P,k)\}.$$

Then

$$\sum_{a \bmod W_S} \mu_S(a; P, k) = \pi((k+1)P - 1) - \pi(kP) - R_S(P,k).$$

If $P > 2\max S$, all atoms in this signed measure lie in unit classes modulo W_S.

However, fixed CRT classwise dominance

$$\mu_S(a; P, k) \geq 0 \qquad \textit{for every unit } a \bmod W_S$$

is false as a route to the LPF-tail inequality. The remaining CRT route must therefore be a character-averaged or cross-residue signed dispersion theorem, not a unit-cell matching argument.

Status and reference. The identity is just the partition of the signed row measure by residue class modulo W_S. If $P > 2\max S$, then every high prime $q \in (P/2, P)$ is outside S. A row prime $n > 2\max S$ is also not divisible by any $\ell \in S$. The S-wheel residual has cofactor m not deleted by any $\ell \in S$. Hence both the prime atoms and the residual atoms are unit classes modulo W_S.

The classwise nonnegativity assertion is not a theorem. The machine certificate is archived in

```
docs/monograph/prime-matrix-phi-lpf-crt-signed-
residue-projection-gate-audit.md.
```

For $P \le 1009$, the stable finite ledger records

```
30-wheel:  negative rows=976, negative unit cells=998
210-wheel:  negative rows=37115, negative unit cells=67547
2310-wheel:  negative rows=45472, negative unit cells=151197
30030-wheel:  negative rows=39964, negative unit cells=107093.
```

At the same time the total signed surplus remains positive in the finite ledger:

```
30-wheel stable_total_surplus=3872506
210-wheel stable_total_surplus=3969154
2310-wheel stable_total_surplus=4021124
30030-wheel stable_total_surplus=4065143.
```

Representative negative unit cells are

W_S	P	k	a	#prime	#residual	$\mu_S(a)$
30	313	183	11	0	4	-4
210	463	448	167	0	3	-3
2310	97	92	2027	0	1	-1
30030	157	145	22831	0	1	-1

Thus CRT projection gives a useful exact ledger, but it does not by itself break the parity obstruction. The narrowed hard point is

```
CharacterAveragedSameRowCRTDispersionForLPFTail
OR SameRowReciprocalWindowTypeIIDispersionForLPFTail
OR SquarePhaseEndpointLowerBound.
```

No unconditional row/column closure follows from fixed CRT classwise projection alone. □

Proposition 7.100 (Phi–LPF exactness does not bypass the parity barrier)**.** *The Phi–LPF identities above are combinatorially exact identities for $N_P(k)$. They do not, by themselves, constitute a positive lower bound for $N_P(k)$. In the top strict band $k \asymp P$, the row starts at $x = kP \asymp P^2$ and has length $P \asymp \sqrt{x}$. Therefore any closure that proceeds only by an ordinary short-interval theorem $[x, x + x^\theta]$ requires $\theta \le 1/2$ with an admissible constant. Known unconditional inputs of Baker–Harman–Pintz 0.525 type, and the newer 0.52 type refinement recorded in the review certificate, remain above this square-root scale and leave an infinite top k-band.*

Thus the non-circular continuation after the parity-capacity bound is not another unsigned rewriting of the same count. It must supply one of the following genuinely new inputs: a square-root-scale interval theorem on the relevant endpoints, a special square-phase/CRT structural lower bound proving $\Delta\Phi_{1/2}(P,k) > C_{\sqrt{2P}}(P,k) = |F(P,k)|$ in the exact wheel limit, or a signed transport/dispersion estimate that breaks the linear-sieve parity obstruction. Without such an input, the remaining object is a reciprocal prime-pair saturation PDEC.

Status and reference. The assertion is a route-control statement. The exact identities express the row prime count as a survivor term minus a hole term; positivity is equivalent to proving that the survivor term dominates. The exponent calculation

$$P \ge C(kP)^\theta \quad \Longrightarrow \quad k \le C^{-1/\theta} P^{(1-\theta)/\theta}$$

shows that every fixed $\theta > 1/2$ leaves k near P uncovered for arbitrarily large P. The review certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-combinatorial-
exactness-parity-barrier-review-router.md.
```

It records

```
status=review_absorbed_phi_lpf_exactness_is_not_positivity
accepted_review_core=true
current_frontier_after_review=
PuncturedSqrtWheelExactForestHolePositivityOr
SignedDispersionOrSpecialSquarePhaseLowerBound.
```

This proposition deliberately does not prove the row/column theorem; it fixes the admissible proof obligations after the Phi–LPF exact-count phase. □

Proposition 7.101 (H_P Cramer-local route reset)**.** *The H_P strict top band cannot be treated as a solved finite-CRT or Phi–LPF bookkeeping problem. In the hard range $k \asymp P$, one has*

$$x = kP \asymp P^2, \qquad \textit{row length} = P \asymp \sqrt{x}.$$

Thus the direct positivity assertion is a Legendre/Cramer-local short-interval problem at square-root scale. All Phi–LPF, least-prime-factor, CRT-wheel, and Eratosthenes-sieve identities developed above remain valid exact reductions, but they form a reduction library rather than a positive lower-bound theorem. Moving the same unsigned survivor-minus-hole identity between equivalent interfaces is no longer counted as a closure step.

The admissible continuation is therefore reset to one of the following inputs:

an exact external square-root-scale theorem match,

a new DI/BFI/Kuznetsov or automorphic dispersion proof for the present full-S non-AP WFD object,

or a genuinely new square-phase structural lower bound beyond the linear-sieve parity barrier.

The Friedlander–Iwaniec, Deshouillers–Iwaniec/Bombieri–Friedlander–Iwaniec, Maynard/GPY, and automorphic L-function technologies are retained as high-risk analytic tool classes. None of them is imported as a ready-made proof of H_P unless the theorem statement is matched exactly to the current weights, windows, moduli, projections, and required error savings.

Status and reference. This is a proof-control proposition, not a new prime-distribution theorem. The exponent calculation in the preceding proposition shows that every ordinary short-interval result with exponent $\theta > 1/2$ leaves an infinite top band. The latest 0.52-type improvement therefore changes the numerical frontier but not the square-root obstruction. The route-reset certificate is recorded in

```
docs/monograph/prime-matrix-hp-cramer-local-route-
reset-router.md.
```

It records

```
status=hp_cramer_local_route_reset_after_parity_barrier_review
review_core_accepted=true
hp_direct_unconditional_closure_claim_allowed=false
current_frontier_after_reset=
ExactExternalSqrtScaleOrFullSNonAPWFDKLSTheoremMatch
OR NewAutomorphicDispersionProof
OR SpecialSquarePhaseStructuralLowerBoundBeyondParity.
```

Accordingly, the monograph may prove conditional chains and exact reductions, but it must not state the unconditional H_P theorem until one of these positive analytic inputs is independently supplied. □

Proposition 7.102 (Full-S non-AP WFD theorem-match matrix)**.** *Let*

$$W_{\mathrm{full}}(C,S,H)=\sum_{c\sim C}\lambda_c\sum_{0<|h|\le H}\omega_h\sum_{\substack{s\sim S\\(s,c)=1}}\beta_s e_c(a_h s+b_h\overline{s})$$

with

$$X\asymp P^2,\qquad C\asymp P/\log^{O(1)}P,\qquad S\asymp P,\qquad H\le P/\log^{O(1)}P.$$

Assume λ_c is well-factorable, β_s is divisor-bounded, and ω_h is smooth. The current external-input screen for the H_P route requires an item-by-item match of object, weights, window, moduli, smoothing/projection convention, saving strength, and conclusion. Under that screen the available primary-source labels do not close the present object: BHP/Li short intervals miss the square-root top band and give no WFD estimate; Friedlander–Iwaniec is a parity-sensitive model for a different polynomial object; BFI is an arithmetic-progression theorem unless an AP-source lift is proved; DI/Kuznetsov supplies the spectral technology class but not a ready-made corollary for the c-dependent completed, uncentered no-projection full-S weights; and Maynard/GPY has the wrong conclusion type.

The only exact match in the current corpus is the separate FullS–KLS-ext contract, and that match is conditional on accepting it as a new black-box theorem. It is not a line-by-line derivation from the cited DI/BFI/Maynard sources. Therefore the active external frontier is precisely

```
FullSNonAPWFDKLSTheoremInput
OR APSourceLift
OR NCBLKActualBlockNonConcentration.
```

Status and reference. The theorem-match matrix certificate is recorded in

```
docs/monograph/prime-matrix-fulls-nonap-wfd-
theorem-match-matrix-router.md.
```

It records

```
status=fulls_nonap_wfd_primary_sources_screened_exact_contract_or_new_theorem_remains
ready_made_primary_source_match_found=false
primary_source_derivation_closed=false
unconditional_hp_closure_reached=false
next_direct_attack_target=FullSNonAPWFDKLSTheoremInput
OR APSourceLift OR NCBLKActualBlockNonConcentration.
```

This proposition closes only the citation discipline: FI/DI/BFI/Maynard and automorphic technologies must be matched theorem-by-theorem before they can be used. It does not prove the unconditional H_P, row/column, or three-claim theorem. $\square$

Proposition 7.103 (True remainder after the full-S theorem-match screen)**.** *The three labels left by the theorem-match screen are not three independent final obstructions:*

```
FullSNonAPWFDKLSTheoremInput OR APSourceLift
OR NCBLKActualBlockNonConcentration.
```

The AP-source lift is filtered out by the existing no-go certificate: the current object is a non-AP, uncentered, no-projection WFD object, and cannot be silently reclassified as a BFI arithmetic-progression discrepancy. The NCBLK label is also not kept as an opaque terminal. Through the BWFD/BSC/KFLS chain, branch alignment, exact source-entropy reduction, range closure, and source anti-atom reduction, it becomes an actual-source support/capacity theorem for the noncanonical full-S source. The generic full-S WFD anti-atom is refuted by the moving-delta model, so the remaining source statement must be about the actual noncanonical source, not the formal template.

Consequently the current no-black-box or primary-source-specialized remainder is

```
(ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
 OR ActualNoncanonicalFullSFactorSupportCapacityTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the independent FullS–KLS-ext contract is accepted as an external black-box theorem, the corresponding conditional basis is

```
AcceptedFullSKLSExt
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The true-remainder cut certificate is recorded in

```
docs/monograph/prime-matrix-fulls-theorem-match-
true-remainder-cut-router.md.
```

It records

```
status=ap_lift_filtered_ncblk_expanded_true_remainder_external_or_actual_source_core_open
unconditional_closure_reached=false
current_true_remainder_basis=
(ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR ActualNoncanonicalFullSFactorSupportCapacityTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes a routing ambiguity only. It does not prove the exact primary-source KLS theorem match, the actual noncanonical source support/capacity theorem, or the independent D-structure/Rankin promotion gate. □

Theorem 7.104 (Q1/Q2 transport latest non-circular synchronization)**.** *The adjacent-prime Q_1, Q_2 transport branch is not an independent pure-CRT terminal for the row/column proof. The existing Q_2-stage ladder shows that endpoint-stable replay is impossible: in the full Q_2-wheel, the two endpoint copies are respectively divisible by Q_1 and Q_2. A persistent closed carrier block without prime endpoints is a ColumnCRT/PDEC formal unit; a controlled fresh endpoint tail is SAE; and an uncontrolled aperture explosion or support motion must be submitted as an explicit moving-family/PDEC schema. Hence the non-circular residue of the Q1/Q2 branch is exact-source dispersion, external spectral input, or the already registered seed/PDEC/new-joint strict frontier:*

```
((NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource
  OR ExternalDIBFIKuznetsovDispersionTheoremMatch)
 OR AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the existing seed/PDEC saturation synchronizers are imported, the strict internal branch is further reduced to

```
NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-q1q2-transport-latest-noncycle-sync-router.md.
```

It records

```
q2_endpoint_stable_replay_impossible=true
persistent_closed_carrier_routes_to_columncrt_pdec=true
controlled_fresh_endpoint_tail_routes_to_sae=true
unnamed_aperture_explosion_forbidden=true
pure_crt_global_phase_contradiction_blocked=true
q1q2_branch_reduced_to_exact_source_or_external=true
q1q2_transport_defect_or_stable_short_return_proved_as_global_contradiction=false
row_column_unconditional_closed=false.
```

The stable-replay CRT contradiction is therefore closed only for that subcase. The proof still lacks a theorem forcing the global counterexample chain into an exact-source nonconcentration contradiction, a new joint constructor, or an accepted external dispersion theorem. Thus the row/column theorem remains open. □

Theorem 7.105 (Exact-UV fiber latest non-circular synchronization)**.** *The target*

```
NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource
```

is not a pure CRT position or phase-rigidity problem. It is an actual-source rank/no-collapse problem inside one formal unit. The formal implication now recorded is

```
ActualPreCauchySourceDomainAbsoluteEntropyLedger
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger
implies NonterminalExactUVFiberAperiodicityEstimateForActualPreCauchySource.
```

After atomization, the self-contained internal basis is

```
((AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward
  AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger
  AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger)
 AND (ActualNoncanonicalPrimitiveEmitterSourceTableLedger
      AND CompleteEmitterTraceKeyBudgetLedger
      AND SignLocalFactorRefinementNoCancellationLedger
      AND OverBudgetOrUnregisteredReturnLedger)
 AND FixedKeyExactUVLocalMultiplicity01Ledger)
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The next direct attack is

```
AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward,
```

with the actual source table, fixed-key local multiplicity, new joint formula, and external dispersion route kept in parallel.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-exactuv-fiber-latest-noncycle-sync-router.md.
```

It records

```
preterminal_fiber_atomization_imported=true
deterministic_source_atom_implication_closed=true
source_entropy_reduced_to_signed_rows=true
complete_key_reduced_to_actual_source_table=true
fixed_pair_fiber_formal_inequality_closed=true
map_rank_equivalent_to_bounded_incidence=true
nonterminal_exactuv_fiber_aperiodicity_proved=false
row_column_unconditional_closed=false.
```

Thus this step closes only the deterministic source-atom implication and the misleading pure-CRT exit. It does not prove the source entropy, the complete key partition, fixed-key local multiplicity, the model tail, rate preservation, or D-structure/Rankin promotion. The row/column theorem remains open. □

Theorem 7.106 (Inverse-alignment $P+1$ row reduction check)**.** *In the row multiplier notation used in this manuscript, the row $x = P$ is the one-indexed $P+1$ row and is the square-anchored interval*

$$P^2 + r, \qquad 1 \le r < P.$$

The older inverse-alignment files already prove the equivalence

```
NoZeroRowAtXEqualsP_PlusOneRowAfterSquare
if and only if PrimeInFirstHalfAfterPrimeSquareForEveryPrimeP.
```

They also prove that $X_0(P) > P$ is equivalent to the absence of zero rows for all $1 \le x \le P$. However, the old files do not prove that the single assertion $x = P$ is nonzero implies that every $1 \le x < P$ is nonzero. Such an implication would require a separate non-circular transfer:

```
EarlyZero(x<P)
implies ZeroAtXEqualsP OR NamedReturn(PDEC/SAE/ColumnCRT/source-rank).
```

Thus the new branch target is

```
AcyclicEarlyZeroToSquareAnchorPhaseTransferOrNamedReturn,
```

while the exact-UV source-rank frontier still keeps

```
AcyclicSeedPrimitiveRowSignedCoefficientLawBeforePushforward
```

as a parallel direct attack.

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-inverse-alignment-pplus1-row-reduction-check-router.md.
```

It records

```
old_minrep_equivalence_closed=true
pplus1_row_is_x_equals_p=true
x_equals_p_no_cover_equivalent_to_first_half_prime_square=true
pplus1_nonzero_suffices_for_all_early_rows_proved=false
row_column_unconditional_closed=false.
```

The obstruction is phase-theoretic: different early multipliers x give different covering phases $\rho_q(x) = -xP \bmod q$. The square anchor $x = P$ is one special phase line and does not automatically dominate the whole family $1 \le x < P$. Therefore this check sharpens the old route but does not close the row/column theorem. □

Theorem 7.107 (Early-to-square phase-transfer split)**.** *The abstract transfer target from an early zero row to the square anchor is not kept as an unnamed obligation. Suppose that an early zero row $x_0 < P$ exists. Moving from x_0 to P, either every row remains zero up to $x = P$, in which case the square-anchor row is itself zero, or there is a first break row $y \le P$. At that row $y - 1$ is still zero and y has a nonempty released-column set. The local phase motion is*

$$\rho_q(t) = -tP \pmod q, \qquad \rho_q(t+1) = \rho_q(t) - P \pmod q.$$

Thus the first break is a unit phase-slip boundary defect. Under the previous early-zero phase-defect schema it is a named PDEC/SAE/ColumnCRT/LocalSurvivor return, not a fourth exit. Under the additional condition that $x = P$ is not a zero row, the transfer branch is reduced to

```
FirstBreakPhaseSlipNamedReturnExclusion.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-early-to-square-phase-transfer-split-router.md.
```

It records

```
contiguous_zero_block_dichotomy_closed=true
persist_to_square_implies_square_zero=true
first_break_release_set_nonempty=true
unit_phase_slip_formula_closed=true
phase_slip_schema_admission_imported=true
transfer_proved_as_contradiction=false
row_column_unconditional_closed=false.
```

This closes the unnamed-transfer ambiguity, but it does not exclude the first-break named terminal. The row/column theorem remains open until that terminal is excluded, or the parallel signed-source/source-rank frontier is closed. □

Theorem 7.108 (First-break phase-slip LCM width barrier). *Let the first break from an early zero block occur at row $y \le P$, so that row $y-1$ is still zero and row y has a nonempty released-column set. Fix a released formal unit and let Λ be its active carrier prime labels. If the same released phase pattern reappears at row $y+d$ with the same carrier labels, then for every $q \in \Lambda$*

$$\rho_q(y+d) = \rho_q(y), \qquad \rho_q(t) = -tP \pmod q,$$

and hence $dP \equiv 0 \pmod q$. Since $q < P$ is prime and $q \neq P$, this is equivalent to $q \mid d$. Therefore

$$\operatorname{lcm}(\Lambda) \mid d.$$

The remaining row support before the square anchor has width $H = P - y$. Thus if $\operatorname{lcm}(\Lambda) > H$, no nonzero fixed-carrier replay can occur before the square anchor. If $\operatorname{lcm}(\Lambda) \le H$, the replay is a small-LCM fixed-residue event and must be routed to ColumnCRT/PDEC. If the carrier labels move, the branch is a moving-carrier phase-slip PDEC/SAE return. Consequently the previous first-break terminal is reduced to

```
SmallLCMColumnCRTPDECExclusion
AND NonreplaySparseFirstBreakSAESummability
AND MovingCarrierPhaseSlipPDECExclusion.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-phase-slip-lcm-barrier-router.md.
```

It records

```
fixed_carrier_lcm_replay_period_closed=true
square_support_width_sharpened_to_p_minus_y=true
lcm_exceeds_width_no_fixed_replay=true
firstbreak_phase_slip_named_return_exclusion_proved=false
row_column_unconditional_closed=false.
```

This is a non-circular sharpening of the first-break named terminal: it uses only the forward phase motion from row y to the square anchor. It does not exclude the small-LCM ColumnCRT/PDEC branch, the nonreplay sparse SAE branch, or the moving-carrier PDEC branch. The inverse-alignment return therefore becomes

```
NoZeroRowAtXEqualsP_PlusOneRowAfterSquare
AND SmallLCMColumnCRTPDECExclusion
AND NonreplaySparseFirstBreakSAESummability
AND MovingCarrierPhaseSlipPDECExclusion.
```

The row/column theorem remains open. □

Theorem 7.109 (First-break small-LCM rank-pressure compression)**.** *In the fixed small-LCM branch of the first-break phase-slip return, the replay width is positive, $H = P - y \geq 1$. If $y = P$, there is no nonzero replay row before the square anchor and the branch returns to the square-anchor/SAE boundary. Let Λ be the active carrier-prime label set. Since these labels are distinct primes,*

$$\operatorname{lcm}(\Lambda) = \prod_{q \in \Lambda} q \leq H.$$

Consequently for every threshold $B > 1$,

$$|\{q \in \Lambda : q > B\}| \leq \left\lfloor \frac{\log H}{\log B} \right\rfloor.$$

In particular, two carriers $q > \sqrt{H}$ cannot occur in the same fixed small-LCM formal unit. Therefore a small-LCM first-break branch cannot be a free superposition of many medium/high carrier phases. Any remaining pressure must be routed to one of three lower-level exits:

```
LowCarrierFixedResidueColumnCRTPDECExclusion
AND LowCarrierNonpersistentSparseSAESummability
AND HighCarrierRankDeficitCapacityBoundOrSingletonSAE.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-small-lcm-rank-pressure-router.md.
```

It records

```
distinct_prime_product_law_closed=true
small_lcm_rank_pressure_closed=true
two_large_carrier_sqrt_barrier_closed=true
small_lcm_branch_excluded=false
row_column_unconditional_closed=false.
```

This closes only the deterministic rank-pressure structure of the small-LCM branch. It does not exclude persistent low-carrier fixed residues, nonpersistent sparse SAE mass, or the high-carrier rank-deficit capacity obligation. The inverse-alignment return is updated to

```
NoZeroRowAtXEqualsP_PlusOneRowAfterSquare
AND LowCarrierFixedResidueColumnCRTPDECExclusion
AND LowCarrierNonpersistentSparseSAESummability
AND HighCarrierRankDeficitCapacityBoundOrSingletonSAE
AND NonreplaySparseFirstBreakSAESummability
AND MovingCarrierPhaseSlipPDECExclusion.
```

The row/column theorem remains open. □

Theorem 7.110 (First-break low-carrier fixed-residue AP skeleton)**.** *In the low-carrier fixed-residue branch, fix a carrier prime $q < P$ and a column residue $a \bmod q$. On a row t, the covering condition is*

$$tP + c \equiv 0 \pmod{q}, \qquad c \equiv a \pmod{q}.$$

Since P is invertible modulo q, this is equivalent to the single row class

$$t \equiv -aP^{-1} \pmod q.$$

Therefore in the post-break row interval $I_y = \{y, \ldots, P-1\}$, whose length is $H = P - y$, one fixed residue cell (q, a) can occupy at most

$$\lceil H/q \rceil$$

row positions. With a fixed low-carrier threshold B, the number of available residue cells is at most $\sum_{q \le B} q$. Thus the low-carrier fixed-residue branch is no longer an unstructured ColumnCRT label; it is a finite AP table. The branch is reduced to

```
LowCarrierResidueAPEnvelopeCapacityComparison
AND DenseLowCarrierResidueTablePDECExclusion
AND SparseLowCarrierResidueCellSAESummability.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-low-carrier-residue-ap-router.md.
```

It records

```
residue_to_row_ap_formula_closed=true
single_residue_ap_envelope_closed=true
low_carrier_residue_table_finite_closed=true
ap_envelope_capacity_comparison_proved=false
low_carrier_fixed_residue_excluded=false
row_column_unconditional_closed=false.
```

This step supplies the exact AP skeleton and the terminal routing. It does not prove that the required counterexample pressure exceeds the AP envelope, nor does it exclude the dense low-carrier residue-table PDEC or the sparse cell SAE branch. The row/column theorem remains open. □

Theorem 7.111 (First-break low-carrier AP exact envelope)**.** *Let $T \subset \{(q, a) : q \le B,\ a \bmod q\}$ be a low-carrier residue table and let $I_y = \{y, \ldots, P-1\}$ have length $H = P - y$. Define*

$$U_T(I_y) = \sum_{(q,a) \in T} \#\{t \in I_y : t \equiv -aP^{-1} \pmod q\}.$$

Then every selected cell satisfies

$$\#\{t \in I_y : t \equiv -aP^{-1} \pmod q\} \le \lceil H/q \rceil.$$

Moreover, for each fixed q,

$$\sum_{a \bmod q} \#\{t \in I_y : t \equiv -aP^{-1} \pmod q\} = H.$$

Consequently

$$U_T(I_y) \le \sum_{(q,a) \in T} \lceil H/q \rceil, \qquad U_T(I_y) \le H \cdot \#\{q : \exists a\ (q, a) \in T\}.$$

Thus the formal supply side of `LowCarrierResidueAPEnvelopeCapacityComparison` *is exact, and the remaining branch is reduced to*

```
ActualLowCarrierRowIncidenceDemandLowerBound
AND LowCarrierAPEnvelopeStrictGapOrDenseTablePDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-low-carrier-ap-envelope-router.md.
```

It records

```
cell_count_formula_closed=true
single_cell_sharp_bound_closed=true
all_residues_exact_mass_closed=true
selected_table_envelope_closed=true
ap_capacity_comparison_proved=false
row_column_unconditional_closed=false.
```

This closes only the exact AP envelope. It does not prove the actual row-incidence demand lower bound, and it does not exclude the dense-table PDEC alternative. The row/column theorem remains open. □

Theorem 7.112 (First-break actual-demand source cut). *In the first-break branch, the imported release-set statement gives only a unit source-side demand:*

$$R_y \neq \emptyset \quad \Longrightarrow \quad D_y \geq 1.$$

This is a genuine actual lower bound, but it is not enough to beat the AP envelope, since a single low-carrier residue cell can already support one incidence whenever it meets I_y. Therefore the gate `ActualLowCarrierRowIncidenceDemandLowerBound` *cannot be closed by the nonemptiness of R_y alone. It is reduced to*

```
FirstBreakReleaseMassAmplificationOrSingletonSAE
AND LowCarrierActualPaymentInjectionWithoutEnvelopeReuse.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-actual-demand-source-cut-router.md.
```

It records

```
firstbreak_release_set_imported=true
unit_release_demand_lower_bound_closed=true
unit_demand_not_gap_sufficient=true
no_envelope_recycling_guard=true
release_mass_amplification_proved=false
low_carrier_payment_injection_proved=false
actual_low_carrier_row_incidence_demand_proved=false
row_column_unconditional_closed=false.
```

The demand lower bound must be proved from the zero-row/first-break source side, not by recycling AP-envelope saturation. If source mass does not amplify, the event is a singleton or sparse SAE; if amplified mass cannot be injected into the low-carrier AP table, it returns to the high-rank, moving-carrier, PDEC, or SAE exits. The row/column theorem remains open. □

Theorem 7.113 (First-break release mass zero-block ladder). *Assume the square-anchor branch does not take over and let y be the first nonzero row after an early zero row $x_0 < P$. Then $[x_0, y-1]$ is a zero-row block of length $L = y - x_0$. For every $t \in [x_0, y-1]$ and every $1 \leq c < P$, there is a carrier $q < P$ with*

$$q \mid tP + c.$$

Hence the source-side cover-obligation mass in the block is exactly

$$L(P-1).$$

The boundary release at row y only gives a unit event; any amplification must therefore come from this zero-block cover ledger. For any threshold $A \geq 1$, the branch is reduced to the two obligations

```
ShortZeroBlockSingletonSAESummability
AND LongZeroBlockCoverMassTransferToAPDemandOrPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-release-mass-zero-block-ladder-router.md.
```

It records

```
contiguous_zero_block_imported=true
boundary_only_amplification_blocked=true
zero_block_cover_obligation_mass_closed=true
block_length_dichotomy_closed=true
short_zero_block_singleton_sae_proved=false
long_zero_block_mass_transfer_proved=false
release_mass_amplification_proved=false
row_column_unconditional_closed=false.
```

This step localizes the only possible amplification source. It does not prove short-block SAE summability, and it does not prove that long-block cover obligations transfer noncircularly into post-break AP demand; failed transfer is routed to persistent cover-history PDEC/ColumnCRT/SAE. The row/column theorem remains open. □

Theorem 7.114 (Long zero-block cover-mass transfer routing)**.** *Assume the long zero-block branch of the first-break ladder. Let $B = [x_0, y-1]$ and let*

$$O_B = \{(t,c) : x_0 \le t < y,\ 1 \le c < P\}.$$

Then $|O_B| = L(P-1)$, where $L = y - x_0$. Each cover obligation $(t,c) \in O_B$ admits a carrier $q < P$ with $q \mid tP + c$, hence a history key

$$\kappa = (q,a), \qquad a \equiv c \pmod q, \qquad t \equiv -aP^{-1} \pmod q.$$

Importing the no-loss return accounting ledger, these obligations cannot disappear into an unnamed sink. They are routed either to a stable low-carrier/residue payment table, or to a named history-switch PDEC/ColumnCRT/MovingCarrier/SAE return. Therefore

```
LongZeroBlockCoverMassTransferToAPDemandOrPDEC
→ ZeroBlockHistoryProjectionNoLossLedger
AND StableLowCarrierPaymentTableOrHistorySwitchPDEC
AND PostBreakAPDemandInjectionFromStableHistory.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-long-zero-block-mass-transfer-router.md.
```

It records

```
zero_block_cover_obligation_mass_imported=true
history_projection_key_defined=true
no_loss_return_accounting_imported=true
stable_table_or_named_switch_route_registered=true
history_switch_pdec_excluded=false
postbreak_ap_demand_injection_proved=false
long_zero_block_mass_transfer_proved=false
row_column_unconditional_closed=false.
```

This step closes only the unnamed-escape route for long zero-block mass. It does not prove that stable history injects into post-break actual demand, and it does not exclude the named history-switch terminals. The row/column theorem remains open. □

Theorem 7.115 (Stable history AP-arrival dichotomy). *Assume the stable low-carrier/residue history branch. For a history key $\kappa = (q, a)$, the associated rows are exactly*

$$t \equiv -aP^{-1} \pmod{q}.$$

Let t_* be the last row in the zero block $B = [x_0, y-1]$ carrying this same key. Then the next row with the same key is $t_* + q$. By maximality of t_*, one has $t_* + q \geq y$. Hence, in the post-break support $I_y = [y, P-1]$, the exact dichotomy is

$$t_* + q \in I_y \iff q \leq P - 1 - t_*, \qquad t_* + q \geq P \iff q \geq P - t_*.$$

Consequently

```
PostBreakAPDemandInjectionFromStableHistory
→ StableHistoryAPSuccessorDichotomyLedger
AND ArrivalCandidateActualDemandInjectionWithoutEnvelopeReuse
AND TerminalNonarrivalLargeStepEscapePDECOrSAE.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-stable-history-ap-arrival-router.md.
```

It records

```
stable_history_route_imported=true
ap_row_class_formula_closed=true
last_prebreak_hit_successor_closed=true
arrival_nonarrival_dichotomy_closed=true
terminal_nonarrival_large_step_registered=true
arrival_candidate_registered=true
arrival_candidate_actual_demand_proved=false
terminal_nonarrival_excluded=false
postbreak_ap_demand_injection_proved=false
row_column_unconditional_closed=false.
```

This step proves only the AP successor and arrival/nonarrival dichotomy. An arrival is still only a candidate for actual demand, and a terminal nonarrival remains a named large-step PDEC/SAE exit. The row/column theorem remains open. □

Theorem 7.116 (Arrival candidate unit incidence and quotient discipline). *Assume an arrival candidate with source history tag (q, a) and*

$$y \leq t_{\text{next}} \leq P - 1, \qquad t_{\text{next}} \equiv -aP^{-1} \pmod{q}.$$

Then it gives one source-tagged low-carrier AP row-incidence

$$(\textit{source tag}, t_{\text{next}}, q, a) \mapsto \text{incidence}(t_{\text{next}}, q, a),$$

without using any AP-envelope saturation. However, raw source-tagged arrivals are not an aggregate demand lower bound. They must be quotiented by

$$\pi : \textit{source-tagged arrivals} \longrightarrow (t_{\text{next}}, q, a),$$

and the usable actual demand is the number of distinct incidences in $\operatorname{im} \pi$. *Multiple sources mapping to the same incidence are not deleted; they are quotient, weighted, duplicate/collision, PDEC, ColumnCRT, or SAE returns. Consequently*

```
ArrivalCandidateActualDemandInjectionWithoutEnvelopeReuse
→ SourceTaggedArrivalUnitIncidenceLedger
AND DistinctArrivalQuotientDemandLowerBoundOrCollisionPDEC
AND ArrivalCollisionOrDuplicatePaymentReturnLedger.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-arrival-candidate-actual-demand-router.md.
```

It records

```
arrival_candidate_imported=true
no_envelope_recycling_guard_imported=true
source_tagged_unit_incidence_closed=true
arrival_quotient_map_defined=true
no_loss_collision_return_imported=true
arrival_collision_return_registered=true
distinct_arrival_quotient_lower_bound_proved=false
arrival_collision_return_excluded=false
arrival_candidate_actual_demand_injection_proved=false
row_column_unconditional_closed=false.
```

This step closes the unit source-tagged incidence injection and the quotient discipline only. It does not prove a large distinct-arrival demand lower bound, and it does not exclude duplicate/collision returns. The row/column theorem remains open. □

Theorem 7.117 (Arrival quotient fiber envelope)**.** *Let A be the set of source-tagged arrivals and*

$$\pi : A \longrightarrow (t_{\text{next}}, q, a)$$

be the arrival quotient map. Fix an image point (t, q, a). Any preimage source row s must lie in the zero block $B = [x_0, y-1]$ and satisfy

$$s \equiv t \pmod q,$$

so the row factor is at most $\lceil L/q \rceil$. Any source column c must satisfy $1 \le c < P$ and $c \equiv a \pmod q$, so the column factor is at most $\lceil (P-1)/q \rceil$. Therefore

$$|\pi^{-1}(t, q, a)| \le \left\lceil \frac{L}{q} \right\rceil \left\lceil \frac{P-1}{q} \right\rceil.$$

Consequently, with

$$F(t, q, a) = \left\lceil \frac{L}{q} \right\rceil \left\lceil \frac{P-1}{q} \right\rceil,$$

one has the weighted image lower bound

$$|\operatorname{im} \pi| \ge \sum_{u \in A} \frac{1}{F(\pi(u))}.$$

Thus

```
DistinctArrivalQuotientDemandLowerBoundOrCollisionPDEC
→ ArrivalQuotientFiberMultiplicityEnvelopeLedger
AND ArrivalRawSourceMassAfterNonarrivalRemoval
AND WeightedArrivalImageLowerBoundFromFiberEnvelope
AND HighFiberArrivalCollisionPDECOrDenseLowCarrierReturn.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-arrival-quotient-fiber-router.md.
```

It records

```
source_tagged_unit_incidence_imported=true
arrival_quotient_map_imported=true
fiber_row_factor_closed=true
fiber_column_factor_closed=true
arrival_fiber_envelope_closed=true
weighted_image_lower_bound_formula_closed=true
raw_arrival_mass_after_nonarrival_removal_proved=false
high_fiber_concentration_registered=true
high_fiber_collision_pdec_excluded=false
distinct_arrival_quotient_lower_bound_proved=false
row_column_unconditional_closed=false.
```

This step closes the fiber envelope and the weighted image lower-bound formula only. It does not prove enough raw arrival mass after terminal nonarrival is removed, and it does not exclude high-fiber collision or dense low-carrier returns. The row/column theorem remains open. □

Theorem 7.118 (Arrival raw-mass layer balance)**.** *Let S be the stable source-tagged history layer after no-loss projection and named history-switch removal. For $u \in S$, let $q(u)$ be its carrier and let $t_*(u) \le y-1$ be the last row in the zero block carrying the same history key. The AP successor dichotomy gives*

$$t_{\text{next}}(u) = t_*(u) + q(u),$$

and partitions

$$S = A \sqcup E,$$

where A is the source-tagged arrival layer and E is the terminal nonarrival layer. Put $H = P - y$. If $q(u) \le H$, then

$$t_*(u) + q(u) \le y - 1 + H = P - 1,$$

so $u \in A$. If $u \in E$, then

$$q(u) \ge P - t_*(u) \ge P - (y-1) = H + 1.$$

Consequently terminal nonarrival removes only the large-step tail $q > H$, and

$$|A| \ge |S_{\le H}|, \qquad S_{\le H} = \{u \in S : q(u) \le H\}.$$

Thus

```
ArrivalRawSourceMassAfterNonarrivalRemoval
→ ArrivalNonarrivalSourceLayerBalanceLedger
AND LowStepStableHistoryAlwaysArrivesLedger
AND RawArrivalMassLowerBoundFromLowStepHistory
AND LowStepStableHistoryMassLowerBoundOrLargeStepTailEscapePDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-arrival-raw-mass-layer-router.md.
```

It records

```
raw_arrival_mass_imported=true
ap_successor_dichotomy_imported=true
source_layer_universe_defined=true
arrival_nonarrival_source_layer_balance_closed=true
low_step_always_arrives_closed=true
terminal_escape_tail_cutoff_closed=true
raw_arrival_lower_bound_formula_closed=true
low_step_stable_history_mass_lower_bound_proved=false
large_step_tail_escape_registered=true
large_step_tail_escape_excluded=false
raw_arrival_mass_after_nonarrival_removal_proved=false
row_column_unconditional_closed=false.
```

This step closes the exact low-step cutoff and layer-balance formula only. It does not prove that $S_{\le H}$ has enough mass, and it does not exclude large-step tail escape. The row/column theorem remains open. □

Theorem 7.119 (Low-step/tail capacity envelope)**.** *Let $B = [x_0, y-1]$, $L = y-x_0$, and $H = P-y$. In the stable history layer, write*

$$S = S_{\le H} \sqcup S_{>H}, \qquad S_{\le H} = \{u \in S : q(u) \le H\}.$$

The preceding layer shows that $S_{\le H}$ is contained in the arrival layer. For $u \in S_{>H}$ to be terminal nonarrival, its last zero-block hit $t_*(u)$ must satisfy

$$t_*(u) + q(u) \ge P, \qquad t_*(u) \in B.$$

Hence, for a fixed $q > H$,

$$t_*(u) \in T_q := B \cap [P-q, y-1], \qquad |T_q| \le \min(L, q-H).$$

For each fixed q and $t \in T_q$, the source column must satisfy

$$c \equiv -tP \pmod q, \qquad 1 \le c < P,$$

so there are at most $\lceil (P-1)/q \rceil$ such columns. Therefore the non-duplicated large-step tail satisfies the explicit envelope

$$|S_{>H}| \le C_{\mathrm{tail}} := \sum_{H<q<P} |B \cap [P-q, y-1]| \left\lceil \frac{P-1}{q} \right\rceil,$$

and consequently, outside duplicate or tail-saturation returns,

$$|S_{\le H}| \ge |S| - C_{\mathrm{tail}}.$$

Thus

```
LowStepStableHistoryMassLowerBoundOrLargeStepTailEscapePDEC
→ StableHistoryLowTailMassPartitionLedger
AND LargeStepTailTerminalWindowEnvelopeLedger
AND LowStepStableMassLowerBoundFromTotalMinusTailEnvelope
AND StableTotalMinusTailEnvelopeGapOrLargeStepTailSaturationPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-lowstep-tail-capacity-router.md.
```

It records

```
lowstep_mass_imported=true
raw_layer_balance_imported=true
stable_low_tail_partition_closed=true
terminal_tail_row_window_closed=true
tail_column_multiplicity_closed=true
large_step_tail_envelope_closed=true
lowstep_mass_from_total_minus_tail_closed=true
large_step_tail_no_small_lcm_replay_imported=true
stable_total_minus_tail_envelope_gap_proved=false
large_step_tail_saturation_pdec_excluded=false
lowstep_stable_history_mass_lower_bound_proved=false
row_column_unconditional_closed=false.
```

This step closes the terminal tail row-window and column-multiplicity envelope only. It does not prove that $|S| - C_{\text{tail}}$ is large enough, and it does not exclude tail saturation PDEC/SAE. The row/column theorem remains open. □

Theorem 7.120 (Stable tail gap functional)**.** *Let $O_B = \{(t,c) : x_0 \le t < y,\ 1 \le c < P\}$ be the zero-block obligation domain. Then*

$$|O_B| = L(P-1).$$

By no-loss accounting, after history-switch and duplicate returns are named,

$$O_B = S \sqcup R_{\text{named}},$$

where S is the stable source layer and R_{named} is the named return layer. Hence

$$|S| \ge L(P-1) - R_{\text{named}}.$$

Combining this with the tail envelope C_{tail} from the preceding theorem gives the explicit low-step lower bound

$$|S_{\le H}| \ge L(P-1) - R_{\text{named}} - C_{\text{tail}}.$$

Thus

```
StableTotalMinusTailEnvelopeGapOrLargeStepTailSaturationPDEC
→ ZeroBlockCoverObligationMassLedger
AND StableSourceTotalAfterNamedReturnsLedger
AND ExplicitStableTailGapFunctionalLedger
AND PositiveStableTailGapOrNamedReturnMassPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-stable-tail-gap-router.md.
```

It records

```
tail_gap_imported=true
zero_block_obligation_mass_imported=true
no_loss_accounting_imported=true
stable_total_after_named_returns_closed=true
tail_envelope_imported=true
explicit_tail_gap_functional_closed=true
positive_stable_tail_gap_proved=false
named_return_mass_pdec_excluded=false
stable_total_minus_tail_envelope_gap_proved=false
row_column_unconditional_closed=false.
```

This step closes only the explicit gap functional. It does not prove that the gap is positive or sufficient, and it does not exclude named return mass or tail saturation PDEC/SAE. The row/column theorem remains open. □

Theorem 7.121 (Tail gap normalization)**.** *Let $H = P - y$ and write every large-step tail carrier as $q = H + r$. Then*

$$P - q = y - r.$$

For the zero block $B = [x_0, y-1]$ of length L, the terminal row window from the preceding theorem becomes

$$B \cap [P-q, y-1] = B \cap [y-r, y-1], \qquad |B \cap [P-q, y-1]| = \min(L, r).$$

Consequently the tail envelope has the one-dimensional form

$$C_{\mathrm{tail}} = \sum_{\substack{q=H+r<P \\ q \text{ prime}}} \min(L,r)\left\lceil \frac{P-1}{H+r} \right\rceil,$$

and is dominated by the integer envelope

$$C_{\mathrm{tail}} \le \sum_{1\le r<P-H} \min(L,r)\left\lceil \frac{P-1}{H+r} \right\rceil.$$

Thus the remaining gap separates as

$$G' = L(P-1) - C_{\mathrm{tail}}, \qquad G = G' - R_{\mathrm{named}}.$$

Hence

```
PositiveStableTailGapOrNamedReturnMassPDEC
→ TailIndexChangeOfVariablesLedger
AND ExactPrimeTailEnvelopeOneDimensionalLedger
AND StableTailGapNamedReturnSeparationLedger
AND NormalizedTailGapPositiveOrDenseTailReturnPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-normalization-router.md.
```

It records

```
positive_gap_imported=true
explicit_gap_functional_imported=true
tail_index_change_of_variables_closed=true
exact_prime_tail_envelope_closed=true
all_integer_dominating_envelope_closed=true
named_return_separation_closed=true
normalized_positive_gap_proved=false
dense_tail_return_pdec_excluded=false
positive_stable_tail_gap_proved=false
row_column_unconditional_closed=false.
```

This step closes only the $q = H + r$ normalization and the one-dimensional tail functional. It does not prove the normalized gap positive, and it does not exclude dense tail, tail saturation, or named-return mass PDEC/SAE. The row/column theorem remains open. □

Theorem 7.122 (Integer margin split for the tail gap)**.** *In the notation of the preceding theorem, define the all-integer tail envelope*

$$C_{\mathrm{all}} := \sum_{1\le r<y} \min(L,r)\left\lceil \frac{P-1}{P-y+r} \right\rceil$$

and the integer margin

$$G_{\mathrm{int}} = L(P-1) - C_{\mathrm{all}}.$$

Since the prime tail is a sub-sum of the integer tail,

$$C_{\mathrm{tail}} \le C_{\mathrm{all}}, \qquad L(P-1) - C_{\mathrm{tail}} \ge G_{\mathrm{int}}.$$

Therefore $G_{\mathrm{int}} > R_{\mathrm{named}}$ implies

$$L(P-1) - C_{\mathrm{tail}} - R_{\mathrm{named}} > 0.$$

Moreover, if P is odd and $y \le \lfloor P/2 \rfloor$, then

$$P - y + r > (P-1)/2, \qquad \left\lceil \frac{P-1}{P-y+r} \right\rceil \le 2$$

for every $1 \le r < y$. Since $1 \le L \le y$,

$$C_{\text{all}} \le 2 \sum_{1 \le r < y} \min(L, r) = L(2y - L - 1),$$

and hence

$$G_{\text{int}} \ge L(P - 2y + L) > 0.$$

Thus a pure tail envelope cannot saturate the zero-block obligation in the early-half support branch. The remaining obstruction must be a late-support dense tail or named-return mass. Hence

```
NormalizedTailGapPositiveOrDenseTailReturnPDEC
→ NormalizedIntegerTailMarginFunctionalLedger
AND PrimeTailDominatedByIntegerTailEnvelopeLedger
AND EarlyHalfSupportTailCannotSaturateLemma
AND IntegerMarginPositiveBranchCriterionLedger
AND IntegerTailMarginPositiveOrLateSupportDenseTailNamedReturnPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-integer-margin-router.md.
```

It records

```
normalized_tail_gap_imported=true
one_dimensional_tail_imported=true
integer_margin_functional_closed=true
prime_tail_dominated_by_integer_tail_closed=true
positive_branch_criterion_closed=true
early_half_support_tail_cannot_saturate_closed=true
integer_margin_positive_globally_proved=false
late_support_dense_tail_named_return_pdec_excluded=false
normalized_positive_gap_proved=false
row_column_unconditional_closed=false.
```

This step closes only the integer-margin domination and the early-half pure-tail non-saturation branch. It does not prove the global integer margin positive after named returns, and it does not exclude late-support dense-tail or named-return mass PDEC/SAE. The row/column theorem remains open. □

Theorem 7.123 (Late-support collar formula for the tail gap)**.** *Assume the late-support branch $y > (P-1)/2$, and write*

$$P = 2m + 1, \qquad H = P - y, \qquad D = 2y - P, \qquad E = y - m - 1 = (D-1)/2.$$

For $r > E$ and $r < y - 1$, one has $H + r > m$ and $H + r < P - 1$, hence

$$\left\lceil \frac{P-1}{H+r} \right\rceil = 2.$$

At the endpoint $r = y - 1$, one has $H + r = P - 1$, hence the ceiling is 1. Thus every excess above two-unit tail multiplicity is concentrated in the short core $1 \le r \le E$. Put

$$X_{\text{core}} := \sum_{1 \le r \le E} \min(L, r) \left(\left\lceil \frac{P-1}{P-y+r} \right\rceil - 2 \right).$$

Then the all-integer tail envelope decomposes exactly as

$$C_{\mathrm{all}} = L(2y - L - 1) + X_{\mathrm{core}} - \min(L, y - 1),$$

and therefore

$$G_{\mathrm{int}} = L(P - 1) - C_{\mathrm{all}} = L(L - D) + \min(L, y - 1) - X_{\mathrm{core}}.$$

Consequently

$$L(L - D) + \min(L, y - 1) > X_{\mathrm{core}} + R_{\mathrm{named}}$$

implies the positive gap $G > 0$. Hence

```
IntegerTailMarginPositiveOrLateSupportDenseTailNamedReturnPDEC
→ LateSupportExcessCoordinateLedger
AND RegularTailTwoUnitEndpointDefectLedger
AND LateCoreExcessFunctionalLedger
AND LateCollarMarginExactFormulaLedger
AND DeepLateShortCollarOrCoreExcessNamedReturnPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-late-collar-router.md.
```

It records

```
late_dense_imported=true
late_coordinate_closed=true
regular_tail_endpoint_defect_closed=true
late_core_excess_functional_closed=true
late_margin_exact_formula_closed=true
positive_late_margin_criterion_closed=true
deep_late_collar_excluded=false
core_excess_named_return_pdec_excluded=false
late_dense_tail_named_return_proved=false
row_column_unconditional_closed=false.
```

This step closes only the exact late-support collar formula. It does not exclude the deep-late short collar, and it does not control X_{core} or R_{named}. The row/column theorem remains open. □

Theorem 7.124 (Deep-late collar and core quotient layers)**.** *Let $x_0 = y - L$, $H = P - y$, and $D = 2y - P$. Then the deep-late short collar condition is equivalent to a self-mirror collar containment:*

$$L \le D \iff y - L \ge y - D \iff x_0 \ge H \iff [x_0, y - 1] \subseteq [H, y - 1].$$

If $L > D$, set

$$M = L(L - D) + \min(L, y - 1).$$

The late-collar formula gives

$$G_{\mathrm{int}} = M - X_{\mathrm{core}}.$$

Therefore $X_{\mathrm{core}} + R_{\mathrm{named}} < M$ implies the positive gap $G > 0$. If a counterexample remains in the cross-collar branch, then necessarily

$$X_{\mathrm{core}} + R_{\mathrm{named}} \ge M.$$

Moreover, with $h = H + r$ and

$$k(h) = \left\lceil \frac{P - 1}{h} \right\rceil - 2,$$

the core excess decomposes by quotient layers

$$I_k = \{h : \lceil (P-1)/h \rceil = k+2\},$$

namely

$$X_{\mathrm{core}} = \sum_{k\geq 1} k \sum_{h\in I_k} \min(L, h-H).$$

Hence

```
DeepLateShortCollarOrCoreExcessNamedReturnPDEC
→ DeepLateCollarMirrorContainmentLedger
AND CrossCollarPositiveMarginCriterionLedger
AND CoreExcessQuotientLayerDecompositionLedger
AND CoreExcessLayerConcentrationOrNamedReturnPDEC
AND SelfMirrorDeepLateCollarOrCoreLayerExcessNamedReturnPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-deep-collar-layer-router.md.
```

It records

```
deep_collar_imported=true
mirror_collar_equivalence_closed=true
cross_collar_positive_margin_closed=true
core_layer_decomposition_closed=true
core_or_named_consumption_closed=true
self_mirror_collar_excluded=false
core_layer_concentration_excluded=false
deep_late_closed=false
row_column_unconditional_closed=false.
```

This step closes only the collar-containment equivalence, the cross-collar margin criterion, and the quotient-layer decomposition of X_{core}. It does not exclude the self-mirror collar branch or quotient core-layer concentration. The row/column theorem remains open. □

Theorem 7.125 (Sieved defect form of the self-mirror tail gap). *Let*

$$w(q) = \min(L, q-H)\left\lceil \frac{P-1}{q} \right\rceil, \qquad H < q < P.$$

Write

$$C_{\mathrm{all}} = \sum_{H<q<P} w(q), \qquad C_{\mathrm{tail}} = \sum_{\substack{H<q<P \\ q \text{ prime}}} w(q),$$

and

$$D_{\mathrm{np}} = \sum_{\substack{H<q<P \\ q \text{ not prime}}} w(q).$$

Then

$$C_{\mathrm{tail}} = C_{\mathrm{all}} - D_{\mathrm{np}}.$$

Let $z = \lfloor \sqrt{P-1} \rfloor$ *and*

$$W_z = \prod_{\ell\leq z,\ \ell \text{ prime}} \ell.$$

For $z < q < P$,

$$q \text{ prime} \iff (q, W_z) = 1.$$

Therefore

$$C_{\text{tail}} = \sum_{\substack{H<q\le z\\ q \text{ prime}}} w(q) + \sum_{\substack{z<q<P\\ (q,W_z)=1}} w(q).$$

Combining this with the late-collar formula gives the true sieved gap

$$G = L(L-D) + \min(L, y-1) - X_{\text{core}} + D_{\text{np}} - R_{\text{named}}.$$

In the self-mirror branch $L \le D$, this becomes

$$G = L(L-D) + L - X_{\text{core}} + D_{\text{np}} - R_{\text{named}}.$$

Consequently, if

$$L(L-D) + \min(L, y-1) - X_{\text{core}} + D_{\text{np}} > R_{\text{named}},$$

then the positive gap is closed. If not, the remaining obstruction is a weighted $\sqrt{P}$-rough CRT over-density or named-return mass:

```
SelfMirrorDeepLateCollarOrCoreLayerExcessNamedReturnPDEC
→ PrimeTailNonprimeDefectExactLedger
AND SqrtRoughPrimeTailIdentityLedger
AND EndpointParityNonprimeDefectLowerBoundLedger
AND SievedTailGapMarginFunctionalLedger
AND SievedPositiveGapCriterionLedger
AND WeightedRoughTailCRTDefectOrNamedReturnPDEC
AND SelfMirrorSievedTailGapOrWeightedRoughCRTDefectPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-sieved-defect-router.md.
```

It records

```
self_mirror_imported=true
nonprime_defect_exact_closed=true
sqrt_rough_prime_tail_identity_closed=true
endpoint_parity_defect_lower_bound_closed=true
sieved_margin_functional_closed=true
sieved_positive_gap_criterion_closed=true
weighted_rough_crt_defect_excluded=false
self_mirror_sieved_gap_proved=false
row_column_unconditional_closed=false.
```

This step removes the purely integer-tail saturation interpretation. It does not yet prove that D_{np} always dominates X_{core} and R_{named}. The remaining branch is the weighted rough CRT defect, and the row/column theorem remains open. □

Theorem 7.126 (Least-prime-factor deletion form of the weighted rough tail)**.** *Keep*

$$w(q) = \min(L, q-H)\left\lceil \frac{P-1}{q} \right\rceil, \qquad z = \lfloor \sqrt{P-1} \rfloor.$$

Separate the finite small tail

$$D_{\text{small}} = \sum_{\substack{H<q\le z\\ q \text{ not prime}}} w(q).$$

For $z < q < P$, every non-prime q has a unique least prime factor $\ell \le z$. Hence the large non-prime defect has the disjoint decomposition

$$D_{\text{large}} = \sum_{\ell \le z} B_\ell, \qquad B_\ell = \sum_{\substack{z<q<P \\ \ell=\text{lpf}(q)}} w(q),$$

and

$$D_{\text{np}} = D_{\text{small}} + D_{\text{large}}.$$

If

$$W_{<\ell} = \prod_{p<\ell,\ p \text{ prime}} p,$$

then each cell has the CRT form

$$B_\ell = \sum_{\substack{z/\ell<n<P/\ell \\ (n,W_{<\ell})=1}} w(\ell n).$$

Equivalently, with $W_u = \prod_{p \le u} p$ and

$$S_u = \sum_{\substack{z<q<P \\ (q,W_u)=1}} w(q),$$

the prefix sieve telescopes by

$$S_{\ell^-} - S_\ell = B_\ell, \qquad S_0 - S_z = \sum_{\ell \le z} B_\ell.$$

Let

$$\Phi = L(L-D) + \min(L, y-1) - X_{\text{core}}.$$

The true gap is therefore

$$G = \Phi + D_{\text{small}} + \sum_{\ell \le z} B_\ell - R_{\text{named}}.$$

Thus $G > 0$ follows from

$$\Phi + D_{\text{small}} + \sum_{\ell \le z} B_\ell > R_{\text{named}}.$$

If a counterexample remains, then

$$\sum_{\ell \le z} B_\ell \le R_{\text{named}} - \Phi - D_{\text{small}},$$

which is a disjoint least-prime-factor deletion debt. Hence

```
SelfMirrorSievedTailGapOrWeightedRoughCRTDefectPDEC
→ SmallTailFiniteNonprimeDefectLedger
AND LargeTailLeastPrimeFactorPartitionLedger
AND RoughPrefixDeletionTelescopingLedger
AND LeastPrimeFactorCRTDeletionCellLedger
AND SievedGapLPFDeletionFunctionalLedger
AND DyadicLPFDeletionDebtOrNamedReturnPDEC
AND LPFDeletionDebtOrRoughPrefixOverdensityPDEC.
```

Status and reference. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-deletion-router.md.
```

It records

```
weighted_rough_target_imported=true
small_tail_finite_nonprime_defect_closed=true
large_tail_lpf_partition_closed=true
rough_prefix_deletion_telescoping_closed=true
lpf_crt_deletion_cell_closed=true
lpf_sieved_gap_functional_closed=true
dyadic_lpf_deletion_debt_excluded=false
lpf_deletion_debt_proved_impossible=false
row_column_unconditional_closed=false.
```

This step removes overlap ambiguity from the weighted rough branch by assigning each large non-prime tail point to its unique least prime factor. It does not prove that the resulting dyadic deletion layers always pay the gap. The row/column theorem remains open. □

Theorem 7.127 (LPF dyadic cofactor interval reduction)**.** *The remaining least-prime-factor deletion debt can be localized to explicit dyadic/quotient/cofactor CRT cells. More precisely, the open hard point*

```
LPFDeletionDebtOrRoughPrefixOverdensityPDEC
```

is reduced to

```
DyadicLPFDeletionLayerPartitionLedger
AND DyadicDebtLocalizationForAnyBudgetVectorLedger
AND RampSaturatedTailWeightSplitLedger
AND QuotientLayerCofactorIntervalLedger
AND CofactorIntervalEndpointFormulaLedger
AND RoughCofactorIntervalCRTSupportLedger
AND DyadicRoughCofactorIntervalDebtOrNamedReturnPDEC
AND DyadicRoughCofactorIntervalDebtOrColumnCRTPDEC.
```

Reduction ledger. For each dyadic least-prime-factor layer let

$$B_Y = \sum_{\substack{Y<\ell\le 2Y\\ \ell \text{ prime}}} B_\ell, \qquad B_{\text{tot}} = \sum_Y B_Y.$$

Thus, for any proposed layer budget vector A_Y, if

$$B_{\text{tot}} \le T < \sum_Y A_Y,$$

then some layer satisfies $B_Y < A_Y$. This is only finite localization; it is not an analytic lower bound.

Within one layer write

$$w(q) = u(q)v(q), \qquad u(q) = \min(L, q-H), \qquad v(q) = \left\lceil \frac{P-1}{q} \right\rceil.$$

The u-factor splits the support into the ramp region $H < q < H+L$ and the saturated region $q \ge H+L$. Next decompose by

$$Q_j = \{q : \lceil (P-1)/q \rceil = j\}.$$

Put $A = P-1$. Then

$$q_{j,\min} = \lfloor A/j \rfloor + 1, \qquad q_{j,\max} = \begin{cases} A, & j=1,\\ \lfloor A/(j-1) \rfloor, & j\ge 2. \end{cases}$$

After intersecting with $z < q < P$, the ramp cell has

$$q_{\min} = \max(z+1, H+1, q_{j,\min}), \qquad q_{\max} = \min(P-1, H+L-1, q_{j,\max}),$$

and the saturated cell has

$$q_{\min} = \max(z+1, H+L, q_{j,\min}), \qquad q_{\max} = \min(P-1, q_{j,\max}).$$

For fixed ℓ and $q = \ell n$, this gives the integer cofactor interval

$$n_{\min} = \lceil q_{\min}/\ell \rceil, \qquad n_{\max} = \lfloor q_{\max}/\ell \rfloor.$$

The LPF condition becomes $(n, W_{<\ell}) = 1$. Hence each cell has the explicit CRT form

$$B_{Y,\sigma,j} = \sum_{\substack{Y<\ell\le 2Y \\ \ell \text{ prime}}} \sum_{\substack{n_{\min}\le n\le n_{\max} \\ (n,W_{<\ell})=1}} u(\ell n) j,$$

and the dyadic layer reassembles exactly as

$$B_Y = \sum_{\sigma\in\{\text{ramp,sat}\}} \sum_j B_{Y,\sigma,j}.$$

The remaining open task is to exclude insufficient rough-cofactor support in these cells, or to route that insufficiency to a named ColumnCRT/PDEC/SAE exit. The row/column theorem is therefore still not unconditionally closed. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-dyadic-cofactor-router.md.`

□

Theorem 7.128 (Cofactor-LPF cover reduction)**.** *The remaining dyadic rough-cofactor support debt is equivalent to an explicit cofactor least-prime-factor cover debt. The open hard point*

```
DyadicRoughCofactorIntervalDebtOrColumnCRTPDEC
```

is reduced to

```
CofactorCellWeightedEnvelopeLedger
AND RoughSupportQuotaCriterionLedger
AND CofactorLeastPrimeFactorPartitionLedger
AND CofactorLPFCRTCellLedger
AND CofactorCoverPrimeProductWidthLedger
AND LocalCofactorLPFCoverDebtOrFiniteAtomPDEC
AND CofactorLPFCoverDebtOrProductWidthColumnCRTPDEC.
```

Reduction ledger. Fix a dyadic/quotient/cofactor cell $C = (Y, \sigma, j)$. Put

$$\alpha_{\ell,n} = j\, u(\ell n).$$

Let

$$F_C = \sum_{\ell\in Y} \sum_{n\in I_{\ell,j,\sigma}} \alpha_{\ell,n},$$

$$B_C = \sum_{\ell\in Y} \sum_{\substack{n\in I_{\ell,j,\sigma} \\ (n,W_{<\ell})=1}} \alpha_{\ell,n}, \qquad E_C = F_C - B_C.$$

Here F_C is the full integer envelope, B_C is the rough cofactor support, and E_C is the mass deleted by smaller cofactor factors. Therefore, for a local budget A_C,

$$B_C < A_C \iff E_C > F_C - A_C.$$

Thus rough support debt is exactly cofactor small-factor over-cover.

For every deleted cofactor, let $r = \operatorname{lpf}(n) < \ell$. The least-prime-factor partition is disjoint:

$$E_C = \sum_{\ell \in Y} \sum_{r < \ell} E_{r \leftarrow \ell}, \qquad E_{r \leftarrow \ell} = \sum_{\substack{n \in I_{\ell,j,\sigma} \\ r = \operatorname{lpf}(n)}} \alpha_{\ell,n}.$$

Each component has the CRT form

$$n = rm, \qquad (m, W_{<r}) = 1, \qquad \lceil n_{\min}/r \rceil \le m \le \lfloor n_{\max}/r \rfloor.$$

This step uses only the unique least prime factor of an integer; it does not use any short-interval prime existence input.

If a local cover debt uses the distinct cofactor primes R_C, its complete phase word has period

$$M_R = \prod_{r \in R_C} r.$$

When M_R exceeds the cell support width, exact replay cannot be a local short-period drift; it must be recorded as a finite atom, a ColumnCRT/PDEC exit, or a deeper moving-support exit. This does not yet exclude the local cofactor-LPF cover debt. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-cover-router.md.
```

□

Theorem 7.129 (Cofactor-LPF dyadic pressure reduction)**.** *The cofactor-LPF cover debt can be reduced to a product-width dichotomy and a dyadic lower-prime pressure problem. The open hard point*

```
CofactorLPFCoverDebtOrProductWidthColumnCRTPDEC
```

is reduced to

```
CofactorCoverExcessThresholdLedger
AND ActiveCofactorPrimeProductDichotomyLedger
AND DyadicCofactorPrimePressurePartitionLedger
AND OverfullDyadicRLayerLocalizationLedger
AND FixedRToMIntervalEndpointLedger
AND RLayerRoughMCRTSupportLedger
AND SmallProductActiveCoverConcentrationPDEC
AND DyadicCofactorLPFPressureOrSmallProductColumnCRTPDEC.
```

Reduction ledger. For a fixed cell C, put

$$H_C = F_C - A_C.$$

The previous ledger gives

$$B_C < A_C \iff E_C > H_C.$$

Thus a local counterexample requires explicit over-cover:

$$E_C > H_C.$$

Let

$$R_C = \{r \text{ prime} : E_r(C) > 0\}, \qquad M_C = \prod_{r \in R_C} r.$$

If M_C exceeds the support width of C, the full cover word has a period larger than the cell itself; exact replay must be treated as a product-width ColumnCRT/PDEC or finite-atom exit. If not, the cover is confined to a small-product active set.

Now decompose the active r-primes dyadically:

$$E_C = \sum_Z E_Z(C), \qquad M_C = \prod_Z M_Z.$$

For any proposed dyadic allowance U_Z, if

$$\sum_Z U_Z \le H_C$$

while $E_C > H_C$, then some dyadic layer satisfies

$$E_Z(C) > U_Z.$$

This is a finite pigeonhole localization of over-cover pressure.

Finally, for fixed r, the cofactor interval becomes

$$n = rm, \qquad m_{\min} = \lceil n_{\min}/r \rceil, \qquad m_{\max} = \lfloor n_{\max}/r \rfloor,$$

with the residual CRT condition

$$(m, W_{<r}) = 1.$$

Thus the remaining pressure is a lower-prime rough-m CRT support problem, not an invocation of the original row-gap statement. This reduction does not exclude dyadic r-layer overpressure or small-product active-cover concentration. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-dyadic-pressure-router.md.
```

□

Theorem 7.130 (Cofactor-LPF single-r pressure reduction)**.** *The small-product dyadic r-pressure branch can be localized to a single cofactor prime and then to a fixed (r, ℓ) rough-m CRT cell. The open hard point*

```
DyadicCofactorLPFPressureOrSmallProductColumnCRTPDEC
```

is reduced to

```
ActivePrimeCardinalityFromProductLedger
AND SmallZFiniteAtomBoundaryLedger
AND SingleCofactorPrimePressureLocalizationLedger
AND FixedRSourceEllPartitionLedger
AND FixedREllRoughMCRTCellLedger
AND FixedPairProductWidthColumnCRTExitLedger
AND SingleRSmallProductPressurePDEC
AND SingleCofactorPrimePressureOrFixedPairMCRTColumnCRTPDEC.
```

Reduction ledger. In a dyadic r-layer $Z < r \le 2Z$, let

$$R_Z(C) = \{r : Z < r \le 2Z,\ E_r(C) > 0\}, \qquad M_Z = \prod_{r \in R_Z(C)} r.$$

The boundary $Z < 2$ is a finite small-prime atom. For $Z \ge 2$, if the small-product branch gives $M_Z \le W_C$, where W_C is the cell support width, then each active r is $> Z$, hence

$$|R_Z(C)| \le K_Z := \lfloor \log W_C / \log Z \rfloor.$$

If the dyadic layer is over budget,

$$E_Z(C) > U_Z,$$

then some active cofactor prime satisfies

$$E_r(C) > U_Z/K_Z.$$

Thus the pressure is not merely a layer total; it localizes to a single r.

For fixed r,

$$E_r(C) = \sum_{\substack{\ell \in Y \\ \ell > r}} E_{r \leftarrow \ell}(C).$$

Applying the same finite localization to the ℓ-sources sends any remaining overpressure to one fixed pair (r, ℓ). On this pair,

$$q = \ell r m, \qquad m_{\min} = \lceil n_{\min}/r \rceil, \qquad m_{\max} = \lfloor n_{\max}/r \rfloor,$$

and

$$(m, W_{<r}) = 1, \qquad \alpha_{\ell, rm} = j\, u(\ell r m).$$

This is a fixed-pair rough-m CRT cell. The reduction does not exclude single-r or fixed-pair pressure; it only removes the remaining small-product dyadic ambiguity. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-cofactor-lpf-single-r-pressure-router.md.
```

□

Theorem 7.131 (Fixed-pair second-LPF descent reduction)**.** *The fixed-pair rough-m pressure branch admits a strict scale descent. The open hard point*

```
SingleCofactorPrimePressureOrFixedPairMCRTColumnCRTPDEC
```

is reduced to

```
FixedPairPressureImportedLedger
AND FixedPairMSupportStrictDescentLedger
AND MEqualsOneFiniteAtomLedger
AND SecondCofactorLeastPrimeFactorPartitionLedger
AND SecondLPFRoughTCRTCellLedger
AND SecondLPFProductWidthColumnCRTExitLedger
AND ResidualSupportWidthStrictDecreaseNoCycleLedger
AND SecondLPFDescentPressureOrTripleMCRTColumnCRTPDEC.
```

Reduction ledger. On a fixed pair (r, ℓ), the cell is

$$q = \ell rm, \qquad m_{\min} \le m \le m_{\max}, \qquad (m, W_{<r}) = 1,$$

with weight

$$\beta_m = j\, u(\ell rm).$$

The case $m = 1$ is the single point $q = \ell r$, hence a finite atom.

For $m > 1$, let $s = \operatorname{lpf}(m)$. Since $(m, W_{<r}) = 1$, one has

$$s \ge r.$$

Thus

$$m = st, \qquad (t, W_{<s}) = 1,$$

with

$$t_{\min} = \lceil m_{\min}/s \rceil, \qquad t_{\max} = \lfloor m_{\max}/s \rfloor.$$

This gives the disjoint second-LPF partition

$$E_{r\leftarrow\ell} = E_{m=1} + \sum_{s\ge r} E_{s\leftarrow r,\ell}.$$

Let $w_m = m_{\max} - m_{\min} + 1$ and

$$w_t = t_{\max} - t_{\min} + 1.$$

For fixed s,

$$w_t \le \lceil w_m/s \rceil \le \lceil w_m/r \rceil.$$

In the non-finite branch $r \ge 2$, so $w_t < w_m$ whenever $w_m > 1$; if $w_m \le 1$, the cell is already finite. Therefore repeated descent cannot cycle at the same support scale.

If the active second-level primes S have product

$$M_S = \prod_{s\in S} s$$

larger than w_m, the second-level phase word has period exceeding the original m-support and must be registered as a ColumnCRT/PDEC or finite atom exit. This step does not exclude the resulting second-LPF/triple rough-t pressure; it records a strict no-cycle reduction. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-fixed-pair-second-lpf-descent-router.md.`

□

Theorem 7.132 (Iterated LPF rank-budget reduction)**.** *The second-LPF/triple rough residual branch is not a free recurrent failure mode. The open hard point*

`SecondLPFDescentPressureOrTripleMCRTColumnCRTPDEC`

is reduced to

```
SecondLPFTriplePressureImportedLedger
AND IteratedLPFOrderedRoughResidualChainLedger
AND LPFSupportProductReciprocityInvariantLedger
AND LPFDepthRankBudgetLedger
AND IteratedLPFCRTWordCellLedger
AND IteratedLPFProductWidthColumnCRTExitLedger
AND TerminalResidualFiniteAtomLedger
AND IteratedLPFWellFoundedNoCycleLedger
AND RankBudgetedIteratedLPFMovingFamilyOrColumnCRTPDEC.
```

Rank-budget ledger. After the fixed-pair second-LPF step, fix (r, ℓ, s). The cell has

$$q = \ell r m, \qquad m = st, \qquad s = \operatorname{lpf}(m) \geq r, \qquad (t, W_{<s}) = 1.$$

Starting from the residual $n_0 = t$, recursively define, whenever $n_i > 1$,

$$a_i = \operatorname{lpf}(n_i), \qquad n_i = a_i n_{i+1}, \qquad (n_{i+1}, W_{<a_i}) = 1.$$

The roughness condition gives an ordered LPF word

$$r \leq s \leq a_1 \leq a_2 \leq \cdots,$$

with repetitions allowed.

Let

$$A_h = s \prod_{i \leq h} a_i.$$

If the original m-support has width w_m, then after fixing this LPF word the residual support has width at most

$$\left\lceil \frac{w_m}{A_h} \right\rceil.$$

Thus each added least-prime-factor enlarges the composite CRT period while shrinking the residual support by the same product scale. If the active word modulus product exceeds w_m, the same phase word can only contribute an isolated atom in the original support and is registered as a ColumnCRT/PDEC or finite-atom exit.

On the non-finite branch $r \geq 2$. If no product-width exit has occurred, then $A_h \leq w_m$. Since every factor in the word is at least r, the total factor depth d, including the initial second-level factor s, satisfies

$$d \leq \lfloor \log_r w_m \rfloor.$$

Consequently the triple branch cannot hide an infinite same-scale LPF tower. The remaining obstruction is a rank-budgeted iterated-LPF moving family, not an unstructured triple pressure. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-iterated-lpf-rank-budget-router.md.
```

□

Theorem 7.133 (LPF word-entropy and first-moving-coordinate reduction)**.** *The rank-budgeted iterated-LPF moving-family branch admits a finite word partition and a first-motion reduction. The open hard point*

```
RankBudgetedIteratedLPFMovingFamilyOrColumnCRTPDEC
```

is reduced to

```
RankBudgetedMovingFamilyImportedLedger
AND IteratedLPFWordSignaturePartitionLedger
AND LPFWordEntropyFiniteCapLedger
AND AggregatePressureToSingleLPFWordLedger
AND FixedLPFWordColumnCRTExitLedger
AND FirstMovingLPFCoordinateLedger
AND MovingCoordinateSupportReciprocityLedger
AND NoAnonymousRankBudgetedMovingFamilyLedger
AND FirstMovingLPFCoordinatePressureOrWordMotionColumnCRTPDEC.
```

Word-motion ledger. By uniqueness of least prime factors, each residual lies in a unique ordered word

$$\omega = (s, a_1, \dots, a_d), \qquad r \le s \le a_1 \le \cdots \le a_d,$$

with product

$$A(\omega) = s \prod_{i \le d} a_i.$$

If $A(\omega) > W$, where W is the original m-support width, the word has already crossed the product-width boundary and is registered as a ColumnCRT/PDEC or finite atom. Otherwise

$$A(\omega) \le W.$$

Since all factors in the non-finite branch are at least $r \ge 2$, the total factor depth obeys

$$d + 1 \le \lfloor \log_r W \rfloor.$$

Thus the active word set $\Omega(P, C)$ is finite and has an explicit entropy budget $H(W, r)$.

The pressure decomposes over disjoint word cells:

$$E(\Omega) = \sum_{\omega \in \Omega} E_\omega.$$

If $E(\Omega) > U$, then one word satisfies

$$E_\omega > \frac{U}{|\Omega|}.$$

Hence any remaining excess must be carried by a single LPF word, not by an anonymous moving-family reservoir.

If that pressure-carrying word has stable prime coordinates and stable phase residues along the family, it is a fixed MCRT word and is routed to the fixed word ColumnCRT/PDEC or finite-atom exit. If not, there is a first moving coordinate μ. Let A_{prefix} be the product of the stable word prefix before μ. After μ is imposed, the residual support width is bounded by

$$\left\lceil \frac{W}{A_{\text{prefix}} \mu} \right\rceil.$$

When $A_{\text{prefix}} \mu > W$, this is again a product-width ColumnCRT/PDEC or finite-atom exit. Otherwise the remaining obstruction is the named first-moving LPF coordinate pressure. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-lpf-word-entropy-motion-router.md.`

□

Theorem 7.134 (First-moving LPF coordinate reduction)**.** *The first-moving LPF coordinate branch has a dyadic product-width reduction. The open hard point*

```
FirstMovingLPFCoordinatePressureOrWordMotionColumnCRTPDEC
```

is reduced to

```
FirstMovingLPFCoordinatePressureImportedLedger
AND StablePrefixProductSupportLedger
AND FirstMovingCoordinateEffectiveWidthLedger
AND LowMovingCoordinateFiniteAtomLedger
AND FirstMovingCoordinateDyadicPartitionLedger
AND MovingCoordinateActiveProductWidthExitLedger
AND MovingCoordinateActiveCountBoundLedger
AND SingleMovingCoordinatePressureLocalizationLedger
AND FixedMovingCoordinateDegeneratesToColumnCRTLedger
AND NoAnonymousFirstMovingCoordinatePoolLedger
AND SingleMovingLPFCoordinateDriftOrPrefixColumnCRTPDEC.
```

Coordinate ledger. Let μ be the first LPF coordinate that moves in the pressure-carrying word. The word prefix before μ is stable; write its product as

$$A_{\text{prefix}}.$$

If W is the original m-support width, define the effective coordinate width

$$W_{\text{prefix}} = \left\lfloor \frac{W}{A_{\text{prefix}}} \right\rfloor.$$

When $\mu > W_{\text{prefix}}$, the composite period

$$A_{\text{prefix}}\mu$$

already exceeds W, so the branch is a product-width ColumnCRT/PDEC or a finite atom. The low coordinate case $\mu \le 2$ is also finite.

For the remaining coordinates, partition into dyadic layers

$$B < \mu \le 2B, \qquad B = 2^b, \qquad B \ge 2, \qquad B \le W_{\text{prefix}}.$$

Fix such a layer and let M_B be its active coordinate set. If

$$\prod_{\mu \in M_B} \mu > W_{\text{prefix}},$$

then the active coordinate word has period exceeding the effective support and is again a Column-CRT/PDEC or finite-atom exit. Otherwise, since every $\mu > B$,

$$|M_B| \le \left\lfloor \frac{\log W_{\text{prefix}}}{\log B} \right\rfloor.$$

Thus the active coordinate pool has a finite logarithmic capacity.

The layer pressure decomposes over disjoint coordinate cells:

$$E_B = \sum_{\mu \in M_B} E_\mu.$$

If $E_B > U_B$, then some coordinate satisfies

$$E_\mu > \frac{U_B}{|M_B|}.$$

If that coordinate is stable along the family, the supposed first motion degenerates to a fixed word coordinate and is routed to the prefix ColumnCRT/PDEC or finite-atom exit. If it is not stable, the remaining obstruction is a single moving LPF coordinate drift. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-first-moving-lpf-coordinate-router.md.`

□

Theorem 7.135 (Single-moving LPF coordinate scale-escape reduction)**.** *The single moving LPF coordinate drift has a scale-escape reduction. The open hard point*

SingleMovingLPFCoordinateDriftOrPrefixColumnCRTPDEC

is reduced to

SingleMovingLPFCoordinateDriftImportedLedger
AND SingleMovingCoordinateStablePrefixLedger
AND SingleMovingCoordinateDyadicScaleLedger
AND BoundedScaleDriftDegeneratesToFixedCoordinateLedger
AND UnboundedCoordinateScaleEscapeLedger
AND PostMovingCoordinateSupportDescentLedger
AND SameScaleCoordinateCycleExcludedLedger
AND SparseCoordinateDriftSAERegistrationLedger
AND NoAnonymousSingleCoordinateDriftLedger
AND ScaleEscapingSingleCoordinateDescentOrSparseDriftSAEColumnCRTPDEC.

Scale-escape ledger. Import the remaining branch in which a single LPF coordinate μ moves with the family. The prefix before μ is stable; with original support width W, set

$$W_{\text{prefix}} = \left\lfloor \frac{W}{A_{\text{prefix}}} \right\rfloor.$$

Assign to μ its dyadic scale

$$B(\mu) = 2^{\lfloor \log_2 \mu \rfloor}, \qquad B(\mu) \le \mu \le 2B(\mu).$$

If $B(\mu)$ is bounded on an infinite subfamily, then μ ranges over only finitely many primes. By the infinite pigeonhole principle, an infinite subfamily has fixed μ. That branch is no longer a moving coordinate drift; it is the already named fixed-coordinate ColumnCRT/PDEC or finite-atom exit. Therefore any genuine single-coordinate drift must satisfy

$$B(\mu) \to \infty$$

on a cofinal subfamily.

After imposing the coordinate μ, the residual support width satisfies

$$W_{\text{after}} \le \left\lceil \frac{W_{\text{prefix}}}{\mu} \right\rceil \le \left\lceil \frac{W_{\text{prefix}}}{B(\mu)} \right\rceil.$$

On every non-finite active layer $B(\mu) \ge 2$, so this gives a strict support descent before any same-scale cycle can recur. Thus the anonymous single-coordinate drift is removed: bounded scales degenerate to fixed coordinates, true moving scales descend, and nonpersistent drift is registered as a sparse drift/SAE exit.

This does not yet exclude scale-escaping coordinate descent, and it does not prove global summability of the sparse drift/SAE mass. The row/column theorem therefore remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-single-moving-lpf-coordinate-scale-escape-router.md.`

□

Theorem 7.136 (Scale-escape support-clock reduction). *The scale-escaping single-coordinate descent has an integer support clock. The open hard point*

`ScaleEscapingSingleCoordinateDescentOrSparseDriftSAEColumnCRTPDEC`

is reduced to

```
ScaleEscapingSingleCoordinateDescentImportedLedger
AND ScaleEscapeIntegerSupportClockLedger
AND ScaleEscapeHalvingClockDescentLedger
AND FiniteDepthScaleEscapePerFiberLedger
AND TerminalWidthOneFiniteAtomLedger
AND ScaleLadderProductWidthColumnCRTExitLedger
AND PersistentScaleLadderSignatureRegistrationLedger
AND SparseScaleLadderSAERegistrationLedger
AND NoCyclicScaleEscapeDescentLedger
AND NoAnonymousScaleEscapeDescentLedger
AND PersistentScaleLadderSignatureOrSparseScaleLadderSAEColumnCRTPDEC.
```

Support-clock ledger. Let W_i be the effective residual support width after i scale-escaping coordinates have been imposed. Define the integer clock

$$K(W) = \lceil \log_2 \max(W, 1) \rceil .$$

Thus $K(W) = 0$ exactly at terminal width $W \leq 1$, which is a finite atom or an already named boundary.

For a genuine scale-escaping coordinate at step i, its dyadic scale B_i satisfies $B_i \geq 2$. The previous scale-escape ledger gives

$$W_{i+1} \leq \left\lceil \frac{W_i}{B_i} \right\rceil \leq \left\lceil \frac{W_i}{2} \right\rceil .$$

If $W_i \geq 2$, then

$$K(W_{i+1}) \leq K(W_i) - 1.$$

Hence a fixed counterexample fiber has at most $K(W_0)$ scale-escape steps. No scale-escape descent can cycle back to the same support-clock layer.

If the descent reaches $W \leq 1$, the branch is terminal. If the dyadic scale ladder satisfies

$$\prod_i B_i > W_0,$$

then the composite period exceeds the original support and the branch is a product-width Column-CRT/PDEC or finite-atom exit. Therefore an infinite obstruction that is not absorbed by terminal width, product width, or nonpersistent sparse mass must carry a named ordered scale-ladder signature with its phase data.

This removes the anonymous scale-escape descent and the possibility of an internal scale-escape cycle. It does not yet exclude persistent scale-ladder signatures, and it does not prove global summability of the sparse scale-ladder SAE mass. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-scale-escape-support-clock-router.md.`

□

Theorem 7.137 (Scale-ladder word-entropy reduction). *The persistent scale-ladder signature branch has a finite dyadic word entropy budget. The open hard point*

```
PersistentScaleLadderSignatureOrSparseScaleLadderSAEColumnCRTPDEC
```

is reduced to

```
PersistentScaleLadderSignatureImportedLedger
AND ScaleLadderDyadicWordPartitionLedger
AND ScaleLadderProductBudgetLedger
AND ScaleLadderWordEntropyFiniteCapLedger
AND AggregatePersistentPressureToSingleScaleWordLedger
AND FixedScaleLadderMCRTColumnCRTExitLedger
AND FirstMovingScaleLadderPhaseCoordinateLedger
AND SparseScaleLadderSAECarriedForwardLedger
AND NoAnonymousPersistentScaleLadderSignatureLedger
AND FirstMovingScaleLadderPhaseDriftOrSparseScaleLadderSAEColumnCRTPDEC.
```

Scale-word ledger. Write the persistent dyadic scale ladder as

$$B_i = 2^{b_i}, \qquad b_i \geq 1, \qquad \sigma = (b_1, \ldots, b_d).$$

Its scale product is

$$A_\sigma = \prod_i B_i = 2^{\sum_i b_i}.$$

If $A_\sigma > W_0$, the composite period already exceeds the initial support and the branch is a product-width ColumnCRT/PDEC or finite atom. Otherwise

$$\sum_i b_i \leq K_0 = \lceil \log_2 \max(W_0, 1) \rceil .$$

The number of ordered positive-integer scale words is therefore finite:

$$N_{\text{ladder}}(K_0) = 1 + \sum_{n=1}^{K_0} 2^{n-1} \leq 2^{K_0}.$$

The persistent pressure decomposes over these scale-word cells,

$$E_{\text{total}} = \sum_\sigma E_\sigma.$$

If $E_{\text{total}} > U$, then at least one scale word satisfies

$$E_\sigma > \frac{U}{N_{\text{ladder}}(K_0)}.$$

Thus a persistent scale-ladder obstruction cannot remain an anonymous signature reservoir.

For the pressure-carrying word σ, if all actual prime coordinates and phase residues stabilize on an infinite subfamily, the branch is a fixed MCRT word and is routed to ColumnCRT/PDEC or a finite atom. If not, there is a first moving prime coordinate or phase residue inside the fixed scale word. That is the remaining named hard point. The sparse scale-ladder SAE outlet is only carried forward here; its global summability is not proved. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-scale-ladder-word-entropy-router.md.
```

□

Theorem 7.138 (Scale-ladder finite-slot lock). *Inside a fixed dyadic scale word, persistent first-moving phase drift is impossible. The open hard point*

```
FirstMovingScaleLadderPhaseDriftOrSparseScaleLadderSAEColumnCRTPDEC
```

is reduced to

```
FirstMovingScaleLadderPhaseDriftImportedLedger
AND FixedScaleWordSlotLedger
AND FinitePrimeChoicesPerScaleSlotLedger
AND FiniteResidueChoicesPerPrimeSlotLedger
AND FiniteActualScaleLadderAtomSetLedger
AND InfinitePigeonholeStableActualScaleLadderLedger
AND NoPersistentFirstMovingScaleLadderPhaseDriftLedger
AND StableActualScaleLadderMCRTColumnCRTExitLedger
AND SparseScaleLadderSAECarriedForwardAfterFiniteSlotLockLedger
AND NoAnonymousFirstMovingScaleLadderPhaseDriftLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderColumnCRTPDEC.
```

Finite-slot ledger. Fix the pressure-carrying dyadic scale word

$$\sigma = (b_1, \ldots, b_d),$$

so each b_i is fixed. The actual prime in slot i must lie in the finite dyadic set

$$Q_i = \{q \text{ prime} : 2^{b_i} \le q < 2^{b_i+1}\}.$$

For a fixed $q \in Q_i$, the phase residue has only q possible values. Hence the number of actual prime-residue ladder atoms under σ is finite, with the crude bound

$$N_{\text{actual}}(\sigma) \le \prod_i \sum_{q \in Q_i} q < \infty.$$

If a fixed scale word carries persistent pressure along an infinite subfamily, the infinite pigeonhole principle gives an infinite subfamily on which the entire actual prime-residue ladder atom is fixed. Such a stable actual ladder is a fixed MCRT cell and is routed to the ColumnCRT/PDEC or finite-atom exit.

Therefore a first moving prime coordinate or phase residue cannot be a persistent obstruction inside a fixed scale word. It is either fixed on an infinite subfamily, or it is nonpersistent and is registered as sparse scale-ladder SAE. This does not prove global summability of that sparse mass, and it does not exclude the stable actual ladder ColumnCRT/PDEC exit. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-scale-ladder-finite-slot-lock-router.md.
```

□

Theorem 7.139 (Stable-ladder Fourier/PDEC bridge)**.** *The stable actual ladder ColumnCRT exit can be written as an explicit finite Fourier/PDEC input. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderColumnCRTPDEC
```

is reduced to

```
StableActualLadderOrSparseSAEImportedLedger
AND StableActualLadderFiniteGroupLedger
AND StableActualLadderZeroMeanCellFunctionLedger
AND StableActualLadderExactExcessIdentityLedger
AND StableActualLadderFourierPDECBridgeLedger
AND StableActualLadderNontrivialCharacterLowerBoundLedger
AND NoAnonymousStableActualLadderColumnCRTExitLedger
AND SparseScaleLadderSAECarriedForwardAfterFourierBridgeLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderFourierPDECCap.
```

Finite-group Fourier ledger. A stable actual ladder fixes a finite list of prime-residue pairs

$$((q_i, a_i))_{i=1}^{d}.$$

Work on the finite product group

$$G = \prod_{i=1}^{d} \mathbb{Z}/q_i\mathbb{Z}, \qquad N = |G| = \prod_{i=1}^{d} q_i,$$

with

$$\tau(n) = (n \bmod q_i)_i, \qquad a = (a_i)_i.$$

This product-group form does not require the displayed primes to be distinct. Define the zero-mean cell function

$$F_a(g) = 1_{\{a\}}(g) - \frac{1}{N}.$$

Then for every support set S the stable-ladder excess is exactly

$$E_a(S) = \sum_{n \in S} F_a(\tau(n)) = |\{n \in S : \tau(n) = a\}| - \frac{|S|}{N}.$$

Expanding F_a in the character group $\widehat{G}$, the trivial character coefficient vanishes. Hence

$$E_a(S) = \sum_{\chi \in \widehat{G},\ \chi \neq 1} \widehat{F}_a(\chi)\, S_\chi, \qquad S_\chi = \sum_{n \in S} \chi(\tau(n)).$$

For the point-mass-minus-average function one has $|\widehat{F}_a(\chi)| = 1/N$ for nontrivial χ. Therefore, if $E_a(S) > 0$, then some nontrivial character satisfies

$$|S_\chi| \geq \frac{N\, E_a(S)}{N-1}.$$

Thus a stable actual ladder cannot remain an anonymous ColumnCRT outlet: any positive ladder excess forces a named nontrivial Fourier correlation. What remains is the explicit stable-ladder Fourier/PDEC cap, or the separate global summability of the sparse scale-ladder SAE outlet. This theorem does not prove either remaining input, and the row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-fourier-pdec-router.md.
```

□

Theorem 7.140 (Stable-ladder pivot-fiber PDEC localization)**.** *The stable-ladder Fourier/PDEC cap has a primitive one-coordinate fiber form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderFourierPDECCap
```

is reduced to

```
StableActualLadderFourierCapImportedLedger
AND StableLadderCharacterFactorizationLedger
AND StableLadderNontrivialPivotCoordinateLedger
AND StableLadderComplementFiberPartitionLedger
AND StableLadderGlobalCharacterToPivotFiberLocalizationLedger
AND StableLadderPrimitivePivotCharacterCorrelationLedger
AND NoAnonymousStableLadderFourierCapLedger
AND SparseScaleLadderSAECarriedForwardAfterPivotFiberLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderPrimitiveFiberPDECCap.
```

Pivot-fiber ledger. Let

$$G = \prod_{i=1}^{d} \mathbb{Z}/q_i\mathbb{Z}, \qquad N = |G| = \prod_{i=1}^{d} q_i.$$

Every character of G factors as

$$\chi(g) = \prod_{i=1}^{d} \chi_i(g_i).$$

If χ is nontrivial, choose a pivot coordinate j with $\chi_j \neq 1$. Write

$$G_{-j} = \prod_{i \neq j} \mathbb{Z}/q_i\mathbb{Z}, \qquad |G_{-j}| = \frac{N}{q_j}.$$

For $h \in G_{-j}$, split the support set by the complementary coordinates,

$$S_h = \{n \in S : \tau_{-j}(n) = h\},$$

and set

$$A_h = \sum_{n \in S_h} \chi_j(n \bmod q_j).$$

Then the global character sum decomposes exactly as

$$S_\chi = \sum_{h \in G_{-j}} \chi_{-j}(h)\, A_h.$$

Since $|\chi_{-j}(h)| = 1$, the triangle inequality gives

$$|S_\chi| \leq \sum_{h \in G_{-j}} |A_h| \leq |G_{-j}| \max_{h \in G_{-j}} |A_h|.$$

Thus if $|S_\chi| \geq \Lambda$, some complementary-coordinate fiber satisfies

$$|A_h| \geq \frac{\Lambda}{|G_{-j}|}.$$

Taking the lower bound from the preceding Fourier bridge,

$$\Lambda = \frac{N\, E_a(S)}{N-1},$$

one obtains the primitive fiber threshold

$$|A_h| \geq \frac{q_j\, E_a(S)}{N-1}.$$

Equivalently, the global stable-ladder Fourier cap is witnessed by a single nontrivial coordinate character on one fixed complementary fiber. On the cyclic factor $\mathbb{Z}/q_j\mathbb{Z}$, such a character has the form

$$r \mapsto \exp(2\pi i c_j r/q_j), \qquad c_j \not\equiv 0 \pmod{q_j}.$$

So the remaining cap is a primitive pivot-fiber PDEC statement, not an anonymous global Fourier outlet.

This theorem does not prove that primitive cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem therefore remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-pivot-fiber-pdec-router.md.`

□

Theorem 7.141 (Stable-ladder residue-count PDEC localization). *The primitive pivot-fiber character cap has an equivalent residue-count imbalance form. The open hard point*

`SparseScaleLadderSAESummabilityOrStableActualLadderPrimitiveFiberPDECCap`

is reduced to

```
StableActualLadderPrimitiveFiberPDECImportedLedger
AND StableLadderPivotFiberResidueCountVectorLedger
AND StableLadderPivotFiberZeroMeanDeviationLedger
AND StableLadderPivotFiberCharacterToResidueDeviationLedger
AND StableLadderPivotFiberResidueImbalanceLocalizationLedger
AND StableLadderPivotFiberResidueImbalanceThresholdLedger
AND NoAnonymousPrimitiveFiberCharacterPDECExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueCountLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderResidueCountPDECCap.
```

Residue-count ledger. Keep the pivot coordinate j, the modulus q_j, and the complementary fiber S_h from the preceding reduction. Let

$$L = |S_h|, \qquad M_r = |\{n \in S_h : n \equiv r \pmod{q_j}\}|, \qquad r \in \mathbb{Z}/q_j\mathbb{Z},$$

and define the centered residue-count deviations

$$D_r = M_r - \frac{L}{q_j}.$$

Then $\sum_r D_r = 0$. For the nontrivial pivot character χ_j, the primitive fiber sum is

$$A_h = \sum_{n \in S_h} \chi_j(n \bmod q_j) = \sum_r M_r \chi_j(r).$$

Since $\sum_r \chi_j(r) = 0$, the average part vanishes and

$$A_h = \sum_r D_r \chi_j(r).$$

Thus

$$|A_h| \le \sum_r |D_r| \le q_j \max_r |D_r|.$$

The preceding pivot-fiber theorem gives the lower bound

$$|A_h| \ge \frac{q_j\, E_a(S)}{N-1}.$$

Consequently at least one residue class r satisfies

$$\left| M_r - \frac{L}{q_j} \right| \ge \frac{E_a(S)}{N-1}.$$

Therefore the primitive fiber character outlet is not anonymous: it is a specific residue-count imbalance inside a fixed complementary-coordinate fiber.

This theorem does not prove the residue-count PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-residue-count-pdec-router.md.`

□

Theorem 7.142 (Stable-ladder positive-surplus PDEC localization). *The signed residue-count imbalance cap has a positive-surplus form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderResidueCountPDECCap
```

is reduced to

```
StableActualLadderResidueCountPDECImportedLedger
AND StableLadderResidueDeviationSignDichotomyLedger
AND StableLadderResidueDeviationZeroSumTransferLedger
AND StableLadderPositiveResidueSurplusLocalizationLedger
AND StableLadderPositiveResidueSurplusThresholdLedger
AND NoAnonymousSignedResidueImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPositiveSurplusLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderPositiveResidueSurplusPDECCap.
```

Positive-surplus ledger. Keep the fixed complementary fiber and write

$$D_r = M_r - \frac{L}{q_j}, \qquad \sum_r D_r = 0.$$

The preceding residue-count localization gives a residue r with

$$|D_r| \geq \delta, \qquad \delta = \frac{E_a(S)}{N-1}.$$

If $D_r \geq \delta$, the desired positive surplus is already present. If $D_r \leq -\delta$, then zero sum gives

$$\sum_{s \neq r} D_s = -D_r \geq \delta.$$

There are $q_j - 1$ residue classes with $s \neq r$, so one of them satisfies

$$D_s \geq \frac{\delta}{q_j - 1}.$$

Thus in all cases there is a residue class s with

$$M_s - \frac{L}{q_j} \geq \frac{E_a(S)}{(N-1)(q_j-1)}.$$

Hence an absolute signed residue imbalance can be routed to a positive overload residue class, with only the explicit $q_j - 1$ dilution in the threshold. This theorem does not prove the positive residue-surplus PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-positive-surplus-pdec-router.md.`

□

Theorem 7.143 (Stable-ladder occupancy dichotomy). *The positive residue-surplus cap has an integer occupancy form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderPositiveResidueSurplusPDECCap
```

is reduced to

```
StableActualLadderPositiveResidueSurplusImportedLedger
AND StableLadderOverloadedCellIntegerOccupancyLedger
AND StableLadderPositiveSurplusSingletonOrPairDichotomyLedger
AND StableLadderSingletonSurplusAtomRegistrationLedger
AND StableLadderSameCellPairCongruenceLedger
AND StableLadderSameCellPairPeriodMultipleLedger
AND NoAnonymousPositiveResidueSurplusExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOccupancyDichotomyLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderSingletonSurplusOrCellPairPeriodPDECCap.
```

Occupancy-dichotomy ledger. Import the positive overload residue class from the preceding localization:

$$M_s - \frac{L}{q_j} \geq \eta, \qquad \eta = \frac{E_a(S)}{(N-1)(q_j-1)} > 0.$$

Here M_s is a nonnegative integer occupancy number inside the fixed complementary fiber. Hence $M_s \geq 1$. There are exactly two cases:

$$M_s = 1 \qquad \text{or} \qquad M_s \geq 2.$$

The first case is recorded as a singleton surplus atom. This theorem does not prove summability of all such singleton atoms.

In the second case, two distinct points $n_1 < n_2$ lie in the same complete stable-ladder cell. Let

$$W = \mathrm{lcm}_i(q_i)$$

be the period of the ladder coordinates. Equality of all cell coordinates gives

$$n_2 - n_1 \equiv 0 \pmod{q_i} \quad \text{for every } i,$$

and therefore

$$W \mid (n_2 - n_1), \qquad n_2 - n_1 > 0.$$

Consequently any upstream width bound $H < W$ rules out the pair branch at once; without such a bound the remaining obstruction is the explicit same-cell period-pair PDEC cap.

Thus the positive residue-surplus outlet is no longer anonymous: it is either a singleton surplus atom or a same-cell period-pair outlet. The theorem does not prove the singleton summability cap, it does not prove the same-cell period-pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-occupancy-dichotomy-router.md.
```

□

Theorem 7.144 (Stable-ladder singleton/pair width reduction)**.** *The singleton-or-period-pair outlet has a low-fiber/long-width form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderSingletonSurplusOrCellPairPeriodPDECCap
```

is reduced to

```
StableActualLadderSingletonOrPairPeriodImportedLedger
AND StableLadderSingletonSurplusLowFiberQuotaLedger
AND StableLadderSameCellPairSupportWidthDichotomyLedger
AND StableLadderShortSupportPairExclusionLedger
AND StableLadderLongWidthSameCellPeriodPairRegistrationLedger
AND NoAnonymousSingletonOrPairPeriodExitLedger
AND SparseScaleLadderSAECarriedForwardAfterSingletonPairWidthLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderLowFiberSingletonSurplusOrLongWidthCellPairPDECCap.
```

Singleton/pair width ledger. For the singleton branch, combine $M_s = 1$ with the positive overload

$$M_s - \frac{L}{q_j} \geq \eta > 0.$$

This gives

$$1 - \frac{L}{q_j} > 0, \qquad L < q_j.$$

Thus a singleton surplus atom cannot live in a high pivot fiber $L \geq q_j$. It is a low-fiber singleton event and remains to be summed or excluded there.

For the pair branch, the preceding occupancy theorem gives two points $n_1 < n_2$ in the same stable-ladder cell. With

$$W = \mathrm{lcm}_i(q_i),$$

there is an integer $t \geq 1$ such that

$$n_2 - n_1 = tW \geq W.$$

If H is the diameter of any support carrying both points, then

$$H \geq n_2 - n_1 \geq W.$$

Consequently the short-support case $H < W$ excludes the pair branch. The only remaining pair obstruction is a named long-width same-cell period-pair PDEC cap.

Therefore the singleton-or-period-pair outlet is not anonymous: it is either a low-fiber singleton outlet or a long-width same-cell period-pair outlet. This theorem does not prove the low-fiber singleton summability cap, it does not prove the long-width period-pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-singleton-pair-width-router.md.
```

□

Theorem 7.145 (Stable-ladder low-fiber phase reduction)**.** *The low-fiber singleton outlet has an isolated-or-anti-pivot form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderLowFiberSingletonSurplusOrLongWidthCellPairPDECCap
```

is reduced to

```
StableActualLadderLowFiberSingletonOrLongPairImportedLedger
AND StableLadderLowFiberOccupancyOneOrMultiDichotomyLedger
AND StableLadderFiberIsolatedSingletonAtomLedger
AND StableLadderRepeatedPivotModulusForcesIsolationLedger
AND StableLadderComplementFiberAntiPivotPairLedger
AND StableLadderComplementFiberPeriodAndAntiPivotPhaseLedger
AND StableLadderComplementFiberPairWidthDichotomyLedger
AND StableLadderLongFullCellPairCarriedForwardLedger
AND NoAnonymousLowFiberSingletonExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLowFiberPhaseLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrAntiPivotComplementPairOrLongFullCellPairPDECCa
```

Low-fiber phase ledger. The preceding theorem leaves the low-fiber singleton branch with

$$M_s = 1, \qquad 1 \leq L < q_j.$$

Hence

$$L = 1 \qquad \text{or} \qquad 2 \leq L < q_j.$$

The case $L = 1$ is a singleton atom isolated inside its fixed complementary fiber.

Assume next that $2 \le L < q_j$. Take the point in the singleton residue class s, and take another point in the same complementary fiber. These two points have identical residues in every coordinate $i \neq j$, but their pivot residues modulo q_j are distinct because $M_s = 1$. Let

$$W_{-j} = \mathrm{lcm}_{i\neq j}(q_i),$$

with $W_{-j} = 1$ when the complementary list is empty. Then the two points $n_1 < n_2$ satisfy

$$W_{-j} \mid (n_2 - n_1), \qquad q_j \nmid (n_2 - n_1).$$

If $q_j \mid W_{-j}$, the fixed complementary fiber already fixes the pivot residue modulo q_j, so the alternative $2 \le L < q_j$ is impossible and the branch collapses to $L = 1$.

If H is the support diameter carrying this complementary-fiber pair, then

$$H \ge n_2 - n_1 \ge W_{-j}.$$

Thus $H < W_{-j}$ excludes the anti-pivot complement-pair branch; otherwise it is a named complement-pair PDEC cap. The previous long full-cell pair branch, with $H \ge W = \mathrm{lcm}_i(q_i)$, is carried forward unchanged.

Therefore the low-fiber singleton outlet is no longer anonymous: it is either a fiber-isolated singleton atom, or a complement-period aligned but pivot anti-aligned pair. This theorem does not prove summability of isolated singletons, it does not prove the anti-pivot complement-pair cap, it does not prove the long full-cell pair cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-low-fiber-phase-router.md.
```

□

Theorem 7.146 (Stable-ladder pair quotient-phase reduction). *The anti-pivot complement-pair outlet and the long full-cell pair outlet have a single quotient-phase form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrAntiPivotComplementPairOrLongFullCellPairPDECCap
```

is reduced to

```
StableActualLadderIsolatedSingletonOrPairImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardLedger
AND StableLadderComplementBasePeriodLedger
AND StableLadderPairDifferenceQuotientNormalizationLedger
AND StableLadderPivotPhaseQuotientPeriodLedger
AND StableLadderAntiPivotPairNonzeroQuotientPhaseLedger
AND StableLadderFullCellPairZeroQuotientPhaseLedger
AND StableLadderPairQuotientWidthEnvelopeLedger
AND StableLadderQuotientNoWrapOrFullPeriodDichotomyLedger
AND NoAnonymousAntiPivotOrFullPairExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPairQuotientPhaseLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrPairQuotientPhaseCyclePDECCap.
```

Pair quotient-phase ledger. The isolated singleton atom is carried forward as its own summability/cap outlet. For the pair outlets, use the complementary-coordinate period

$$B = W_{-j} = \mathrm{lcm}_{i\neq j}(q_i),$$

with $B = 1$ for an empty complementary list. Every complement-fiber pair has a unique quotient

$$d = n_2 - n_1 = tB, \qquad t \ge 1.$$

Set

$$g = \gcd(q_j, B), \qquad R = \frac{q_j}{g}.$$

Then the pivot phase $d \bmod q_j$ is determined exactly by $t \bmod R$. Indeed $q_j \mid tB$ if and only if $R \mid t$.

Therefore the anti-pivot complement-pair branch is precisely the nonzero quotient phase branch

$$t \not\equiv 0 \pmod R,$$

whereas the full-cell pair branch is the zero quotient phase branch

$$t \equiv 0 \pmod R, \qquad t = Ru.$$

If H is the support diameter, then

$$1 \le t \le \left\lfloor \frac{H}{B} \right\rfloor.$$

Thus the quotient interval either has no complete pivot phase period, $\lfloor H/B \rfloor < R$, or contains a complete quotient phase cycle, $\lfloor H/B \rfloor \ge R$. This is the remaining explicit quotient-phase PDEC/cap input.

The two pair outlets are therefore no longer separate anonymous exits; they are the nonzero and zero residues of the same quotient variable modulo R. This theorem does not prove summability of isolated singletons, it does not prove the quotient phase cycle cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-pair-quotient-phase-router.md.`

□

Theorem 7.147 (Stable-ladder quotient short-arc/mean reduction)**.** *The quotient-phase cycle outlet has a non-cyclic short-arc/mean decomposition. The open hard point*

`SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrPairQuotientPhaseCyclePDECCap`

is reduced to

```
StableActualLadderQuotientPhaseCycleImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterArcMeanLedger
AND StableLadderQuotientSpanParameterLedger
AND StableLadderQuotientNoWrapZeroPhaseExclusionLedger
AND StableLadderQuotientShortArcClusterLedger
AND StableLadderFullCycleEuclideanDecompositionLedger
AND StableLadderFullCycleFormalPhaseMeanLedger
AND StableLadderFullCycleResidualTailArcLedger
AND NoAnonymousQuotientPhaseCycleExitLedger
AND SparseScaleLadderSAECarriedForwardAfterArcMeanLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientShortArcClusterOrPhaseCycleMeanPDECCap.
```

Short-arc/mean ledger. Keep the notation of the pair quotient-phase reduction:

$$B = W_{-j}, \qquad g = \gcd(q_j, B), \qquad R = q_j/g, \qquad d = n_2 - n_1 = tB.$$

If H is the support diameter, set

$$T = \lfloor H/B \rfloor.$$

Then every quotient-pair witness satisfies $1 \le t \le T$.

If $T < R$, the set $\{1, \ldots, T\}$ is a proper non-wrapping arc in $\mathbf{Z}/R\mathbf{Z}$, and it contains no zero phase. Hence the full-cell branch is impossible inside the no-wrap case; any actual pair in that case is registered as a quotient short-arc cluster.

If $T \geq R$, write

$$T = aR + s, \qquad a = \lfloor T/R \rfloor \geq 1, \qquad 0 \leq s < R.$$

The first aR formal quotient parameters decompose into a complete cycles, so each pivot phase occurs exactly a times in the formal quotient parameter interval. The residual s parameters again form a proper no-wrap tail arc. Thus a persistent actual excess cannot hide in an anonymous quotient cycle: after subtracting the complete-cycle formal mean it must appear either as a phase-cycle actual-mean cap problem, or as a tail short-arc cluster.

This theorem does not prove the short-arc cluster cap, it does not prove the phase-cycle actual-mean cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-phase-arc-mean-router.md.
```

□

Theorem 7.148 (Stable-ladder quotient arc Fourier reduction)**.** *The quotient short-arc cluster and the phase-cycle actual-mean outlet have a single finite Fourier form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientShortArcClusterOrPhaseCycleMeanPDECCap
```

is reduced to

```
StableLadderQuotientShortArcOrMeanImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterQuotientFourierLedger
AND StableLadderQuotientCircleGroupLedger
AND StableLadderQuotientActualPhaseLoadMeasureLedger
AND StableLadderQuotientArcDiscrepancyFunctionalLedger
AND StableLadderQuotientDirichletKernelIdentityLedger
AND StableLadderQuotientNontrivialFrequencyLowerBoundLedger
AND StableLadderPhaseMeanCapAsPointArcLedger
AND NoAnonymousShortArcOrPhaseMeanExitLedger
AND SparseScaleLadderSAECarriedForwardAfterQuotientFourierLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientArcFourierPDECCap.
```

Quotient arc Fourier ledger. Let

$$C_R = \mathbf{Z}/R\mathbf{Z}, \qquad R = q_j/\gcd(q_j, B), \qquad d = n_2 - n_1 = tB.$$

All pivot phases of the quotient-pair witnesses are determined by $t \bmod R$. Let $\mu(r)$ be the number of actual quotient witnesses with $t \equiv r \pmod R$. Subtract the relevant actual-load model or complete cycle mean and write

$$\nu(r) = \mu(r) - \mathrm{model}(r), \qquad \sum_{r \in C_R} \nu(r) = 0.$$

This centering is the formal-to-actual guard: the model is not used as actual load unless the actual-counting ledger has supplied it.

For any arc $A \subset C_R$, set

$$\Delta(A) = \sum_{r \in A} \nu(r).$$

A quotient short-arc cluster is exactly a proper arc with $\Delta(A) > 0$. The phase-cycle actual-mean cap is the point-arc case $A = \{r_0\}$.

Use the Fourier transform

$$\widehat{f}(h) = \sum_{r \in C_R} f(r) e^{-2\pi i h r / R}.$$

Since $\widehat{\nu}(0) = 0$, Fourier inversion gives the exact identity

$$\Delta(A) = \frac{1}{R} \sum_{h=1}^{R-1} \widehat{\nu}(h) \widehat{\mathbf{1}_A}(-h).$$

Let

$$\Lambda_R(A) = \sum_{h=1}^{R-1} \left| \widehat{\mathbf{1}_A}(h) \right|.$$

If $\Delta(A) > 0$, the triangle inequality forces a non-trivial frequency

$$|\widehat{\nu}(h)| \geq \frac{R\Delta(A)}{\Lambda_R(A)} \qquad (1 \leq h \leq R-1).$$

For $A = \{r_0\}$, one has $\Lambda_R(A) = R - 1$. Thus both the short-arc cluster and the phase-cycle actual-mean outlet are no longer anonymous: they are quotient arc Fourier/PDEC inputs.

This theorem does not prove the quotient arc Fourier/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-arc-fourier-router.md.`

□

Theorem 7.149 (Stable-ladder quotient Fourier lift reduction)**.** *The quotient arc Fourier outlet lifts to an endpoint bilinear Fourier outlet on the original pair variables. The open hard point*

`SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientArcFourierPDECCap`

is reduced to

```
StableLadderQuotientArcFourierImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterFourierLiftLedger
AND StableLadderNontrivialQuotientFrequencyForcesRGreaterOneLedger
AND StableLadderQuotientCoprimeFactorizationLedger
AND StableLadderQuotientInverseLiftLedger
AND StableLadderQuotientCharacterToPivotDifferenceLedger
AND StableLadderLiftedCharacterNontrivialityLedger
AND StableLadderEndpointBilinearPhaseFactorizationLedger
AND StableLadderCenteredModelKernelSeparatedLedger
AND NoAnonymousQuotientArcFourierExitLedger
AND SparseScaleLadderSAECarriedForwardAfterFourierLiftLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientEndpointBilinearFourierPDECCap.
```

Fourier lift ledger. Let $d = n_2 - n_1 = tB$, and put

$$g = \gcd(q_j, B), \qquad B = gB_0, \qquad q_j = gR.$$

Then $\gcd(B_0, R) = 1$. If the quotient Fourier branch contains a non-trivial frequency h, then $1 \leq h \leq R - 1$, so $R > 1$. Choose

$$uB_0 \equiv 1 \pmod{R}.$$

Since $d/g = tB_0$, one has

$$t \equiv u(d/g) \pmod{R}.$$

Set $\beta \equiv hu \pmod{R}$. For every actual pair witness,

$$e_R(ht) = e_R(hu\,d/g) = e_{q_j}(\beta d).$$

Here $\beta \not\equiv 0 \pmod{R}$, because $h \not\equiv 0 \pmod{R}$ and u is invertible modulo R. Thus the lifted character is non-trivial on the quotient phase.

Finally, using $d = n_2 - n_1$,

$$e_{q_j}(\beta(n_2 - n_1)) = e_{q_j}(\beta n_2)\overline{e_{q_j}(\beta n_1)}.$$

The centered coefficient from the quotient arc Fourier certificate is therefore an actual endpoint bilinear phase sum minus its explicit model or Dirichlet-kernel term. The formal model is not being counted as actual load; it is separated as a known kernel.

This theorem does not prove the endpoint bilinear Fourier/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-quotient-fourier-lift-router.md.
```

□

Theorem 7.150 (Stable-ladder endpoint bilinear balance reduction). *The endpoint bilinear Fourier outlet has a finite matrix balance form. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrQuotientEndpointBilinearFourierPDECCap
```

is reduced to

```
StableLadderEndpointBilinearFourierImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterBilinearBalanceLedger
AND StableLadderEndpointPairMatrixRegisteredLedger
AND StableLadderEndpointCenteredKernelTotalZeroLedger
AND StableLadderEndpointMarginalBalancedDecompositionLedger
AND StableLadderEndpointBilinearPhasePairingLedger
AND StableLadderEndpointBilinearTriangleTrichotomyLedger
AND StableLadderEndpointRowMarginalFourierExitLedger
AND StableLadderEndpointColumnMarginalFourierExitLedger
AND StableLadderEndpointBalancedCoreEnergyLowerBoundLedger
AND NoAnonymousEndpointBilinearFourierExitLedger
AND SparseScaleLadderSAECarriedForwardAfterBilinearBalanceLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointMarginalFourierOrBalancedBilinearEnergy
```

Endpoint bilinear balance ledger. Let $G = \mathbf{Z}/q_j\mathbf{Z}$. For the actual pair witnesses supplied by the previous quotient-Fourier lift, record the endpoint residues

$$x \equiv n_1 \pmod{q_j}, \qquad y \equiv n_2 \pmod{q_j}.$$

Let $M(x, y)$ be the actual witness-counting matrix. Subtract the explicit model or Dirichlet-kernel term already separated by the previous ledger and write

$$K(x, y) = M(x, y) - \mathrm{Model}(x, y), \qquad \sum_{x,y} K(x, y) = 0.$$

This is only a centered kernel identity; the model is not counted as actual load.

On the active endpoint domains X, Y, define

$$\rho(x) = \sum_{y \in Y} K(x, y), \qquad \sigma(y) = \sum_{x \in X} K(x, y).$$

The centered ANOVA decomposition is

$$K(x,y)=\frac{\rho(x)}{|Y|}+\frac{\sigma(y)}{|X|}+K_0(x,y),$$

where

$$\sum_{y\in Y}K_0(x,y)=0,\qquad \sum_{x\in X}K_0(x,y)=0.$$

If $X=Y=G$, non-trivial characters annihilate the pure row and column pieces. If a proper endpoint aperture remains, these pieces are exactly row or column marginal Fourier outlets.

For a non-trivial endpoint character

$$\phi_\beta(z)=e^{2\pi i\beta z/q_j},$$

the previous hard point has the form

$$S_\beta=\sum_{x\in X,y\in Y}K(x,y)\overline{\phi_\beta(x)}\phi_\beta(y).$$

Substituting the decomposition gives

$$S_\beta=S_{\mathrm{row}}(\beta)+S_{\mathrm{col}}(\beta)+S_{\mathrm{bal}}(\beta).$$

Thus if $|S_\beta|\ge\eta$, then at least one of the three alternatives

$$|S_{\mathrm{row}}(\beta)|\ge\eta/3,\qquad |S_{\mathrm{col}}(\beta)|\ge\eta/3,\qquad |S_{\mathrm{bal}}(\beta)|\ge\eta/3$$

holds. The first two alternatives are endpoint marginal Fourier/PDEC inputs. For the balanced alternative, Cauchy–Schwarz gives

$$|S_{\mathrm{bal}}(\beta)|\le\sqrt{|X||Y|}\,\|K_0\|_{\mathrm{HS}},$$

and therefore

$$\|K_0\|_{\mathrm{HS}}^2\ge\frac{\eta^2}{9|X||Y|}.$$

The anonymous endpoint bilinear cap is therefore replaced by endpoint marginal Fourier caps or a balanced-core energy cap.

This theorem does not prove the endpoint marginal Fourier/PDEC cap, it does not prove the balanced bilinear energy/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-bilinear-balance-router.md.`

□

Theorem 7.151 (Stable-ladder endpoint energy packet reduction). *The endpoint marginal Fourier outlet and the balanced bilinear energy outlet have a common dyadic packet form. The open hard point*

`SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointMarginalFourierOrBalancedBilinearEnergyPDEC`

is reduced to

StableLadderEndpointMarginalOrBalancedEnergyImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterEnergyPacketLedger
AND StableLadderEndpointMarginalFourierParsevalVarianceLedger
AND StableLadderEndpointMarginalVarianceDyadicPacketLedger
AND StableLadderBalancedCoreEnergyImportedLedger
AND StableLadderBalancedCoreEnergyDyadicCellPacketLedger
AND StableLadderEndpointEnergyPacketSignedHalfLedger
AND StableLadderEndpointDyadicEnergyPacketRegisteredLedger
AND NoAnonymousEndpointMarginalOrBalancedEnergyExitLedger
AND SparseScaleLadderSAECarriedForwardAfterEndpointEnergyPacketLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointDyadicEnergyPacketPDECCap.

Endpoint energy packet ledger. Consider first a row endpoint marginal ρ on $\mathbf{Z}/q_j\mathbf{Z}$. If the endpoint marginal Fourier outlet gives a non-trivial frequency with

$$|\widehat{\rho}(\beta)| \geq \eta,$$

then Parseval gives

$$\sum_x |\rho(x)|^2 \geq \frac{\eta^2}{q_j}.$$

The same argument applies to a column marginal σ. Hence every marginal Fourier outlet is already a one-dimensional endpoint variance outlet.

The balanced branch from the previous ledger is already an energy outlet:

$$\|K_0\|_{\mathrm{HS}}^2 \geq E,$$

where K_0 has zero row and column marginals. Thus both possible outlets are finite L^2-energy statements: either on endpoint residues or on endpoint cells.

For any finite real-valued deviation function F on a finite support Ω, let

$$E_F = \sum_{z \in \Omega} |F(z)|^2.$$

Partition the non-zero values of $|F|$ into dyadic layers. If L is the number of non-empty layers, then one layer with scale λ satisfies

$$\lambda^2 \, \#\{z : \lambda < |F(z)| \leq 2\lambda\} \geq \frac{E_F}{L}.$$

Applied to $F = \rho$ or $F = \sigma$, this gives a one-dimensional endpoint marginal packet. Applied to $F = K_0$, this gives a two-dimensional balanced endpoint-cell packet.

Finally split the selected layer by sign. Since the deviations are real, at least one sign contributes half the selected square energy. Therefore the remaining obstruction can be registered as a signed dyadic endpoint energy packet, carrying its dimension, scale, support, sign, and energy lower bound.

This theorem does not prove the endpoint dyadic energy packet/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-energy-packet-router.md.`

□

Theorem 7.152 (Stable-ladder endpoint packet autocorrelation reduction)**.** *The endpoint dyadic energy packet outlet reduces to either a singleton packet atom or a non-zero endpoint displacement autocorrelation. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointDyadicEnergyPacketPDECCap
```

is reduced to

```
StableLadderEndpointDyadicEnergyPacketImportedLedger
AND StableLadderIsolatedSingletonCarriedForwardAfterPacketAutocorrelationLedger
AND StableLadderEndpointPacketFiniteGroupLedger
AND StableLadderEndpointPacketSignedSupportLedger
AND StableLadderEndpointPacketSingletonAtomLedger
AND StableLadderEndpointPacketNonzeroPairCountLedger
AND StableLadderEndpointPacketDisplacementPigeonholeLedger
AND StableLadderEndpointPacketWeightedAutocorrelationLedger
AND StableLadderEndpointPacketDimensionPreservedLedger
AND NoAnonymousEndpointDyadicEnergyPacketExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPacketAutocorrelationLedger
AND SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointPacketSingletonAtomSAEOrEndpointDisplac
```

Endpoint packet autocorrelation ledger. The previous ledger produces a signed dyadic packet on a finite endpoint group

$$H = \mathbf{Z}/q_j\mathbf{Z} \quad \text{or} \quad H = (\mathbf{Z}/q_j\mathbf{Z})^2.$$

It has a scale λ, a sign $\epsilon \in \{\pm 1\}$, and support

$$S = \{z \in H : \lambda < \epsilon F(z) \leq 2\lambda\}.$$

Here F is either an endpoint marginal deviation or the balanced endpoint cell deviation.

If $|S| = 1$, the packet is a singleton endpoint atom. This branch is not closed here; it is merely no longer anonymous.

Assume now that $|S| = m \geq 2$. Define

$$C_S(\delta) = \#\{z \in S : z + \delta \in S\}.$$

Every ordered pair of distinct elements of S contributes one non-zero displacement, hence

$$\sum_{\delta \neq 0} C_S(\delta) = m(m-1).$$

Since H is finite, there is a non-zero displacement δ such that

$$C_S(\delta) \geq \frac{m(m-1)}{|H| - 1}.$$

On the signed dyadic support, all products have the same sign and satisfy

$$F(z)F(z+\delta) \geq \lambda^2.$$

Therefore the weighted autocorrelation obeys

$$A_F(\delta) = \sum_{z, z+\delta \in S} F(z)F(z+\delta) \geq \lambda^2 C_S(\delta).$$

For dimension one this is an endpoint-residue displacement; for dimension two it is an endpoint-cell displacement. Thus the packet has become an explicit same-sign displacement autocorrelation.

This theorem does not prove the endpoint packet singleton atom/SAE branch, it does not prove the endpoint displacement autocorrelation/PDEC cap, it does not prove summability of isolated singletons, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-packet-autocorrelation-router.md.`

□

Theorem 7.153 (Stable-ladder endpoint displacement orbit reduction)**.** *The endpoint displacement autocorrelation outlet reduces to a cyclic adjacency packet on one translation orbit. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableActualLadderIsolatedSingletonOrEndpointPacketSingletonAtomSAEOrEndpointDisplacemen
```

is reduced to

```
StableLadderEndpointSingletonOrDisplacementAutocorrelationImportedLedger
AND StableLadderEndpointSingletonAtomSAEUnifiedLedger
AND StableLadderEndpointDisplacementAutocorrelationImportedLedger
AND StableLadderEndpointNonzeroDisplacementOrderLedger
AND StableLadderEndpointDisplacementOrbitPartitionLedger
AND StableLadderEndpointAutocorrelationOrbitDecompositionLedger
AND StableLadderEndpointWeightedOrbitPigeonholeLedger
AND StableLadderEndpointOrbitCyclicAdjacencyPacketLedger
AND StableLadderEndpointOrbitDimensionPreservedLedger
AND NoAnonymousEndpointDisplacementAutocorrelationExitLedger
AND SparseScaleLadderSAECarriedForwardAfterDisplacementOrbitLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointTranslationOrbitAdjacencyPDECCap.
```

Endpoint displacement orbit ledger. The singleton branches from the previous ledger are first unified as endpoint singleton atoms. This unification does not prove their summability; it only prevents two names for the same single-point obstruction.

Now consider the displacement autocorrelation branch. Let H be the finite endpoint group and let $\delta \neq 0$ be the displacement. Put

$$r = \operatorname{ord}_H(\delta) > 1.$$

The translation $T_\delta z = z + \delta$ partitions H into cycles

$$O = a + \langle \delta \rangle = \{a + t\delta : t \in \mathbf{Z}/r\mathbf{Z}\}.$$

The set of all cycles is $\Omega = H/\langle \delta \rangle$, with $|\Omega| = |H|/r$.

The weighted autocorrelation from the previous ledger decomposes exactly along these cycles:

$$A_F(\delta) = \sum_{O \in \Omega} A_O(\delta),$$

where

$$A_O(\delta) = \sum_{t \in \mathbf{Z}/r\mathbf{Z}} F(a + t\delta) F(a + (t+1)\delta)\, \mathbf{1}_{a+t\delta \in S}\, \mathbf{1}_{a+(t+1)\delta \in S}.$$

Hence, if $A_F(\delta)$ is the active lower-bound obstruction, then some orbit satisfies

$$A_O(\delta) \geq \frac{A_F(\delta)}{|\Omega|}.$$

On that orbit the obstruction is a signed dyadic adjacency packet on the cycle $\mathbf{Z}/r\mathbf{Z}$. In dimension one this is an endpoint-residue cycle; in dimension two it is an endpoint-cell cycle. Thus the full-group autocorrelation has been reduced to one explicit CRT translation orbit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the endpoint translation orbit adjacency/PDEC cap, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-displacement-orbit-router.md.

□

Theorem 7.154 (Stable-ladder endpoint orbit run-boundary reduction)**.** *The endpoint translation-orbit adjacency outlet reduces to a long same-sign arc outlet or to a boundary-flux outlet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointTranslationOrbitAdjacencyPDECCap
```

is reduced to

```
StableLadderEndpointTranslationOrbitAdjacencyImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterOrbitRunLedger
AND StableLadderEndpointOrbitSignedCycleModelLedger
AND StableLadderEndpointOrbitDyadicEdgeCountLedger
AND StableLadderEndpointOrbitRunPartitionLedger
AND StableLadderEndpointOrbitAdjacencyRunBoundaryIdentityLedger
AND StableLadderEndpointOrbitRunBoundaryBudgetDichotomyLedger
AND StableLadderEndpointOrbitLongSameSignArcRegistrationLedger
AND StableLadderEndpointOrbitBoundaryFluxRegistrationLedger
AND NoAnonymousEndpointTranslationOrbitAdjacencyExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOrbitRunBoundaryLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitLongSameSignArcSAEOrEndpointOrbit
```

Endpoint orbit run-boundary ledger. The endpoint singleton atom outlet and the sparse scale-ladder SAE outlet are carried forward. Consider the translation-orbit adjacency branch from the previous ledger. On the selected orbit write

$$C_r = \mathbf{Z}/r\mathbf{Z}, \qquad z_t = a + t\delta.$$

Let u_t be the active indicator of $z_t \in S$, and let $\sigma_t = \operatorname{sgn} F(z_t)$. A same-sign active edge is

$$e_t = 1 \quad\Longleftrightarrow\quad u_t = u_{t+1} = 1, \quad \sigma_t = \sigma_{t+1}.$$

Let $E = \sum_t e_t$.

Because the packet is dyadic, there is a scale λ such that each active endpoint satisfies $\lambda \leq |F(z_t)| < 2\lambda$. Therefore the weighted adjacency mass W and the unweighted same-sign edge count obey

$$\lambda^2 E \leq W \leq 4\lambda^2 E.$$

Thus the capacity part of the translation-orbit adjacency obstruction may be tracked by E, up to an absolute dyadic constant.

Partition the active same-sign consecutive vertices into maximal cyclic runs. Let n be the number of active vertices and let b be the number of cuts, equivalently the number of runs unless the whole cycle is one same-sign run. If the whole cycle is one same-sign run, then the long-arc outlet is already present with length r. Otherwise the exact run identity is

$$E = n - b.$$

Consequently, for every boundary budget $B \geq 1$, either $b > B$, which is registered as the boundary-flux outlet, or $b \leq B$, in which case a pigeonhole bound gives a same-sign run of length

$$L_{\max} \geq \frac{n}{b} \geq \frac{E}{B}.$$

The anonymous translation-orbit adjacency outlet has therefore been replaced by two explicit controlled exits: a long same-sign arc on C_r, or many cycle cuts carrying boundary flux.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the long same-sign arc SAE outlet, it does not prove the boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-run-boundary-router.md.
```

□

Theorem 7.155 (Stable-ladder endpoint orbit signed Fourier reduction)**.** *The endpoint long-arc/boundary-flux outlet reduces to a full-cycle mean atom or to a signed Fourier cap on the endpoint orbit. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitLongSameSignArcSAEOrEndpointOrbitBoun
```

is reduced to

```
StableLadderEndpointOrbitRunBoundaryImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSignedFourierLedger
AND StableLadderEndpointOrbitSignedIndicatorSequenceLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomRegistrationLedger
AND StableLadderEndpointOrbitLongArcCenteredDiscrepancyLedger
AND StableLadderEndpointOrbitArcDirichletKernelLedger
AND StableLadderEndpointOrbitBoundaryDerivativeSupportLedger
AND StableLadderEndpointOrbitBoundaryDerivativeFourierLedger
AND StableLadderEndpointOrbitSignedFourierCapRegistrationLedger
AND NoAnonymousEndpointOrbitLongArcBoundaryExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOrbitSignedFourierLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit signed Fourier ledger. The endpoint singleton atom outlet and sparse scale-ladder SAE outlet are carried forward. On the endpoint orbit $C_r = \mathbf{Z}/r\mathbf{Z}$, define

$$g_t = \sigma_t u_t \in \{-1, 0, 1\}, \qquad \bar{g} = \frac{1}{r}\sum_t g_t, \qquad \nu_t = g_t - \bar{g}.$$

First suppose the long same-sign arc outlet is active. Take I to be the maximal cyclic run and let $\epsilon \in \{\pm 1\}$ be its sign, so $g_t = \epsilon$ for every $t \in I$. If $I = C_r$, this is registered as a full-cycle mean atom. Otherwise $\epsilon \sum_t g_t \le r - 1$, because the maximal run has at least one endpoint outside I not equal to ϵ. Hence

$$\Delta_\epsilon(I) = \epsilon \sum_{t\in I} \nu_t = |I| - \frac{|I|}{r}\epsilon \sum_t g_t \ge \frac{|I|}{r} > 0.$$

Since ν has zero mean, Fourier inversion gives

$$\Delta_\epsilon(I) = \frac{\epsilon}{r}\sum_{h\neq 0} \widehat{g}(h)\,\widehat{\mathbf{1}_I}(-h).$$

Thus, with $\Lambda_r(I) = \sum_{h\neq 0} |\widehat{\mathbf{1}_I}(h)|$, some nonzero frequency satisfies

$$|\widehat{g}(h)| \ge \frac{r\,\Delta_\epsilon(I)}{\Lambda_r(I)}.$$

Now suppose the boundary-flux outlet is active with b cuts. Put

$$Dg_t = g_{t+1} - g_t.$$

Each cut gives a nonzero cyclic difference, so

$$\sum_t |Dg_t|^2 \geq b.$$

Parseval and the identity $\widehat{Dg}(h) = (e^{2\pi ih/r} - 1)\widehat{g}(h)$ give

$$\sum_{h\neq 0} |(e^{2\pi ih/r} - 1)\widehat{g}(h)|^2 = r \sum_t |Dg_t|^2 \geq rb.$$

Therefore some $h \neq 0$ satisfies

$$|(e^{2\pi ih/r} - 1)\widehat{g}(h)| \geq \sqrt{\frac{rb}{r-1}}.$$

Both active branches have become signed Fourier data on the endpoint orbit, except for the explicit full-cycle mean atom.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the signed Fourier PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-signed-fourier-router.md.
```

□

Theorem 7.156 (Stable-ladder endpoint orbit conductor-character reduction)**.** *The endpoint-orbit signed Fourier outlet reduces to a primitive conductor character packet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi
```

is reduced to

```
StableLadderEndpointOrbitSignedFourierImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterConductorLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardLedger
AND StableLadderEndpointOrbitNonzeroFrequencyConductorLedger
AND StableLadderEndpointOrbitFrequencyKernelQuotientLedger
AND StableLadderEndpointOrbitKernelFiberCollapseLedger
AND StableLadderEndpointOrbitPrimitiveConductorCharacterPacketLedger
AND StableLadderEndpointOrbitDerivativeMultiplierAbsorbedLedger
AND NoAnonymousEndpointOrbitSignedFourierExitLedger
AND SparseScaleLadderSAECarriedForwardAfterOrbitConductorLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit conductor-character ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. Consider the signed Fourier branch on $C_r = \mathbf{Z}/r\mathbf{Z}$. Let h be the nonzero frequency. Put

$$d = \gcd(h, r), \qquad m = r/d, \qquad h_0 = h/d.$$

Then $m > 1$ and $\gcd(h_0, m) = 1$. The character factors through the conductor quotient:

$$e^{-2\pi iht/r} = e^{-2\pi ih_0(t \bmod m)/m}.$$

Its kernel in C_r is

$$\{0, m, 2m, \ldots, (d-1)m\},$$

of size d, and the quotient is C_m.

Collapse the signed load along these kernel fibers:

$$G_s = \sum_{u=0}^{d-1} g_{s+um}, \qquad s \in \mathbf{Z}/m\mathbf{Z}.$$

Then the Fourier coefficient has the exact primitive-conductor form

$$\widehat{g}(h) = \sum_{s \bmod m} G_s \, e^{-2\pi i h_0 s/m}.$$

Because h_0 is coprime to m, this is a primitive additive character packet on C_m.

If the previous lower bound came from the derivative Fourier branch, then

$$|(e^{2\pi i h/r} - 1)\widehat{g}(h)| \geq M$$

implies

$$|\widehat{g}(h)| \geq M/2,$$

since $|e^{2\pi i h/r} - 1| \leq 2$. Thus the long-arc Fourier and boundary derivative-Fourier branches both enter the same primitive conductor-character outlet, up to an absolute constant already recorded in the ledger.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the primitive conductor-character PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-conductor-character-router.md.
```

□

Theorem 7.157 (Stable-ladder endpoint orbit standard-character reduction)**.** *The primitive conductor-character outlet reduces to a standard first-harmonic packet on the same conductor. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPr
```

is reduced to

```
StableLadderEndpointOrbitPrimitiveConductorCharacterImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterStandardCharacterLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterStandardCharacterLedger
AND StableLadderEndpointOrbitConductorUnitAutomorphismLedger
AND StableLadderEndpointOrbitStandardPhaseCoordinateLedger
AND StableLadderEndpointOrbitAutomorphicLoadPermutationLedger
AND StableLadderEndpointOrbitFirstHarmonicIdentityLedger
AND StableLadderEndpointOrbitLoadNormsAndSupportPreservedLedger
AND NoAnonymousPrimitiveConductorFrequencyExitLedger
AND SparseScaleLadderSAECarriedForwardAfterStandardCharacterLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit standard-character ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. Consider the remaining primitive conductor packet on $C_m = \mathbf{Z}/m\mathbf{Z}$:

$$\sum_{s \bmod m} G_s e^{-2\pi i h_0 s/m}, \qquad (h_0, m) = 1.$$

Let u be the inverse of h_0 modulo m. Put

$$a = h_0 s \bmod m, \qquad S_a = G_{ua \bmod m}.$$

Since multiplication by h_0 is a permutation of C_m, this is an exact change of variables. Hence

$$\sum_{s \bmod m} G_s e^{-2\pi i h_0 s/m} = \sum_{a \bmod m} S_a e^{-2\pi i a/m}.$$

The permutation preserves support size, L^1 and L^2 mass, the total mean, and every lower-bound constant in the Fourier outlet. Thus the primitive frequency h_0 is not a separate terminal obstruction: it is only a coordinate choice on the conductor group. The remaining outlet is the standard first-harmonic conductor packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the standard first-harmonic PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-standard-character-router.md.
```

□

Theorem 7.158 (Stable-ladder endpoint orbit axis-lobe reduction)**.** *The standard first-harmonic conductor outlet reduces to an axis-lobe weighted surplus packet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt
```

is reduced to

```
StableLadderEndpointOrbitStandardFirstHarmonicImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterAxisLobeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterAxisLobeLedger
AND StableLadderEndpointOrbitComplexFirstHarmonicAmplitudeLedger
AND StableLadderEndpointOrbitAxisProjectionDichotomyLedger
AND StableLadderEndpointOrbitAxisSignChoiceLedger
AND StableLadderEndpointOrbitTrigonometricLobeWeightLedger
AND StableLadderEndpointOrbitAxisLobeWeightedSurplusPacketLedger
AND NoAnonymousStandardFirstHarmonicExitLedger
AND SparseScaleLadderSAECarriedForwardAfterAxisLobeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit axis-lobe ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The remaining standard first-harmonic packet has the form

$$F = \sum_{a \bmod m} S_a e^{-2\pi i a/m}, \qquad |F| \geq M.$$

Its coordinate projections are

$$\Re F = \sum_a S_a \cos(2\pi a/m), \qquad \Im F = -\sum_a S_a \sin(2\pi a/m).$$

Since $|F|^2 = (\Re F)^2 + (\Im F)^2$, one of these two real projections has absolute value at least $M/\sqrt{2}$. Define

$$\phi_0(a) = \cos(2\pi a/m), \qquad \phi_1(a) = -\sin(2\pi a/m).$$

Thus for some $j \in \{0,1\}$ and some sign $\epsilon \in \{\pm 1\}$,

$$\sum_a S_a\, \epsilon\phi_j(a) \geq M/\sqrt{2}.$$

Split the chosen trigonometric weight into positive and negative lobes:

$$w_a = \max(\epsilon\phi_j(a), 0), \qquad v_a = \max(-\epsilon\phi_j(a), 0).$$

Then w and v are supported on opposite axis half-circles and

$$\sum_a S_a(w_a - v_a) \geq M/\sqrt{2}.$$

Hence the complex first-harmonic cap has been converted to a real axis-lobe weighted surplus cap with explicit axis, sign, and half-circle weights.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the axis-lobe weighted-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-axis-lobe-router.md.
```

□

Theorem 7.159 (Stable-ladder endpoint orbit single-lobe reduction)**.** *The axis-lobe weighted-surplus outlet reduces to a single-lobe signed surplus packet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitAx
```

is reduced to

```
StableLadderEndpointOrbitAxisLobeWeightedSurplusImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSingleLobeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterSingleLobeLedger
AND StableLadderEndpointOrbitTwoLobeSurplusDecompositionLedger
AND StableLadderEndpointOrbitHalfThresholdLossLedger
AND StableLadderEndpointOrbitSingleLobeSignChoiceLedger
AND StableLadderEndpointOrbitSingleLobeHalfCircleSupportLedger
AND StableLadderEndpointOrbitSingleLobeSignedSurplusPacketLedger
AND NoAnonymousAxisLobeDifferenceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterSingleLobeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit single-lobe ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The axis-lobe outlet has the form

$$\sum_a S_a(w_a - v_a) \geq L, \qquad w_a \geq 0, \quad v_a \geq 0.$$

Put

$$A = \sum_a S_a w_a, \qquad B = \sum_a S_a v_a.$$

Then $A - B \geq L$. If $A \geq L/2$, choose $W = w$ and $\eta = +1$. If $A < L/2$, then $-B > L/2$, so choose $W = v$ and $\eta = -1$. In either case,

$$\sum_a \eta S_a W_a \geq L/2.$$

The chosen W is one of the two trigonometric lobe weights, hence

$$0 \le W_a \le 1$$

and its support is a single axis half-circle. Thus the two-lobe difference has been converted, with only an absolute factor 2, into a single-lobe signed surplus packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the single-lobe signed-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-single-lobe-router.md.
```

□

Theorem 7.160 (Stable-ladder endpoint orbit single-arc reduction)**.** *The single-lobe signed-surplus outlet reduces to a single-arc signed surplus packet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi
```

is reduced to

```
StableLadderEndpointOrbitSingleLobeSignedSurplusImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterSingleArcLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterSingleArcLedger
AND StableLadderEndpointOrbitSingleLobeLayerCakeIdentityLedger
AND StableLadderEndpointOrbitLobeSuperlevelArcSupportLedger
AND StableLadderEndpointOrbitLayerCakeArcPigeonholeLedger
AND StableLadderEndpointOrbitSingleArcSignedSurplusPacketLedger
AND NoAnonymousSingleLobeWeightExitLedger
AND SparseScaleLadderSAECarriedForwardAfterSingleArcLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit single-arc ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The single-lobe outlet has the form

$$\sum_a \eta S_a W_a \ge L, \qquad 0 \le W_a \le 1.$$

The layer-cake identity gives, pointwise,

$$W_a = \int_0^1 \mathbf{1}_{\{W_a \ge t\}}\, dt.$$

Therefore

$$\sum_a \eta S_a W_a = \int_0^1 \sum_{a: W_a \ge t} \eta S_a \, dt.$$

If every level set had signed mass below L, the integral would be below L. Hence for some $t \in [0, 1]$,

$$\sum_{a: W_a \ge t} \eta S_a \ge L.$$

Because W is a single trigonometric lobe supported on an axis half-circle, each nonempty superlevel set

$$A_t = \{a \bmod m : W_a \ge t\}$$

is a cyclic arc inside that half-circle. Thus the continuous lobe weight has been replaced by one explicit single-arc signed surplus packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the single-arc signed-surplus PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-single-arc-router.md.
```

□

Theorem 7.161 (Stable-ladder endpoint orbit arc endpoint-potential reduction)**.** *The single-arc signed-surplus outlet reduces to either the full-cycle mean atom outlet or an endpoint-potential gap packet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi
```

is reduced to

```
StableLadderEndpointOrbitSingleArcSignedSurplusImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterArcEndpointPotentialLedger
AND StableLadderEndpointOrbitArcSignedLoadSequenceLedger
AND StableLadderEndpointOrbitArcMeanContributionDichotomyLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterArcEndpointPotentialLedger
AND StableLadderEndpointOrbitCenteredArcSurplusLedger
AND StableLadderEndpointOrbitCenteredPrefixPotentialLedger
AND StableLadderEndpointOrbitArcEndpointPotentialGapLedger
AND StableLadderEndpointOrbitArcEndpointPotentialPacketLedger
AND NoAnonymousSingleArcInteriorSurplusExitLedger
AND SparseScaleLadderSAECarriedForwardAfterArcEndpointPotentialLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit arc endpoint-potential ledger. The endpoint singleton atom outlet and the sparse scale-ladder SAE outlet are carried forward. The single-arc outlet has the form

$$\sum_{a \in A} X_a \geq L, \qquad X_a = \eta S_a,$$

where $A = [u, v)$ is a cyclic arc in C_m. Put

$$\mu = \frac{1}{m} \sum_{a \in C_m} X_a, \qquad Y_a = X_a - \mu.$$

Then

$$\sum_{a \in A} X_a = |A|\mu + \sum_{a \in A} Y_a.$$

If the mean contribution satisfies $|A|\mu \geq L/2$, then, since $|A| \leq m$,

$$\sum_{a \in C_m} X_a = m\mu \geq L/2,$$

and the packet is registered as a full-cycle mean atom outlet. Otherwise the centered part satisfies

$$\sum_{a \in A} Y_a \geq L/2.$$

Define the centered prefix potential on a lift of the orbit by

$$F(0) = 0, \qquad F(j+1) = F(j) + Y_j.$$

Because $\sum_{C_m} Y_a = 0$, this potential is cyclic, $F(m) = 0$. For the arc $A = [u, v)$,

$$\sum_{a \in A} Y_a = F(v) - F(u).$$

Thus the non-mean branch forces a pair of arc endpoints whose centered potential gap is at least $L/2$. The anonymous single-arc interior surplus has therefore been replaced by the named endpoint-potential PDEC/cap outlet, or by the full-cycle mean atom outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the endpoint-potential PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-arc-endpoint-potential-router.md.
```

□

Theorem 7.162 (Stable-ladder endpoint orbit potential-variation reduction)**.** *The endpoint-potential outlet reduces to a signed-variation packet on actual orbit increments. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitAr
```

is reduced to

```
StableLadderEndpointOrbitArcEndpointPotentialImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPotentialVariationLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPotentialVariationLedger
AND StableLadderEndpointOrbitCenteredPotentialIncrementLedger
AND StableLadderEndpointOrbitEndpointGapDirectedArcLedger
AND StableLadderEndpointOrbitPositiveVariationLowerBoundLedger
AND StableLadderEndpointOrbitNegativeVariationLowerBoundLedger
AND StableLadderEndpointOrbitSignedVariationPacketLedger
AND NoAnonymousEndpointPotentialGapExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPotentialVariationLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit potential-variation ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The endpoint-potential outlet has the form

$$F(v) - F(u) \geq G, \qquad G = L/2.$$

The centered potential increments are

$$Y_j = F(j+1) - F(j), \qquad \sum_{j \in C_m} Y_j = 0.$$

Let $I = [u, v)$ and $J = [v, u)$ be the two directed cyclic arcs. Then

$$\sum_{j \in I} Y_j = F(v) - F(u) \geq G$$

and, by the zero total increment condition,

$$\sum_{j \in J} Y_j = F(u) - F(v) \leq -G.$$

Hence the positive variation on I and the negative variation on J obey

$$\sum_{j\in I}(Y_j)_+ \geq G, \qquad \sum_{j\in J}(Y_j)_- \geq G.$$

Thus the anonymous two-endpoint potential gap has been converted into a signed-variation obligation carried by actual adjacent orbit increments.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the signed-variation PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-potential-variation-router.md.
```

□

Theorem 7.163 (Stable-ladder endpoint orbit variation run–boundary reduction)**.** *The signed-variation outlet reduces to an increment-run surplus outlet or to a variation-boundary flux outlet. The open hard point*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSi
```

is reduced to

```
StableLadderEndpointOrbitSignedVariationImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterVariationRunBoundaryLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterVariationRunBoundaryLedger
AND StableLadderEndpointOrbitVariationSignedSideChoiceLedger
AND StableLadderEndpointOrbitPositiveIncrementEdgeSetLedger
AND StableLadderEndpointOrbitVariationRunPartitionLedger
AND StableLadderEndpointOrbitVariationRunBoundaryBudgetDichotomyLedger
AND StableLadderEndpointOrbitIncrementRunSurplusPacketLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxPacketLedger
AND NoAnonymousSignedVariationExitLedger
AND SparseScaleLadderSAECarriedForwardAfterVariationRunBoundaryLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit variation run–boundary ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, and the sparse scale-ladder SAE outlet are carried forward. The signed-variation outlet gives a directed support K, a sign $\sigma \in \{\pm 1\}$, and

$$\sum_{j\in K}(\sigma Y_j)_+ \geq G.$$

Let

$$E_+ = \{j \in K : \sigma Y_j > 0\}.$$

Partition E_+ into maximal consecutive runs $R_1, \dots, R_b$ along the directed orbit. Given any boundary budget $B \geq 1$, there are two cases. If $b > B$, the packet has high sign-switch boundary flux and is registered as the variation-boundary flux PDEC/cap outlet. If $b \leq B$, then by the pigeonhole principle one run satisfies

$$\sum_{j\in R_h} \sigma Y_j \geq G/B.$$

This is the increment-run surplus outlet. Thus the anonymous signed variation has been converted into a named long same-sign increment run or a named boundary-flux packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the increment-run surplus SAE outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-variation-run-boundary-router.md.`

∎

Theorem 7.164 (Stable-ladder endpoint orbit increment-run amplitude drift ledger)**.** *The endpoint-orbit increment-run surplus outlet admits the following non-cyclic refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitIn
⇒ StableLadderEndpointOrbitIncrementRunSurplusImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND StableLadderEndpointOrbitIncrementRunSameSignCoordinateLedger
AND StableLadderEndpointOrbitIncrementRunMassLowerBoundLedger
AND StableLadderEndpointOrbitIncrementRunAmplitudeThresholdDichotomyLedger
AND StableLadderEndpointOrbitIncrementEdgeSpikePacketLedger
AND StableLadderEndpointOrbitLongBoundedIncrementDriftPacketLedger
AND NoAnonymousIncrementRunSurplusExitLedger
AND SparseScaleLadderSAECarriedForwardAfterIncrementRunAmplitudeDriftLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit increment-run amplitude–drift ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The low-boundary branch from the previous ledger gives a consecutive run R and a sign σ such that, with

$$Z_j = \sigma Y_j > 0 \qquad (j \in R),$$

one has

$$\sum_{j \in R} Z_j \geq H, \qquad H = G/B.$$

Fix any amplitude threshold $\Lambda > 0$. If

$$\max_{j \in R} Z_j \geq \Lambda,$$

then the packet contains a named increment edge-spike atom. Otherwise

$$0 < Z_j < \Lambda \qquad (j \in R),$$

and the mass lower bound forces

$$|R| \geq H/\Lambda.$$

For thresholds with large H/Λ, this is a long bounded-increment monotone drift packet. Thus the anonymous increment-run surplus outlet has been converted into either an increment edge-spike atom or a long bounded-increment drift packet, while the high-boundary variation-flux outlet remains a named carried-forward outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the increment edge-spike SAE outlet, it does not prove the long bounded-increment drift PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-increment-run-amplitude-drift-router.md.`

□

Theorem 7.165 (Stable-ladder endpoint orbit edge-spike mean–singleton ledger). *The endpoint-orbit increment edge-spike outlet is absorbed by the named singleton and full-cycle mean outlets:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitIn
⇒ StableLadderEndpointOrbitIncrementEdgeSpikeImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitLongBoundedIncrementDriftCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND StableLadderEndpointOrbitEdgeSpikeCenteredLoadExpansionLedger
AND StableLadderEndpointOrbitEdgeSpikeHalfThresholdDichotomyLedger
AND StableLadderEndpointOrbitEdgeSpikeSingletonAtomAbsorptionLedger
AND StableLadderEndpointOrbitEdgeSpikeFullCycleMeanAtomAbsorptionLedger
AND NoAnonymousIncrementEdgeSpikeExitLedger
AND SparseScaleLadderSAECarriedForwardAfterEdgeSpikeMeanSingletonLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit edge-spike mean–singleton ledger. The long bounded-increment drift outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The edge-spike branch gives an index j, a sign σ, and a threshold $\Lambda > 0$ such that

$$\sigma Y_j \geq \Lambda.$$

Here Y_j is the centered point load used in the previous ledgers, so

$$Y_j = X_j - \mu,$$

where X_j is the uncentered signed point load and μ is the full-cycle mean. Hence

$$\sigma X_j - \sigma\mu \geq \Lambda.$$

The elementary half-threshold dichotomy gives

$$\sigma X_j \geq \Lambda/2 \qquad \text{or} \qquad -\sigma\mu \geq \Lambda/2.$$

In the first case the spike is already an endpoint singleton atom. In the second case it is already a full-cycle mean atom. Thus the edge-spike outlet does not remain as an independent anonymous terminal; it is absorbed by the two already named outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the long bounded-increment drift PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-edge-spike-mean-singleton-router.md.`

□

Theorem 7.166 (Stable-ladder endpoint orbit long-drift dyadic plateau ledger). *The endpoint-orbit long bounded-increment drift outlet admits the following non-cyclic dyadic refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLongBoundedIncrementDriftImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterLongDriftDyadicPlateauLedger
AND StableLadderEndpointOrbitLongDriftPositiveBoundedRunLedger
AND StableLadderEndpointOrbitLongDriftDyadicAmplitudePartitionLedger
AND StableLadderEndpointOrbitLongDriftAmplitudeDepthBudgetDichotomyLedger
AND StableLadderEndpointOrbitLongDriftHeavyDyadicBandLedger
AND StableLadderEndpointOrbitDyadicBandPlateauRunPartitionLedger
AND StableLadderEndpointOrbitDyadicPlateauRunBoundaryDichotomyLedger
AND StableLadderEndpointOrbitComparableAmplitudePlateauDriftPacketLedger
AND StableLadderEndpointOrbitAmplitudeDepthPacketLedger
AND NoAnonymousLongBoundedIncrementDriftExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLongDriftDyadicPlateauLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit long-drift dyadic plateau ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The long bounded-increment drift branch gives a consecutive run R such that

$$0 < Z_j < \Lambda \qquad (j \in R), \qquad \sum_{j \in R} Z_j \geq H.$$

For $l \geq 0$, set

$$A_l = \{j \in R : 2^{-(l+1)}\Lambda \leq Z_j < 2^{-l}\Lambda\}.$$

Since R is finite, only finitely many A_l are nonempty. Given a depth budget D, if the number of nonempty dyadic bands is greater than D, the packet is registered as an amplitude-depth PDEC/cap outlet. Otherwise, by pigeonhole, some band satisfies

$$\sum_{j \in A_l} Z_j \geq H/D.$$

Partition this heavy dyadic band into maximal consecutive plateau runs inside R. Given a plateau boundary budget B, if the number of plateau runs is greater than B, this is a boundary-flux outlet. Otherwise some plateau run P_s satisfies

$$\sum_{j \in P_s} Z_j \geq H/(DB).$$

All increments in P_s lie in the same dyadic band and are comparable within a factor less than two. Thus the anonymous long bounded-increment drift outlet has been converted into a comparable-amplitude plateau drift, an amplitude-depth packet, or a boundary-flux packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the comparable-amplitude plateau drift PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-long-drift-dyadic-plateau-router.md.`

□

Theorem 7.167 (Stable-ladder endpoint orbit plateau-ramp ledger). *The comparable-amplitude plateau-drift outlet admits the following non-cyclic ramp refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitCo
⇒ StableLadderEndpointOrbitComparableAmplitudePlateauDriftImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauRampLedger
AND StableLadderEndpointOrbitComparablePlateauSameSignLoadLedger
AND StableLadderEndpointOrbitPlateauLengthBudgetDichotomyLedger
AND StableLadderEndpointOrbitShortPlateauEdgeSpikeAbsorptionLedger
AND StableLadderEndpointOrbitLongPlateauMonotoneRampPacketLedger
AND NoAnonymousComparableAmplitudePlateauDriftExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPlateauRampLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit plateau-ramp ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The comparable-amplitude plateau branch gives a consecutive arc P_s, a sign σ, and a scale $a > 0$ such that

$$Z_j = \sigma Y_j > 0, \qquad a \le Z_j < 2a \qquad (j \in P_s), \qquad \sum_{j \in P_s} Z_j \ge H_0.$$

Fix a length budget $L_0 \ge 1$. If $|P_s| \le L_0$, then

$$\max_{j \in P_s} Z_j \ge H_0 / L_0.$$

This is an edge-spike at the induced threshold. By the already recorded centered-load expansion $Y_j = X_j - \mu$ and the half-threshold dichotomy, this short-plateau branch is absorbed by the endpoint singleton atom outlet or by the full-cycle mean atom outlet.

It remains to consider $|P_s| > L_0$. On this branch the prefix potential

$$S(t) = \sum_{i \le t} \sigma Y_i$$

increases monotonically along the plateau arc and its total increase across P_s is at least H_0. Thus the anonymous comparable-amplitude plateau drift has been converted into a long monotone plateau-ramp potential packet. The next direct task is to exclude this ramp potential packet, or to show that it necessarily enters the already named amplitude-depth or boundary-flux controlled exits.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the plateau-ramp potential PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-ramp-router.md.`

□

Theorem 7.168 (Stable-ladder endpoint orbit plateau-return mirror ledger)**.** *The plateau-ramp potential outlet admits the following non-cyclic return-mirror refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPl
⇒ StableLadderEndpointOrbitPlateauRampPotentialImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauReturnMirrorLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauReturnMirrorLedger
```

```
AND StableLadderEndpointOrbitPositivePlateauRampImportedModelLedger
AND StableLadderEndpointOrbitFullCycleZeroSumReturnObligationLedger
AND StableLadderEndpointOrbitNegativeReturnMassLowerBoundLedger
AND StableLadderEndpointOrbitReturnMassDyadicAmplitudePartitionLedger
AND StableLadderEndpointOrbitReturnAmplitudeDepthDichotomyLedger
AND StableLadderEndpointOrbitHeavyReturnDyadicBandLedger
AND StableLadderEndpointOrbitReturnPlateauRunPartitionLedger
AND StableLadderEndpointOrbitReturnRunBoundaryDichotomyLedger
AND StableLadderEndpointOrbitOppositeSignPlateauRampPairPacketLedger
AND NoAnonymousPlateauRampPotentialExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPlateauReturnMirrorLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit plateau-return mirror ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The plateau-ramp branch gives a consecutive arc P and a sign σ such that

$$Z_j = \sigma Y_j > 0 \qquad (j \in P), \qquad \sum_{j \in P} Z_j \geq H_0.$$

Because Y is the centered orbit load, the full-cycle identity gives

$$\sum_{\text{cycle}} \sigma Y_j = 0.$$

Thus the complementary arc must carry return mass. With

$$W_j = (-\sigma Y_j)_+$$

on the complement of P, we have

$$\sum W_j \geq H_0.$$

Partition this return mass into dyadic amplitude bands. Given a depth budget D, if the number of nonempty return bands is greater than D, the packet enters the amplitude-depth PDEC/cap outlet. Otherwise one return band carries mass at least H_0/D. Partition that heavy return band into maximal consecutive return plateau runs. Given a boundary budget B, if the number of return runs is greater than B, the packet enters the variation-boundary flux outlet. Otherwise some negative return plateau Q satisfies

$$\sum_{j \in Q} W_j \geq H_0/(DB).$$

Consequently a plateau-ramp potential cannot remain as a one-sided anonymous growth packet: it is either an amplitude-depth packet, a boundary-flux packet, or a paired positive-ramp/negative-return mirror packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the opposite-sign plateau-ramp pair PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-return-mirror-router.md.`

□

Theorem 7.169 (Stable-ladder endpoint orbit plateau-pair phase ledger)**.** *The opposite-sign plateau-ramp pair outlet admits the following non-cyclic phase refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitOp
⇒ StableLadderEndpointOrbitOppositeSignPlateauRampPairImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPlateauPairPhaseLedger
AND StableLadderEndpointOrbitOppositeSignPlateauPairModelLedger
AND StableLadderEndpointOrbitPlateauPairCyclicGapDecompositionLedger
AND StableLadderEndpointOrbitPlateauPairGapBudgetDichotomyLedger
AND StableLadderEndpointOrbitNearContactOppositeSignBoundaryPacketLedger
AND StableLadderEndpointOrbitPhaseSeparatedBipolarPlateauPairPacketLedger
AND NoAnonymousOppositeSignPlateauRampPairExitLedger
AND SparseScaleLadderSAECarriedForwardAfterPlateauPairPhaseLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit plateau-pair phase ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The opposite-sign plateau pair gives two disjoint consecutive arcs P and Q, with a sign σ, such that

$$\sigma Y_j > 0 \quad (j \in P), \qquad -\sigma Y_j > 0 \quad (j \in Q).$$

The two arcs determine two cyclic gaps g_1, g_2 between them. Fix a gap budget E. If

$$\min(g_1, g_2) \le E,$$

then the positive and negative plateau packets meet through a short bridge; this is recorded as a near-contact opposite-sign boundary packet and is carried by the variation-boundary flux outlet. Otherwise

$$g_1 > E, \qquad g_2 > E,$$

so the two packets form a genuinely phase-separated bipolar plateau pair.

Thus the opposite-sign pair is no longer anonymous. It is either a near-contact boundary packet, or a phase-separated bipolar packet with explicit cyclic separation.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the phase-separated bipolar plateau pair PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-plateau-pair-phase-router.md.`

□

Theorem 7.170 (Stable-ladder endpoint orbit bipolar-shelf ledger)**.** *The phase-separated bipolar plateau-pair outlet admits the following non-cyclic bridge refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitPh
⇒ StableLadderEndpointOrbitPhaseSeparatedBipolarPlateauPairImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterBipolarShelfLedger
AND StableLadderEndpointOrbitPhaseSeparatedBipolarPairModelLedger
AND StableLadderEndpointOrbitBipolarPrefixPotentialCoordinateLedger
AND StableLadderEndpointOrbitBipolarBridgeDecompositionLedger
AND StableLadderEndpointOrbitBridgeCancellationOrShelfDichotomyLedger
AND StableLadderEndpointOrbitBridgeCancellationPacketLedger
AND StableLadderEndpointOrbitLongPotentialShelfPacketLedger
AND NoAnonymousPhaseSeparatedBipolarPlateauPairExitLedger
AND SparseScaleLadderSAECarriedForwardAfterBipolarShelfLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit bipolar-shelf ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The phase-separated bipolar pair gives a positive plateau P, a negative plateau Q, and a sign σ, with both cyclic gaps between them greater than E. Define the directed prefix potential

$$S(t) = \sum_{i \le t} \sigma Y_i.$$

The arc P raises S by at least the packet height H, while Q returns S by at least the same scale. Consider the bridge from the end of P to the start of Q. If this bridge changes S downward by at least $H/2$ before the packet reaches Q, then the bridge itself is a named bridge-cancellation packet. Otherwise S remains displaced by at least $H/2$ throughout a bridge of length greater than E, giving a long potential shelf. The bridge from Q back to P is treated in the same way with the sign reversed.

Thus a phase-separated bipolar pair is not an anonymous terminal. It must produce either explicit bridge cancellation or a long potential shelf, unless it has already entered amplitude-depth or variation-boundary flux.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the long potential shelf PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-bipolar-shelf-router.md.
```

□

Theorem 7.171 (Stable-ladder endpoint orbit long-potential-shelf static-bias ledger)**.** *The long-potential-shelf outlet admits the following non-cyclic internal refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLongPotentialShelfImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterShelfStaticBiasLedger
AND StableLadderEndpointOrbitLongPotentialShelfModelLedger
AND StableLadderEndpointOrbitShelfPotentialFloorLedger
AND StableLadderEndpointOrbitShelfIncrementBalanceLedger
AND StableLadderEndpointOrbitShelfOscillationBoundaryFluxDichotomyLedger
AND StableLadderEndpointOrbitShelfPositiveDriftAmplitudeDepthDichotomyLedger
AND StableLadderEndpointOrbitShelfNegativeCancellationDichotomyLedger
AND StableLadderEndpointOrbitStaticPotentialShelfBiasPacketLedger
AND NoAnonymousLongPotentialShelfExitLedger
AND SparseScaleLadderSAECarriedForwardAfterShelfStaticBiasLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit long-potential-shelf static-bias ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A long potential shelf gives an interval I of length $L > E$ and a sign σ such that

$$\sigma S(t) \ge H/2 \qquad (t \in I).$$

Writing $d_t = S(t+1) - S(t)$, the internal motion on I has only three named ways to avoid becoming static. If the boundary switches or total variation exceed the active budget, it is a variation-boundary flux packet. If same-sign drift keeps increasing the potential through a dyadic depth threshold, it is an amplitude-depth packet. If opposite-sign increments cancel a fixed part of the height before the opposite plateau is reached, it is a bridge-cancellation packet.

If none of these three alternatives occurs, the shelf has a one-sided potential floor, low oscillation, no deep same-sign drift, and no early opposite-sign cancellation. This remaining case is precisely the named static potential shelf bias packet. Thus a long potential shelf is no longer an anonymous terminal.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the static potential shelf bias PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-long-potential-shelf-static-bias-router.md.`

□

Theorem 7.172 (Stable-ladder endpoint orbit static-shelf area-moment ledger)**.** *The static-potential-shelf-bias outlet admits the following non-cyclic area refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt
⇒ StableLadderEndpointOrbitStaticPotentialShelfBiasImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterStaticShelfAreaLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterStaticShelfAreaLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterStaticShelfAreaLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterStaticShelfAreaLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterStaticShelfAreaLedger
AND StableLadderEndpointOrbitStaticShelfBiasModelLedger
AND StableLadderEndpointOrbitStaticShelfAreaLowerBoundLedger
AND StableLadderEndpointOrbitShelfSummationByPartsLedger
AND StableLadderEndpointOrbitShelfEndpointChargeReturnLedger
AND StableLadderEndpointOrbitStaticShelfAreaMomentPacketLedger
AND NoAnonymousStaticPotentialShelfBiasExitLedger
AND SparseScaleLadderSAECarriedForwardAfterStaticShelfAreaLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit static-shelf area-moment ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A static potential shelf bias gives an interval I and a sign σ such that

$$\sigma S(t) \geq H/2 \qquad (t \in I), \qquad |I| = L > E.$$

Therefore the rectangular potential area satisfies

$$A(I) := \sum_{t \in I} \sigma S(t) \geq HL/2.$$

Discrete summation by parts writes $A(I)$ as endpoint charges plus a sawtooth/triangular weighted increment moment in the increments $d_t = S(t+1) - S(t)$. If the endpoint charges already pay this area, the packet returns to bridge cancellation, amplitude depth, or variation-boundary flux. If they do not, the remaining obstruction is exactly a static shelf area-moment packet.

Thus a static potential shelf bias is not an anonymous terminal. It is either paid by already named endpoint exits, or it leaves a named area-moment PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the static shelf area-moment PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-static-shelf-area-moment-router.md.`

□

Theorem 7.173 (Stable-ladder endpoint orbit static-shelf centroid-moment ledger)**.** *The static-shelf area-moment outlet admits the following non-cyclic collar/core refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt
⇒ StableLadderEndpointOrbitStaticShelfAreaMomentImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterStaticShelfCentroidLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterStaticShelfCentroidLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterStaticShelfCentroidLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterStaticShelfCentroidLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterStaticShelfCentroidLedger
AND StableLadderEndpointOrbitStaticShelfAreaMomentModelLedger
AND StableLadderEndpointOrbitStaticShelfBoundaryCollarSplitLedger
AND StableLadderEndpointOrbitStaticShelfCollarChargeReturnLedger
AND StableLadderEndpointOrbitStaticShelfInteriorCoreMassLedger
AND StableLadderEndpointOrbitStaticShelfInteriorCentroidMomentPacketLedger
AND NoAnonymousStaticShelfAreaMomentExitLedger
AND SparseScaleLadderSAECarriedForwardAfterStaticShelfCentroidLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit static-shelf centroid-moment ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A static shelf area moment is an area moment not paid by the endpoint charges. Choose $0 < \eta < 1/4$ and decompose the shelf interval as

$$I = C_{\mathrm{left}}(\eta L) \cup K_\eta \cup C_{\mathrm{right}}(\eta L),$$

where the two collars have length ηL and K_η is the interior core. If the weighted increment moment is mainly supported on the collars, then it is still endpoint charge and returns to bridge cancellation, amplitude depth, or variation-boundary flux.

Otherwise a fixed proportion of the weighted moment lies in K_η. The interior signed centroid

$$c_K = \frac{\sum_{t \in K_\eta} t\, w(t)\sigma d_t}{\sum_{t \in K_\eta} w(t)\sigma d_t}$$

is then well-defined after the usual dyadic sign choice and is separated from both endpoints by ηL. This packet can no longer be explained by collar charging; it is the named static shelf interior-centroid moment outlet.

Thus a static shelf area moment is not an anonymous terminal. It is either paid by already named endpoint exits, or it leaves a named interior-centroid PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the static shelf interior-centroid moment PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-static-shelf-centroid-moment-router.md.`

□

Theorem 7.174 (Stable-ladder endpoint orbit static-shelf centroid phase-lock ledger). *The static-shelf interior-centroid moment outlet admits the following non-cyclic phase-lock refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt
⇒ StableLadderEndpointOrbitStaticShelfInteriorCentroidMomentImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterCentroidPhaseLockLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterCentroidPhaseLockLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterCentroidPhaseLockLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterCentroidPhaseLockLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterCentroidPhaseLockLedger
AND StableLadderEndpointOrbitInteriorCentroidCoordinateModelLedger
AND StableLadderEndpointOrbitInteriorCentroidScaleWindowLedger
AND StableLadderEndpointOrbitInteriorCentroidDriftReturnLedger
AND StableLadderEndpointOrbitInteriorCentroidOppositeSignBridgeReturnLedger
AND StableLadderEndpointOrbitInteriorCentroidSameSignStackReturnLedger
AND StableLadderEndpointOrbitInteriorCentroidPhaseLockPacketLedger
AND NoAnonymousStaticShelfInteriorCentroidMomentExitLedger
AND SparseScaleLadderSAECarriedForwardAfterCentroidPhaseLockLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit static-shelf centroid phase-lock ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. The interior centroid moment gives, for each surviving stable-ladder window j, an interior signed mass and centroid

$$m_j = \sum_{t \in K_j} w_j(t)\sigma_j d_t, \qquad c_j = \frac{\sum_{t \in K_j} t\, w_j(t)\sigma_j d_t}{m_j}.$$

Writing $K_j = [a_j, a_j + L_j]$ after normalization gives an interior phase

$$\theta_j = \frac{c_j - a_j}{L_j} \in [\eta, 1-\eta].$$

Partition the compact interval $[\eta, 1-\eta]$ into fixed phase cells and compare successive stable-ladder windows. If the θ_j have persistent macroscopic drift, the signed mass must cross an internal phase cut and the packet returns to the variation-boundary flux outlet. If opposite signs occupy the same or adjacent phase cells with comparable mass, the packet returns to the bridge-cancellation outlet. If one sign stacks in a phase cell past the available potential budget, the packet returns to the amplitude-depth outlet.

If none of these three payments occurs, the remaining centroid mass is locked in finitely many interior phase cells with stable sign and scale. This is not an anonymous centroid-moment terminal; it is the named static-shelf interior centroid phase-lock PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the static shelf interior-centroid phase-lock PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-static-shelf-centroid-phase-lock-router.md.`

□

Theorem 7.175 (Stable-ladder endpoint orbit locked phase-cell pressure ledger). *The static-shelf interior-centroid phase-lock outlet admits the following non-cyclic CRT-slot refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitSt
⇒ StableLadderEndpointOrbitInteriorCentroidPhaseLockImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterLockedPhaseCellLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterLockedPhaseCellLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterLockedPhaseCellLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterLockedPhaseCellLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterLockedPhaseCellLedger
AND StableLadderEndpointOrbitLockedInteriorPhaseCellModelLedger
AND StableLadderEndpointOrbitLockedPhaseCellFiniteSlotLedger
AND StableLadderEndpointOrbitLockedPhaseCellActualResidueWordLedger
AND StableLadderEndpointOrbitLockedPhaseCellSparseOrStableSubsequenceLedger
AND StableLadderEndpointOrbitLockedPhaseCellColumnPressurePacketLedger
AND NoAnonymousInteriorCentroidPhaseLockExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLockedPhaseCellLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit locked phase-cell pressure ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A phase-lock packet places surviving centroid mass in finitely many compact interior phase cells $B_r \subset [\eta, 1-\eta]$. Refine the bookkeeping by the finite slot

$$\omega = (r, \text{sign}, \text{dyadic mass band}, \text{actual residue word}).$$

If no slot ω persists through an infinite stable-ladder subsequence, the mass is sparse in the scale ladder and returns to the sparse SAE outlet.

Otherwise an infinite pigeonhole step gives a persistent actual residue word inside a fixed interior phase cell and dyadic mass band. Its CRT-cylinder projection is therefore a fixed column or a narrow family of columns carrying the locked sign and mass. This is not an anonymous phase-lock terminal; it is the named locked interior phase-cell column-pressure PDEC/cap outlet. Width collapse and mean-only collapse are still carried by the singleton and full-cycle mean atom outlets respectively.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the locked interior phase-cell column-pressure PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-locked-phase-cell-pressure-router.md.`

□

Theorem 7.176 (Stable-ladder endpoint orbit locked-column capacity ledger). *The locked phase-cell column-pressure outlet admits the following non-cyclic column-capacity refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLockedPhaseCellColumnPressureImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterLockedColumnCapacityLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterLockedColumnCapacityLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterLockedColumnCapacityLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterLockedColumnCapacityLedger
```

```
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterLockedColumnCapacityLedger
AND StableLadderEndpointOrbitLockedColumnFamilyModelLedger
AND StableLadderEndpointOrbitLockedColumnFiberQuotaLedger
AND StableLadderEndpointOrbitLockedColumnLoadQuotaDichotomyLedger
AND StableLadderEndpointOrbitLockedColumnUnderQuotaReturnLedger
AND StableLadderEndpointOrbitLockedColumnCapacityDefectPacketLedger
AND NoAnonymousLockedPhaseCellColumnPressureExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLockedColumnCapacityLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit locked-column capacity ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A locked phase-cell column-pressure packet gives a fixed column or narrow column family C in the CRT cylinder with signed load

$$L(C) = \sum_{\omega \mapsto C} \text{signed\_mass}(\omega).$$

The same local CRT data give a column fiber quota

$$Q(C) = \text{allowed\_fiber\_count}(C) \cdot \text{scale\_weight}\,.$$

If $|L(C)| \le Q(C)$, the column family does not carry excess load; then it cannot pay the preceding locked-mass obligation except through the already named singleton, full-cycle mean, sparse, bridge-cancellation, amplitude-depth, or variation-boundary flux exits. If $|L(C)| > Q(C)$, the fixed column family carries more signed load than its CRT fiber quota permits. This is the named locked column capacity-defect PDEC/cap outlet.

Thus locked column pressure is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit column-capacity defect.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the locked column capacity-defect PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-locked-column-capacity-router.md.
```

□

Theorem 7.177 (Stable-ladder endpoint orbit locked-column residue-shadow ledger)**.** *The locked column capacity-defect outlet admits the following non-cyclic residue-shadow refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLockedColumnCapacityDefectImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterLockedColumnResidueShadowLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterLockedColumnResidueShadowLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterLockedColumnResidueShadowLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterLockedColumnResidueShadowLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterLockedColumnResidueShadowLedger
AND StableLadderEndpointOrbitLockedColumnResidueShadowModelLedger
AND StableLadderEndpointOrbitLockedColumnResidueShadowQuotaLedger
AND StableLadderEndpointOrbitLockedColumnOverfullShadowPigeonholeLedger
AND StableLadderEndpointOrbitLockedColumnResidueShadowNamedReturnSplitLedger
AND StableLadderEndpointOrbitLockedColumnResidueShadowImbalancePacketLedger
AND NoAnonymousLockedColumnCapacityDefectExitLedger
AND SparseScaleLadderSAECarriedForwardAfterLockedColumnResidueShadowLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit locked-column residue-shadow ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A locked column capacity defect gives a column family C with $|L(C)| > Q(C)$. Decompose C into CRT residue shadows C_s, and write

$$L(C) = \sum_s L_s, \qquad Q(C) = \sum_s Q_s.$$

After fixing the sign of the excess, if every residue shadow satisfied $|L_s| \leq Q_s$, then the signed load of the whole column family would be bounded by the sum of the shadow quotas, contradicting the column capacity defect. Hence at least one residue shadow is overfull.

If that shadow degenerates to an isolated atom, it is carried by the singleton outlet. If it only contributes a full-cycle mean component, it is carried by the full-cycle mean outlet. If opposite shadow signs cancel, if one sign stacks across scales, or if the overfull shadow migrates through the shadow boundary, the packet returns respectively to bridge cancellation, amplitude depth, or variation-boundary flux. If none of those named payments occurs, the remaining packet is an explicit locked column residue-shadow imbalance PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the locked column residue-shadow imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-locked-column-residue-shadow-router.md.`

□

Theorem 7.178 (Stable-ladder endpoint orbit residue-shadow dual-row ledger)**.** *The locked column residue-shadow imbalance outlet admits the following non-cyclic dual-row refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitLo
⇒ StableLadderEndpointOrbitLockedColumnResidueShadowImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowLedger
AND StableLadderEndpointOrbitResidueShadowDualRowMapLedger
AND StableLadderEndpointOrbitResidueShadowDualRowFiberLedger
AND StableLadderEndpointOrbitResidueShadowDualRowAverageReturnLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPressurePacketLedger
AND NoAnonymousLockedColumnResidueShadowImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A locked column residue-shadow imbalance is represented by a column residue shadow C_s with signed excess $E(C_s)$. The CRT dual coordinate sends this shadow to a dual-row shadow R_s, preserving the signed excess:

$$s \mapsto R_s, \qquad E(R_s) = E(C_s).$$

Thus the column shadow cannot remain an anonymous one-sided column object.

If the dual-row fiber has no actual row pressure, the preserved excess is absorbed by row-fiber averaging and returns to the already named singleton, full-cycle mean, sparse, bridge-cancellation, amplitude-depth, or variation-boundary flux exits. If the averaging fails to absorb the excess, the remaining packet is an explicit residue-shadow dual-row pressure PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-shadow dual-row pressure PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-router.md.`

□

Theorem 7.179 (Stable-ladder endpoint orbit residue-shadow dual-row capacity ledger)**.** *The residue-shadow dual-row pressure outlet admits the following non-cyclic row-window capacity refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPressureImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowCapacityLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowCapacityLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowCapacityLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowCapacityLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowCapacityLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPressureModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowWindowLedger
AND StableLadderEndpointOrbitResidueShadowDualRowCapacityQuotaLedger
AND StableLadderEndpointOrbitResidueShadowDualRowLoadQuotaDichotomyLedger
AND StableLadderEndpointOrbitResidueShadowDualRowUnderQuotaReturnLedger
AND StableLadderEndpointOrbitResidueShadowDualRowCapacityDefectPacketLedger
AND NoAnonymousResidueShadowDualRowPressureExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowCapacityLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row capacity ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A residue-shadow dual-row pressure packet gives a finite dual-row window R with signed load

$$H(R) = \sum_{\omega \mapsto R} \text{signed\_mass}(\omega).$$

The same local CRT row-fiber data give a row-window quota

$$B(R) = \text{allowed\_row\_fiber\_count}(R) \cdot \text{scale\_weight}\,.$$

If $|H(R)| \le B(R)$, the dual-row window does not carry capacity pressure. Then it cannot pay the preceding forced pressure except through the already named singleton, full-cycle mean, sparse, bridge-cancellation, amplitude-depth, or variation-boundary flux exits. If $|H(R)| > B(R)$, the window carries more signed load than its CRT row-fiber quota permits. This is the named residue-shadow dual-row capacity-defect PDEC/cap outlet.

Thus residue-shadow dual-row pressure is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit dual-row capacity defect.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-shadow dual-row capacity-defect PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-capacity-router.md.`

□

Theorem 7.180 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell ledger)**.** *The residue-shadow dual-row capacity-defect outlet admits the following non-cyclic CRT phase-cell refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowCapacityDefectImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellQuotaLedger
AND StableLadderEndpointOrbitResidueShadowDualRowOverfullPhaseCellPigeonholeLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellNamedReturnSplitLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellImbalancePacketLedger
AND NoAnonymousResidueShadowDualRowCapacityDefectExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A residue-shadow dual-row capacity-defect packet gives a finite dual-row window R whose signed load exceeds its row-fiber quota. Partition R into CRT phase-cells R_θ. The load and quota decompose as

$$H(R) = \sum_\theta H_\theta, \qquad B(R) = \sum_\theta B_\theta.$$

After fixing the sign of the excess, if every phase-cell satisfied $|H_\theta| \le B_\theta$, then the whole window would satisfy $|H(R)| \le B(R)$. This contradicts the imported capacity defect. Therefore some phase-cell is overfull.

An overfull phase-cell is not left anonymous. If it degenerates to a singleton atom, it returns to the endpoint singleton outlet. If it only records a full-cycle mean, it returns to the full-cycle mean outlet. If the cell is paid by opposite-sign phase-cells, by same-sign amplitude stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the overfull cell, the remaining packet is the explicit residue-shadow dual-row phase-cell imbalance PDEC/cap outlet.

Thus residue-shadow dual-row capacity defect is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit phase-cell imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-shadow dual-row phase-cell imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-router.md
```

□

Theorem 7.181 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom ledger)**.** *The residue-shadow dual-row phase-cell imbalance outlet admits the following non-cyclic atom refinement inside a fixed CRT phase-cell:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomQuotaLedger
AND StableLadderEndpointOrbitResidueShadowDualRowOverfullPhaseCellAtomPigeonholeLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomNamedReturnSplitLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomImbalancePacketLedger
AND NoAnonymousResidueShadowDualRowPhaseCellImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A residue-shadow dual-row phase-cell imbalance packet gives a fixed CRT phase-cell θ with signed load exceeding its quota. Partition θ into actual residue/source atoms a. The load and quota decompose as

$$H_\theta = \sum_a h_{\theta,a}, \qquad B_\theta = \sum_a b_{\theta,a}.$$

Let $\sigma = \operatorname{sign}(H_\theta)$. If every atom satisfied $\sigma h_{\theta,a} \le b_{\theta,a}$, then summing over a would give

$$|H_\theta| = \sum_a \sigma h_{\theta,a} \le \sum_a b_{\theta,a} = B_\theta,$$

contradicting the imported overfull phase-cell. Hence at least one sign-aligned atom satisfies $\sigma h_{\theta,a} > b_{\theta,a}$.

This over-quota atom is not left anonymous. If it is a singleton atom, it returns to the endpoint singleton outlet. If it only records a full-cycle mean, it returns to the full-cycle mean outlet. If the atom is paid by an opposite-sign atom, by same-sign amplitude stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the atom, the remaining packet is the explicit residue-shadow dual-row phase-cell atom imbalance PDEC/cap outlet.

Thus residue-shadow dual-row phase-cell imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit atom imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-shadow dual-row phase-cell atom imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-rout
```

□

Theorem 7.182 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core ledger)**.** *The residue-shadow dual-row phase-cell atom imbalance outlet admits the following non-cyclic signed-core refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedDecompositionLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreLowerBoundLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreNamedReturnSplitLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreImbalancePacketLedger
AND NoAnonymousResidueShadowDualRowPhaseCellAtomImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A phase-cell atom imbalance packet gives an atom a and a sign $\sigma = \operatorname{sign}(h_a)$ with $\sigma h_a > b_a$. Split the actual internal contributions of the atom into the σ-aligned core and the opposite-sign cancellation mass:

$$\sigma h_a = C_a^+ - C_a^-, \qquad C_a^+ \geq 0, \quad C_a^- \geq 0.$$

Since $\sigma h_a > b_a$, the same identity gives

$$C_a^+ > b_a + C_a^-.$$

Thus the same-sign core exceeds the atom quota even after accounting for the opposite-sign cancellation mass.

The opposite-sign part is not ignored. If it is a material internal or mirror payment, it returns to the bridge-cancellation outlet. If the same-sign core degenerates to a singleton atom or a full-cycle mean atom, it returns to those outlets. If the same-sign core is paid by cross-scale same-sign stacking or by boundary migration, it returns respectively to the amplitude-depth or variation-boundary flux outlets. If none of these named returns pays the core, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core imbalance PDEC/cap outlet.

Thus residue-shadow dual-row phase-cell atom imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit signed-core imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-shadow dual-row phase-cell atom signed-core imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.183 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core imbalance outlet admits the following finite support-slice refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceL
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceLedg
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceQuotaDebtAllocationLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSlicePigeonholeLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceNamedReturnSplitLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceImbalancePacketLedger
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A signed-core imbalance packet gives

$$C_a^+ > b_a + C_a^-.$$

Partition the same-sign core into finitely many actual CRT/source support slices s, and allocate the atom quota plus cancellation debt over the same slices:

$$C_a^+ = \sum_s C_s, \qquad b_a + C_a^- = \sum_s d_s.$$

If every support slice satisfied $C_s \le d_s$, then summing over s would give $C_a^+ \le b_a + C_a^-$, contradicting the imported signed-core imbalance. Therefore some support slice satisfies $C_s > d_s$.

This overfull support slice is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the slice, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice imbalance PDEC/cap outlet.

Thus residue-shadow dual-row phase-cell atom signed-core imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit support-slice imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the support-slice imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.184 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice imbalance outlet admits the following finite residue-fiber refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberQuotaDebtAllocationLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPigeonholeLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberNamedReturnSplitLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberImbalancePacketLedger
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A support-slice imbalance packet gives, for one actual support slice s,

$$C_s > d_s.$$

Partition this support slice into finitely many actual CRT residue fibers ρ, and allocate the local quota plus cancellation debt over the same fibers:

$$C_s = \sum_\rho C_{s,\rho}, \qquad d_s = \sum_\rho d_{s,\rho}.$$

If every residue fiber satisfied $C_{s,\rho} \le d_{s,\rho}$, then summing over ρ would give $C_s \le d_s$, contradicting the imported support-slice imbalance. Therefore some residue fiber satisfies $C_{s,\rho} > d_{s,\rho}$.

This overfull residue fiber is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the fiber, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber imbalance PDEC/cap outlet.

Thus support-slice imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit residue-fiber imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the residue-fiber imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.185 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber imbalance outlet admits the following finite phase-word refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberImbalanceImportedLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordQuotaDebtAllocat
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordPigeonho
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordNamedReturnSplit
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordImbalancePacketL
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordLe
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word ledger. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A residue-fiber imbalance packet gives, for one actual support slice s and one actual residue fiber ρ,

$$C_{s,\rho} > d_{s,\rho}.$$

Partition this residue fiber into finitely many actual CRT phase words ω, and allocate the local quota plus cancellation debt over the same phase words:

$$C_{s,\rho} = \sum_{\omega} C_{s,\rho,\omega}, \qquad d_{s,\rho} = \sum_{\omega} d_{s,\rho,\omega}.$$

If every phase word satisfied $C_{s,\rho,\omega} \le d_{s,\rho,\omega}$, then summing over ω would give $C_{s,\rho} \le d_{s,\rho}$, contradicting the imported residue-fiber imbalance. Therefore some phase word satisfies $C_{s,\rho,\omega} > d_{s,\rho,\omega}$.

This overfull phase word is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the phase word, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word imbalance PDEC/cap outlet.

Thus residue-fiber imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit phase-word imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the phase-word imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.186 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word imbalance outlet admits the following finite word-slot refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordImbalanceImported
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotModelLedger
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotQuotaDebtAll
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotPige
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotNamedReturnS
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotImbalancePac
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot ledg
The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A phase-word imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, and one actual phase word ω,

$$C_{s,\rho,\omega} > d_{s,\rho,\omega}.$$

Partition this phase word into finitely many actual CRT word slots τ, and allocate the local quota plus cancellation debt over the same slots:

$$C_{s,\rho,\omega} = \sum_{\tau} C_{s,\rho,\omega,\tau}, \qquad d_{s,\rho,\omega} = \sum_{\tau} d_{s,\rho,\omega,\tau}.$$

If every word slot satisfied $C_{s,\rho,\omega,\tau} \leq d_{s,\rho,\omega,\tau}$, then summing over τ would give $C_{s,\rho,\omega} \leq d_{s,\rho,\omega}$, contradicting the imported phase-word imbalance. Therefore some word slot satisfies

$$C_{s,\rho,\omega,\tau} > d_{s,\rho,\omega,\tau}.$$

This overfull word slot is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the word slot, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot imbalance PDEC/cap outlet.

Thus phase-word imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit phase-word-slot imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the phase-word-slot imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign`

□

Theorem 7.187 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot imbalance outlet admits the following finite source-atom refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotImbalanceImpo
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMo
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomQu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomNa
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomIm
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotImbalanceExitLedger
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot sour
The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A phase-word-slot imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, and one actual word slot τ,

$$C_{s,\rho,\omega,\tau} > d_{s,\rho,\omega,\tau}.$$

Partition this word slot into finitely many actual endpoint/source atoms α, and allocate the local quota plus cancellation debt over the same atoms:

$$C_{s,\rho,\omega,\tau} = \sum_{\alpha} C_{s,\rho,\omega,\tau,\alpha}, \qquad d_{s,\rho,\omega,\tau} = \sum_{\alpha} d_{s,\rho,\omega,\tau,\alpha}.$$

If every source atom satisfied $C_{s,\rho,\omega,\tau,\alpha} \le d_{s,\rho,\omega,\tau,\alpha}$, then summing over α would give $C_{s,\rho,\omega,\tau} \le d_{s,\rho,\omega,\tau}$, contradicting the imported phase-word-slot imbalance. Therefore some source atom satisfies

$$C_{s,\rho,\omega,\tau,\alpha} > d_{s,\rho,\omega,\tau,\alpha}.$$

This overfull source atom is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the source atom, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom imbalance PDEC/cap outlet.

Thus phase-word-slot imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit source-atom imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the phase-word-slot source-atom imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.188 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed–core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom imbalance outlet admits the following finite multiplicity-fiber refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomImb
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomImbalanceExitLed
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot sour
The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are carried forward. A source-atom imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, one actual word slot τ, and one actual endpoint/source atom α,

$$C_{s,\rho,\omega,\tau,\alpha} > d_{s,\rho,\omega,\tau,\alpha}.$$

If the actual occurrence multiplicity of this source atom already exceeds the local cap used by the load contract, the packet is not anonymous: it is the source-atom multiplicity-cap PDEC/cap outlet. Otherwise partition the source atom into finitely many actual multiplicity fibers μ, and allocate the local quota plus cancellation debt over the same fibers:

$$C_{s,\rho,\omega,\tau,\alpha} = \sum_{\mu} C_{s,\rho,\omega,\tau,\alpha,\mu}, \qquad d_{s,\rho,\omega,\tau,\alpha} = \sum_{\mu} d_{s,\rho,\omega,\tau,\alpha,\mu}.$$

If every multiplicity fiber satisfied $C_{s,\rho,\omega,\tau,\alpha,\mu} \leq d_{s,\rho,\omega,\tau,\alpha,\mu}$, then summing over μ would give $C_{s,\rho,\omega,\tau,\alpha} \leq d_{s,\rho,\omega,\tau,\alpha}$, contradicting the imported source-atom imbalance. Hence, unless the multiplicity-cap outlet is active, some multiplicity fiber satisfies

$$C_{s,\rho,\omega,\tau,\alpha,\mu} > d_{s,\rho,\omega,\tau,\alpha,\mu}.$$

This overfull multiplicity fiber is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the fiber, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber imbalance PDEC/cap outlet.

Thus source-atom imbalance is not an anonymous terminal. It is either absorbed by existing exits, becomes a multiplicity-cap packet, or leaves an explicit multiplicity-fiber imbalance packet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the source-atom multiplicity-fiber imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign`

□

Theorem 7.189 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed–core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-unit ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber imbalance outlet admits the following finite signed-occurrence-unit refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMul
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFibe
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot sour
The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward. A multiplicity-fiber imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, one actual word slot τ, one actual endpoint/source atom α, and one actual multiplicity fiber μ,

$$C_{s,\rho,\omega,\tau,\alpha,\mu} > d_{s,\rho,\omega,\tau,\alpha,\mu}.$$

Partition this fiber into finitely many actual signed occurrence/load units η, and allocate the local quota plus cancellation debt over the same units:

$$C_{s,\rho,\omega,\tau,\alpha,\mu} = \sum_{\eta} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu} = \sum_{\eta} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta}.$$

If every signed occurrence unit satisfied $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta}$, then summing over η would give $C_{s,\rho,\omega,\tau,\alpha,\mu} \le d_{s,\rho,\omega,\tau,\alpha,\mu}$, contradicting the imported multiplicity-fiber imbalance. Therefore some signed occurrence unit satisfies

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta}.$$

This overfull signed unit is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the signed unit, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit imbalance PDEC/cap outlet.

Thus multiplicity-fiber imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit signed-occurrence-unit imbalance packet, while the source-atom multiplicity-cap outlet remains a separate open exit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the signed-occurrence-unit imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.190 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-unit incidence-cell ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit imbalance outlet admits the following finite incidence-cell refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMul
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFibe
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot sour
The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward. A signed-occurrence-unit imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, one actual word slot τ, one actual endpoint/source atom α, one actual multiplicity fiber μ, and one actual signed occurrence unit η,

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta}.$$

Partition this signed occurrence unit into finitely many actual incidence cells ι, and allocate the local quota plus cancellation debt over the same cells:

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta} = \sum_{\iota} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu,\eta} = \sum_{\iota} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}.$$

If every incidence cell satisfied $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}$, then summing over ι would give $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta}$, contradicting the imported signed-occurrence-unit imbalance. Therefore some incidence cell satisfies

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}.$$

This overfull incidence cell is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the cell, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit incidence-cell imbalance PDEC/cap outlet.

Thus signed-occurrence-unit imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit incidence-cell imbalance packet, while the source-atom multiplicity-cap outlet remains a separate open exit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the incidence-cell imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.191 (Stable-ladder endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-unit incidence-cell primitive-witness ledger)**.** *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit incidence-cell imbalance outlet admits the following finite primitive-witness refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMul
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFibe
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Endpoint orbit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot sour
The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward. An incidence-cell imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, one actual word slot τ, one actual endpoint/source atom α, one actual multiplicity fiber μ, one actual signed occurrence unit η, and one actual incidence cell ι,

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}.$$

Partition this incidence cell into finitely many actual primitive witnesses κ. Each κ records the actual row, column, carrier prime, residue representative, endpoint side, and orientation, so the cell is no longer treated as an anonymous capacity bucket. Allocate the local quota plus cancellation debt over the same witnesses:

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} = \sum_{\kappa} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} = \sum_{\kappa} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}.$$

If every primitive witness satisfied $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} \leq d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}$, then summing over κ would give $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota} \leq d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota}$, contradicting the imported incidence-cell imbalance. Therefore some primitive witness satisfies

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}.$$

This overfull primitive witness is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the witness, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit incidence-cell primitive-witness imbalance PDEC/cap outlet.

Thus incidence-cell imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit primitive-witness imbalance packet, while the source-atom multiplicity-cap outlet remains a separate open exit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the primitive-witness imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-firstbreak-tail-gap-stable-ladder-endpoint-orbit-residue-shadow-dual-row-phase-cell-atom-sign
```

□

Theorem 7.192 (Stable-ladder primitive-witness CRT-coordinate-atom ledger). *The residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit incidence-cell primitive-witness imbalance outlet admits the following finite CRT-coordinate-atom refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMul
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidue
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceR
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSlice
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResi
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSl
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreOverfullSupportSliceResidueFiberPhaseWordSlotSour
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND NoAnonymousResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFibe
AND SparseScaleLadderSAECarriedForwardAfterResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSl
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Primitive-witness CRT-coordinate-atom ledger. The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward. A primitive-witness imbalance packet gives, for one actual support slice s, one actual residue fiber ρ, one actual phase word ω, one actual word slot τ, one actual endpoint/source atom α, one actual multiplicity fiber μ, one actual signed occurrence unit η, one actual incidence cell ι, and one actual primitive witness κ,

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}.$$

Partition this primitive witness into finitely many canonical CRT coordinate atoms χ. Each χ records the actual row, column, carrier prime q, residue a, endpoint side, orientation, and the congruence equation

$$N_{\text{row,column,side}} \equiv a \pmod{q}.$$

Allocate the local quota plus cancellation debt over the same coordinate atoms:

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} = \sum_{\chi} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} = \sum_{\chi} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi}.$$

If every coordinate atom satisfied $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi}$, then summing over χ would give $C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa}$, contradicting the imported primitive-witness imbalance. Therefore some CRT coordinate atom satisfies

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi}.$$

This overfull CRT coordinate atom is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the atom, the remaining packet is the explicit residue-shadow dual-row phase-cell atom signed-core support-slice residue-fiber phase-word-slot source-atom multiplicity-fiber signed-occurrence-unit incidence-cell primitive-witness CRT-coordinate-atom imbalance PDEC/cap outlet.

Thus primitive-witness imbalance is not an anonymous terminal. It is either absorbed by existing exits or leaves an explicit CRT-coordinate-atom imbalance packet, while the source-atom multiplicity-cap outlet remains a separate open exit.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the CRT-coordinate-atom imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-primitive-witness-crt-coordinate-atom-router.md.`

□

Theorem 7.193 (CRT-coordinate canonical-congruence-equation ledger). *The CRT-coordinate-atom imbalance outlet admits the following finite canonical-congruence-equation refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMul
AND StableLadderEndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMu
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationAtomModelLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationNormalFormLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationIDStabilityLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationQuotaDebtAllocationLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationAtomPigeonholeLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationAtomNamedReturnSplitLedger
AND StableLadderEndpointOrbitCRTCoordinateCanonicalCongruenceEquationAtomImbalancePacketLedger
AND NoAnonymousCRTCoordinateAtomImbalanceExitAfterCanonicalCongruenceEquationLedger
AND SparseScaleLadderSAECarriedForwardAfterCRTCoordinateCanonicalCongruenceEquationAtomLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

CRT-coordinate canonical-congruence-equation ledger. The source-atom multiplicity-cap outlet is again carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward.

A CRT-coordinate-atom imbalance packet gives, for one actual coordinate atom χ,

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi}.$$

The coordinate atom may still contain several presentations of the same underlying congruence equation. Put each presentation into a primitive affine normal form and define a canonical equation atom ϵ by the stable tuple

$$\epsilon = (\text{equation\_id}, \text{normal\_form}, \text{row}, \text{column}, q, a, \text{side}, \text{orientation}, \text{phase\_boundary\_key}).$$

The equation identifier is the hash-stable normal form together with the actual row-column coordinate, carrier prime q, residue a, endpoint side, and orientation. Hence different carrier primes, residues, rows, columns, sides, or boundary phases cannot silently pay each other inside this ledger.

Allocate the local quota plus cancellation debt over the same equation atoms:

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi} = \sum_{\epsilon} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi} = \sum_{\epsilon} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon}.$$

If every canonical congruence equation atom satisfied

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon},$$

then summing over ϵ would contradict the imported CRT-coordinate-atom imbalance. Therefore some canonical congruence equation atom is overfull.

This overfull equation atom is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the atom, the remaining packet is the explicit canonical-congruence-equation atom imbalance PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the canonical-congruence-equation atom imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-crt-coordinate-canonical-equation-router.md.`

□

Theorem 7.194 (Canonical-equation phase-residue-evaluation ledger)**.** *The canonical-congruence-equation atom imbalance outlet admits the following finite phase-residue-evaluation refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitCanonicalCongruenceEquationAtomImbalanceImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationAtomModelLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationRowPhaseLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationColumnPhaseLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationCarrierResidueEvaluationLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationIDStabilityLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationQuotaDebtAllocationLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationAtomPigeonholeLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationAtomNamedReturnSplitLedger
AND StableLadderEndpointOrbitCanonicalCongruenceEquationPhaseResidueEvaluationAtomImbalancePacketLedger
AND NoAnonymousCanonicalCongruenceEquationAtomImbalanceAfterPhaseResidueEvaluationLedger
AND SparseScaleLadderSAECarriedForwardAfterPhaseResidueEvaluationAtomLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Canonical-equation phase-residue-evaluation ledger. The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward.

A canonical-congruence-equation atom imbalance packet gives

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon}.$$

The atom ϵ fixes the congruence equation, but the present ledger also materializes its actual CRT phase evaluation. Define a phase-residue evaluation atom ζ by

$$\zeta = (\text{evaluation\_id}, \text{equation\_id}, q, a, \text{row} \mod q, \text{column} \mod q, N_{\text{row,column,side}} \mod q, \text{side}, \text{orientation}, \text{phase\_bo}$$

Thus the row phase, column phase, carrier prime, target residue, evaluated residue, side, orientation, and boundary tag are all fixed before any payment is allowed. Different row phases, column phases, evaluated residues, or boundary phases cannot be silently merged.

Allocate the local quota plus cancellation debt over the same evaluation atoms:

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon} = \sum_{\zeta} C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta}, \qquad d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon} = \sum_{\zeta} d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta}.$$

If every phase-residue evaluation atom satisfied

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta},$$

then summing over ζ would contradict the imported canonical-equation imbalance. Therefore some phase-residue evaluation atom is overfull.

This overfull phase-residue atom is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the atom, the remaining packet is the explicit phase-residue-evaluation atom imbalance PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue-evaluation atom imbalance PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-canonical-equation-phase-residue-evaluation-router.md.`

□

Theorem 7.195 (Phase-residue Hall-defect ledger)**.** *The phase-residue-evaluation atom imbalance outlet admits the following finite Hall-defect refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueEvaluationAtomImbalanceImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterPhaseResidueHallDefectLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationLoadTokenLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationQuotaDebtSlotLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationPaymentGraphLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationHallDefectNormalFormLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationHallDefectPigeonholeLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationHallDefectPacketLedger
AND StableLadderEndpointOrbitPhaseResidueEvaluationHallDefectNamedReturnSplitLedger
AND NoAnonymousPhaseResidueEvaluationImbalanceAfterHallDefectLedger
AND SparseScaleLadderSAECarriedForwardAfterPhaseResidueHallDefectLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue Hall-defect ledger. The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward.

A phase-residue-evaluation atom imbalance packet gives

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta} > d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta}.$$

Expand the left side into a finite set L_ζ of actual load tokens and the right side into a finite set D_ζ of quota/debt slots. Form the payment graph

$$G_\zeta = (L_\zeta, D_\zeta, E_\zeta),$$

where an edge is allowed only when it preserves the evaluation identifier, the endpoint side, the orientation, and the phase-boundary key. Thus a token cannot be paid by a slot from a different row-column phase, carrier-residue evaluation, side, orientation, or boundary phase.

If every subset $S \subseteq L_\zeta$ satisfied

$$|S| \le |N_{G_\zeta}(S)|,$$

Hall's theorem would give an injective matching from all load tokens into the quota/debt slots. This would imply

$$C_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta} \le d_{s,\rho,\omega,\tau,\alpha,\mu,\eta,\iota,\kappa,\chi,\epsilon,\zeta},$$

contradicting the imported phase-residue imbalance. Therefore a Hall defect subset exists:

$$\exists S \subseteq L_\zeta : \qquad \Delta_H(S) = |S| - |N_{G_\zeta}(S)| > 0.$$

This overfull Hall defect is not left anonymous. If it is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the defect, the remaining packet is the explicit phase-residue Hall-defect PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue Hall-defect PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-hall-defect-router.md.`

□

Theorem 7.196 (Phase-residue critical-Hall-cut ledger)**.** *The phase-residue Hall-defect outlet admits the following finite critical-Hall-cut refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueHallDefectImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterCriticalHallCutLedger
AND StableLadderEndpointOrbitPhaseResidueHallDefectFiniteFamilyLedger
AND StableLadderEndpointOrbitPhaseResidueHallDefectMinimalChoiceLedger
AND StableLadderEndpointOrbitPhaseResidueHallDefectConnectedCoreLedger
AND StableLadderEndpointOrbitPhaseResidueHallDefectProperSubsetHallOKLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutBoundaryLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutMarginLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutPacketLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutNamedReturnSplitLedger
AND NoAnonymousPhaseResidueHallDefectAfterCriticalCutLedger
AND SparseScaleLadderSAECarriedForwardAfterCriticalHallCutLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue critical-Hall-cut ledger. The source-atom multiplicity-cap outlet is carried forward as an independent PDEC/cap outlet. The endpoint singleton atom outlet, the full-cycle mean atom outlet, the bridge-cancellation outlet, the amplitude-depth outlet, the variation-boundary flux outlet, and the sparse scale-ladder SAE outlet are also carried forward.

A phase-residue Hall defect provides a finite nonempty family

$$\mathcal{F} = \{S \subseteq L_\zeta : \ |S| > |N_{G_\zeta}(S)|\}.$$

Choose a canonical member S_* by minimizing first $|S|$, then maximizing the defect $\Delta_H(S) = |S| - |N_{G_\zeta}(S)|$, and finally using the stable hash of the finite token set as a tie-breaker. Put

$$B_* = N_{G_\zeta}(S_*), \qquad \Delta_* = |S_*| - |B_*| > 0.$$

By the minimality of $|S_*|$, every proper subset $T \subsetneq S_*$ satisfies

$$|T| \le |N_{G_\zeta}(T)|.$$

If S_* decomposed into disconnected components in the induced graph on $S_* \cup B_*$, then the neighborhoods of those components would be disjoint and the positive total defect would force at least one component to have positive defect. That component would be a smaller member of $\mathcal{F}$, contradicting the choice of S_*. Hence the defect can be taken as a connected critical cut.

Thus an arbitrary Hall defect is not a terminal object. It is replaced by a finite critical cut with explicit load side S_*, payment boundary B_*, and indivisible margin Δ_*. If this cut is a singleton atom or only a full-cycle mean atom, it returns to those outlets. If it is paid by opposite-sign quota debt, by same-sign cross-scale stacking, or by boundary migration, it returns respectively to the bridge-cancellation, amplitude-depth, or variation-boundary flux outlets. If none of these named returns pays the cut, the remaining packet is the explicit phase-residue critical-Hall-cut PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue critical-Hall-cut PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-critical-hall-cut-router.md.
```

□

Theorem 7.197 (Phase-residue unit-defect critical-cut ledger)**.** *The phase-residue critical-Hall-cut outlet admits the following unit-defect refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueCriticalHallCutImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterUnitDefectCriticalCutLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutProperSubsetHallOKImportedLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutSingletonCaseUnitDefectLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutSingleDeletionBoundLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutUnitMarginLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutSingleDeletionBoundarySaturationLedger
AND NoMultiUnitPhaseResidueCriticalHallCutDefectLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalHallCutBoundaryRedundancyLedger
AND StableLadderEndpointOrbitPhaseResidueUnitDefectCriticalHallCutPacketLedger
AND NoAnonymousPhaseResidueCriticalHallCutAfterUnitDefectLedger
AND SparseScaleLadderSAECarriedForwardAfterUnitDefectCriticalCutLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue unit-defect critical-cut ledger. All parallel outlets from the critical-Hall-cut ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

Let S_* be the critical Hall cut and put

$$B_* = N_{G_\varsigma}(S_*), \qquad \Delta_* = |S_*| - |B_*| > 0.$$

The previous ledger gives

$$|T| \le |N_{G_\varsigma}(T)| \qquad (T \subsetneq S_*).$$

If $|S_*| = 1$, then the positive defect forces $B_* = \emptyset$, so $\Delta_* = 1$. If $|S_*| \ge 2$, choose any $s \in S_*$. Since $S_* \setminus \{s\}$ is a proper subset,

$$|S_*| - 1 \le |N_{G_\varsigma}(S_* \setminus \{s\})| \le |B_*| = |S_*| - \Delta_*.$$

Thus $\Delta_* \le 1$. Since Δ_* is a positive integer, one has $\Delta_* = 1$. The same chain also gives equality

$$N_{G_\varsigma}(S_* \setminus \{s\}) = B_* \qquad (s \in S_*),$$

because the left-hand neighborhood is contained in B_* and has the same cardinality.

Therefore no multi-unit capacity defect can remain hidden inside a critical Hall cut. Any surviving critical-cut obstruction is a unit unpaid phase-residue token with a boundary that remains visible after deleting any single source token, or it returns to one of the named outlets above.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue unit-defect critical-Hall-cut PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-unit-defect-critical-cut-router.md.
```

□

Theorem 7.198 (Phase-residue near-perfect matching circuit ledger)**.** *The phase-residue unit-defect critical-Hall-cut outlet admits the following near-perfect matching circuit refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueUnitDefectCriticalHallCutImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterNearPerfectMatchingCircuitLedger
AND StableLadderEndpointOrbitPhaseResidueUnitDefectMarginImportedLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalCutProperSubsetHallImportedForMatchingLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalCutDeletionBoundarySaturationImportedForMatchingLedger
AND StableLadderEndpointOrbitPhaseResidueEverySingleDeletionHallOKLedger
AND StableLadderEndpointOrbitPhaseResidueEverySingleDeletionPerfectMatchingLedger
AND StableLadderEndpointOrbitPhaseResidueCriticalCutMaximumMatchingSizeLedger
AND StableLadderEndpointOrbitPhaseResidueEverySourceCanBeUnmatchedLedger
AND StableLadderEndpointOrbitPhaseResidueBoundaryDoubleCoverLedger
AND StableLadderEndpointOrbitPhaseResidueNearPerfectMatchingCircuitPacketLedger
AND NoAnonymousUnitDefectCriticalHallCutAfterMatchingCircuitLedger
AND SparseScaleLadderSAECarriedForwardAfterNearPerfectMatchingCircuitLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue near-perfect matching circuit ledger. All parallel outlets from the unit-defect critical-cut ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

Let S_* be the unit-defect critical cut and put

$$B_* = N_{G_\varsigma}(S_*), \qquad |S_*| = |B_*| + 1.$$

The previous ledgers give both the proper-subset Hall condition

$$|T| \le |N_{G_\varsigma}(T)| \qquad (T \subsetneq S_*)$$

and the deletion-boundary saturation

$$N_{G_\varsigma}(S_* \setminus \{s\}) = B_* \qquad (s \in S_*).$$

Fix $s \in S_*$. Every subset $U \subseteq S_* \setminus \{s\}$ is a proper subset of S_*, hence it satisfies Hall's inequality. Since $|S_* \setminus \{s\}| = |B_*|$, Hall's theorem gives a perfect matching

$$M_s : S_* \setminus \{s\} \longrightarrow B_*.$$

Thus the maximum matching size of the critical cut is exactly $|B_*| = |S_*| - 1$, and every $s \in S_*$ can be realized as the unique unmatched source token in some maximum matching.

There is also a boundary-degree consequence. If some $b \in B_*$ had a unique neighbor s in S_*, then $b \notin N_{G_\varsigma}(S_* \setminus \{s\})$, contradicting the deletion saturation above. Hence every boundary slot is supported by at least two source tokens.

Therefore a surviving unit-defect obstruction is not an anonymous one-unit capacity failure. It is a near-perfect matching circuit: every single-source deletion is perfectly payable, the whole source side has exactly one extra token, and every boundary slot has redundant support. If this circuit is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue near-perfect matching circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue near-perfect matching circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-near-perfect-matching-circuit-router.md.
```

□

Theorem 7.199 (Phase-residue alternating exchange circuit ledger)**.** *The phase-residue near-perfect matching circuit outlet admits the following alternating exchange circuit refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueNearPerfectMatchingCircuitImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterAlternatingExchangeCircuitLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterAlternatingExchangeCircuitLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterAlternatingExchangeCircuitLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterAlternatingExchangeCircuitLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterAlternatingExchangeCircuitLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterAlternatingExchangeCircuitLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueBaseOmittedSourceAndMatchingChoiceLedger
AND StableLadderEndpointOrbitPhaseResidueEveryDeletionMatchingImportedForExchangeLedger
AND StableLadderEndpointOrbitPhaseResidueMatchingSymmetricDifferenceGraphLedger
AND StableLadderEndpointOrbitPhaseResidueAlternatingComponentDecompositionLedger
AND StableLadderEndpointOrbitPhaseResidueUniqueSourceDefectAlternatingPathLedger
AND StableLadderEndpointOrbitPhaseResidueEverySourceExchangeReachableLedger
AND StableLadderEndpointOrbitPhaseResidueBoundaryDoubleCoverCarriedForwardAfterExchangeLedger
AND StableLadderEndpointOrbitPhaseResidueAlternatingExchangeCircuitPacketLedger
AND NoAnonymousNearPerfectMatchingCircuitAfterAlternatingExchangeLedger
AND SparseScaleLadderSAECarriedForwardAfterAlternatingExchangeCircuitLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue alternating exchange circuit ledger. All parallel outlets from the near-perfect matching circuit ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a near-perfect family

$$M_s : S_* \setminus \{s\} \longrightarrow B_* \qquad (s \in S_*),$$

where every M_s is a perfect matching onto B_*. Fix a base source $s_0 \in S_*$ and write $M_0 = M_{s_0}$. For $s \neq s_0$, form the matching symmetric difference

$$H_s = M_0 \triangle M_s.$$

Every boundary vertex $b \in B_*$ is matched once by M_0 and once by M_s. After common edges cancel in the symmetric difference, its boundary degree is therefore 0 or 2. On the source side, every vertex other than s and s_0 is either used by both matchings or by neither after cancellation, hence also has degree 0 or 2. The source s is used by M_0 but not by M_s, while s_0 is used by M_s but not by M_0, so these are the only degree-one vertices of H_s.

Thus each connected component of H_s is an alternating even cycle, except for the single alternating open path whose endpoints are s and s_0. This path is the exchange route that moves the unique unmatched source from s to the base missing source s_0. Since s was arbitrary, every source token is exchange-reachable to the same base omission.

Consequently a surviving near-perfect obstruction is not merely a family of unrelated one-source deletions. It carries an explicit alternating exchange field: all source omissions communicate through symmetric-difference paths over the same boundary set. If this exchange circuit is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue alternating exchange circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue alternating exchange circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-alternating-exchange-circuit-router.md.`

□

Theorem 7.200 (Phase-residue rooted directed exchange path ledger)**.** *The phase-residue alternating exchange circuit outlet admits the following rooted directed exchange path refinement:*

$$
\begin{aligned}
&\mathit{SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe}\\
&\Rightarrow \mathit{StableLadderEndpointOrbitPhaseResidueAlternatingExchangeCircuitImportedLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointSingletonAtomSAECarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeRootBaseMatchingImportedLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueAlternatingPathFamilyImportedForRootedDirectionLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangePathM0MsEdgeColoringLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangePathRootOrientationLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueSimpleDirectedExchangePathNormalFormLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeToggleTransferLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueEverySourceRootReachableByDirectedExchangeLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueRootedDirectedExchangePathPacketLedger}\\
&\mathit{AND}\ \mathit{NoAnonymousAlternatingExchangeCircuitAfterRootedDirectedPathLedger}\\
&\mathit{AND}\ \mathit{SparseScaleLadderSAECarriedForwardAfterRootedDirectedExchangePathLedger}\\
&\mathit{AND}\ \mathit{SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb}
\end{aligned}
$$

Phase-residue rooted directed exchange path ledger. All parallel outlets from the alternating exchange circuit ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives a base missing source s_0, a base matching $M_0 = M_{s_0}$, and for every $s \neq s_0$ an alternating path P_s from s to s_0 inside $M_0 \triangle M_s$. The edges of P_s have a canonical two-colouring: they are either M_0-only edges or M_s-only edges, and the colours alternate along the path.

Orient the M_0-only edges from source to boundary and the M_s-only edges from boundary to source. The endpoint s is incident to an M_0-only edge, while the endpoint s_0 is incident to an M_s-only edge. Hence the whole alternating path is directed from s to s_0:

$$s \longrightarrow b_1 \longrightarrow u_1 \longrightarrow \cdots \longrightarrow s_0.$$

If a repeated vertex is present, deleting the closed alternating subcycle leaves a simple directed alternating path with the same endpoints. Thus every source token is root-reachable by a concrete directed exchange path.

The same path records the matching transfer. Toggling the path component by symmetric difference replaces the M_0-edges on that component by the M_s-edges, so the base missing-source state and the s-missing state are connected by an explicit exchange move.

Therefore a surviving alternating exchange obstruction is not an unoriented exchange graph. It carries a root, a two-coloured edge word, a root-directed path from every source, and an explicit toggle rule. If this rooted directed exchange path packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue rooted directed exchange path circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue rooted directed exchange path circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-rooted-directed-exchange-path-router.md.`

□

Theorem 7.201 (Phase-residue exchange toggle word ledger)**.** *The phase-residue rooted directed exchange path outlet admits the following exchange toggle word refinement:*

$$\begin{aligned}
&\mathit{SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe}\\
&\Rightarrow \mathit{StableLadderEndpointOrbitPhaseResidueRootedDirectedExchangePathImportedLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueRootedDirectedExchangePathImportedForToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueAlternatingVertexWordNormalFormLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeWordSourceDistinctnessLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeWordBoundaryDistinctnessLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeWordLocalPivotCellLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueOrderedToggleProductLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueNoInternalBoundaryReuseInToggleWordLedger}\\
&\mathit{AND}\ \mathit{StableLadderEndpointOrbitPhaseResidueExchangeToggleWordPacketLedger}\\
&\mathit{AND}\ \mathit{NoAnonymousRootedDirectedExchangePathAfterToggleWordLedger}\\
&\mathit{AND}\ \mathit{SparseScaleLadderSAECarriedForwardAfterExchangeToggleWordLedger}\\
&\mathit{AND}\ \mathit{SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb}
\end{aligned}$$

Phase-residue exchange toggle word ledger. All parallel outlets from the rooted directed exchange path ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives, for every source s, a simple root-directed alternating path from s to the base missing source s_0. Write this path as the finite alternating vertex word

$$P_s = (u_0 = s, b_1, u_1, \dots, b_r, u_r = s_0).$$

Since the path is simple, the source vertices u_i are pairwise distinct inside the word, and the boundary vertices b_i are also pairwise distinct. Thus the same boundary payment slot cannot be internally replayed within one exchange word.

Each two-step block

$$u_{i-1} \longrightarrow b_i \longrightarrow u_i$$

is a local pivot cell: the edge (u_{i-1}, b_i) is the M_0-edge and the edge (u_i, b_i) is the M_s-edge. The local toggle replaces the former by the latter. The whole path toggle is therefore the ordered finite product of these local pivots.

Consequently a surviving rooted directed exchange path obstruction is not an anonymous directed path. It carries an alternating vertex word, source and boundary distinctness, a list of local pivot cells, and an ordered toggle product. If this exchange toggle word packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange toggle word circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange toggle word circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-toggle-word-router.md.`

□

Theorem 7.202 (Phase-residue exchange prefix defect ladder ledger). *The phase-residue exchange toggle word outlet admits the following prefix defect ladder refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeToggleWordImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangePrefixDefectLadderLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeToggleWordImportedForPrefixLadderLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixSourceSetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixBoundarySetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixUnitDefectIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixNestedLadderLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixOneBoundaryIncrementLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixToggleStateTransportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixDefectLadderPacketLedger
AND NoAnonymousExchangeToggleWordAfterPrefixDefectLadderLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangePrefixDefectLadderLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange prefix defect ladder ledger. All parallel outlets from the exchange toggle word ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger writes every remaining exchange route as an alternating word

$$P_s = (u_0 = s, b_1, u_1, \ldots, b_r, u_r = s_0).$$

For every nonempty prefix $1 \le t \le r$, define

$$U_t = \{u_0, \ldots, u_t\}, \qquad B_t = \{b_1, \ldots, b_t\}.$$

The word is simple on both source vertices and boundary vertices. Hence

$$|U_t| = t + 1, \qquad |B_t| = t, \qquad |U_t| - |B_t| = 1.$$

Thus each prefix carries an exact unit source surplus over its boundary slots.

Moreover U_t and B_t are nested in t. Passing from $t-1$ to t adds precisely one new source vertex and one new boundary slot, so the boundary side cannot be internally replayed inside the same word. The ordered prefix product of local pivots transports the missing-source state from u_0 to u_t.

Consequently a surviving exchange toggle word obstruction is not an unordered list of pivots. It contains a nested prefix ladder of exact unit defects and an explicit state-transport rule along the ladder. If this prefix defect ladder packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange prefix defect ladder circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange prefix defect ladder circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-prefix-defect-ladder-router.md.`

□

Theorem 7.203 (Phase-residue exchange endpoint telescoping charge ledger)**.** *The phase-residue exchange prefix defect ladder outlet admits the following endpoint telescoping charge refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangePrefixDefectLadderImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixDefectLadderImportedForEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePivotBoundaryOperatorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeInternalSourceCancellationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrefixChargeTelescopingLedger
AND StableLadderEndpointOrbitPhaseResidueNoInteriorAnonymousDebtAfterEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointTelescopingChargePacketLedger
AND NoAnonymousExchangePrefixDefectLadderAfterEndpointTelescopingChargeLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeEndpointTelescopingChargeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange endpoint telescoping charge ledger. All parallel outlets from the exchange prefix defect ladder ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives the alternating route

$$P_s = (u_0 = s, b_1, u_1, \dots, b_r, u_r = s_0)$$

and its nested prefix defect ladder. For the i-th local pivot

$$u_{i-1} \longrightarrow b_i \longrightarrow u_i$$

define the source-side boundary operator

$$d_i = [u_i] - [u_{i-1}].$$

This records the one-step transport of the missing-source state from u_{i-1} to u_i.

Every internal source u_i with $1 \le i < r$ appears with positive sign in d_i and with negative sign in d_{i+1}. Hence the internal source charges cancel in the sum:

$$\sum_{i=1}^{r} d_i = [u_r] - [u_0] = [s_0] - [s].$$

The same computation on the first t pivots gives

$$\sum_{i=1}^{t} d_i = [u_t] - [u_0],$$

which is exactly the prefix state transport already recorded by the prefix ladder.

Consequently a surviving prefix defect ladder obstruction cannot hide a net charge at an interior source vertex. Interior debt either cancels with the adjacent pivot, is paid by one of the named outlets, or is forced to the endpoint telescoping charge $[s_0] - [s]$. If that endpoint charge packet is not paid by a named outlet, it is the explicit phase-residue exchange endpoint telescoping charge circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange endpoint telescoping charge circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-endpoint-telescoping-charge-router.md.`

□

Theorem 7.204 (Phase-residue exchange root-star charge ledger)**.** *The phase-residue exchange endpoint telescoping charge outlet admits the following root-star charge refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeEndpointTelescopingChargeImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeRootStarChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointTelescopingChargeImportedForRootStarLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCommonRootEndpointLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeFamilyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootChargeMultiplicityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNonrootUnitOutputChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootStarTotalChargeZeroLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousEndpointChargeDistributionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootStarChargePacketLedger
AND NoAnonymousEndpointTelescopingChargeAfterRootStarLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeRootStarChargeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange root-star charge ledger. All parallel outlets from the exchange endpoint telescoping charge ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives, for every non-root source s, an endpoint charge

$$c_s = [s_0] - [s],$$

where the root s_0 is the same base missing source for the whole exchange family. Summing over all $s \neq s_0$ gives

$$C = \sum_{s \neq s_0} c_s = (|S_*| - 1)[s_0] - \sum_{s \neq s_0} [s].$$

Thus the root s_0 carries root multiplicity $|S_*| - 1$, while every non-root source contributes exactly one negative endpoint charge.

The total coefficient of C is zero:

$$(|S_*| - 1) - (|S_*| - 1) = 0.$$

This root-star field therefore does not create net mass; it rigidly fixes the distribution of the remaining endpoint pressure. The endpoint obstruction is not an anonymous family of endpoint charges. It is a unique directed star into the common root, with one outgoing unit from each non-root source.

If this root-star charge packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange root-star charge circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange root-star charge circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-root-star-charge-router.md.`

□

Theorem 7.205 (Phase-residue exchange normalized root-mean dipole ledger). *The phase-residue exchange root-star charge outlet admits the following normalized root-mean dipole refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeRootStarChargeImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootStarChargeImportedForNormalizedRootMeanDipoleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootMeanNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNonrootMeanMeasureLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalizedRootMeanDipoleFieldLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalizedRootMeanDipoleZeroMeanLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootToNonrootMeanDeviationCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueNoArtificialRootMultiplicityAfterNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalizedRootMeanDipolePacketLedger
AND NoAnonymousRootStarChargeAfterNormalizedRootMeanDipoleLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeNormalizedRootMeanDipoleLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange normalized root-mean dipole ledger. All parallel outlets from the exchange root-star charge ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives the root-star charge

$$C = (|S_*| - 1)[s_0] - \sum_{s \neq s_0} [s].$$

If $|S_*| = 1$, there is no non-root mean to form and the packet is carried by the endpoint singleton atom/SAE outlet. Otherwise define

$$\mu_* = \frac{1}{|S_*| - 1} \sum_{s \neq s_0} [s].$$

Then

$$C = (|S_*| - 1)([s_0] - \mu_*).$$

Thus the remaining root-star packet is equivalent, up to the positive scalar $|S_*|-1$, to the normalized dipole

$$D = [s_0] - \frac{1}{|S_*| - 1} \sum_{s \neq s_0} [s].$$

The total coefficient of D is zero:

$$1 - \frac{|S_*| - 1}{|S_*| - 1} = 0.$$

Therefore the obstruction is no longer a root multiplicity amplification. It is the single deviation coordinate measuring the common root against the mean of all non-root sources.

If this normalized root-mean dipole packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange normalized root-mean dipole circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange normalized root-mean dipole circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-normalized-root-mean-dipole-router.md.`

□

Theorem 7.206 (Phase-residue exchange root-source pair-average contrast ledger)**.** *The phase-residue exchange normalized root-mean dipole outlet admits the following root-source pair-average contrast refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeNormalizedRootMeanDipoleImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalizedRootMeanDipoleImportedForPairAverageLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairFanLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairContrastFamilyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUniformPairWeightLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePairAverageContrastIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairUnitBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousNonrootMeanAfterPairAverageLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairAverageContrastPacketLedger
AND NoAnonymousNormalizedRootMeanDipoleAfterPairAverageLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeRootSourcePairAverageContrastLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange root-source pair-average contrast ledger. All parallel outlets from the exchange normalized root-mean dipole ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives the normalized dipole

$$D = [s_0] - \frac{1}{|S_*| - 1} \sum_{s \neq s_0} [s].$$

The degenerate case $|S_*| = 1$ is already carried by the endpoint singleton atom/SAE outlet. For $|S_*| > 1$, define the root-source pair contrast

$$e_s = [s_0] - [s] \qquad (s \neq s_0).$$

Each e_s has one positive unit at the common root and one negative unit at the selected non-root source, hence has zero total coefficient. Averaging these pair contrasts gives the exact identity

$$\frac{1}{|S_*|-1}\sum_{s\neq s_0} e_s = [s_0] - \frac{1}{|S_*|-1}\sum_{s\neq s_0}[s] = D.$$

Thus the non-root mean is not an anonymous cloud. It is the uniform average of a named fan of root-source unit contrasts, with fixed weight $(|S_*|-1)^{-1}$ on every non-root source.

If this root-source pair-average contrast packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange root-source pair-average contrast circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange root-source pair-average contrast circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-root-source-pair-average-contrast-router.md.`

□

Theorem 7.207 (Phase-residue exchange root-source pair witness localization ledger)**.** *The phase-residue exchange root-source pair-average contrast outlet admits the following single-pair witness localization refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairAverageContrastImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePairAverageContrastImportedForWitnessLocalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeLinearWitnessFunctionalLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairWitnessValueListLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePairWitnessMeanIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrientedWitnessSignLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeAverageToSinglePairMaxLocalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNamedSinglePairWitnessLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousAverageOnlyPairWitnessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairWitnessLocalizationPacketLedger
AND NoAnonymousRootSourcePairAverageContrastAfterWitnessLocalizationLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeRootSourcePairWitnessLocalizationLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange root-source pair witness localization ledger. All parallel outlets from the exchange root-source pair-average contrast ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives

$$D = \frac{1}{|S_*|-1}\sum_{s\neq s_0} e_s, \qquad e_s = [s_0]-[s].$$

The degenerate case $|S_*| = 1$ is already carried by the endpoint singleton atom/SAE outlet. For $|S_*| > 1$, let Λ be any imported linear phase/capacity witness for the present circuit and put

$$a_s = \Lambda(e_s) = \Lambda([s_0] - [s]).$$

Linearity gives

$$\Lambda(D) = \frac{1}{|S_*| - 1} \sum_{s \neq s_0} a_s.$$

If $\Lambda(D) \neq 0$, orient the witness by $\sigma = \operatorname{sign}(\Lambda(D))$. A finite average cannot exceed all of its entries, so there is a non-root source s_* with

$$\sigma a_{s_*} \geq \frac{1}{|S_*| - 1} \sum_{s \neq s_0} \sigma a_s = |\Lambda(D)|.$$

Thus a linear average witness cannot remain anonymous at the whole fan level. It localizes without loss to one named root-source pair.

If this named root-source pair witness packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange root-source pair witness localization circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange root-source pair witness localization circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only proves that such a linear average witness localizes to one pair. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-root-source-pair-witness-localization-router.md.`

□

Theorem 7.208 (Phase-residue exchange oriented root-source witness gap ledger)**.** *The phase-residue exchange root-source pair witness localization outlet admits the following oriented endpoint-gap refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairWitnessLocalizationImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourcePairWitnessLocalizationImportedForOrientedGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNamedRootSourcePairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeWitnessOrientationCarriedForwardLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootWitnessScoreLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSourceWitnessScoreLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrientedWitnessGapCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeWitnessGapLowerBoundLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousPairWitnessAfterOrientedGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrientedRootSourceWitnessGapPacketLedger
AND NoAnonymousRootSourcePairWitnessLocalizationAfterOrientedGapLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeOrientedRootSourceWitnessGapLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange oriented root-source witness gap ledger. All parallel outlets from the root-source pair witness localization ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a named source s_* and an oriented imported linear witness $\sigma\Lambda$ with

$$\sigma\Lambda([s_0]-[s_*]) \geq |\Lambda(D)|\,.$$

Define the two endpoint scores

$$R=\sigma\Lambda([s_0]), \qquad T=\sigma\Lambda([s_*]).$$

By linearity, the oriented endpoint gap is

$$G=R-T=\sigma\Lambda([s_0]-[s_*]).$$

Hence

$$G \geq |\Lambda(D)|\,,$$

and if the imported average witness is nonzero then $G>0$. Thus the named pair witness is no longer an anonymous pair-localization object; it is a specified oriented root-source endpoint gap.

If this oriented gap packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange oriented root-source witness gap circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange oriented root-source witness gap circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only records the lossless oriented endpoint-gap coordinate once such a witness has been imported. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-oriented-root-source-witness-gap-router.md.`

□

Theorem 7.209 (Phase-residue exchange centered root-source score dipole ledger). *The phase-residue exchange oriented root-source witness gap outlet admits the following centered two-endpoint score refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeOrientedRootSourceWitnessGapImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrientedWitnessGapImportedForCenteredScoreDipoleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeWitnessScoreMidpointLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourceScoreCenteringLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredRootHalfGapSurplusLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredSourceHalfGapDeficitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredTwoPointScoreZeroMeanLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredHalfGapLowerBoundLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousCommonScoreOffsetAfterCenteringLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredRootSourceScoreDipolePacketLedger
AND NoAnonymousOrientedRootSourceWitnessGapAfterCenteredScoreDipoleLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeCenteredRootSourceScoreDipoleLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange centered root-source score dipole ledger. All parallel outlets from the oriented root-source witness gap ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies two endpoint scores and an oriented gap

$$R = \sigma\Lambda([s_0]), \qquad T = \sigma\Lambda([s_*]), \qquad G = R - T \geq |\Lambda(D)|\,.$$

Set

$$M = \frac{R+T}{2}.$$

Then

$$R = M + \frac{G}{2}, \qquad T = M - \frac{G}{2}.$$

Equivalently, after subtracting the common score offset M,

$$U = R - M = \frac{G}{2}, \qquad V = T - M = -\frac{G}{2}.$$

Thus

$$U + V = 0, \qquad U - V = G, \qquad U \geq \frac{|\Lambda(D)|}{2}.$$

If the imported average witness is nonzero then $U > 0$ and $V < 0$. The common score offset no longer belongs to the hard point; the remaining packet is a centered two-point root-source score dipole.

If this centered dipole packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange centered root-source score dipole circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange centered root-source score dipole circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only centers the already imported oriented endpoint scores. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-centered-root-source-score-dipole-router.md.`

□

Theorem 7.210 (Phase-residue exchange signed root-source half-gap atom ledger)**.** *The phase-residue exchange centered root-source score dipole outlet admits the following signed half-gap atom refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCenteredRootSourceScoreDipoleImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
```

```
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCenteredScoreDipoleImportedForHalfGapAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeHalfGapAmplitudeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeHalfGapAmplitudeLowerBoundLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootPositiveSourceNegativeSupportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedHalfGapAtomFactorizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedHalfGapAtomZeroMassLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedHalfGapAtomTotalVariationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousUnequalEndpointScoreAfterHalfGapAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedRootSourceHalfGapAtomPacketLedger
AND NoAnonymousCenteredRootSourceScoreDipoleAfterHalfGapAtomLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeSignedRootSourceHalfGapAtomLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange signed root-source half-gap atom ledger. All parallel outlets from the centered root-source score dipole ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies centered endpoint scores

$$U = \frac{G}{2}, \qquad V = -\frac{G}{2}, \qquad U + V = 0, \qquad G \geq |\Lambda(D)|\,.$$

Define the half-gap amplitude

$$A = \frac{G}{2} = U = -V.$$

Then

$$A \geq \frac{|\Lambda(D)|}{2},$$

and, if the imported average witness is nonzero, $A > 0$. The centered score vector on the two named endpoints factors as

$$W = A([s_0] - [s_*]).$$

This signed atom has zero total mass and total variation

$$\mathrm{mass}(W) = 0, \qquad \|W\|_{\mathrm{TV}} = 2A = G.$$

Thus the remaining packet contains neither a common offset nor unequal endpoint magnitudes; it is one positive amplitude multiplying the fixed root-positive/source-negative support.

If this signed half-gap atom packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange signed root-source half-gap atom circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange signed root-source half-gap atom circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only factors the already centered endpoint scores. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-signed-root-source-half-gap-atom-router.md.`

□

Theorem 7.211 (Phase-residue exchange oriented root-source transport edge ledger)**.** *The phase-residue exchange signed root-source half-gap atom outlet admits the following oriented transport-edge refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeSignedRootSourceHalfGapAtomImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedHalfGapAtomImportedForTransportEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeFluxLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeBoundaryIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeDivergenceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeMassConservationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeFluxLowerBoundLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousSignedAtomAfterTransportEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrientedRootSourceTransportEdgePacketLedger
AND NoAnonymousSignedRootSourceHalfGapAtomAfterTransportEdgeLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeOrientedRootSourceTransportEdgeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange oriented root-source transport edge ledger. All parallel outlets from the signed root-source half-gap atom ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a signed atom

$$W = A([s_0]-[s_*]), \qquad A \geq \frac{|\Lambda(D)|}{2}, \qquad \mathrm{mass}(W) = 0.$$

Orient a single edge from the source endpoint to the root endpoint,

$$e = s_* \to s_0,$$

and assign it flux $F = A$. With the boundary convention $\partial e = [s_0] - [s_*]$, one has

$$\partial(Fe) = A([s_0]-[s_*]) = W.$$

Equivalently, the endpoint divergence is

$$\mathrm{div}(s_0) = +A, \qquad \mathrm{div}(s_*) = -A, \qquad \mathrm{div}(s_0) + \mathrm{div}(s_*) = 0.$$

Thus the remaining packet is no longer an anonymous signed atom. It is a named source-to-root transport edge with positive flux $F = A$, and the flux inherits the same lower bound $F \geq |\Lambda(D)|/2$.

If this transport-edge packet is paid by one of the named outlets it returns there; otherwise it is the explicit phase-residue exchange oriented root-source transport edge circuit PDEC/cap outlet.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange oriented root-source transport edge circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported signed atom as a directed edge boundary. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-phase-residue-exchange-oriented-root-source-transport-edge-router.md.`

□

Theorem 7.212 (Phase-residue exchange transport-edge incidence column ledger). *The phase-residue exchange oriented root-source transport-edge outlet admits the following single-column incidence refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeOrientedRootSourceTransportEdgeImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeImportedForIncidenceColumnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeTailHeadCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeIncidenceVectorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeBoundaryMatrixColumnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeFluxCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeMatrixBoundaryIdentityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeIncidenceColumnZeroSumLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeTwoEndpointSupportLedger
AND StableLadderEndpointOrbitPhaseResidueNoHiddenCycleInsideSingleIncidenceColumnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeIncidenceColumnPacketLedger
AND NoAnonymousOrientedRootSourceTransportEdgeAfterIncidenceColumnLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeTransportEdgeIncidenceColumnLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange transport-edge incidence column ledger. All parallel outlets from the oriented transport-edge ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a directed edge and flux

$$e = s_* \to s_0, \qquad F = A, \qquad \partial(Fe) = A([s_0] - [s_*]) = W.$$

Define the single incidence column

$$b_e = [s_0] - [s_*],$$

put this column into the boundary matrix as $B_{\cdot,e} = b_e$, and take the one active flux coordinate $f_e = A$. Then the matrix boundary identity is

$$Bf = Ab_e = A([s_0] - [s_*]) = W.$$

In coordinates,

$$b_e(s_0) = +1, \qquad b_e(s_*) = -1, \qquad b_e(v) = 0 \quad (v \notin \{s_0, s_*\}),$$

and therefore $\sum_v b_e(v) = 0$. In the nondegenerate case the support is exactly the two endpoints; the degenerate case is returned to the singleton outlet. A single boundary-matrix column has no internal vertex reuse or hidden cycle. Thus any remaining cyclic pressure must be named as the transport-edge incidence-column circuit PDEC/cap outlet, not as an anonymous directed edge.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange transport-edge incidence-column circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported directed edge as a single boundary-matrix incidence column. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-transport-edge-incidence-column-router.md.`

□

Theorem 7.213 (Phase-residue exchange incidence-column Kronecker stencil ledger)**.** *The phase-residue exchange transport-edge incidence-column outlet admits the following endpoint Kronecker-stencil refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeTransportEdgeIncidenceColumnImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeIncidenceColumnImportedForKroneckerStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootKroneckerDeltaCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSourceKroneckerDeltaCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerDeltaDifferenceStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilSignedCoefficientLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilEndpointCRTCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootSourceCoordinateDistinctOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilZeroSumLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeIncidenceColumnEqualsKroneckerStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilFluxScalingLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousBoundaryMatrixColumnAfterKroneckerStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeIncidenceColumnKroneckerStencilPacketLedger
AND NoAnonymousTransportEdgeIncidenceColumnAfterKroneckerStencilLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeIncidenceColumnKroneckerStencilLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange incidence-column Kronecker stencil ledger. All parallel outlets from the transport-edge incidence-column ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a single incidence column and flux coordinate

$$b_e = [s_0] - [s_*], \qquad B_{\cdot,e} = b_e, \qquad f_e = A, \qquad Bf = W.$$

Remove the boundary-matrix-column abstraction by introducing endpoint unit coordinates

$$\delta_{s_0}(v) = 1_{v=s_0}, \qquad \delta_{s_*}(v) = 1_{v=s_*}.$$

Then

$$k = \delta_{s_0} - \delta_{s_*} = b_e$$

as a coordinate vector, and the scaled stencil is

$$Ak = A(\delta_{s_0} - \delta_{s_*}) = A([s_0] - [s_*]) = W.$$

The two endpoints inherit the upstream primitive-witness CRT-coordinate atom: one may record this stencil as coefficient $+1$ at the root CRT word $\mathrm{crt}(s_0)$ and coefficient -1 at the source CRT word crt(s_*). If $s_0 = s_*$, the stencil is zero and the packet returns to the singleton/degenerate outlet; otherwise it is a two-point signed Kronecker stencil with total coefficient sum zero.

Thus the remaining packet is no longer an anonymous boundary-matrix column. It is a named difference of two endpoint unit coordinates with inherited CRT words. Any remaining obstruction must be named as the incidence-column Kronecker-stencil circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange incidence-column Kronecker-stencil circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported incidence column as an endpoint Kronecker stencil. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-incidence-column-kronecker-stencil-router.md.`

□

Theorem 7.214 (Phase-residue exchange Kronecker-stencil signed CRT pair ledger). *The phase-residue exchange incidence-column Kronecker-stencil outlet admits the following signed CRT word-pair refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeIncidenceColumnKroneckerStencilImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilImportedForSignedCRTPairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootCRTWordCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSourceCRTWordCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootCanonicalCongruenceEquationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSourceCanonicalCongruenceEquationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrderedRootSourceCRTWordPairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTWordPairSupportDictionaryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTWordPairCoefficientBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCRTWordPairCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairEqualsKroneckerStencilLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairFluxScalingLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousKroneckerDeltaAfterSignedCRTPairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilSignedCRTPairPacketLedger
AND NoAnonymousIncidenceColumnKroneckerStencilAfterSignedCRTPairLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeKroneckerStencilSignedCRTPairLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange Kronecker-stencil signed CRT pair ledger. All parallel outlets from the incidence-column Kronecker-stencil ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the endpoint stencil

$$k = \delta_{s_0} - \delta_{s_*}, \qquad Ak = W,$$

and the upstream primitive-witness CRT-coordinate atom supplies endpoint CRT words. Record them as

$$r_0 = \mathrm{crt}(s_0), \qquad r_* = \mathrm{crt}(s_*), \qquad \Pi = (r_0, r_*).$$

For every active prime coordinate p_i, these words carry the canonical congruence equations

$$s_0 \equiv (r_0)_i \pmod{p_i}, \qquad s_* \equiv (r_*)_i \pmod{p_i}.$$

Define the finite signed CRT-word dictionary

$$C_\Pi(r_0) = +1, \qquad C_\Pi(r_*) = -1, \qquad C_\Pi(r) = 0 \quad (r \notin \{r_0, r_*\}).$$

If $r_0 = r_*$, the signed dictionary cancels and the packet returns to the singleton/degenerate outlet. Otherwise C_Π is exactly the two-word signed support dictionary corresponding to the Kronecker stencil. Hence

$$AC_\Pi = A(\delta_{s_0} - \delta_{s_*}) = W.$$

Thus the remaining packet is no longer an anonymous delta-function stencil. It is a named ordered pair of CRT words with coefficients $+1$ at the root word and -1 at the source word. Any remaining obstruction must be named as the Kronecker-stencil signed CRT pair circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange Kronecker-stencil signed CRT pair circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported Kronecker stencil as an ordered signed CRT word pair. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-kronecker-stencil-signed-crt-pair-router.md.`

□

Theorem 7.215 (Phase-residue exchange signed CRT pair primitive atom ledger)**.** *The phase-residue exchange Kronecker-stencil signed CRT pair outlet admits the following two-word primitive-atom refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeKroneckerStencilSignedCRTPairImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairImportedForPrimitiveAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairSupportSizeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPositiveNegativeAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomZeroMassLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomTotalVariationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomFluxWeightLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairEqualsPrimitiveAtomLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousSignedCRTDictionaryAfterPrimitiveAtomLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomPacketLedger
AND NoAnonymousSignedCRTPairDictionaryAfterPrimitiveAtomLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeSignedCRTPairPrimitiveAtomLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange signed CRT pair primitive atom ledger. All parallel outlets from the Kronecker-stencil signed CRT pair ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the ordered pair and finite signed dictionary

$$\Pi = (r_0, r_*), \qquad C_\Pi(r_0) = +1, \qquad C_\Pi(r_*) = -1, \qquad AC_\Pi = W.$$

In the collision case $r_0 = r_*$, the coefficients cancel and the packet returns to the singleton/degenerate outlet. In the nondegenerate case the packet has exactly two support words,

$$\operatorname{supp}(C_\Pi) = \{r_0, r_*\},$$

with one positive root atom and one negative source atom. Its scalar invariants are

$$|\operatorname{supp}(C_\Pi)| = 2, \qquad \sum_r C_\Pi(r) = 0, \qquad \sum_r |C_\Pi(r)| = 2.$$

After flux scaling the two weights are $+A$ at the root word and $-A$ at the source word, still satisfying $AC_\Pi = W$. The orientation is retained as source-to-root; the pair is not replaced by an unordered set.

Thus the remaining packet is no longer an anonymous finite signed CRT dictionary. It is a two-word signed support primitive atom with named support size, zero mass, total variation, flux weights, and orientation. Any remaining obstruction must be named as the signed CRT pair primitive atom circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange signed CRT pair primitive atom circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported signed CRT dictionary as a two-word primitive atom. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-signed-crt-pair-primitive-atom-router.md.
```

□

Theorem 7.216 (Phase-residue exchange primitive atom endpoint charge ledger)**.** *The phase-residue exchange signed CRT pair primitive atom outlet admits the following endpoint-charge refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeSignedCRTPairPrimitiveAtomImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
```

```
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrimitiveAtomImportedForEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRootPositiveEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSourceNegativeEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeOrderedEndpointChargePairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointPositiveNegativeMassEqualityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointAbsoluteFluxLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeEqualsPrimitiveAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousPrimitiveAtomAfterEndpointChargeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePrimitiveAtomEndpointChargePacketLedger
AND NoAnonymousSignedCRTPairPrimitiveAtomAfterEndpointChargeLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangePrimitiveAtomEndpointChargeLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange primitive atom endpoint charge ledger. All parallel outlets from the signed CRT pair primitive atom ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a two-word primitive atom with flux weights

$$\mathrm{supp}(C_\Pi) = \{r_0, r_*\}, \qquad C_\Pi(r_0) = +1, \qquad C_\Pi(r_*) = -1, \qquad AC_\Pi = W.$$

Rewrite the scaled atom as a named endpoint-charge pair

$$q_0 = (r_0, +A), \qquad q_* = (r_*, -A).$$

The pair is ordered as (q_0, q_*), equivalently as a source-to-root charge transfer. Its conservation identities are

$$(+A) + (-A) = 0, \qquad |+A| = |-A| = A, \qquad |+A| + |-A| = 2A.$$

In the collision case $r_0 = r_*$, the two endpoint charges cancel at the same CRT word and the packet returns to the singleton/degenerate outlet. Otherwise the packet is exactly the original primitive atom with its two charges named explicitly.

Thus the remaining packet is no longer an anonymous two-point primitive atom. It is a root/source endpoint-charge packet with named signs, equal positive and negative mass, zero net charge, fixed absolute flux, and retained orientation. Any remaining obstruction must be named as the primitive atom endpoint-charge circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange primitive atom endpoint charge circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported primitive atom as a named endpoint-charge packet. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-primitive-atom-endpoint-charge-router.md.`

□

Theorem 7.217 (Phase-residue exchange endpoint charge Kirchhoff cell ledger)**.** *The phase-residue exchange primitive atom endpoint-charge outlet admits the following local Kirchhoff-cell refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangePrimitiveAtomEndpointChargeImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeImportedForKirchhoffCellLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeLocalKirchhoffCellLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeRootPositiveDivergenceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeSourceNegativeDivergenceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeKirchhoffBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargePositiveNegativeDivergenceEqualityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeBoundaryVariationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellEqualsEndpointChargePacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousEndpointChargeAfterKirchhoffCellLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeKirchhoffCellPacketLedger
AND NoAnonymousEndpointChargePacketAfterKirchhoffCellLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeEndpointChargeKirchhoffCellLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange endpoint charge Kirchhoff cell ledger. All parallel outlets from the primitive atom endpoint-charge ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the ordered endpoint-charge packet

$$q_0 = (r_0, +A), \qquad q_* = (r_*, -A), \qquad AC_\Pi = W.$$

Rewrite this packet as a local CRT-word Kirchhoff cell K. In the nondegenerate case,

$$K = \{r_0, r_*\}, \qquad \operatorname{div}_K(r_0) = +A, \qquad \operatorname{div}_K(r_*) = -A.$$

The local conservation and boundary-variation identities are

$$\sum_{r \in K} \operatorname{div}_K(r) = 0, \qquad \operatorname{div}_K^+(K) = \operatorname{div}_K^-(K) = A, \qquad \sum_{r \in K} |\operatorname{div}_K(r)| = 2A.$$

In the collision case $r_0 = r_*$, the two divergences cancel at the same CRT word and the packet returns to the singleton/degenerate outlet. Otherwise the cell is exactly the endpoint-charge packet with its local divergence law named explicitly. The source-to-root orientation is retained as metadata; the packet is not replaced by an anonymous unordered pair.

Thus the remaining packet is no longer an anonymous endpoint-charge pair. It is a two-point local Kirchhoff cell with named root/source divergences, zero total divergence, equal positive and negative divergence mass, fixed boundary variation, and retained orientation. Any remaining obstruction must be named as the endpoint-charge Kirchhoff-cell circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange endpoint charge Kirchhoff cell circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported endpoint-charge packet as a named local Kirchhoff cell. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-endpoint-charge-kirchhoff-cell-router.md.`

□

Theorem 7.218 (Phase-residue exchange Kirchhoff cell cut-potential ledger)**.** *The phase-residue exchange endpoint-charge Kirchhoff-cell outlet admits the following normalized cut-potential refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeEndpointChargeKirchhoffCellImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellImportedForCutPotentialLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialSupportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialValuesLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialZeroMeanLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialUnitOscillationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialDivergencePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialEqualsKirchhoffCellLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousKirchhoffCellAfterCutPotentialLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialPacketLedger
AND NoAnonymousKirchhoffCellAfterCutPotentialLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeKirchhoffCellCutPotentialLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange Kirchhoff cell cut-potential ledger. All parallel outlets from the endpoint-charge Kirchhoff-cell ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a nondegenerate two-point local cell

$$K = \{r_0, r_*\}, \qquad \operatorname{div}_K(r_0) = +A, \qquad \operatorname{div}_K(r_*) = -A, \qquad \sum_{r\in K} \operatorname{div}_K(r) = 0.$$

Register the normalized cut-potential ϕ_K on this two-point support by

$$\phi_K(r_0) = \frac{1}{2}, \qquad \phi_K(r_*) = -\frac{1}{2}.$$

It has zero two-point mean and unit source-to-root oscillation:

$$\phi_K(r_0) + \phi_K(r_*) = 0, \qquad \phi_K(r_0) - \phi_K(r_*) = 1.$$

The divergence pairing is therefore a named positive quantity,

$$\langle \operatorname{div}_K, \phi_K \rangle = (+A)\frac{1}{2} + (-A)\left(-\frac{1}{2}\right) = A.$$

In the collision case $r_0 = r_*$, the oscillation and pairing degenerate and the packet returns to the singleton/degenerate outlet. Otherwise the orientation is retained by the sign of the potential gap from source to root.

Thus the remaining packet is no longer an anonymous two-point Kirchhoff cell. It is a normalized cut-potential certificate with named support, zero mean, unit oscillation, positive divergence pairing, and retained orientation. Any remaining obstruction must be named as the endpoint-charge Kirchhoff-cell cut-potential circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange endpoint charge Kirchhoff cell cut-potential circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported Kirchhoff cell as a normalized cut-potential pairing certificate. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-kirchhoff-cell-cut-potential-router.md.
```

∎

Theorem 7.219 (Phase-residue exchange cut-potential dual-norm ledger). *The phase-residue exchange Kirchhoff-cell cut-potential outlet admits the following L^1-L^∞ dual-norm saturation refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeKirchhoffCellCutPotentialImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeCutPotentialDualNormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialImportedForDualNormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialDivergenceL1NormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialLInfinityNormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialHolderDualBoundLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialPairingValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialDualNormSaturationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialPolarSignAlignmentLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialExtremalSupportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeDualNormEqualsCutPotentialLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeDualNormCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeDualNormOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousCutPotentialAfterDualNormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialDualNormPacketLedger
AND NoAnonymousCutPotentialAfterDualNormLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeCutPotentialDualNormLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange cut-potential dual-norm ledger. All parallel outlets from the Kirchhoff-cell cut-potential ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies

$$\phi_K(r_0) = \frac{1}{2}, \qquad \phi_K(r_*) = -\frac{1}{2}, \qquad \langle \mathrm{div}_K, \phi_K \rangle = A.$$

Together with

$$\operatorname{div}_K(r_0) = +A, \qquad \operatorname{div}_K(r_*) = -A,$$

this gives the exact norm identities

$$\|\operatorname{div}_K\|_1 = 2A, \qquad \|\phi_K\|_\infty = \frac{1}{2}, \qquad \|\operatorname{div}_K\|_1 \|\phi_K\|_\infty = A.$$

The direct pairing calculation is

$$\langle \operatorname{div}_K, \phi_K \rangle = (+A)\frac{1}{2} + (-A)\left(-\frac{1}{2}\right) = A.$$

Thus the L^1-L^∞ Holder bound is attained with equality. The equality condition is named explicitly: the positive divergence sits at the positive potential extremum and the negative divergence sits at the negative potential extremum. In the collision case $r_0 = r_*$, the potential gap and the polar certificate degenerate and the packet returns to the singleton/degenerate outlet.

Thus the remaining packet is no longer an anonymous cut-potential pairing. It is a dual-norm saturation certificate with named L^1 divergence mass, named L^∞ potential radius, exact Holder equality, polar sign alignment, extremal support, and retained orientation. Any remaining obstruction must be named as the cut-potential dual-norm saturation circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange cut-potential dual-norm saturation circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported cut-potential as a named dual-norm saturation certificate. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-cut-potential-dual-norm-router.md.
```

□

Theorem 7.220 (Phase-residue exchange dual-norm complementary-slackness ledger)**.** *The phase-residue exchange cut-potential dual-norm outlet admits the following primal/dual complementary-slackness refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCutPotentialDualNormImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeComplementarySlacknessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeDualNormImportedForComplementarySlacknessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCalibratedEdgeOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessUnitPotentialDropLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessPrimalFluxLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessPrimalCostLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessDualValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessZeroDualityGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessSaturatedEdgeSupportLedger
AND StableLadderEndpointOrbitPhaseResidueNoHiddenPositiveCostCycleAfterComplementarySlacknessLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessEqualsDualNormLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousDualNormAfterComplementarySlacknessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessPacketLedger
AND NoAnonymousDualNormAfterComplementarySlacknessLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeComplementarySlacknessLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange dual-norm complementary-slackness ledger. All parallel outlets from the cut-potential dual-norm ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the polar two-point data

$$\operatorname{div}_K(r_0) = +A, \qquad \operatorname{div}_K(r_*) = -A, \qquad \phi_K(r_0) = \frac{1}{2}, \qquad \phi_K(r_*) = -\frac{1}{2},$$

together with

$$\|\operatorname{div}_K\|_1 = 2A, \qquad \|\phi_K\|_\infty = \frac{1}{2}, \qquad \langle \operatorname{div}_K, \phi_K \rangle = A.$$

Register the nondegenerate object as the oriented edge

$$e = (r_* \to r_0), \qquad \partial(Ae) = A([r_0] - [r_*]) = \operatorname{div}_K.$$

Give this local edge unit cost. The cut potential is calibrated on the edge:

$$\operatorname{cost}(e) = 1, \qquad \phi_K(r_0) - \phi_K(r_*) = 1 = \operatorname{cost}(e).$$

Hence the primal and dual values coincide,

$$\text{PrimalCost} = A \operatorname{cost}(e) = A, \qquad \text{DualValue} = \langle \operatorname{div}_K, \phi_K \rangle = A,$$

and the duality gap is zero:

$$\text{PrimalCost} - \text{DualValue} = 0.$$

Thus all nonzero local flux is supported on an edge on which the potential drop equals the edge cost. Inside this two-point single-edge certificate there is no additional positive-cost cycle with the same boundary. In the collision case $r_0 = r_*$, the edge, potential drop, and zero-gap certificate degenerate and the packet returns to the singleton/degenerate outlet.

Thus the remaining packet is no longer an anonymous dual-norm saturation equality. It is a calibrated transport-edge certificate with named edge orientation, unit potential drop, primal cost, dual value, zero duality gap, saturated support, and retained source-to-root orientation. Any remaining obstruction must be named as the dual-norm complementary-slackness circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange dual-norm complementary-slackness circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported dual-norm saturation certificate as a named zero-gap complementary-slackness certificate. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-phase-residue-exchange-dual-norm-complementary-slackness-router.md.`

□

Theorem 7.221 (Phase-residue exchange active-facet normal-cone ledger)**.** *The phase-residue exchange complementary-slackness outlet admits the following single active-facet normal-cone refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeActiveFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeComplementarySlacknessImportedForActiveFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetDualFeasibilityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetEqualityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetNormalVectorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetNormalConeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetFlowNormalConeMembershipLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetGapFormulaLedger
AND StableLadderEndpointOrbitPhaseResidueNoInactiveConstraintCarriesFluxAfterActiveFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetSaturatedSupportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetEqualsComplementarySlacknessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousComplementarySlacknessAfterActiveFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetNormalConePacketLedger
AND NoAnonymousComplementarySlacknessAfterActiveFacetLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeActiveFacetLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange active-facet normal-cone ledger. All parallel outlets from the complementary-slackness ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the calibrated local edge

$$e = (r_* \to r_0), \qquad \partial(Ae) = A([r_0] - [r_*]) = \mathrm{div}_K,$$

with unit cost and zero duality gap:

$$\mathrm{cost}(e) = 1, \qquad \phi_K(r_0) - \phi_K(r_*) = 1, \qquad \mathrm{PrimalCost} = \mathrm{DualValue} = A.$$

Register the corresponding dual feasible constraint

$$\phi_K(r_0) - \phi_K(r_*) \leq 1.$$

Since the local nondegenerate branch has positive flux $A > 0$ and zero gap, this constraint is active:

$$\phi_K(r_0) - \phi_K(r_*) = 1.$$

The active facet has normal vector

$$n_e = [r_0] - [r_*],$$

and its local normal cone is the positive ray

$$N_e = \{\lambda n_e : \lambda \geq 0\}.$$

The imported divergence is exactly

$$\mathrm{div}_K = An_e \in N_e.$$

Equivalently, the complementary-slackness gap is the named nonnegative slack

$$A\bigl(1 - (\phi_K(r_0) - \phi_K(r_*))\bigr) = 0.$$

If the constraint were strict while $A > 0$, this quantity would be positive, contradicting the imported zero-gap certificate. Thus the nonzero local flux cannot be carried by an inactive dual constraint. In the collision case $r_0 = r_*$, or in the zero-flux case, the normal vector and normal cone certificate degenerate and the packet returns to the singleton/degenerate outlet.

Thus the remaining packet is no longer an anonymous complementary-slackness equality. It is a single active Lipschitz facet with named normal vector, positive normal-cone membership, inactive-constraint exclusion, saturated support, and retained source-to-root orientation. Any remaining obstruction must be named as the active-facet normal-cone circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange complementary-slackness active-facet normal-cone circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported complementary-slackness certificate as a named active-facet normal-cone certificate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-complementary-slackness-active-facet-router.md.`

□

Theorem 7.222 (Phase-residue exchange normal-cone ray-coordinate ledger). *The phase-residue exchange active-facet normal-cone outlet admits the following positive ray-coordinate refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetNormalConeImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActiveFacetImportedForRayCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeRayGeneratorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeRayCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConePositiveCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeUniqueCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeNoTransverseComponentLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeCoordinateBoundaryPairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeCoordinateTotalVariationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateEqualsActiveFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateCollisionOrSingletonExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousActiveFacetAfterRayCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNormalConeRayCoordinatePacketLedger
AND NoAnonymousActiveFacetAfterRayCoordinateLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeNormalConeRayCoordinateLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange normal-cone ray-coordinate ledger. All parallel outlets from the active-facet normal-cone ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies the active constraint and its normal cone:

$$\phi_K(r_0) - \phi_K(r_*) = 1, \qquad n_e = [r_0] - [r_*], \qquad N_e = \{\lambda n_e : \lambda \geq 0\},$$

together with

$$\mathrm{div}_K = A n_e \in N_e, \qquad AC_\Pi = W.$$

In the nondegenerate branch $r_0 \neq r_*$ and $A > 0$. The vector n_e has two-point support and is nonzero, so the one-dimensional cone N_e has a unique coordinate on this generator. Consequently

$$\mathrm{div}_K = \lambda n_e, \qquad \lambda = A > 0.$$

There is no transverse normal-cone component: any element of N_e is already a scalar multiple of n_e. The coordinate also preserves the boundary pairing and total variation:

$$\langle \mathrm{div}_K, \phi_K \rangle = \lambda(\phi_K(r_0) - \phi_K(r_*)) = A, \qquad \|\mathrm{div}_K\|_1 = 2\lambda = 2A.$$

If $r_0 = r_*$, or if $A = 0$, the ray generator or its coordinate degenerates and the packet returns to the singleton/degenerate outlet. Thus the remaining packet is no longer an anonymous active-facet normal cone. It is a named positive ray-coordinate packet with generator n_e, unique coordinate A, no transverse component, retained source-to-root orientation, and the same signed CRT packet $AC_\Pi = W$.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported active-facet normal-cone certificate as a named positive ray-coordinate certificate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-router.md.`

□

Theorem 7.223 (Phase-residue exchange ray-coordinate scalar-load ledger). *The phase-residue exchange normal-cone ray-coordinate outlet admits the following scalar-load refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeNormalConeRayCoordinateImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateImportedForScalarLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinatePositiveScalarLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateRankOneSupportLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateScalarPairingEqualsLoadLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateHalfTotalVariationEqualsLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateUnitSaturationRatioLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateNoResidualVectorGeometryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinatePDECCapReducedToScalarLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadCollisionOrZeroExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousRayCoordinateAfterScalarLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeRayCoordinateScalarLoadPacketLedger
AND NoAnonymousRayCoordinateAfterScalarLoadLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeRayCoordinateScalarLoadLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange ray-coordinate scalar-load ledger. All parallel outlets from the ray-coordinate ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies

$$\mathrm{div}_K = \lambda n_e, \qquad \lambda = A > 0, \qquad n_e = [r_0] - [r_*],$$

with no transverse component and with the two scalar readings

$$\langle \mathrm{div}_K, \phi_K \rangle = A, \qquad \|\mathrm{div}_K\|_1 = 2A.$$

Thus the same one-dimensional positive quantity is seen both as the boundary pairing load and as half of the boundary total variation:

$$\mathrm{PairingLoad} = A, \qquad \mathrm{VariationLoad} = \frac{1}{2}\,\|\mathrm{div}_K\|_1 = A.$$

Consequently the saturation ratio is exactly

$$\frac{\mathrm{PairingLoad}}{\mathrm{VariationLoad}} = 1.$$

After the ray-coordinate step, there is no remaining vector geometry: no transverse normal-cone component, no second direction, and no hidden local phase parameter. The nondegenerate packet is therefore a scalar payment problem for A along the already fixed source-to-root generator n_e, with the same signed CRT packet $AC_\Pi = W$.

If $r_0 = r_*$, or if $A = 0$, this scalar load degenerates and returns to the singleton/degenerate outlet. Otherwise any unresolved obstruction must be the named scalar-load circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate scalar-load circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported ray-coordinate certificate as a named scalar-load payment certificate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-router.md.`

□

Theorem 7.224 (Phase-residue exchange scalar-load unit-normalization ledger)**.** *The phase-residue exchange scalar-load outlet admits the following unit-normalization refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadImportedForUnitNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitGeneratorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitPairingLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitVariationLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitSaturationRatioLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadScaleFactorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadDivFactorizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadTotalWeightLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadNoResidualScaleFreedomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadPDECCapReducedToUnitNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitNormalizationPacketLedger
AND NoAnonymousScalarLoadScaleAfterUnitNormalizationLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeScalarLoadUnitNormalizationLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange scalar-load unit-normalization ledger. All parallel outlets from the scalar-load ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The scalar-load ledger supplies a single positive load A along the already fixed generator $n_e = [r_0] - [r_*]$:

$$\mathrm{div}_K = A n_e, \qquad \langle \mathrm{div}_K, \phi_K \rangle = A, \qquad \frac{1}{2} \|\mathrm{div}_K\|_1 = A.$$

Separate the unit shape from the magnitude by setting

$$u_e = n_e = [r_0] - [r_*].$$

Because the active facet has $\phi_K(r_0) - \phi_K(r_*) = 1$, this unit shape has the two normalized readings

$$\langle u_e, \phi_K \rangle = 1, \qquad \frac{1}{2} \|u_e\|_1 = 1.$$

Therefore

$$\mathrm{div}_K = A u_e, \qquad \mathrm{TotalWeight} = A, \qquad AC_\Pi = W,$$

and the unit saturation ratio is again 1. Thus the remaining obstruction cannot hide in a scale convention, in a split of the load into several weights, or in a residual normalization parameter. The only nondegenerate scalar quantity left is the total weight A on the fixed unit two-point certificate.

If $r_0 = r_*$, or if $A = 0$, the unit shape or the positive scale degenerates and returns to the singleton/degenerate outlet. Otherwise any unresolved obstruction must be the named scalar-load unit-normalization circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate scalar-load unit-normalization circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder

SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported scalar-load certificate as a unit-shape plus total weight certificate. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.225 (Phase-residue exchange unit-face-value ledger)**.** *The phase-residue exchange scalar-load unit-normalization outlet admits the following unit-face-value refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeScalarLoadUnitNormalizationImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeUnitFaceValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitNormalizationImportedForUnitFaceValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueUnitShapeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValuePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueVariationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueOneLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueSignedDictionaryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueTotalAmountLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValuePaymentValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueNoCapacityMultiplierLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueNoDenominationSplitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValuePDECCapReducedToPriceOnePaymentLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousUnitNormalizationAfterUnitFaceValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValuePacketLedger
AND NoAnonymousCapacityFaceValueAfterUnitFaceValueLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeUnitFaceValueLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange unit-face-value ledger. All parallel outlets from the unit-normalization ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger supplies a fixed unit two-point certificate

$$u_e = n_e = [r_0] - [r_*], \qquad \langle u_e, \phi_K \rangle = 1, \qquad \frac{1}{2}\,\|u_e\|_1 = 1,$$

and a positive total amount A such that

$$\mathrm{div}_K = A u_e, \qquad W = A C_\Pi.$$

Thus the same unit object has face value 1 in both readings. The signed dictionary has

$$C_\Pi(r_0) = +1, \qquad C_\Pi(r_*) = -1, \qquad C_\Pi(r) = 0 \quad (r \notin \{r_0, r_*\}),$$

so the payment value is exactly

$$\mathrm{PaymentValue} = A \cdot 1 = A.$$

There is no remaining capacity multiplier, no hidden unit re-scaling, and no split into several denominations. The unresolved nondegenerate packet is a price-one payment problem for the single amount A on the fixed oriented unit certificate.

If $r_0 = r_*$, or if $A = 0$, the unit-face-value certificate degenerates and returns to the singleton/degenerate outlet. Otherwise any unresolved obstruction must be the named unit-face-value circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate scalar-load unit-normalization unit-face-value circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported unit-normalization certificate as a price-one payment certificate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.226 (Phase-residue exchange signed-amount-coordinate ledger)**.** *The phase-residue exchange unit-face-value outlet admits the following signed-amount-coordinate refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitFaceValueImportedForSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinatePriceOneLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinatePositiveEndpointAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateNegativeEndpointAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateSignedMassLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateEndpointMassBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateHalfL1EqualsAmountLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateRecoveryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateSupportCardinalityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateNoAmountSlotSplitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateNoOrientationFlipLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinatePDECCapReducedToEndpointSignedMassLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateCollisionOrZeroExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousUnitFaceValueAfterSignedAmountCoordinateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinatePacketLedger
AND NoAnonymousAmountPoolAfterSignedAmountCoordinateLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeSignedAmountCoordinateLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange signed-amount-coordinate ledger. All parallel outlets from the unit-face-value ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger fixes a price-one signed dictionary:

$$C_\Pi(r_0) = +1, \qquad C_\Pi(r_*) = -1, \qquad C_\Pi(r) = 0 \quad (r \notin \{r_0, r_*\}),$$

with amount $A > 0$, face value 1, and payment value A. Therefore the actual signed mass is not an additional object:

$$W_+ = A\delta_{r_0}, \qquad W_- = A\delta_{r_*}, \qquad W = W_+ - W_- = A(\delta_{r_0} - \delta_{r_*}) = AC_\Pi.$$

Its positive and negative masses are both A, its net mass is 0, and

$$\|W\|_1 = 2A, \qquad \frac{1}{2}\|W\|_1 = A.$$

Thus the amount coordinate can be recovered from either endpoint mass or from the half-ℓ^1 reading. In the nondegenerate case the support is exactly $\{r_0, r_*\}$, so there is no third payment slot, no anonymous amount pool, and no orientation flip by which the obstruction can be absorbed.

If $r_0 = r_*$, or if $A = 0$, the signed amount coordinate degenerates and returns to the singleton/degenerate outlet. Otherwise any unresolved obstruction must be the named signed-amount-coordinate circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate scalar-load unit-normalization unit-face-value signed-amount-coordinate circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported unit-face-value certificate as a two-endpoint signed amount coordinate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.227 (Phase-residue exchange phase-pairing ledger). *The phase-residue exchange signed-amount-coordinate outlet admits the following phase-pairing refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangePhasePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSignedAmountCoordinateImportedForPhasePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingUnitGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingPositiveEndpointLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingNegativeEndpointLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingSignedMassValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingEqualsAmountLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingEqualsHalfL1Ledger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingEqualsPaymentValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingConstantOffsetCancelsLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingNoPhaseRescaleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingNoSignMismatchLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingTwoEndpointSupportLockedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingPDECCapReducedToCalibratedPairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingCollisionOrZeroExitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingOrientationLedger
AND StableLadderEndpointOrbitPhaseResidueNoAnonymousSignedAmountAfterPhasePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingPacketLedger
AND NoAnonymousPhasePairingAfterSignedAmountCoordinateLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangePhasePairingLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange phase-pairing ledger. All parallel outlets from the signed-amount-coordinate ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger gives the two-endpoint signed mass

$$W = A(\delta_{r_0} - \delta_{r_*}), \qquad \frac{1}{2}\|W\|_1 = A.$$

The imported unit-face-value normalization also gives

$$\langle \delta_{r_0} - \delta_{r_*}, \phi_K \rangle = \phi_K(r_0) - \phi_K(r_*) = 1.$$

Therefore the phase pairing of the actual signed mass is fixed:

$$\langle W, \phi_K \rangle = A\big(\phi_K(r_0) - \phi_K(r_*)\big) = A.$$

This is the same A already read as amount, payment value, and half total variation. Since the signed mass has net mass 0, adding a constant to ϕ_K cancels. Since the unit phase gap is 1, there is no hidden phase rescaling. Since the positive endpoint is r_0 and the negative endpoint is r_*, there is no sign-mismatch outlet inside this packet.

If $r_0 = r_*$, or if $A = 0$, the phase pairing degenerates and returns to the singleton/degenerate outlet. Otherwise any unresolved obstruction must be the named phase-pairing circuit PDEC/cap outlet or one of the parallel outlets.

This theorem does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the phase-residue exchange active-facet normal-cone ray-coordinate scalar-load unit-normalization unit-face-value signed-amount-coordinate phase-pairing circuit PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness; it only rewrites the imported signed-amount-coordinate certificate as a calibrated phase-pairing certificate. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.228 (Phase-residue exchange circuit-materialization ledger)**.** *The phase-residue exchange phase-pairing outlet admits the following circuit-materialization refinement:*

SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
$\Rightarrow$ *StableLadderEndpointOrbitPhaseResidueExchangePhasePairingImportedLedger*
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeCircuitMaterializationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePhasePairingImportedForCircuitMaterializationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationSameFormalUnitKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationSourceAtomSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationOccurrenceUnitSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationCRTCoordinateSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationCanonicalCongruenceSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationPhaseEvaluationSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationSignedMassSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationCalibratedPairingSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCircuitMaterializationSameActualObjectGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoObjectSwitchAfterPhasePairingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualCircuitMaterializationDefectReturnLedger

```
AND StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCircuitCapacityGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCircuitCapacityInputLedger
AND NoAnonymousPhasePairingCircuitAfterMaterializationGateLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeCircuitMaterializationLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtom
```

Phase-residue exchange circuit-materialization ledger. All parallel outlets from the phase-pairing ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The previous ledger fixes one calibrated two-endpoint value:

$$W = A(\delta_{r_0} - \delta_{r_*}), \qquad \phi_K(r_0) - \phi_K(r_*) = 1, \qquad \langle W, \phi_K \rangle = A.$$

Thus amount, payment value, half total variation, and phase pairing are all the same actual-load candidate A. What remains cannot be an unstructured "circuit" whose slots may be silently exchanged. The source atom, signed occurrence unit, CRT coordinate, canonical congruence equation, phase-residue evaluation, signed mass, and calibrated pairing must all be attached to the same formal-unit key. Moreover, to use the value A in a capacity comparison, this common formal key must be materialized as one actual CRT/fiber object.

If any slot fails to materialize on that same actual object, the branch is not a hidden phase-pairing circuit; it is the named actual-circuit materialization PDEC/cap outlet. If all slots do materialize, the remaining obstruction is the materialized-circuit capacity PDEC/cap outlet, computed on the same load A rather than on a formal envelope or on a different object.

This theorem does not prove actual-circuit materialization, it does not prove the materialized-circuit capacity bound, it does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness. It only replaces the anonymous phase-pairing circuit by the same-object materialization gate and the materialized-capacity gate. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.229 (Phase-residue exchange actual-object-predicate ledger). *The phase-residue exchange actual-circuit materialization outlet admits the following actual-object-predicate refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtomSign
⇒ StableLadderEndpointOrbitPhaseResidueExchangeActualCircuitMaterializationImportedLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeActualObjectPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualCircuitMaterializationImportedForPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectTupleSchemaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCanonicalHashLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSourceIncidencePredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectOccurrenceIncidencePredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCRTRepresentativePredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCongruenceEvaluationPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectPhaseEndpointPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSignedMassPredicateLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCalibratedPairingPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingActualObjectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeDuplicateActualObjectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotMismatchReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectIncidencePredicatePacketLedger
AND NoAnonymousActualCircuitMaterializationAfterObjectPredicateLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeActualObjectPredicateLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtom
```

Phase-residue exchange actual-object-predicate ledger. All parallel outlets from the actual-circuit materialization ledger are carried forward: source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The imported materialization ledger says that the source, occurrence, CRT coordinate, congruence, phase evaluation, signed mass, and pairing slots share one formal-unit key. This is made into an actual object by the tuple

$$O = (s, u, c, E, \Phi, W, \langle W, \phi_K \rangle).$$

The tuple is admissible only when s is the source atom specified by the key, u lies in its signed occurrence incidence cell, c is the primitive CRT witness coordinate for u, the canonical congruence equation $E(c)$ holds, the phase evaluation Φ returns the same endpoints r_0, r_* with unit phase gap, the signed mass is

$$W = A(\delta_{r_0} - \delta_{r_*}),$$

and the calibrated pairing is

$$\langle W, \phi_K \rangle = A.$$

The hash of the actual object is the hash of this formal key and tuple. Hence a missing tuple, two different tuples for the same key, or any slot mismatch is a named actual-object incidence predicate PDEC/cap return, not a hidden materialization.

If the actual-object predicates all hold, the branch proceeds only to the materialized-circuit capacity gate. Otherwise it has already returned through the explicit incidence-predicate outlet.

This theorem does not prove the actual-object incidence predicate outlet, it does not prove the materialized-circuit capacity bound, it does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness. It only replaces anonymous actual materialization by the seven-slot actual-object predicate. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.230 (Phase-residue exchange materialized-capacity-slack ledger)**.** *The phase-residue exchange materialized-circuit capacity outlet admits the following same-object slack refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtomSign
⇒ StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCircuitCapacityImportedLedger
AND StableLadderEndpointOrbitActualObjectIncidencePredicateCarriedForwardAfterCapacitySlackLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeCapacitySlackLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeCapacitySlackLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeCapacitySlackLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeCapacitySlackLedger
```

```
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeCapacitySlackLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeCapacitySlackLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCapacityImportedForSlackLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackSameActualObjectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackActualLoadLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackCapacityValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackFormulaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackPositiveAbsorptionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackNegativeDefectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackBoundaryEqualityAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoFormalEnvelopeAfterCapacitySlackLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoLoadSwitchAfterCapacitySlackLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCapacitySlackPacketLedger
AND NoAnonymousMaterializedCapacityAfterSlackLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeCapacitySlackLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtom
```

Phase-residue exchange materialized-capacity-slack ledger. All parallel outlets from the actual-object predicate ledger are carried forward: the actual-object incidence predicate outlet itself, source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

On the materialized branch, the load and the capacity must be read from the same actual object O. The load is already fixed by the phase-pairing ledger:

$$L(O) = A = \langle W, \phi_K \rangle = \text{amount} = \text{payment value} = \frac{1}{2}\|W\|_1.$$

Let $C(O)$ denote the capacity attached to the same materialized circuit. The only admissible comparison is the same-object slack

$$\Sigma(O) = C(O) - A.$$

If $\Sigma(O) > 0$, this object is absorbed by capacity. If $\Sigma(O) < 0$, the branch is a genuine over-capacity PDEC/cap. If $\Sigma(O) = 0$, the branch is a boundary equality atom and must be registered or excluded as such; equality is not a strict contradiction by itself.

Thus formal envelopes, external substitute loads, and load switches to a different actual object are not legal exits from the materialized capacity branch.

This theorem does not prove the materialized-capacity slack bound, it does not prove the actual-object incidence predicate outlet, it does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness. It only replaces anonymous materialized capacity by the same-object slack inequality. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.231 (Phase-residue exchange materialized-capacity-slack-sign ledger)**.** *The phase-residue exchange materialized-capacity slack outlet admits the following sign refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtomSign
⇒ StableLadderEndpointOrbitPhaseResidueExchangeMaterializedCapacitySlackImportedLedger
AND StableLadderEndpointOrbitActualObjectIncidencePredicateCarriedForwardAfterSlackSignLedger
```

```
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterExchangeSlackSignLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackImportedForSignLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacitySlackSignTrichotomyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangePositiveSlackAbsorptionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackDefectPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityAtomPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoStrictContradictionFromEqualityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoSlackSignMergeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSlackSignPacketLedger
AND NoAnonymousMaterializedCapacitySlackAfterSignLedger
AND SparseScaleLadderSAECarriedForwardAfterExchangeSlackSignLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtom
```

Phase-residue exchange materialized-capacity-slack-sign ledger. All parallel outlets from the capacity-slack ledger are carried forward: the actual-object incidence predicate outlet, source-atom multiplicity cap, endpoint singleton atom, full-cycle mean atom, bridge cancellation, amplitude depth, variation-boundary flux, and sparse scale-ladder SAE.

The imported slack ledger has already reduced the materialized capacity comparison to

$$\Sigma(O) = C(O) - A$$

on a single actual object O. Since $\Sigma(O)$ is a real number, the branch has exactly three sign cases. If $\Sigma(O) > 0$, the object is strictly inside capacity and this branch is absorbed. If $\Sigma(O) < 0$, the object carries a genuine over-capacity defect and the branch is the named materialized negative-slack PDEC/cap outlet. If $\Sigma(O) = 0$, the object is a critical boundary equality atom. Equality cannot be used as a strict contradiction; it must be registered and either excluded as a persistent atom or routed through a named return.

Thus the sign of the slack is no longer an anonymous escape, and the positive, negative, and zero cases cannot be merged after the capacity comparison.

This theorem does not prove the negative-slack PDEC/cap outlet, it does not exclude boundary equality atoms, it does not prove the actual-object incidence predicate outlet, it does not prove endpoint singleton atom/SAE summability, it does not prove the full-cycle mean atom/SAE outlet, it does not prove the source-atom multiplicity-cap PDEC/cap outlet, it does not prove the bridge-cancellation PDEC/cap outlet, it does not prove the amplitude-depth PDEC/cap outlet, it does not prove the variation-boundary flux PDEC/cap outlet, and it does not prove global summability of the sparse scale-ladder SAE outlet. It also does not prove the existence of the linear phase/capacity witness. It only replaces anonymous materialized capacity slack by the sign trichotomy. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.232 (Phase-residue exchange negative-unit-defect ledger)**.** *The negative-slack outlet of the materialized-capacity-slack-sign ledger admits the following unit-defect refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeSlackSignImportedForNegativeUnitDefectLedger
AND StableLadderEndpointOrbitActualObjectIncidencePredicateCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterNegativeUnitDefectLedger
```

```
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitBoundaryEqualityAtomCarriedForwardAfterNegativeUnitDefectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackImportedForUnitDefectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackSameActualObjectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackAmountUnitCountLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackCapacityValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacityValueUnitIntegralityGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackDeficitFormulaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNegativeSlackPositiveDeficitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitNegativeDefectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoInfinitesimalNegativeSlackLedger
AND NoAnonymousRealNegativeSlackAfterUnitDefectLedger
AND SparseScaleLadderSAECarriedForwardAfterNegativeUnitDefectLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange negative-unit-defect ledger. The previous sign ledger leaves the negative branch

$$\Sigma(O) = C(O) - A < 0$$

and the zero branch as a separate boundary-equality atom. On the negative branch define the deficit

$$D(O) = A - C(O) = -\Sigma(O) > 0.$$

The load A is the price-one signed amount/payment value already imported by the unit-face-value and signed-amount-coordinate ledgers, and $C(O)$ is still read from the same materialized actual object. If $C(O)$ is not a same-object unit integer capacity count, the branch is not a real-valued "small negative slack"; it is the named capacity-value unit-integrality PDEC/cap outlet. If that unit-integrality gate passes, then $D(O)$ is a positive integer, hence $D(O) \geq 1$. Thus the negative slack outlet is refined to either a capacity-value integrality defect or a one-unit-or-larger negative capacity defect.

This theorem does not prove capacity-value unit-integrality, it does not exclude the resulting unit negative defect, it does not exclude the boundary equality atom, it does not prove the actual-object incidence predicate outlet, and it does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, or sparse scale-ladder summability. It also does not prove the existence of the linear phase/capacity witness. It only removes the possibility that negative slack remains an anonymous infinitesimal real defect. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.233 (Phase-residue exchange unit-defect-witness ledger)**.** *The unit negative defect outlet of the negative-unit-defect ledger admits the following witness refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitResidueShadowDualRowPhaseCellAtomSign
⇒ StableLadderEndpointOrbitPhaseResidueExchangeNegativeUnitDefectImportedForWitnessLedger
AND StableLadderEndpointOrbitActualObjectIncidencePredicateCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitCapacityValueUnitIntegralityCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitBoundaryEqualityAtomCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterUnitDefectWitnessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitNegativeDefectImportedLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeDemandUnitSetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCapacityAdmittedUnitSetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitCapacityAssignmentIncidenceGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitDefectCardinalityGapLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeUnitDefectNonemptyComplementLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingCapacityUnitWitnessLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitWitnessCoordinateTupleLedger
AND NoAnonymousAggregateUnitNegativeDefectAfterWitnessLedger
AND SparseScaleLadderSAECarriedForwardAfterUnitDefectWitnessLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange unit-defect-witness ledger. The previous ledger leaves a unit negative defect

$$D(O) = A - C(O) \geq 1.$$

Let U_A be the price-one demand-unit set with $|U_A| = A$, and let U_C be the capacity-admitted unit set read from the same materialized actual object with $|U_C| = C(O)$. If U_C is not materialized as a same-object assigned subset of U_A, the branch is the named unit-capacity assignment incidence PDEC/cap. If that assignment gate passes, then

$$|U_A \setminus U_C| = A - C(O) = D(O) \geq 1,$$

so a missing capacity unit $u_* \in U_A \setminus U_C$ exists. This witness must carry the source atom, occurrence unit, CRT coordinate, canonical congruence equation, phase endpoint, and signed amount coordinate inherited from the actual object predicate chain.

Thus the unit negative defect is no longer an anonymous aggregate shortage: it is either an assignment-incidence defect or a concrete missing-unit witness.

This theorem does not prove the unit-capacity assignment incidence outlet, it does not exclude the missing capacity unit witness, it does not prove capacity-value unit-integrality, it does not exclude the boundary equality atom, it does not prove the actual-object incidence predicate outlet, and it does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, or sparse scale-ladder summability. It also does not prove the existence of the linear phase/capacity witness. It only replaces the aggregate unit negative defect by a local witness route. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.234 (Phase-residue exchange missing-unit-coordinate-lock ledger)**.** *The missing capacity unit witness outlet of the unit-defect-witness ledger admits the following coordinate-lock refinement:*

```
SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrbitRe
⇒ StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitWitnessImportedForCoordinateLockLedger
AND StableLadderEndpointOrbitActualObjectIncidencePredicateCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitCapacityValueUnitIntegralityCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitUnitCapacityAssignmentIncidenceCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitBoundaryEqualityAtomCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitMultiplicityCapCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointSingletonAtomSAECarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitFullCycleMeanAtomCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitBridgeCancellationCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitAmplitudeDepthCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitVariationBoundaryFluxCarriedForwardAfterMissingUnitCoordinateLockLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingCapacityUnitWitnessImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitDemandMembershipLedger
```

```
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCapacityExclusionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitSourceAtomSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitOccurrenceSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCRTCoordinateSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCanonicalCongruenceSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitPhaseEndpointSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitSignedAmountSlotLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCanonicalHashLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCoordinateSlotMismatchReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingCapacityUnitWitnessPacketLedger
AND NoAnonymousMovingMissingUnitWitnessAfterCoordinateLockLedger
AND SparseScaleLadderSAECarriedForwardAfterMissingUnitCoordinateLockLedger
AND SparseScaleLadderSAESummabilityOrStableLadderEndpointSingletonAtomSAEOrEndpointOrbitFullCycleMeanAtomSAEOrEndpointOrb
```

Phase-residue exchange missing-unit-coordinate-lock ledger. The previous ledger leaves a missing capacity unit

$$u_* \in U_A \setminus U_C.$$

This witness is not allowed to move between objects or to change slots. It must carry demand membership, capacity exclusion, the source atom, occurrence unit, CRT coordinate, canonical congruence equation, phase endpoint, and price-one signed amount inherited from the same actual-object predicate chain. If any of these slots is absent or switched, the branch is the named missing unit coordinate-slot mismatch PDEC/cap. If all slots are present and agree, they define a canonical missing-unit key and the remaining branch is the canonical missing capacity unit witness outlet.

Thus a missing unit witness is no longer an anonymous movable object: it is either a coordinate-slot mismatch or a canonical local witness.

This theorem does not prove the coordinate-slot mismatch outlet, it does not exclude the canonical missing capacity unit witness, it does not prove unit-capacity assignment incidence, it does not prove capacity-value unit-integrality, it does not exclude the boundary equality atom, it does not prove the actual-object incidence predicate outlet, and it does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, or sparse scale-ladder summability. It also does not prove the existence of the linear phase/capacity witness. It only replaces the movable missing-unit witness by a canonical coordinate route. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.235 (Phase-residue exchange canonical-missing-unit-balance ledger)**.** *The canonical missing capacity unit witness outlet of the missing-unit coordinate-lock ledger admits the following balance refinement:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitImportedForBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitDemandIndicatorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitCapacityIndicatorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitFaceValueLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitCellDeficitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitCompensatorGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitUnpaidDemandCellReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitCompensationMismatchReturnLedger
AND NoAnonymousCanonicalMissingUnitCompensationAfterBalanceLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-missing-unit-balance ledger. The previous ledger leaves a canonical missing capacity unit. Its coordinate tuple defines a canonical key k. Demand membership gives $I_A(k) = 1$; capacity exclusion gives $I_C(k) = 0$; and the signed amount coordinate has already been normalized to one unit. Hence the same canonical cell carries the explicit unit deficit

$$I_A(k) - I_C(k) = 1.$$

If no legal same-object same-key compensator exists, the branch is the canonical unpaid-demand-cell PDEC/cap. If a purported compensator changes the object, key, or slot, or if it lacks a named boundary, bridge, or assignment outlet, the branch is the canonical compensation-mismatch PDEC/cap. Thus the canonical missing unit can no longer hide as an unregistered compensation or phase slip.

This theorem does not prove the canonical unpaid-demand-cell outlet, it does not prove the canonical compensation-mismatch outlet, it does not prove the missing-unit coordinate-slot mismatch outlet, it does not prove unit-capacity assignment incidence, it does not prove capacity-value unit-integrality, it does not exclude the boundary equality atom, and it does not prove the actual-object incidence predicate outlet. It also does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.236 (Phase-residue exchange canonical-unit-payment-conservation ledger)**.** *The canonical unpaid-demand-cell and canonical compensation-mismatch outlets admit the following payment-conservation refinement:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitPaymentConservationImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalMissingUnitBalanceImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitPaymentGraphLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalDemandKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalCapacityKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSameKeyPaymentEdgeObligationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentReturnWhitelistLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoHiddenCrossKeyPaymentEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitPaymentConservationDefectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitCompensatorKeyCollisionReturnLedger
AND NoAnonymousCanonicalUnitPaymentEscapeAfterConservationLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-unit-payment-conservation ledger. The previous ledger fixes a canonical key k with $I_A(k) = 1$ and $I_C(k) = 0$. Put the demand and capacity units of the same actual object into a canonical payment graph indexed by these keys. A direct payment of the cell k must be a same-object, same-key capacity edge. Any non-same-key payment must be one of the already named returns: boundary equality, bridge cancellation, unit-capacity assignment, or slot mismatch.

Hence there is no hidden cross-key payment edge that can pay the same canonical cell. If no same-key capacity edge exists and no named return pays the cell, the branch is the canonical unit payment-conservation defect. If a purported compensator exists but uses another key, the branch is the canonical unit compensator-key collision.

This theorem does not prove the payment-conservation defect outlet, it does not prove the compensator-key-collision outlet, it does not prove the canonical payment graph is globally closed, it does not prove the same-key edge obligation or the return whitelist, and it does not prove unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It also does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.237 (Phase-residue exchange canonical-unit-payment-singleton-cut ledger)**.** *The canonical unit payment-conservation defect and canonical compensator-key collision outlets admit the following singleton-cut refinement:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitPaymentSingletonCutImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalUnitPaymentConservationImportedForSingletonCutLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonDemandOneLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonCapacityZeroLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonCutBalanceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSameKeyCapacityEdgeEmptyOnSingletonCutLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalNamedReturnBoundaryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSingletonHallCutDefectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalKeyCollisionTrichotomyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyInjectivityDefectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalCrossKeyReturnWhitelistLeakReturnLedger
AND NoAnonymousCanonicalPaymentConservationDefectAfterSingletonCutLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-unit-payment-singleton-cut ledger. The previous ledger leaves either an unpaid same-key payment-conservation defect or a compensator using a different canonical key. Take the singleton cut $S = \{k\}$ in the canonical payment graph, where k is the key of the missing unit. The imported balance ledger gives

$$\mathrm{demand}(S) = I_A(k) = 1, \qquad \mathrm{capacity}(S) = I_C(k) = 0.$$

Thus the same-key direct capacity across this singleton cut is empty. If no named return leaves the cut, the residual is an explicit one-unit Hall-cut defect. If a cross-key compensator claims to pay the same canonical cell, the key map is not injective on that cell. If it does not claim the same cell, it must be a named return; otherwise it is a cross-key return-whitelist leak. Therefore the former payment-conservation/key-collision pair is no longer an anonymous outlet.

This theorem does not prove the singleton Hall-cut defect outlet, the key injectivity-defect outlet, or the cross-key return-whitelist leak outlet. It also does not prove the global payment graph closure, the return whitelist, unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.238 (Phase-residue exchange canonical-payment-key-tuple-lock ledger)**.** *The canonical payment key-injectivity defect outlet admits the following tuple-lock refinement:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyTupleLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCanonicalHashImportedForPaymentKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeMissingUnitCanonicalHashImportedForPaymentKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCellTupleSchemaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyDefinedAsTupleLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSameCellProjectionEqualityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalCrossKeySameCellTrichotomyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyTupleAliasDefectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSlotMismatchReturnLedger
AND NoAnonymousCanonicalPaymentKeyInjectivityDefectAfterTupleLockLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-key-tuple-lock ledger. The singleton-cut ledger leaves a possible key-injectivity defect: a purported cross-key compensator claims to pay the same canonical cell. Import the actual-object tuple hash and the missing-unit canonical hash. The payment key is now treated as the canonical tuple determined by these slots, not as an independent label.

Therefore a same-cell claim forces equality of all tuple projections. If the tuple projections are equal but the keys differ, the branch is the canonical payment key tuple-alias defect. If some tuple projection differs, the object does not pay the same canonical cell and the branch is the missing-unit coordinate slot-mismatch outlet. Thus the former key-injectivity defect is no longer an anonymous outlet.

This theorem does not prove the key tuple-alias defect outlet or the missing-unit coordinate slot-mismatch outlet. It also does not prove the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.239 (Phase-residue exchange canonical-payment-key-alias-normalization ledger)**.** *The canonical payment key tuple-alias defect outlet admits the following alias-normalization refinement:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyAliasNormalizationImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalFormalUnitHashStabilityImportedForPaymentKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyDomainSeparatorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeySerializationSchemaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentEqualTupleEqualSerializationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyHashFormulaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAliasNormalizationTrichotomyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeySerializationDriftReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentNoncanonicalKeyLabelResidueReturnLedger
AND NoAnonymousCanonicalPaymentKeyTupleAliasAfterNormalizationLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-key-alias-normalization ledger. The tuple-lock ledger leaves a possible key tuple-alias defect: the same payment tuple is represented by different keys. Import the canonical formal unit hash stability discipline and define the payment key with a fixed domain separator:

$$\text{key} = H(\texttt{canonical\_payment\_key}, \text{canonical\_bytes}(\text{payment tuple})).$$

Equal payment tuples have equal canonical byte strings under this discipline. If the same tuple produces two byte strings, the branch is a canonical payment key serialization-drift defect. If the byte string is the same but a different key is still present, that key is not the canonical hash output and the branch is a noncanonical key-label residue. Thus key tuple aliasing is no longer an anonymous payment outlet.

This theorem does not prove the serialization-drift outlet or the noncanonical key-label residue outlet. It also does not prove the missing-unit coordinate slot-mismatch outlet, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.240 (Phase-residue exchange canonical-payment-serialization-codec-lock ledger)**.** *The canonical payment key serialization-drift outlet is not an independent activity outlet under the canonical payment codec:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSerializationCodecLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentTupleFieldVectorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentFieldTagTotalOrderLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentScalarEncodingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentLengthPrefixInjectiveCodecLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentNullSentinelLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCodecDeterminismLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentEqualFieldVectorEqualBytesLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSerializationDriftFieldOrCodecDichotomyLedger
AND NoIndependentCanonicalPaymentSerializationDriftAfterCodecLockLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-serialization-codec-lock ledger. The alias-normalization ledger leaves the possibility that the same payment tuple has two canonical byte strings. Replace implementation text by a field-vector codec: the payment tuple has a fixed field order, typed field tags, length prefixes, unique scalar encodings, and a fixed null sentinel. Thus canonical bytes are a deterministic function of the payment tuple field vector.

If two records have different canonical byte strings, then their field vectors are different and the branch is already a named slot/object/assignment outlet, not a new key-side outlet. If the field vectors are equal, codec determinism forces the byte strings to be equal. Hence the serialization-drift outlet is removed from the active target; the key-side residue left by this layer is the noncanonical key-label residue.

This theorem does not prove the noncanonical key-label residue outlet. It also does not prove the missing-unit coordinate slot-mismatch outlet, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization

□

Theorem 7.241 (Phase-residue exchange canonical-payment-key-label-admission-lock ledger)**.** *The canonical payment noncanonical key-label residue outlet is not an independent activity outlet under canonical edge admission:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentKeyLabelAdmissionLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAdmittedKeyFormulaLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentExternalLabelErasureLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentEdgeAdmissionPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentReturnWhitelistBindingLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentNoncanonicalLabelNotAdmittedEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentNoncanonicalLabelReturnWhitelistLeakLedger
AND NoIndependentCanonicalPaymentNoncanonicalKeyLabelResidueAfterAdmissionLockLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-key-label-admission-lock ledger. The serialization-codec lock leaves the possibility that a record still carries a noncanonical external key label. Edge admission removes that freedom: an admitted payment edge must use the canonical key

$$H(\texttt{canonical\_payment\_key}, \text{canonical\_bytes}(\text{payment tuple}))$$

and must also pass the object and unit-capacity gates. External labels are erased from canonical graph identity.

Therefore a label different from the admitted key is not an admitted payment edge. If such a label nevertheless acts as a cross-key return, the defect is exactly a canonical cross-key return-whitelist leak. Hence noncanonical key-label residue is removed as an independent outlet.

This theorem does not prove the missing-unit coordinate slot-mismatch outlet, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, unit-capacity assignment incidence, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization

□

Theorem 7.242 (Phase-residue exchange canonical-payment-same-cell-slot-admission ledger)**.** *The missing-unit coordinate slot-mismatch outlet is not an independent activity outlet once same-cell payment admission is enforced:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSameCellSlotAdmissionImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSameCellClaimRequiresSlotVectorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSlotVectorEqualityGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSlotMismatchNotSameCellEdgeLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCountedSlotMismatchAssignmentIncidenceLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentUncountedSlotMismatchSingletonCutLedger
AND NoIndependentMissingUnitCoordinateSlotMismatchAfterSameCellAdmissionLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-same-cell-slot-admission ledger. An admitted compensator for a canonical missing unit cell must be a same-cell edge. Hence it must have the same source atom, occurrence slot, CRT coordinate slot, canonical congruence slot, phase endpoint slot, and signed amount slot as the missing cell.

If any of these slots differ, the candidate is not an admitted payment edge for that missing cell. If it is nevertheless counted as capacity for the missing cell, the branch is a unit-capacity assignment-incidence defect. If it is not counted, the singleton deficit remains: demand one, admitted same-cell capacity zero. Thus the slot-mismatch outlet is removed as an independent outlet.

This theorem does not prove the assignment-incidence outlet, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, capacity-value unit-integrality, actual-object incidence, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.243 (Phase-residue exchange canonical-payment-assignment-incidence-lock ledger). *The unit-capacity assignment-incidence outlet is not an independent activity outlet once counted capacity units are admitted by gate:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentIncidenceLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCountedCapacityUnitLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAdmittedCapacityUnitPredicateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCapacityUnitValueOneLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentActualObjectGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentSameCellGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentWhitelistGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentPartialInjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentBadCountedUnitTrichotomyLedger
AND NoIndependentUnitCapacityAssignmentIncidenceAfterCanonicalAssignmentLockLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-assignment-incidence-lock ledger. After same-cell slot admission, a counted capacity unit must pass the admitted unit predicate. This predicate is the conjunction of the actual-object gate, the unit-value-one gate, the same-cell gate, and the return-whitelist gate. Units passing these gates form a partial injection into the demand-unit set; repeated occupation or external labels do not create new capacity.

If a counted unit fails the actual-object gate, the branch is the actual-object incidence predicate outlet. If it fails the unit-value-one gate, the branch is the capacity-value unit-integrality outlet. If it fails the whitelist gate, the branch is the cross-key return-whitelist leak. If the bad count is simply removed, the admitted same-cell capacity remains too small and the singleton Hall-cut deficit persists. Hence assignment incidence is no longer an independent outlet.

This theorem does not prove actual-object incidence, capacity-value unit-integrality, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.244 (Phase-residue exchange canonical-payment-capacity-unit-value-lock ledger). *The capacity-value unit-integrality outlet is not an independent activity outlet once admitted capacity units are normalized to one-unit indicators:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCapacityUnitValueLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentIncidenceImportedForCapacityUnitValueLockLedge
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSignedAmountUnitAtomLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentUnitValueOneNormalizationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCapacityAsUnitIndicatorSumLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCapacityIntegerSumLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentNonunitValueActualObjectReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentBadUnitValueRemovalLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentBadUnitValueSingletonCutLedger
AND NoIndependentCapacityValueUnitIntegralityAfterCapacityUnitValueLockLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFi
```

Phase-residue exchange canonical-payment-capacity-unit-value-lock ledger. After the assignment-incidence lock, counted capacity units are exactly the admitted units. The unit-value-one gate normalizes every admitted signed amount atom to one unit. Therefore the capacity value is a finite sum of zero-one unit indicators, hence a nonnegative integer.

If a proposed capacity unit has a non-unit signed amount, it has not supplied the required actual admitted object and returns to the actual-object incidence predicate outlet. If that bad unit is removed from capacity, the same canonical cell still has one unit of demand and no same-cell admitted capacity, so the singleton Hall-cut deficit remains. Thus capacity-value unit-integrality is no longer an independent outlet.

This theorem does not prove actual-object incidence, the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.245 (Phase-residue exchange canonical-payment-actual-object-field-packet-lock ledger). *The actual-object incidence-predicate outlet is not an independent anonymous activity outlet once actual objects are locked to their canonical field packet:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentActualObjectFieldPacketLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCapacityUnitValueImportedForActualObjectFieldPacketLockL
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectPredicateRouterImportedForFieldPacketLockLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectTupleSchemaImportedForFieldPacketLockLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCanonicalHashImportedForFieldPacketLockLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSourceFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectOccurrenceFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCRTFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectCongruenceFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectPhaseEndpointFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSignedMassFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectPairingFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectMissingFieldPacketLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectDuplicateFieldPacketLedger
AND NoIndependentActualObjectIncidencePredicateAfterFieldPacketLockLedger.
```

Phase-residue exchange canonical-payment-actual-object-field-packet-lock ledger. The earlier actual-object predicate certificate fixes an actual object as a canonical tuple and hash. Its fields are source incidence, occurrence incidence, CRT representative, congruence evaluation, phase endpoint, signed mass, and calibrated pairing, together with missing-object and duplicate-object returns.

Thus a failure of actual-object incidence cannot remain an unnamed predicate failure. If the object is absent, the missing-field outlet is active; if it is duplicated, the duplicate-field outlet is active; otherwise some named tuple field fails. If no named field fails, the actual-object incidence predicate is satisfied. Hence the single actual-object outlet has been replaced by named field-packet outlets.

This theorem does not prove that the field-packet outlets are impossible. It does not prove the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, or exclusion of the boundary equality atom. It does not prove endpoint singleton atom/SAE, full-cycle mean atom/SAE, source-atom multiplicity cap, bridge cancellation, amplitude depth, variation-boundary flux, sparse scale-ladder summability, or the linear phase/capacity witness. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.246 (Phase-residue exchange canonical-payment-actual-object-slot-mismatch-unification ledger)**.** *The seven named actual-object field failures reduce to a single slot-mismatch outlet, with missing and duplicate objects kept as separate existence outlets:*

```
ActualObjectSourceIncidenceFieldPDECCap OR ActualObjectOccurrenceIncidenceFieldPDECCap OR ActualObjectCRTRepresentativeFi
OR ActualObjectCongruenceEvaluationFieldPDECCap OR ActualObjectPhaseEndpointFieldPDECCap OR ActualObjectSignedMassFieldPD
OR ActualObjectCalibratedPairingFieldPDECCap
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentActualObjectSlotMismatchUnificationImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectFieldPacketLockImportedForSlotMismatchUnificationLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotVectorLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotMismatchUnionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectFieldFailureToSlotMismatchLedger
AND NoIndependentActualObjectNamedFieldExitAfterSlotMismatchUnificationLedger.
```

Phase-residue exchange canonical-payment-actual-object-slot-mismatch-unification ledger. The actual object has a fixed slot vector consisting of source, occurrence, CRT coordinate, congruence, phase endpoint, signed mass, and calibrated pairing. A failure of any one of these field predicates says exactly that the corresponding coordinate of the vector is not the required coordinate for the canonical object.

Thus the seven named field failures are not seven independent mechanisms. They are the disjoint coordinates of one slot-mismatch predicate. Missing objects and duplicate objects are not slot mismatches, so they remain as the two object existence outlets.

This theorem does not prove missing-object, duplicate-object, or slot-mismatch exclusion. It does not prove the singleton Hall-cut defect, the cross-key return-whitelist leak, the global payment graph closure, the return whitelist, or exclusion of the boundary equality atom. It does not prove the endpoint parallel exits or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.247 (Phase-residue exchange canonical-payment-actual-object-admission-trichotomy ledger)**.** *The remaining actual-object admission defects are not independent outlets:*

```
ActualObjectMissingFieldPDECCap OR ActualObjectDuplicateFieldPDECCap OR ActualObjectSlotMismatchPDECCap
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentActualObjectAdmissionTrichotomyImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotMismatchUnificationImportedForAdmissionTrichotomyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentAssignmentIncidenceImportedForActualObjectAdmissionTrich
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectMissingNotAdmittedCapacityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotMismatchNotSameCellCapacityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectDuplicateCollisionOrWhitelistLeakLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectBadEdgeRemovalSingletonCutLedger
AND NoIndependentActualObjectAdmissionDefectAfterTrichotomyLedger.
```

Phase-residue exchange canonical-payment-actual-object-admission-trichotomy ledger. If an actual object is missing, the proposed edge is not an admitted capacity edge. If the actual object has a slot mismatch, it is not a same-cell admitted capacity edge. In both cases deleting the bad edge leaves the same singleton demand cell unpaid. If an actual object is duplicated, a same-key duplicate does not create new capacity, while a cross-key return is exactly the cross-key whitelist-leak outlet.

Thus missing object, duplicate object, and slot mismatch do not supply independent capacity mechanisms. They return to the singleton Hall-cut or cross-key whitelist-leak outlets.

This theorem does not prove the singleton Hall-cut defect or the cross-key return-whitelist leak. It does not prove payment graph closure, return whitelist, boundary equality exclusion, endpoint parallel exits, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.248 (Phase-residue exchange canonical-payment-singleton-cut-return-admission-lock ledger)**.** *The broad singleton Hall-cut outlet is not an independent return mechanism:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonCutReturnAdmissionLockImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectAdmissionTrichotomyImportedForSingletonReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSingletonCutRouterImportedForReturnAdmissionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSingletonCutBalanceImportedForReturnAdmissionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSingletonSameKeyCapacityEmptyImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSingletonReturnAdmissionPartitionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSingletonSameKeyReturnNoNewCapacityLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSingletonCrossKeyReturnWhitelistGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSingletonLegalReturnNamedBoundaryLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSingletonNoReturnHallAtomRegistrationLedger
AND NoIndependentBroadCanonicalSingletonHallCutAfterReturnAdmissionLockLedger.
```

Phase-residue exchange canonical-payment-singleton-cut-return-admission-lock ledger. The previous singleton-cut ledger gives the single key cut balance $I_A(k)-I_C(k)=1$ and emptiness of the same-key capacity edge. The actual-object admission trichotomy says that missing, mismatched, or duplicate actual objects do not create an independent admitted capacity edge.

Thus every attempted return from the singleton cut is exhausted by a finite partition. A same-key duplicate adds no capacity. A bad actual-object edge has already returned to the admission trichotomy. A cross-key return which is not whitelisted is the cross-key return-whitelist leak. A legal named return is carried by the boundary or endpoint return outlets. If none of these returns exists, the broad singleton Hall-cut defect is precisely the narrower singleton no-return Hall atom.

This theorem does not prove the no-return Hall atom, the cross-key return-whitelist leak, boundary equality exclusion, endpoint parallel exits, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.249 (Phase-residue exchange canonical-payment-singleton-no-return-endpoint-projection ledger). *The singleton no-return Hall atom is not an independent payment-side outlet:*

EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentSingletonNoReturnEndpointProjectionImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalSingletonNoReturnHallAtomImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectFieldPacketImportedForNoReturnProjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnSingletonUnitKeyLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnPhaseEndpointProjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnSignedMassUnitProjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnSourceMultiplicityGateLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnEndpointSingletonAtomReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeNoReturnSourceMultiplicityCapReturnLedger
AND NoIndependentCanonicalSingletonNoReturnHallAtomAfterEndpointProjectionLedger.

Phase-residue exchange canonical-payment-singleton-no-return-endpoint-projection ledger. A no-return Hall atom is still a single canonical demand unit. The actual object field-packet ledger fixes its source, occurrence, phase endpoint, and signed mass coordinates. Therefore the atom cannot remain an anonymous payment-side deficit after projection to the endpoint orbit.

If the endpoint projection has no companion source or occurrence, it is the endpoint singleton atom outlet. If it has such a companion, the obstruction is the source-atom multiplicity cap outlet. Thus the no-return Hall atom is carried by already named endpoint/source outlets rather than by a separate payment escape.

This theorem does not prove endpoint singleton SAE, source-atom multiplicity cap exclusion, cross-key return-whitelist exclusion, boundary equality exclusion, endpoint parallel exits, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.250 (Phase-residue exchange canonical-payment-cross-key-return-whitelist-slot-projection ledger). *The cross-key return-whitelist leak is not an independent payment-side outlet:*

EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeCanonicalPaymentCrossKeyReturnWhitelistSlotProjectionImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCanonicalCrossKeyReturnWhitelistLeakImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeActualObjectSlotVectorImportedForCrossKeyReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnUnitPairLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnNotSameActualObjectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnSlotChangePartitionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnSourceMultiplicityReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnCRTBoundaryReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnPhaseOrbitReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeCrossKeyReturnSignedPairingReturnLedger
AND NoIndependentCanonicalCrossKeyReturnWhitelistLeakAfterSlotProjectionLedger.

Phase-residue exchange canonical-payment-cross-key-return-whitelist-slot-projection ledger. The actual-object slot vector is fixed. A non-whitelisted cross-key return cannot remain an anonymous payment edge: if it is not the same actual object, then at least one named slot changes.

A source or occurrence change is the source-atom multiplicity outlet. A CRT or congruence change is carried by the boundary equality outlet. A phase endpoint change is carried by bridge, amplitude-depth, or variation-boundary flux outlets. A signed-mass or pairing change is carried by the full-cycle mean or amplitude-depth outlets. Hence the cross-key return-whitelist leak is projected into already named source, boundary, or endpoint exits.

This theorem does not prove source multiplicity, boundary equality exclusion, endpoint orbit exits, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in `docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.251 (Phase-residue exchange boundary-equality-critical-endpoint-projection ledger). *The boundary equality atom is not an independent outlet:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityFiberS
⇒ StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityCriticalEndpointProjectionImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityAtomImportedForCriticalEndpointProjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeSlackSignImportedForBoundaryEqualityProjectionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityZeroSlackActualObjectLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityCriticalFacetLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityFirstVariationPartitionLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityFullCycleMeanReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityAmplitudeDepthReturnLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityVariationFluxReturnLedger
AND NoIndependentBoundaryEqualityAtomAfterCriticalEndpointProjectionLedger.
```

Phase-residue exchange boundary-equality-critical-endpoint-projection ledger. The slack-sign ledger identifies the boundary equality atom with $\Sigma(O) = 0$ for the same actual object. This is a critical active-facet condition, not a strict contradiction and not an anonymous residual outlet.

The critical equality is projected through the endpoint model. If the endpoint phase is constant across the cycle, the obstruction is the full-cycle mean atom. If the equality is held by a nonzero depth imbalance, it is the amplitude-depth outlet. If it is held by boundary transport, it is the variation-boundary flux outlet. Hence the boundary equality atom is carried by already named endpoint critical exits.

This theorem does not prove full-cycle mean, amplitude-depth, variation-boundary flux, endpoint singleton, source multiplicity, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.252 (Phase-residue exchange endpoint-dynamic-signed-depth-flux-packet ledger). *The bridge, amplitude-depth, and variation-boundary endpoint outlets are one signed-depth/flux packet:*

```
EndpointOrbitBridgeCancellationPDECCap OR EndpointOrbitAmplitudeDepthPDECCap OR EndpointOrbitVariationBoundaryFluxPDECCap
⇒ StableLadderEndpointOrbitPhaseResidueExchangeEndpointDynamicSignedDepthFluxPacketImportedLedger
AND StableLadderEndpointOrbitPhaseResidueExchangeBoundaryEqualityProjectionImportedForDynamicPacketLedger
AND StableLadderEndpointOrbitDynamicSignedEndpointMeasureLedger
AND StableLadderEndpointOrbitDynamicPacketZeroMeanReductionLedger
AND StableLadderEndpointOrbitSignedDepthFluxDecompositionLedger
AND StableLadderEndpointOrbitBridgeCancellationAsSignedDepthFluxComponentLedger
AND StableLadderEndpointOrbitAmplitudeDepthAsSignedDepthFluxComponentLedger
AND StableLadderEndpointOrbitVariationFluxAsSignedDepthFluxComponentLedger
AND StableLadderEndpointOrbitSignedDepthFluxPacketRegistrationLedger
AND NoIndependentEndpointBridgeAmplitudeVariationAfterSignedDepthFluxPacketLedger.
```

Phase-residue exchange endpoint-dynamic-signed-depth-flux-packet ledger. After the payment and boundary-equality outlets have been projected away, the remaining endpoint dynamics live in one signed endpoint measure. The full-cycle mean is kept as a separate outlet, so the dynamic residual may be treated as a zero-mean signed-depth/flux object.

The opposite-sign component is the bridge-cancellation outlet. The same-sign positive depth component is the amplitude-depth outlet. The boundary transport component is the variation-boundary flux outlet. Therefore these three names are not independent mechanisms; they are the three projections of one endpoint signed-depth/flux packet.

This theorem does not prove the signed-depth/flux packet, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.253 (Phase-residue exchange endpoint-signed-depth-flux-cycle-skeleton ledger). *The signed-depth/flux packet is not an anonymous endpoint outlet:*

```
EndpointOrbitSignedDepthFluxPacketPDECCap
⇒ StableLadderEndpointOrbitSignedDepthFluxPacketImportedForCycleSkeletonLedger
AND StableLadderEndpointOrbitSignedDepthFluxFiniteSupportSkeletonLedger
AND StableLadderEndpointOrbitSignedDepthFluxUnitAtomExpansionLedger
AND StableLadderEndpointOrbitSignedDepthFluxZeroMeanBalanceLedger
AND StableLadderEndpointOrbitSignedDepthFluxPairingGraphLedger
AND StableLadderEndpointOrbitSignedDepthFluxAcyclicLeafReturnLedger
AND StableLadderEndpointOrbitSignedDepthFluxAlternatingCycleResidualLedger
AND NoIndependentEndpointSignedDepthFluxPacketAfterCycleSkeletonLedger.
```

Phase-residue exchange endpoint-signed-depth-flux-cycle-skeleton ledger. The previous ledger registers one endpoint signed-depth/flux packet. Inside the finite CRT endpoint support this packet has a finite signed support skeleton. Unit normalization and face-value normalization expand it into positive and negative unit atoms.

The full-cycle mean outlet is kept separate, so the packet residual has zero signed mass. The positive and negative unit atoms therefore induce a finite endpoint pairing graph. If this graph is acyclic, finite leaf stripping exposes an unmatched endpoint unit, terminal source occurrence, or constant endpoint phase; these are exactly the endpoint singleton, source multiplicity, and full-cycle mean outlets already carried forward.

Thus a nonempty residual signed-depth/flux packet that is not returned to those named outlets must contain an alternating endpoint transport cycle. This theorem does not prove that alternating cycle, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability is impossible. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.254 (Phase-residue exchange endpoint-alternating-cycle-monodromy-lock ledger). *The alternating endpoint transport cycle is not an anonymous cycle outlet:*

```
EndpointOrbitSignedDepthFluxAlternatingTransportCyclePDECCap
⇒ StableLadderEndpointOrbitAlternatingTransportCycleImportedForMonodromyLockLedger
AND StableLadderEndpointOrbitAlternatingCycleOrderedSupportLedger
AND StableLadderEndpointOrbitAlternatingCycleSignedEdgeIncidenceLedger
AND StableLadderEndpointOrbitAlternatingCycleBoundaryZeroLedger
AND StableLadderEndpointOrbitAlternatingCycleCRTPhaseIncrementWordLedger
AND StableLadderEndpointOrbitAlternatingCycleTelescopingMonodromyDichotomyLedger
AND StableLadderEndpointOrbitAlternatingCycleZeroMonodromyCirculationReturnLedger
AND StableLadderEndpointOrbitAlternatingCycleNonzeroMonodromyResidualLedger
AND NoIndependentEndpointAlternatingTransportCycleAfterMonodromyLockLedger.
```

Phase-residue exchange endpoint-alternating-cycle-monodromy-lock ledger. The previous cycle-skeleton ledger leaves only a finite alternating endpoint transport cycle. Such a cycle has an ordered endpoint word $v_0 \to v_1 \to \cdots \to v_0$, alternating signed edge incidence, and zero incidence boundary. If the boundary were nonzero, the obstruction would have already returned to the leaf, singleton, source-multiplicity, or full-mean outlets.

Each edge also has a CRT endpoint phase increment. Summing the increments around the ordered cycle gives a single monodromy value. The telescoping dichotomy is sharp: zero monodromy is a pure circulation with no endpoint load defect, while nonzero monodromy is the only remaining cycle obstruction.

This theorem does not prove nonzero monodromy, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability impossible. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.255 (Phase-residue exchange endpoint-alternating-cycle-pivot-phase-slip ledger)**.** *The nonzero alternating-cycle monodromy is not an anonymous global outlet:*

```
EndpointOrbitSignedDepthFluxAlternatingCycleMonodromyPDECCap
⇒ StableLadderEndpointOrbitAlternatingCycleMonodromyImportedForPivotPhaseSlipLedger
AND StableLadderEndpointOrbitAlternatingCycleMonodromyCRTVectorLedger
AND StableLadderEndpointOrbitAlternatingCycleNonzeroCRTCoordinateLocalizationLedger
AND StableLadderEndpointOrbitAlternatingCyclePivotPrimeSelectionLedger
AND StableLadderEndpointOrbitAlternatingCyclePivotPrimePhaseSlipLedger
AND StableLadderEndpointOrbitAlternatingCycleAllCoordinatesZeroReturnLedger
AND NoIndependentEndpointAlternatingCycleMonodromyAfterPivotPhaseSlipLedger.
```

Phase-residue exchange endpoint-alternating-cycle-pivot-phase-slip ledger. The monodromy value from the previous ledger lives in a finite CRT coordinate space. Hence it is represented by its prime-modulus coordinate vector. If every coordinate is zero, the monodromy is zero and the obstruction returns to the pure-circulation branch already excluded as an endpoint load defect.

For nonzero monodromy, at least one prime coordinate is nonzero. Choose the least such prime as the canonical pivot. This removes anonymous coordinate choice: the remaining obstruction is precisely a nonzero pivot-prime phase slip on the endpoint alternating cycle.

This theorem does not prove pivot phase-slip, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability impossible. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.256 (Phase-residue exchange endpoint-pivot LCM-support barrier)**.** *The pivot-prime phase-slip is not an anonymous single outlet:*

```
EndpointOrbitSignedDepthFluxAlternatingCyclePivotPrimePhaseSlipPDECCap
⇒ StableLadderEndpointOrbitAlternatingCyclePivotPhaseSlipImportedForLCMSupportBarrierLedger
AND StableLadderEndpointOrbitPivotPrimePhaseMotionFormulaLedger
AND StableLadderEndpointOrbitPivotPrimeFixedReplayLCMPeriodLedger
AND StableLadderEndpointOrbitPivotPrimeSupportWidthLedger
AND StableLadderEndpointOrbitPivotPrimeLCMExceedsSupportNoReplayLedger
AND StableLadderEndpointOrbitPivotPrimeSmallLCMBranchLedger
AND StableLadderEndpointOrbitPivotPrimeMovingLabelReturnLedger
AND NoIndependentEndpointPivotPrimePhaseSlipAfterLCMSupportBarrierLedger.
```

Phase-residue exchange endpoint-pivot LCM-support barrier. Let q be the canonical pivot prime from the previous ledger, and let $\sigma \not\equiv 0 \pmod q$ be the pivot phase-slip amount. If the same pivot label replays with formal endpoint displacement d, exact replay requires $d\sigma \equiv 0 \pmod q$. Since q is prime and σ is nonzero modulo q, one has $q \mid d$. For a fixed pivot label set Λ, exact replay therefore forces $\mathrm{lcm}(\Lambda) \mid d$.

The endpoint alternating cycle has finite support. Write W for the support width in which a fixed-label replay could occur. If $\mathrm{lcm}(\Lambda) > W$, there is no nonzero fixed-label replay inside the support. If $\mathrm{lcm}(\Lambda) \leq W$, the obstruction is a small-LCM/fixed-residue ColumnCRT/PDEC branch. If the chain changes the pivot label to evade this barrier, it is a moving-pivot PDEC/SAE branch, not a fixed pivot replay.

This theorem does not prove the small-LCM branch, the nonreplay sparse branch, the moving-pivot branch, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability impossible. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.257 (Phase-residue exchange endpoint-pivot small-LCM rank pressure)**.** *The endpoint pivot small-LCM branch is not an anonymous broad ColumnCRT outlet:*

```
EndpointOrbitAlternatingCyclePivotPrimeSmallLCMColumnCRTPDECExclusion
⇒ StableLadderEndpointOrbitPivotPrimeSmallLCMImportedForRankPressureLedger
AND StableLadderEndpointOrbitPivotPrimePositiveReplayWidthGuardLedger
AND StableLadderEndpointOrbitPivotPrimeDistinctPrimeProductLawLedger
AND StableLadderEndpointOrbitPivotPrimeSmallLCMRankPressureLedger
AND StableLadderEndpointOrbitPivotPrimeTwoLargeCarrierSqrtBarrierLedger
AND StableLadderEndpointOrbitPivotPrimeShortWindowLCMDisciplineImportedLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierFixedResidueRouteLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierNonpersistentSparseRouteLedger
AND StableLadderEndpointOrbitPivotPrimeHighCarrierRankDeficitRouteLedger
AND NoIndependentEndpointPivotPrimeSmallLCMAfterRankPressureLedger.
```

Phase-residue exchange endpoint-pivot small-LCM rank pressure. In the small-LCM branch from the previous ledger, a fixed pivot label set Λ satisfies $L = \mathrm{lcm}(\Lambda) \leq W$, where W is the finite endpoint support width. Repeated use of the same pivot prime does not increase L, and is therefore a fixed-residue reuse branch. For the distinct pivot primes, $L = \prod_{q\in\Lambda} q$.

For any threshold $B > 1$, let r_B be the number of distinct pivot primes with $q > B$. Then $B^{r_B} \leq \prod_{q>B} q \leq L \leq W$, so

$$r_B \leq \left\lfloor \frac{\log W}{\log B} \right\rfloor .$$

Taking $B = \sqrt{W}$ shows that two pivot carriers larger than $\sqrt{W}$ cannot occur in one fixed small-LCM unit.

Thus the small-LCM branch has only three non-anonymous continuations: pressure from low carriers must enter fixed-residue ColumnCRT/PDEC, nonpersistent low carriers enter sparse SAE accounting, and pressure from high carriers is rank-limited and must be discharged by a high-carrier rank-deficit or singleton SAE argument.

This theorem does not prove those three continuations, the nonreplay sparse branch, the moving-pivot branch, endpoint singleton, full-cycle mean, source multiplicity, or sparse scale-ladder summability impossible. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.258 (Phase-residue exchange endpoint-pivot low-carrier AP envelope)**.** *The endpoint pivot low-carrier fixed-residue branch is not an anonymous ColumnCRT outlet:*

```
EndpointOrbitAlternatingCyclePivotPrimeLowCarrierFixedResidueColumnCRTPDECExclusion
⇒ StableLadderEndpointOrbitPivotPrimeLowCarrierFixedResidueImportedForAPEnvelopeLedger
AND StableLadderEndpointOrbitPivotPrimeBoundaryCarrierDegeneratesToColumnCRTOrEndpointSingletonLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierResidueToRowAPFormulaLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierSingleResidueSharpEnvelopeLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierAllResiduesExactMassLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierSelectedTableExactEnvelopeLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierResidueTableFiniteLedger
AND StableLadderEndpointOrbitPivotPrimeLowEffectiveModColumnCRTImportedLedger
AND StableLadderEndpointOrbitPivotPrimeDenseLowCarrierResidueTableRouteLedger
AND StableLadderEndpointOrbitPivotPrimeSparseLowCarrierResidueCellRouteLedger
AND StableLadderEndpointOrbitPivotPrimeActualDemandLowerBoundStillOpenLedger
AND NoIndependentEndpointPivotPrimeLowCarrierFixedResidueAfterAPEnvelopeLedger.
```

Phase-residue exchange endpoint-pivot low-carrier AP envelope. Let q be a low carrier pivot prime in the fixed-residue branch. If $q = P$, then P is not invertible modulo q; this is a boundary carrier, so it is registered as a ColumnCRT boundary degeneration or as an already named endpoint singleton/source-multiplicity exit, not as an AP row-family.

Assume $q < P$. In the endpoint support window write t for the normalized row/phase coordinate and H for the window length. A fixed residue $a \bmod q$ imposes

$$tP + a \equiv 0 \pmod q.$$

Since $(P, q) = 1$, this is equivalent to the single arithmetic progression

$$t \equiv -aP^{-1} \pmod q.$$

Thus a single cell (q, a) contributes at most $\lceil H/q \rceil$ points in the endpoint window. For fixed q, summing over all $a \bmod q$ gives exactly H, because each endpoint row selects one residue $a \equiv -tP \pmod q$. Hence any selected low-carrier residue table has the exact AP envelope obtained by summing these cell bounds, with the additional bound H times the number of active pivot primes.

If such a low-dimensional table persistently approaches or exceeds the envelope needed by the counterexample chain, it is a dense residue-table PDEC/ColumnCRT route. If the cells are nonpersistent or sparse, they enter endpoint sparse cell SAE accounting. The remaining open work is to prove the endpoint actual demand lower bound, the strict AP-envelope gap or dense-table exclusion, and the sparse-cell SAE summability; the other endpoint exits also remain open. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.259 (Phase-residue exchange endpoint-pivot actual demand source cut)**.** *The endpoint pivot actual low-carrier row-incidence demand is reduced to a source-side amplification problem and a non-recycling payment-injection problem:*

```
EndpointOrbitAlternatingCyclePivotPrimeActualLowCarrierRowIncidenceDemandLowerBound
⇒ StableLadderEndpointOrbitPivotPrimeActualDemandImportedForSourceCutLedger
AND StableLadderEndpointOrbitPivotPrimeUnitIncidenceDemandLedger
AND StableLadderEndpointOrbitPivotPrimeUnitDemandNotEnvelopeGapLedger
AND StableLadderEndpointOrbitPivotPrimeNoAPEnvelopeRecyclingDemandGuardLedger
AND StableLadderEndpointOrbitPivotPrimeReleaseMassAmplificationRouteLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierPaymentInjectionRouteLedger
AND NoIndependentEndpointPivotPrimeActualDemandAfterSourceCutLedger.
```

Phase-residue exchange endpoint-pivot actual demand source cut. The previous theorem has already provided the AP exact envelope for a low carrier residue table. Therefore the actual-demand lower bound cannot be deduced by saying that the envelope is nearly saturated; that would recycle the supply estimate as a demand estimate.

If the endpoint fixed-residue demand branch is active, there is at least one source-tagged endpoint row/cell incidence. This closes the unit incidence ledger, but it is not enough for a capacity contradiction: a single low carrier AP cell already has envelope large enough to accommodate one hit whenever it intersects the endpoint window. Hence the remaining demand must come from source-side amplification along the endpoint orbit, not from a formal envelope count.

The source-side amplification must also be injected non-cyclically into the same low carrier AP table. If the source mass cannot amplify, the branch is accounted for by endpoint singleton, full-cycle mean, source-multiplicity, or sparse SAE exits. If amplified pressure cannot be injected into the same AP table, it returns to dense-table PDEC, sparse-cell SAE, high-carrier rank deficit, or moving-pivot exits.

This theorem does not prove source-side amplification, low-carrier payment injection, AP strict gap, dense-table exclusion, sparse-cell summability, or the other endpoint exits. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.260 (Phase-residue exchange endpoint-pivot release-mass support ladder). *The endpoint pivot release-mass amplification branch is reduced to a finite source-support ladder:*

```
EndpointOrbitAlternatingCyclePivotPrimeReleaseMassAmplificationOrEndpointSingletonSAE
⇒ StableLadderEndpointOrbitPivotPrimeReleaseMassImportedForSupportLadderLedger
AND StableLadderEndpointOrbitPivotPrimeUnitReleaseNotAmplificationLedger
AND StableLadderEndpointOrbitPivotPrimeReleaseSourceFiniteSupportLedger
AND StableLadderEndpointOrbitPivotPrimeSourceSupportMassLedger
AND StableLadderEndpointOrbitPivotPrimeSourceSupportThresholdDichotomyLedger
AND StableLadderEndpointOrbitPivotPrimeBoundedSourceSupportRouteLedger
AND StableLadderEndpointOrbitPivotPrimeLongSourceSupportTransferRouteLedger
AND NoIndependentEndpointPivotPrimeReleaseMassAmplificationAfterSupportLadderLedger.
```

Phase-residue exchange endpoint-pivot release-mass support ladder. The preceding source-cut theorem shows that a single source-tagged endpoint incidence is only a unit demand and cannot by itself create an AP-envelope gap. Thus amplification must come from the number of source-tagged unit atoms in the endpoint signed-depth/flux skeleton.

That skeleton has finite support and unit-atom expansion. Let M be the number of active source-tagged unit atoms in the relevant endpoint packet. For any threshold A, either $M < A$ or $M \geq A$. In the bounded-support case there is no comparable AP demand mass; the branch must be charged to endpoint singleton, full-cycle mean, source-multiplicity, or sparse SAE exits. In the long-support case, the only remaining amplification route is to transfer that source support non-cyclically into the same low-carrier AP table. Failure of this transfer is a named dense-table, sparse-cell, high-carrier, moving-pivot, or endpoint exit, not an anonymous loss.

This theorem does not prove bounded-support SAE, long-support mass transfer, payment injection, AP strict gap, dense-table exclusion, sparse-cell summability, or the other endpoint exits. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.261 (Phase-residue exchange endpoint-pivot low-carrier payment-injection lock). *The endpoint pivot low-carrier payment-injection branch is reduced to a same-AP-table canonical key lock and two named failure exits:*

```
EndpointOrbitAlternatingCyclePivotPrimeLowCarrierActualPaymentInjectionWithoutEnvelopeReuse
⇒ StableLadderEndpointOrbitPivotPrimeLowCarrierPaymentInjectionImportedForLockLedger
AND StableLadderEndpointOrbitPivotPrimePaymentNoEnvelopeReuseCarriedForwardLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierSameAPTableKeyTupleLedger
AND StableLadderEndpointOrbitPivotPrimeLowCarrierActualPaymentDomainLedger
AND StableLadderEndpointOrbitPivotPrimeCanonicalAssignmentIncidenceImportedLedger
AND StableLadderEndpointOrbitPivotPrimeNoHiddenCrossKeyPaymentImportedLedger
AND StableLadderEndpointOrbitPivotPrimeNoLossReturnAccountingImportedLedger
AND StableLadderEndpointOrbitPivotPrimeActualPaymentStitchingDisciplineImportedLedger
AND StableLadderEndpointOrbitPivotPrimeSameAPTablePaymentInjectionLockLedger
AND StableLadderEndpointOrbitPivotPrimeCrossTableSwitchRouteLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicatePaymentCollisionRouteLedger
AND NoIndependentEndpointPivotPrimeLowCarrierPaymentInjectionAfterLockLedger.
```

Phase-residue exchange endpoint-pivot low-carrier payment-injection lock. The actual-demand source cut has already forbidden AP-envelope reuse as a demand source, and the low-carrier AP-envelope theorem has fixed the table fields. A legal endpoint payment unit must therefore be an actual source-support unit attached to the same endpoint packet, the same pivot prime q, the same residue a, the same row class $t \equiv -aP^{-1} \pmod q$, and the same support window H. These fields define the same-AP-table canonical payment key.

The imported canonical assignment and payment-conservation ledgers rule out two anonymous escapes. If a claimed payment changes the pivot prime, residue, row AP class, support window, endpoint packet, or source atom, then it is no longer a payment into the same low-carrier AP table; it is a cross-table switch, hence a PDEC/moving-pivot return. If several source units claim the same already paid canonical slot, the extra count is not new capacity; it is a duplicate-payment collision or must be charged to sparse SAE accounting. The no-loss return accounting keeps every unpaid obligation as a named return.

This theorem does not prove that same-AP-table payment injection exists or has sufficient mass. It also does not exclude the cross-table switch, duplicate collision, bounded-support, long-support transfer, AP strict-gap, dense-table, sparse-cell, high-rank, nonreplay, moving-pivot, or endpoint parallel exits. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization
```

□

Theorem 7.262 (Phase-residue exchange endpoint-pivot duplicate-payment slot accounting)**.** *The endpoint pivot duplicate-payment branch is reduced to same-slot source multiplicity or duplicate sparse accounting:*

```
EndpointOrbitAlternatingCyclePivotPrimeDuplicatePaymentCollisionOrSparseSAE
⇒ StableLadderEndpointOrbitPivotPrimeDuplicatePaymentImportedForSlotAccountingLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateSameCanonicalSlotKeyLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateCapacityUnitValueImportedLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateAssignmentPartialInjectionImportedLedger
AND StableLadderEndpointOrbitPivotPrimeActualObjectDuplicateCollisionImportedLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateCountNotNewCapacityLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateSameSlotMultiplicityRouteLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateSparseRouteLedger
AND NoIndependentEndpointPivotPrimeDuplicatePaymentCollisionAfterSlotAccountingLedger.
```

Phase-residue exchange endpoint-pivot duplicate-payment slot accounting. The previous payment-injection lock has fixed a canonical same-AP-table slot. The capacity-unit value ledger says that an admitted slot has value one, and the canonical assignment ledger gives only a partial injection into the demand units. Therefore the second and later source units mapped to the same canonical slot do not create additional AP-table capacity.

If such extra source units persist in the counterexample chain, they are not an anonymous payment supply; they are a same-slot source-multiplicity defect and must enter the corresponding multiplicity cap/PDEC outlet. If they do not persist, their contribution is a duplicate sparse family and must be charged to SAE summability. The actual-object duplicate-collision ledger supplies the object-level guard: field duplication or object mismatch is already a named collision/whitelist return.

This theorem does not prove the same-slot source-multiplicity cap/PDEC outlet, and it does not prove duplicate sparse SAE summability. It also does not exclude same-AP-table injection, cross-table switch, bounded or long support, strict AP gap, dense-table, sparse-cell, high-rank, nonreplay, moving-pivot, or endpoint parallel exits. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phase-residue-exchange-active-facet-normal-cone-ray-coordinate-scalar-load-unit-normalization`

□

Theorem 7.263 (Phase-residue exchange endpoint-pivot duplicate same-slot multiplicity cap import)**.** *The endpoint pivot duplicate same-slot source-multiplicity outlet is not a new endpoint-specific capacity outlet. It imports into the existing source-atom multiplicity cap:*

```
EndpointOrbitAlternatingCyclePivotPrimeDuplicateSameSlotSourceMultiplicityCapPDEC
⇒ StableLadderEndpointOrbitPivotPrimeDuplicateSameSlotMultiplicityImportedLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateSameSlotKeyCarriesSourceAtomLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateExtraUnitSameSourceAtomLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateMultiplicityImportedToExistingSourceAtomCapLedger
AND StableLadderEndpointOrbitPivotPrimeDuplicateMultiplicityNoNewPaymentCapacityLedger
AND NoIndependentEndpointPivotPrimeDuplicateSameSlotMultiplicityAfterCapImportLedger
AND EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityCapPDECCap.
```

Endpoint-pivot duplicate same-slot multiplicity cap import. The previous theorem fixed the canonical AP-table slot. Its key includes the source-support atom. Thus any later source unit on that slot is the same source atom, not a new anonymous payment source. Since the slot capacity is one, the extra unit is not new capacity.

The only persistent interpretation is therefore multiplicity of the same source-support atom at the same slot. This is exactly the previously registered source-atom multiplicity cap outlet, so the endpoint-specific duplicate-same-slot label is removed and replaced by the source-atom multiplicity cap named in the display above.

This theorem does not prove the source-atom multiplicity cap/PDEC outlet, and it does not prove duplicate sparse SAE summability. It also leaves open same-AP-table injection, cross-table switch, bounded or long support, strict AP gap, dense-table, sparse-cell, high-rank, nonreplay, moving-pivot, endpoint singleton, full-cycle mean, and sparse-scale SAE exits. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-endpoint-pivot-duplicate-same-slot-multiplicity-cap-import-router.md.`

□

Theorem 7.264 (Source-atom multiplicity cap ExactUV fixed-key bridge)**.** *The source-atom multiplicity cap is reduced to the existing ExactUV source-rank atoms:*

```
EndpointOrbitResidueShadowDualRowPhaseCellAtomSignedCoreSupportSliceResidueFiberPhaseWordSlotSourceAtomMultiplicityCapPDECCap
⇒ StableLadderEndpointOrbitSourceAtomMultiplicityCapImportedForExactUVFixedKeyBridgeLedger
AND StableLadderEndpointOrbitSourceAtomMultiplicitySameSourceKeyFieldLedger
AND StableLadderEndpointOrbitSourceAtomMultiplicityActualEmitterSourceTableGateLedger
AND StableLadderEndpointOrbitSourceAtomMultiplicityCompleteKeyPartitionGateLedger
AND StableLadderEndpointOrbitSourceAtomMultiplicityFixedKeyLocalMultiplicityGateLedger
AND NoIndependentSourceAtomMultiplicityCapAfterExactUVFixedKeyBridgeLedger
AND ActualNoncanonicalPrimitiveEmitterSourceTableLedger
AND CompletePrimitiveEmitterKeyPartitionLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger.
```

Source-atom multiplicity cap ExactUV fixed-key bridge. The multiplicity-fiber certificate registered this cap as excessive local multiplicity of one source atom. The preceding endpoint duplicate theorem imported same-slot duplicate source units into that cap, so it is now a shared source-rank obstruction, not an endpoint-local capacity supply.

The ExactUV source-rank synchronization has already recorded the deterministic implication: an actual primitive emitter source table, a complete key partition, and fixed-key exact-UV local multiplicity $O(1)$ together rule out fixed-key source collapse. Therefore a persistent source-atom multiplicity cap must be paid by those three source-rank atoms. If any of the three is missing, the failure is a named source-table, key-partition, or fixed-key multiplicity gate.

This theorem does not prove the actual emitter source table, the complete key partition, or fixed-key local multiplicity $O(1)$. It also leaves open duplicate sparse SAE, payment injection, cross-table switch, bounded or long support, strict AP gap, dense-table, sparse-cell, high-rank, nonreplay, moving-pivot, singleton, full-cycle mean, and sparse-scale SAE exits. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-source-atom-multiplicity-cap-exactuv-fixed-key-bridge-router.md.`

$\square$

Theorem 7.265 (LPF ownership sieve source declaration)**.** *For every $N \geq 2$, the ascending least-prime-factor sieve gives an exact ownership partition of the composites up to N. If $p \leq \sqrt{N}$ is prime, the new objects owned by the p-layer are exactly*

$$\{pm \leq N : \ m \geq p, \ \textit{and } m \textit{ has no prime factor } < p\}.$$

Consequently

$$\pi(N) = N - 1 - \sum_{p \leq \sqrt{N}} \#\{n \leq N : \ n \textit{ is composite and } \operatorname{lpf}(n) = p\}.$$

Within the strict source-declaration chain this closes only the unsigned ownership field of the pre-Cauchy declaration line:

```
PreCauchyConstructorDeclarationLineForActualNoncanonicalEmitter
⇒ LeastPrimeFactorOwnershipUnsignedSourceDeclarationLedger
AND ExplicitAlphaDeltaPrimitiveConstructorRuleForActualNoncanonicalEmitter
AND ConstructorDomainCleanCoreMembershipLedger
AND ConstructorFormulaEmitsUVKeySignLocalFactorRowsLedger
AND ConstructorFormulaFailureReturnTagsLedger
AND SourceDeclarationNoDownstreamRecoveryAndNamedReturnLedger.
```

LPF ownership sieve source declaration. Every composite $n \leq N$ has a unique least prime factor $p \leq \sqrt{n} \leq \sqrt{N}$. Writing $n = pm$, one has $m \geq p$, and no prime below p divides m. Conversely, if $m \geq p$, $pm \leq N$, and no smaller prime divides m, then the least prime factor of pm is exactly p. Thus the sets displayed above are disjoint and their union is precisely the composite set in $[2, N]$. Removing those composites from the $N - 1$ integers $2, \ldots, N$ gives the displayed formula for $\pi(N)$.

This is a pre-Cauchy unsigned ownership declaration: it uses only divisibility and the ascending sieve order. It does not emit a signed α/δ primitive coefficient, an orientation, a local factor, a complete key, or a fixed-key ExactUV multiplicity bound. Those remain the next hard fields, starting with `ExplicitAlphaDeltaPrimitiveConstructorRuleForActualNoncanonicalEmitter`. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-lpf-ownership-sieve-source-declaration-router.md.`

$\square$

Theorem 7.266 (LPF candidate-row map for the alpha-side rule)**.** *The deterministic alpha-side row-emission map has a non-posthoc candidate-row part supplied by least-prime-factor ownership:*

```
DeterministicAlphaPrimitiveRowEmissionMapLedger
⇒ LPFOwnershipAlphaCandidateRowEmissionMapLedger
AND ActualNoncanonicalPrimitiveSummandSignedWeightExpressionBeforePushforward.
```

LPF candidate-row map for the alpha-side rule. The previous LPF ownership theorem assigns every composite candidate to a unique least-prime-factor layer p and cofactor m. Hence the source of a candidate row is no longer a downstream choice from the payment graph: it is indexed before Cauchy by the pair (p, m). The existing unsigned skeleton certificates then attach this candidate to the carry-shell, P-column anchor, phase rule, and layered-wheel coordinates.

This does not make the candidate row an actual signed alpha primitive row. For that, the proof must still give the primitive summand signed weight expression before pushforward, together with the local factor, key, sign, and failure return. Thus the new direct hard point is `ActualNoncanonicalPrimitiveSummandSignedWeightExpressionBeforePushforward`. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-lpf-candidate-row-map-alpha-rule-router.md.
```

□

Theorem 7.267 (Phi-recursive LPF ownership)**.** *Let $p_1 = 2, p_2 = 3, \ldots$ be the primes and define*

$$\Phi(x, p) = \#\{1 \le m \le x : \text{ every prime factor of } m \text{ is at least } p\},$$

with $m = 1$ included. Then

$$\Phi(x, p_k) = \Phi(x, p_{k+1}) + \Phi(\lfloor x/p_k \rfloor, p_k).$$

Consequently the number of composites newly owned by the p-layer up to N is

$$c_N(p) = \Phi(\lfloor N/p \rfloor, p) - 1,$$

and

$$\pi(N) = N - 1 - \sum_{p \le \sqrt{N}} \big(\Phi(\lfloor N/p \rfloor, p) - 1\big).$$

Phi-recursive LPF ownership. The p_k-rough integers $m \le x$ split into two disjoint classes. If $p_k \nmid m$, then all prime factors of m are at least p_{k+1}. If $p_k \mid m$, write $m = p_k m'$; then $m' \le \lfloor x/p_k \rfloor$ is again p_k-rough. This proves the displayed recursion, with terminal value $\Phi(x, p_k) = 1$ when $p_k > x$.

For the sieve layer p, a newly removed composite is exactly $pm \le N$ where m is p-rough. The rough cofactor $m = 1$ gives the prime p itself rather than a composite, so it is subtracted. If $p > \sqrt{N}$, then $\lfloor N/p \rfloor < p$, hence $\Phi(\lfloor N/p \rfloor, p) = 1$ and the layer contributes zero. Summing the disjoint least-prime-factor layers and removing them from $2, \ldots, N$ gives the formula for $\pi(N)$.

This strengthens the unsigned LPF ownership ledger and the candidate-row source capacity ledger, but it still does not emit signed coefficients, orientations, local factors, or the primitive summand expression before pushforward. The next direct hard point therefore remains

```
ActualNoncanonicalPrimitiveSummandSignedWeightExpression
BeforePushforward.
```

The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-recursive-lpf-ownership-router.md.
```

□

Theorem 7.268 (Phi-LPF support stripping for the signed kernel). *The unsigned support and capacity part of the noncircular signed coefficient kernel is already determined by the Phi-recursive least-prime-factor ledger. For each N, the primitive candidate support keys are*

$$(p,m), \qquad p \le \sqrt{N}, \quad m \le \lfloor N/p \rfloor, \quad m > 1, \quad m \text{ has no prime factor below } p.$$

The map $(p,m) \mapsto pm$ is a bijection from these keys to the composites up to N. Hence the remaining signed kernel problem is not support selection or capacity counting, but the following signed law on the already determined keys:

PhiLPFBucketSignedCoefficientLawBeforePushforward.

Phi-LPF support stripping for the signed kernel. The previous Phi-recursive LPF theorem gives

$$c_N(p) = \Phi(\lfloor N/p \rfloor, p) - 1$$

for the p-layer. The subtraction removes $m = 1$, so the remaining cofactors $m > 1$ give exactly the composite candidates owned by the least-prime-factor layer p. If a composite $n \le N$ has least prime factor p, then $n = pm$, $m \ge p$, and m is p-rough. Conversely such a pair has least prime factor p. Thus the keys are disjoint and exhaust the unsigned support.

Therefore a noncircular signed coefficient emission kernel cannot still treat row support, row capacity, or the $p > \sqrt{N}$ zero-mass cutoff as open signed data. What remains is the genuinely signed field: for every Phi-LPF support key one must give the signed coefficient, sign and local factor, alpha/delta side, branch key, exact (u,v), named return tags, and the prepushforward sum identity. This is precisely the new direct hard point displayed above. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-support-stripped-signed-kernel-router.md.`

□

Theorem 7.269 (Phi-LPF signed transport reduction). *Let p be the least-prime-factor owner bucket and let $a_p(m)$ denote the still unproved prepushforward signed coefficient attached to the Phi-LPF key (p,m). The Phi recursion lifts to the signed layer only as a support partition:*

$$\sum_{\substack{m \le x \\ m > 1 \\ m \ p\text{-rough}}} a_p(m) = \sum_{\substack{m \le x \\ m > 1 \\ m \ p^+\text{-rough}}} a_p(m) + \sum_{\substack{m' \le \lfloor x/p \rfloor \\ m' \ p\text{-rough}}} a_p(pm'),$$

where p^+ is the next prime after p. Hence Phi-LPF closes the signed sum domain and the finite partition, but a noncircular proof still needs a forward law for how the signed coefficient, orientation, local factor, α/δ side, branch key and return tag change under the rough cofactor multiplication $m' \mapsto pm'$. The next direct hard point is

*PhiLPFRoughCofactorMultiplicationSignedTransportLaw
BeforePushforward.*

Phi-LPF signed transport reduction. The unsigned Phi recursion partitions the p-rough cofactors into those not divisible by p, which are p^+-rough, and those of the form pm' with m' again p-rough. Multiplying an arbitrary already-defined coefficient function a_p by the characteristic functions of these two disjoint sets gives the displayed finite signed-sum identity.

This identity is only a partition identity. It does not determine the value of $a_p(pm')$, nor does it determine the orientation parity, local-factor multiplier, branch transition, exact (u,v), nonzero condition, or named return. Therefore the earlier hard point

`PhiLPFBucketSignedCoefficientLawBeforePushforward`

is sharpened to either the rough-cofactor signed transport law above or an equivalent pointwise signed value table on the Phi-LPF support. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-bucket-signed-transport-router.md.`

□

Theorem 7.270 (Phi-LPF signed transport unit seed). *The rough-cofactor signed transport law cannot be started from the Phi count alone. In the unsigned formula*

$$c_N(p) = \Phi(\lfloor N/p \rfloor, p) - 1,$$

the subtracted unit cofactor $m = 1$ is the prime row p, not a composite support key. But in the signed transport split, the divisible branch contains the preimage $m' = 1$, which maps to the genuine composite key (p, p), namely p^2. Thus any recursive signed transport proof must first provide either a virtual unit seed or the square-base signed coefficient for (p, p), together with a guard forbidding prime-row leakage. The next direct hard point is

`PhiLPFUnitCofactorVirtualSeedOrSquareBaseSignedCoefficient`
`BeforePushforward.`

Phi-LPF signed transport unit seed. The Phi-LPF support ledger excludes $m = 1$ because $p \cdot 1 = p$ is prime. For every $p \leq \sqrt{N}$, however, $p \cdot p = p^2 \leq N$, and p is p-rough. Hence the unit preimage in the transport recursion is not a support row itself, but it is exactly the predecessor of the first square-base support key (p, p).

Consequently a transport rule of the form $a_p(qm)$ cannot bootstrap from the unsigned Phi count. It must declare the square-base signed coefficient, orientation, local factor, nonzero condition and return tag before any later rough-prime step update can be meaningful. Even after this seed is supplied, one still needs the step local-factor update law and the ordered-factorization coherence law, unless one instead submits the full pointwise signed value table on Phi-LPF support. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-signed-transport-unit-seed-router.md.`

□

Theorem 7.271 (Phi-LPF square-base diagonal source). *In the Phi-LPF signed transport seed interface, the virtual key $(p, 1)$ is not a composite support row. It is the prime row p removed by the -1 in $c_N(p) = \Phi(\lfloor N/p \rfloor, p) - 1$. Moreover, for an owner bucket p, no integer m with $1 < m < p$ is p-rough. Hence the first genuine composite support key is the square-base diagonal root (p, p). The virtual unit can only be a predecessor of this root, not an independent signed source. The next direct hard point is therefore*

`PhiLPFSquareBaseDiagonalRootSignedSourceDeclaration`
`BeforePushforward.`

Phi-LPF square-base diagonal source. If $1 < m < p$, then the least prime factor of m is less than p, so m is not p-rough. Thus the composite support of the LPF owner bucket p has no real cofactor below p. The key (p, p) is present whenever $p \leq \sqrt{N}$, since $p^2 \leq N$ and p has no prime factor below p. Therefore the only predecessor below this support root is the virtual unit $(p, 1)$, which refers to the prime row and cannot carry a composite signed coefficient.

Consequently the earlier seed alternative is sharpened: a proof must provide, before pushforward, the source tuple, basis word, signed coefficient, orientation/local factor, alpha/delta branch, exact (u, v), prime-row leak guard and return tag for the square-base root (p, p). This theorem does not prove that declaration, nor the rough-cofactor step update or ordered factorization coherence. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-square-base-diagonal-source-router.md.`

□

Theorem 7.272 (Phi-LPF square-base source packet reduction)**.** *The square-base diagonal root does not create a private signed source lane. After the LPF root key (p, p) and the prime-row leak guard for $(p, 1)$ are fixed, every remaining field needed for a signed source declaration is a field of the common pre-Cauchy actual source declaration packet or one of its known downstream lanes. Thus the square-base-specific hard point is reduced to*

`PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket`

together with the built-in signed pairing and ExactUV entropy/fiber subfields.

Phi-LPF square-base source packet reduction. The previous theorem supplies only the domain facts: the key is the first genuine composite support key in the LPF owner bucket, and the virtual unit is not a support row. These facts determine the support address and forbid prime-row leakage, but they do not emit a source tuple, primitive basis word, signed coefficient, orientation/local factor, alpha/delta branch, exact (u, v), prepushforward identity, or return tag.

Those remaining fields are exactly the fields demanded by the common pre-Cauchy actual source declaration packet: declaration rows, primitive rows, basis-word/coefficient identity, prepushforward equality, ExactUV entropy/fiber control, no downstream recovery and named returns. Hence square-base geometry removes no signed obligation by itself. The current corpus still lacks the common packet, the built-in signed coefficient pairing closed form, and the ExactUV entropy/fiber pair; the row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-square-base-source-packet-reduction-router.md.`

□

Theorem 7.273 (Phi-LPF source packet cycle guard sync)**.** *Once the Phi-LPF square-base route has reduced its remaining source obligation to the common pre-Cauchy packet, that packet cannot be used as a self-contained non-circular proof. The existing signed lane forms the cycle*

$$\begin{gathered}\mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}\\ \to \mathtt{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows}\\ \to \mathtt{branch\ trace} \to \mathtt{signed\ payload} \to \mathtt{origin\ identity}\\ \to \mathtt{PreCauchyActualNoncanonicalEmitterSourceDeclarationPacket}.\end{gathered}$$

Phi-LPF support, capacity and square-root facts do not by themselves create a primitive signed payload or trace formula. The existing post-antisplit convergence certificates then absorb the old new-primitive and terminal exits into the source-rank/no-collapse package and the pointwise primitive kernel table.

Phi-LPF source packet cycle guard sync. The preceding Phi-LPF certificates fix the unsigned ownership and the first composite support key. They also rule out prime-row leakage from $(p, 1)$. These facts are domain and support facts. They contain no orientation parity, local factor, signed coefficient, branch trace payload, or prepushforward signed identity.

Therefore the immediate common-packet self-proof is blocked, and the already registered downstream convergence moves the current non-circular priority to the pointwise kernel atoms: the alpha row anchor/phase emission formula, independent pre-Cauchy arithmetic identity, and same-unit ExactUV rank/multiplicity certificate. A same-set PDEC scope match, a pointwise Phi-LPF signed value table, ExactUV entropy/fiber and rough-cofactor transport coherence remain independent obligations. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-source-packet-cycle-guard-sync-router.md.`

□

Theorem 7.274 (Phi-LPF pointwise signed value table frontier)**.** *After the Phi-LPF support and capacity reductions, the remaining direct Phi-LPF signed route is not a support-counting problem. It is exactly the need to submit a pointwise signed value table on the already fixed Phi-LPF bucket support:*

`PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupport`
`BeforePushforward.`

This table must be a new prepushforward signed artifact. If one tries to derive it by expanding the existing signed-source/source-origin chain, the chain returns to the registered signed-source fixed point. Thus the retained non-circular alternatives are seed cycle-cut, same-set PDEC, or a new explicit joint constructor formula, together with the independent ExactUV, rough cofactor coherence, rate/model and D-structure gates.

Phi-LPF pointwise signed value table frontier. The Phi-LPF certificates have already fixed the unsigned bucket key (p, m), the bucket capacity, the prime-row leak guard and the square-base root. These are domain facts. They do not assign orientation, local factor, signed coefficient, alpha/delta side, exact (u, v), or a prepushforward signed sum identity.

The rough-cofactor transport certificates show that a recursive signed law would still need a signed transport update and ordered factorization coherence. The square-base packet reduction shows that the base case has no private signed lane, and the cycle guard shows that the common packet cannot self-prove without returning to the signed-source cycle. Therefore the only direct Phi-LPF bypass is an actual pointwise signed table on the fixed support. Expanding that table through the current source-origin machinery again reaches the registered fixed point, so it must be supplied as a genuine new artifact or replaced by one of the controlled non-circular exits. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-pointwise-signed-value-table-frontier-router.md.`

□

Theorem 7.275 (Phi-LPF rough cofactor ordered factorization coherence)**.** *The ordered factorization coherence part of the Phi-LPF rough-cofactor transport obligation is a support-level consequence of least-prime-factor peeling. Fix an owner prime p. Every p-rough cofactor m in the Phi-LPF support has a unique nondecreasing prime word*

$$m = q_1 q_2 \cdots q_t, \qquad p \le q_1 \le \cdots \le q_t,$$

obtained by repeatedly removing the least prime factor. Each prefix remains p-rough and satisfies $pq_1 \cdots q_j \le pm \le N$. *Hence the recursive rough-cofactor path stays in the same owner bucket and has no permutation ambiguity.*

Phi-LPF rough cofactor ordered factorization coherence. Existence and uniqueness of the nondecreasing word are exactly the fundamental theorem of arithmetic written in least-prime-factor order. Since m is p-rough, no factor q_j is below p. Products of factors all at least p are again p-rough, and every prefix product is bounded by the final cofactor m, so its composite image remains inside the same Phi-LPF owner bucket. This closes the ordered path/coherence field.

This argument is intentionally unsigned. It does not assign a signed coefficient, orientation parity, local factor multiplier, alpha/delta side, or ExactUV value. Therefore the latest direct remaining transport input is the step local-factor update law, with the pointwise Phi-LPF signed value table and the non-circular seed/PDEC/new-joint exits still open. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-rough-cofactor-ordered-factorization-coherence-router.md.`

□

Theorem 7.276 (Phi-LPF step local-factor update frontier)**.** *After the Phi-LPF ordered cofactor path has been fixed, the remaining step-update obligation is no longer an ordering or support problem. It is equivalent to a pre-pushforward signed multiplier table on the ordered edges*

$$(p, \mathrm{prefix}, q, \mathrm{prefix} \cdot q)$$

inside the same owner bucket. Each edge must supply the signed local-factor/truncation multiplier, the orientation-parity increment, the alpha/delta branch update, the ExactUV transition, a nonzero-or-return tag, and the path-product identity giving the signed coefficient of the final Phi-LPF support key.

Consequently the current direct frontier is

$$\mathrm{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}.$$

The pointwise Phi-LPF signed value table remains a parallel direct input, and complete branch traces or primitive signed-origin identities remain conditional generators. This is a reduction of the step-update interface, not an unconditional row/column proof.

Phi-LPF step local-factor update frontier. The preceding theorem gives a unique least-prime-factor word for every p-rough cofactor and proves that every prefix stays in the same owner bucket. Once that word is fixed, any recursive signed transport law can only change from one prefix to the next by multiplying by an edge-local signed factor and by updating the accompanying orientation, branch, and ExactUV data. Thus a complete step-update law is the same data as such an edge table together with the identity that the ordered product of its edge multipliers gives the signed coefficient before pushforward.

The Phi-LPF count and the least-prime-factor path do not determine this signed table. In particular, the first edge with unit prefix cannot be recovered from the prime row removed by the Phi formula, and downstream payment or terminal capacity data cannot be used to reconstruct a pre-pushforward multiplier. The router therefore records the precise new frontier and keeps the row/column theorem open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-step-local-factor-update-frontier-router.md.`

□

Theorem 7.277 (Phi-LPF first-edge slab frontier)**.** *The pre-pushforward edge multiplier table splits into a first-edge seed slab and an internal transition slab. For an owner prime p, if the least-prime factor word of a p-rough cofactor begins with q, then the first edge is*

$$(p, 1, q, q),$$

and the remaining tail is q-rough. Hence the number of support keys whose first edge is (p, q) is exactly

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

This closes the unsigned first-edge fiber ledger. It does not give the semiprime first-edge signed seed, nor does it give the signed local transition for later edges with non-unit prefix. Thus the current edge-table frontier is the pair consisting of a semiprime first-edge signed seed table and an internal prime-adjoin signed transition law.

Phi-LPF first-edge slab frontier. The ordered LPF word writes every support cofactor as $m = qr$, where q is the least prime factor of m, $q \geq p$, and all prime factors of r are at least q. Conversely any q-rough $r \leq N/(pq)$ gives a unique support key pqr whose first edge is (p, q). This proves the stated Phi fiber formula and gives a disjoint first-edge slab decomposition.

The formula is purely unsigned. The signed seed attached to the semiprime edge cannot be read from the removed prime row, and an internal signed transition cannot be obtained by dividing unknown pointwise coefficients, because the predecessor may be zero or may have to return through a named exit. The row/column theorem therefore remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-first-edge-slab-frontier-router.md.`

□

Theorem 7.278 (Phi-LPF semiprime seed diagonal frontier)**.** *The semiprime first-edge seed table has a forced diagonal/offdiagonal split. The diagonal first edges are exactly (p, p), hence the square-base roots already handled by the square-base source-packet reduction. They do not form a private signed source. The offdiagonal first edges (p, q), $p < q$, remain the genuine semiprime first-seed signed table. For $N = 10000$ the certificate counts 25 diagonal seed types and 2600 offdiagonal seed types, with occurrence mass $3302 + 5468 = 8770$.*

Phi-LPF semiprime seed diagonal frontier. Every first-edge seed has ordered prime pair $p \leq q$. The equality case is $p = q$, so it is exactly the square-base key (p, p). The previous square-base packet reduction proves that this key has no separate private signed escape; it must be supplied by the common pre-Cauchy source packet and its downstream signed/ExactUV fields. The strict inequality case $p < q$ is disjoint from that diagonal packet and therefore remains as an offdiagonal ordered semiprime seed table.

This is again an unsigned support split. It does not supply the offdiagonal signed seed values, does not close the common packet, and does not close the internal prime-adjoin transition law. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-semiprime-seed-diagonal-frontier-router.md.`

□

Theorem 7.279 (Phi-LPF offdiagonal semiprime seed tuple-fields frontier)**.** *For every offdiagonal first seed (p, q) with $p < q$, the unsigned source fields are fixed before pushforward. Each occurrence is uniquely represented as*

$$(p, q, t), \qquad p < q, \qquad t \text{ } q\text{-rough},$$

and its composite value is pqt. Hence the occurrence mass for the ordered type (p, q) is exactly

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

Thus the offdiagonal seed table has no remaining LPF/Phi support ambiguity: owner prime, first rough prime, tail fiber, and diagonal exclusion are closed. For $N = 10000$ the certificate counts 2600 offdiagonal seed types and 5468 tuple occurrences, split as 2600 tail-one occurrences and 2868 nonunit-tail occurrences.

Phi-LPF offdiagonal semiprime seed tuple-fields frontier. Given an offdiagonal first edge, the owner bucket supplies p, while the least prime factor of the rough cofactor supplies the strict first prime $q > p$. Writing the remaining cofactor as t, the LPF order implies that all prime factors of t are at least q. Conversely any such q-rough $t \le N/(pq)$ gives exactly one occurrence in the offdiagonal first-edge fiber. This proves both the tuple bijection and the displayed Phi mass formula.

The closed fields are unsigned. They do not emit the pre-pushforward signed seed value, the orientation parity or branch side, the local factor, the ExactUV fixed pair, or the return tag. The next direct hard point is therefore

```
PhiLPFOffDiagonalOrderedSemiprimeSourceTupleSignedSeedFormula
BeforePushforward,
```

with the orientation, ExactUV, internal-transition, and common-packet gates remaining parallel. The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-offdiagonal-semiprime-seed-tuple-fields-router.md.
```

□

Theorem 7.280 (Phi-LPF offdiagonal pure semiprime seed atom frontier)**.** *The offdiagonal source-tuple signed-seed formula has a forced first-atom split. For every ordered type (p, q) with $p < q$, the unique pure semiprime pair atom is the tail-one occurrence*

$$(p, q, 1), \qquad \text{with value } pq.$$

All nonunit-tail occurrences are continuations of this first atom, not new first seeds. Their unsigned continuation mass for a fixed (p, q) is

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right) - 1.$$

For $N = 10000$, the certificate counts 2600 pure pair atoms and 2868 tail-lift occurrences, together recovering the 5468 offdiagonal occurrences of the previous tuple-field ledger.

Phi-LPF offdiagonal pure semiprime seed atom frontier. In the tuple-field ledger, an occurrence is (p, q, t) with $p < q$ and t q-rough. The first seed is attached to the first edge from the unit cofactor to q. Hence the seed itself is represented by the unique tail-one tuple $(p, q, 1)$. If $t > 1$, the same first seed has already been chosen, and the remaining q-rough tail can only be accounted for by subsequent prime-adjoin transitions. Subtracting the tail-one tuple from the Phi fiber gives the displayed $\Phi - 1$ continuation mass.

This proves only the unsigned atom/tail split. It does not provide the signed value of the pure pair atom, nor the orientation parity, ExactUV return, or internal transition law. The next direct hard point is therefore

```
PhiLPFOffDiagonalPureSemiprimePairSignedSeedAtom
BeforePushforward.
```

The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-offdiagonal-pure-semiprime-seed-atom-router.md.
```

□

Theorem 7.281 (Phi-LPF pure pair Ferrers support frontier)**.** *The unsigned support of the off-diagonal pure pair atoms is a Ferrers bipartite prime graph. Its left vertices are the owner primes $p \le \sqrt{N}$, its right vertices are primes q, and*

$$(p, q) \text{ is an edge} \quad \Longleftrightarrow \quad p < q \le N/p.$$

If $p < r$, then the r-neighbourhood is contained in the p-neighbourhood. Consequently the edge count and all left and right degrees are fixed by the prime table and the values $\lfloor N/p \rfloor$. For $N = 10000$, the certificate has 25 left owner layers, 668 right prime vertices, and 2600 edges, with the Ferrers nesting and degree identities verified.

Phi-LPF pure pair Ferrers support frontier. The pure pair atom has value pq, with p the LPF owner and q the strictly larger first rough prime. The condition $pq \le N$ is exactly $q \le N/p$, so the displayed edge rule is equivalent to the previous pure atom ledger. If $p < r$, then any q adjacent to r satisfies $q > r > p$ and $q \le N/r < N/p$, hence it is adjacent to p. This proves the Ferrers nesting. Summing the neighbour sizes over p, or the right degrees over q, gives the same edge count.

This closes only support and degree accounting. A Ferrers support graph does not emit signed seed values, local factors, orientation parity, or ExactUV return tags. The remaining direct hard point is

```
PhiLPFOffDiagonalTwoPrimeInteractionSignedKernel
BeforePushforward.
```

The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-pure-pair-ferrers-support-router.md.
```

□

Theorem 7.282 (Phi-LPF two-prime ordered no-swap frontier)**.** *The two-prime signed interaction kernel cannot be obtained from the product symmetry $pq = qp$. In the LPF owner source domain, a distinct offdiagonal semiprime product has exactly one canonical edge (p, q) with $p < q$. The reverse ordered edge (q, p) is not in the same source domain, because q is not the least prime factor of pq. For $N = 10000$, the certificate has 2600 ordered edges, 2600 distinct unordered semiprime products, no reverse edges, and no duplicate products.*

Phi-LPF two-prime ordered no-swap frontier. Let $n = pq$ with $p < q$ prime. Its least prime factor is p, so the LPF owner bucket records the source edge as (p, q). The reversed pair (q, p) would place the same integer in owner bucket q, but $p < q$ divides n, so q is not the least prime factor. Thus the reversed edge is not an admissible pre-Cauchy source row. Product commutativity only identifies the integer value; it does not provide a second signed source that could define or cancel the first one.

This removes the swap-symmetry exit from the two-prime kernel. The proof still must give an edge-local signed interaction formula, or a named return, for the canonical edge itself. The next direct hard point is

```
PhiLPFEdgeLocalTwoPrimeSignedInteractionFormulaOrReturn
BeforePushforward.
```

The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-two-prime-ordered-no-swap-router.md.
```

□

Theorem 7.283 (Phi-LPF edge-local two-prime field-cut frontier)**.** *The edge-local two-prime formula-or-return has no remaining unsigned support ambiguity. For every canonical edge (p, q), $p < q$, the LPF/Ferrers ledger fixes the owner prime p, the second prime q, the product pq, the LPF bucket, the row and column rank-degree data, and the fact that this product has exactly one pure-pair source atom. For $N = 10000$, the certificate has 2600 canonical edges, 2600 unique edge labels, and 2600 unique products, with the LPF, row-degree, column-degree, and atom-multiplicity audits all closed.*

Phi-LPF edge-local two-prime field-cut frontier. The no-swap theorem leaves only the canonical edge (p, q). Its product is pq, and because $p < q$, the least prime factor is p. The Ferrers support theorem gives the row rule $p < q \le N/p$, so the row degree is $\pi(\lfloor N/p \rfloor) - \pi(p)$. For a fixed right prime q, the admissible owners are the primes $p < q$ with $p \le N/q$, so the column degree is $\pi(\min(q-1, \lfloor N/q \rfloor))$. These formulae determine the closed edge label, and the product uniqueness of the LPF owner domain gives atom multiplicity one.

This label is deliberately unsigned. It contains no signed seed value, local factor, orientation parity, alpha/delta side, ExactUV fixed pair, return tag, or pre-Cauchy source row. Thus the direct remaining hard point is

```
PhiLPFEdgeLocalTwoPrimeSignedAtomFieldsOrNamedReturnTag
BeforePushforward.
```

The row/column theorem remains open. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-edge-local-two-prime-field-cut-router.md.
```

□

Theorem 7.284 (Phi-LPF edge-local signed atom trace-sync frontier)**.** *After the unsigned two-prime edge label is fixed, the remaining signed atom fields must all live on one pre-Cauchy trace key. The signed value, local factor, orientation and branch side, alpha/delta payload, exact (u, v), and source row cannot be assembled from separate downstream ledgers. If any field is missing, inconsistent, zero, over budget, or read only after pushforward, the failure is a named return or PDEC exit, not an anonymous signed gap. For $N = 10000$, the certificate records 2600 edge trace packets and 15600 open signed slots.*

Phi-LPF edge-local signed atom trace-sync frontier. The preceding field-cut fixes the unsigned edge label. A signed edge-local formula is stronger: it must attach the signed payload and all side data to the same source row before Cauchy/Phi/payment pushforward. Thus the same trace key must carry the signed coefficient, the local-factor product, the orientation and alpha/delta side, the exact (u, v) key, and the return tag. If two of these fields come from different source keys, the asserted edge-local formula has split into a named same-key defect. If a field is absent or zero, the return tag records that failure.

The existing branch-trace and atomic-trace routes are not independent proofs in the current corpus: the signed-lane cycle has already synchronized them back to the payload/origin/common-packet loop. Therefore the new direct input is a primitive atomic signed payload or trace artifact, with terminal descent, same-set PDEC scope, the pointwise signed table, ExactUV, model, rate, and D-structure gates remaining parallel. The row/column theorem remains open. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-edge-local-signed-atom-trace-sync-router.md.`

□

Theorem 7.285 (Phi-LPF new-payload alignment to the source atom). *The Phi-LPF edge-local exit*

$$\mathtt{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact}$$

is not an independent terminal proof point in the present archive. Once the edge-local signed atom fields have been synchronized to one pre-Cauchy trace key, a genuine new payload artifact must carry the actual pre-Cauchy source-rank/no-collapse package

$$\mathtt{ActualPreCauchySourceDomainAbsoluteEntropyLedger},$$
$$\mathtt{CompletePrimitiveEmitterKeyPartitionLedger},$$
$$\mathtt{FixedKeyExactUVLocalMultiplicity01Ledger}.$$

Otherwise it is a renaming inside the signed-lane cycle, or it exits through terminal descent, PDEC scope, or a pointwise Phi-LPF signed value table.

Phi-LPF new-payload source-atom alignment sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-new-payload-source-`
`atom-alignment-sync-router.md.`

The Phi-LPF/Ferrers ledger fixes only unsigned support data: owner, product, least-prime-factor bucket, tuple or fiber label, and capacity. Those fields do not determine the signed value, local factor, orientation, alpha/delta side, or return tag. The preceding trace-sync theorem therefore requires these signed fields to arrive from the same pre-Cauchy trace key.

The strict new-primitive alignment theorem has already identified the payload needed to break the signed-lane cycle: it must name the actual pre-Cauchy source object and control the source entropy, complete-key partition, and fixed-key exact-(u, v) local multiplicity. Hence the direct Phi-LPF new-payload exit is absorbed into that same source-rank package. These atoms, as well as the ExactUV, model, rate-preservation, and D-structure/Rankin gates, remain open; no unconditional row/column theorem is claimed here. □

Theorem 7.286 (Phi-LPF candidate capacity versus signed source entropy). *The Phi-recursive least-prime-factor ledger closes the unsigned candidate-row capacity part of the source-entropy problem, but it does not close actual signed source entropy. The source-support atom splits as*

$$\begin{aligned} &\mathtt{PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger}\\ &\Longrightarrow \mathtt{PhiLPFCandidateRowCapacityLowerBoundLedger}\\ \textit{and}\quad &\mathtt{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward}. \end{aligned}$$

The first ledger is paid by LPF ownership and the Φ-recursion; the second still requires a pre-pushforward signed summand expression and the same-formal-unit row-mass/no-heavy-row ledger.

Phi-LPF source entropy signed-survival frontier. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-source-entropy-
signed-survival-router.md.
```

The LPF bucket identity assigns every composite candidate to a unique least prime factor. The Φ-recursion gives the exact capacity

$$c_N(p) = \Phi(\lfloor N/p \rfloor, p) - 1$$

for the p-owner layer, and summing these capacities recovers the prime counting identity. Thus the unsigned candidate support is no longer the missing object.

However, an unsigned candidate row is not an actual signed row. It carries no signed weight, local factor, orientation, alpha/delta side, or return tag. To produce source-domain entropy, enough Phi-LPF candidate rows must survive as nonzero signed rows in the same formal unit, and their row masses must satisfy the normalization/no-heavy-row ledger. The current direct missing formula is

```
ActualNoncanonicalPrimitiveSummandSignedWeightExpressionBeforePushforward.
```

This theorem therefore closes only the candidate-capacity side of the support ledger and keeps the signed survival and row-mass ledgers open. No unconditional row/column theorem is claimed here. □

Theorem 7.287 (Phi-LPF signed survival reduces to row-level origin)**.** *The signed-survival side of the Phi-LPF source-entropy ledger does not stop at an unnamed signed expression. In the present archive the synchronization is*

$$\begin{gathered} \mathtt{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward} \\ \to \mathtt{ActualNoncanonicalPrimitiveSummandSignedWeightExpressionBeforePushforward} \\ \to \mathtt{PrimitiveSummandSignedCoefficientOriginIdentityBeforePushforward} \\ \to \mathtt{RowLevelCleanCoreOriginalCoefficientGenerationTableForActualNoncanonicalPrimitiveSummands}. \end{gathered}$$

Thus the current direct hard point is the row-level clean-core origin table, not the unsigned Phi-LPF bucket count.

Phi-LPF signed-survival origin-table sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-signed-survival-
origin-table-sync-router.md.
```

The Phi-LPF bucket identity still closes the unsigned candidate capacity and the candidate-row owner address. It does not name the signed coefficient source for any actual noncanonical primitive summand. The strict primitive summand expression router shows that a signed expression must be a pre-Cauchy signed-coefficient origin identity, and the origin-identity router then reduces that identity to the row-level clean-core original generation table.

If that row table is supplied, it simultaneously records the source tuple, signed coefficient, sign/local factor, exact (u, v) branch key, and prepushforward alpha/delta summation identity. At present the table is not proved; nonzero signed survival, row-mass/no-heavy-row, complete-key, fixed-key multiplicity, ExactUV, and D-structure/Rankin gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.288 (Row-origin table reduces to Phi-LPF bucket signed law)**.** *The row-level clean-core origin table has a further non-circular synchronization with the LPF/Phi support-stripped signed-kernel ledger. In the current archive,*

$$\begin{gathered}
\mathtt{RowLevelCleanCoreOriginalCoefficientGenerationTableForActualNoncanonicalPrimitiveSummands} \\
\to \mathtt{AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedWithSignedRowEmitterAndPrepushforwardSumIdenti} \\
\to \mathtt{NoncircularPreCauchySignedCoefficientEmissionKernelForActualNoncanonicalPrimitiveRows} \\
\to \mathtt{PhiLPFPrimitiveRowSupportAndCapacityLedger} \wedge \mathtt{PhiLPFBucketSignedCoefficientLawBeforePushforward}
\end{gathered}$$

The first factor in the last line is already an unsigned LPF/Phi support and capacity ledger. Hence the latest direct hard point is the bucket-level signed coefficient law, not another row-counting or owner-address problem.

Row-origin table Phi-LPF bucket-law sync. The certificate is recorded in

```
docs/monograph/prime-matrix-row-origin-table-phi-lpf-
bucket-law-sync-router.md.
```

The row-level table cannot be recovered from the signed-source fixed point: source identities, basis words, coefficient assignments, and row emitters refer back to each other unless an independent Cauchy-before-payment source is submitted. The fixed-point cut therefore requires a non-circular signed coefficient emission kernel.

The LPF/Phi bucket ledger removes the unsigned part of that kernel. Each candidate has a unique support key (p, m), with $p = \mathrm{LPF}(pm)$ and m p-rough, and the Phi recursion pays the corresponding capacity. What remains is to assign, before pushforward, a signed coefficient, non-zero local factor, branch key, exact (u, v), and alpha/delta summation identity to those support keys. This is exactly

```
PhiLPFBucketSignedCoefficientLawBeforePushforward.
```

That law is not proved here; row-mass/no-heavy-row, complete-key, fixed-key multiplicity, ExactUV, and D-structure/Rankin inputs also remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.289 (Phi-LPF bucket signed law enters the transport stack)**.** *The latest Phi-LPF bucket signed-law hard point is not an indivisible terminal input. If it is not replaced by a direct pointwise signed value table, the existing archive synchronizes it with the rough-cofactor signed transport stack:*

$$\begin{gathered}
\mathtt{PhiLPFBucketSigne} \\
\to \mathtt{PhiLPFRoughCofactorMultip} \\
\to \mathtt{PhiLPFUnitCofactorVirtualSeedOrSquareBaseSignedCoefficientBeforePushforward} \wedge \mathtt{PhiLPFRoughCofac}
\end{gathered}$$

The ordered-coherence factor is already a closed LPF path statement. The remaining recursive signed data are the base/source-packet atoms and the step-edge signed multiplier table.

Phi-LPF latest bucket transport-stack sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
bucket-transport-stack-sync-router.md.
```

The Phi recursion partitions the rough cofactor domain, but it does not tell how a signed coefficient, orientation, local factor, branch key, or exact UV output changes across a cofactor multiplication step. The unit seed boundary also matters: the virtual cofactor 1 is the prime row removed by the $\Phi - 1$ composite count, so it cannot be used as a composite signed source. The square-base route removes the private root escape and returns the base data to the common source-packet cycle guard.

Consequently the recursive route now requires, in the same formal unit, the base/source-packet three atoms

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows
```

and the edge table

```
PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward.
```

A direct pointwise signed table on the Phi-LPF buckets remains a parallel bypass. ExactUV, model, rate, and D-structure gates remain independent. No unconditional row/column theorem is claimed here. □

Theorem 7.290 (Phi-LPF latest edge multiplier slab sync)**.** *The latest rough-cofactor transport hard point*

```
PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward
```

is synchronized with the existing first-edge slab frontier. In the latest basis it may be replaced by

```
PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward
AND PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward.
```

The first-edge fiber formula

$$\mathrm{mass}(p, q) = \Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right)$$

pays only the unsigned q-rough continuation mass. It does not emit the semiprime signed seed, the internal local-factor transition, orientation data, alpha/delta payload, or an ExactUV return.

Phi-LPF latest edge multiplier slab sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-edge-
multiplier-slab-sync-router.md.
```

The preceding transport-stack theorem identifies the edge multiplier table as the active recursive signed edge factor. The earlier first-edge slab theorem then applies to that exact factor: each LPF ordered path begins with the unique edge $(p, 1, q, q)$, and all later edges have non-unit prefix. Thus the edge table splits into the first-seed slab and the internal transition slab.

This substitution only updates the latest frontier. The Phi fiber formula counts the first-edge support keys, but it is unsigned. It gives no signed coefficient, nonzero local factor, branch key, exact (u, v), or return tag. The base/source-packet atoms, pointwise signed-table bypass, ExactUV, model, rate, and D-structure gates therefore remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.291 (Phi-LPF latest semiprime seed diagonal sync)**.** *The latest semiprime first-edge signed seed hard point has a forced diagonal/offdiagonal synchronization:*

```
PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward
→ PhiLPFDiagonalSquareBaseFirstSeedCommonPacketSourceBeforePushforward
AND PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward.
```

The diagonal part is the square-base root (p, p). Its private signed escape has already been removed, so in the latest basis it is carried by the common source-packet atoms. The new seed-side hard point is therefore the offdiagonal ordered semiprime first-seed signed table, paired with the internal prime-adjoin transition law.

Phi-LPF latest semiprime seed diagonal sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
semiprime-seed-diagonal-sync-router.md.
```

For a first edge $(p, 1, q, q)$, LPF order gives $p \leq q$. Hence the support types split disjointly into $p = q$ and $p < q$. The equality case is precisely the diagonal square-base key already routed to the common pre-Cauchy source packet; the latest edge basis already carries that packet through the source-packet three atoms.

Consequently replacing the latest FIRST-SEED hard point leaves only the offdiagonal ordered semiprime seed table as new seed data. The replacement is unsigned at the support level: it supplies no offdiagonal signed seed value, no local factor, no orientation, no ExactUV return, and no internal transition. Those fields remain open, together with ExactUV, model, rate and D-structure gates. No unconditional row/column theorem is claimed here. □

Theorem 7.292 (Phi-LPF latest offdiagonal seed tuple sync)**.** *The latest offdiagonal ordered semiprime seed hard point has a forced tuple-field synchronization. The input*

$$\mathit{PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward}$$

is reduced to the closed unsigned tuple ledger

$$\mathit{PhiLPFOffDiagonalSemiprimeSourceTupleFieldLedgerBeforePushforward}$$

together with the still-open signed tuple formula, orientation law, ExactUV return ledger, internal prime-adjoin transition law, and the carried source atoms.

Phi-LPF latest offdiagonal seed tuple sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
offdiagonal-seed-tuple-sync-router.md.
```

The previous latest synchronization identifies the active seed input as the offdiagonal table. The earlier tuple-fields theorem applies to that exact input: each occurrence is uniquely represented as

$$(p, q, t), \qquad p < q, \qquad t \ q\text{-rough},$$

and the mass of the ordered type (p, q) is

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

Thus LPF/Phi has exhausted its unsigned contribution: owner prime, first rough prime, tail fiber, and occurrence capacity are fixed.

No signed data follows from that unsigned ledger. The remaining direct input is the source-tuple signed seed formula, together with orientation/branch-side data, ExactUV return data, the internal prime-adjoin transition, and the carried source atoms. The pointwise signed-table bypass, complete trace routes, ExactUV, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.293 (Phi-LPF latest offdiagonal pure-pair atom sync)**.** *The latest offdiagonal source-tuple signed seed formula has a forced first-atom synchronization. Its first seed atom is the pure semiprime pair*

$$(p, q, 1), \qquad p < q, \qquad \text{value } pq,$$

while every nonunit tail occurrence is a continuation lift with mass

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right) - 1.$$

Thus the latest direct seed-side hard point is the pure-pair signed seed atom, not the full q-rough tail family.

Phi-LPF latest offdiagonal pure-pair atom sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
offdiagonal-pure-pair-atom-sync-router.md.
```

The previous latest synchronization reduces the offdiagonal seed table to the source tuple signed seed formula. The pure-atom theorem applies to that input: for a fixed ordered type (p, q), the unique tuple with tail $t = 1$ is the first seed atom. If $t > 1$, the occurrence has already passed the first seed and belongs to the internal prime-adjoin continuation ledger.

This closes only the unsigned atom location and the tail-lift mass formula. It does not produce the pure-pair signed value, orientation parity, branch side, local factor, ExactUV return tag, or the internal transition law. The carried source atoms and the global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.294 (Phi-LPF latest pure-pair Ferrers support sync)**.** *The latest pure-pair signed seed atom has a forced Ferrers support synchronization. Its unsigned support is the bipartite prime graph with*

$$p \le \sqrt{N}, \qquad q \text{ prime}, \qquad p < q \le N/p.$$

The left neighborhoods are nested as p increases, and the left degrees, right degrees, and edge count are determined by the prime table and $\lfloor N/p \rfloor$. The latest signed hard point is therefore the two-prime interaction signed kernel on those edges.

Phi-LPF latest pure-pair Ferrers support sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
pure-pair-ferrers-support-sync-router.md.
```

The preceding latest synchronization identifies the active input as the pure-pair atom. The Ferrers support theorem applies to that input: the edge condition $p < q \le N/p$ is exactly the condition that the pure pair pq lies in the N-truncated offdiagonal LPF bucket. If p increases, the upper bound N/p decreases, so the right-neighborhoods are nested.

This is a support and degree ledger only. It fixes which two-prime edges are available, but does not give the signed kernel, orientation, local factor, ExactUV return tag, internal transition, or the carried source atoms. The pointwise signed-table bypass and global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.295 (Phi-LPF latest two-prime no-swap sync)**.** *The latest two-prime signed kernel has a forced LPF-owner no-swap synchronization. The canonical source edge is (p, q) with $p < q$. Although $pq = qp$ as an integer, the reverse edge (q, p) is not in the same LPF owner source domain and cannot supply a second signed source row or a cancellation partner. The latest direct hard point is therefore the edge-local two-prime signed interaction formula or a named return.*

Phi-LPF latest two-prime no-swap sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
two-prime-no-swap-sync-router.md.
```

The previous latest synchronization reduces the pure-pair atom to a signed kernel on Ferrers edges. The ordered no-swap theorem applies to that kernel: the LPF owner is the smaller prime p, so the only source-domain edge for a distinct semiprime is the canonical ordered edge (p, q). Product symmetry is erased before signed data is emitted.

This removes only the swap-symmetry pseudo-exit. It does not provide the edge-local signed formula, orientation parity, branch side, local factor, ExactUV return tag, internal transition, or the carried source atoms. The pointwise signed-table bypass and global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.296 (Phi-LPF latest edge-local field-cut sync)**.** *The latest edge-local two-prime signed interaction formula admits a forced field cut into a closed unsigned edge-label ledger and an open signed atom field table. For each canonical edge (p, q), the LPF/Phi/Ferrers data fix the owner, second prime, product pq, row and column degree labels, and multiplicity-one atom label. The remaining direct hard point is the signed atom field table or a named return tag.*

Phi-LPF latest edge-local field-cut sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
edge-local-field-cut-sync-router.md.
```

The previous latest synchronization has already reduced the two-prime kernel to the edge-local formula-or-return. The existing field-cut certificate applies to that same input. It closes only unsigned data: LPF owner p, the admissible second prime q, product pq, Ferrers rank and degree fields, and atom multiplicity one.

Unsigned LPF/Phi/Ferrers labels do not determine signed seed values, local factors, orientation parity, branch side, ExactUV return tags, internal transition compatibility, or the carried source atoms. Those fields remain open, together with the pointwise signed-table bypass and the global ExactUV/model/rate/D-structure gates. No unconditional row/column theorem is claimed here. □

Theorem 7.297 (Phi-LPF latest signed atom trace-sync)**.** *The latest edge-local signed atom field problem reduces to a same-trace-key payload problem or to named returns. Unsigned LPF/Phi/Ferrers labels give the edge domain, but a signed value, local factor, orientation side, alpha/delta side, ExactUV fixed pair, and source row must be emitted by one common pre-Cauchy trace key. After the trace self-proof cycle is cut, the productive direct hard point is a new primitive atomic signed payload or trace formula.*

Phi-LPF latest signed atom trace-sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-
signed-atom-trace-sync-router.md.
```

The latest field-cut leaves only signed atom fields or a named return tag as the direct edge-local problem. The trace-sync certificate applies to that same input: if any signed field is missing, conflicts with the edge key, has a zero local factor in the wrong place, exceeds the ExactUV budget, or is read from a different key, the case is a named return/PDEC rather than an anonymous proof gap.

The existing branch-trace and atomic-trace routes are already registered as a self-proof cycle in the current corpus, so they cannot close the signed field table by themselves. The remaining productive input is a new primitive payload/trace formula, or the controlled alternatives of terminal descent, same-set PDEC, or a pointwise signed table. The ExactUV/model/rate/D-structure gates remain required. No unconditional row/column theorem is claimed here. □

Theorem 7.298 (Phi-LPF latest new-payload source-atom alignment)**.** *The latest productive new-payload exit is not an independent terminal proof point. If*

$$NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact$$

is not merely a renaming inside the signed-lane cycle, then it must carry the actual pre-Cauchy source-rank/no-collapse package: source-domain entropy, complete emitter-key partition, and fixed-key ExactUV local multiplicity.

Phi-LPF latest new-payload source-atom alignment sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-payload-
source-atom-alignment-sync-router.md.
```

The latest trace-sync theorem leaves a new primitive payload/trace formula as the productive direct exit. The strict source-atom alignment theorem applies to that same exit: a payload which does not declare the actual source object before Cauchy/Phi/payment, and does not control source entropy, complete keys, and fixed exact-(u, v) local multiplicity, cannot break the signed-lane cycle.

Consequently the direct hard point is now `ActualPreCauchySourceDomainAbsoluteEntropyLedger`, together with the complete-key and fixed-key ExactUV multiplicity atoms, or else one of the named alternatives: terminal descent, same-set PDEC, or pointwise signed table. The ExactUV/model/rate/D-structure gates remain required. No unconditional row/column theorem is claimed here. □

Theorem 7.299 (Phi-LPF latest source-entropy downstream cycle sync)**.** *The latest source-domain entropy atom is not paid by the unsigned Phi-LPF bucket count alone. After importing the strict downstream chain, the remaining non-cyclic source-side input is either*

$$AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput$$

or a controlled terminal-descent alternative, together with the parallel complete-key, fixed-key ExactUV, PDEC, pointwise table, ExactUV/model/rate, and D-structure gates.

Phi-LPF latest source-entropy downstream cycle sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-source-
entropy-downstream-cycle-sync-router.md.
```

The preceding latest source-atom theorem leaves `ActualPreCauchySourceDomainAbsoluteEntropyLedger` as the direct source-side atom. The Phi-LPF bucket identity and LPF ownership ledger pay the unsigned candidate-row capacity, but the imported source-entropy certificate separates this capacity from nonzero signed survival, row-mass normalization, and the pre-pushforward signed coefficient expression. Hence the candidate bucket count cannot by itself become actual signed source entropy.

The strict downstream chain then sends source-domain entropy through the primitive signed coefficient law, basis-weight source, internal basis expansion, and basis alphabet. Continuing from the basis alphabet returns to the signed coordinate/source dependency cycle, and the cycle guard records that this raw cycle is not a proof. Therefore the productive non-cyclic input is the cycle-cut primitive basis/coefficient source input, or else a named terminal descent/PDEC/pointwise-table alternative with the parallel source and ExactUV gates still open. No unconditional row/column theorem is claimed here. □

Theorem 7.300 (Phi-LPF latest cycle-cut and terminal unified frontier). *The latest cycle-cut/terminal alternative is not a final non-cyclic stopping point. In the current strict corpus it reduces to the first productive joint field*

$$\mathtt{PreCauchyJointWordCoefficientEmitterDeclarationLineForActualNoncanonicalSourceTuple},$$

with the remaining joint rows, pre-pushforward identity, no-return ledger, canonical-lock, independent bridge, PDEC/external, ExactUV/model/rate, and D-structure gates still open.

Phi-LPF latest cycle-cut terminal unified sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-cyclecut-
terminal-unified-sync-router.md.
```

The previous theorem leaves the two-way alternative cycle-cut primitive basis/coefficient source input or terminal descent. The strict unified-frontier certificate applies to this same alternative: the cycle-cut branch is saturated to PDEC or a new joint formula, terminal descent is registered as a terminal-source-pair-joint-terminal macrocycle, and the same-set PDEC internal branch is not an independent non-cyclic source in the current corpus.

Thus the latest internal productive atom is the pre-Cauchy joint declaration line. That atom is only the first field of a larger joint package: rows formula, word/coefficient identity, and no-downstream-return ledger remain open, as do canonical-lock, independent source bridge, PDEC/external inputs, complete/fixed-key, ExactUV, model, rate, and D-structure gates. No unconditional row/column theorem is claimed here. □

Theorem 7.301 (Phi-LPF latest antisplit downstream sync). *The latest joint declaration line does not close by the ordinary constructor route. The non-cyclic internal route must pass through atomic antisplit rows and is reduced to*

$$\mathtt{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows}.$$

In parallel, the ExactUV incidence gate remains the pair

$$\mathtt{ActualEmitterSourceDomainEntropyLedger} \quad \text{and} \quad \mathtt{ExactUVMapFixedPairPolylogFiberBoundLedger}.$$

Phi-LPF latest antisplit downstream sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-antisplit-
downstream-sync-router.md.
```

The ordinary joint declaration line synchronizes to an explicit constructor rule, but the ordinary constructor route has already been registered as a return to the signed-source fixed point. Thus it cannot serve as a non-cyclic proof source. The strict antisplit downstream certificate instead requires an atomic declaration whose rows already contain the word/coefficient pairing and pre-pushforward identity.

For those atomic rows the remaining signed field is the built-in signed coefficient pairing closed form. This does not automatically prove the ExactUV incidence side: actual source-domain entropy and fixed exact-pair polylog fiber control remain separate inputs. Canonical-lock, independent bridge, PDEC/external inputs, complete/fixed-key, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.302 (Phi-LPF latest built-in pairing trace sync)**.** *The latest built-in signed pairing node reduces first to*

$$\mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}.$$

However, the existing signed/payload lane makes this branch-trace route a cycle back to the common source packet and then to the same built-in pairing node. Hence the current non-cyclic productive exit is a new primitive atomic signed payload/trace artifact, or a controlled terminal descent/PDEC exit, with the ExactUV source-entropy and fixed-fiber gates still open.

Phi-LPF latest built-in pairing trace sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-builtin-pairing-
trace-sync-router.md.
```

The previous theorem leaves `BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows` as the signed internal hard point. The strict built-in-pairing frontier shows that this closed form cannot be supplied by the unsigned Phi-LPF bucket capacity or by the unsigned carry-shell skeleton: the signed coefficient requires orientation and local-factor data before Cauchy/Phi/payment. Thus the intermediate target is the exact atomic branch trace.

The strict signed-lane cycle certificate then applies to that intermediate target. Continuing along the existing route gives

$$\texttt{common packet} \to \texttt{built-in pairing} \to \texttt{branch trace} \to \texttt{payload} \to \texttt{origin} \to \texttt{common packet},$$

so no node on that loop can be reused as a non-cyclic proof of the same node. After this cycle guard the productive internal exit is `NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact`, with terminal descent and same-set PDEC as named alternatives. The ExactUV side still requires actual source-domain entropy and fixed exact-pair polylog fiber control; the latter remains reduced to complete primitive emitter key partition and fixed-key local multiplicity. No unconditional row/column theorem is claimed here. □

Theorem 7.303 (Phi-LPF latest trace-exit source-rank convergence)**.** *The latest new primitive payload/trace exit is not an independent terminal closure point. In the current corpus it synchronizes to the source-rank and no-collapse package, and the internal source-rank routes meet at the pointwise primitive alpha/delta kernel table. The current direct hard point is*

$$\mathit{AlphaRowAnchorPhaseEmissionFormulaLedger},$$

with the pre-Cauchy arithmetic identity, same-unit ExactUV rank/multiplicity, PDEC, pointwise Phi-LPF signed table, ExactUV entropy/fiber, and rough-cofactor transport/coherence still open.

Phi-LPF latest trace-exit source-rank convergence sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-trace-exit-
source-rank-convergence-sync-router.md.
```

The preceding trace-sync theorem leaves a new primitive atomic signed payload/trace artifact as the productive exit after the signed-lane cycle is removed. The existing new-payload/source-atom alignment certificate applies to exactly that exit: a genuine new artifact must carry the same pre-Cauchy actual source object, source-domain entropy, complete key partition, and fixed-key exact-UV local multiplicity. Otherwise it is only a renamed node in the signed-lane cycle or a named terminal/PDEC exit.

The post-antisplit convergence and Phi-LPF source-packet guard certificates then identify the common non-posterior variable table for these source-rank and no-collapse obligations: the pointwise same-formal-unit primitive alpha/delta kernel table. That table leaves alpha row anchor/phase emission as the first direct field, with the arithmetic-identity and rank/multiplicity gates in parallel. This is a frontier synchronization only; no unconditional row/column theorem is claimed here. □

Theorem 7.304 (Phi-LPF latest alpha terminal three-atoms sync). *The latest alpha row anchor/phase frontier is not the deepest current non-cyclic frontier. The existing post-antisplit alpha terminal-leaf and post-alpha three-atoms certificates synchronize it to the terminal atoms*

$$\mathtt{AcyclicTerminalCanonicalLockToCanonicalSourceBoundary},\quad \mathtt{A1CleanBranchCanonicalSourceAdmission},$$

Among these, the current narrow direct attack is

$$\mathtt{IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore}.$$

Phi-LPF latest alpha terminal three-atoms sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-alpha-terminal-
three-atoms-sync-router.md.
```

The latest source-rank convergence theorem left the alpha row anchor/phase formula as the first direct field of the pointwise primitive kernel table. The post-antisplit alpha terminal-leaf certificate applies to the same frontier: separate attacks on the alpha, weight, and rank legs are fixed points, and the old joint-constructor route returns to the terminal leaf.

The post-alpha non-cyclic synchronization then removes the old new-joint, branch-trace, signed-payload, and signed-lane routes as self-returning dependencies. After those returns are removed, the strict internal frontier is the three terminal atoms above. The moving-atom atom is the narrowest direct next target because it asks for an exclusion mechanism for actual noncanonical clean-core moving atoms that does not pass through ExactUV, pair-mass, or terminal self-return. ExactUV incidence, rate preservation, D-structure/Rankin, PDEC, and Phi-LPF pointwise/transport gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.305 (Phi-LPF moving-atom source-bucket preimage sync). *The independent nonterminal moving-atom frontier has no anonymous unsigned source-preimage escape. If an actual noncanonical clean-core moving atom is fed by composite pre-Cauchy source support, then that support decomposes into disjoint Phi-LPF owner buckets (p, m), where $p = \mathrm{LPF}(pm)$ and m is p-rough. The $m = 1$ term is the prime-row correction in the Phi formula and is not a composite moving-atom source. The remaining direct hard point is therefore*

$$\mathtt{LPFMovingAtomSignedPreimageMassInjectionLedger},$$

with the pointwise Phi-LPF signed table, signed survival/row-mass, rate packet, model-gap, rate-preservation, and D-structure gates still open.

Phi-LPF moving-atom source-bucket preimage sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-moving-atom-source-
bucket-preimage-sync-router.md.
```

The Phi-LPF ownership certificates prove the exact finite partition

$$C(N) = \sum_{p \le \sqrt{N}} \big(\Phi(\lfloor N/p \rfloor, p) - 1\big), \qquad \pi(N) = N - 1 - C(N),$$

where the subtracted unit is precisely the prime-row term. The new audit uses $N = 10000$ as a consistency check: the bucket sum is 8770, matching the composite count, and the identity gives $\pi(10000) = 1229$.

This imports only unsigned ownership and capacity. It removes the possibility that a moving atom is supported by an unowned source object or by the virtual unit/prime-row correction. It does not create the signed coefficient, local factor, alpha/delta branch, same-formal-unit no-heavy-row normalization, or log-power rate preservation needed to turn a final same-(u, v) atom into a contradiction. Thus the moving-atom frontier is sharpened to the signed preimage mass-injection ledger rather than closed. No unconditional row/column theorem is claimed here. □

Theorem 7.306 (Phi-LPF moving-atom signed-injection split)**.** *The ledger*

$$\mathtt{LPFMovingAtomSignedPreimageMassInjectionLedger}$$

contains no further unsigned counting freedom after the Phi-LPF source-bucket preimage partition is imported. Once an exact prepushforward same-(u, v) signed-sum identity is supplied, the passage from a large final same-(u, v) atom to a positive or negative sign lane of at least half the absolute mass is finite algebra. The current direct hard point is therefore

$$\mathtt{PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward},$$

together with nonzero signed-row survival and same-formal-unit row-mass/no-heavy-row normalization.

Phi-LPF moving-atom signed-injection split. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-moving-atom-signed-
injection-split-router.md.
```

The previous theorem fixes every composite source preimage inside a unique LPF-owned bucket. Hence signed injection cannot ask for more unsigned support or capacity. It can only ask for the signed weights on those support keys and for the identity proving that their prepushforward signed sum is the target same-(u, v) contribution.

If that identity has the form $S = \sum_{e \in F} w_e$ and $|S| \geq T$, then one of the positive and negative sign lanes has absolute mass at least $T/2$. This is the only loss in this reduction and is independent of the sieve counting. The missing ingredient is therefore the pointwise Phi-LPF signed value table, or an equivalent bucket signed law, plus signed survival and row-mass normalization in the same formal unit. This is a frontier synchronization, not an unconditional row/column proof. □

Theorem 7.307 (Phi-LPF pointwise signed table to orientation law)**.** *The Phi-LPF pointwise signed value table is not the final independent signed hard point. In the existing strict signed-value route it reduces through signed alpha weights, primitive summand signed expressions, and pre-Cauchy origin identities to the row-level clean-core original generation table. The internal signed slot/value-map route then returns to that same row-level table, so it is a fixed point rather than a proof. The first non-cyclic field is*

$$\mathtt{PrimitiveOrientationLocalFactorProductLawBeforePushforward}.$$

Phi-LPF pointwise signed table orientation-law sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-pointwise-signed-table-
orientation-law-sync-router.md.
```

The latest moving-atom signed-injection split leaves the pointwise Phi-LPF signed table as the direct hard point. The pointwise-alpha table certificates then show that all Phi atoms, variation charges, and integration fields can only verify a signed weight after it has been supplied. The generating field is the signed alpha weight, which reduces to a primitive summand signed expression and then to a pre-Cauchy signed-coefficient origin identity. That identity is equivalent to a row-level clean-core original generation table.

The signed-value cycle certificates show that expanding this row-level table through signed slots, assignments, value maps, and basis-word origin identities returns to the same row-level table. Therefore the non-cyclic input must be a pre-Cauchy signed coefficient emission kernel. Its first orientation-sensitive field is the product of a primitive orientation bit, local factors, and the truncation weight, together with a prepushforward signed-sum identity and named return tags. Unsigned LPF/Phi ownership, CRT skeletons, and ExactUV support are even geometric data; they locate candidate rows but do not determine this odd signed orientation. No unconditional row/column theorem is claimed here. □

Theorem 7.308 (Phi-LPF orientation trace payload source-rank sync)**.** *The current Phi-LPF orientation hard point has a narrower non-cyclic form. A primitive orientation/local-factor law would have to be generated by an actual noncanonical primitive branch trace before Cauchy, Phi, payment, or terminal return. In the current internal corpus, however, the visible part of such a trace only supplies coordinate data, while the signed payload part returns to the row-level/source-packet cycle. Any genuinely new primitive payload or trace artifact must therefore include the source-rank/no-collapse package, and the direct attack target becomes*

$$\texttt{ActualPreCauchySourceDomainAbsoluteEntropyLedger}.$$

Phi-LPF orientation trace payload source-rank synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-orientation-trace-payload-
source-rank-sync-router.md.
```

The previous Phi-LPF step reduced the pointwise signed table to

$$\texttt{PrimitiveOrientationLocalFactorProductLawBeforePushforward}.$$

The existing orientation-to-trace router shows that this law is not an isolated Mobius or parity shadow: it must be produced by a complete actual noncanonical primitive branch trace carrying the orientation bit, local factor product, signed coefficient, exact (u, v), branch key, and return tag.

The branch-trace cycle audit then separates the trace into visible coordinate fields and signed payload fields. The coordinate fields can be followed through anchors, $D_0/K/\Omega$, phase, and word coordinates, but they do not emit the odd signed payload. Expanding that payload through the current assignment and origin maps returns to the row-level source table. Thus trace format alone is not an independent proof.

The remaining way to add a genuine primitive payload/trace artifact is to declare the same pre-Cauchy actual source object and to control exact-UV fiber collapse. Existing source-rank certificates reduce this package to source domain absolute entropy, complete primitive emitter key partition, and fixed-key exact-UV local multiplicity. LPF/Phi bucket identities have already fixed the unsigned support ownership; they do not generate signed orientation. This is a frontier synchronization only, not an unconditional row/column closure. □

Theorem 7.309 (Phi-LPF latest source-entropy to built-in pairing sync)**.** *The current source-domain entropy hard point is not an isolated stopping point. After importing the existing source-entropy downstream, cycle-cut/terminal unified, and antisplit downstream certificates, the internal signed route is reduced to*

BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows.

In parallel, the ExactUV incidence side remains

ActualEmitterSourceDomainEntropyLedger and ExactUVMapFixedPairPolylogFiberBoundLedger.

Phi-LPF latest source-entropy to built-in pairing sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-source-
entropy-to-builtin-pairing-sync-router.md.
```

The preceding orientation/trace synchronization leaves `ActualPreCauchySourceDomainAbsoluteEntropyLedger` as the direct source-side target. The imported source-entropy downstream certificate shows that expanding this target through the primitive signed coefficient law, basis source, internal basis, and basis alphabet returns to the seed coordinate/source cycle. That cycle is a guard against self-proof, not a proof of source entropy.

The cycle-cut and terminal unified certificate then applies to the remaining cycle-cut or terminal alternative: the cycle-cut branch is saturated to PDEC or a joint declaration, terminal descent is a macrocycle in the current corpus, and the same-set PDEC internal branch is not a self-contained non-cyclic closure. The antisplit downstream certificate removes the ordinary joint constructor route as another fixed point and forces the internal signed route through atomic antisplit rows. For those rows the first missing signed field is the built-in word/coefficient pairing closed form. This synchronization does not prove that closed form, the ExactUV entropy/fiber pair, complete or fixed keys, terminal/PDEC/external alternatives, model gaps, rate preservation, or D-structure acceptance. No unconditional row/column theorem is claimed here. ☐

Theorem 7.310 (Phi-LPF latest signed macrocycle and ExactUV source table sync)**.** *The latest signed route is a complete diagnostic cycle, not a non-cyclic proof. Following the current certificates sends*

source entropy $\to$ *built-in pairing* $\to$ *new payload* $\to$ *alpha* $\to$ *moving atom* $\to$ *pointwise signed table* $\to$

After this cycle is removed, the ExactUV side reduces the next internal source obligation to

PreCauchyConstructorDeclarationLineForActualNoncanonicalEmitter,

with fixed-key exact-UV local multiplicity still in parallel.

Phi-LPF latest signed macrocycle ExactUV source-table sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-signed-
macrocycle-exactuv-source-table-sync-router.md.
```

The previous theorem leaves built-in signed pairing as the signed-side hard point. The imported built-in trace certificate sends that node to a new primitive payload/trace exit, the trace-exit certificate sends that exit to the pointwise primitive kernel, the alpha terminal synchronization sends the kernel to the clean-core moving atom, and the Phi-LPF moving-atom certificates send the moving-atom obstruction to a pointwise signed table. The signed table then reduces to the orientation/local-factor law, and the orientation trace certificate returns to source-domain entropy. Hence the full signed route is a closed diagnostic cycle; no node on it can be reused as its own non-cyclic proof.

The LPF/Phi bucket identity has still done real work: it removes unsigned ownership, support, and capacity freedom, including the moving-atom source preimage. What remains is not an unsigned counting problem. The ExactUV atomization certificate splits the parallel incidence requirement into signed row law, registered complete key, and fixed-key local multiplicity. The fixed-pair fiber inequality is only formal unless the actual complete key partition and fixed-key $O(1)$ multiplicity are supplied. The complete-key certificate then reduces registered keys to an actual noncanonical primitive emitter source table, and the source-table certificate makes the first productive field the pre-Cauchy constructor declaration line. This is still an open input, as are fixed-key multiplicity, PDEC/external alternatives, model gaps, rate preservation, and D-structure acceptance. No unconditional row/column theorem is claimed here. □

Theorem 7.311 (Phi-LPF latest pre-Cauchy alpha-terminal sync)**.** *The latest pre-Cauchy declaration hard point is not a new terminal endpoint. Following the strict productive alpha-side chain sends it through constructor formula, explicit alpha/delta rule, deterministic alpha row emission, anchor/phase formula, signed lift, signed weight law, independent identity, and actual moving-block/NC-BLK. That branch reaches the global terminal gate*

$$\mathtt{PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve} \quad \text{and} \quad \mathtt{ExplicitModelGapAndFiniteDPRCLedger}.$$

The declaration itself, the delta/pairing/nonzero sibling fields, fixed-key ExactUV multiplicity, signed row law, rate preservation, and D-structure gates remain open.

Phi-LPF latest pre-Cauchy alpha-terminal sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-precauchy-
alpha-terminal-sync-router.md.
```

The previous source-table synchronization leaves `PreCauchyConstructorDeclarationLineForActualNoncanonical` The strict declaration-line certificate reduces this to the actual constructor formula line. The constructor formula line reduces to the explicit alpha/delta primitive constructor rule; its first productive branch is the alpha-side primitive rule, then the deterministic alpha row map, then the anchor/phase formula. The anchor/phase formula is still unsigned, so the next obligation is the signed coefficient lift, which reduces to the alpha signed weight law and then to an independent noncanonical pre-Cauchy arithmetic identity. The identity taxonomy shows this is not a fifth source class; it returns to actual moving-block/NC-BLK. The moving-block router sends that object to the global PDEC/CleanKLS capacity gate together with the explicit model-gap ledger.

This is only a route synchronization. It does not prove the global terminal gate, the model-gap ledger, fixed-key ExactUV multiplicity, signed row law, or the remaining alpha/delta sibling fields. The Phi-LPF bucket identity remains an unsigned ownership/support/capacity identity, not a source of signed pre-Cauchy weights. No unconditional row/column theorem is claimed here. □

Theorem 7.312 (Phi-LPF latest terminal saturation to new-joint formula)**.** *The terminal gate reached by the latest Phi-LPF productive alpha branch is itself saturated by the current internal frontier. The* `PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve` *gate splits into a same-set PDEC scope arm or a self-contained Kuznetsov/DLS arm. The Kuznetsov/DLS arm returns to the strict acyclic terminal family, and the PDEC scope arm is saturated, in the current internal corpus, to the need for a new explicit actual joint alpha/delta formula. Thus the current strict internal non-cyclic target is*

$$\mathtt{NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact}.$$

Phi-LPF latest terminal saturation to new-joint sync. The certificate is recorded in `docs/monograph/prime-matrix-phi-lpf-latest-terminal-saturation-to-new-joint-sync-router.md.`

The previous theorem leaves the PDEC/CleanKLS gate with the model-gap ledger still parallel. The terminal hardpoint split imports two internal arms: same-set PDEC scope matching and self-contained Kuznetsov/DLS. The Kuznetsov/DLS synchronization attacks the formal spectral layer down to its current end and shows that it returns to the strict acyclic terminal family rather than producing a self-contained contradiction. The PDEC scope saturation certificate shows that the same-set PDEC arm remains an admissible new scope input or conditional external line, but it is not an already proved internal non-cyclic exit. The remaining internal breaker is the new explicit joint alpha/delta formula artifact.

The existing joint-constructor route is not enough: declaration, alpha-side, same-row, row-level, and signed-emitter expansions return to the signed-source fixed point. The terminal descent alternative is also a macrocycle in the current corpus. Therefore this theorem is only a saturation synchronization; it does not supply the new formula, ExactUV, fixed-key multiplicity, signed row law, model gap, rate preservation, or D-structure acceptance. No unconditional row/column theorem is claimed here. □

Theorem 7.313 (Phi-LPF latest new-joint to antisplit atom)**.** *The latest internal new-joint target cannot be discharged by the old split constructor route. That route expands through alpha/delta, alpha-side, row-level, and signed-source data, and returns to a registered fixed point. Consequently the next strict non-cyclic target is the anti-split same-row atom*

$$\mathit{NonSplitActualJointPrimitiveWordCoefficientFormulaBeforeAlphaSideProjection}.$$

It must give, before alpha-side projection, a single primitive row carrying the basis word, signed coefficient, alpha/delta pairing payload, exact (u, v), branch key, local factor, prepushforward identity, and no-split certificate.

Phi-LPF latest new-joint to antisplit atom sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-new-joint-antisplit-atom-sync-router.md.`

The previous synchronization leaves `NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact`. The strict new-joint anti-split atom certificate shows that following the old constructor path is a non-proof cycle: it reaches the row-level/signed-source fixed point rather than producing a new actual joint formula. Hence a genuine formula must be anti-split and same-row, with the word and coefficient produced together before pushforward.

The already recorded downstream also shows the next place to drill if this atom is expanded: atomic rows with built-in signed pairing, plus the ExactUV entropy/fiber split,

`BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows` and `ActualEmitterSourceDomainEntro`

These are not proved here. The LPF/Phi bucket identity remains an unsigned ownership and capacity identity, so it does not supply signed joint coefficients. Fixed-key ExactUV, signed row law, model gap, rate preservation, and D-structure acceptance also remain open. No unconditional row/column theorem is claimed. □

Theorem 7.314 (Phi-LPF latest antisplit downstream to built-in pairing)**.** *The latest anti-split atom has a strict downstream reduction. A non-split same-row formula must first become an atomic* *pre-Cauchy rows formula with built-in word/coefficient pairing, and the first signed obstruction in that atomic rows formula is*

$$\mathit{BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows}.$$

In parallel, the ExactUV bounded-multiplicity gate is the conjunction of

$$\mathit{ActualEmitterSourceDomainEntropyLedger} \quad \text{and} \quad \mathit{ExactUVMapFixedPairPolylogFiberBoundLedger}.$$

Phi-LPF latest new-joint antisplit downstream sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
antisplit-downstream-sync-router.md.
```

The previous theorem leaves the anti-split same-row atom. The strict downstream firewall says that a merely named non-split declaration is not enough: to avoid the old split fixed point it must contain atomic rows, a word/coefficient same-source identity, prepushforward equality, and a no-return firewall. The atomic-rows certificate then localizes the remaining signed content to the built-in pairing closed form for each row.

The old signed-value/origin-table route is explicitly blocked, because it returns to the signed-source fixed point. Thus the current productive internal target is the built-in pairing closed form, with source-domain entropy and fixed-pair fiber control as the parallel ExactUV target. Complete/fixed-key budgets, signed row law, model gap, rate preservation, and D-structure acceptance remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.315 (Phi-LPF latest built-in pairing to branch-trace exit)**.** *The current latest built-in signed pairing target is the same target handled by the existing built-in-pairing trace synchronization. Hence it reduces first to*

$$\mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}.$$

The existing signed payload, origin, and common-packet continuation of that trace is a self-proof cycle, so the non-cyclic exit is a new primitive trace or payload artifact, or one of the controlled exits

$$\mathit{AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate} \quad \text{or} \quad \mathit{AcyclicSameSetScopeMatchForD}$$

Phi-LPF latest current built-in pairing to branch trace sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
builtin-pairing-trace-sync-router.md.
```

The current latest target is `BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows`. The existing built-in trace synchronization has the same input target, so it can be imported without changing the theorem. It shows that unsigned LPF/Phi bucket data and the old signed-value/origin-table chain cannot produce the orientation and local-factor odd data needed for signed coefficients. The immediate formal object is therefore the exact atomic branch trace.

That trace cannot be justified by the existing payload/origin/common-packet subline: those nodes return to built-in pairing and form the signed-lane self-proof cycle. Removing the cycle leaves a new primitive signed payload-or-trace formula, or the two controlled exits listed above. The ExactUV parallel target remains the source-entropy/fixed-fiber pair, with fixed fiber further reduced to complete-key and fixed-key multiplicity ledgers. No unconditional row/column theorem is claimed here. □

Theorem 7.316 (Phi-LPF latest new-joint trace-exit to source-rank kernel)**.** *The current new-joint trace-exit*

$$\mathtt{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact}$$

does not create a new unnamed terminal endpoint. If it is not merely a renaming of the signed-lane cycle, it must supply the actual source-rank and ExactUV no-collapse package. The existing source-atom and post-antisplit convergence certificates send that package to the pointwise primitive alpha/delta kernel table. Hence the latest direct non-cyclic target is

$$\mathtt{AlphaRowAnchorPhaseEmissionFormulaLedger},$$

with independent pre-Cauchy arithmetic identity and same-unit ExactUV rank/multiplicity still in parallel.

Phi-LPF latest new-joint trace-exit source-rank sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
trace-exit-source-rank-sync-router.md.
```

The previous theorem leaves a new primitive signed payload-or-trace formula as the productive exit after deleting the branch-trace self-proof cycle. The latest new-payload/source-atom synchronization shows that such a payload cannot be just a visible trace label: to break the cycle it must declare the same actual pre-Cauchy source object and prove source-domain entropy, complete key partition, and fixed-key exact-UV local multiplicity. The older trace-exit convergence, the post-antisplit source-rank convergence, and the source-packet cycle guard all have the same absorber: the pointwise primitive alpha/delta kernel table.

That kernel table is not yet proved. Its first productive field is the alpha row anchor and phase emission formula; in parallel it still requires the independent arithmetic identity, same-unit exact-UV rank/multiplicity, the terminal and same-set PDEC exits, the pointwise Phi-LPF signed table, ExactUV entropy/fiber control, rough-cofactor transport/coherence, model-gap control, rate preservation, and D-structure acceptance. The LPF/Phi bucket identity remains an unsigned ownership/support/capacity identity and does not produce signed payloads or alpha rows. No unconditional row/column theorem is claimed here. □

Theorem 7.317 (Phi-LPF latest new-joint alpha to terminal three atoms)**.** *The current new-joint alpha frontier is the same alpha target handled by the existing alpha-terminal synchronization. Its local unsigned branches do not produce signed source mass, and its signed branches return to the terminal capacity/model gate. The old joint and branch-trace continuations are absorbed by the signed-lane cycle. After those returns are deleted, the strict terminal frontier is the three-atom alternative*

$$\mathtt{AcyclicTerminalCanonicalLockToCanonicalSourceBoundary} \quad or \quad \mathtt{A1CleanBranchCanonicalSourceAdmission}$$

The latest direct target is the independent nonterminal moving-atom exclusion.

Phi-LPF latest new-joint alpha terminal three-atoms sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
alpha-terminal-three-atoms-sync-router.md.
```

The preceding theorem leaves `AlphaRowAnchorPhaseEmissionFormulaLedger`. The alpha local frontier certificate shows that the source-tuple, carry-shell, and phase-wheel branches are only unsigned geometric indexing, while the signed-lift and overload branches return to the global

PDEC/CleanKLS capacity gate, the model gap ledger, and D-structure acceptance. The existing latest alpha-terminal certificate has the same input target and therefore imports without a theorem switch.

Following the old joint route does not create a new proof endpoint: the new joint, exact branch trace, signed payload, origin identity, and common source packet form the signed-lane self-proof cycle already removed from the frontier. Thus the remaining strict terminal alternatives are canonical lock, A1 clean branch admission, and clean-core moving atom exclusion. The most direct actual-load pressure is now the independent nonterminal moving-atom exclusion, with canonical-lock, A1 admission, same-set PDEC, ExactUV, pointwise signed tables, rough-cofactor transport, model gap, rate preservation, and D-structure still open. No unconditional row/column theorem is claimed here. □

Theorem 7.318 (Phi-LPF latest new-joint moving atom to signed table)**.** *The current latest new-joint moving-atom target*

`IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore`

is not a new unsigned endpoint. The existing Phi-LPF moving-atom source-bucket preimage certificate removes ownerless, prime-row, and virtual-unit source escapes, and the signed-injection split removes the remaining unsigned counting freedom. Hence the latest direct non-cyclic target is

`PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward`.

Phi-LPF latest new-joint moving-atom signed-table sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-new-joint-moving-atom-signed-table-sync-router.md`.

The previous theorem leaves the independent nonterminal moving-atom exclusion. That target has the same moving-atom input already handled by the LPF/Phi source-bucket preimage router: any actual source must be LPF-owned and cannot escape through an ownerless point, a prime row, or a virtual unit. The signed-injection split then shows that the remaining obstruction is not an unsigned bucket count but a same-formal-unit signed coefficient problem before pushforward.

Thus the productive next target is the pointwise signed coefficient value table on Phi-LPF bucket support. A full closure would still need nonzero signed-row survival, same-formal-unit row-mass normalization and no-heavy-row control, or the parallel rate/PDEC and ExactUV/cofactor exits. These are not proved in this synchronization step. No unconditional row/column theorem is claimed here. □

Theorem 7.319 (Phi-LPF latest new-joint pointwise table to origin identity)**.** *The latest pointwise Phi-LPF signed value table target*

`PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward`

is not supplied by the LPF support and capacity ledger. If it is to be used as a genuine prepushforward signed artifact, its first non-formal field is a pointwise signed weight, then an actual primitive summand signed expression, and finally the source-origin identity

`PrimitiveSummandSignedCoefficientOriginIdentityBeforePushforward`.

Phi-LPF latest new-joint pointwise-table origin sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
pointwise-table-origin-sync-router.md.
```

The previous theorem leaves the pointwise Phi-LPF signed coefficient table. The pointwise-table frontier certificate blocks two shortcuts: unsigned LPF/Phi ownership data do not assign signs or local factors, and common source packet expansion re-enters the signed-lane self-proof cycle. The strict pointwise signed-weight and primitive-summand routers then identify the first actual field: the signed value must come from a primitive summand expression before pushforward, and that expression must be backed by a pre-Cauchy source-tuple origin identity.

Reverse recovery from zero-row covers, payment skeletons, canonical scoped formulae, or external spectral estimates is already blocked. Continuing the existing source-coordinate expansion reaches the registered non-proof cycle, leaving only a seed cycle-cut input, same-set PDEC scope match, or a new explicit joint alpha/delta formula as non-cyclic exits. Signed-row survival, row-mass/no-heavy-row control, ExactUV, rough-cofactor transport, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.320 (Phi-LPF latest new-joint origin identity to cycle-cut triad)**.** *The latest primitive summand source-origin identity target does not produce a new terminal proof endpoint. Along the recorded strict downstream chain it is equivalent to a row-level clean-core origin generation table; that table requires an acyclic pre-Cauchy source seed with a signed row emitter and a prepushforward sum identity. The existing emitter expansion enters the signed coordinate-source dependency cycle. Hence the latest non-cyclic target is the triad*

```
AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact.
```

Phi-LPF latest new-joint origin-identity cycle-cut sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
origin-identity-cyclecut-sync-router.md.
```

The primitive summand origin identity is not just a label: the strict origin router reduces it to a row-level clean-core original generation table for actual noncanonical primitive summands. The row-level router then requires an acyclic pre-Cauchy source seed carrying a signed row emitter, sign/local-factor law, exact key output, and prepushforward sum identity. Formal-unit containers, unsigned skeletons, zero-row covers, payment reversal, and source loops do not generate those signed rows.

The signed row emitter already lies on the registered coordinate-source cycle: signed coordinates, coefficient assignments, origin identities, row emitters, basis alphabets, and word constructors define one another. That cycle cannot count as a proof. The latest nonrecursive breaker therefore leaves only an acyclic seed cycle-cut primitive source, a same-set PDEC scope match, or a new explicit joint alpha/delta formula, with terminal WFD, signed survival and row-mass, ExactUV, model, rate, and D-structure gates still open. No unconditional row/column theorem is claimed here. □

Theorem 7.321 (Phi-LPF latest new-joint triad to explicit constructor)**.** *The latest three-way non-cyclic frontier*

```
AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
```

does not remain a wide terminal frontier in the current strict corpus. The seed cycle-cut branch is saturated to PDEC scope or a new joint formula; the same-set PDEC branch is internally saturated to a new joint formula or a conditional external line; and the new-joint branch has first productive field at the pre-Cauchy joint declaration line. The declaration line itself then reduces to the explicit constructor rule

```
ExplicitJointAlphaDeltaPrimitiveWordCoefficientConstructorRule
ForActualNoncanonicalSourceTuple.
```

Phi-LPF latest triad constructor sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
triad-constructor-sync-router.md.
```

The previous theorem leaves the cycle-cut/PDEC/new-joint triad. The strict seed-cycle-cut saturation router removes the sequential cycle-cut route unless a genuinely new source input is supplied; its remaining internal alternatives are same-set PDEC scope or a new joint formula. The strict PDEC scope saturation router then shows that same-set PDEC is not an already proved self-contained internal exit: it remains a conditional PDEC/external line or returns to the new-joint requirement.

The strict unified frontier and joint-emitter field-atom routers agree on the first productive new-joint field: a pre-Cauchy declaration line for the same actual noncanonical source tuple emitting both the primitive word and the signed coefficient. The existing joint declaration/constructor synchronization then shows that such a declaration is still not the formula. It must be replaced by an explicit joint alpha/delta primitive constructor rule with the same source tuple, basis word, signed coefficient, exact (u, v), branch key, sign/local factor, timestamp, no-leak discipline, and failure-return tags. The anti-split boundary is carried forward, so the rule cannot fall back to the old split alpha/delta route. This theorem does not prove the constructor, ExactUV, complete or fixed-key multiplicity, signed survival, row-mass, terminal exits, model gap, rate preservation, or D-structure acceptance. No unconditional row/column theorem is claimed. □

Theorem 7.322 (Phi-LPF latest constructor to built-in pairing)**.** *The latest explicit joint constructor target is not a valid stopping point for the non-cyclic route. In the current strict corpus, the ordinary expansion of the constructor through alpha-side, same-row, and row-level signed-source data returns to the signed-source fixed point. After deleting that route, the productive replacement must be anti-split atomic rows, whose first signed field is*

```
BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows.
```

The ExactUV side remains the parallel target

```
ActualEmitterSourceDomainEntropyLedger
AND ExactUVMapFixedPairPolylogFiberBoundLedger.
```

Phi-LPF latest constructor anti-split downstream sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-antisplit-downstream-sync-router.md.
```

The previous theorem leaves the explicit joint alpha/delta primitive constructor rule. The strict direct-attack certificate for that rule says that the ordinary constructor continuation is a non-proof route: it returns through joint alpha-side, same-row, row-level generation, and signed-source data to the registered fixed point. Therefore the formula cannot be accepted merely as a named constructor.

The strict anti-split downstream certificate imports the necessary replacement: the declaration must already contain atomic pre-Cauchy rows with built-in word/coefficient pairing and prepush-forward identity. The atomic rows frontier then identifies the next signed field as the built-in coefficient/pairing closed form for each atomic joint row. This theorem does not prove that closed form, source-domain entropy, fixed-pair fiber control, complete or fixed-key multiplicity, signed survival, row-mass, terminal exits, model gap, rate preservation, or D-structure acceptance. No unconditional row/column theorem is claimed. □

Theorem 7.323 (Phi-LPF latest constructor built-in trace exit)**.** *The constructor-side built-in pairing target has the same input as the recorded latest built-in trace synchronization. Thus it first reaches the exact atomic branch trace,*

$$\mathit{ExactAtomicJointBranchTraceSignedCoefficientFormulaOrReturn}.$$

However, the present branch-trace continuation through signed payload, origin identity, and common packet is the signed-lane self-proof cycle. After that cycle is removed, the current non-cyclic target is

$$\mathit{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact},$$

with terminal descent and same-set PDEC retained as controlled exits.

Phi-LPF latest constructor built-in trace sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-builtin-trace-sync-router.md.
```

The previous theorem leaves `BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows`. The strict built-in pairing frontier says that this closed form must be backed by an exact atomic branch trace. The strict branch-trace payload frontier then shows that following the existing trace line leads to a signed payload constructor, and the signed-lane cycle certificate identifies the continuation through payload, origin identity, common source packet, and built-in pairing as a closed dependency cycle.

Therefore the branch trace is not a final proof object in the current corpus. The non-cyclic route must submit a new primitive atomic signed payload or trace formula, or else use a controlled terminal-descent or same-set PDEC exit. The ExactUV side remains the same source-domain entropy and fixed-pair fiber conjunction, with complete-key and fixed-key multiplicity still open. Model gap, rate preservation, and D-structure acceptance also remain open. No unconditional row/column theorem is claimed. □

Theorem 7.324 (Phi-LPF latest constructor trace source-rank exit)**.** *The constructor-side trace exit*

$$\mathit{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact}$$

is not an independent terminal object in the current non-cyclic route. If it is not merely a renaming inside the signed-lane cycle, it must carry the same actual pre-Cauchy source-rank/no-collapse package already recorded for the latest trace-exit line. That package is absorbed by the same formal-unit pointwise primitive alpha/delta kernel table. Consequently the current direct target is

$$\mathit{AlphaRowAnchorPhaseEmissionFormulaLedger},$$

with the pre-Cauchy arithmetic identity and same-table ExactUV rank/multiplicity certificate still parallel.

Phi-LPF latest constructor trace source-rank sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-trace-source-rank-sync-router.md.
```

The previous theorem leaves a new primitive atomic signed payload or trace formula after deleting the branch-trace, payload, origin-identity, and common source-packet self-proof cycle. The existing latest new-payload/source-atom alignment certificate says that such a payload cannot be supplied by unsigned LPF or Phi labels. It must instead declare an actual pre-Cauchy source-rank and no-collapse package: source-domain entropy, complete primitive emitter key partition, and fixed-key ExactUV local multiplicity.

The post-antisplit source-rank convergence certificate and the earlier trace-exit/source-rank synchronization use the same absorber: the pointwise same-formal-unit primitive alpha/delta kernel table with nonzero rank. The strict pointwise-kernel frontier then identifies the first missing actual field as the alpha row anchor and phase emission formula. The LPF/Phi bucket identity is useful here only as an owner, support, capacity, and root ledger; it does not generate a primitive signed payload or trace formula. Terminal descent, same-set PDEC, pointwise signed Phi-LPF value table, ExactUV entropy/fiber, rough-cofactor transport, model gap, rate preservation, and D-structure acceptance remain open. No unconditional row/column theorem is claimed. ☐

Theorem 7.325 (Phi-LPF latest constructor alpha terminal three-atoms exit)**.** *The constructor-side alpha row anchor target*

$$\mathtt{AlphaRowAnchorPhaseEmissionFormulaLedger}$$

does not remain a local alpha-formula endpoint. Its unsigned carry-shell, source-tuple, and phase-wheel branches are already synchronized to geometric or terminal ledgers, while signed lift and overload return to the terminal capacity/model front. After the already removed joint/trace/payload cycle is excluded, the strict current frontier is the three-atom terminal front

$$\mathtt{AcyclicTerminalCanonicalLockToCanonicalSourceBoundary} \quad or \quad \mathtt{A1CleanBranchCanonicalSourceAdmission}$$

The narrowest direct target is

$$\mathtt{IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore}.$$

Phi-LPF latest constructor alpha terminal-three-atoms sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-new-joint-constructor-alpha-terminal-three-atoms-sync-router.md.`

The previous theorem synchronizes the constructor-side trace/source-rank exit to the pointwise primitive alpha/delta kernel table and leaves the alpha row anchor/phase formula as the first direct field. The existing alpha terminal synchronization has the same target. Its local alpha-row frontier separates the unsigned geometric branches from the signed branch: carry-shell congruence, source tuple binding, and phase-wheel compatibility do not produce pre-Cauchy signed coefficients, and the signed-lift/overload branches feed the terminal capacity, model, and D-structure ledgers.

Returning to a new joint formula, branch trace, or signed payload does not give an independent proof, because that route has already been identified as the signed-lane dependency cycle. The current strict terminal front is therefore canonical lock, A1 canonical-source admission, or actual clean-core moving atom exclusion. Among these, the moving-atom branch is the closest actual-load phase obstruction, but it still requires an independent nonterminal exclusion not factored through ExactUV, pair-mass, terminal return, or the joint constructor loop. The parallel PDEC, ExactUV, pointwise signed table, transport/coherence, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. ☐

Theorem 7.326 (Phi-LPF latest constructor moving-atom signed-table exit)**.** *The constructor-side independent moving-atom target*

$$\mathit{IndependentNonterminalMovingAtomExclusionForActualNoncanonicalCleanCore}$$

has no ownerless nonterminal escape in the LPF/Phi source model. If such a clean-core moving atom is sourced by an actual composite preimage, then its prepushforward source is owned by a unique LPF bucket. After the signed injection split removes the remaining unsigned counting freedom, the current direct target is

$$\mathit{PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward}.$$

Phi-LPF latest constructor moving-atom signed-table sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-moving-atom-signed-table-sync-router.md.
```

The previous theorem leaves the independent nonterminal exclusion of an actual noncanonical clean-core moving atom. The LPF/Phi source-bucket preimage router eliminates the ownerless, prime-row, and virtual-unit source exits: every actual composite source preimage must sit in a unique LPF-owned (p, m) bucket, with the cofactor m rough relative to p. This uses the least-prime-factor bucket identity as an owner and support ledger, not as a signed coefficient theorem.

The signed-injection split then shows that the remaining obstruction is not an unsigned count of source buckets. It is a same-formal-unit signed table before pushforward, or an equivalent complete bucket signed law/primitive summand signed expression. Even after such a table is supplied, nonzero signed-row survival and row-mass/no-heavy-row control remain necessary, as do the rate-bearing PDEC/CleanKLS packet, ExactUV entropy/fiber, rough-cofactor transport, model, rate, and D-structure gates. No unconditional row/column theorem is claimed. □

Theorem 7.327 (Phi-LPF latest constructor pointwise-origin exit)**.** *The constructor-side pointwise signed-table target*

$$\mathit{PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward}$$

is not supplied by the LPF/Phi owner and support identities. In the non-cyclic constructor chain, such a table must first supply pointwise signed weights, then primitive summand signed expressions, and finally the source origin identity

$$\mathit{PrimitiveSummandSignedCoefficientOriginIdentityBeforePushforward}.$$

Phi-LPF latest constructor pointwise-origin sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-pointwise-origin-sync-router.md.
```

The previous theorem moves the constructor branch to a same-formal-unit pointwise signed table on LPF/Phi bucket support. The LPF/Phi bucket identity fixes ownership, support, capacity, and square-root boundary data. It does not determine the signed coefficient, local factor, orientation, or alpha/delta side attached to a prepushforward summand.

The existing pointwise-origin router therefore applies without changing the theorem target: the first verifiable field of the pointwise table is a skeleton-row signed weight, that weight requires a primitive summand signed expression, and that expression is actual only after a primitive summand source-origin identity is supplied. Expanding that identity through the current source-coordinate corpus returns to the known signed-source dependency cycle, so the acyclic exits remain seed cycle-cut, same-set PDEC scope, or a new explicit joint constructor artifact. Signed-row survival, row-mass, ExactUV entropy/fiber, rough-cofactor transport, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. □

Theorem 7.328 (Phi-LPF latest constructor origin-cyclecut exit)**.** *The constructor-side source-origin target*

$$\mathtt{PrimitiveSummandSignedCoefficientOriginIdentityBeforePushforward}$$

does not close as an isolated atom. It must expand to a row-level clean-core origin generation table and then to an acyclic signed-row emitter. The current source-coordinate expansion of that emitter is a registered dependency cycle, so the non-cyclic exits are seed cycle-cut, same-set PDEC scope, or a new explicit joint constructor artifact.

Phi-LPF latest constructor origin-cyclecut sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-origin-cyclecut-sync-router.md.
```

The previous theorem reduces the constructor pointwise signed table to a primitive summand source-origin identity. The primitive-origin and row-level origin routers identify what such an identity must contain: a clean-core original coefficient generation table for each actual noncanonical primitive summand, generated by a pre-Cauchy source seed with a signed-row emitter and a prepushforward summation identity.

The unsigned routes remain blocked: formal-unit support, LPF/Phi ownership, zero-row covering, and payment skeletons do not generate signed origins. The coordinate-source guard records that expanding the signed emitter along the current basis/coefficient/source corpus returns to the known source cycle. Thus the constructor branch now has only the registered acyclic exits: seed cycle-cut input, same-set PDEC scope, or a new explicit joint constructor artifact. Terminal WFD, signed-row survival, row-mass, ExactUV entropy/fiber, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. □

Theorem 7.329 (Phi-LPF latest constructor triad-unified exit)**.** *The constructor-side trichotomy*

$$\mathtt{AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput} \vee \mathtt{AcyclicSameSetScopeMatchForDirectPDE}$$

does not provide a terminal black box. In the strict unified frontier, the cycle-cut and same-set PDEC branches are internally saturated, and the new joint branch has first productive field

$$\mathtt{PreCauchyJointWordCoefficientEmitterDeclarationLineForActualNoncanonicalSourceTuple}.$$

Phi-LPF latest constructor triad-unified sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-triad-unified-sync-router.md.
```

The previous theorem reduces the constructor branch to three registered acyclic exits: seed cycle-cut input, same-set PDEC scope, or a new explicit joint constructor artifact. The strict unified frontier shows that these exits are not independent terminal formulae inside the current corpus. Seed cycle-cut saturation leaves same-set PDEC scope or a new joint formula; same-set PDEC saturation leaves a new joint formula or an external spectral input; and a new joint formula first has to materialize as a pre-Cauchy joint word/coefficient emitter declaration line.

That declaration line is still not the full formula. It must be accompanied by joint primitive-summand rows, a prepushforward word/coefficient identity, and a no-downstream-recovery ledger. The parallel canonical-lock, independent-bridge, PDEC/external spectral, terminal WFD, signed-survival, row-mass, ExactUV, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. □

Theorem 7.330 (Phi-LPF latest constructor joint-declaration antisplit exit)**.** *The constructor-side joint declaration target*

PreCauchyJointWordCoefficientEmitterDeclarationLineForActualNoncanonicalSourceTuple

does not close by the ordinary declaration route. That route synchronizes to an explicit constructor rule and then returns to the signed-source fixed point. The non-cyclic route must pass through an anti-split atomic joint-row declaration, whose first signed obstruction is

BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows.

Phi-LPF latest constructor joint-declaration antisplit sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-joint-declaration-antisplit-sync-router.md.
```

The previous theorem puts the constructor branch at the pre-Cauchy joint word/coefficient emitter declaration line. The ordinary joint-declaration synchronization only recovers the explicit joint alpha/delta constructor rule; following that rule through the existing alpha-side and row-level origin corpus returns to the signed-source fixed point. Hence that route is not an acyclic proof.

The anti-split firewall forces the declaration to carry atomic joint rows with built-in word/coefficient pairing before alpha-side projection. The atomic rows router and source-declaration downstream router agree that the first remaining signed field is the built-in signed coefficient/pairing closed form. ExactUV source-domain entropy and fixed-pair fiber bounds remain parallel requirements; they are not produced by the pairing formula. Canonical lock, independent bridge, PDEC/external spectral input, signed survival, row-mass, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. □

Theorem 7.331 (Phi-LPF latest constructor macrocycle cut)**.** *The latest constructor anti-split branch cannot close by feeding the built-in signed coefficient/pairing target through the already registered constructor downstream chain. That chain returns to the same built-in target. After this macrocycle is deleted as a self-proof, the next productive mandatory signed gate is*

RowLevelCleanCoreOriginalCoefficientGenerationTableForActualNoncanonicalPrimitiveSummands.

Phi-LPF latest constructor macrocycle cut signed-survival sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-new-joint-
constructor-macrocycle-cut-signed-survival-sync-router.md.
```

Starting from `BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows`, the registered constructor chain goes through the new-payload trace exit, the source-rank/alpha kernel absorber, the clean-core moving atom split, the Phi–LPF pointwise signed table, the primitive-summand origin identity, the cycle-cut/PDEC/new-joint triad, and the joint-declaration anti-split layer. The last layer returns to the built-in signed coefficient/pairing target. Thus the chain is a diagnostic macrocycle, not a non-cyclic proof.

The latest retained basis still contains the signed-survival and row-mass side gates independently of this loop. The Phi–LPF signed-survival router has already reduced the signed expression side to the row-level clean-core origin generation table. This does not prove that table, nor does it prove nonzero signed row survival, row-mass normalization, ExactUV entropy/fiber, complete or fixed-key ledgers, model, rate, or D-structure gates. No unconditional row/column theorem is claimed. □

Theorem 7.332 (Phi-LPF latest constructor row-origin bucket-law sync)**.** *After the constructor macrocycle cut, the row-level clean-core origin table does not leave an unsigned support problem. The LPF/Phi bucket identity already supplies the support keys and capacities. The remaining direct signed field is*

$$PhiLPFBucketSignedCoefficientLawBeforePushforward.$$

Phi-LPF latest constructor row-origin bucket-law synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
row-origin-bucket-law-sync-router.md.
```

The row-level origin generation table would have to be produced by a pre-Cauchy noncanonical source seed with a signed-row emitter and a prepushforward sum identity. The registered signed-source fixed-point breaker shows that the present internal origin/emitter/value-map chain returns to the same row-level table, so it cannot be used as a proof.

Once this fixed point is cut, the noncircular signed coefficient kernel splits into a closed unsigned component and an open signed component. The closed component is the LPF/Phi support-capacity ledger: each support key has an LPF owner p, a p-rough cofactor m, and the exact bucket capacity coming from the Phi recursion. What remains is the bucket signed coefficient law: signed coefficient, sign/local factor, branch key, exact (u, v), and the prepushforward alpha/delta sum identity for those same keys. Row-mass, signed survival, ExactUV, complete/fixed-key, terminal/PDEC, model, rate, and D-structure gates remain open. No unconditional row/column theorem is claimed. □

Theorem 7.333 (Phi-LPF latest constructor bucket transport-stack sync)**.** *The constructor-side bucket signed-law target*

$$PhiLPFBucketSignedCoefficientLawBeforePushforward$$

is not a terminal non-cyclic input. If it is not replaced by a direct pointwise signed value table, the existing rough-cofactor transport stack reduces it to

$$PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward$$

together with the source-packet three atoms. The constructor-side signed survival and row-mass/no-heavy-row gates remain parallel requirements.

Phi-LPF latest constructor bucket transport-stack synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
bucket-transport-stack-sync-router.md.
```

The previous theorem leaves the bucket signed coefficient law after the row-origin fixed point is cut and after LPF/Phi support and capacity have been paid. The bucket-transport certificate applies to that same signed law: a recursive route must give rough-cofactor signed transport, unless a complete pointwise signed table is supplied directly.

The recursive transport stack then separates its unsigned and signed parts. The ordered LPF factorization path is closed, and the virtual-unit/square-base private escape has already been reduced to the common source packet. What is left on the transport side is the step signed multiplier table; what is left on the source-packet side is the alpha anchor/phase formula, the independent pre-Cauchy arithmetic identity, and the same-unit ExactUV rank/multiplicity certificate. None of these fields is generated by the LPF/Phi bucket count.

This substitution only advances the latest frontier. It does not prove the edge multiplier table, the source-packet three atoms, nonzero signed survival, same-formal-unit row-mass normalization, complete/fixed keys, ExactUV entropy/fiber, terminal/PDEC alternatives, model gap, rate preservation, or D-structure acceptance. No unconditional row/column theorem is claimed here. □

Theorem 7.334 (Phi-LPF latest constructor edge multiplier slab sync)**.** *The constructor-side edge multiplier target*

$$\mathtt{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}$$

is not an atomic black box. The LPF ordered path splits it into the semiprime first-edge signed seed table and the internal prime-adjoin signed transition law:

$$\mathtt{PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward} \quad \text{and} \quad \mathtt{PhiLPFInternalPrimeAdjoinSignedTr}$$

The first-edge Phi fiber formula pays only unsigned occurrence mass.

Phi-LPF latest constructor edge multiplier slab synchronization. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-constructor-`
`edge-multiplier-slab-sync-router.md.`

The previous theorem leaves the step signed multiplier table as the transport side of the constructor bucket route. The first-edge slab certificate applies to that same table: every LPF ordered path has a unique first edge $(p, 1, q, q)$, and every later edge has non-unit prefix. Hence the table splits into a first-seed slab and an internal transition slab.

For the first edge, LPF/Phi gives the exact unsigned fiber

$$\mathrm{mass}(p, q) = \Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

This counts q-rough continuations. It does not emit a signed seed value, orientation bit, local factor, branch key, exact (u, v), or return tag. The internal transition likewise requires a nonzero predecessor or a named return; it cannot be reconstructed by dividing by an unknown final signed coefficient.

Thus the latest edge-side hard points are the semiprime first-edge signed seed table and the internal prime-adjoin transition law. The source-packet three atoms, nonzero signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV entropy/fiber, pointwise signed table, terminal/PDEC alternatives, model gap, rate preservation, and D-structure acceptance remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.335 (Phi-LPF latest constructor semiprime seed diagonal sync)**.** *The constructor-side semiprime first-edge signed seed table has a forced diagonal/offdiagonal split:*

$$\mathtt{PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward} \to \mathtt{PhiLPFDiagonalSquareBaseFirstSeedComm}$$

The diagonal part has no private signed exit; it is carried by the common source-packet atoms. The new seed-side hard point is the offdiagonal ordered semiprime first-seed signed table.

Phi-LPF latest constructor semiprime seed diagonal synchronization. The certificate is recorded in `docs/monograph/prime-matrix-phi-lpf-latest-constructor-semiprime-seed-diagonal-sync-router.md.`

The previous theorem leaves the semiprime first-edge signed seed table. LPF order gives $p \le q$, so the seed types split disjointly into $p = q$ and $p < q$. The equality case is exactly the square-base diagonal root already routed away from a private signed lane. In the constructor latest basis this diagonal obligation is therefore carried by the source-packet three atoms.

The remaining new seed data are the offdiagonal ordered semiprime first-edge signed values. The support split and the Phi occurrence counts do not provide those values, their orientations, local factors, branch keys, exact (u, v) returns, or named failure tags. The internal prime-adjoin transition law also remains paired with the seed table, and signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV entropy/fiber, pointwise signed table, terminal/PDEC alternatives, model gap, rate preservation, and D-structure acceptance remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.336 (Phi-LPF latest constructor offdiagonal seed tuple sync)**.** *The constructor-side offdiagonal first-seed table has a forced source-tuple payload split. Its unsigned part is the ledger of owner prime, first rough prime, rough tail, and Phi tail-fiber mass. After this split the active constructor seed hard point is the source-tuple signed seed formula, together with orientation, ExactUV return, internal transition, and the carried source atoms.*

Phi-LPF latest constructor offdiagonal seed tuple synchronization. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-constructor-offdiagonal-seed-tuple-sync-router.md.`

The preceding constructor seed reduction leaves the offdiagonal ordered semiprime first-seed table. The imported tuple-fields certificate gives a unique source tuple (p, q, t) for each offdiagonal occurrence, with $p < q$ and t q-rough, and the occurrence count is exactly

$$\sum_{p<q} \Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

Thus LPF/Phi pays only unsigned support, owner, first rough prime, rough tail, and fiber mass.

None of these unsigned fields determines the pre-pushforward signed seed value, the orientation parity, the branch side, the local factor, the exact (u, v) fixed pair, or the return tag. Therefore the latest constructor basis replaces the offdiagonal seed black box by the source-tuple signed seed formula and its paired orientation/ExactUV/internal-transition obligations, while retaining the source-packet three atoms, signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV entropy/fiber, pointwise signed table, terminal/PDEC alternatives, model gap, rate preservation, and D-structure acceptance. No unconditional row/column theorem is claimed here. □

Theorem 7.337 (Phi-LPF latest constructor pure-pair atom sync)**.** *The constructor-side source-tuple signed seed formula has a forced first-atom synchronization. Its first seed atom is the pure semiprime pair*

$$(p, q, 1), \qquad p < q, \qquad \text{value } pq,$$

while every nonunit tail occurrence is a continuation lift with mass

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right) - 1.$$

Thus the latest constructor seed-side hard point is the pure-pair signed seed atom, not the full q-rough tail family.

Phi-LPF latest constructor pure-pair atom synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
pure-pair-atom-sync-router.md.
```

The preceding constructor tuple synchronization reduces the offdiagonal seed table to the source-tuple signed seed formula. The pure-atom theorem applies to that input: for a fixed ordered type (p, q), the unique tuple with tail $t = 1$ is the first seed atom. If $t > 1$, the occurrence has already passed the first seed and belongs to the internal prime-adjoin continuation ledger.

This closes only the unsigned atom location and the tail-lift mass formula. It does not produce the pure-pair signed value, orientation parity, branch side, local factor, ExactUV return tag, or the internal transition law. The carried source atoms, signed survival, row-mass/no-heavy-row, complete/fixed keys, and the global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.338 (Phi-LPF latest constructor pure-pair Ferrers support sync)**.** *The constructor-side pure-pair signed seed atom has a forced Ferrers support synchronization. Its unsigned support is the bipartite prime graph with*

$$p \le \sqrt{N}, \qquad q \text{ prime}, \qquad p < q \le N/p.$$

The left neighborhoods are nested as p increases, and the left degrees, right degrees, and edge count are determined by the prime table and $\lfloor N/p \rfloor$. The latest signed hard point is therefore the two-prime interaction signed kernel on those edges.

Phi-LPF latest constructor pure-pair Ferrers support synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
pure-pair-ferrers-support-sync-router.md.
```

The preceding constructor pure-pair synchronization identifies the active input as the pure-pair atom. The Ferrers support theorem applies to that input: the edge condition $p < q \le N/p$ is exactly the condition that the pure pair pq lies in the N-truncated offdiagonal LPF bucket. If p increases, the upper bound N/p decreases, so the right-neighborhoods are nested.

This is a support and degree ledger only. It fixes which two-prime edges are available, but does not give the signed kernel, orientation, local factor, ExactUV return tag, internal transition, or the carried source atoms. Signed survival, row-mass/no-heavy-row, complete/fixed keys, the pointwise signed-table bypass, and global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.339 (Phi-LPF latest constructor two-prime no-swap sync)**.** *The constructor-side two-prime signed kernel has a forced LPF-owner no-swap synchronization. The canonical source edge is (p, q) with $p < q$. Although $pq = qp$ as an integer, the reverse edge (q, p) is not in the same LPF owner source domain and cannot supply a second signed source row or a cancellation partner. The latest direct hard point is therefore the edge-local two-prime signed interaction formula or a named return.*

Phi-LPF latest constructor two-prime no-swap synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
two-prime-no-swap-sync-router.md.
```

The preceding constructor Ferrers synchronization reduces the pure-pair atom to a signed kernel on Ferrers edges. The ordered no-swap theorem applies to that kernel: the LPF owner is the smaller prime p, so the only source-domain edge for a distinct semiprime is the canonical ordered edge (p, q). Product symmetry is erased before signed data is emitted.

This removes only the swap-symmetry pseudo-exit. It does not provide the edge-local signed formula, orientation parity, branch side, local factor, ExactUV return tag, internal transition, or the carried source atoms. Signed survival, row-mass/no-heavy-row, complete/fixed keys, the pointwise signed-table bypass, and global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.340 (Phi-LPF latest constructor edge-local field-cut sync)**.** *The constructor edge-local signed formula has a forced unsigned field-cut. For every canonical edge (p, q), the LPF, Phi, and Ferrers ledgers fix the owner p, the second prime q, the product pq, the row and column degree data, and the multiplicity-one label. These unsigned fields do not emit a signed seed, local factor, orientation parity, ExactUV return tag, or source-row coefficient. The latest direct hard point is therefore the signed atom field table or a named return tag.*

Phi-LPF latest constructor edge-local field-cut synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
edge-local-field-cut-sync-router.md.
```

The preceding constructor no-swap synchronization leaves only the canonical edge-local formula. The edge-local field-cut theorem applies to that input: LPF ownership fixes the smaller prime, the Phi bucket fixes the admissible cofactor, and the Ferrers support ledger fixes the rank-degree and multiplicity-one edge label. Hence the remaining constructor obligation is not another support or capacity count.

This field-cut does not prove the signed atom fields, orientation parity, ExactUV return, internal transition, carried source atoms, signed survival, or same-unit row-mass/no-heavy-row ledger. Complete/fixed keys, the pointwise signed-table bypass, and global ExactUV/model/rate/D-structure gates remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.341 (Phi-LPF latest constructor signed atom trace-sync)**.** *The constructor-side signed atom fields have a forced same-trace-key and named return synchronization. The unsigned edge label only supplies the input domain. The signed value, local factor, orientation side, alpha/delta side, ExactUV key, and source row must be emitted on one pre-Cauchy trace key; otherwise the branch is a named return or PDEC. The latest productive hard point is therefore a new primitive atomic signed payload or trace artifact.*

Phi-LPF latest constructor signed atom trace synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
signed-atom-trace-sync-router.md.
```

The preceding constructor field-cut leaves exactly the signed atom field table or a named return. The signed atom trace-sync theorem applies to that table: all signed fields must be produced by the same pre-Cauchy trace key, and every missing, conflicting, zero-factor, over-budget, or cross-key read has a named return.

This closes only the routing discipline. It does not provide the new payload, the carried source atoms, signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV source/fiber, model, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.342 (Phi-LPF latest constructor new-payload source-atom alignment)**.** *The constructor-side new primitive payload is not an independent terminal object. If* `NewPrimitiveAtomicSignedPayloadOrTraceFo` *is not merely a signed-lane rename or a controlled exit, then it must carry the actual pre-Cauchy source-rank/no-collapse package: source-domain entropy, complete primitive emitter key partition, and fixed-key ExactUV local multiplicity. The latest direct hard point is source-domain absolute entropy.*

Phi-LPF latest constructor new-payload source-atom alignment sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
new-payload-source-atom-alignment-sync-router.md.
```

The preceding constructor trace-sync leaves the new payload as the productive exit. The strict new-payload/source-atom alignment theorem applies to that exit: a true new payload must declare an actual pre-Cauchy source object before payment or pushforward, and it must prevent source collapse by the entropy, complete-key, and fixed-key multiplicity ledgers.

This does not prove source-domain entropy, the carried source atoms, row-mass/no-heavy-row, signed survival, complete/fixed keys, ExactUV, model, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.343 (Phi-LPF latest constructor source-entropy payload-loop cut)**.** *The constructor-side source-entropy hard point cannot be discharged by the current constructor chain itself. If* `ActualPreCauchySourceDomainAbsoluteEntropyLedger` *is expanded along the existing strict source-entropy downstream and cycle-cut/terminal unified frontier, it returns to* `PreCauchyJointWordCoefficientEmitter` *That same joint declaration line is already the upstream constructor input used to reach built-in pairing, new payload, and then source entropy. Hence the current internal route is a payload/source-entropy self-dependency, not a non-circular proof.*

Phi-LPF latest constructor source-entropy payload-loop cut sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
source-entropy-payload-loop-cut-sync-router.md.
```

The previous constructor new-payload/source-atom theorem leaves source-domain entropy as the direct hard point. The strict downstream source-entropy theorem sends that hard point to cycle-cut or terminal descent. The unified cycle-cut/terminal/PDEC frontier then sends the remaining internal branch back to the pre-Cauchy joint declaration line.

But the constructor chain has already used this joint declaration line upstream: joint declaration produces built-in pairing, built-in pairing produces the new primitive payload frontier, and the new payload frontier demands source-domain entropy. Thus the internal chain reads

$$\text{joint declaration} \to \text{built-in pairing} \to \text{new payload} \to \text{source entropy} \to \text{cycle/terminal} \to \text{joint declaration.}$$

This closes a dependency loop only. It does not prove source-domain entropy or give a global contradiction. A non-circular continuation must provide a fresh independent joint declaration outside this payload loop, or pass through a canonical lock, independent source bridge, PDEC/external spectral input, pointwise signed table, ExactUV source/fiber, complete/fixed-key, signed survival, row-mass, model, rate, and D-structure gates. No unconditional row/column theorem is claimed here. □

Theorem 7.344 (Phi-LPF latest constructor fresh-joint identity taxonomy)**.** *The fresh joint declaration left by the payload-loop cut is not a renamed copy of the old constructor declaration line.*

Its freshness condition forbids the already-used route from joint declaration to built-in pairing and payload. Consequently any such declaration must enter as an independent noncanonical pre-Cauchy arithmetic identity. Under the existing strict source taxonomy, the self-contained content of that identity is the actual noncanonical moving-block/NC-BLK input.

Phi-LPF latest constructor fresh-joint identity taxonomy sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
fresh-joint-identity-taxonomy-sync-router.md.
```

The payload-loop cut requires a declaration line outside the constructor payload loop. The strict joint-emitter field atomization shows that this line is the first productive field: without it, the joint rows formula, the word/coefficient identity, the prepushforward sum identity, and the named return ledger cannot begin.

Freshness removes the old constructor antisplit route. Thus the declaration line must be justified as an independent noncanonical pre-Cauchy arithmetic identity, not by the downstream payload it is supposed to support. The strict identity taxonomy has already exhausted the canonical, generic WFD, AP/external, and actual-source classes. After the non-self-contained classes are removed, the remaining self-contained mathematical input is `ActualNoncanonicalMovingBlockSpreadNCBLKForCounterexample`

This does not prove that moving-block/NC-BLK input. It also does not prove the joint rows formula, the word/coefficient identity, the prepushforward identity, the no-downstream return ledger, signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV, model, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.345 (Phi-LPF latest constructor moving-block terminal sync)**.** *The constructor-side moving-block/NC-BLK input left by the fresh-joint taxonomy is not a new unnamed terminal. In the strict moving-block router, a registered low-dimensional signature enters a PDEC/SAE/ColumnCRT/sparse terminal, while the no-signature branch enters the early-zero terminal package. The global terminal split then sends this package to*

PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve and ExplicitModelGapAndFiniteDPRCLedger.

Thus the constructor fresh-joint branch is synchronized to the same global terminal and model-gap gate as the productive pre-Cauchy alpha branch.

Phi-LPF latest constructor moving-block terminal synchronization. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
moving-block-terminal-sync-router.md.
```

The previous theorem reduced the fresh joint declaration to `ActualNoncanonicalMovingBlockSpreadNCBLKForCoun` The strict moving-block certificate applies exactly to that input and closes the possibility that it remains an unnamed self-contained exit. Its output is the named terminal pair consisting of the PDEC/CleanKLS capacity gate and the explicit model-gap ledger. The global PDEC/sparse terminal split supplies the same terminal capacity gate, and the latest pre-Cauchy alpha-terminal sync confirms that the productive alpha branch reaches this same pair.

This synchronization does not prove the PDEC/CleanKLS capacity theorem, the explicit model-gap and finite-DPRC ledger, the joint rows formula, the word/coefficient identity, the no-downstream return ledger, source entropy, ExactUV, signed survival, row-mass/no-heavy-row, complete/fixed keys, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.346 (Phi-LPF latest constructor terminal hardpoint split). *The constructor-side terminal pair reached after moving-block synchronization has a finer non-cyclic split. The PDEC/CleanKLS component splits into a same-set PDEC scope-matching arm and a self-contained Kuznetsov/DLS large-sieve arm. The explicit model-gap ledger splits off the finite $P < 2003$ DPRC certificate and leaves the high-segment model gap; that high-segment gap is factorized into the harmonic-window upper ledger and the dynamic rough-skeleton lower ledger. Thus the current explicit terminal basis is*

$$
\begin{gathered}
(\mathtt{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\
\text{or}\quad \mathtt{SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks}) \\
\text{and}\quad \mathtt{HarmonicWindowAlpha043PGe3001Upper0850Ledger} \\
\text{and}\quad \mathtt{DynamicRoughSkeletonAlpha043PGe3001Lower401Ledger}.
\end{gathered}
$$

Phi-LPF latest constructor terminal hardpoint split sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
terminal-hardpoint-split-sync-router.md.
```

The previous synchronization imports the active constructor terminal pair `PDEC_CAP_OR_INTERNAL_CleanKLS_Large` and `ExplicitModelGapAndFiniteDPRCLedger`. The strict terminal hardpoint certificate splits the first component into the scoped PDEC arm and the clean Kuznetsov/DLS arm. The explicit model-gap certificate removes the finite DPRC segment below 2003, and the high-segment factorization reduces the remaining model margin to two tail ledgers: the harmonic window bound and the rough skeleton lower bound.

This is a synchronization and factorization of open inputs, not a proof of any of them. The self-contained Kuznetsov/DLS inequality, same-set PDEC scope, harmonic window, rough skeleton, rate preservation, D-structure, and constructor sibling fields remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.347 (Phi-LPF latest constructor Kuznetsov terminal-cycle sync). *The constructor-side Kuznetsov/DLS arm is not an independent terminal exit. The strict Kuznetsov/DLS synchronization closes the KZ-A through KZ-D formal spectral spine, but its KZ-E saving reduces to an acyclic NC-BLK/source anti-atom. That object, if it fails, returns to the moving-atom/global terminal family. Hence the constructor KZ arm returns to*

$$
\begin{gathered}
\mathtt{AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn} \\
\text{and}\quad \mathtt{PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve\_FOR\_AcyclicNoncanonicalTerminalFamily} \\
\text{and}\quad \mathtt{HarmonicWindowAlpha043PGe3001Upper0850Ledger} \\
\text{and}\quad \mathtt{DynamicRoughSkeletonAlpha043PGe3001Lower401Ledger}.
\end{gathered}
$$

Phi-LPF latest constructor Kuznetsov terminal-cycle sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
kuznetsov-terminal-cycle-sync-router.md.
```

The previous theorem left the self-contained Kuznetsov/DLS inequality as the first constructor-side terminal hard point. The strict KZ/DLS terminal synchronization applies to that exact atom. It shows that the formal spectral layers have already been normalized, while the remaining KZ-E dispersion saving is not available as a new self-contained proof. Its failure mode is the acyclic NC-BLK/source anti-atom, and the existing NC-BLK synchronization sends that mode back to the acyclic terminal family and the model ledger.

The model ledger is then read through the already imported high-segment factorization, leaving the harmonic-window and rough-skeleton tail inputs. This does not prove the terminal family, the nonrecursive breaker, the new-joint formula, same-set PDEC scope, harmonic window, rough skeleton, rate, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.348 (Phi-LPF latest constructor terminal-family saturation)**.** *The constructor-side acyclic noncanonical terminal family left by the KZ return is not a remaining unnamed terminal. In the current strict corpus it saturates to the same two explicit alternatives as the global CRT terminal frontier: same-set PDEC scope matching or a new explicit actual joint alpha/delta constructor formula. Thus the constructor terminal-family contribution is replaced by*

$$\begin{gathered}
\mathtt{AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn} \\
\textit{and} \quad (\mathtt{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \\
\textit{or} \quad \mathtt{NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact}) \\
\textit{and} \quad \mathtt{HarmonicWindowAlpha043PGe3001Upper0850Ledger} \\
\textit{and} \quad \mathtt{DynamicRoughSkeletonAlpha043PGe3001Lower401Ledger} \\
\textit{and} \quad \mathtt{DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance}.
\end{gathered}$$

Phi-LPF latest constructor terminal-family saturation sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
terminal-family-saturation-sync-router.md.
```

The preceding theorem sends the constructor KZ arm back to `PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_`
The strict terminal-family saturation theorem then expands that family into its three arms. The canonical-lock arm is scoped to the canonical-source case, the direct PDEC arm remains a same-set scope-matching input, and the CleanKLS/DLS arm returns to the terminal cycle. The nonrecursive breaker and the seed-cycle-cut continuation have already been synchronized to the signed-source fixed point, while the global CRT terminal synchronization gives the same PDEC-scope or new-joint frontier.

Therefore this constructor branch no longer contains a separate terminal-family black box. It is now a named frontier consisting of source seed, same-set PDEC scope or new joint formula, and the harmonic/skeleton/D-structure tail ledgers. This does not prove the source seed, the PDEC scope certificate, the new joint formula, the harmonic window, the rough skeleton, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.349 (Phi-LPF latest constructor terminal new-joint macrocycle sync)**.** *The new-joint alternative left by constructor terminal-family saturation is not a fresh free exit. When it is expanded through the current constructor new-joint downstream, its first productive field is the pre-Cauchy joint declaration line; the anti-split route then reduces the signed content to built-in signed coefficient pairing. The built-in route is already registered as a constructor signed macrocycle. Therefore the new-joint alternative is absorbed into the row-level signed-side gate*

$$\begin{gathered}
\mathtt{RowLevelCleanCoreOriginalCoefficientGenerationTableForActualNoncanonicalPrimitiveSummands} \\
\textit{and} \quad \mathtt{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward} \\
\textit{and} \quad \mathtt{SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger}.
\end{gathered}$$

Phi-LPF latest constructor terminal new-joint macrocycle sync. The certificate is recorded in `docs/monograph/prime-matrix-phi-lpf-latest-constructor-terminal-new-joint-macrocycle-sync-router.md.`

The previous theorem leaves a PDEC-scope or new-joint alternative. The constructor triad-unified certificate sends the new-joint alternative to `PreCauchyJointWordCoefficientEmitterDeclarationLineForActual` The declaration anti-split certificate rejects the ordinary declaration route as a signed-source fixed point and sends the non-circular anti-split route to `BuiltInSignedCoefficientPairingClosedFormForAtomicJoin`

The constructor macrocycle certificate then identifies the loop

$$\text{built-in} \to \text{payload} \to \text{alpha kernel} \to \text{moving atom} \to \text{signed table} \to \text{origin} \to \text{triad} \to \text{declaration} \to \text{built-in}.$$

The source-entropy/payload route also returns to the declaration line. Thus the current new-joint branch cannot be counted as a new proof unless it pays the row-level clean-core origin table and the signed survival/row-mass side gates, or exits through PDEC scope, canonical lock, independent bridge, external spectral input, terminal WFD, or a direct pointwise signed table.

This synchronization does not prove the row-level origin table, signed survival, row-mass normalization, PDEC scope, ExactUV, harmonic/skeleton tail ledgers, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.350 (Phi-LPF latest constructor row-level signed-source fixed-point sync)**.** *The row-level alternative left by the constructor new-joint macrocycle cut is not a terminal proof atom. In the current archive it synchronizes as*

$$\begin{aligned}
&\mathtt{RowLevelCleanCoreOriginalCoefficientGenerationTableForActualNoncanonicalPrimitiveSummands}\\
\to\ &\mathtt{AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedWithSignedRowEmitterAndPrepushforwardSumIdenti}\\
&\to \textit{signed-source fixed point}\\
&\to \mathtt{NoncircularPreCauchySignedCoefficientEmissionKernelForActualNoncanonicalPrimitiveRows}.
\end{aligned}$$

Thus the latest constructor signed-side gate is

$$\begin{aligned}
&\mathtt{NoncircularPreCauchySignedCoefficientEmissionKernelForActualNoncanonicalPrimitiveRows}\\
&\quad \textit{and} \quad \mathtt{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward}\\
&\quad \textit{and} \quad \mathtt{SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger}.
\end{aligned}$$

Phi-LPF latest constructor row-level signed-source fixed-point sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-constructor-row-level-signed-source-fixed-point-sync-router.md.`

The preceding constructor theorem leaves the row-level clean-core origin table as the mandatory signed-side input. The strict row-level router shows that this table must be generated before Cauchy/payment by a seed equipped with a signed-row emitter and a prepushforward summation identity. However, the signed-source fixed-point breaker records that the current internal expansion through the emitter, basis words, signed coordinate slots, assignments, value maps, and origin identity returns to the same row-level table. The seed coordinate/source cycle guard records the same obstruction: a raw cycle is not a proof of signed coefficients.

Consequently the row-level table name is removed as a coarse frontier atom. The remaining non-circular input is a pre-Cauchy signed coefficient emission kernel that does not read the downstream row table, origin identity, payment/Phi projection, or zero-row cover. This theorem does not prove that kernel, signed survival, row-mass normalization, PDEC scope, ExactUV/key ledgers, harmonic/skeleton ledgers, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.351 (Phi-LPF latest constructor noncircular kernel bucket signed-law sync)**.** *The non-circular signed-emission kernel left by the latest constructor row-level fixed-point cut is not a support-counting problem. After importing the Phi-LPF support-stripped kernel certificate, its unsigned support component is paid by least-prime-factor ownership and the Φ-recursion:*

$$\begin{aligned}&\mathtt{NoncircularPreCauchySignedCoefficientEmissionKernelForActualNoncanonicalPrimitiveRows}\\ &\quad\to \mathtt{PhiLPFPrimitiveRowSupportAndCapacityLedger}\\ &\quad \text{and}\quad \mathtt{PhiLPFBucketSignedCoefficientLawBeforePushforward}.\end{aligned}$$

The first factor is the LPF/Phi support ledger. Hence the latest constructor signed-side gate becomes

$$\begin{gathered}\mathtt{PhiLPFBucketSignedCoefficientLawBeforePushforward}\\ \text{and}\quad \mathtt{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward}\\ \text{and}\quad \mathtt{SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger}.\end{gathered}$$

Phi-LPF latest constructor noncircular kernel bucket signed-law sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
noncircular-kernel-bucket-signed-law-sync-router.md.
```

The strict non-circular-kernel router keeps the signed coefficient source before Cauchy/payment and forbids recovery from the row table, origin identity, payment/Phi projection, or zero-row cover. The Phi-LPF support-stripping certificate then separates the kernel into an unsigned support-capacity ledger and a signed law on the resulting support keys. The support keys are (p, m), where p is the least prime factor of pm and m is p-rough; their capacity is governed by $\Phi(\lfloor N/p\rfloor, p) - 1$, with the $p > \sqrt{N}$ layers contributing zero mass.

Thus the latest constructor problem is no longer to locate or count candidate rows. It is to assign, before pushforward, a signed coefficient, sign/local factor, branch key, exact (u, v), and alpha/delta summation identity to each Phi-LPF bucket. This theorem does not prove that bucket signed law, signed survival, row-mass normalization, PDEC scope, ExactUV/key ledgers, harmonic/skeleton ledgers, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.352 (Phi-LPF latest constructor bucket transport-stack rebase sync)**.** *The latest constructor bucket signed-law frontier imports the existing transport stack without changing the theorem target. If one does not submit a direct pointwise signed value table on Phi-LPF buckets, then the remaining recursive route is*

$$\begin{gathered}\mathtt{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}\\ \text{and}\quad \mathtt{AlphaRowAnchorPhaseEmissionFormulaLedger}\\ \text{and}\quad \mathtt{IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger}\\ \text{and}\quad \mathtt{SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows}.\end{gathered}$$

The signed survival and row-mass side gates remain attached to this constructor frontier.

Phi-LPF latest constructor bucket transport-stack rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
bucket-transport-stack-rebase-sync-router.md.
```

The preceding theorem leaves `PhiLPFBucketSignedCoefficientLawBeforePushforward` as the latest constructor hard point. The existing bucket transport stack already shows that this law either is supplied directly as a pointwise bucket signed value table or must be transported through rough cofactor multiplication. The ordered factorization path is a closed LPF path fact, while the unit/square-base boundary and common-packet self-proof routes have already been removed.

Therefore the recursive transport branch reduces to an ordered-edge signed multiplier table together with the source-packet three atoms: alpha anchor phase emission, an independent pre-Cauchy arithmetic identity, and same-unit ExactUV rank multiplicity. This theorem does not prove those atoms, the edge multiplier table, signed survival, row-mass normalization, PDEC scope, ExactUV/key ledgers, harmonic/skeleton ledgers, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.353 (Phi-LPF latest constructor edge multiplier slab rebase sync)**.** *The latest constructor edge-multiplier frontier reuses the first-edge slab decomposition. The ordered-edge signed multiplier table splits as*

$$\begin{gathered}\mathit{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}\\ \to \mathit{PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward}\\ \text{and}\quad \mathit{PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward}.\end{gathered}$$

The first term controls prefix-one semiprime seed edges; the second controls all internal prime-adjoin edges.

Phi-LPF latest constructor edge multiplier slab rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
edge-multiplier-slab-rebase-sync-router.md.
```

The preceding rebase theorem leaves the ordered-edge signed multiplier table as the latest constructor hard point. The existing first-edge slab certificate has the same target input and can be imported without changing the theorem target. It separates the first edge $(p, 1, q, q)$ from the internal edges $(p, \text{prefix}, q, \text{prefix}\, q)$ with $\text{prefix} > 1$.

The Phi fiber formula pays only the unsigned occurrence mass of first-edge q-rough continuations. It does not supply signed seed values, local factors, orientation, branch keys, ExactUV payloads, or named returns. Therefore the latest direct signed input is the semiprime first-edge signed seed table, while the internal prime-adjoin transition law remains a paired obligation. This theorem does not prove either signed table, the source-packet three atoms, signed survival, row-mass normalization, PDEC scope, ExactUV/key ledgers, harmonic/skeleton ledgers, rate preservation, D-structure, or constructor sibling fields. No unconditional row/column theorem is claimed here. □

Theorem 7.354 (Phi-LPF latest constructor semiprime seed diagonal rebase sync)**.** *The latest rebase of the constructor first-seed frontier imports the existing semiprime diagonal/offdiagonal split. Thus*

$$\mathit{PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward}$$

reduces, in the rebased constructor basis, to the offdiagonal ordered semiprime first-seed table together with the carried source-packet atoms:

$$\begin{gathered}\mathit{PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward}\\ \text{and}\quad \mathit{AlphaRowAnchorPhaseEmissionFormulaLedger}\\ \text{and}\quad \mathit{IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger}\\ \text{and}\quad \mathit{SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows}.\end{gathered}$$

Phi-LPF latest constructor semiprime seed diagonal rebase sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-constructor-semiprime-seed-diagonal-rebase-sync-router.md.`

The preceding edge-multiplier rebase leaves the semiprime first-edge signed seed table. The existing semiprime diagonal certificate has the same input: LPF order gives $p \le q$, so the first-seed support splits disjointly into the diagonal case $p = q$ and the offdiagonal case $p < q$. The diagonal square-base lane has already been stripped of any private signed exit and is carried only by the common source-packet atoms.

Consequently the latest seed-side signed hard point is the offdiagonal ordered semiprime first-seed table. The internal prime-adjoin transition law remains paired with it, and the source atoms, signed survival, row-mass/no-heavy-row, complete/fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate preservation, and D-structure inputs remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.355 (Phi-LPF latest constructor offdiagonal seed tuple rebase sync)**.** *The latest offdiagonal first-seed frontier imports the existing source-tuple field split. The LPF/Phi identities determine the owner prime p, the first rough prime q, the q-rough tail, and the tail-fiber mass, but they do not determine the pre-pushforward signed seed. Hence the current direct hard point is*

$$\mathit{PhiLPFOffDiagonalOrderedSemiprimeSourceTupleSignedSeedFormulaBeforePushforward}.$$

Phi-LPF latest constructor offdiagonal seed tuple rebase sync. The certificate is recorded in

`docs/monograph/prime-matrix-phi-lpf-latest-constructor-offdiagonal-seed-tuple-rebase-sync-router.md.`

The preceding rebase leaves `PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward`. The tuple-field certificate has the same input. Each offdiagonal occurrence has a unique tuple (p, q, t), with $p < q$ and t q-rough, and its unsigned mass is paid by

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right).$$

This exhausts the LPF/Phi support information.

The signed seed value, orientation parity, branch side, local factor, ExactUV fixed pair, and return tag are not functions of that unsigned tuple ledger. The internal prime-adjoin transition and the carried source atoms also remain paired obligations. Thus this rebase narrows the open problem, but it does not prove the signed formula, signed survival, row-mass/no-heavy-row, complete and fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.356 (Phi-LPF latest constructor pure-pair atom rebase sync)**.** *The latest source-tuple signed-seed frontier imports the pure-pair atom split. For each ordered offdiagonal pair $p < q$, the tuple with tail $t = 1$ is the unique pure semiprime first-seed atom. Every nonunit tail occurrence has mass*

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right) - 1$$

and is an internal-transition lift, not a new first seed. The current direct hard point is therefore

$$\mathit{PhiLPFOffDiagonalPureSemiprimePairSignedSeedAtomBeforePushforward}.$$

Phi-LPF latest constructor pure-pair atom rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
pure-pair-atom-rebase-sync-router.md.
```

The preceding rebase leaves the offdiagonal source-tuple signed formula. The pure-pair atom certificate has the same input and separates the first atom $(p, q, 1)$ from all nonunit q-rough tails. This is a support and routing fact: the tail-lift mass is paid by the Phi ledger, while its signed transition must be supplied by the internal prime-adjoin law.

Thus the open signed seed has been narrowed to the pure semiprime pair atom. This theorem does not prove its signed value, orientation parity, branch side, ExactUV return tag, internal transition, carried source atoms, signed survival, row-mass/no-heavy-row, complete and fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.357 (Phi-LPF latest constructor pure-pair Ferrers support rebase sync)**.** *The latest pure-pair atom frontier imports the Ferrers support ledger. The unsigned edge set is*

$$p \le \sqrt{N}, \qquad p < q \le N/p, \qquad p, q \text{ prime}.$$

Its nesting, left degrees, right degrees, and edge count are fixed by the prime table and $\lfloor N/p \rfloor$. Hence the current direct hard point is the two-prime signed interaction kernel

$$\mathit{PhiLPFOffDiagonalTwoPrimeInteractionSignedKernelBeforePushforward}.$$

Phi-LPF latest constructor pure-pair Ferrers support rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
pure-pair-ferrers-support-rebase-sync-router.md.
```

The preceding rebase leaves the pure semiprime pair signed atom. The Ferrers support certificate has the same input: the pure pair pq lies below N exactly when $q \le N/p$, and increasing p only shrinks the admissible right-neighbourhood. Thus support, degree, and edge-count data are already paid by the LPF/Phi prime table ledger.

This does not emit the signed two-prime kernel. Orientation parity, branch side, ExactUV return, internal transition, carried source atoms, signed survival, row-mass/no-heavy-row, complete and fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate, and D-structure inputs remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.358 (Phi-LPF latest constructor two-prime no-swap rebase sync)**.** *The latest two-prime signed-kernel frontier imports the LPF-owner no-swap ledger. For a distinct semiprime pq with $p < q$, the pre-Cauchy source domain has only the canonical edge (p, q). The reversed edge (q, p) is not in the same LPF owner bucket. Hence product symmetry gives no second signed source row, and the current direct hard point is*

$$\mathit{PhiLPFEdgeLocalTwoPrimeSignedInteractionFormulaOrReturnBeforePushforward}.$$

Phi-LPF latest constructor two-prime no-swap rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
two-prime-no-swap-rebase-sync-router.md.
```

The preceding rebase leaves the two-prime signed interaction kernel. The ordered no-swap certificate has the same input: the least prime factor of pq is p, so the owner bucket records (p, q). The reversed ordered pair would put the same integer into owner bucket q, contradicting LPF ownership. Thus $pq = qp$ is only product commutativity, not an additional signed source or cancellation partner.

This synchronization does not provide the edge-local signed interaction formula. Orientation parity, branch side, ExactUV return, internal transition, carried source atoms, signed survival, row-mass/no-heavy-row, complete and fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate, and D-structure inputs remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.359 (Phi-LPF latest constructor edge-local field-cut rebase sync)**.** *The latest edge-local signed-formula frontier imports the field-cut ledger for canonical LPF owner edges. For each edge (p, q), the owner, product, LPF bucket, Ferrers rank and degree fields, and multiplicity-one edge label are closed unsigned data. They do not emit the signed atom field table. Hence the current direct hard point is*

$$\mathit{PhiLPFEdgeLocalTwoPrimeSignedAtomFieldsOrNamedReturnTagBeforePushforward}.$$

Phi-LPF latest constructor edge-local field-cut rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
edge-local-field-cut-rebase-sync-router.md.
```

The preceding rebase leaves the edge-local signed interaction formula. The field-cut certificate has the same input and exhausts the canonical edge's closed unsigned ledger: LPF ownership fixes the bucket, the product fixes the integer label, the Ferrers table fixes the rank and degree data, and multiplicity one prevents an additional unsigned copy from appearing.

None of these fields specifies the signed value, local factor, orientation, ExactUV return, or source-row coefficient. The synchronization therefore narrows the hard point to signed atom fields or a named return tag. Orientation parity, branch side, ExactUV return, internal transition, carried source atoms, signed survival, row-mass/no-heavy-row, complete and fixed keys, ExactUV source/fiber, PDEC scope, harmonic/skeleton ledgers, rate, and D-structure inputs remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.360 (Phi-LPF latest constructor signed atom trace rebase sync)**.** *The latest signed-atom-fields frontier imports the signed atom trace-sync ledger. The unsigned edge label gives only the input domain. The signed value, local factor, orientation side, ExactUV fixed pair, and source row must lie in one pre-Cauchy trace key or enter a named return matrix. Thus the current direct hard point is*

$$\mathit{NewPrimitiveAtomicSignedPayloadOrTraceFormulaArtifact}.$$

Phi-LPF latest constructor signed atom trace rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
signed-atom-trace-rebase-sync-router.md.
```

The preceding rebase leaves signed atom fields or a named return tag. The trace-sync certificate has that same input and closes the routing rule: all signed slots of a canonical edge must share one pre-Cauchy trace key. Missing slots, conflicting slots, zero factors, budget overflow, or cross-key reads are named returns rather than anonymous exits.

This is still not a formula for the signed payload. It removes a routing ambiguity and leaves a new primitive atomic signed payload or trace artifact as the productive hard point. Terminal descent, same-set PDEC, the pointwise Phi-LPF signed table, carried source atoms, signed survival, row-mass/no-heavy row, complete and fixed keys, ExactUV source/fiber, model, rate, and D-structure inputs remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.361 (Phi-LPF latest constructor new-payload source-atom alignment rebase sync)**.** *The latest new-payload frontier imports the strict source-atom alignment ledger. An unsigned label and a signed-lane trace do not produce a new signed payload. If the payload is genuinely new, it must carry an actual pre-Cauchy source-rank/no-collapse package. Hence the current direct hard point is*

$$\mathtt{ActualPreCauchySourceDomainAbsoluteEntropyLedger}.$$

Phi-LPF latest constructor new-payload source-atom alignment rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
new-payload-source-atom-alignment-rebase-sync-router.md.
```

The preceding rebase leaves a new primitive atomic signed payload or trace artifact. The strict alignment certificate has the same input: if this object is not merely a signed-lane renaming or a named terminal/PDEC/pointwise-table exit, then before any payment or Phi pushforward it must declare the same actual source object and carry source-domain entropy, a complete key partition, and a fixed-key ExactUV local multiplicity bound.

This theorem only performs that alignment. It does not prove source entropy, complete keys, fixed-key multiplicity, the carried source atoms, row mass, signed survival, the pointwise Phi-LPF signed table, ExactUV source/fiber, model, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.362 (Phi-LPF latest constructor source-entropy payload-loop cut rebase sync)**.** *The latest source-domain entropy frontier imports the constructor payload-loop cut ledger. Expanding source entropy through the strict downstream chain returns to the joint declaration line already used upstream to produce built-in pairing, new payload, and source entropy. This raw loop is not a noncircular proof. Hence the current direct hard point is*

$$\mathtt{FreshIndependentPreCauchyJointDeclarationLineOutsideConstructorPayloadLoop}.$$

Phi-LPF latest constructor source-entropy payload-loop cut rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
source-entropy-payload-loop-cut-rebase-sync-router.md.
```

The preceding rebase leaves actual pre-Cauchy source-domain absolute entropy. The strict downstream ledger expands this entropy to the cycle-cut or terminal descent alternatives, and the unified frontier routes those alternatives back to the joint declaration line. But the constructor chain has already used that same line to reach built-in pairing, new payload, and source entropy. The result is a dependency loop, not an independent source of signed mass.

The loop can only be broken by a fresh joint declaration outside the constructor payload loop, or by canonical lock, an independent source bridge, PDEC/external input, the pointwise Phi-LPF signed table, or the remaining key, ExactUV, signed-survival, row-mass, model, rate, and D-structure ledgers. No unconditional row/column theorem is claimed here. □

Theorem 7.363 (Phi-LPF latest constructor fresh-joint identity taxonomy rebase sync)**.** *The fresh joint declaration left by the latest payload-loop cut imports the strict identity taxonomy. Freshness only deletes the old constructor joint-declaration to payload route; it does not create a new source class. A valid independent declaration must enter the noncanonical pre-Cauchy identity taxonomy, whose current actual-source remainder is*

$$\mathit{ActualNoncanonicalMovingBlockSpreadNCBLKForCounterexampleBranchAndReturn}.$$

Phi-LPF latest constructor fresh-joint identity taxonomy rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
fresh-joint-identity-taxonomy-rebase-sync-router.md.
```

The preceding rebase leaves a fresh independent joint declaration outside the constructor payload loop. The strict joint-field ledger says that such a line must precede rows, word/coefficient identity, and no-downstream return ledgers. The identity taxonomy then exhausts the canonical, generic WFD, AP/external, and actual noncanonical cases; after the forbidden constructor route is removed, the remaining actual noncanonical case is moving-block/NC-BLK.

This theorem does not prove moving-block/NC-BLK, joint rows, identity, return ledger, signed survival, row mass, complete or fixed keys, ExactUV, model, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.364 (Phi-LPF latest constructor moving-block terminal rebase sync)**.** *The actual moving-block/NC-BLK remainder left by the fresh-joint taxonomy is not a new unnamed constructor exit. Rebased against the strict actual moving-block router, the older constructor moving-block terminal certificate, the global PDEC/sparse terminal split, and the pre-Cauchy alpha terminal sync, it reduces to*

$$\mathit{PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve} \quad \text{and} \quad \mathit{ExplicitModelGapAndFiniteDPRCLedger}.$$

Phi-LPF latest constructor moving-block terminal rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
moving-block-terminal-rebase-sync-router.md.
```

The preceding rebase leaves `ActualNoncanonicalMovingBlockSpreadNCBLKForCounterexampleBranchAndReturn`. The strict moving-block router prevents that input from becoming an anonymous terminal. The older constructor terminal certificate has the same moving-block input, while the global PDEC/sparse split and the pre-Cauchy alpha terminal sync route the materialized terminal alternatives to the same PDEC/CleanKLS capacity gate and retain the explicit model-gap ledger.

This theorem removes only the unnamed moving-block exit. It does not prove the PDEC/CleanKLS capacity gate, the explicit model gap and finite DPRC ledger, joint rows, word/coefficient identity, no-downstream return, source entropy, ExactUV, signed survival, row mass, complete or fixed keys, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.365 (Phi-LPF latest constructor terminal hardpoint and high-model rebase sync)**.** *The wide pair*

$$\mathit{PDEC\_CAP\_OR\_INTERNAL\_CleanKLS\_LargeSieve} \quad \text{and} \quad \mathit{ExplicitModelGapAndFiniteDPRCLedger}$$

left by the latest constructor terminal rebase is not a terminal atom. It rebases to the two strict terminal arms

$$\mathit{AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate} \quad \text{or} \quad \mathit{SelfContainedKuznetsovDLSLargeSieve}$$

and to the high-segment model-gap ledger

$$\mathit{HighSegmentModelGapAlpha043C3AnalyticLedger}.$$

Phi-LPF latest constructor terminal hardpoint and high-model rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
terminal-hardpoint-high-model-rebase-sync-router.md.
```

The strict PDEC/CleanKLS hardpoint certificate splits the wide terminal gate into the direct PDEC scope-matching arm and the self-contained Kuznetsov/DLS large-sieve arm. Separately, the strict global-scope ledger splits the explicit model-gap/DPRC input into a closed finite segment below 2003 and the remaining high-segment analytic model-gap ledger. The moving-block DPRC compatibility certificate ensures that this replacement does not introduce a new DPRC object.

This theorem proves only the rebase of the wide terminal pair. It does not prove the PDEC scope match, the self-contained Kuznetsov/DLS inequality, the high-segment model gap, joint rows, word/coefficient identity, no-downstream return, source entropy, ExactUV, signed survival, row mass, complete or fixed keys, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.366 (Phi-LPF latest constructor Kuznetsov and high-model rebase sync). *The latest constructor pair*

$$\mathit{SelfContainedKuznetsovDLSLargeSieveInequalityForAcyclicCleanBlocks} \quad \text{and} \quad \mathit{HighSegmentModelGapAlph}$$

does not close as a pair of black boxes. The Kuznetsov/DLS atom rebases to the acyclic NC-BLK/source anti-atom, while the high-model ledger rebases to the self-contained beta-sieve and sawtooth tail package.

Phi-LPF latest constructor Kuznetsov and high-model rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
kuznetsov-high-model-rebase-sync-router.md.
```

The strict Kuznetsov atom ledger closes the KZ-A smoothing, KZ-B trace specialization, KZ-C Bessel decay, and KZ-D spectral large-sieve spine. It does not prove the KZ-E log-saving step; that step is reduced to `AcyclicNCBLKActualBlockNonconcentrationOrStrengthenedSourceAntiAtom`. The constructor-specific KZ cycle ledger agrees that this is not an independent terminal exit. Separately, the strict high-model tail ledger sends the high-segment model gap to the self-contained Rosser–Iwaniec beta-sieve appendix, the 99% main-coefficient error, and the exact sawtooth remainder bound.

This theorem only performs the rebase. It does not prove the NC-BLK/source anti-atom, the beta-sieve appendix, the coefficient error, the sawtooth bound, PDEC scope, terminal family, nonrecursive breaker, new-joint constructor, joint rows, source entropy, ExactUV, signed survival, row mass, key ledgers, rate, or D-structure inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.367 (Phi-LPF latest constructor post-KZ-E source-admission rebase sync). *The latest constructor route after the NC-BLK/source anti-atom and high-tail rebase does not stop at the NC-BLK name. Importing the strict source-root, PDEC, new-joint, KZ no-cycle, and post-KZ-E direct source-bridge ledgers rebases the current internal non-circular KZ-E route to*

$$\mathit{A1CleanBranchCanonicalSourceAdmission}.$$

The high-segment model side is still the self-contained beta-sieve/sawtooth tail package. Thus the current direct target is

```
A1CleanBranchCanonicalSourceAdmission
AND SelfContainedRosserIwaniecBetaSieveWeightConstructionAppendix
    THEN ExactSawtooth.
```

Phi-LPF latest constructor post-KZ-E source-admission rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
post-kze-source-admission-rebase-sync-router.md.
```

The NCBLK/source anti-atom frontier sends the actual route to a forward source-root packet or a named terminal return. The documented source-root subroutes return to the terminal capacity cycle; the post-source-root PDEC audit saturates the direct PDEC arm; the post-PDEC new-joint audit removes the old signed-lane self-proof route; and the KZ no-cycle gate forbids using the existing KZ-E route through NCBLK/source-root projection. The post-KZ-E source-bridge certificate then shows that, without an accepted external no-projection DI/BFI/Kuznetsov input, the internal route must prove admission of the clean A1 branch to the canonical Rosser–Iwaniec/Buchstab source before Cauchy and dispersion.

The LPF/Phi bucket identity is used only as an unsigned owner/support/capacity ledger: it prevents a signed source from being read backward out of bucket capacity, and it does not prove canonical source admission. The high-model tail ledger is re-applied separately, so the self-contained beta-sieve appendix, the 99% main-coefficient error, and the exact residue-weighted sawtooth bound remain open. This theorem is a non-circular frontier rebase; it does not prove source admission, beta-sieve/sawtooth, rate preservation, D-structure/Rankin, or the row/column theorem. □

Theorem 7.368 (Phi-LPF latest constructor post-source-admission macrocycle rebase sync)**.** *The latest constructor source-admission target is absorbed as a scoped branch boundary, not as an independent global contradiction. The canonical Rosser–Iwaniec/Buchstab branch is internally covered only as a canonical source branch; the generic noncanonical branch remains external or terminal. Following the internal A1/T1/signed-lift/PDEC/new-joint/KZ/KZ-E chain returns to a macrocycle. Thus the current non-circular constructor frontier is*

```
(AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
 OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
 OR NewPrimitiveJointPayloadArtifactOutsideSignedLaneCycle
 OR ExactExternalDIBFIKuznetsovNoProjectionCertificate_FOR_NONCIRCULAR_KZ_ONLY)
AND SelfContainedRosserIwaniecBetaSieveWeightConstructionAppendix
AND BetaSieveMainCoefficientNinetyNinePercentExplicitErrorAlpha043PGe100000
AND ExactResidueWeightedFloorSawtoothTenPercentBound.
```

Phi-LPF latest constructor post-source-admission macrocycle rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
post-source-admission-macrocycle-rebase-sync-router.md.
```

The source-admission branch absorption ledger shows that `A1CleanBranchCanonicalSourceAdmission` is a branch-boundary discipline: it permits the canonical RIW/Buchstab source branch to use the closed internal chain, but it does not eliminate the generic noncanonical complement. The post-source-admission macrocycle ledger then follows the remaining internal path through T1 mismatch, signed-lift registration, the alpha signed-weight downstream, PDEC/CleanKLS, new-joint saturation, KZ no-cycle, and the KZ-E direct source bridge, returning to the same A1 gate. This identifies a cycle, not a contradiction.

Therefore the next productive inputs must be outside the cycle: a seed cycle-cut primitive source, a direct same-set PDEC scope certificate, a new primitive joint/payload artifact outside the signed-lane cycle, or an accepted external no-projection KZ/DI/BFI certificate. The large-threshold plus finite verification strategy is retained only with its proper scope: it can close registered finite bridges, but it does not replace the $P \geq 100000$ Rosser–Iwaniec beta-sieve construction, the 99% main-coefficient ledger, or the exact residue-weighted sawtooth bound. Rate preservation and D-structure/Rankin remain open. No unconditional row/column theorem is claimed here. □

Theorem 7.369 (Phi-LPF latest constructor cycle-cut and antisplit downstream rebase sync)**.** *The latest constructor seed-cycle-cut target is not an independent exit in the current internal corpus. The strict seed-cycle-cut branch is saturated to PDEC/new-joint alternatives; the cycle-cut, terminal-descent, and PDEC internal branches unify to a pre-Cauchy joint declaration line; and the ordinary joint declaration/constructor route returns to the signed-source fixed point. Hence the non-circular internal downstream target is*

```
BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
AND ActualEmitterSourceDomainEntropyLedger
AND ExactUVMapFixedPairPolylogFiberBoundLedger,
```

together with the beta-sieve/sawtooth tail package.

Phi-LPF latest constructor cycle-cut and antisplit downstream rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
cyclecut-antisplit-downstream-rebase-sync-router.md.
```

The seed-cycle-cut saturation ledger rejects both sequential directions: one cannot first generate primitive basis words and then attach signed coefficients, nor first declare coefficients and then recover their primitive source words. The unified cycle-cut/terminal ledger sends the remaining productive content to a joint declaration line, while the antisplit downstream ledger shows that the ordinary declaration line merely re-enters the signed-source fixed point. A non-circular declaration must instead carry atomic rows with built-in word/coefficient pairing, whose first signed gap is the built-in signed coefficient closed form. ExactUV does not follow from that pairing; it remains the source-domain entropy and fixed-pair fiber bound.

The high-threshold plus finite-verification strategy again has only finite scope: it removes registered finite bridges, but it does not prove the $P \geq 100000$ beta-sieve lower-weight construction, the 99% main coefficient ledger, or the exact sawtooth bound. This theorem is a frontier rebase only; it does not prove the built-in pairing, ExactUV, beta-sieve, RatePreservation, D-structure/Rankin, or the row/column theorem. □

Theorem 7.370 (Phi-LPF latest constructor common signed-table rebase sync)**.** *The latest constructor downstream triple*

```
BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows
AND ActualEmitterSourceDomainEntropyLedger
AND ExactUVMapFixedPairPolylogFiberBoundLedger
```

does not consist of three independent black-box obligations. Importing the built-in pairing trace guard, the trace-exit source-rank convergence, the ExactUV fiber atomization, the source-entropy signed-row atomization, the fixed-pair fiber atomization, the complete-key source-table boundary, and the Phi-LPF recursive ownership ledger rebases the current internal obligation to one same-formal-unit pre-Cauchy primitive-emitter table:

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows
AND PhiLPFBucketSignedCoefficientLawBeforePushforward
AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger
AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
AND FixedKeyExactUVLocalMultiplicityO1Ledger.
```

The beta-sieve/sawtooth tail package is still carried in parallel.

Phi-LPF latest constructor common signed-table rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
common-signed-table-rebase-sync-router.md.
```

The built-in pairing trace ledger shows that the signed coefficient pairing cannot be proved from the existing signed-lane cycle; a productive trace exit must submit a new primitive trace/payload or a controlled terminal/PDEC return. The trace-exit convergence ledger then sends any genuine new primitive trace to the same pointwise primitive alpha/delta kernel table. Independently, the ExactUV fiber ledger splits the fiber obligation into source entropy, complete keys, and fixed-key local multiplicity. The source-entropy atomization requires pre-Cauchy signed rows, row-mass/no-heavy-row, and a support lower bound; the fixed-pair fiber atomization requires a complete key partition and fixed-key $O(1)$ multiplicity.

The LPF/Phi recursive ownership identity is used at exactly its legitimate strength. It closes the unsigned least-prime-factor ownership, support and capacity ledger, the exact prime-counting identity, and the $p > \sqrt{N}$ zero-mass layer. It does not create signed coefficients, orientations, local factors, row mass, complete keys, or fixed-key multiplicity. Therefore the remaining internal load is the common signed table displayed above, together with the Rosser–Iwaniec beta-sieve appendix, the 99% main coefficient ledger, the exact sawtooth bound, RatePreservation, and D-structure/Rankin. No unconditional row/column theorem is claimed here. □

Theorem 7.371 (Phi-LPF latest constructor common-table transport-stack rebase sync)**.** *In the common signed-table frontier, the coarse atom*

$$\mathtt{PhiLPFBucketSignedCoefficientLawBeforePushforward}$$

is not left as a terminal name. Importing the Phi-LPF bucket transport stack, the rough-cofactor signed transport split, the step local-factor update frontier, and the source-packet cycle guard replaces it by the finer alternative

$$\mathtt{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}$$

or by a direct pointwise signed value table on Phi-LPF bucket support. Consequently the current internal table obligation is

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows
AND PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward
AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger
AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
AND FixedKeyExactUVLocalMultiplicityO1Ledger,
```

again with the beta-sieve/sawtooth tail package carried in parallel.

Phi-LPF latest constructor common-table transport-stack rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
common-table-transport-stack-rebase-sync-router.md.
```

The Phi-LPF bucket signed transport ledger records the exact support split given by the Phi recursion. This split is still unsigned: it partitions the rough cofactor domain, but does not determine signs, orientations, local factors, or alpha/delta branch transitions. To upgrade the support split to a signed recursion one must either submit a pointwise signed value table on the Phi-LPF bucket support, or give a rough-cofactor signed transport law.

The ordered LPF path and square-base private exits have already been accounted for. After the path is fixed, the signed transport law reduces to a step signed multiplier table for each ordered edge. The square-base/common packet route returns to the source-packet cycle guard, so it contributes only the source three atoms rather than a new self-contained signed formula. Thus the bucket signed-law name has been removed from the latest constructor frontier, but the edge multiplier, source three atoms, row mass/support, complete and fixed keys, beta-sieve/sawtooth, RatePreservation, and D-structure/Rankin inputs remain unproved. No unconditional row/column theorem is claimed here. □

Theorem 7.372 (Phi-LPF latest constructor common-table edge multiplier slab rebase sync)**.** *In the latest common-table transport frontier, the atom*

$$\mathit{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward}$$

is not a terminal black box. Importing the first-edge slab decomposition, the ordered LPF path forces this edge multiplier table to split into

```
PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward
AND PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward.
```

Therefore the current common-table internal obligation is

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows
AND PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward
AND PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward
AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger
AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger,
```

again with the beta-sieve/sawtooth tail package, RatePreservation, and D-structure/Rankin carried in parallel.

Phi-LPF latest constructor common-table edge multiplier slab rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
common-table-edge-multiplier-slab-rebase-sync-router.md.
```

The LPF/Phi bucket identity has already paid the unsigned ownership and capacity. For a first edge (p, q), the exact mass is the q-rough continuation fiber

$$\Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right),$$

but this number is only an occurrence count. It does not supply the signed seed value, branch orientation, local factor, row mass, complete key, fixed key multiplicity, or ExactUV payload. For later edges the same obstruction appears as the internal prime-adjoin transition law: one needs a nonzero predecessor or a named return, not a posterior division by an unknown signed value.

Thus the edge multiplier name has been removed from the latest common-table frontier. The remaining active load is the first-seed table together with the internal transition law, source three atoms, row mass/support, complete and fixed keys, and the beta-sieve/sawtooth tail. The previous large-threshold plus finite-verification attempts close only finite bridges and do not replace these structural inputs. No unconditional row/column theorem is claimed here. □

Theorem 7.373 (Phi-LPF latest constructor common-table semiprime seed diagonal rebase sync)**.** *In the latest common-table frontier, the atom*

PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward

splits by the forced diagonal/off-diagonal dichotomy. The diagonal square-base lane (p,p) has no private signed exit; it returns to the common source-packet atoms. Hence the current internal table obligation is

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows
AND PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward
AND PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward
AND SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger
AND PrimitiveRowSupportLowerBoundBeforeExactUVProjectionLedger
AND RegisteredCompletePrimitiveEmitterKeyPartitionPolylogLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger,
```

with the beta-sieve/sawtooth tail package still carried in parallel.

Phi-LPF latest constructor common-table semiprime seed diagonal rebase sync. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-latest-constructor-
common-table-semiprime-seed-diagonal-rebase-sync-router.md.
```

The semiprime first edge has only two cases. When $p = q$, the edge is the square-base root already handled by the source-packet cycle guard; it cannot serve as a new signed source without returning to the three common atoms. When $p < q$, no existing ledger supplies the prepushforward signed seed table. Thus the first-seed obligation is reduced, but not proved, and the paired internal prime-adjoin transition, row mass/support, complete and fixed keys, and the beta-sieve/sawtooth tail remain active. No unconditional row/column theorem is claimed here. □

Theorem 7.374 (Phi-LPF endpoint interval difference)**.** *For every interval $2 \le A \le B$, the Phi-LPF bucket identity gives the exact endpoint difference*

$$\pi(B) - \pi(A-1) = (B - A + 1) - \sum_{p \le \sqrt{B}} \left(\Phi\left(\left\lfloor \frac{B}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{A-1}{p} \right\rfloor, p \right) \right).$$

In particular, for the closed aligned interval $[kP, kP+P]$,

$$\pi(kP+P) - \pi(kP-1) = P + 1 - \sum_{p \le \sqrt{kP+P}} \left(\Phi\left(\left\lfloor \frac{kP+P}{p} \right\rfloor, p \right) - \Phi\left(\left\lfloor \frac{kP-1}{p} \right\rfloor, p \right) \right).$$

Phi-LPF endpoint interval difference. The certificate is recorded in

```
docs/monograph/prime-matrix-phi-lpf-endpoint-
interval-difference-router.md.
```

The composite integers in $[A, B]$ are partitioned by their least prime factor. For a fixed least prime factor p, writing the integer as pm leaves a p-rough cofactor m, so the number of such composites is

$$\Phi\left(\left\lfloor \frac{B}{p} \right\rfloor, p\right) - \Phi\left(\left\lfloor \frac{A-1}{p} \right\rfloor, p\right).$$

Summing over $p \leq \sqrt{B}$ counts every composite in the interval exactly once, and subtracting from the interval length gives the displayed prime count.

This identity is exact and useful for local audit and finite verification, but it is not by itself a positivity theorem. To prove that an interval contains a prime one still needs a structural upper bound on the summed LPF bucket increments. In the unrestricted aligned-interval problem, a CRT construction can choose k so that every integer in $[kP, kP + P]$ is composite; the certificate records the sample $P = 5$, $k = 8166$, where $[40830, 40835]$ contains no prime. This sample is outside the strict row range $1 < k < P$, and therefore cannot be used as a counterexample to the strict row problem. Thus the endpoint formula must be paired with extra phase, capacity, or signed information before it can support a global contradiction. No unconditional row/column theorem is claimed here. □

Theorem 7.375 (Strict $1 < k < P$ endpoint correction and author residue)**.** *The unrestricted CRT prime-free sample in the endpoint-difference ledger does not apply to the strict row domain $1 < k < P$. In that strict domain,*

$$\pi(kP + P) - \pi(kP - 1) = \pi(kP + P - 1) - \pi(kP),$$

because the endpoints kP and $(k + 1)P$ are composite. Hence the strict row count is the exact Phi–LPF endpoint difference

$$\pi(kP + P - 1) - \pi(kP) = (P - 1) - \sum_{p \leq \sqrt{kP+P-1}} \Delta\Phi_p.$$

The positivity of this integer is still open in the present corpus: it is equivalent to $\sum \Delta\Phi_p < P - 1$, equivalently to the existence of a full-root uncovered slot, equivalently to a prime in the strict row.

Strict $1 < k < P$ endpoint correction and author residue. The certificate is recorded in

```
docs/monograph/prime-matrix-k-less-p-endpoint-correction-
author-residue-router.md.
```

It imports the strict row endpoint-difference certificate and the final external/internal closure split. The CRT sample has $P = 5$ and $k = 8166$, so $1 < k < P$ is false. Therefore it is only an unrestricted warning. For strict rows, every point $kP + a$, $1 \leq a < P$, lies below P^2. If such a point has no prime divisor at most $\sqrt{kP + P - 1}$, then it cannot be composite and must be prime. Thus the full-root uncovered-slot condition is not an extra consequence of the Phi–LPF identity; it is the strict row prime assertion itself.

The same certificate records a finite sweep up to $P \leq 2003$, with all strict rows positive and minimum prime count 1. This is a regression and structure audit, not an infinite-tail proof. Consequently the author-side ordinary work for the external-lemma version is sealed, while the remaining non-author input is `DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance`. The no-black-box external version still requires an exact primary-source FullS non-AP WFD KLS theorem match, or a new automorphic/dispersion proof for the same object. The self-contained version still requires the signed-source fine package and `SelfContainedDStructureTailLog4FiniteRankinProofPackage`. No unconditional row/column theorem is promoted by this correction. □

Theorem 7.376 (External-lemma and self-contained closure split)**.** *The present proof ledger has two different closure notions, and they cannot be identified. In the external-lemma version, the conditional package*

AcceptedFullSKLSExtExternalContract
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance

closes the row/column counterexample chain. In the self-contained version, the corresponding theorem-level package is

ActualNoncanonicalCleanCoreMovingAtomExclusion
AND SelfContainedDStructureTailLog4FiniteRankinProofPackage.

The first package is a conditional external/referee-input theorem. It is not a current unconditional theorem, because the final D-structure/Tail-log4/finite Rankin promotion package has not been independently accepted. The second package is not proved in the current corpus.

External-lemma and self-contained closure split. The certificate is recorded in

`docs/monograph/prime-matrix-dual-closure-external-internal-hardpoint-router.md.`

It imports four already established boundary facts. First, the Phi–LPF endpoint difference formula is an exact counting identity, not a positivity theorem. Second, the worst strict top band is the Cramer-local scale $x \asymp P^2$, interval length $\asymp \sqrt{x}$, so a direct H_P closure cannot be obtained by rearranging Eratosthenes-equivalent identities. Third, FullS–KLS-ext has been matched as an external black-box contract to the non-AP full-S WFD object, but it is not a derivation from the currently cited DI/BFI primary theorems. Fourth, the D-structure/Tail-log4/finite Rankin promotion package has an author-side dossier and a pass-or-return Rankin subledger, but independent acceptance is still absent.

Therefore the external-lemma version is logically closed only under the displayed external package. Without that package, the no-black-box external route still requires an exact primary-source FullS non-AP WFD KLS theorem match, or a new automorphic/dispersion proof for the same object.

For the self-contained route, DI/BFI/Kuznetsov estimates live after completion and after coefficient production, so they cannot emit the pre-Cauchy noncanonical signed source. The fine source-side package currently begins with the alpha-row anchor, the independent pre-Cauchy arithmetic identity, same-unit ExactUV rank multiplicity, the off-diagonal semiprime first-seed signed table, the internal prime-adjoin signed transition table, same-formal-unit row-mass normalization, primitive-row support, complete key partition, and fixed-key local multiplicity. These are genuine inputs, not consequences of the unsigned Phi–LPF bucket identity. Hence the split is a closure firewall: it closes the logical classification of the remaining routes while preserving `row_column_unconditional_closed=false`. □

Theorem 7.377 (D-structure/Rankin author-side remainder split)**.** *For the external FullS–KLS contract version, the ordinary author-side remainder is empty. The remaining promotion condition is the non-author input*

DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.

It cannot be removed by additional author-side dossier work. If one wants a no-black-box external version, one must supply either

$$\texttt{ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch} \quad \text{or} \quad \texttt{NewAutomorphicDispersionProof}.$$

If one wants a self-contained replacement for the D-structure/Rankin promotion gate, one must prove the conjunction of the D-structure/Structured-EHPD reduction package, the Tail-log4 adapter with exact constants, the reproducible finite-verification archive, and the full Rankin pass-or-return ledger with all downstream returns.

D-structure/Rankin author-side remainder split. The certificate is recorded in

```
docs/monograph/prime-matrix-dstructure-rankin-author-
remainder-split-router.md.
```

It imports the author-side completion ledger, the final guard gate, the D-structure/Rankin promotion boundary, the dual closure split, and the strict $1 < k < P$ endpoint correction. These inputs agree on a single boundary: author-side completion has sealed the conditional external line, while the promotion package remains a referee/independent-acceptance gate.

Thus the next author-side tasks are only replacement tasks. The no-black-box external replacement must match the exact non-AP full-S WFD object in the primary external theorem variables, or prove the same object by a new automorphic/dispersion argument. The self-contained replacement must turn four checkable interfaces into in-text proofs: D-structure definitions and the A/B reduction, Tail-log4 adaptation with constants and theorem numbers, finite verification reproducibility, and the complete Rankin pass-or-return ledger. Until those packages are supplied, the final promotion condition remains present and no unconditional row/column theorem is promoted. □

Theorem 7.378 (Two replacement-line terminal attack)**.** *The remaining alternatives have been separated into two non-renamable theorem inputs. The no-black-box external line is*

```
ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR NewAutomorphicDispersionProof.
```

This means an item-by-item theorem match, or a new proof, for the exact full-S non-AP WFD Kloosterman large-sieve object with

$$C \asymp P/\log^{O(1)} P, \qquad S \asymp P, \qquad H \leq P/\log^{O(1)} P,$$

well-factorable modulus weights, divisor-bounded S-weights, c-dependent completed residue weights, no AP-source lift, no hidden centering/projection loss, and saving at the natural WFD scale by $\log^{-A} P$ for every fixed A.

The internal self-contained line is

```
FineSignedSourcePackage
AND SelfContainedTailLog4RKSLogReciprocalKloostermanFixedSaving.
```

The second factor is the D-structure/Rankin replacement package reduced to the Tail-log4/RKS-log reciprocal Kloosterman fixed logarithmic saving; the first is the pre-Cauchy signed-source package for the same formal unit. The current corpus proves neither line unconditionally.

Two replacement-line terminal attack. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-
terminal-attack-router.md.
```

It imports the dual external/internal closure split, the full-S non-AP WFD theorem-match matrix, the true-remainder cut, the D-structure/Rankin author-remainder split, and the strict self-contained replacement hardpoint.

For the no-black-box external line, the certificate checks the available families by object, weights, window, modulus range, smoothing/projection, error saving, constants, and conclusion. BFI and Maynard supply AP well-factorable distribution inputs; without an AP-source lift they do not become the present uncentered non-AP object. DI/Kuznetsov is the right post-completion spectral technology, but the current source still requires a proof for c-dependent completed residue weights and a de-completion step with no hidden projection loss. Friedlander–Iwaniec remains a parity-breaking technology model, not a ready theorem for W_{full}. Hence the external line is not closed by citation names; it is closed only by the displayed exact primary-source match or by a new automorphic/dispersion theorem for the same object.

For the self-contained line, the D/AB structured shell, finite verification format, and Rankin pass-or-return ledger have been reduced to a checkable author-side package. The true analytic residue is the Tail-log4/RKS-log bilinear or multilinear reciprocal Kloosterman saving, with enough fixed logarithmic saving to survive the downstream loss ledger. A pointwise single-frequency Baker-type input does not replace this coherent average. Together with the fine signed-source package, this is the current internal frontier. The external-lemma version remains conditionally closed only under `AcceptedFullSKLSExtExternalContract` and `DStructureTailLog4FiniteRankinFullLedgerIndependentAccept` therefore `row_column_unconditional_closed=false` is preserved. □

Theorem 7.379 (RNRS/ExactUV synchronization of the two replacement lines). *After importing the RNRS/Rudnev reciprocal-energy transfer, the internal replacement-line obstruction is no longer the Tail-log4/RKS-log reciprocal Kloosterman saving. That author-side analytic core is closed in the ledger. The internal self-contained source-side normal form is now*

```
ExactCleanCoreFullSNonAPWFDSourceEntropy,
```

with support form

```
ActualNoncanonicalExactUVSupportLowerBound,
```

and next layer

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

The no-black-box external normal form is

```
ModulusDependentCompletedFullSKLSInput
OR ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR NewAutomorphicDispersionProof.
```

Thus the two replacement lines remain open, but their current hard points are ExactUV/source entropy on the internal side and completed modulus-dependent FullS–KLS on the external side.

RNRS/ExactUV synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-after-
rnrs-exactuv-sync-router.md.
```

It imports the previous two-line split, the RKS-log RNRS transfer closure, the RKS23 final promotion audit, the ExactUV support attack, the ExactUV support terminal attack, the clean-core support-incidence attack, and the clean-core terminal normal-form certificate.

The RNRS transfer gives the required fixed logarithmic saving in the RKS-log collar by reducing the balanced reciprocal Kloosterman block to reciprocal additive energy and then applying the Rudnev point-plane incidence energy input. Hence the previous open label `SelfContainedTailLog4RKSLogReciprocalK` is superseded as the latest internal blocker. This does not create an independent promotion-acceptance event, and it does not prove the source-side theorem.

The remaining internal source obstruction is exact clean-core source entropy: ordinary squarefree abundance, K4/K6 incidence, canonical support transfer, and generic WFD support models have all been filtered out. What remains is a noncanonical exact-layer statement: the clean-core layer must admit enough Buchstab products, the actual alpha/delta coefficients on those products must be nonzero in the pre-Cauchy source, and every thin or rejected block must return to a named exit.

On the external side, the standard form is not a generic DI/BFI citation. It is the completed, modulus-dependent FullS–KLS input with c-dependent completed residue weights, no AP-source lift, no hidden centering or projection loss, and de-completion errors still inside the saved budget. The older external-lemma conditional theorem remains valid only under the explicit package

`AcceptedFullSKLSExtExternalContract` and `DStructureTailLog4FiniteRankinFullLedgerIndependentAcc`

Therefore the synchronization narrows the current true hard points while preserving `row_column_unconditional_cl`

□

Theorem 7.380 (Exact-layer/completed-KLS attack of the two replacement lines)**.** *After the RNRS/ExactUV synchronization, the internal replacement line reduces further from*

`CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn`

to the origin-ledger chain

```
CleanCoreExactCoefficientPathPartitionNoCancellationAndThinReturn
CleanCorePreCauchyCoefficientSourceLawAndReturn
CleanCoreOriginalCoefficientGenerationLedgerAndReturn.
```

The no-black-box external line reduces from the completed FullS–KLS input to

`CDependentResidueWeightSpectralCancellationInput,`

or else to an exact primary-source theorem match or a new automorphic dispersion theorem for the same object. The external-lemma version remains conditional on the accepted FullS–KLS contract and the independent D-structure/Rankin acceptance gate. The internal self-contained version and the no-black-box external version are not yet unconditionally closed.

Exact-layer/completed-KLS attack of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-exact-layer-
completed-kls-attack-router.md.
```

It imports the RNRS/ExactUV synchronization, the clean-core support-incidence attack, the clean-core layer-transfer path router, the path-source firewall, the pre-Cauchy source-law atomization, the completed-weight spectral-gap router, the three-final-atoms attack, and the D-structure/Rankin promotion audit.

On the internal side, layer admission is not a plain squarefree count, a K4/K6 incidence statement, a canonical-source transfer, or a generic WFD template. It must be proved before Cauchy, Type/Fourier, completion, and pushforward, on the actual clean-core alpha/delta coefficients in one formal unit. The required ledger lists every summand with its source class, branch key, u/v map, sign, and local factor; proves a polylogarithmic path budget and same-key nonzero/sign-split discipline; and returns thin, rejected, missing-source, or cancellation failures to named PDEC/SAE/ColumnCRT/CleanK or external spectral exits. This is the current minimal internal author-side task.

On the external side, completing the full-S variable modulo c removes the length obstruction but leaves completed residue weights

$$B_{c,x} = \sum_k \beta_{x+kc}.$$

These weights depend on the modulus c, are not assumed flat or centered, and must be handled together with well-factorable λ_c and smooth ω_h over the c, h families. Pointwise Weil plus L^2, the ordinary large sieve, AP-source lift, or hidden projection/centering does not match the required uncentered non-AP target. A usable external theorem must also keep gcd, smoothing, endpoint, and de-completion losses inside the $B(A)$ budget and deliver NaturalWFDScale/$\log^A P$ for every fixed $A > 0$.

Thus the replacement lines have been narrowed to two explicit interfaces: the internal origin-generation ledger, or the external c-dependent completed-residue spectral cancellation theorem, with the final D-structure/Rankin promotion discipline still carried. No unconditional row/column closure is asserted at this layer. □

Theorem 7.381 (Deep terminal synchronization of the two replacement lines)**.** *The exact-layer/completed-KLS frontier does not stop at the two labels*

```
CleanCoreOriginalCoefficientGenerationLedgerAndReturn
CDependentResidueWeightSpectralCancellationInput.
```

Importing the existing downstream ledgers synchronizes the no-black-box external line to a same-object DI/BFI/Kuznetsov theorem match, and synchronizes the internal line to a common pre-Cauchy signed table whose first active signed-seed atom is

```
PhiLPFOffDiagonalOrderedSemiprimeFirstSeedSignedTableBeforePushforward.
```

This synchronization removes another layer of relabelling but does not close the row/column theorem.

Deep terminal synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-deep-
terminal-sync-router.md.
```

On the external side, the c-dependent completed-residue input is first identified with the finite-Fourier/BWFD/BSC/KFLS core. If it is not accepted as an external theorem, it enters the NC-BLK branch. The NC-BLK branch is a source-entropy or external-dispersion branch, not a permission to import the canonical RIW/Buchstab source silently. The exact-entropy route further reduces to support and capacity compatibility, and then to a strengthened source anti-atom. Since the unrestricted generic anti-atom is already ruled out by the moving-delta model, the non-cyclic no-black-box external target is

```
ExternalDIBFIKuznetsovDispersionTheoremMatch.
```

Such a match must still be for the same completed full-S non-AP WFD object, with $B_{c,x} = \sum_k \beta_{x+kc}$, well-factorable λ_c, smooth ω_h, no AP-source lift, no hidden centering or projection, and de-completion and endpoint losses inside the logarithmic saving budget.

On the internal side, the origin ledger is not allowed to read signed coefficients backward from payment, Φ-LPF support, or a downstream row table. The row-origin ledger therefore goes through the non-circular pre-Cauchy signed coefficient emission kernel. The Φ-LPF bucket identity pays only unsigned ownership, support, capacity, and the $p > \sqrt{N}$ zero-mass layers. It does not produce signs, local factors, branch orientations, row mass, complete keys, fixed-key multiplicity, or a short-interval positivity theorem. After importing the common-table rebase and the semiprime diagonal split, the current internal basis is the off-diagonal first-seed signed table, the internal prime-adjoin signed transition law, the source three-atom package, row-mass/support and key ledgers, the beta-sieve/sawtooth tail package, rate preservation, and the D-structure/Rankin acceptance gate.

Thus this layer is a genuine non-cyclic synchronization: old labels are replaced by strictly downstream obligations, and signed-lane self-proof is blocked. It is not an unconditional proof. The external theorem match, the internal signed tables and side ledgers, the beta-sieve/sawtooth tail, rate preservation, and D-structure/Rankin promotion remain open. □

Theorem 7.382 (Latest true-remainder synchronization of the two replacement lines)**.** *After the deep terminal synchronization, neither replacement line may be left at its previous coarse label. The no-black-box external line is reduced to a same-object Full-S non-AP WFD theorem match or to an actual noncanonical source support/capacity theorem. The internal self-contained line is reduced past the off-diagonal seed labels to the acyclic NC-BLK/source anti-atom and the self-contained beta-sieve/sawtooth tail package, with the source, joint, signed-survival, row-mass, key, rate, and D-structure gates still carried. The external lemma version is only conditionally closed under the accepted FullS-KLS external contract plus D-structure/Rankin acceptance.*

Latest true-remainder synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-latest-
true-remainder-sync-router.md.
```

On the external side, the theorem-match matrix checks the present object against ordinary short-interval inputs, FI parity-breaking models, BFI AP dispersion, DI/Kuznetsov spectral large-sieve technology, Maynard/GPY machinery, and the exact FullS-KLS external contract. The ready-made primary sources do not directly cover the uncentered, no-projection, completed full-S non-AP WFD object. After the true-remainder cut removes AP-source lift and the generic source anti-atom, the no-black-box external basis is

```
(ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
OR NewAutomorphicDispersionProof)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If one accepts an external lemma, the conditional package is instead

```
AcceptedFullSKLSExtExternalContract
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is a conditional external-input theorem, not a derivation from the primary DI/BFI sources.

On the internal side, the Φ-LPF/off-diagonal seed route has been stripped through tuple fields, pure-pair atoms, Ferrers support, no-swap ordering, edge-local labels, trace keys, new payload, and source entropy. The payload/source-entropy route returns to the joint declaration line and is therefore a self-proof loop unless a fresh independent declaration is supplied. The fresh-declaration taxonomy reduces the remaining black box to actual moving-block NC-BLK. The moving-block terminal split and the strict Kuznetsov/DLS audit then show that KZ-A–D bookkeeping is exhausted, while KZ-E well-factorable dispersion returns to the acyclic NC-BLK/source anti-atom or to the corresponding external contract. The high-model tail is still the self-contained beta-sieve/sawtooth package.

Consequently the latest internal self-contained basis contains the acyclic NC-BLK/source anti-atom or its named non-cyclic alternatives, the self-contained Rosser–Iwaniec beta-sieve construction, the explicit main coefficient and sawtooth tail bounds, the joint emitter rows and identity ledgers, actual source entropy and ExactUV fiber control, signed row survival, row-mass normalization, complete and fixed-key ledgers, rate preservation, and D-structure/Rankin acceptance. The Φ-LPF identity remains a precise unsigned support/capacity ledger; it does not produce signed coefficients, KZ-E log-saving, or short-interval positivity. Thus this synchronization removes another layer of relabelling and a constructor payload loop, but it does not prove the row/column theorem unconditionally. □

Theorem 7.383 (Atomized hard package for the two replacement lines)**.** *The latest true-remainder package can be sharpened further. Internally, the NC-BLK/source anti-atom is not a terminal name but reduces to a forward pre-Cauchy source-root packet or to the global terminal/PDEC-scope route. The beta-sieve tail has its finite lower-weight construction and pointwise dominance closed, and its continuous $B = 3$ main-term surplus closed; the remaining beta-sieve atom is a uniform discrete prime-sum error for $P \geq 100000$. The exact sawtooth atom is equivalent to a quadratic-arc, then signed near-square strip, discrepancy; near-square strip failures enter the global terminal/PDEC-CleanKLS gate or the canonical terminal absorption boundary. The external line remains a FullS-KLS/no-projection theorem-match or accepted external-contract line.*

Atomized hard package for the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-atomized-
hard-package-router.md.
```

The NCBLK/source anti-atom frontier imports the generic WFD anti-atom no-go, the actual-source seed split, the hypothetical zero-row seed no-go, the independent pre-Cauchy identity taxonomy, the moving-block return, and the source-declaration downstream split. Hence a productive internal proof must first supply a forward acyclic pre-Cauchy source-root packet, with primitive summand rows, branch/sign/local-factor data, source entropy, fixed ExactUV fiber control, and named return partition. If this packet is not supplied, the branch returns to global PDEC/sparse terminal exclusion or same-set PDEC scope; it is not a new fourth endpoint.

The beta-sieve tail is no longer a single appendix black box. The explicit descending prime-word lower weights have been fixed with squarefree support, sign, and $d < P$; the Buchstab-tree parity-pruning argument proves pointwise lower-bound dominance. The 99% coefficient atom is therefore reduced to two micro-inputs. The continuous input is closed because $\alpha = 0.43$ gives $s = 1/\alpha \in (2, 3)$, where the linear lower-sieve formula gives the required one-percent surplus. The remaining self-contained beta atom is the uniform discrete prime-sum error

```
B3DiscretePrimeSumUniformErrorPGe100000.
```

The exact floor remainder is also not left as a formal sawtooth. For $d > 1$, with $t = P \bmod d$, the floor error is exactly the indicator that $\pm t^2 \bmod d$ lies in the interval $(0, t)$, minus t/d. Lifting $P = hd + t$ turns this into a signed near-square strip condition. The near-square strip terminal-admission and canonical-absorption ledgers show that persistent strip failure enters the existing PDEC/CleanKLS or canonical terminal boundary. Thus the current internal atomized basis is governed by the forward source-root packet, the discrete beta prime-sum error, the strict terminal/PDEC-CleanKLS scope, the joint/source/signed/key ledgers, rate preservation, and D-structure/Rankin acceptance. This sharpens the worklist, but does not close the theorem unconditionally. □

Theorem 7.384 (B=3 deep synchronization of the two replacement lines)**.** *In the atomized internal basis, the label `B3DiscretePrimeSumUniformErrorPGe100000` is not the latest frontier. The existing*

B=3 ledgers reduce it to an external Dusart/Mertens route which reaches the D-structure/Rankin gate, and to a fully self-contained route whose remaining analytic inputs are an explicit zero-free-region theta envelope and a Meissel–Mertens constant interval. This reduces the beta-sieve hard point but does not close the source-root, PDEC/CleanKLS, FullS theorem-match, or D-structure/Rankin gates.

B=3 deep synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-b3-
deep-sync-router.md.
```

The discrete prime-sum certificate already rejects finite checkpoint evidence as a uniform proof and replaces the discrete atom by a prime-word Stieltjes ledger and an alternating boundary remainder. The prime-harmonic/Mertens envelope then splits the Stieltjes side into a finite $x < 286$ step ledger and a high-x explicit reciprocal-prime Mertens envelope. Accepting the Dusart/Rosser–Schoenfeld type external Mertens estimate closes that high-x envelope as an external input. On the boundary side, the 20000-anchor promotion and the delay-kernel bounded-variation multiplier close the one-percent $B = 3$ budget for the external Mertens route. Therefore this route has already reached

```
DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the proof is required to be fully self-contained, the external Mertens tail cannot be used. The self-contained Mertens-tail ledger reduces that tail to the finite $10372 \leq x < 20000$ prime-reciprocal step ledger, partial summation from a theta envelope, an explicit zero-free-region theta bound from $x \geq 20000$, and a Meissel–Mertens constant interval at the same anchor. Thus the self-contained B=3 replacement is

```
B3LowPrimeStepFiniteLedgerXLt286PGe100000
AND B3FinitePrimeReciprocalStepLedger286To10371PGe100000
AND B3FinitePrimeReciprocalStepLedger10372To19999PGe100000
AND PrimeReciprocalPartialSummationFromThetaEnvelopeClosed
AND SelfContainedExplicitZetaZeroFreeRegionThetaEnvelopeXGe20000
AND SelfContainedMeisselMertensConstantIntervalLedgerAt20000
AND B3RosserFaceDictionaryClosedAlpha043
AND B3Anchor20000BoundaryVariationBudgetClosedAlpha043.
```

Consequently the current internal attack list should no longer carry the coarse $B = 3$ discrete label. It should carry the forward source-root packet, the self-contained explicit-PNT/Mertens tail inputs if no external Mertens theorem is accepted, the strict PDEC/CleanKLS terminal scope, and the D-structure/Rankin gate or its self-contained replacement. This is a proper narrowing of the non-circular frontier, not an unconditional closure. □

Theorem 7.385 (Rate-bearing Mertens tail synchronization of the two replacement lines)**.** *After the B=3 deep synchronization, the self-contained explicit PNT/Mertens tail is itself superseded by the strict rate-bearing Mertens synchronization. The direct internal Dusart theta/PNT envelope, the unsmoothed Perron constant layer, and the Meissel–Mertens B_1 interval have all been imported as self-contained certificates. Hence the active internal replacement line no longer carries the PNT/Mertens tail as a hard point; it returns to the source-root, PDEC/CleanKLS, rate-preservation, and D-structure/Rankin gates.*

Rate-bearing Mertens tail synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-rate-tail-
mertens-closed-sync-router.md.
```

The strict rate-bearing tail ledger imports the later analytic certificates: the direct internal Dusart theta/PNT envelope is closed by the P5.1 self-contained synchronization, the unsmoothed Perron constant ledger is closed with finite-T reserve $C = 12128$, and the Meissel–Mertens B_1 interval is closed by an Euler-product interval certificate. Together with the finite reciprocal-prime step ledgers and the partial-summation interface, these inputs remove the old self-contained Mertens-tail atom from the active basis.

Thus the internal replacement line is condensed to

```
ForwardAcyclicPreCauchySourceRootPacketOrNamedReturn
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance,
```

up to the already named terminal/source alternatives and source/joint/key side ledgers. The external replacement line is unchanged: an external lemma version is conditional on the accepted FullS-KLS contract and D-structure acceptance, while a no-black-box version still requires the same-object Full-S non-AP WFD theorem match, an actual source-capacity theorem, or a new automorphic/dispersion proof. This step closes an analytic tail package inside the frontier; it does not close the row/column theorem unconditionally. □

Theorem 7.386 (Source-root/no-cycle synchronization of the two replacement lines)**.** *After the rate-bearing Mertens synchronization, the internal replacement line should no longer carry* `ForwardAcyclicPreCauc` *as an active proof atom. The existing strict source-root, PDEC-scope, KZ no-cycle, and KZ-E source-bridge certificates replace that atom by the fresh A1 source-admission, same-set PDEC, or new-joint interfaces, together with the rate-bearing PDEC, rate-preservation, and D-structure/Rankin gates.*

Source-root/no-cycle synchronization of the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-source-
root-nocycle-sync-router.md.
```

The strict forward source-root certificate shows that the source-root route returns to a signed/source-rank/alpha terminal cycle; it is not an independent non-circular proof outlet. The direct same-set PDEC branch has also been audited: its canonical tools are available, but the acyclic noncanonical branch still lacks a same formal-unit scope certificate for the bad windows, the $U_{\mathrm{CRT}}/L_{\mathrm{PDEC}}$ comparison, and mass pushforward. The remaining KZ/DLS branch cannot reuse the NC-BLK/source-root projection. With that reuse forbidden, KZ-E is reduced to a direct well-factorable dispersion log-saving, and the self-contained route then reduces to A1 clean-branch canonical source admission.

Consequently the source-root part of the internal replacement line is now

```
A1CleanBranchCanonicalSourceAdmission
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact.
```

The full internal basis must still carry

```
PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage),
```

plus the source/joint/key ledgers already recorded in the machine certificate. The external lemma version is still only conditional on `AcceptedFullSKLSExtExternalContract` and independent D-structure/Rankin acceptance. A no-black-box external version still needs the same-object Full-S non-AP WFD theorem match, an actual source-capacity theorem, or a new automorphic/dispersion proof. Thus this synchronization removes an obsolete activity label; it does not close the theorem unconditionally. □

Theorem 7.387 (A1 source-admission absorption in the two replacement lines)**.** *In the current two replacement lines, the label `A1CleanBranchCanonicalSourceAdmission` is not an independent global contradiction. It is a scoped canonical/generic branch statement. After importing the strict source-admission absorption and post-admission macrocycle certificates, the active internal basis must replace that label by outside-cycle seed, PDEC, or payload inputs, while preserving the rate-bearing PDEC, rate-preservation, and D-structure/Rankin gates.*

A1 source-admission absorption in the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-source-
admission-absorption-sync-router.md.
```

The A1 canonical branch-admission audit states a branch discipline rather than a numerical contradiction: the canonical RIW/Buchstab source branch may be treated by the internal support chain, but the generic or noncanonical well-factorable branch must be externalized or returned. Thus placing `A1CleanBranchCanonicalSourceAdmission` as a standalone OR outlet would silently upgrade a scoped branch statement to a global exclusion.

The post-admission macrocycle certificate also shows that continuing inside the A1/PDEC/new-joint/KZ/KZ-E loop gives no descent. A non-circular advance must instead provide one of the outside-cycle inputs

```
AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput
OR AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewPrimitiveJointPayloadArtifactOutsideSignedLaneCycle
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact.
```

The remaining carrier part is unchanged:

```
PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

External FullS-KLS and no-projection Kuznetsov/DI/BFI inputs remain conditional unless their theorem match is supplied on the same completed object with the same weights, windows, projections, and error budget. This step removes another pseudo-terminal label; it does not prove the outside cycle input, the rate packet, or the promotion gate. ☐

Theorem 7.388 (Seed/payload saturation in the two replacement lines)**.** *After the A1 source-admission absorption, the labels `AcyclicSeedCycleCutPrimitiveBasisAndCoefficientSourceInput` and `NewPrimitiveJointPayloadArtifactOutsideSignedLaneCycle` are not standalone non-circular proof outlets. The first is saturated by the strict seed-cycle-cut frontier into same-set PDEC or a new joint formula. The second must be instantiated as a pre-Cauchy atomic signed payload/trace artifact and then must pay the source-rank/no-collapse atoms.*

Seed/payload saturation in the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-seed-
payload-saturation-sync-router.md.
```

The strict seed-cycle-cut certificate rejects sequential word/coefficient splitting and shows that continuing through terminal descent returns to a macrocycle. Therefore the seed-cycle-cut label only transfers the burden to

```
AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact.
```

It is not a separate contradiction.

The signed-lane certificate closes the dependency cycle through common source declaration, built-in signed pairing, branch trace, signed payload, origin identity, and back to the common packet. Hence an outside signed-lane payload label has content only if it supplies a new pre-Cauchy atomic signed payload/trace. The payload/source-atom alignment and preterminal source-rank certificates then force the source-rank package

```
ActualPreCauchySourceDomainAbsoluteEntropyLedger
AND CompletePrimitiveEmitterKeyPartitionLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger.
```

Thus the internal line is updated to

```
(AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR (ActualPreCauchySourceDomainAbsoluteEntropyLedger
AND CompletePrimitiveEmitterKeyPartitionLedger
AND FixedKeyExactUVLocalMultiplicity01Ledger)
OR AcyclicNoncanonicalTerminalReturnWellFoundedDescentCertificate
OR ExactExternalDIBFIKuznetsovNoProjectionCertificate_FOR_NONCIRCULAR_KZ_ONLY)
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The external FullS-KLS lemma line remains conditional on the accepted external contract and independent D-structure/Rankin acceptance. A no-black-box external line still requires a same-object Full-S theorem match, an actual source-capacity theorem, or a new automorphic/dispersion proof. This step removes two more pseudo-outlets; it does not prove PDEC, the new joint formula, source rank, rate preservation, or D-structure. □

Theorem 7.389 (Source-rank/terminal kernel synchronization in the two replacement lines)**.** *In the current two replacement lines, the source-rank/no-collapse package and the noncanonical terminal descent label are not independent final proof outlets. Their existing strict downstream certificates synchronize them to the same formal-unit pointwise primitive alpha/delta kernel table.*

Source-rank/terminal kernel synchronization in the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-sourcerank-
terminal-kernel-sync-router.md.
```

The source-domain entropy atom has already been reduced to signed row emission, row-mass normalization, and row support before exact-UV projection. The complete-key atom needs an actual pre-Cauchy emitter source table and a trace/key budget. The fixed-key multiplicity bridge likewise needs the same actual source table, complete key, and fixed-key exact-UV local multiplicity. Finally, the terminal descent synchronization sends the terminal leaf chain through the same source-rank/no-collapse package.

Thus the common non-post-hoc object is

```
PointwiseSameFormalUnitPrimitiveAlphaDeltaKernelTableWithNonzeroRankCertificate,
```

which is itself reduced to

```
AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows.
```

Consequently the internal replacement line is updated to

```
(AcyclicSameSetScopeMatchForDirectPDECCapDualCertificate
OR NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR (AlphaRowAnchorPhaseEmissionFormulaLedger
AND IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger
AND SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows)
OR ExactExternalDIBFIKuznetsovNoProjectionCertificate_FOR_NONCIRCULAR_KZ_ONLY)
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

This synchronization removes coarse labels only. It does not prove the alpha row formula, the arithmetic identity, same-unit rank/multiplicity, same-set PDEC, the new joint formula, the rate packet, or D-structure. □

Theorem 7.390 (Terminal-leaf/source-bridge synchronization in the two replacement lines)**.** *After importing the strict three-leg kernel return, the nonrecursive pointwise table contract, and the current terminal-leaf firewall, the source-rank/kernel branch in the two replacement lines no longer has the three pointwise kernel inputs as a standalone frontier. Without a new actual joint constructor formula, the old joint-alpha route returns to the terminal leaf and then to canonical lock or an actual source bridge.*

Terminal-leaf/source-bridge synchronization in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-terminal-leaf-
source-bridge-sync-router.md.
```

The strict three-leg return certificate shows that separately attacking the alpha row formula, the pre-Cauchy arithmetic identity, and the same-unit rank/multiplicity certificate forms a fixed point: the alpha and weight legs return to the PDEC/CleanKLS terminal gate, while the rank leg asks again for the same primitive source table. The nonrecursive pointwise table field contract then pins the missing productive field as an actual joint alpha/delta constructor rule. Existing joint-alpha and same-row certificates show that the old route for that rule returns to the signed-source fixed point unless a genuinely new formula is supplied.

The current terminal leaf has already reduced all materialized PDEC and sparse instances to

```
AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR NoncanonicalFullSComplementLegalClosureMode.
```

The noncanonical legal mode is filtered further by the actual-source bridge:

```
A1CleanBranchCanonicalSourceAdmission
OR ExactCleanCoreFullSNonAPWFDSourceEntropy.
```

Consequently the updated internal replacement-line basis is

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR A1CleanBranchCanonicalSourceAdmission
OR ExactCleanCoreFullSNonAPWFDSourceEntropy)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The external lemma version is still only conditional on

```
AcceptedFullSKLSExtExternalContract
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance,
```

and the no-black-box external version still needs a same-object Full-S theorem match, an actual source-capacity theorem, or a new automorphic/dispersion proof. This theorem is therefore a frontier synchronization, not an unconditional proof of the row/column theorem. □

Theorem 7.391 (Alpha-return/source-bridge synchronization in the two replacement lines)**.** *In the two replacement lines, the terminal source-admission and exact source-entropy labels left by the terminal-leaf synchronization are not standalone non-circular exits. The A1 source-admission label is absorbed by the scoped canonical branch, and the exact source-entropy route returns to the alpha/terminal backedge unless an independent actual-source bridge is supplied before entering that route.*

Alpha-return/source-bridge synchronization in the two replacement lines. The certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-alpha-return-
bridge-sync-router.md.
```

It imports the strict terminal-atoms-to-alpha-return bridge certificate. That certificate shows that

```
A1CleanBranchCanonicalSourceAdmission
```

is only a scoped canonical-branch statement. It does not close the unrestricted noncanonical terminal family. The same certificate also follows the exact source-entropy route through post-Mertens synchronization, the same-formal-unit kernel identity, the pointwise primitive table, the nonrecursive field contract, and the terminal leaf. That route is therefore a backedge, not a descent measure.

The non-circular internal front is consequently

```
NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn.
```

Keeping the parallel rank, model, rate, and promotion gates, the updated internal replacement-line basis is

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR IndependentActualSourceBridgeNotFactoredThroughAlphaReturn)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The external lemma version and the no-black-box external version are unchanged from the preceding synchronization. This step rules out two standalone labels; it does not prove the new joint formula, canonical lock, independent source bridge, exact-UV incidence, rate preservation, or D-structure/Rankin promotion. □

Theorem 7.392 (Source-identity/antiatom synchronization in the two replacement lines)**.** *In the two replacement lines, the independent actual-source bridge left after the alpha-return synchronization is not a standalone terminal proof label. If it is to be used as a non-circular proof outlet, its actual content must be an actual full-S source identity or a strengthened actual-source anti-atom theorem.*

Source-identity/antiatom synchronization in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-source-identity-
antiatom-sync-router.md.
```

It imports the strict independent-source outside-loop attack. That attack does not prove the bridge; it removes the bridge as a coarse name. Once the alpha-return, exact-UV, pair-energy, and old joint-constructor return routes are excluded as proof routes, the remaining non-circular content is

```
ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource.
```

The first alternative requires a pre-Cauchy identification of the actual full-S non-AP source with the canonical RIW/Buchstab decision-tree source. The second alternative requires a strengthened anti-atom theorem for the actual final capacity measure. The Phi-LPF and CRT identities give exact unsigned support and capacity bookkeeping, but they do not by themselves produce signed moving-source anti-atoms or a pre-Cauchy source identity.

Keeping the parallel new-joint, canonical-lock, rank, model, rate, and promotion gates, the updated internal replacement-line basis is

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
OR ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The external lemma version remains conditional on the accepted FullS-KLS contract and independent D-structure/Rankin acceptance. A no-black-box external version still needs a same-object Full-S theorem match, an actual source-capacity theorem, or a new automorphic/dispersion proof. This theorem therefore narrows the true frontier; it is not an unconditional proof of the row/column theorem. □

Theorem 7.393 (Canonical exact-certificate synchronization in the two replacement lines)**.** *In the two replacement lines, the canonical-lock alternative left after the source-identity/antiatom synchronization is not a standalone terminal proof label. It can be used only as an exact same-set canonical promotion certificate; if that certificate is missing, the branch returns to a new actual clean-core source-entropy theorem.*

Canonical exact-certificate synchronization in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-canonical-exact-
certificate-sync-router.md.
```

It imports the strict canonical-lock nonrecursive-exit attack. That attack refines

```
AcyclicTerminalCanonicalLockToCanonicalSourceBoundary
```

to the alternative

```
AcyclicCanonicalExactSameSetPromotionCertificate
OR NewActualCleanCoreFullSNonAPWFDSourceEntropyTheorem.
```

The exact same-set certificate consists of the five source-preserving ledgers

```
AcyclicSeedCanonicalBranchAdmissionBeforeCauchy
AND AcyclicSeedFiniteMeasurableFactorMapAndWeightIdentity
AND AcyclicSeedNoSourceReplacementOrPayloadCreation
AND TerminalCertificateSameSetPushforwardIdentity
AND NoNoncanonicalPayloadSurvivesCanonicalProjection.
```

These ledgers must hold before the downstream Cauchy, payment, and terminal pushforward steps are used as evidence. If any ledger is absent, canonical-lock is not an admissible proof outlet; a direct PDEC/CleanKLS label would again be only a return label, not a source-level proof.

The updated internal replacement-line basis is therefore

```
(NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact
OR AcyclicCanonicalExactSameSetPromotionCertificate
OR NewActualCleanCoreFullSNonAPWFDSourceEntropyTheorem
OR ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

This synchronization removes the canonical-lock coarse label. It does not prove the five same-set ledgers, the new source-entropy theorem, the new joint formula, source identity, strengthened anti-atom, exact-UV incidence, rate preservation, or D-structure/Rankin promotion. □

Theorem 7.394 (New-joint six-field synchronization in the two replacement lines)**.** *In the two replacement lines, the remaining new-joint label is not an admissible coarse proof outlet. After importing the strict terminal obligation audit, it is exactly the six-field actual joint alpha/delta constructor artifact*

```
NewActualJointAlphaDeltaSixFieldConstructorArtifact.
```

The six fields are source tuple domain, joint row and formal unit, basis-word formula, signed-coefficient formula, exact UV/Φ pairing, and budget/failure return. Old joint rules, terminal descent, and pair-energy routes return to fixed points or macrocycles and cannot replace this artifact.

New-joint six-field synchronization in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-new-joint-
sixfield-sync-router.md.
```

It imports the canonical exact-certificate synchronization together with the strict new-joint terminal-obligation certificate. The earlier label `NewExplicitActualJointAlphaDeltaConstructorFormulaArtifact` is therefore sharpened to a pre-Cauchy six-field formula obligation. The formula must emit the source tuple, row/formal-unit key, basis word, signed coefficient, exact UV/Φ pairing, and budget/return data before downstream Cauchy, payment, or terminal extraction evidence is used.

Thus the current internal replacement-line basis is

```
(NewActualJointAlphaDeltaSixFieldConstructorArtifact
OR AcyclicCanonicalExactSameSetPromotionCertificate
OR NewActualCleanCoreFullSNonAPWFDSourceEntropyTheorem
OR ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
```

```
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The external lemma version remains conditional on the accepted FullS–KLS contract and the D-structure/Rankin promotion gate; the no-black-box external version still needs the same-object FullS theorem match, an actual source capacity theorem, or a new automorphic/dispersion proof. This theorem removes the new-joint coarse label only; it is not an unconditional row/column proof. □

Theorem 7.395 (Non-cyclic hard attack on the two replacement lines)**.** *The final hard-attack boundary is non-cyclic. The external-lemma version is author-side closed only in the conditional sense*

```
AcceptedFullSKLSExtExternalContract
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If one requires an absolute no-condition external line, the replacement is

```
(ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
OR NewAutomorphicDispersionProof)
AND (DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
OR SelfContainedDStructureTailLog4FiniteRankinProofPackage).
```

The internal self-contained line requires

```
(NewActualJointAlphaDeltaSixFieldConstructorArtifact
OR AcyclicCanonicalExactSameSetPromotionCertificate
OR NewActualCleanCoreFullSNonAPWFDSourceEntropyTheorem
OR ProveActualFullSNonAPSourceIsCanonicalRIWBuchstab
OR FullSNonAPStrengthenedSourceAntiAtomForActualSource)
AND ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet
AND SelfContainedDStructureTailLog4FiniteRankinProofPackage.
```

No unconditional row/column theorem is promoted by the current corpus.

Non-cyclic hard attack on the two replacement lines. The hard-attack certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-noncycle-
hard-attack-router.md.
```

It combines the FullS–KLS external-match audit, the primary-source theorem match matrix, the true-remainder cut, the D-structure/Rankin author-remainder split, and the new-joint six-field synchronization.

The reason the two lines cannot be collapsed is structural. Euler's product discipline is used here as a source-before-pushforward rule: multiplicative source data must be emitted before payment data is read. Gauss's congruence discipline is used as a same-set CRT rule: formal-unit keys, phases, and canonical promotion must refer to the same object. Riemann's spectral discipline is used as a placement rule: explicit or Kuznetsov-type estimates may bound completed sums only after the relevant signed coefficients have been generated. Under these three disciplines, DI/BFI/Kuznetsov estimates cannot manufacture the missing pre-Cauchy six-field constructor, and Phi–LPF/CRT identities cannot manufacture signed source coefficients from unsigned support.

Therefore the external lemma branch is a closed conditional theorem boundary, not an absolute unconditional theorem; the no-black-box branch remains a same-object theorem-match or new-proof task; and the internal branch remains a conjunction of the source-side alternative, exact-UV, model, PDEC/CleanKLS, rate, and self-contained D-structure/Rankin packages. This closes the non-cyclic classification while preserving `row_column_unconditional_closed=false`. □

Theorem 7.396 (Common unconditional kernel of the two replacement lines)**.** *The common obstruction shared by the absolute external line and the internal self-contained line is not the FullS spectral front and not the six-field source front. It is the self-contained D-structure/Rankin package*

```
SelfContainedDStructureTailLog4FiniteRankinProofPackage.
```

Thus the conditional external lemma basis

```
AcceptedFullSKLSExtExternalContract
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

must not be promoted to an absolute author-side proof. The absolute external author-proof line requires

```
(ExactPrimarySourceFullSNonAPWFDKLSTheoremMatch
OR ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
OR NewAutomorphicDispersionProof)
AND SelfContainedDStructureTailLog4FiniteRankinProofPackage,
```

while the internal self-contained line also requires the source/front gates, exact-UV, model, PDEC/CleanKLS, and rate gates. These gates are not mutually substitutable.

Common unconditional kernel of the two replacement lines. The kernel certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-common-
unconditional-kernel-router.md.
```

It imports the non-cyclic hard-attack boundary, the new-joint six-field synchronization, the FullS–KLS external-match audit, and the D-structure/Rankin author-remainder split. The split shows that the ordinary author-side remainder of the external lemma version is empty only after accepting the external FullS–KLS contract and the independent D-structure/Rankin gate. If those acceptances are not allowed, the D-structure/Rankin gate must be replaced by the self-contained package.

That package has four conjunctive subpackages:

```
SelfContainedDStructureStructuredEHPDDefinitionsAndABReductionProof
AND SelfContainedTailLog4BGOrRKSTailAdapterWithExactTheoremNumbersAndConstants
AND ReproducibleFiniteVerificationArchiveWithHashesAndIndependentRunner
AND SelfContainedFullRankinPassOrReturnLedgerAndDownstreamReturnIntegration.
```

The FullS theorem match can pay only the external spectral front. The six-field/source alternatives can pay only the internal pre-Cauchy source front. The exact-UV, high-model, PDEC/CleanKLS, and rate gates remain internal conjuncts:

```
ActualEmitterExactUVBoundedMultiplicityIncidenceTheorem
AND HighSegmentModelGapAlpha043C3AnalyticLedger
AND PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve_FOR_rate_bearing_packet
AND RatePreservationLedger_FOR_moving_atom_packet.
```

Hence the common kernel boundary is closed, but the common kernel is not proved. The row/column theorem remains unclosed unconditionally. □

Theorem 7.397 (RKS-log final atom in the two replacement lines)**.** *In the common kernel of the two replacement lines, the current active self-contained analytic atom is the Tail-log4 low-spectrum reciprocal Kloosterman input. The structured D-interface, the smooth and middle Tail-log4 ledgers, and the finite Rankin pass-or-return ledger are no longer the active mathematical obstruction. The strict self-contained remainder is*

```
MultilinearReciprocalKloostermanFixedLogSavingForRKSBlocks
OR BakerFrequencyLargeSieveOrDBGAverageReplacement.
```

The external theorem version may close the author-side mathematics after accepting FullS–KLS-ext and the EXT-BG/RKS fixed logarithmic saving input, but that is not a self-contained reproof.

RKS-log final atom in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-rks-log-
final-atom-sync-router.md.
```

It imports the common-kernel certificate, the D-structure/Tail-log4/Rankin compression certificate, the EXT-BG/RKS/Tail-log4 audit, the Structured-EHPD final-interface audit, and the full Rankin ledger inventory. These imports show that the common D-structure package has been compressed to the RKS-log low-spectrum atom:

```
SelfContainedRKSLogReciprocalKloostermanTailLog4Input.
```

The external theorem replacement basis is

```
AcceptedFullSKLSExtExternalContract
AND AcceptEXTBGForRKSLogFixedSaving
AND AuthorSideStructuredEHPDInterfaceAuditClosed
AND ReproducibleFiniteVerificationArchiveWithHashesAndIndependentRunner
AND SelfContainedFullRankinPassOrReturnLedgerAndDownstreamReturnIntegration.
```

The strict author-side basis, with no external theorem acceptance, replaces the last line by the reciprocal Kloosterman reproof or the Baker/DB averaged large-sieve substitute. Burgess multiplicative character estimates cannot replace this step, because they diagonalize product ratios whereas RKS-log is an additive reciprocal-phase problem.

This theorem only fixes the final non-cyclic atom. The internal self-contained line still carries the source/front alternatives, exact-UV, model, PDEC/CleanKLS, and rate gates, and the row/column theorem is not promoted to an unconditional conclusion. □

Theorem 7.398 (RKS-log/RNRS version reconciliation in the two replacement lines)**.** *The RKS-log final atom just isolated is the same Tail-log4/RKS2/RKS3 reciprocal Kloosterman input as the earlier RNRS/Rudnev transfer certificate: the object name matches, the required logarithmic saving is the same* $\log^{-118}$*, and the RNRS dependency direction is acyclic from the strict RKS/Rudnev/RNRS chain into the two replacement-line ledger. Consequently the latest RKS-log open flag is superseded by the RNRS transfer. The current internal active frontier returns to*

```
ExactCleanCoreFullSNonAPWFDSourceEntropy
OR ActualNoncanonicalExactUVSupportLowerBound
OR CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

The external lemma version is still only conditional, and the row/column theorem is not unconditionally closed.

RKS-log/RNRS version reconciliation in the two replacement lines. The reconciliation certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-rks-log-rnrs-
version-reconciliation-router.md.
```

It compares the latest RKS-log final-atom certificate with the strict multilinear reciprocal Kloosterman statement and the RNRS/Rudnev transfer closure. The audit records

```
same_rks_log_object_reconciled=true
same_log118_parameter_reconciled=true
rnrs_imports_exact_statement=true
noncycle_dependency_direction_closed=true
latest_rks_log_open_flag_superseded_by_rnrs=true.
```

Thus the previous RKS-log final-atom theorem is an exact-statement locator, not a remaining active obstruction after RNRS import. The import does not use the two replacement-line conclusion to prove RNRS; it follows the strict RKS23/Rudnev energy path. It also does not pay the FullS theorem-match, source/ExactUV, model, PDEC/CleanKLS, rate, or independent-acceptance gates. Therefore the activity frontier is updated, while unconditional final promotion remains unavailable. □

Theorem 7.399 (ExactUV/source non-cyclic frontier in the two replacement lines)**.** *After the RKS/RNRS reconciliation, the source-side labels*

```
ExactCleanCoreFullSNonAPWFDSourceEntropy,
ActualNoncanonicalExactUVSupportLowerBound,
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn
```

are not independent alternatives. They form one descending source chain. The chain reaches the clean-core origin ledger, but the known origin-ledger–constructor–formula–emitter route returns to the origin ledger and is therefore a source loop. The current non-cyclic internal source frontier is

```
AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
AND ActualNoncanonicalCleanCoreMovingAtomExclusion.
```

The external no-blackbox frontier still requires a same-object Full-S theorem-match, actual source-capacity theorem, or completed-residue dispersion input. The external lemma version remains conditional, and the row/column theorem is not unconditionally closed.

ExactUV/source non-cyclic frontier in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-exactuv-source-
noncycle-frontier-router.md.
```

It imports the RKS/RNRS reconciliation, the exact-layer/completed-KLS attack, the clean-core support-incidence and layer-transfer certificates, the clean-core source-loop cut, the moving-atom sharp-input certificate, and the strict pair-mass and independent-source bridge audits. The audit records

```
rks_log_current_active_obstruction=false
exactuv_entropy_layer_labels_are_ordered_interfaces=true
source_loop_cut_closed=true
acyclic_seed_current_corpus_proved=false
moving_atom_exclusion_current_corpus_proved=false.
```

Thus the ExactUV/source part of the internal proof cannot be closed by cycling among equivalent labels or by recovering a primitive source from a downstream payment skeleton. Any self-contained proof must enter from an acyclic pre-Cauchy source seed and independently exclude the clean-core moving same-(u, v) atom. If one insists on the ExactUV/pair-mass route, the independent input becomes the acyclic seed together with an independent pair L^2/max-atom estimate for that seed. None of these inputs is proved here. □

Theorem 7.400 (Seed/moving-atom/global-terminal synchronization in the two replacement lines)**.** *After the ExactUV/source non-cyclic frontier, the internal source-side remainder*

```
AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
AND ActualNoncanonicalCleanCoreMovingAtomExclusion
```

is synchronized with the strict moving-atom/global-terminal router. The moving atom exclusion is no longer an isolated hard point: if a clean-core moving atom still exists below an acyclic seed, it must enter the global PDEC/sparse terminal ledger and pay the explicit model-gap/finite-DPRC ledger. The updated internal active frontier is

```
AcyclicPreCauchyNoncanonicalPrimitiveSourceSeedAndReturn
AND GlobalPDECorSparseTerminalExclusion
AND ExplicitModelGapAndFiniteDPRCLedger.
```

The fully self-contained version still carries rate preservation and the self-contained D-structure/Tail-log4/finite-Rankin replacement package. The row/column theorem is not unconditionally closed.

Seed/moving-atom/global-terminal synchronization in the two replacement lines. The synchronization certificate is recorded in

```
docs/monograph/prime-matrix-two-replacement-lines-seed-moving-atom-
global-terminal-sync-router.md.
```

It imports the ExactUV/source non-cyclic frontier, the strict moving-atom to global-terminal router, the rate-bearing moving-atom terminal packet frontier, the independent pair-energy attack, and the global terminal family boundary and split certificates. The audit records

```
moving_atom_isolated_hardpoint_removed=true
acyclic_seed_current_corpus_proved=false
global_pdec_sparse_terminal_exclusion_proved=false
explicit_model_gap_and_finite_dprc_ledger_proved=false
rate_preservation_ledger_proved=false
row_column_unconditional_closed=false.
```

Thus this step removes only the isolated status of the moving atom label. It does not prove the acyclic seed, the global terminal exclusion, the model/DPRC ledger, rate preservation, or the self-contained D-structure package. The ExactUV/pair-mass branch also remains non-final, since it reduces to a rate-bearing large-pair packet or returns to the same global terminal ledger. □

Theorem 7.401 (Internal line-by-line audit)**.** *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

Proof. The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. □

Remark 7.402 (No overclaim principle)**.** The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.

Chapter 8

Unconditional K-series Side Theorems for H_P

This chapter records the unconditional, explicit-effective side-theorems for the H_P prime-matrix proposition produced in `docs/monograph/frontier-honest-status-and-true-side-theorems-20260522.md`. Each theorem comes with status, external inputs, and the precise log-saving structure. None of them resolves H_P itself; their value is in providing the strictest available unconditional bounds on the H_P exception sets.

8.1 Setting

Let P be a prime and $A(P)_{i,j} = (i-1)P + j$ for $1 \le i, j \le P$. For $k \in [1, P-1]$ set

$$N_P(k) := \pi(kP + P) - \pi(kP), \qquad M_P(j) := \bigl|\{0 \le m \le P-1 : j + mP \text{ prime}\}\bigr|.$$

The row exception set and column exception set are

$$E(P) := \{k \in [1, P-1] : N_P(k) = 0\}, \qquad E^*(P) := \{j \in [1, P-1] : M_P(j) = 0\}.$$

The propositions $H_P^{\text{row}} \Leftrightarrow E(P) = \varnothing$ and $H_P^{\text{col}} \Leftrightarrow E^*(P) = \varnothing$ remain **Not claimed**: they are equivalent respectively to $g(P^2) \le P$ (a local Cramér-type inequality) and to Linnik constant equal to 2, both 70-year open problems.

8.2 External inputs

External-theorem-closed Statement 8.1 (Montgomery–Vaughan 1973 sharp Brun–Titchmarsh)**.** *For $y \ge 2$ and $x \ge 0$,*

$$\pi(x+y) - \pi(x) \le \frac{2y}{\log y}.$$

For arithmetic progressions, with $\gcd(a, q) = 1$ *and* $q < x$*,*

$$\pi(x; q, a) \le \frac{2x}{\phi(q)\log(x/q)}.$$

External-theorem-closed Statement 8.2 (Rosser–Schoenfeld 1962 explicit PNT bounds)**.**

$$\frac{x}{\log x} < \pi(x) < \frac{1.25506\,x}{\log x}, \quad x \ge 17.$$

External-theorem-closed Statement 8.3 (Dusart 2010 sharper explicit PNT).

$$\pi(x) \geq \frac{x}{\log x - 1}, \quad x \geq 5393.$$

8.3 Theorems K1–K2

Proved-in-text Statement 8.4 (K1: row mean). *$\sum_{k=1}^{P-1} N_P(k) = \pi(P^2) - \pi(P)$ and consequently $\frac{1}{P-1}\sum_{k=1}^{P-1} N_P(k) \sim \frac{P}{2\log P}$ as $P \to \infty$ along primes.*

External-theorem-closed Statement 8.5 (K2: uniform row upper bound). *For every prime $P \geq 5$ and every $k \in [1, P-1]$,*

$$N_P(k) \leq \frac{2P}{\log P}\Big(1 + O\big(\tfrac{1}{\log P}\big)\Big).$$

This follows directly from Montgomery–Vaughan 1973 with $y = P$.

8.4 Theorem K3: trivial PNT-only exception bound

External-theorem-closed Statement 8.6 (K3: PNT-only exception bound). *$|E(P)|/(P-1) \leq 3/4 + o(1)$ as $P \to \infty$.*

Proof. Combine K1 and K2: write $S(P) = \sum_k N_P(k) \sim P^2/(2\log P)$. Since $N_P(k) \leq 2P/\log P$ uniformly,

$$S(P) = \sum_{k \notin E(P)} N_P(k) \leq (P - 1 - |E(P)|) \cdot \frac{2P}{\log P}.$$

Hence $P - 1 - |E(P)| \geq P/4 - o(P)$ and $|E(P)| \leq 3P/4 + o(P)$. □

8.5 Theorem K3' (row, RS1962-effective)

External-theorem-closed Statement 8.7 (K3': explicit row exception bound). *For every prime $P \geq 5$,*

$$|E(P)| < \frac{3P}{4} - 0.37247.$$

Proof. By Montgomery–Vaughan 1973, $N_P(k) \leq 2P/\log P$ for all k. By Rosser–Schoenfeld 1962, $\pi(P^2) > P^2/(2\log P)$ and $\pi(P) < 1.25506\, P/\log P$, so $\pi(P^2) - \pi(P) > P^2/(2\log P) - 1.25506P/\log P$. Combining with $(P - 1 - |E|) \cdot 2P/\log P \geq \pi(P^2) - \pi(P)$:

$$P - 1 - |E(P)| \geq \frac{P}{4} - 0.62753, \qquad |E(P)| \leq \frac{3P}{4} - 0.37247.$$

For integer $|E(P)|$ this is strict. □

8.6 Theorem K3* (column, dual)

External-theorem-closed Statement 8.8 (K3*: explicit column exception bound). *For every prime $P \geq 5$,*

$$|E^*(P)| < \frac{3P}{4}.$$

Proof. Apply Montgomery–Vaughan 1973 in arithmetic progressions: for any $j \in [1, P-1]$, $M_P(j) \le \pi(P^2; P, j) \le 2P^2/((P-1)\log P)$. Sum over j: $\sum_{j=1}^{P-1} M_P(j) = \pi(P^2) - 2$. By Rosser–Schoenfeld, $\pi(P^2) > P^2/(2\log P)$, so

$$(P - 1 - |E^*(P)|) \cdot \frac{2P^2}{(P-1)\log P} \ge \pi(P^2) - 2,$$

giving $P - 1 - |E^*(P)| > (P-1)/4 - 1$ and the claim. □

8.7 Theorem K3"' (row, Dusart-enhanced)

External-theorem-closed Statement 8.9 (K3"': $\log P$-order row improvement)**.** *For every prime $P \ge 79$,*

$$|E(P)| < \frac{3P}{4} - \frac{P}{8\log P}.$$

Proof. Replace Rosser–Schoenfeld's lower bound by Dusart 2010: $\pi(P^2) \ge P^2/(2\log P - 1)$ (valid for $P^2 \ge 5393$, i.e. $P \ge 79$). Then with the same MV1973 row upper bound,

$$P - 1 - |E(P)| \ge \frac{P\log P}{2(2\log P - 1)} - 0.62753 = \frac{P}{4 - 2/\log P} - 0.62753.$$

The geometric-series identity $\frac{1}{1-1/(2\log P)} \ge 1 + \frac{1}{2\log P}$ (valid for $\log P > 1/2$) gives $\frac{P}{4-2/\log P} \ge \frac{P}{4} + \frac{P}{8\log P}$, hence $|E(P)| \le 3P/4 - P/(8\log P) - 0.37247 < 3P/4 - P/(8\log P)$. □

8.8 Theorem K3*-Dusart (column, Dusart-enhanced)

External-theorem-closed Statement 8.10 (K3*-Dusart: $\log P$-order column improvement)**.** *For every prime $P \ge 79$,*

$$|E^*(P)| < \frac{3P}{4} - \frac{P}{8\log P}.$$

Proof. Same skeleton as K3"' with MV1973 in arithmetic progressions and Dusart 2010 as input. The geometric-series argument yields the identical $\log P$-order saving. □

8.9 Theorem K3-united (row + column joint bound)

External-theorem-closed Statement 8.11 (K3-united)**.** *For every prime $P \ge 79$,*

$$|E(P)| + |E^*(P)| < \frac{3P}{2} - \frac{P}{4\log P}.$$

Proof. Direct sum of K3"' and K3*-Dusart. □

8.10 Status table

Theorem	Status	Range
K1 row mean	Proved-in-text	$P \to \infty$
K2 uniform row upper	External-theorem closed (MV1973)	$P \geq 5$
K3 exception PNT-only	External-theorem closed	asymptotic
K3' row effective	External-theorem closed (RS1962+MV1973)	$P \geq 5$
K3* column effective	External-theorem closed (RS1962+MV-AP)	$P \geq 5$
K3"' row Dusart	External-theorem closed (D2010+RS+MV)	$P \geq 79$
K3*-Dusart column	External-theorem closed (D2010+RS+MV-AP)	$P \geq 79$
K3-united joint	External-theorem closed	$P \geq 79$
H_P row $E(P) = \varnothing$	**Not claimed**	70-year open
H_P col $E^*(P) = \varnothing$	**Not claimed**	70-year open

8.11 What this chapter does not do

The K-series gives the strictest unconditional, explicit-effective exception bounds presently available. It does *not* approach H_P: the gap between $\frac{3P}{4} - \frac{P}{8 \log P}$ and 0 is of order P, i.e. the whole linear-sieve barrier. Closing that gap requires either an unconditional resolution of the $h = X^{1/2}$ short-interval second-moment problem, an unconditional Linnik constant ≤ 2, or a non-linear sieve technique bypassing Bombieri 1976 parity barrier. This is recorded honestly in the abstract and in `docs/monograph/frontier-honest-status-and-true-side-theorems-20260522.md`.

8.12 Internal self-contained K3-trivial

The K-series external versions above all use Brun–Titchmarsh-type sieve inputs (Montgomery–Vaughan 1973 or its arithmetic-progression analogue). This section records the strictly weaker, but strictly *internal* version that uses only the Prime Number Theorem (Hadamard 1896, de la Vallée-Poussin 1896) and the trivial bound $\pi(x+y) - \pi(x) \leq y$.

Proved-in-text Statement 8.12 (K3-trivial: PNT-only internal bound)**.** *For every prime P large enough,*

$$|E(P)| \leq P - \frac{P}{2 \log P} + O\Big(\frac{P}{\log^2 P}\Big).$$

Proof. By PNT, $\pi(P^2) - \pi(P) = \frac{P^2}{2 \log P}(1 + o(1))$. By the trivial interval-length bound, $N_P(k) \leq P$ for every k. Hence

$$(P - 1 - |E(P)|) \cdot P \geq \sum_{k \notin E(P)} N_P(k) = \pi(P^2) - \pi(P),$$

giving $P - 1 - |E(P)| \geq \frac{P}{2 \log P}(1 + o(1))$, equivalently $|E(P)| \leq P - \frac{P}{2 \log P} + O(P/\log^2 P)$. □

Proved-in-text Statement 8.13 (K3*-trivial: PNT-only internal column bound)**.** *For every prime P large enough,*

$$|E^*(P)| \leq P - \frac{P}{2 \log P} + O\Big(\frac{P}{\log^2 P}\Big).$$

Proof. Identical to K3-trivial with rows replaced by columns, using $\sum_{j=1}^{P-1} M_P(j) = \pi(P^2) - 1$ and $M_P(j) \le P$. □

Remark 8.14 (Internal vs. external closure)**.** The user-requested "internal-self-contained version" and "external-lemma version" of H_P closure correspond to the K-series above:

- *Internal self-contained (PNT only)*: K3-trivial and K3*-trivial give $|E(P)|, |E^*(P)| \le P(1 - 1/(2\log P))$.
- *External-lemma (RS1962 + MV1973 + Dusart 2010)*: K3"', K3*-Dusart, K3-united give $|E(P)|, |E^*(P)| < 3P/4 - P/(8\log P)$.

Both versions remain at distance $\Theta(P)$ from H_P (which would require $|E(P)| = |E^*(P)| = 0$). Closing either gap is equivalent to unconditional Cramér-local / Linnik = 2, both 70-year open problems. The closure cannot be performed inside the linear-sieve framework (Bombieri 1976 parity barrier; cf. Iwaniec–Friedlander *Opera de Cribro* Theorem 11.4 with $f(1/2) = 0$).

External-theorem-closed Statement 8.15 (K3"" and K3*-three-term, with self-audit correction)**.** *Using Dusart's three-term explicit lower bound*

$$\pi(x) \ge \frac{x}{\log x}\left(1 + \frac{1}{\log x} + \frac{1.8}{\log^2 x}\right), \qquad x \ge 32299,$$

one obtains, for every prime $P \ge 180$,

$$|E(P)|,\ |E^*(P)| < \frac{3P}{4} - \frac{P}{8\log P} - \frac{0.1125\,P}{\log^2 P}.$$

Proof. At $x = P^2$, Dusart gives

$$\pi(P^2) \ge \frac{P^2}{2\log P} + \frac{P^2}{4\log^2 P} + \frac{0.225P^2}{\log^3 P}.$$

Insert this in the K3' counting inequality and keep the same Montgomery–Vaughan row or column upper bound $2P/\log P$, with the Rosser–Schoenfeld subtraction for the lower endpoint. Dividing by $2P/\log P$, the third-term contribution becomes

$$\frac{\log P}{2P} \cdot \frac{0.225P^2}{\log^3 P} = \frac{0.225P}{2\log^2 P} = \frac{0.1125P}{\log^2 P},$$

so

$$P - 1 - |E(P)| \ge \frac{P}{4} + \frac{P}{8\log P} + \frac{0.1125P}{\log^2 P} - 0.62753,$$

and the same argument applies to columns. The companion check is archived in

```
experiments/k3_quadruple_prime_corrected_check.py.
```

□

Remark 8.16 (Self-audit correction)**.** An earlier draft of this theorem stated the third-term coefficient as 0.05625, off by a factor of two from the correct $0.225/2 = 0.1125$. The corrected statement strictly tightens the bound; all previous numerical checks remain valid since the previous looser bound is implied by the new tighter one.

External-theorem-closed Statement 8.17 (K3""-Dusart-upper: further constant tightening). *For every prime $P \geq 60184$,*

$$|E(P)| \leq \frac{3P}{4} - \frac{P}{8\log P} - \frac{0.1125P}{\log^2 P} - 1 + \frac{\log P}{2(\log P - 1.1)}.$$

Proof. Replace Rosser–Schoenfeld's $\pi(P) \leq 1.25506P/\log P$ with Dusart 2010 $\pi(x) \leq x/(\log x - 1.1)$ (valid for $x \geq 60184$). Keeping this upper bound in exact form gives the lower-endpoint contribution $\log P/(2(\log P - 1.1))$ after division by $2P/\log P$. Re-deriving K3"" gives the displayed exact offset. Asymptotically it is $-1/2 + 0.55/\log P + O(1/\log^2 P)$, replacing the -0.37247 term coming from Rosser–Schoenfeld. □

8.13 External frontier theorem stress test

This section records the strongest currently useful external-theorem imports for the row/column endpoint. It does not promote the main theorem; it converts the external literature into exact side bounds and identifies the remaining exponent gap.

External-theorem-closed Statement 8.18 (Baker–Harman–Pintz row-run side theorem). *Assume the Baker–Harman–Pintz 2001 prime-gap theorem: every sufficiently large interval of length $x^{0.525}$ contains a prime. Then, for large prime P, no run of consecutive empty P-rows has length $> P^{0.05+\varepsilon}$.*

Proof. A run of R empty rows gives a prime-free interval of length RP inside the region $x \asymp P^2$. Baker–Harman–Pintz forbids a prime-free interval longer than $x^{0.525+o(1)} = P^{1.05+o(1)}$. Hence $RP \ll P^{1.05+\varepsilon}$, so $R \ll P^{0.05+\varepsilon}$. □

External-theorem-closed Statement 8.19 (Frontier short-interval candidates). *As verified on 2026-05-23, Runbo Li's arXiv:2308.04458v8 preprint claims the exponent* 0.52*; if that claim is accepted, the preceding side theorem improves to $R \ll P^{0.04+\varepsilon}$. Guth–Maynard arXiv:2405.20552v2 and the Gafni–Tao exceptional-interval framework are important for short-interval technology. Le Duc Hieu arXiv:2509.04883v2 adds prime arithmetic-progression abundance in intervals of length x^θ for $\theta > 17/30$. These* 17/30*-scale inputs give rich structure only after thickening a P-row by $P^{2/15+o(1)}$ rows.*

External-theorem-closed Statement 8.20 (External-frontier transfer gates). *A pointwise theorem giving a prime in every interval of length x^θ near $x \asymp P^2$ implies only*

$$R(P) \ll P^{2\theta - 1 + o(1)}$$

for the longest consecutive empty P-row block. Thus every-row closure requires $\theta \leq 1/2$. Similarly, a least-prime-in-AP theorem $p(a \bmod P) \ll P^L$ enters the P^2 square only when $L \leq 2$ with compatible constants, and almost-all x short-interval results do not control the rigid P-spaced row grid without an extra grid-transfer input.

External-theorem-closed Statement 8.21 (Linnik and almost-prime column stress test). *Xylouris' Linnik bounds give a prime in each reduced residue class modulo P by height $P^{5+o(1)}$ in the strongest general published form, and Meng's bounded-cubic-part result gives $P^{4.5+o(1)}$ for prime moduli. Both are far outside the P^2 square. Li–Zhang–Cai give a P_2 almost-prime in each class by exponent* 1.8345*, which enters the square but is the wrong parity object.*

External-theorem-closed Statement 8.22 (Average AP and conditional Linnik near-gates). *The 2025–2026 AP-distribution frontier now includes Stadlmann's smooth-moduli level* $1/2 + 1/40$*, Runbo Li's smooth-moduli minorant level* $10/19$*, Pascadi's weighted level* $5/8 - o(1)$*, and Runbo Li's 2026 bilinear/trilinear large-moduli levels* $9/17$ *and* $17/32$*. At* $x = P^2$ *all of these ranges contain the modulus size* P*, but only through smooth, weighted, minorant, or almost-all mean-value structures. They do not imply the fixed-prime-modulus statement for* $q = P$ *and every reduced residue class. Bruna's 2026 conditional result under generalized Lindelof gives* $p(a \bmod q) \ll_\epsilon q^{2+\epsilon}$*, which is a conditional near-square statement but still does not place a prime inside the strict* P^2 *square unconditionally.*

Remark 8.23 (Current external frontier). The stress-test certificate is recorded in

```
docs/monograph/prime-matrix-external-frontier-theorem-stress-router.md.
```

After importing these external theorems, the remaining non-cyclic breakthrough target is

```
PointwiseShortIntervalPrimeTheoremThetaLeHalf
OR LinnikExponentLeTwoWithSquareWindowConstants
OR GridTransferredShortIntervalSecondMomentAtThetaHalf
OR NonlinearParityBreakingActualSourceConstructor.
```

No known imported theorem closes the row/column theorem unconditionally.

8.14 External Frontier Residual-Gap Audit

External-theorem-closed Statement 8.24 (Residual-gap audit for current external frontier). *As of 2026-05-23, every imported external theorem leaves a quantified residual gate. A pointwise short-interval theorem at exponent* θ *near* $x \asymp P^2$ *gives only an empty-row-run exponent*

$$2\theta - 1.$$

Thus Baker–Harman–Pintz $\theta = 0.525$ *leaves* 0.05*, and Runbo Li's arXiv v8* $\theta = 0.52$ *would still leave* 0.04 *if accepted. The Guth–Maynard/Hieu* $17/30$ *inputs leave a row thickening* $P^{2/15+o(1)}$*. On the column side, a least-prime-in-AP theorem* $p(a \bmod P) \ll P^L$ *enters the* P^2 *square only if* $L \le 2$*; Meng's prime-modulus-compatible* $L = 4.5$ *still overshoots by* $P^{2.5}$*. Bruna's GLH result reaches* $2+\varepsilon$*, but remains conditional and outside the strict square by* P^ε*. Li–Zhang–Cai's* P_2*-almost-prime exponent* 1.8345 *enters the square but is the wrong parity object. Average AP distribution results, including Pascadi's* $5/8 - o(1)$*, cover* $q = P$ *in range but not as a fixed-prime-modulus zero-exception theorem.*

Proof. The transformations are direct scale substitutions. A block of R empty P-rows gives a prime-free interval of length RP at $x \asymp P^2$, so $RP \ll x^\theta = P^{2\theta+o(1)}$ gives $R \ll P^{2\theta-1+o(1)}$. A column class modulo P is filled inside the $P \times P$ square only if the first prime in that class is $\le P^2$, i.e. $L \le 2$ with compatible constants. Mean-value AP theorems average over moduli or weights and therefore do not imply a zero-exception theorem for the single prime modulus P. The P_2 result has the right location but the wrong object. Hence the audit closes only side bounds and misuse exclusions, not the row/column theorem. □

Remark 8.25 (Residual basis). The current non-cyclic residual basis is

```
PointwiseShortIntervalPrimeTheoremThetaLeHalf
OR GridTransferredShortIntervalSecondMomentAtThetaHalf
OR LinnikExponentLeTwoWithSquareWindowConstants
OR MeanValueAPToFixedPrimeModulusZeroExceptionTransfer
OR ConditionalLinnikTwoPlusEpsilonToUnconditionalLinnikLeTwoWithConstants
OR NonlinearParityBreakingActualSourceConstructor.
```

The machine-readable certificate is archived as

```
docs/monograph/prime-matrix-external-frontier-residual-gap-audit.md.
```

8.15 Prime-Square Half-Scale Specialization Audit

External-theorem-closed Statement 8.26 (Prime-square endpoints do not auto-drop 0.52 to 1/2)**.** *As of the 2026-05-23 external-source check, the known pointwise short-interval input at exponent* 0.52*, when specialized to*

$$X = P^2,$$

gives an interval length $X^{0.52} = P^{1.04}$*. It does not by itself give a prime in every interval of length* $P = X^{1/2}$ *beginning or ending at a prime square. The primality of* P *gives square-phase rigidity, not a known unconditional half-scale theorem.*

Proof. The scale conversion is literal. Baker–Harman–Pintz gives $X^{0.525}$, hence $P^{1.05}$ at $X = P^2$. Runbo Li's arXiv:2308.04458v8 claim gives $X^{0.52} = P^{1.04}$. Both exponents remain longer than P.

The special endpoint P^2 does supply useful structure. For $1 \le r < P$, the prime P divides neither $P^2 + r$ nor $P^2 - r$. For every prime $q < P$, the right and left square anchors have the fixed forbidden residues

$$q \mid P^2 + r \iff r \equiv -P^2 \pmod q, \qquad q \mid P^2 - r \iff r \equiv P^2 \pmod q.$$

Thus a survivor avoiding all prime divisors $q < P$ is prime. This converts the problem into a special square-phase long-block exclusion, but it is not a published pointwise short-interval theorem at exponent 1/2. □

Remark 8.27 (Half-scale residual basis)**.** The right-hand target remains

$$\pi(P^2 + P) - \pi(P^2) > 0,$$

and the left top-row target remains

$$\pi(P^2 - 1) - \pi(P^2 - P) > 0.$$

The existing finite scan through $P \le 200000$ records no right-window failure, but this is finite evidence only. The new certificate is archived in

```
docs/monograph/prime-matrix-prime-square-halfscale-specialization-audit.md.
```

After this audit the relevant non-cyclic residual basis is

```
SquarePhaseSpecialPhaseLongBlockPDECExclusion
OR TwoSidedSquarePhaseLayeredWheelSurvivorLowerBound
OR PrimeSquareEndpointNoExceptionalPhaseTheorem
OR PrimeInFirstHalfAfterPrimeSquareForEveryPrimeP
OR PuncturedWheel6EndpointCapacityInequalityOrReciprocalPrimePairWheel6SaturationPDEC
OR ExactExternalSqrtScaleOrFullSNonAPWFDKLSTheoremMatch
OR NewAutomorphicDispersionProof.
```

The boundary status is

```
prime_square_halfscale_auto_drop_closed=false
square_phase_attack_surface_identified=true
external_lemma_version_unconditional_closed=false
internal_self_contained_closed=false
row_column_unconditional_closed=false.
```

8.16 Prime-Square $P^2 \pm 1$ Sandwich Audit

External-theorem-closed Statement 8.28 ($P^2 \pm 1$ endpoints do not localize 0.52 to half-scale)**.** *Applying a pointwise short-interval theorem with exponent* 0.52 *at the two adjacent endpoints* $X = P^2 - 1$ *and* $X = P^2 + 1$ *does not force a prime into either inner half-scale window*

$$(P^2 - P, P^2), \qquad (P^2, P^2 + P).$$

Indeed

$$(P^2 \pm 1)^{0.52} = P^{1.04}\big(1 + O(P^{-2})\big),$$

and the absolute change caused by the ± 1 *shift is only* $O(P^{-0.96})$*. Thus the endpoint shift changes phase, not the power scale.*

Proof. For fixed $\theta = 0.52$,

$$(P^2 \pm 1)^{\theta} = P^{2\theta}\left(1 \pm P^{-2}\right)^{\theta} = P^{1.04}\left(1 + O(P^{-2})\right).$$

The difference between $(P^2 \pm 1)^{\theta}$ and $P^{1.04}$ is $O(P^{1.04-2}) = O(P^{-0.96})$, while the target length is P.

Consequently the right endpoint theorem gives at best a container of the form

$$(P^2 + 1,\ P^2 + 1 + (P^2 + 1)^{0.52}],$$

not the subinterval $(P^2, P^2 + P)$. A guaranteed prime may still lie in the outer tail

$$[P^2 + P,\ P^2 + P^{1.04} + O(1)].$$

Similarly, a left endpoint theorem gives a container ending at $P^2 - 1$, but the guaranteed prime may lie in

$$[P^2 - P^{1.04} + O(1),\ P^2 - P].$$

Thus the endpoint applications provide thick containers only. To deduce a prime in the inner P-window one would need an additional outer-tail exclusion or a lower bound for primes in the container that exceeds an upper bound for primes in the outer tail. Current Brun–Titchmarsh type bounds leave the outer-tail capacity on the order of $P^{1.04}/\log P$, so the single-prime lower bound from a short-interval theorem cannot win. $\square$

Remark 8.29 ($P^2 \pm 1$ residual basis)**.** The factorization $P^2 - 1 = (P-1)(P+1)$ concerns the endpoint itself and does not determine where the next or previous prime in a $P^{1.04}$-thick container lies. The endpoint $P^2 + 1$ supplies the same square-adjacent phase information already recorded by the square-phase routers, and $P \nmid P^2 \pm r$ for $1 \le r < P$ removes only the local $q = P$ obstruction. The new certificate is archived in

```
docs/monograph/prime-matrix-prime-square-pm1-sandwich-audit.md.
```

After this audit the non-cyclic residual basis is

```
PM1OuterTailExclusionForTheta052Containers
OR PrimeSquareNearestPrimeWithinPOnAtLeastOneSide
OR TwoSidedSquarePhaseInnerWindowLocalization
OR SquarePhaseSpecialPhaseLongBlockPDECExclusion
OR PuncturedWheel6EndpointCapacityInequalityOrReciprocalPrimePairWheel6SaturationPDEC
OR ExactExternalSqrtScaleOrGridTransferredThetaHalfSecondMoment
OR NewSameObjectSignedDispersionOrAutomorphicProof.
```

The boundary status is

```
pm1_sandwich_halfscale_closed=false
pm1_sandwich_no_go_closed=true
prime_square_halfscale_auto_drop_closed=false
external_lemma_version_unconditional_closed=false
internal_self_contained_closed=false
row_column_unconditional_closed=false.
```

8.17 Prime-Power Slope Sandwich Audit

External-theorem-closed Statement 8.30 (Prime-power exponent interpolation does not localize the P^2 half-window)**.** *Let $\theta = 0.52 = 13/25$. Applying a θ-short-interval input at $P^{50/26} = P^{25/13}$ gives a length*

$$\left(P^{25/13}\right)^{13/25} = P,$$

which matches the formal half-window length

$$\left(P^{50/25}\right)^{1/2} = P.$$

However, this is only a length coincidence. The lower container is centered near $P^{25/13}$, far below P^2, while the upper container near $P^{50/24} = P^{25/12}$ remains far above P^2. Hence the two endpoint applications do not force a prime into the P^2 half-window.

Proof. The lower endpoint is

$$X_- = P^{50/26} = P^{25/13}.$$

Its guaranteed short-interval radius under the $\theta = 13/25$ input is

$$X_-^\theta = P.$$

But its distance from the target center is

$$P^2 - P^{25/13} = P^2(1 - P^{-1/13}) \asymp P^2,$$

so the ratio between the gap and the guaranteed radius is $\asymp P$. The container around X_- cannot reach the P^2 row.

The upper endpoint is

$$X_+ = P^{50/24} = P^{25/12}.$$

Its guaranteed radius is

$$X_+^\theta = P^{(25/12)(13/25)} = P^{13/12}.$$

Yet

$$P^{25/12} - P^2 = P^2(P^{1/12} - 1) \asymp P^{25/12},$$

again a factor P larger than the guaranteed radius. Thus the upper container also cannot touch the P^2 half-window.

The general obstruction is simple. If $X = P^a$, a X^θ short-interval theorem gives length $P^{a\theta}$. Location at the prime-square endpoint requires $a = 2$, while radius P requires $a\theta = 1$. Both equations hold simultaneously only when $\theta = 1/2$. For $\theta = 13/25$, the radius condition gives $a = 25/13$, which is exactly not the P^2 location. $\square$

Remark 8.31 (Slope-sandwich residual basis)**.** The certificate is archived in

```
docs/monograph/prime-matrix-prime-power-slope-sandwich-audit.md.
```

The current residual basis is

```
PrimeSquareEndpointLocalizationNotExponentInterpolation
OR ThetaEqualsHalfOrPrimeSquareSpecificPointwiseTheorem
OR P2CenteredContainerPrimeLowerBound
OR OuterScaleGapBridgeBetweenP25Over13AndP2
OR SameObjectSignedDispersionOrAutomorphicEndpointProof.
```

The boundary status is

```
prime_power_slope_sandwich_no_go_closed=true
exponent_length_coincidence_closed=true
lower_container_reaches_p2=false
upper_container_reaches_p2=false
prime_square_halfscale_closed=false
row_column_unconditional_closed=false.
```

8.18 Legendre-Frontier External Theorem Audit

External-theorem-closed Statement 8.32 (Latest Legendre-frontier inputs are near-gates, not half-scale closure)**.** *The 2026 external literature contains useful Legendre-adjacent inputs, but none gives the unconditional Prime Matrix half-scale statement. Chamberland–Straub prove, under RH, prime existence between consecutive powers*

$$x^{2+\delta} \quad \textit{and} \quad (x+1)^{2+\delta}$$

for $\delta > 0$*, not the* $\delta = 0$ *square case. Campbell proves, in an arXiv preprint, that every interval between consecutive squares contains an integer with at most three prime factors counted with multiplicity; this has the right location but the wrong parity object. Guth–Maynard's 2026 Annals zero-density technology gives short-interval PNT at exponent* $17/30$*, still above* $1/2$*.*

Proof. For the larger-powers input, put $X = x^{2+\delta}$. The interval length is

$$(x+1)^{2+\delta} - x^{2+\delta} = (2+\delta)x^{1+\delta} + O(x^{\delta}),$$

so its exponent in X is

$$\frac{1+\delta}{2+\delta} = \frac{1}{2} + \frac{\delta}{2(2+\delta)}.$$

Every fixed $\delta > 0$ is therefore strictly above $1/2$, and $\delta = 0$ is exactly the Legendre/prime-square half-scale case. Moreover the theorem is RH-conditional, so it cannot be imported as an unconditional closure.

The P3 result is located in the exact square interval, but it produces an almost-prime object, not a prime. This is a useful parity diagnostic: linear sieve methods can reach the square interval with P_3 objects while still failing to isolate primes. Thus the missing transfer is an object-sensitive parity-breaking step, not another same-set counting identity.

Finally, Guth–Maynard's exponent $17/30$ gives, after $X = P^2$, a window of length $P^{17/15} = P \cdot P^{2/15}$. It controls a thickened block of $P^{2/15}$ rows, not a single P-row. Hence each external input narrows a frontier but leaves the half-scale prime-square target open. □

Remark 8.33 (Legendre-frontier residual basis)**.** The certificate is archived in

```
docs/monograph/prime-matrix-legendre-frontier-external-audit.md.
```

The current non-cyclic residual basis is

```
DeltaZeroLegendreOrPrimeSquareHalfscaleTheorem
OR P3ToPrimeParityBreakingTransferOrObjectSensitiveSignedSieve
OR ThetaLeHalfPointwiseShortIntervalPrimeTheorem
OR GridTransferredThetaHalfSecondMoment
OR PrimeSquareSpecialPhaseNoOuterTailTheorem
OR NewSameObjectSignedDispersionOrAutomorphicProof.
```

The boundary status is

```
legendre_frontier_external_inputs_imported=true
rh_larger_powers_delta_zero_closed=false
p3_to_prime_transfer_closed=false
prime_square_halfscale_closed=false
row_column_unconditional_closed=false.
```

8.19 Short-Interval Transference Parity Audit

External-theorem-closed Statement 8.34 (Short-interval transference does not localize the first prime-square row). *Short-interval arithmetic-progression and transference inputs are genuine external technology, but they do not by themselves close the Prime Matrix Phi-LPF parity gate. Le Duc Hieu's 2025 preprint proves k-term prime arithmetic progressions in intervals* $[x, x + x^{\theta}]$ *for* $\theta > 17/30$*. Green–Tao/W-trick transference supplies a mechanism for prime patterns after small-prime normalization. BDH-type inputs supply mean-square or average control. None of these statements is a pointwise assertion that every prime-square endpoint* P^2 *contains a prime in the first or last length-*P *half-window.*

Proof. Put $X = P^2$. A short interval of length X^{θ} becomes a Prime Matrix window of length $P^{2\theta}$. The target half-scale row is exactly $\theta = 1/2$, because $X^{1/2} = P$. Guth–Maynard's 17/30 exponent gives

$$X^{17/30} = P^{17/15} = P \cdot P^{2/15},$$

and any $\theta > 17/30$ gives an even thicker block

$$P^{2\theta} = P \cdot P^{2\theta-1}.$$

Thus Hieu's short-interval prime-AP input lives in a block with positive row thickness $P^{2\theta-1}$, not in a single Prime Matrix row.

Pattern abundance in the full container

$$(P^2, P^2 + P^{2\theta}]$$

does not force a prime in the initial subinterval $(P^2, P^2 + P]$. For $\theta > 1/2$, the outer tail

$$[P^2 + P, P^2 + P^{2\theta}]$$

has length comparable to the full container, so the whole pattern count can be absorbed outside the first row unless an additional localization theorem is proved.

The W-trick removes logarithmic small-prime biases and enables dense-model transference, but Phi-LPF closure requires avoiding every residue cover from all $q < P$, uniformly at each prime-square endpoint. Similarly, BDH-type mean-square control does not preclude a sparse exceptional set containing all prime-square phases. Therefore these inputs narrow the residual basis but do not break the same-object parity obstruction. □

Remark 8.35 (Transference residual basis)**.** The certificate is archived in

```
docs/monograph/prime-matrix-short-interval-transference-parity-audit.md.
```

The current residual basis is

```
ThetaLeHalfUniformShortIntervalPrimeTheorem
OR APPatternLocalizationInsidePrimeSquareHalfWindow
OR BDHNoExceptionalPrimeSquarePhaseTheorem
OR WTrickToFullPhiLPFObjectSensitiveSieve
OR MaynardClusterAnchoredAtEveryPrimeSquare
OR SameObjectSignedDispersionOrAutomorphicEndpointProof.
```

The boundary status is

```
short_interval_transference_inputs_imported=true
prime_pattern_to_first_row_transfer_closed=false
w_trick_phi_lpf_parity_closed=false
bdh_pointwise_all_rows_closed=false
prime_square_halfscale_closed=false
row_column_unconditional_closed=false.
```

8.20 Almost-All Short-Interval Exceptional-Spine Audit

External-theorem-closed Statement 8.36 (Almost-all short intervals do not remove the prime-square spine)**.** *Recent almost-all short-interval inputs are much stronger in scale than the Prime Matrix half-window, but they remain almost-all statements. Runbo Li's almost-all prime interval theorem reaches intervals of length $x^{1/21.5+\epsilon}$, and the newer working-paper direction records $x^{1/22+\epsilon}$ left intervals. Gafni–Tao's exceptional-interval framework gives pointwise short-interval PNT above* $17/30$ *and almost-all control down to* $2/15$*. The Matomaki–Radziwill–Shao–Tao–Teravainen higher-uniformity theorem supplies deep almost-all control for Λ and related functions. None of these inputs asserts that every prime-square endpoint P^2 is non-exceptional.*

Proof. Set $X = P^2$. A theorem at scale X^θ becomes a Prime Matrix window of length $P^{2\theta}$. Hence

$$\theta = \frac{2}{43} \quad \Longrightarrow \quad X^\theta = P^{4/43},$$

$$\theta = \frac{1}{22} \quad \Longrightarrow \quad X^\theta = P^{1/11},$$

and

$$\theta = \frac{2}{15} \quad \Longrightarrow \quad X^\theta = P^{4/15}.$$

All these lengths are smaller than $P = X^{1/2}$. Thus, if they were pointwise at every $X = P^2$, they would more than suffice for the half-window target.

The obstruction is not scale but exceptional-set geometry. In a dyadic block $[X, 2X]$, the prime-square endpoint spine has size

$$\#\{P^2 \in [X, 2X] : P \text{ prime}\} \asymp \frac{X^{1/2}}{\log X},$$

and therefore density about $1/(X^{1/2}\log X)$. An almost-all theorem permits a zero-density exceptional set, and such a set can in principle contain this entire arithmetic spine unless the theorem supplies additional spine-disjointness information. The Prime Matrix row/column statement needs every endpoint, not density-one endpoints. Therefore the required upgrade is a pointwise theorem on the prime-square spine or an object-sensitive signed sieve that forbids the exceptional set from carrying all P^2 phases. □

Remark 8.37 (Almost-all residual basis)**.** The certificate is archived in

```
docs/monograph/prime-matrix-almost-all-exceptional-spine-audit.md.
```

The current residual basis is

```
ExceptionalPrimeSquareSpineDisjointness
OR PointwiseEndpointUniformityAtEveryPrimeSquare
OR AlmostAllToAllRowsUpgradeWithArithmeticSpineRepulsion
OR NoPrimeSquareExceptionalPhaseForGafniTaoBounds
OR PhiLPFObjectSensitiveSignedSieveOnSparseSpine
OR ThetaLeHalfPointwiseShortIntervalPrimeTheorem.
```

The boundary status is

```
almost_all_short_interval_inputs_imported=true
scale_stronger_than_halfwindow_if_pointwise=true
exceptional_prime_square_spine_excluded=false
pointwise_every_prime_square_endpoint_closed=false
phi_lpf_parity_closed=false
prime_square_halfscale_closed=false
row_column_unconditional_closed=false.
```

8.21 Parity-Breaking Candidate Source Audit

External-theorem-closed Statement 8.38 (Five-gate audit for parity-breaking candidates)**.** *As of 2026-05-23, no currently imported parity-breaking candidate supplies a direct unconditional row/column closure for the Prime Matrix endpoint. The audit requires five simultaneous gates:*

```
PrimeObjectNotP2AlmostPrime
SquareScaleWindowOrP2ColumnCompatibility
RigidPointwiseGridOrFixedPrimeModulusZeroException
SameObjectNonlinearActualSourceConstructorBeforeProjection
UnconditionalPublishedOrIndependentlyAcceptedInput.
```

Li–Zhang–Cai's P_2 arithmetic-progression result enters the P^2 square but has the wrong parity object. Friedlander–Iwaniec supplies a genuine nonlinear prime-producing model, but for the different $X^2 + Y^4$ object. BFI/DI/Kuznetsov/Maynard technologies supply dispersion, spectral, or multidimensional-sieve tools, but not a fixed $q = P$ zero-exception theorem for every reduced residue class and not the present same-object actual-source constructor. Ford–Maynard prime-producing sieve theory is retained as design guidance for a future constructor, not as a ready-made Prime Matrix theorem.

Proof. The verification is a gate-by-gate theorem-match audit. A valid import must produce primes, not merely P_2 almost-primes; it must land inside the strict P^2 square; it must be pointwise on the rigid P-row grid or on every nonzero class modulo the fixed prime P; and it must emit the same actual source before projection or averaging losses. The candidate list fails at least one of these gates in every case. Thus the parity audit narrows the remaining target to a same-object nonlinear prime-source constructor or a fixed-prime-modulus zero-exception transfer, while preserving the boundary

```
direct_closure_candidate_count=0
external_lemma_version_unconditional_closed=false
internal_self_contained_closed=false
row_column_unconditional_closed=false.
```

□

Remark 8.39 (Parity-breaking residual basis)**.** The machine-readable certificate is archived as

```
docs/monograph/prime-matrix-parity-breaking-obstruction-audit.md.
```

After this audit the non-cyclic residual basis is refined to

```
PrimeObjectNotP2AlmostPrime
OR RigidPointwiseGridOrFixedPrimeModulusZeroException
OR SameObjectNonlinearActualSourceConstructorBeforeProjection
OR MeanValueAPToFixedPrimeModulusZeroExceptionTransfer
OR PointwiseShortIntervalPrimeTheoremThetaLeHalf
OR LinnikExponentLeTwoWithSquareWindowConstants.
```

This is a misuse-exclusion and frontier-narrowing theorem, not a proof of the row/column theorem.

8.22 P_2-to-Prime Transfer Atom

Proved-in-text Statement 8.40 (Composite P_2 witnesses split into small-factor cofactor AP fibers)**.** *Let P be prime and $1 \le a < P$. Suppose*

$$n \equiv a \pmod P, \qquad \Omega(n) = 2, \qquad n < P^2.$$

Then

$$n = rm, \qquad r < P, \qquad m \equiv ar^{-1} \pmod P$$

for a prime r. More generally, if $n \le P^\sigma$ with $\sigma < 2$, then $r \le P^{\sigma/2}$. In particular, Li–Zhang–Cai's P_2 exponent 1.8345 *would place every composite P_2 witness in fibers with $r \le P^{0.91725}$ and $m \equiv ar^{-1}$* $\pmod P$*.*

Proof. Write the composite P_2 witness as $n = uv$ with primes $u \le v$. Since $a \not\equiv 0 \pmod P$, neither factor is divisible by P. From $uv = n < P^2$ we get $u < P$. Taking $r = u$ and $m = v$ gives

$$rm \equiv a \pmod P, \qquad (r, P) = 1,$$

and hence $m \equiv ar^{-1} \pmod P$. If $n \le P^\sigma$, then $r = u \le \sqrt{n} \le P^{\sigma/2}$. This is a closed elementary transfer atom; it does not prove that such composite fibers are sparse enough to force a prime in every fixed class. $\square$

Remark 8.41 (P_2-to-prime transfer remains open)**.** The audit generated by

```
docs/monograph/prime-matrix-p2-to-prime-transfer-atom-audit.md
```

checks the fiber identity for all prime moduli $P \le 997$. In the sample there are no cofactor-AP identity failures and no small-factor-bound failures. The same audit also records that composite P_2 witnesses are not a small error term in the square window: for $P = 997$, the number of composite P_2 witnesses in nonzero classes below P^2 is 208680, while the corresponding prime count is 78059, a ratio 2.673362. Thus the remaining non-cyclic target is

```
SmallFactorCofactorAPCompositeFiberDominanceBound
OR PrimeBeforeCompositeP2SelectorInEveryFixedClass
OR FixedPrimeModulusZeroExceptionTransferForPrimeObjects
OR SameObjectNonlinearActualSourceConstructorBeforeProjection.
```

The status remains

```
p2_to_prime_transfer_closed=false
row_column_unconditional_closed=false.
```

External-theorem-closed Statement 8.42 (Least-P_2 prime-selector route is false)**.** *The natural selector upgrade*

```
PrimeBeforeCompositeP2SelectorInEveryFixedClass
```

is false. Even after discarding trivial witnesses $n \le P$, the least P_2 witness in a reduced class can be composite. The smallest audited counterexample is

$$P = 3, \qquad a = 1, \qquad X = \lfloor P^{1.8345} \rfloor = 7, \qquad 4 \equiv 1 \pmod 3, \qquad 4 = 2^2.$$

Thus Li–Zhang–Cai's least-P_2 theorem cannot be promoted to a prime theorem by asserting that its chosen witness is prime.

Proof. The displayed example satisfies $4 > P$, $4 \le X$, $4 \equiv 1 \pmod 3$, and $\Omega(4) = 2$. Since no integer $n > P$ with $n \equiv 1 \pmod 3$ lies between 3 and 4, this is the first post-modulus P_2 witness in that class, and it is composite. Hence the proposed universal selector statement is refuted by a finite counterexample. Larger diagnostic scans show the same phenomenon is not an isolated small-prime artifact: for $P = 997$, 641 of the 996 nonzero classes have a composite first P_2 witness after P below $\lfloor P^{1.8345} \rfloor$. $\square$

Remark 8.43 (Selector route after rejection)**.** The selector-rejection audit is archived as

```
docs/monograph/prime-matrix-p2-selector-route-rejection-audit.md.
```

After deleting the false least-P_2 selector route, the P_2-to-prime frontier becomes

```
SmallFactorCofactorAPCompositeFiberDominanceBound
OR NonleastPrimeSelectorRequiresAdditionalDistributionInput
OR FixedPrimeModulusZeroExceptionTransferForPrimeObjects
OR SameObjectNonlinearActualSourceConstructorBeforeProjection.
```

The deletion of a false route is real progress in proof hygiene, but it is not an unconditional prime-existence theorem.

External-theorem-closed Statement 8.44 (Support-only P_2-to-prime transfer is parity-blind)**.** *Let $X = \lfloor P^{1.8345} \rfloor$. In the audited prime moduli*

$$P \in \{101, 199, 499, 997, 2003, 5003\},$$

the composite P_2 objects $n \le X$, $P \nmid n$, $\Omega(n) = 2$, and n non-prime already meet every nonzero residue class modulo P. From $P = 499$ onward in the audit, every nonzero residue class contains strictly more composite P_2 objects than prime objects below X. For $P = 5003$ the minimum residue-wise excess

$$\#\{n \le X : n \equiv a \pmod P,\ \Omega(n) = 2,\ n \text{ composite}\} - \#\{q \le X : q \equiv a \pmod P,\ q \text{ prime}\}$$

over $1 \le a < P$ is 95, and the total composite-P_2/prime count ratio is 2.826571.

Proof. This is a finite certificate generated by

```
experiments/prime_matrix_composite_p2_support_saturation_audit.py.
```

The script sieves to $\max_P \lfloor P^{1.8345} \rfloor$, classifies each integer by primality and by the capped value of $\Omega(n)$, and then counts prime and composite-P_2 objects in every nonzero residue class for each audited modulus. The recorded minimums certify full composite-P_2 support for all audited rows and strict residue-wise composite dominance beginning at $P = 499$. $\square$

Remark 8.45 (Residual gate after support saturation)**.** The support-saturation audit removes the weaker route

```
ResidueSupportOnlyP2ToPrimeTransfer.
```

After this removal the P_2 frontier is object-sensitive:

```
ObjectSensitivePrimeMinusCompositeP2SeparationInput
OR SmallFactorCofactorAPCompositeFiberDominanceBound
OR NonleastPrimeSelectorRequiresAdditionalDistributionInput
OR FixedPrimeModulusZeroExceptionTransferForPrimeObjects
OR SameObjectNonlinearActualSourceConstructorBeforeProjection.
```

Thus P_2 support coverage by itself is not a parity-breaking mechanism. This is a proof-hygiene advance only; it is not a proof of the row/column prime-existence target.

8.23 K6 joint density and row-run rigidity

Corollary 8.46 (K6: joint density and consecutive-block bound)**.** *For every prime P large enough:*

1. *Maximum length of a consecutive prime-free row block is $\leq P^{0.05+\varepsilon}$ (Baker–Harman–Pintz 2001).*
2. *Total row-exception density satisfies $|E(P)| < 3P/4 - P/(8\log P) - 0.1125P/\log^2 P$ (K3"" corrected).*

Hence the exception set $E(P)$, even at maximum permissible density $3P/4$, is forced into short consecutive clusters of length $\leq P^{0.05+\varepsilon}$.

Remark 8.47 (Honest boundary after K3"" / K6)**.** K3"" refines K3"' by a $P/\log^2 P$ order saving, but the *leading* $3P/4$ constant is unchanged. The gap to H_P is still $\Theta(P)$. Eliminating the constant $3/4$ requires sharpening Brun–Titchmarsh below the Selberg constant 2, which cannot be done in linear-sieve framework alone (parity barrier).

8.24 Theorem K3-united-three-term and final K-series summary

External-theorem-closed Statement 8.48 (K3-united-three-term)**.** *For every prime $P \geq 180$,*

$$|E(P)| + |E^*(P)| < \frac{3P}{2} - \frac{P}{4\log P} - \frac{0.225\,P}{\log^2 P}.$$

Proof. Sum of K3"" and K3*-three-term. □

Proved-in-text Statement 8.49 (K3-trivial-enhanced: internal self-contained with Li expansion)**.** *For sufficiently large prime P,*

$$|E(P)| \leq P - \frac{P}{2\log P} - \frac{P}{4\log^2 P} + O\Big(\frac{P}{\log^3 P}\Big).$$

Proof. Standard expansion of $\mathrm{Li}(x)$ gives $\mathrm{Li}(P^2) = \frac{P^2}{2\log P} + \frac{P^2}{4\log^2 P} + O(P^2/\log^3 P)$, and de la Vallée-Poussin's 1899 error term gives $\pi(P^2) = \mathrm{Li}(P^2) + O(P^2 e^{-c\sqrt{\log P}})$ where the exponential decay dominates any inverse-log error. Hence $\pi(P^2) - \pi(P) = \frac{P^2}{2\log P} + \frac{P^2}{4\log^2 P} + O(P^2/\log^3 P)$. Combining with the trivial bound $N_P(k) \leq P$ gives the stated $P/(4\log^2 P)$-order saving over K3-trivial. No sieve input is used. □

Proved-in-text Statement 8.50 (K3*-trivial-enhanced)**.** *For sufficiently large prime P,*

$$|E^*(P)| \leq P - \frac{P}{2\log P} - \frac{P}{4\log^2 P} + O\Big(\frac{P}{\log^3 P}\Big).$$

Proof. Identical to K3-trivial-enhanced with rows replaced by columns and $M_P(j) \leq P$ in place of $N_P(k) \leq P$. □

Proved-in-text Statement 8.51 (K3-trivial-three-term-Li: internal three-term Li expansion)**.** *For sufficiently large prime P,*

$$|E(P)| \leq P - \frac{P}{2\log P} - \frac{P}{4\log^2 P} - \frac{P}{4\log^3 P} + O\Big(\frac{P}{\log^4 P}\Big).$$

Proof. Expand $\mathrm{Li}(x) = (x/\log x)\sum_{k\geq 0} k!/\log^k x$ to order $k = 2$. At $x = P^2$ the $k = 2$ contribution is $(P^2/(2\log P)) \cdot 2/(4\log^2 P) = P^2/(4\log^3 P)$. Hence $\pi(P^2) - \pi(P) = \frac{P^2}{2\log P} + \frac{P^2}{4\log^2 P} + \frac{P^2}{4\log^3 P} + O(P^2/\log^4 P)$. The subtracted $\pi(P) = O(P/\log P)$ is absorbed in the $O(P^2/\log^4 P)$ term for sufficiently large P. Trivial $N_P(k) \leq P$ then yields the stated bound. Still no sieve input; the expansion is internal to PNT. □

Proved-in-text Statement 8.52 (K3*-trivial-three-term-Li)**.** *For sufficiently large prime P,*

$$|E^*(P)| \leq P - \frac{P}{2\log P} - \frac{P}{4\log^2 P} - \frac{P}{4\log^3 P} + O\Big(\frac{P}{\log^4 P}\Big).$$

Proof. Identical to K3-trivial-three-term-Li, column version with $M_P(j) \leq P$. □

External-theorem-closed Statement 8.53 (K3-united-three-term-corrected (factor-of-2 correction))**.** *For every prime $P \geq 180$,*

$$|E(P)| + |E^*(P)| < \frac{3P}{2} - \frac{P}{4\log P} - \frac{0.225\,P}{\log^2 P}.$$

Proof. Sum of K3""-corrected and K3*-three-term-corrected, both with the correct third-term coefficient 0.1125. The joint coefficient is twice that, namely 0.225. □

Proved-in-text Statement 8.54 (K3-trivial-K-term-Li (general internal asymptotic family))**.** *For every nonneg integer K and sufficiently large prime P,*

$$|E(P)| \leq P - \sum_{k=0}^{K} \frac{k!}{2^{k+1}} \cdot \frac{P}{\log^{k+1} P} + O\Big(\frac{P}{\log^{K+2} P}\Big),$$

and the same bound holds for $|E^*(P)|$.

Proof. $\mathrm{Li}(x) = (x/\log x)\sum_{k\geq 0} k!/\log^k x$. At $x = P^2$ the k-th term contributes $P^2 k!/(2^{k+1}\log^{k+1} P)$. Sum to $k = K$ and combine with $\pi(P^2) - \pi(P)$ and the trivial bound $N_P(k) \leq P$. The de la Vallée-Poussin 1899 exponential remainder is absorbed into the $O(P/\log^{K+2} P)$ tail. Identical column-side proof with $M_P(j) \leq P$. □

Remark 8.55 (Internal-version saturation)**.** The coefficient sequence $c_k = k!/2^{k+1}$ grows factorially, so $\mathrm{Li}(x)$ is a *divergent* asymptotic series with optimal truncation $K^* \approx 2\log P$. Beyond K^* further terms increase the error rather than reduce it. Hence $K = 2$ or $K = 3$ already extracts the dominant useful information in this internal chain; further K gives diminishing returns in any range of interest.

The leading constant remains 1 for every truncation K, so the gap to H_P (which would require leading constant 0) is $\Theta(P)$ regardless of K. Closing this gap requires either Brun–Titchmarsh-type sieve (taking the leading constant from 1 to 3/4 in the external chain) or breaking the Bombieri 1976 parity barrier ($3/4 \to 0$, 70-year open).

Remark 8.56 (Final K-series summary)**.** The K-series side theorems established in this chapter provide the strictest currently available unconditional, explicit-effective bounds on the H_P exception sets. They split into two parallel chains:

Internal self-contained chain (PNT only, no sieve input):

$$|E(P)|,\ |E^*(P)| \leq P - \frac{P}{2\log P} - \frac{P}{4\log^2 P} - \frac{P}{4\log^3 P} + O\Big(\frac{P}{\log^4 P}\Big).$$

External-lemma chain (PNT + Brun–Titchmarsh + Dusart 2010):

$$|E(P)|,\ |E^*(P)| < \frac{3P}{4} - \frac{P}{8\log P} - \frac{0.1125\,P}{\log^2 P}.$$

Joint chain (row + column + BHP 2001 long-block bound):

$$|E(P)| + |E^*(P)| < \frac{3P}{2} - \frac{P}{4\log P} - \frac{0.225\,P}{\log^2 P},$$

$$\text{max consecutive empty rows} \leq P^{0.05+\varepsilon}.$$

Both chains close to their respective unconditional limits inside the linear-sieve framework. The residual $\Theta(P)$ gap to H_P is precisely the parity barrier (Bombieri 1976; $f(1/2) = 0$). H_P itself remains **Not claimed**, equivalent to Cramér-local and Linnik-2, both 70-year open problems.

8.25 On whether $X = P^2$ specialness can reduce the BHP exponent

Remark 8.57 (BHP exponent at $X = P^2$: strict analysis)**.** A natural question: can the BHP 2001 short-interval prime-gap exponent 0.525 be reduced to 0.5 when restricted to $X = P^2$, exploiting the special factor structure $X = P \cdot P$?

Direct answer: No. The BHP proof (zero-density estimates + Heath-Brown identity + exponential sums) is uniform in X and does not access the factor structure of X. The "pre-sieved P" condition on row $I_k = (kP, kP + P]$, namely $\gcd(n, P) = 1$ for all $n \in I_k$, is automatically encoded in the sieve framework with $z = P$ and contributes no extra cancellation.

Indirect path ("almost all P"): The set $\{P^2 : P \leq \sqrt{X}\}$ has cardinality $\sim 2\sqrt{X}/\log X$, density zero in $[1, X]$. Jia 1996 / Baker–Harman 1996 give an unconditional almost-all short-interval prime theorem with exceptional set $\ll X^{0.535}$. To deduce "H_P for almost all primes P" from this, one would need the exceptional-set exponent reduced below 1/2, since $|\{P^2 \leq X\}| \sim 2\sqrt{X}/\log X = X^{1/2-o(1)}$. The required sharpening $0.535 \to 0.5$ is itself an open problem of the same strength as a 1/2-exponent BHP improvement.

Open speculation: Whether the subset $\{P^2\}$ admits additional cancellation (e.g. via $\mathbb{Z}[i]$ factorisation when $P \equiv 1 \pmod 4$), or via $L_K(s)$ for quadratic extensions $K = \mathbb{Q}(\sqrt{-P})$) is a genuine research question allied with Friedlander–Iwaniec 1998 / Maynard 2013-style breakthroughs, but no known technique resolves it inside the linear-sieve framework.

8.26 Maynard 2013 close reading and rigorous embedding of H_P

Remark 8.58 (Why Maynard 2013 is the most promising real-frontier route)**.** Among the three real-frontier breakthrough routes (Friedlander–Iwaniec 1998 cubic forms; $L_K(s)$ for quadratic extensions; Maynard 2013 multidimensional GPY), Maynard 2013 is selected for atomic close reading because: (i) the paper is 31 pages of well-organized combinatorial-sieve material; (ii) the multidimensional Selberg–GPY weights apply to any admissible k-tuple, structurally compatible with H_P rows; (iii) it does not require Riemann Hypothesis or analytic continuation of automorphic L-functions.

External-theorem-closed Statement 8.59 (Maynard 2013 main theorem)**.** *There exists $m \geq 0$ such that for every admissible k-tuple $\mathcal{H} = \{h_1, \ldots, h_k\}$ with $k \geq k_0(m)$,*

$$\liminf_{n\to\infty} \#\big(\{n + h_i : h_i \in \mathcal{H}\} \cap \mathcal{P}\big) \geq m + 1.$$

Proposition 8.60 (H_P embedded in Maynard framework)**.** *Take $\mathcal{H} = \{0, 1, \ldots, P-1\}$ and $N = kP \sim P^2$. Maynard's sieve level $R = N^{\theta/2} = P^\theta$. The Maynard ratio*

$$\frac{S_2}{S_1} \sim \frac{k\theta}{2} \cdot \frac{J_k(F)}{I_k(F)} > \frac{\theta \log P}{2}(1 - o(1)).$$

The critical sieve parameter is $s = \log R/\log z = \theta$, where $z = \sqrt{N} = P$. Unconditional Bombieri–Vinogradov gives $\theta < 1/2$; the Elliott–Halberstam conjecture (EH) gives $\theta \leq 1$. In both cases $s \leq 1 \leq 2$, so $f(s) = 0$ in the linear sieve sense.

Remark 8.61 (Why Maynard cannot close H_P in this session)**.** Two independent reasons prevent closure inside Maynard's framework:

Statistical vs. pointwise gap: Maynard's output is $\liminf_n$, i.e. "infinitely many n have ≥ 2 primes among $\{n+h_i\}$." H_P requires "every $k \in [1, P-1]$ has at least one prime in I_k." No general statistical bound implies a pointwise bound; this is a framework-essential gap, not a constant gap.

Critical sieve parameter: Even at the EH limit $\theta = 1$, the sieve parameter $s = 1 \leq 2$, so $f(1) = 0$. The Bombieri 1976 parity barrier remains binding. EH itself is unproved.

Remark 8.62 (The Maynard/EH row-exception estimate is not derived here)**.** A tempting statement would be that Maynard's method, perhaps under EH, forces all but $O(P^{1-\delta})$ of the H_P rows to contain a bounded-gap prime pair. That statement is not a consequence of the Maynard theorem alone. Maynard gives a prime-cluster conclusion for infinitely many translates of an admissible tuple, whereas H_P asks for every prescribed row I_k. Passing from the former to a uniform row-exception estimate would require an additional distribution theorem for the specific row family.

Thus the only accepted conclusion at this point is negative: Maynard 2013 supplies a powerful statistical sieve architecture, but it does not by itself close H_P, even conditionally on EH.

Remark 8.63 (Maynard is not H_P)**.** H_P remains **Not claimed**. Any usable Maynard-derived row statement would need an extra, explicitly stated statistical-to-row upgrade; without that input the possible exceptional rows could still contain all of the obstruction.

The honest closure of this frontier is the rigorous diagnosis recorded above: Maynard 2013, the best-positioned of the three real-frontier routes, structurally cannot close H_P in a single-session ambient. Future progress requires either an unconditional resolution of EH plus a statistical-to-pointwise upgrade, or a non-sieve technique breaking the Bombieri 1976 parity barrier outright.

8.27 Ford–Maynard prime-producing sieve theorem-match for the Phi–LPF tail

External-theorem-closed Statement 8.64 (Ford–Maynard prime-producing sieve framework). *Ford and Maynard develop a general framework for non-negative sequences a_n, b_n, supported on $x/2 < n \leq x$, with $w_n = a_n - b_n$. If w_n satisfies Type-I estimates in a divisor range and Type-II bilinear estimates in a prescribed range, their framework gives optimal constants $C^-(\gamma, \theta, \nu)$ and $C^+(\gamma, \theta, \nu)$ controlling the prime sum of a_n in terms of the comparison prime sum of b_n. The same paper also shows that a genuinely positive Type-II range is necessary in general.*

Proposition 8.65 (H_P row sequence embedded in the Ford–Maynard interface). *In the strict top band put $x \asymp P^2$ and*

$$I_{P,k} = (kP, (k+1)P), \qquad H = |I_{P,k}| = P = x^{1/2}.$$

A formal Ford–Maynard target sequence is

$$a_{P,k}(n) = \frac{x}{H} 1_{kP<n<(k+1)P},$$

optionally multiplied by fixed wheel unit filters, with $b_{P,k}$ a smooth local-density comparison sequence of the same total mass. Then

$$\sum_p a_{P,k}(p) > 0 \quad \Longleftrightarrow \quad \pi((k+1)P - 1) - \pi(kP) > 0.$$

Thus the row-prime problem is formally a prime-producing sequence problem. The formal embedding, however, supplies no Type-I or Type-II estimate.

Theorem 8.66 (Ford–Maynard theorem-match obstruction for the Phi–LPF tail). *The 30-wheel Phi–LPF tail has the balanced scale required for a Type-II attack: in the top band $x \asymp P^2$ one has*

$$q \asymp m \asymp P \asymp x^{1/2}.$$

But the support is the same-row reciprocal graph

$$I_q(P, k) = [\max(q, \lfloor kP/q \rfloor + 1), \min(2P - 1, \lfloor((k + 1)P - 1)/q \rfloor)],$$

with fibres

$$\#I_q(P, k) \leq 2, \qquad \#\{q : m \in I_q(P, k)\} \leq 2, \qquad \#\{a : kP < qra < (k + 1)P\} \leq 1.$$

Consequently the Ford–Maynard framework does not apply by citation alone. A genuine application to H_P requires the following row-uniform inputs:

```
FMTypeIShortRowDivisorSwitchEstimate
FMTypeIISameRowReciprocalGraphBilinearDispersion
FMLocalDensityForWheelRowComparisonSequence
FMPointwiseUniformAllRowsUpgrade.
```

The existing CRT audit also rules out replacing these inputs by fixed unit-cell non-negativity; any CRT continuation must be character averaged or otherwise genuinely dispersive.

Status and reference. The scale $q, m \asymp P$ is immediate from the already recorded LPF-tail representation. The three fibre bounds are the deterministic reciprocal-window bounds in the Type-II obligation certificate. The Ford–Maynard hypotheses are estimates for the target sequence $w_n = a_n - b_n$, with arbitrary divisor-bounded coefficients in the Type-II range. The certificate

```
docs/monograph/prime-matrix-phi-lpf-ford-maynard-
embedding-obligation-audit.md
```

records the theorem-match table. It marks the row sequence and prime sum target as formally matched, but the Type-I short-row divisor estimate, same-row reciprocal-graph Type-II dispersion, local-density comparison, and pointwise all-row upgrade as unproved. Therefore the Ford–Maynard embedding is complete as a conditional schema, not as an unconditional proof of H_P. ☐

External-theorem-closed Statement 8.67 (Conditional Ford–Maynard schema for H_P)**.** *If, for every sufficiently large prime P and every strict row k, the normalized row sequence above satisfies the Ford–Maynard Type-I, Type-II, local-density, and positive C^- hypotheses uniformly, then H_P follows for those rows.*

Remark 8.68 (Current status of the selected route)**.** This is the present non-circular compression of the hard point. It does not assert that the required Type-I/Type-II estimates are known. The most direct remaining line is

`FMTypeIISameRowReciprocalGraphBilinearDispersion`,

together with the accompanying short-row Type-I and local-density checks. The external lemma version and the internal self-contained version both remain unconditionally open.

8.28 Kloosterman gateway for the same-row reciprocal graph

External-theorem-closed Statement 8.69 (DFI, Bettin–Chandee, and Wright Kloosterman technologies)**.** *The relevant external Kloosterman-fraction technology includes Duke–Friedlander–Iwaniec's bilinear forms with Kloosterman fractions, Bettin–Chandee's trilinear Kloosterman fractions, and Wright's 2026 partially fixed-moduli refinement for unbalanced convolution discrepancies. These are genuine frontier tools for bilinear or trilinear inverse-fraction phases and AP convolution averages.*

Proposition 8.70 (Three frequency entrances from the reciprocal graph)**.** *The same-row Phi–LPF tail admits three formal entrances into frequency analysis.*

Endpoint sawtooth. *Solving the row condition for one variable uses*

$$\lfloor ((k+1)P-1)/u \rfloor - \lfloor kP/u \rfloor$$

and hence sawtooth terms with phases of the form

$$e(hkP/u).$$

These are reciprocal phases, not modular-inverse Kloosterman fractions.

Product-window Fourier. *Keeping the product condition $kP < uv < (k+1)P$ before solving for either variable gives additive bilinear phases of the form $e(tuv/Y)$. A further Cauchy–Poisson or completion step would be needed before an inverse-fraction Kloosterman theorem can be cited.*

CRT character average. *The signed residue measure $\sum_a \mu_S(a; P, k)\chi(a)$ is compatible with character averaging, but the fixed unit-cell non-negativity route is already false. It still requires same-row signed dispersion.*

Theorem 8.71 (Kloosterman theorem-match gate for the Phi–LPF reciprocal graph). *The external Kloosterman sources above do not directly close*

`FMTypeIISameRowReciprocalGraphBilinearDispersion`.

DFI estimates bilinear inverse-fraction phases $e(a\bar{m}/n)$; Bettin–Chandee estimates trilinear inverse-fraction phases $e(\theta a\bar{m}/n)$; Wright's 2026 theorem estimates averaged AP convolution discrepancies with a Siegel–Walfisz input. The present object is a fixed pointwise product-window row with q prime, $m = ra$, $r = \mathrm{LPF}(m)$, a r-rough, and one-point (q, r)-fibres. Therefore a direct citation would miss a necessary completion step.

The narrowed hard point is

```
ReciprocalGraphToKloostermanCompletionIdentity
AND CompletedKloostermanMeanForPrimeQAndLPFShellWeights
AND SawtoothTailLogSavingForThinReciprocalFibres.
```

Status and reference. The endpoint and product-window identities are deterministic rewritings of the already established reciprocal graph

$$I_q(P, k) = [\max(q, \lfloor kP/q \rfloor + 1), \min(2P - 1, \lfloor ((k + 1)P - 1)/q \rfloor)].$$

They show exactly where the external Kloosterman technology could enter and exactly why it cannot yet be cited: the phases first obtained are reciprocal or additive product phases, whereas the DFI/BC theorems control inverse-modulus Kloosterman fractions after a completion step. Wright's theorem is closer to a dispersion package with partially fixed moduli, but it remains an averaged AP convolution theorem, not a pointwise fixed-row result for H_P.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-reciprocal-
graph-kloosterman-gateway-audit.md.
```

It also records that the 2026 Dong–Robles–Zeindler bilinear Kloosterman-fraction preprint is withdrawn and is not accepted as a source. No unconditional row/column closure follows from the Kloosterman gateway audit. □

8.29 Sawtooth reciprocal tail gateway

Proposition 8.72 (Unweighted reciprocal phase benchmark). *Consider the unweighted model sum*

$$S(A; N) = \sum_{N<n\leq 2N} e(A/n), \qquad A = hkP, \qquad N = P.$$

The classical second derivative estimate gives

$$S(A; N) \ll \sqrt{A/N} + \sqrt{N^3/A} = \sqrt{hk} + \frac{P}{\sqrt{hk}},$$

up to an absolute constant. Moreover, if the high-q reciprocal graph is nonempty, then $q, m > P/2$ forces $qm > P^2/4$, and hence $k+1 > P/4$. Thus the active rows are P-scale. For $h \leq H = (\log P)^B$, the unweighted finite sawtooth modes have power saving against the trivial P-scale total, while the Vaaler tail contributes $O(P/H)$.

Remark 8.73 (What this does and does not close)**.** This removes the unweighted real-reciprocal phase as the main obstruction inside

`SawtoothTailLogSavingForThinReciprocalFibres`.

It does not close the Phi–LPF problem. The actual object has a prime q, the constraint $m = ra \in I_q(P, k)$, $r = \mathrm{LPF}(m) \geq 7$, and an r-rough quotient a. None of these weights is present in the unweighted second-derivative model.

External-theorem-closed Statement 8.74 (Current external match for the sawtooth gate)**.** *The relevant external tools now separate cleanly. The classical second derivative estimate matches the unweighted real phase $e(A/u)$. DFI, Bettin–Chandee, and Wright remain relevant only after an inverse-fraction or dispersion completion. The 2025/2026 Shao–Shparlinski–Wijaya results on Kloosterman sums over square-free and smooth integers are useful frontier evidence for weighted Kloosterman parameters, but they work after finite-field Kloosterman completion; they do not directly estimate the real reciprocal phase $e(A/q)$ with prime-q and Phi–LPF row weights.*

Theorem 8.75 (Sawtooth gate compression)**.** *The sawtooth tail gate is not closed, but it is now compressed to*

```
PrimeQLPFShellWeightedReciprocalPhaseSaving
AND WeightExtractionFromLPFShellToBilinearKloostermanOrVaughanTypeII
AND UniformFiniteHTruncationWithHPolylog.
```

In particular, the unweighted reciprocal-phase benchmark is closed, whereas the weighted Phi–LPF sawtooth phase is still open.

Status and reference. The first assertion is the preceding second-derivative calculation and the elementary comparability $k + 1 > P/4$ on active high-q rows. The Vaaler truncation bookkeeping is deterministic once the total thin-fibre weight is $O(P)$. The missing part is cancellation for the finite Fourier modes against the prime-q, LPF-shell, and rough quotient weights. This is precisely the named weighted reciprocal phase saving above. The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-sawtooth-
reciprocal-tail-gateway-audit.md.
```

It records the finite unweighted prime-q diagnostics and the external theorem-match table. No unconditional row/column closure follows. ☐

8.30 Finite-H truncation closure

Proposition 8.76 (Choice among the three manuscript claims)**.** *Among the three current manuscript fronts, the Prime Matrix row/column Phi–LPF line has the shortest fully closable subgate. The two-point sieve line still requires the Buchstab transfer* `BMD=>TLI` *without a hidden denominator or parity gap, and the RH contradiction-field line remains an independent referee package for controlled exits. By contrast*

`UniformFiniteHTruncationWithHPolylog`

is a deterministic truncation ledger once the reciprocal thin-fibre mass bound is fixed.

Theorem 8.77 (Polylogarithmic finite-H truncation)**.** *For every strict row and every $A > 0$, choose*

$$H = \lceil (\log P)^{A+2} \rceil.$$

The total high-q reciprocal candidate mass satisfies

$$W_{\rm int}(P,k) \le 2\pi(P) < 2P.$$

Since the two endpoint sawtooth terms have absolute mass at most $4P$, the Vaaler truncation tail contributes

$$O(P/H) = O(P/(\log P)^{A+2}) = O(P/(\log P)^A).$$

Thus

`UniformFiniteHTruncationWithHPolylog`

is closed unconditionally.

Proof. Each prime $q \in (P/2, P)$ contributes at most two integer m-values to the reciprocal window $I_q(P,k)$. Hence the total absolute counting mass is at most $2\pi(P)$, and the two endpoint sawtooth expansions have total absolute mass at most $4P$. Vaaler's finite Fourier approximation leaves a pointwise endpoint error $O(1/H)$; summing against the nonnegative LPF-shell counting weights gives the claimed $O(P/H)$ tail. The finite modes left after truncation have $|h| \le H$ and only the harmonic coefficient cost $O(\log H)$. The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-finite-h-
truncation-closure-audit.md.
```

□

Remark 8.78 (Updated true remaining mouth). The deterministic finite-H gate is no longer part of the hard point. The current sawtooth branch is reduced to

```
PrimeQLPFShellWeightedReciprocalPhaseSaving
AND WeightExtractionFromLPFShellToBilinearKloostermanOrVaughanTypeII.
```

The recent Kloosterman-frontier inputs of Milicevic–Qin–Wu, Pascadi, and Shao–Shparlinski–Wijaya remain useful only after a finite-field or inverse-fraction completion; they do not by themselves close the prime-q, LPF-shell weighted real reciprocal phase.

8.31 LPF weight extraction norm closure

Proposition 8.79 (Next fastest gate among the three claims). *After the finite-H truncation is closed, the quickest remaining fully deterministic subgate is again on the row/column Phi–LPF line:*

`LPFShellWeightBoundedCoefficientExtraction`.

The two-point sieve line still requires the Buchstab transfer `BMD=>TLI` *without a hidden denominator or parity gap, and the RH contradiction-field line remains an independent referee package. The LPF extraction gate is only a support and norm statement, so it can be closed before any new prime distribution theorem is proved.*

Theorem 8.80 (LPF-shell weights as bounded coefficients). *For the 30-wheel residual in a fixed row,*

$$R_{30}(P,k) = \sum_{\substack{P/2<q<P \\ q \text{ prime}}} \sum_{\substack{r\ge 7 \\ r \text{ prime}}} \sum_{\substack{a\ge r \\ P^-(a)\ge r}} \beta_{P,k}(q,r,a)\, \mathbf{1}_{kP<qra<(k+1)P},$$

where $\beta_{P,k}(q,r,a) \in \{0,1\}$. Moreover, since $q > P/2$ and $r \geq 7$, each fixed (q,r) has at most one quotient a in the row. The projected prime-q coefficient

$$b_{P,k}(q) = \#\{m \in I_q(P,k) : m \text{ composite, } \mathrm{LPF}(m) \geq 7\}$$

satisfies

$$0 \leq b_{P,k}(q) \leq \#I_q(P,k) \leq 2$$

and

$$\sum_q b_{P,k}(q) = R_{30}(P,k) \leq W_{\mathrm{int}}(P,k) \leq 2\pi(P) < 2P.$$

Thus LPF-shell extraction causes no coefficient blow-up.

Proof. The identity is the ordered LPF factorization $m = ra$, with $r = \mathrm{LPF}(m)$ and a r-rough. Conversely every such triple gives exactly one residual cofactor $m = ra$. The bound for fixed (q,r) follows from

$$\frac{(k+1)P - kP}{qr} = \frac{P}{qr} < 1.$$

The projected bound follows from the already established reciprocal window estimate $\#I_q(P,k) \leq 2$. Summing over prime q gives the total-mass estimate. This avoids the unnecessary full Mobius expansion of the rough condition over all primes $< r$, whose exact form is true but has a useless exponential ℓ^1 risk.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-weight-
extraction-norm-closure-audit.md.
```

□

Remark 8.81 (Updated true remaining mouth after weight extraction)**.** The LPF weight is no longer a separate coefficient-norm obstruction. The current row/column sawtooth branch is reduced to

```
PrimeQBoundedLPFCoefficientReciprocalPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. It is the sharper phase/completion problem left after deterministic finite truncation and bounded coefficient extraction have both been removed.

8.32 Boolean prime-q projection closure

Proposition 8.82 (Next deterministic prime-q gate)**.** *After the LPF-shell weights have been extracted with bounded norm, the next fully deterministic subgate is*

PrimeQBooleanProjectionForLPFShellResidual.

It is again on the row/column Phi–LPF line. The two-point sieve and RH fronts do not have an equally small deterministic gate at this stage. This gate sharpens the projected coefficient but does not prove any exponential-sum cancellation.

Theorem 8.83 (Projected LPF residual coefficients are Boolean). *For every prime $q \in (P/2, P)$, define*

$$b_{P,k}(q) = \#\{m \in I_q(P,k) : m \text{ composite}, \ \mathrm{LPF}(m) \geq 7\}.$$

Then

$$b_{P,k}(q) \in \{0,1\}.$$

Consequently the finite sawtooth modes reduce to exponential sums over a subset $Q_{P,k} \subset \{q : P/2 < q < P, \ q \text{ prime}\}$, rather than over a prime sequence with multiplicity.

Proof. The reciprocal window $I_q(P,k)$ has at most two integer points because $P/q < 2$. If two points occur, they are consecutive integers. On the other hand, every residual cofactor counted by $b_{P,k}(q)$ is composite and has least prime factor at least 7; in particular it is odd. Two consecutive integers contain at most one odd integer. Hence at most one point of $I_q(P,k)$ can be counted.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-boolean-
q-projection-closure-audit.md.
```

□

Remark 8.84 (Updated true remaining mouth after Boolean projection). The current row/column sawtooth branch is now reduced to

```
PrimeQBooleanSubsetReciprocalPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This sharpening uses the elementary 2-wheel parity obstruction in a positive way: it removes double prime-q multiplicity. It still does not break the global Phi–LPF parity barrier, because cancellation over the resulting prime subset $Q_{P,k}$ remains unproved.

8.33 Reciprocal matching graph closure

Proposition 8.85 (Next deterministic reciprocal graph gate). *After the prime-q projection is Boolean, the next deterministic subgate is*

`ReciprocalResidualGraphIsPartialMatching`.

It again belongs to the row/column Phi–LPF line. It sharpens the support geometry of the residual graph, but it is not a replacement for a prime-q reciprocal phase estimate.

Theorem 8.86 (The residual reciprocal graph is a partial matching). *Let $G_{P,k}$ be the bipartite graph whose left vertices are primes $q \in (P/2, P)$, whose right vertices are residual cofactors m, and whose edges are the pairs*

$$kP < qm < (k+1)P, \qquad m \in I_q(P,k), \qquad m \text{ composite}, \qquad \mathrm{LPF}(m) \geq 7.$$

Then every left vertex and every right vertex has degree at most 1. Hence $G_{P,k}$ is a partial matching.

Proof. The left degree bound is the Boolean prime-q projection theorem above. For the right degree bound, fix a residual cofactor m. The possible integer q's lie in

$$\frac{kP}{m} < q < \frac{(k+1)P}{m},$$

with the additional restrictions $P/2 < q < P$, $q \le m$, and q prime. Since residual edges have $m > P/2$, this interval has length $P/m < 2$. Thus it contains at most two integers, and if two occur they are consecutive. But $q > P/2 > 2$, so any prime q is odd; two consecutive integers cannot both be odd primes. Therefore at most one prime q can connect to the fixed m.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-matching-
graph-closure-audit.md.
```

□

Remark 8.87 (Updated true remaining mouth after matching closure)**.** The current row/column sawtooth branch is now reduced to

```
PrimeQMatchingSubsetReciprocalPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This removes bidirectional multiplicity from the reciprocal residual graph. The hard point is still a phase/completion theorem for a moving matched prime subset, not a closed unconditional Phi–LPF proof.

8.34 Matched displacement phase closure

Proposition 8.88 (Next deterministic phase normal form gate)**.** *After the residual reciprocal graph is a partial matching, the next deterministic subgate is*

$$\mathtt{MatchedDisplacementPhaseNormalForm}.$$

It again belongs to the row/column Phi–LPF line. This gate changes the shape of the remaining exponential phase, but it is not a cancellation theorem.

Theorem 8.89 (Matched displacement phase normal form)**.** *Let (q, m) be an edge of the residual matching graph $G_{P,k}$, and put*

$$d = qm - kP.$$

Then

$$1 \le d < P,$$

and for every integer h,

$$e\left(\frac{hkP}{q}\right) = e\left(-\frac{hd}{q}\right).$$

Moreover m is one of the two clipped floor endpoints in the q-side window $I_q(P, k)$, and q is one of the two clipped floor endpoints in the reverse m-side window.

Proof. The edge condition gives $kP < qm < (k + 1)P$, hence

$$1 \le qm - kP < P.$$

Thus $d = qm - kP$ lies in the desired interval. Since $kP = qm - d$,

$$\frac{hkP}{q} = hm - \frac{hd}{q}.$$

The first term is an integer, so it disappears from the additive character and gives the stated identity.

For the endpoint assertion, the q-side integer window has length less than 2, because its uncut real interval has length $P/q < 2$. Hence every integer point in it is either the lower endpoint, the upper endpoint, or both. The reverse m-side window has length less than 2 as well, because residual edges have $m > P/2$. Hence its matched prime q is also a lower endpoint, an upper endpoint, or both.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-matched-
displacement-phase-closure-audit.md.
```

□

Remark 8.90 (Updated true remaining mouth after phase normal form)**.** The large-numerator reciprocal phase has now been replaced by a matched displacement phase over labelled edges (q, m, d):

$$e(hkP/q) = e(-hd/q), \qquad 1 \leq d < P.$$

The current row/column sawtooth branch is reduced to

```
PrimeQMatchedDisplacementPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining problem is to prove cancellation for the moving displacement phase, or to complete it to an external Kloosterman/Vaughan Type-II estimate without losing the fixed row P, k.

8.35 Floor-residue branch phase closure

Proposition 8.91 (Next deterministic floor-residue branch gate)**.** *After the matched displacement normal form is established, the next deterministic subgate is*

$$\texttt{PrimeQFloorResidueBranchNormalForm}.$$

It removes the artificial q-side clipping alternatives from the residual phase. It is still a normal-form statement, not a phase-saving theorem.

Theorem 8.92 (Residual edges are floor-residue branch edges)**.** *Let (q, m) be a residual edge with $P/2 < q < P$, q prime,*

$$kP < qm < (k+1)P, \qquad m \text{ composite}, \qquad \mathrm{LPF}(m) \geq 7.$$

Then the q-side endpoint supporting m is never the boundary cap $m = q$ or $m = 2P - 1$. Hence every residual edge lies on a lower and/or upper floor endpoint:

$$m = \left\lfloor \frac{kP}{q} \right\rfloor + 1 \quad \textit{or} \quad m = \left\lfloor \frac{(k+1)P - 1}{q} \right\rfloor.$$

Writing

$$\rho_q \equiv kP \pmod q, \qquad \sigma_q \equiv (k+1)P - 1 \pmod q, \qquad 0 \leq \rho_q, \sigma_q < q,$$

the matched displacement $d = qm - kP$ satisfies the exact branch formulae

$$d = q - \rho_q \quad \textit{on the lower branch}, \qquad d = P - 1 - \sigma_q \quad \textit{on the upper branch}.$$

If an edge is both lower and upper, the two formulae give the same integer d.

Proof. The q-side product window has real length $P/q < 2$. If the lower cap $m = q$ were responsible for a residual edge, then the only possible integer points at that cap are q and, at most, $q+1$. The first is prime, and the second is even. Neither can be a composite cofactor with least prime factor at least 7.

For the upper cap, the point $m = 2P - 1$ cannot occur in any row with $k < P$, because every integer $q > P/2$ satisfies

$$q(2P-1) \geq \frac{P+1}{2}(2P-1) > P^2 \geq (k+1)P.$$

The only adjacent point allowed by the length < 2 window is $2P - 2$, which is even. Hence no residual edge is supported by the upper boundary cap either.

Thus residual edges on the q-side come only from the lower and upper floor endpoints. On the lower branch,

$$qm = q\left(\left\lfloor \frac{kP}{q} \right\rfloor + 1\right) = kP - \rho_q + q,$$

so $d = q - \rho_q$. On the upper branch,

$$qm = q\left\lfloor \frac{(k+1)P-1}{q} \right\rfloor = (k+1)P - 1 - \sigma_q,$$

so $d = P - 1 - \sigma_q$. A singleton endpoint is simultaneously lower and upper, and then both computations evaluate the same $qm - kP$.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-floor-
residue-branch-phase-closure-audit.md.
```

□

Remark 8.93 (Updated true remaining mouth after floor-residue split)**.** The matched displacement phase has now been split into lower/upper floor-residue branches:

$$d = q - (kP \bmod q) \quad \text{or} \quad d = P - 1 - \big(((k+1)P - 1) \bmod q\big).$$

The current row/column sawtooth branch is reduced to

```
PrimeQFloorResidueBranchPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is to prove cancellation for these LPF-selected branch phases, or to complete them to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k. The 2026 Kloosterman-fraction near-miss arXiv:2601.00292 is recorded only as withdrawn and is not used as an external input.

8.36 Parity-selected branch phase closure

Proposition 8.94 (Next deterministic branch selector gate)**.** *After the floor-residue branch normal form is established, the next deterministic subgate is*

$$\texttt{TwoPointFloorWindowParitySelector}.$$

It removes the last endpoint-side choice in two-point q-windows. It is not a phase-saving theorem.

Theorem 8.95 (Two-point floor windows are parity selected). *Let (q, m) be a residual edge and set*

$$L = \left\lfloor \frac{kP}{q} \right\rfloor + 1, \qquad U = \left\lfloor \frac{(k+1)P - 1}{q} \right\rfloor.$$

After the boundary-cap exclusions above, $m \in \{L, U\}$, and on every residual edge

$$U - L \in \{0, 1\}.$$

If $U = L$, then the edge is simultaneously lower and upper, and the two floor-residue formulae give the same $d = qm - kP$. If $U = L + 1$, then the edge is lower exactly when L is odd, and upper exactly when U is odd.

Proof. The real product interval has length $P/q < 2$. Since a residual edge exists, the integer floor window is nonempty; hence it contains either one point or two consecutive points, so $U - L \in \{0, 1\}$.

If $U = L$, the unique endpoint is both lower and upper. Both formulae compute the same integer $d = qm - kP$, so there is no branch ambiguity.

If $U = L + 1$, the two candidates are consecutive integers. A residual cofactor has least prime factor at least 7, hence is odd. Therefore only the odd one of the two consecutive endpoints can be the residual cofactor. This proves the lower/upper parity selector.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-parity-
selected-branch-phase-closure-audit.md.
```

□

Remark 8.96 (Updated true remaining mouth after parity selection). The endpoint side in every two-point floor window is now deterministic: the LPF condition decides whether the odd endpoint is a residual cofactor, but not which side it would occupy. The current row/column sawtooth branch is reduced to

```
PrimeQParitySelectedFloorResidueBranchPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is cancellation for the parity-selected branch phase, or a completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

8.37 Unique odd candidate projection closure

Proposition 8.97 (Next deterministic q-candidate projection gate). *After parity-selected branch closure, the next deterministic subgate is*

$$\texttt{PrimeQUniqueOddCandidateProjection}.$$

It replaces the residual branch graph by a single-valued candidate map on prime q. It is still not a phase-saving theorem.

Theorem 8.98 (Residual edges are an LPF subset of one odd candidate). *For each prime $q \in (P/2, P)$, define the clipped reciprocal window*

$$J_q(P, k) = \left[\max\left(q, \left\lfloor \frac{kP}{q} \right\rfloor + 1\right), \min\left(2P - 1, \left\lfloor \frac{(k+1)P - 1}{q} \right\rfloor\right)\right].$$

Then $J_q(P,k)$ contains at most one odd integer. Denote it, when it exists, by $\omega_{P,k}(q)$. The residual edges are exactly

$$(q,m) \quad \textit{with} \quad m = \omega_{P,k}(q), \qquad m \text{ composite}, \qquad \mathrm{LPF}(m) \geq 7.$$

On such a residual q, if

$$D(q) = q\,\omega_{P,k}(q) - kP,$$

then $1 \leq D(q) < P$ and

$$e(hkP/q) = e(-hD(q)/q)$$

for every integer h.

Proof. The uncut product interval has length $P/q < 2$. Clipping by q and $2P-1$ cannot increase the number of integer points, so $J_q(P,k)$ contains at most two consecutive integers. Hence it contains at most one odd integer.

Every residual cofactor has least prime factor at least 7, hence is odd. Therefore any residual edge over the fixed q must use the unique odd integer $\omega_{P,k}(q)$, if it exists. Conversely, if that odd candidate exists, is composite, and has least prime factor at least 7, then by construction it lies in $J_q(P,k)$, so it gives exactly one residual edge. This proves the edge equivalence.

The displacement and phase identity are the matched-displacement identity already proved, applied to $m = \omega_{P,k}(q)$.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-unique-
odd-candidate-projection-closure-audit.md.
```

□

Remark 8.99 (Updated true remaining mouth after odd-candidate projection)**.** The residual object is now a q-subset of the single-valued odd-candidate map

$$q \mapsto \omega_{P,k}(q).$$

The current row/column sawtooth branch is reduced to

```
PrimeQUniqueOddCandidateLPFSubsetPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is cancellation over the q's for which $\omega_{P,k}(q)$ is composite with least prime factor at least 7, or a completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

8.38 Unique odd candidate LPF partition closure

Proposition 8.100 (The next deterministic LPF-partition gate)**.** *After the unique odd-candidate projection, the next deterministic subgate is*

$$\texttt{UniqueOddCandidateFiveWayLPFPartition}.$$

It is a pointwise partition of the candidate map $q \mapsto \omega_{P,k}(q)$. It is not a phase-saving theorem.

Theorem 8.101 (Five-way LPF partition of the unique odd candidate)**.** *For each prime $q \in (P/2, P)$, exactly one of the following five alternatives holds:*

$$\begin{array}{ll} (0) & \omega_{P,k}(q) \text{ does not exist,} \\ (\mathrm{p}) & \omega_{P,k}(q) \text{ is prime,} \\ (3) & \mathrm{LPF}(\omega_{P,k}(q)) = 3, \\ (5) & \mathrm{LPF}(\omega_{P,k}(q)) = 5, \\ (\mathrm{R}) & \omega_{P,k}(q) \text{ is composite and} \mathrm{LPF}(\omega_{P,k}(q)) \geq 7. \end{array}$$

Moreover the residual edges are exactly the last cell. Equivalently, on the candidate support,

$$\mathbf{1}_{(\omega_{P,k}(q),30)=1} = \mathbf{1}_{\omega_{P,k}(q) \text{ prime}} + \mathbf{1}_{\omega_{P,k}(q) \text{ composite, LPF}\geq 7}.$$

Thus a residual edge is the same as

$$\omega_{P,k}(q) \text{ exists}, \qquad (\omega_{P,k}(q), 30) = 1, \qquad \omega_{P,k}(q) \text{ composite.}$$

Proof. The previous theorem gives at most one odd candidate $\omega_{P,k}(q)$ for the fixed prime q. If no such candidate exists we are in cell (0). Otherwise the candidate is odd. If it is prime we are in cell (p). If it is composite and has least prime factor < 7, that least prime factor cannot be 2, so it is 3 or 5. These give cells (3) and (5). The remaining composite candidates have least prime factor at least 7, giving cell (R).

The cells are mutually exclusive by definition of the least prime factor, and they are exhaustive by the preceding paragraph. The residual edge equivalence is the unique odd-candidate theorem with the identity

$$\text{composite and LPF} \geq 7 \iff \text{composite and } (\omega_{P,k}(q), 30) = 1$$

for odd candidates.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-unique-
odd-candidate-lpf-partition-closure-audit.md.
```

□

Corollary 8.102 (Exact four-term candidate phase decomposition)**.** *For every integer h, the candidate phase sum decomposes exactly as*

$$S_{\mathrm{cand}}(h) = S_{\mathrm{prime}}(h) + S_{\mathrm{LPF}=3}(h) + S_{\mathrm{LPF}=5}(h) + S_{\mathrm{res}}(h),$$

where each term uses the same phase

$$e(-hD(q)/q), \qquad D(q) = q\,\omega_{P,k}(q) - kP.$$

In particular, the remaining residual term is the composite part of the 30*-wheel survivor candidate set.*

Remark 8.103 (Updated true remaining mouth after the LPF partition)**.** The small-prime cells have now been separated pointwise. The current row/column sawtooth branch is reduced to

```
PrimeQUniqueOddCandidateWheel30CompositeSurvivorPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is a signed or oscillatory separation of composite 30-wheel survivors from prime survivors, or a completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

8.39 Wheel-30 composite LPF descent closure

Proposition 8.104 (Next deterministic LPF descent gate). *After the* 30*-wheel composite survivor partition, the next deterministic subgate is*

`Wheel30CompositeLPFDescentUniqueRoughQuotient.`

It replaces the composite part of the unique odd-candidate map by a prime q, small least-prime-factor r, and rough quotient candidate. It is still not a phase-saving theorem.

Theorem 8.105 (Unique rough quotient for fixed q, r). *Let $q \in (P/2, P)$ be prime and let $r \geq 7$ be prime. Define*

$$\alpha_{P,k}(q, r) = \left\lfloor \frac{kP}{qr} \right\rfloor + 1.$$

If a is an integer satisfying

$$kP < qra < (k+1)P,$$

then $a = \alpha_{P,k}(q, r)$. In particular, for fixed q, r there is at most one possible quotient a.

Proof. The real interval for a has length

$$\frac{(k+1)P - kP}{qr} = \frac{P}{qr} < \frac{2}{r} < 1.$$

Therefore it contains at most one integer. The least integer strictly larger than $kP/(qr)$ is $\lfloor kP/(qr) \rfloor + 1$, proving the claim. □

Theorem 8.106 (Residual edges as a prime-q, small-r rough quotient graph). *For a residual edge over q, write $m = \omega_{P,k}(q)$. Then*

$$m = ra, \qquad r = \mathrm{LPF}(m), \qquad 7 \leq r \leq \sqrt{2P-1}, \qquad a \geq r, \qquad \mathrm{LPF}(a) \geq r.$$

Conversely, for primes $q \in (P/2, P)$ and

$$7 \leq r \leq \sqrt{2P-1},$$

put $\alpha = \alpha_{P,k}(q, r)$. Then q contributes a residual edge with least prime factor r if and only if

$$\alpha \leq \left\lfloor \frac{(k+1)P - 1}{qr} \right\rfloor, \qquad \alpha \geq r, \qquad q \leq r\alpha \leq 2P - 1, \qquad \mathrm{LPF}(\alpha) \geq r.$$

On this edge,

$$D(q, r) = qr\alpha - kP, \qquad 1 \leq D(q, r) < P,$$

and

$$e(hkP/q) = e(-hD(q, r)/q)$$

for every integer h.

Proof. Let $m = \omega_{P,k}(q)$ be residual. By the previous partition, m is composite and has least prime factor at least 7. Put $r = \mathrm{LPF}(m)$ and $a = m/r$. Since r is the least prime factor of a composite integer, $a \geq r$ and $r^2 \leq m \leq 2P - 1$, giving $r \leq \sqrt{2P-1}$. Also no prime below r divides a, so $\mathrm{LPF}(a) \geq r$. The product-window condition

$$kP < qm < (k+1)P$$

becomes $kP < qra < (k+1)P$, hence $a = \alpha_{P,k}(q,r)$ by the previous theorem.

Conversely, assume the displayed conditions for q, r, α. Let $m = r\alpha$. The upper bound on α and the definition of α give

$$kP < qm = qr\alpha < (k+1)P.$$

The clipping inequalities give $q \le m \le 2P-1$. Since $\alpha \ge r$ and $\mathrm{LPF}(\alpha) \ge r$, the least prime factor of $m = r\alpha$ is $r \ge 7$, so m is exactly a residual cofactor. The displacement and phase identity are the matched displacement identity applied to this m.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-wheel30-
composite-lpf-descent-closure-audit.md.
```

□

Remark 8.107 (Updated true remaining mouth after LPF descent)**.** The residual object is now a 0/1 graph on prime q, small prime $r \le \sqrt{2P-1}$, and the unique rough quotient $\alpha_{P,k}(q,r)$. The current row/column sawtooth branch is reduced to

```
PrimeQSmallRoughQuotientCandidatePhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is cancellation or signed separation on this rough quotient graph, or a completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

8.40 Rough quotient second LPF split closure

Proposition 8.108 (Next deterministic second-LPF split gate)**.** *After the prime-q, small-r rough quotient graph has been formed, the next deterministic subgate is*

$$\mathtt{RoughQuotientSecondLPFSplitAndUniqueBeta}.$$

It splits the quotient $\alpha_{P,k}(q,r)$ into a prime quotient cell and a second-LPF composite quotient cell. It is still not a phase-saving theorem.

Theorem 8.109 (Second-LPF split of the rough quotient)**.** *Let a residual edge be written in the previous normal form as*

$$m = r\alpha, \qquad r = \mathrm{LPF}(m), \qquad \alpha = \alpha_{P,k}(q,r), \qquad \mathrm{LPF}(\alpha) \ge r.$$

Then exactly one of the following alternatives holds:

$$\alpha \text{ is prime},$$

or

$$\alpha = s\beta, \qquad s = \mathrm{LPF}(\alpha) \ge r, \qquad \beta \ge s, \qquad \mathrm{LPF}(\beta) \ge s.$$

In the second case, for fixed q, r, s the only possible value of β is

$$\beta_{P,k}(q,r,s) = \left\lfloor \frac{kP}{qrs} \right\rfloor + 1.$$

Proof. Since $\mathrm{LPF}(\alpha) \ge r$, either α is prime or it is composite. In the composite case put $s = \mathrm{LPF}(\alpha)$ and $\beta = \alpha/s$. Then $s \ge r$, $\beta \ge s$, and no prime below s divides β, so $\mathrm{LPF}(\beta) \ge s$.

For fixed q, r, s, any such β must satisfy

$$kP < qrs\beta < (k+1)P.$$

The real interval has length

$$\frac{P}{qrs} < \frac{2}{rs} \le \frac{2}{49} < 1,$$

so it contains at most one integer. The least integer strictly larger than $kP/(qrs)$ is $\lfloor kP/(qrs) \rfloor + 1$, proving the uniqueness formula. □

Corollary 8.110 (Two-cell iterated rough quotient graph)**.** *The residual graph decomposes exactly into two disjoint cells:*

$$\text{semiprime-}\alpha \text{ cell} \quad \sqcup \quad \text{second-LPF rough quotient cell.}$$

The first cell has $m = r\alpha$ with α prime. The second has

$$m = rs\beta, \qquad \beta = \beta_{P,k}(q, r, s).$$

Both cells keep the matched displacement phase

$$e(-hD/q),$$

with $D = qr\alpha - kP$ in the first cell and $D = qrs\beta - kP$ in the second cell.

Remark 8.111 (Updated true remaining mouth after the second-LPF split)**.** The rough quotient is no longer a single black-box rough integer. It has been split into a semiprime quotient cell and a second-LPF rough quotient cell. The current row/column sawtooth branch is reduced to

```
PrimeQIteratedRoughQuotientTwoCellPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is cancellation or signed separation across these two iterated quotient cells, or a completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-rough-
quotient-second-lpf-split-closure-audit.md.
```

8.41 Complete rough factor tree closure

Proposition 8.112 (Next deterministic complete-tree gate)**.** *After the second-LPF split, the next deterministic subgate is*

$$\texttt{CompleteRoughFactorTreeAndPrefixSingletons}.$$

It exhausts the purely factor-theoretic LPF descent. It is still not a phase-saving theorem.

Theorem 8.113 (Complete rough factor tree normal form)**.** *Let (q, m) be a residual edge in the 30-wheel survivor cell. Then*

$$m = p_1 p_2 \cdots p_t, \qquad 7 \le p_1 \le p_2 \le \cdots \le p_t, \qquad t \ge 2,$$

is its unique complete rough-prime factor chain. For every proper prefix

$$G_j = p_1 p_2 \cdots p_j, \qquad 1 \le j < t,$$

the suffix $A_j = m/G_j$ is the unique possible integer and is given by

$$A_j = \left\lfloor \frac{kP}{qG_j} \right\rfloor + 1.$$

Proof. The existence and uniqueness of the displayed nondecreasing prime factor chain is the ordinary unique factorization theorem, together with the residual condition $\mathrm{LPF}(m) \ge 7$ and the fact that m is composite.

Fix a proper prefix G_j. Since the edge condition is

$$kP < qm < (k+1)P,$$

the suffix $A_j = m/G_j$ must lie in

$$\frac{kP}{qG_j} < A_j < \frac{(k+1)P}{qG_j}.$$

This real interval has length

$$\frac{P}{qG_j} < \frac{2}{G_j} \le \frac{2}{7} < 1,$$

because $q > P/2$ and $G_j \ge p_1 \ge 7$. Hence it contains at most one integer. The least integer strictly larger than $kP/(qG_j)$ is

$$\left\lfloor \frac{kP}{qG_j} \right\rfloor + 1,$$

which proves the formula. □

Corollary 8.114 (Deterministic LPF descent is exhausted)**.** *Further splitting by least prime factors only reveals deeper prefixes of the same complete factor chain. No additional deterministic quotient branch remains after the complete rough factor tree has been formed.*

Remark 8.115 (Updated true remaining mouth after the complete tree)**.** The residual object is now a set of complete rough-prime leaves with matched displacement

$$D = qm - kP$$

and phase $e(-hD/q)$. The current row/column sawtooth branch is reduced to

```
PrimeQCompleteRoughFactorTreeLeafPhaseSaving
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is signed cancellation or separation over the complete rough-factor leaves, or an exact completion to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-complete-
rough-factor-tree-closure-audit.md.
```

8.42 Complete leaf phase collapse

Proposition 8.116 (Next deterministic phase-collapse gate). *After the complete rough factor tree has been formed, the next deterministic subgate is*

$$\texttt{CompleteLeafPhaseCollapseToPrimeQSupport}.$$

It removes the possibility of extracting cancellation from the internal LPF factor chain at fixed q.

Theorem 8.117 (Complete leaf phase depends only on q). *Let (q, m) be any complete rough-factor leaf edge and put*

$$D = qm - kP.$$

Then

$$D \equiv -kP \pmod q.$$

Consequently, for every integer h,

$$e(-hD/q) = e(hkP/q).$$

Thus the matched displacement phase of a complete leaf is independent of the factor chain of m.

Proof. The congruence is immediate from $D = qm - kP$, since $qm \equiv 0 \pmod q$. For the phase identity,

$$e(-hD/q) = e(-h(qm - kP)/q) = e(-hm)\, e(hkP/q) = e(hkP/q),$$

because h and m are integers. □

Corollary 8.118 (No internal factor-tree oscillation). *The complete rough factor tree only changes whether a prime q lies in the residual support. It does not create additional oscillation inside the leaves over a fixed q. Therefore complete-leaf phase saving is reduced to signed control of the actual prime-q support set.*

Remark 8.119 (Updated true remaining mouth after phase collapse). The current row/column sawtooth branch is reduced to

```
PrimeQSupportSetReciprocalPhaseSavingBeyondParity
AND CompletionToExternalKloostermanOrVaughanTypeII.
```

This is still not an unconditional proof of the Phi–LPF target. The remaining task is an object-sensitive theorem for the actual prime-q support set, or an exact completion of that support-set reciprocal phase to an external Kloosterman/Vaughan Type-II estimate without losing pointwise control in P, k.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-complete-
leaf-phase-collapse-audit.md.
```

8.43 External theorem match for the prime-q support phase

Proposition 8.120 (Fastest current non-cyclic target). *Among the current three coauthored frontier lines, the fastest non-cyclic continuation after complete leaf phase collapse is the row/column Phi–LPF prime-q support phase. Its present object is*

$$\sum_{q \in S(P,k)} e(hkP/q),$$

where q ranges over primes in $(P/2, P)$ and $S(P, k)$ is the actual residual support produced by the Phi–LPF construction.

Theorem 8.121 (Direct external-name closure is not available)**.** *The current external Kloosterman and Type-II candidates do not, by name alone, prove cancellation for*

$$\sum_{q \in S(P,k)} e(hkP/q).$$

Wright's 2026 trilinear Kloosterman-fraction input, the Milićević–Qin–Wu bilinear Kloosterman input, Pascadi's composite modulus Type-II input, the Shao–Shparlinski–Wijaya smooth or square-free parameter input, and the Ford–Maynard Type-I/II sieve framework all require an additional same-object bridge. That bridge must convert the present $0/1$ prime-q support predicate into a completed bilinear or trilinear Kloosterman convolution with the required coefficient, dyadic-range, and Siegel–Walfisz-type hypotheses, and it must transfer the resulting averaged estimate back to every fixed row (P, k).

Proof. The previous section reduced every complete leaf phase to $e(hkP/q)$. Thus the remaining information is the actual support set $S(P, k)$, not an already completed Kloosterman family. The listed external inputs estimate completed or organised bilinear, trilinear, composite-modulus, smooth-parameter, or Type-I/II families. No certificate in the present manuscript supplies the missing same-object conversion, the Siegel–Walfisz factor, or the uniform fixed-(P, k) transfer. Therefore those theorems remain candidate inputs, but none directly closes the Phi–LPF support phase. □

Remark 8.122 (Updated true remaining mouth after theorem matching)**.** The latest narrow mouth is

```
QSupportToCompletedBilinearOrTrilinearKloostermanConvolutionWithSWFactor
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target. The external lemma version and the internal self-contained version both remain open until the same-object convolution bridge, fixed-row transfer, and prime-q support phase saving are proved.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
external-theorem-match-audit.md.
```

8.44 LPF bucket normal form for the prime-q support

Theorem 8.123 (Prime-q support has a unique LPF bucket)**.** *Fix P, k and a prime $q \in (P/2, P)$. Let $\omega_{P,k}(q)$ denote the unique odd integer, if it exists, in the clipped q-cofactor window*

$$\max(q, \lfloor kP/q \rfloor + 1) \le m \le \min(2P - 1, \lfloor ((k + 1)P - 1)/q \rfloor).$$

Then

$$q \in S(P, k)$$

if and only if $\omega_{P,k}(q)$ exists, is composite, and has least prime factor at least 7. Equivalently, there is a unique prime $r \ge 7$ and an integer β such that

$$\omega_{P,k}(q) = r\beta, \qquad r = \mathrm{LPF}(\omega_{P,k}(q)), \qquad \beta \ge r, \qquad \mathrm{LPF}(\beta) \ge r.$$

Proof. Since $q > P/2$, the unclipped interval has length $P/q < 2$. Thus the clipped window contains at most two consecutive integers and hence at most one odd candidate. The residual Phi–LPF support is exactly the composite cofactor support left after removing the $2, 3, 5$ forced-composite classes and the prime cofactor class. Thus membership is equivalent to being composite with least prime factor at least 7. Writing r for that least prime factor gives the displayed factorisation, and uniqueness follows from uniqueness of least prime factor. □

Corollary 8.124 (Rough-Mobius LPF bucket identity). *For the same odd candidate ω,*

$$1_{S(P,k)}(q) = \sum_{\substack{7 \le r \le \sqrt{\omega} \\ r \text{ prime} \\ r \mid \omega}} 1_{P^-(\omega) \ge r}\, 1_{\omega/r \ge r},$$

where

$$1_{P^-(\omega) \ge r} = \sum_{\substack{d \mid \omega \\ P^+(d) < r}} \mu(d).$$

Each non-zero term gives a sparse product-window triple

$$kP < q\, r\, \beta < (k+1)P, \qquad \beta = \omega/r, \qquad P^-(\beta) \ge r.$$

Remark 8.125 (What the LPF bucket does and does not close). The LPF bucket normal form is a genuine deterministic advance: it removes any remaining ambiguity in the q-support predicate. It also shows why the next bridge is still nontrivial. The product-window triples form a sparse same-row graph; filling them into a dense dyadic convolution would introduce non-object terms. Therefore the external Kloosterman inputs still require a same-object bridge to a completed bilinear or trilinear family, together with a Siegel–Walfisz or replacement uniformity input for the rough β-factor.

The latest narrow mouth is

```
SparseLPFBucketProductWindowGraphToCompletedKloostermanConvolutionBridge
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
lpf-bucket-completion-bridge-audit.md.
```

8.45 Reverse prime selector for fixed rough cofactors

Theorem 8.126 (Fixed rough cofactor has a unique reverse prime selector). *Fix a support triple from the preceding LPF bucket normal form and put*

$$m = r\beta.$$

For fixed P, k, m, the admissible prime q must lie in

$$\max(P/2 + 1, \lfloor kP/m \rfloor + 1) \le q \le \min(P - 1, m, \lfloor ((k+1)P - 1)/m \rfloor).$$

This reverse window contains at most two consecutive integers and therefore at most one admissible prime q.

Proof. The product inequality $kP < qm < (k+1)P$ gives

$$\lfloor kP/m \rfloor + 1 \le q \le \lfloor ((k+1)P - 1)/m \rfloor.$$

The original clipped cofactor condition also imposes

$$P/2 < q < P, \qquad q \le m.$$

Combining these inequalities gives the displayed reverse window. Since every support cofactor satisfies $m > P/2$, the uncut interval has length $P/m < 2$. Hence the integer window has at most two consecutive points. As $q > P/2 > 2$ is prime, it is odd; two consecutive integers cannot both be odd primes. Thus there is at most one admissible prime. □

Corollary 8.127 (Actual support equals the reverse selector graph)**.** *The prime-q support graph is equivalently obtained by ranging over rough cofactors $m = r\beta$ and selecting the unique odd prime in the reverse window above, when it exists. The phase becomes*

$$e(hkP/Q_{P,k}(r\beta)),$$

where $Q_{P,k}(m)$ is this floor-defined reverse prime selector.

Remark 8.128 (Reverse selector is not a completed Kloosterman family)**.** The reverse selector closes another deterministic ambiguity in the same object. It also makes the completion obstruction sharper. The current phase is a floor-prime-selector phase over rough cofactors, not a completed bilinear or trilinear Kloosterman family. Filling the other integer in a two-point reverse window would add non-object terms and lose pointwise equivalence.

The latest narrow mouth is

```
FloorPrimeSelectorToCompletedKloostermanConvolutionBridge
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
reverse-prime-selector-completion-bridge-audit.md.
```

8.46 Floor prime selector as a unique odd LPF test

Theorem 8.129 (Reverse floor-prime selector is an LPF primality atom)**.** *Fix P, k and a rough cofactor $m = r\beta$ from the preceding reverse-selector normal form. Put*

$$L = \max(P/2 + 1, \lfloor kP/m \rfloor + 1), \qquad U = \min(P - 1, m, \lfloor ((k+1)P - 1)/m \rfloor).$$

Define

$$Q_{\mathrm{odd}}(P, k, m) = \begin{cases} L, & L \text{ odd}, \\ L + 1, & L \text{ even}, \end{cases}$$

and regard it as undefined if $Q_{\mathrm{odd}} > U$. Then the reverse prime selector chooses a value if and only if Q_{odd} is defined and

$$\mathrm{LPF}(Q_{\mathrm{odd}}) = Q_{\mathrm{odd}}.$$

In that case the chosen prime is exactly Q_{odd}.

Proof. The preceding reverse-window theorem gives $U - L < 2$ after clipping, so the window contains at most two consecutive integers. Hence it contains at most one odd integer, and that integer is precisely the displayed $Q_{\rm odd}$ when $Q_{\rm odd} \le U$. Every admissible prime q lies above $P/2 > 2$, hence is odd. Therefore any selected prime must be this unique odd integer. Finally, for an integer $n > 1$, n is prime exactly when its least prime factor is n itself. Applying this with $n = Q_{\rm odd}$ proves the equivalence. □

Corollary 8.130 (Actual support equals the LPF-prime selector graph)**.** *The prime-q support after fixing $m = r\beta$ is obtained by the single pointwise rule*

$$(P, k, m) \mapsto Q_{\rm odd}(P, k, m)$$

followed by the primality test

$$\mathrm{LPF}(Q_{\rm odd}) = Q_{\rm odd}.$$

Thus the remaining phase is

$$e(hkP/Q_{\rm odd}(P, k, r\beta))$$

on the selected atoms and is absent otherwise.

Remark 8.131 (LPF atomization is not completion)**.** This closes the deterministic floor-prime selector ambiguity: the selector is a unique odd candidate plus an LPF primality test. It does not turn the pointwise selector into a completed Kloosterman family. A completion would still have to preserve the same object while handling the rough β-weights and the fixed-row P, k uniformity.

The latest narrow mouth is

```
OddCandidateLPFPrimeSelectorToCompletedKloostermanConvolutionBridge
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
floor-prime-lpf-selector-audit.md.
```

8.47 Dynamic sqrt-sieve form of the floor prime selector

Theorem 8.132 (LPF-prime selector equals a dynamic sqrt-sieve)**.** *In the notation of the preceding theorem, suppose $Q_{\rm odd}$ is defined. Since $P/2 < Q_{\rm odd} < P$, the condition*

$$\mathrm{LPF}(Q_{\rm odd}) = Q_{\rm odd}$$

is equivalent to

$$Q_{\rm odd} \not\equiv 0 \pmod{\ell} \qquad \textit{for every prime } \ell \le \sqrt{P-1}.$$

Consequently the fixed-cofactor prime selector is exactly the dynamic $\sqrt{P}$-sieve selector on the single candidate $Q_{\rm odd}$.

Proof. If $Q_{\rm odd}$ is composite, it has a prime divisor $\ell \le \sqrt{Q_{\rm odd}}$. Since $Q_{\rm odd} < P$, this divisor satisfies $\ell \le \sqrt{P-1}$, and the displayed congruence fails. Conversely, if a prime $\ell \le \sqrt{P-1}$ divides $Q_{\rm odd}$, then $Q_{\rm odd}$ is composite because $Q_{\rm odd} > P/2 > 2$. Thus surviving all these residue exclusions is equivalent to primality, hence to the LPF identity above. □

Corollary 8.133 (Odd composite candidates have a unique sieve bucket)**.** *Every rejected odd candidate Q_{odd} is assigned to the unique least prime $\ell \leq \sqrt{P-1}$ dividing Q_{odd}. Hence the selector splits into disjoint outcomes:*

$$\textit{empty window}, \qquad \textit{even singleton}, \qquad \ell\textit{-rejection}, \qquad \sqrt{P}\textit{-sieve survivor}.$$

Remark 8.134 (Dynamic sieve is exact, not a fixed-wheel proof)**.** This is another deterministic compression of the same object. It shows that the remaining primality selector is a finite CRT exclusion family, but the family grows with $\sqrt{P}$. A fixed finite wheel leaves rough composite survivors and therefore cannot replace this dynamic selector without reintroducing the original prime-selection problem. The Kloosterman frontier still requires a same-object bridge from this dynamic sieve selector to a completed convolution, together with the rough-β and fixed-row uniformity inputs.

The latest narrow mouth is

```
DynamicSqrtSieveSelectorToCompletedKloostermanConvolutionBridge
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
dynamic-sqrt-sieve-selector-audit.md.
```

8.48 Dynamic primorial unit form of the selector

Theorem 8.135 (Dynamic sqrt-sieve equals a primorial unit-class test)**.** *Let*

$$W_P = \prod_{\substack{\ell \leq \sqrt{P-1} \\ \ell \text{ prime}}} \ell.$$

For every defined Q_{odd} in the preceding section, the dynamic sqrt-sieve condition is equivalent to

$$\gcd(Q_{\mathrm{odd}}, W_P) = 1.$$

Equivalently,

$$1_{\gcd(Q_{\mathrm{odd}}, W_P)=1} = \sum_{\substack{d \mid W_P \\ d \mid Q_{\mathrm{odd}}}} \mu(d).$$

Proof. The prime factors of W_P are exactly the primes $\ell \leq \sqrt{P-1}$. Thus $\gcd(Q_{\mathrm{odd}}, W_P) = 1$ is precisely the assertion that no such ℓ divides Q_{odd}, which is the dynamic sqrt-sieve condition. Since W_P is squarefree, the standard Mobius identity $\sum_{d \mid g} \mu(d) = 1$ if $g = 1$ and 0 otherwise, applied to $g = \gcd(Q_{\mathrm{odd}}, W_P)$, gives the displayed finite sum. $\square$

Corollary 8.136 (Nonunit rejection has a least-prime bucket)**.** *Every nonunit candidate has a unique least prime divisor $\ell \mid W_P$. Therefore the non-surviving part of the selector is partitioned by the same least-prime buckets as the dynamic sqrt-sieve.*

Remark 8.137 (The primorial modulus is dynamic)**.** The primorial unit-class form is exact, but it is not a static-modulus completion. The modulus W_P grows with P, and freezing it would leave rough composite candidates that are removed only by later primes. Hence this step closes the CRT/Mobius normal form of the selector while leaving the analytic bridge to a completed Kloosterman family open.

The latest narrow mouth is

```
DynamicPrimorialUnitSelectorToCompletedKloostermanConvolutionBridge
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
dynamic-primorial-unit-selector-audit.md.
```

8.49 Ramanujan expansion of the dynamic unit class

Theorem 8.138 (Dynamic unit class has an exact Ramanujan expansion)**.** *Let $W = W_P$ be the squarefree primorial modulus in the preceding section. For every integer n,*

$$1_{\gcd(n,W)=1} = \frac{\varphi(W)}{W} \sum_{d|W} \frac{\mu(d)}{\varphi(d)} c_d(n),$$

where

$$c_d(n) = \sum_{\substack{a \bmod d \\ (a,d)=1}} e(an/d)$$

is the Ramanujan sum. Equivalently, since W is squarefree, the right hand side factors as the product over primes $\ell \mid W$ of

$$\frac{\ell-1}{\ell}\left(1 - \frac{c_\ell(n)}{\ell-1}\right).$$

Proof. For a prime ℓ, $c_\ell(n) = \ell - 1$ if $\ell \mid n$, and $c_\ell(n) = -1$ otherwise. Hence the displayed local factor is 0 when $\ell \mid n$, and 1 otherwise. Multiplying over the prime factors of the squarefree W gives $1_{\gcd(n,W)=1}$. The divisor-sum formula is the multiplicative expansion of the same local identity, using the multiplicativity of Ramanujan sums and of $\mu(d)/\varphi(d)$ on squarefree divisors of W. □

Remark 8.139 (Why this is still not a completed Kloosterman input)**.** Opening all Ramanujan sums gives exactly

$$\sum_{d|W_P} \varphi(d) = W_P$$

additive modes. Thus the exact expansion is a genuine CRT normal form, but it is not a usable analytic completion by itself: the number of modes is the dynamic primorial W_P, and the phase after expansion still contains the reciprocal factor $e(hkP/Q_{\mathrm{odd}})$ attached to the same thin product-window selector. Therefore this step identifies the exact mode family that a future Kloosterman or Type-II bridge must control; it does not supply that bridge.

The latest narrow mouth is

```
DynamicRamanujanUnitExpansionToUsableKloostermanCompletionBridge
AND UniformRamanujanModeCancellationOrTruncation
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
dynamic-ramanujan-unit-expansion-audit.md.
```

8.50 Full Fourier support of the dynamic unit selector

Theorem 8.140 (No exact Ramanujan mode truncation). *Let $W = W_P$ be squarefree and define*

$$f_W(n) = 1_{\gcd(n,W)=1}$$

on $\mathbf{Z}/W\mathbf{Z}$. Its additive Fourier coefficients are

$$\widehat{f}_W(a) = \frac{1}{W} c_W(a).$$

Moreover, for every $a \bmod W$,

$$c_W(a) = \prod_{\substack{\ell | W \\ \ell | a}} (\ell - 1) \prod_{\substack{\ell | W \\ \ell \nmid a}} (-1),$$

so $\widehat{f}_W(a) \neq 0$ for every $a \bmod W$. Hence any exact additive Fourier representation of f_W uses all W modes; no proper mode truncation reproduces the same selector.

Proof. The Fourier coefficient formula is the standard orthogonality identity for the Ramanujan sum:

$$\sum_{\substack{n \bmod W \\ (n,W)=1}} e(-an/W) = c_W(a).$$

For squarefree W, multiplicativity reduces $c_W(a)$ to the displayed product of the prime local sums. Every local factor is either $\ell - 1$ or -1, so the product is never zero. Therefore the Fourier support is all of $\mathbf{Z}/W\mathbf{Z}$. Omitting any frequency changes a nonzero Fourier coefficient, and by uniqueness of finite Fourier expansion changes the function. $\square$

Remark 8.141 (The truncation exit is closed, the cancellation exit is not). For the dynamic primorial selector the exact number of additive modes is W_P, not $2^{\omega(W_P)}$. The divisor-level Ramanujan form is compact, but opening it to additive characters gives full support. Thus the previous alternative `UniformRamanujanModeCancellationOrTruncation` narrows to full dynamic-spectrum cancellation: exact truncation is unavailable.

The latest narrow mouth is

```
DynamicFullRamanujanSpectrumToUsableKloostermanCompletionBridge
AND UniformCancellationAcrossFullDynamicRamanujanSpectrum
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
full-ramanujan-spectrum-obstruction-audit.md.
```

8.51 Conductor stratification of the full Ramanujan spectrum

Theorem 8.142 (Equal mass on exact conductor layers)**.** *Let $W = W_P$ be squarefree. For a frequency $a \bmod W$, define its exact additive conductor by*

$$q = \frac{W}{\gcd(a, W)}.$$

Then $q \mid W$, there are exactly $\varphi(q)$ frequencies of exact conductor q, and for all such frequencies

$$\frac{c_W(a)}{W} = \mu(q)\frac{\varphi(W)}{W\varphi(q)}.$$

Consequently each exact-conductor layer has total L^1-mass

$$\sum_{\substack{a \bmod W \\ W/\gcd(a,W)=q}} \left|\frac{c_W(a)}{W}\right| = \frac{\varphi(W)}{W}.$$

Proof. Writing $g = \gcd(a, W)$, the primitive frequency on the quotient has conductor $q = W/g$. The standard formula for Ramanujan sums gives

$$c_W(a) = \mu(q)\frac{\varphi(W)}{\varphi(q)}$$

for squarefree W. The number of residues $a \bmod W$ with $W/\gcd(a, W) = q$ is the number of units modulo q, namely $\varphi(q)$. Multiplying that count by the absolute value of the coefficient gives the displayed common L^1-mass. □

Remark 8.143 (Low conductor layers are not a negligible exact tail)**.** The full spectrum has $2^{\omega(W_P)}$ exact-conductor layers, but each layer carries the same total L^1-mass. Thus conductor stratification is a useful organisation of the full dynamic spectrum, not a proof that low conductor layers dominate. Dropping high conductor layers changes the exact selector and removes a fixed fraction of the spectral mass whenever a fixed fraction of layers is removed.

The latest narrow mouth is

```
ConductorStratifiedSpectrumToKloostermanOrTraceFamilyBridge
AND UniformCancellationAcrossAllPrimorialConductorLayers
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
conductor-stratified-ramanujan-spectrum-audit.md.
```

8.52 Complementary conductor pairs

Theorem 8.144 (Complementary conductor pairs do not cancel automatically)**.** *Let $W = W_P$ be squarefree and let $q \mid W$. Put*

$$q^\vee = \frac{W}{q}, \qquad R_m(n) = \frac{c_m(n)}{\varphi(m)}.$$

The involution $q \mapsto q^\vee$ pairs low and high exact-conductor layers with the same L^1-mass $\varphi(W)/W$. Its sign ledger is

$$\mu(q^\vee) = \mu(W)\mu(q).$$

However the complementary pair kernel

$$K_q(n) = \mu(q)R_q(n) + \mu(q^\vee)R_{q^\vee}(n)$$

is not identically zero for any complementary pair.

Proof. The equal L^1-mass is the preceding conductor-layer identity applied to q and $q^\vee$. Since W is squarefree and $(q, q^\vee) = 1$, multiplicativity gives $\mu(q^\vee) = \mu(W)\mu(q)$.

If $\mu(q^\vee) = \mu(q)$, then $R_q(0) = R_{q^\vee}(0) = 1$, so

$$K_q(0) = 2\mu(q) \neq 0.$$

It remains to handle the opposite-sign case. If $q > 1$, choose a prime $\ell \mid q$ and put $n = W/\ell$. Then every prime factor of $q^\vee$ divides n, while exactly the ℓ-factor is missing from q. Hence

$$R_{q^\vee}(n) = 1, \qquad R_q(n) = -\frac{1}{\ell - 1},$$

and $K_q(n) = \mu(q)(-1/(\ell-1) - 1) \neq 0$. If $q = 1$, choose $\ell \mid q^\vee$ and the same $n = W/\ell$; then

$$R_q(n) = 1, \qquad R_{q^\vee}(n) = -\frac{1}{\ell - 1},$$

so $K_q(n) = 1 + 1/(\ell - 1) \neq 0$. Thus the mass mirror is not a pointwise cancellation identity. □

Remark 8.145 (The remaining pair-uniform outlet)**.** The machine audit up to $P \leq 1009$ records 32554 complementary pairs, with zero failures of low/high mass balance and zero identically vanishing pair kernels. The largest checked case has 2048 conductor layers and 1024 complementary pairs.

The latest narrow mouth is

```
ComplementaryConductorPairPhaseMatchingOrTraceFamilyBridge
AND UniformCancellationAcrossComplementaryPrimorialConductorPairs
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
complementary-conductor-pair-audit.md.
```

8.53 Radial tensor form of complementary conductor pairs

Theorem 8.146 (Complementary pair kernels are radial two-cylinder tensors)**.** *Let $W = W_P$ be squarefree, let $q \mid W$, and put $q^\vee = W/q$. For each prime $\ell \mid W$, define*

$$\rho_\ell(n) = \begin{cases} 1, & \ell \mid n, \\ -1/(\ell - 1), & \ell \nmid n. \end{cases}$$

Then the complementary pair kernel from the previous section has the exact local tensor form

$$K_q(n) = \mu(q) \prod_{\ell \mid q} \rho_\ell(n) + \mu(q^\vee) \prod_{\ell \mid q^\vee} \rho_\ell(n).$$

Consequently $K_q(n)$ depends only on $\gcd(n, W)$ and is invariant under $n \mapsto un \bmod W$ for every $u \in (\mathbb{Z}/W\mathbb{Z})^*$. If both q and $q^\vee$ are greater than 1, its divisor-state matrix across the split $q \mid W/q^\vee$ has rank exactly 2. For the endpoint pair $\{1, W\}$ the rank is 1.

Proof. For squarefree m, the normalized Ramanujan sum factors as

$$\frac{c_m(n)}{\varphi(m)} = \prod_{\ell \mid m} \rho_\ell(n).$$

Substituting this formula into

$$K_q(n) = \mu(q) \frac{c_q(n)}{\varphi(q)} + \mu(q^\vee) \frac{c_{q^\vee}(n)}{\varphi(q^\vee)}$$

gives the displayed tensor form. Each factor $\rho_\ell(n)$ depends only on whether $\ell \mid n$. Hence the whole expression depends only on $\gcd(n, W)$, and multiplication by a unit modulo W preserves it.

Now index rows by divisibility states on the prime factors of q and columns by divisibility states on the prime factors of $q^\vee$. The matrix has the form

$$M_{r,c} = u_r + v_c.$$

If both sides are nonempty, the local products on both sides are nonconstant, so the column space is spanned by two independent vectors, for example $u + v_{c_1}\mathbf{1}$ and $u + v_{c_2}\mathbf{1}$ with $v_{c_1} \neq v_{c_2}$. Thus the rank is 2. If one side is empty, the matrix has only one row or one column after the split, so its rank is 1. □

Remark 8.147 (No direct trace input is present)**.** The radial tensor form is a genuine structural reduction, but it is not itself a Kloosterman or trace-function phase. It has no modular inverse phase and is constant on unit orbits. Thus the currently available Kloosterman-fraction, trace-function, and composite-modulus bilinear estimates can only enter after an additional same-object coupling bridge is proved.

The latest narrow mouth is

```
RadialPairKernelToReciprocalTracePhaseCouplingBridge
AND UniformCancellationAcrossComplementaryPrimorialConductorPairsAfterCoupling
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
complementary-pair-radial-tensor-audit.md.
```

8.54 Coupling radial pair kernels to the actual reciprocal phase

Theorem 8.148 (Radial pair coupled floor-reciprocal normal form)**.** *Let $W = W_P$, and let $m = r\beta$ be a residual rough cofactor in the reverse graph. Define*

$$L_m = \max\{P/2 + 1, \lfloor kP/m \rfloor + 1\}, \qquad U_m = \min\{P - 1, m, \lfloor ((k+1)P - 1)/m \rfloor\}.$$

Let $Q_{P,k}(m)$ be the least odd integer in $[L_m, U_m]$, if it exists. Then the actual prime-q reciprocal phase atoms are exactly the terms

$$\mathbf{1}_{(Q_{P,k}(m),W)=1}\, e\left(\frac{hkP}{Q_{P,k}(m)}\right)$$

over residual m for which $Q_{P,k}(m)$ exists. Moreover

$$\mathbf{1}_{(Q,W)=1} = \frac{\varphi(W)}{W} \sum_{\{d,W/d\}} \left[\mu(d)\frac{c_d(Q)}{\varphi(d)} + \mu(W/d)\frac{c_{W/d}(Q)}{\varphi(W/d)}\right],$$

where the sum is over unordered complementary conductor pairs.

Proof. The reverse selector window has length at most two in the previously proved fixed-cofactor normal form. Hence it has at most one odd candidate. Since $Q_{P,k}(m) < P$, the condition that this odd candidate is prime is equivalent to its avoiding every prime $\ell \le \sqrt{P-1}$, namely $(Q_{P,k}(m), W_P) = 1$. Therefore each selected atom carries exactly the displayed denominator $Q_{P,k}(m)$ and phase $e(hkP/Q_{P,k}(m))$.

The Ramanujan expansion of the dynamic unit class gives

$$\mathbf{1}_{(Q,W)=1} = \frac{\varphi(W)}{W} \sum_{d|W} \mu(d)\frac{c_d(Q)}{\varphi(d)}.$$

Grouping the divisor d with its complementary divisor W/d gives the unordered-pair formula. This is only a regrouping of the exact selector, so it changes neither the selected atoms nor their reciprocal phases. □

Remark 8.149 (The completed trace bridge is still missing)**.** The coupled normal form closes the deterministic question of how the radial pair kernels meet the actual phase. It does not provide cancellation. The denominator remains the floor-defined $Q_{P,k}(m)$, and the weights are radial divisor-lattice kernels. Thus the present object is still not a completed Kloosterman, trace-function, or Type-II family.

The latest narrow mouth is

```
FloorRadialReciprocalPhaseToTraceFamilyBridge
AND UniformCancellationAcrossCoupledFloorRadialPairKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
radial-pair-coupled-phase-audit.md.
```

8.55 Recovering the floor denominator as a forward floor cell

Theorem 8.150 (Floor-cell radial support normal form)**.** *Let q be odd with $P/2 < q < P$, and let m be a residual rough cofactor with $P/2 < m < 2P$. Put*

$$I_{P,k}(q) = [\max\{P/2+1, q, \lfloor kP/q \rfloor + 1\}, \min\{2P-1, \lfloor ((k+1)P-1)/q \rfloor\}] .$$

With $Q_{P,k}(m)$ as in the previous theorem, one has

$$Q_{P,k}(m) = q \iff m \in I_{P,k}(q)$$

inside the residual odd-cofactor graph. Consequently the selected phase atoms may be written equivalently as

$$\mathbf{1}_{(q,W_P)=1}\,\mathbf{1}_{m\in I_{P,k}(q)}\, e\!\left(\frac{hkP}{q}\right),$$

with the same complementary radial Ramanujan pair selector acting on the original q-coordinate.

Proof. The reverse condition $Q_{P,k}(m) = q$ means that the odd integer q lies in

$$[\max\{P/2+1, \lfloor kP/m \rfloor + 1\}, \min\{P-1, m, \lfloor ((k+1)P-1)/m \rfloor\}] .$$

Since $m > P/2$, this reverse interval has length at most 2, hence it contains at most one odd integer. Thus membership of the odd integer q is equivalent to being the least odd member.

The lower floor inequality is

$$q \ge \lfloor kP/m \rfloor + 1 \iff q > kP/m \iff m > kP/q \iff m \ge \lfloor kP/q \rfloor + 1.$$

The upper floor inequality is

$$q \le \lfloor ((k+1)P-1)/m \rfloor \iff mq \le (k+1)P-1 \iff m \le \lfloor ((k+1)P-1)/q \rfloor.$$

The remaining caps are exactly $q < P$, $q > P/2$, $m < 2P$, and $q \le m$, which produce the displayed interval $I_{P,k}(q)$. Finally, because $q < P$, the W_P-unit condition is precisely the prime-q selector already proved in the primorial selector normal form. The reciprocal phase is therefore the same phase $e(hkP/q)$, not a new completed trace variable. □

Remark 8.151 (The cell identity is not cancellation)**.** The floor-cell normal form closes another deterministic gap: the floor-defined denominator has been recovered as the original q-coordinate of the product window. This prevents a cyclic renaming of the obstruction. It also shows exactly what remains missing: a theorem completing these floor cells and their coupled radial pair kernels into a genuine trace/Kloosterman or Type-II family with uniform phase saving.

The latest narrow mouth is

```
FloorCellRadialSupportToCompletedTraceFamilyBridge
AND UniformCancellationAcrossFloorCellsWithRadialPairKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
floor-cell-radial-support-audit.md.
```

8.56 Why the floor cells are not already a Type-II family

Theorem 8.152 (Direct Type-II completion from floor cells is not available)**.** *In the floor-cell normal form of the previous theorem, split every residual cofactor as $m = r\beta$, where $r \geq 7$ is prime and β is r-rough. Then, inside the same floor-cell object, the fibres*

$$q \mapsto m, \qquad m \mapsto q, \qquad q \mapsto (r, \beta),$$

and also

$$(q, r) \mapsto \beta, \qquad (q, \beta) \mapsto r, \qquad (r, \beta) \mapsto q$$

are all singleton fibres whenever they are nonempty. Therefore the floor-cell support does not itself provide the long bilinear or trilinear variables required by Type-II Kloosterman or trace-function estimates. The phase remains the graph-supported denominator phase

$$e\left(\frac{hkP}{q}\right) = e\left(-\frac{h(qm - kP)}{q}\right),$$

not a completed inverse-variable Kloosterman sum.

Proof. The interval $I_{P,k}(q)$ has length at most 2 because

$$\frac{(k+1)P - 1}{q} - \frac{kP}{q} < \frac{P}{q} < 2.$$

Hence it contains at most one odd integer. Since residual cofactors in the present graph are odd, each fixed q contributes at most one cofactor m. The previous reverse-cell equivalence gives the reverse statement: each residual m contributes at most one odd q.

Once m is fixed, its least-prime-factor split $m = r\beta$ is unique. Thus the fibres $q \mapsto (r, \beta)$, $(q, r) \mapsto \beta$, and $(q, \beta) \mapsto r$ are singleton. Conversely (r, β) fixes m, and the reverse singleton property fixes q. Therefore none of these same-object decompositions creates a long inner variable.

The displayed phase identity is only the already proved matched displacement congruence $qm - kP = D$. It rewrites the same graph phase; it does not introduce a completed modular inverse variable or a rectangular bilinear coefficient family. □

Remark 8.153 (External estimates still need a new embedding)**.** This theorem does not say that modern trace-function or Kloosterman estimates are irrelevant. It says only that the current floor-cell graph cannot be fed to them directly. Fouvry–Kowalski–Michel–Sawin type trace-function bilinear estimates, Milićević–Qin–Wu arbitrary-modulus Kloosterman estimates, Pascadi's composite-modulus Type-II amplification, Wright's unbalanced Kloosterman-fraction convolution, and Shao–Shparlinski–Wijaya smooth/squarefree-parameter estimates still require a same-object embedding that creates a genuine averaged graph dispersion or trace family.

The latest narrow mouth is

```
SameObjectAveragedGraphDispersionOrTraceEmbedding
AND UniformCancellationAcrossGraphSupportedRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
floor-cell-typeii-fiber-obstruction-audit.md.
```

8.57 Row-averaged mixed-modulus graph

Theorem 8.154 (Row-averaged q-support mixed-modulus normal form)**.** *Combine all rows $1 \le k < P$ in the floor-cell q-support graph. For each admissible edge (q, m), with $q \le m$, $P/2 < q < P$, and $P/2 < m < 2P$, the row is uniquely determined by*

$$k = \left\lfloor \frac{qm}{P} \right\rfloor, \qquad 1 \le k < P,$$

and the displacement is

$$D = qm - kP \equiv qm \pmod{P}, \qquad 1 \le D < P.$$

On this row-averaged graph the phase has the exact form

$$e\left(\frac{hkP}{q}\right) = e\left(-\frac{hD}{q}\right).$$

Thus row averaging creates long q-fibres, but the resulting phase is a mixed-modulus graph phase: the numerator D is a residue modulo P, while the oscillating denominator is q.

Proof. The edge condition in row k is

$$kP < qm < (k+1)P.$$

For a fixed pair (q, m) there is at most one such integer k, namely $k = \lfloor qm/P \rfloor$. The condition $1 \le k < P$ selects exactly the rows present in the row/column problem, and the cap $q \le m$ is the same q-side cofactor cap already present in $I_{P,k}(q)$. Since P is prime and neither q nor the residual composite m is divisible by P, the remainder $D = qm - kP$ is nonzero and lies in $\{1, \dots, P-1\}$.

The phase identity follows from $kP = qm - D$:

$$e(hkP/q) = e(h(qm - D)/q) = e(hm)e(-hD/q) = e(-hD/q).$$

This is an exact normal form. It does not identify the phase with a same-modulus trace function modulo q, because the residue D is defined modulo the external prime P, not modulo q. □

Remark 8.155 (What row averaging gives, and what it still misses)**.** Row averaging is a real gain over the single-row floor-cell ledger: the same q can now support many cofactors m and many rows k. However, the gained fibre length is not yet a Kloosterman or trace-function completion. The obstruction has changed from "no long same-row fibre" to "mixed-modulus row-averaged phase." Current trace-function, arbitrary-modulus Kloosterman, composite Type-II, unbalanced Kloosterman-fraction, and smooth/squarefree-parameter estimates still require a theorem embedding $D = qm \bmod P$ with denominator q into a same-object trace, Kloosterman, or dispersion family.

The latest narrow mouth is

```
MixedModulusRowAveragedGraphToTraceOrKloostermanEmbedding
AND UniformCancellationAcrossRowAveragedMixedModulusRadialGraph
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-mixed-modulus-graph-audit.md.
```

8.58 Row-averaged additive-k support

Theorem 8.156 (Row-averaged phase as a sparse additive k-character)**.** *In the row-averaged q-support graph of the preceding section, write*

$$k = \left\lfloor \frac{qm}{P} \right\rfloor, \qquad D = qm - kP.$$

Then every selected edge satisfies

$$D \equiv -kP \pmod q, \qquad e\left(-\frac{hD}{q}\right) = e\left(\frac{hPk}{q}\right).$$

Thus the mixed-modulus phase can be relabelled as an additive character in the variable $k \bmod q$. The support, however, is not the full interval of k's: it is the image

$$\mathcal{K}_{P,q} = \left\{ \left\lfloor \frac{qm}{P} \right\rfloor : m \textit{ is a selected residual LPF cofactor in the q-bucket} \right\}.$$

Consequently complete-interval additive character cancellation is not available without an additional completion or dispersion theorem for the sparse LPF-induced set $\mathcal{K}_{P,q}$.

Proof. The congruence is immediate from the identity $D = qm - kP$, since

$$D \equiv -kP \pmod q.$$

Therefore, for every integer h,

$$e(-hD/q) = e(hPk/q).$$

Because P and q are distinct primes in the audited range, P is invertible modulo q, so this is a genuine q-modulus additive character in the residue class of k. The final assertion is not a new asymptotic estimate: it is the exact support identity inherited from the selected residual LPF cofactors m. Finite audit data show that most q-buckets have holes between their minimum and maximum selected k's, so replacing $\mathcal{K}_{P,q}$ by a complete interval would change the object. □

Remark 8.157 (What the additive-k relabelling closes)**.** This closes the purely formal phase-relabeling gate left by the row-averaged mixed-modulus normal form. The remaining hard point is now sharper: one needs cancellation on the sparse LPF-induced support $\mathcal{K}_{P,q}$, or a loss-controlled completion of that support which does not change the Phi–LPF selector. Current trace-function, arbitrary-modulus Kloosterman, composite Type-II, unbalanced convolution, and smooth/squarefree-parameter estimates still enter only after such a sparse-support completion or dispersion input is supplied.

The latest narrow mouth is

```
SparseLPFKSupportCompletionOrDispersion
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-support-audit.md.
```

8.59 The additive-k completion tax

Theorem 8.158 (Complete-interval replacement has a non-object correction). *Fix a selected q-bucket in the row-averaged additive-k normal form, and let*

$$\mathcal{K}_{P,q} = \left\{ \left\lfloor \frac{qm}{P} \right\rfloor : m \text{ is a selected residual LPF cofactor in the } q\text{-bucket} \right\}.$$

Let

$$\mathcal{C}_{P,q} = [\min \mathcal{K}_{P,q}, \max \mathcal{K}_{P,q}] \cap \mathbf{Z}$$

and write $\mathcal{H}_{P,q} = \mathcal{C}_{P,q} \setminus \mathcal{K}_{P,q}$. For every $k \in \mathcal{H}_{P,q}$, the product cell

$$I_{P,k}(q) = \left[\max\left\{ \frac{P+1}{2}, q, \left\lfloor \frac{kP}{q} \right\rfloor + 1 \right\}, \min\left\{ 2P-1, \left\lfloor \frac{(k+1)P-1}{q} \right\rfloor \right\} \right]$$

contains no selected residual LPF cofactor. More precisely, since $q > P/2$, it contains at most one odd integer; if that odd integer were a residual LPF cofactor, then k would already belong to $\mathcal{K}_{P,q}$. Therefore replacing $\mathcal{K}_{P,q}$ by the complete interval $\mathcal{C}_{P,q}$ introduces a correction supported only on non-object holes.

Proof. The cell $I_{P,k}(q)$ is exactly the set of cofactors m satisfying

$$kP < qm < (k+1)P$$

together with the existing row/column caps. Its length is $< P/q < 2$, so it contains at most one odd integer. If this odd integer m is a selected residual LPF cofactor, then

$$k = \left\lfloor \frac{qm}{P} \right\rfloor$$

and hence $k \in \mathcal{K}_{P,q}$, contradicting $k \in \mathcal{H}_{P,q}$. Thus each completion hole has either no odd representative or an odd representative which fails the residual LPF cofactor condition. The completed interval is therefore not an object-preserving replacement of the sparse Phi–LPF support. ☐

Remark 8.159 (Finite completion-tax ledger). The audit up to $P \leq 1009$ gives

$$|\mathcal{K}| = 299977, \qquad |\mathcal{C}| = 1602928, \qquad |\mathcal{H}| = 1302951,$$

so the completion correction is about 4.343503 times the true support in this finite ledger. The holes split as 373676 no-odd cells and 929275 non-residual odd candidates; among the latter the prime, LPF 3, and LPF 5 counts are 355919, 409713, and 163643. There are no residual-candidate holes in the audit. This does not prove the Phi–LPF target; it only shows that any complete interval additive-character argument must also control a large, non-object completion correction.

The latest narrow mouth is

```
CompletionCorrectionCancellationOrAbsorptionForSparseLPFKSupport
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-completion-tax-audit.md.
```

8.60 The completion-hole correction phase

Theorem 8.160 (Sparse additive-k sum equals geometric interval minus holes). *With $\mathcal{K}_{P,q}$, $\mathcal{C}_{P,q}$, and $\mathcal{H}_{P,q}$ as above, define*

$$S_K(h) = \sum_{k\in\mathcal{K}_{P,q}} e\left(\frac{hPk}{q}\right), \qquad S_C(h) = \sum_{k\in\mathcal{C}_{P,q}} e\left(\frac{hPk}{q}\right), \qquad S_H(h) = \sum_{k\in\mathcal{H}_{P,q}} e\left(\frac{hPk}{q}\right).$$

Then, for every integer h,

$$S_K(h) = S_C(h) - S_H(h).$$

Moreover, if $q \nmid h$, then the completed interval term is the explicit geometric progression

$$S_C(h) = e\left(\frac{hP\min\mathcal{K}_{P,q}}{q}\right)\frac{1-e(hP|\mathcal{C}_{P,q}|/q)}{1-e(hP/q)}.$$

Thus complete-interval additive cancellation closes only the elementary geometric part; the true Phi–LPF object still requires control of the completion-hole correction $S_H(h)$.

Proof. The first identity follows from the disjoint decomposition

$$\mathcal{C}_{P,q} = \mathcal{K}_{P,q} \sqcup \mathcal{H}_{P,q}.$$

For the second identity, $\mathcal{C}_{P,q}$ is a consecutive integer interval. Since P is invertible modulo the prime q, and $q \nmid h$, the ratio $e(hP/q)$ is not 1. Summing the resulting finite geometric progression gives the displayed formula. □

Remark 8.161 (What remains after the geometric sum)**.** The finite audit at $P \le 1009$, using $h = 1$ as a magnitude diagnostic, verifies the phase identity and the geometric formula with maximum numerical errors $1.641 \cdot 10^{-13}$ and $3.329 \cdot 10^{-11}$. Across the audited q-buckets,

$$\sum |S_C(1)| = 12232.215742, \qquad \sum |S_H(1)| = 56664.077674,$$

and in 5150 of 6020 buckets the hole-correction magnitude is larger than the complete-interval geometric magnitude. These numbers are finite diagnostics only; the rigorous conclusion is the structural one: a proof cannot replace the sparse Phi–LPF support by a complete interval unless it also proves cancellation or absorption for the no-odd, prime-candidate, and small-LPF completion holes.

The latest narrow mouth is

```
HoleCorrectionPhaseCancellationOrAbsorptionForNoOddPrimeSmallLPFCells
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-completion-correction-phase-audit.md.
```

8.61 Five local classes for the hole correction

Theorem 8.162 (Five-class decomposition of the completion-hole phase)**.** *For each $k \in \mathcal{H}_{P,q}$, consider the product cell $I_{P,k}(q)$. Since $q > P/2$, this cell has length < 2. Hence the hole is of one of the following local types:*

empty cell, *even singleton,* *odd prime candidate,* *odd candidate with LPF 3,* *odd candidate with L*

Consequently the hole-correction phase has the exact decomposition

$$S_H(h) = S_\emptyset(h) + S_{\mathrm{even}}(h) + S_{\mathrm{prime}}(h) + S_3(h) + S_5(h),$$

where each term is the same additive character $e(hPk/q)$ restricted to the corresponding local class of holes.

Proof. The cell length bound follows from the defining inequalities

$$kP < qm < (k+1)P$$

and $q > P/2$. Thus $I_{P,k}(q)$ contains either no integer, a single even integer, or a single odd integer. In the last case call the odd candidate m. If m is a composite odd integer with least prime factor $r \geq 7$, then $m = r\beta$, $\beta \geq r$, and every prime factor of β is at least r. This is exactly the residual LPF cofactor condition. Such an m would put $k = \lfloor qm/P \rfloor$ back into $\mathcal{K}_{P,q}$, contradicting $k \in \mathcal{H}_{P,q}$. Therefore an odd candidate hole is either prime or has least prime factor 3 or 5. The displayed phase identity is just the disjoint partition of $\mathcal{H}_{P,q}$ into these local classes. □

Remark 8.163 (Finite class ledger)**.** The audit up to $P \leq 1009$ verifies the class identity with maximum numerical error $9.664 \cdot 10^{-14}$. The 1302951 holes split as

0 empty, 373676 even singletons, 355919 prime candidates, 409713 LPF 3, 163643 LPF 5.

No residual-candidate or other forbidden hole appears. For $h = 1$, the dominant class by q-bucket is usually the even singleton packet (5166 of 6020 buckets). This is a structural refinement, not a cancellation theorem.

The latest narrow mouth is

```
FiveClassHolePhaseCancellationOrAbsorption
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-hole-class-phase-decomposition-audit.md.
```

8.62 No empty holes and the four-class correction

Theorem 8.164 (No empty holes in the row-averaged additive-k completion)**.** *Fix a selected q-bucket in the row-averaged additive-k normal form, and let*

$$M_{P,q} = \{\max(P/2+1, q) \leq m \leq 2P-1 :\ m \in \mathbf{Z}\}.$$

Then the map $m \mapsto \lfloor qm/P \rfloor$ has no skipped values between any two values in its image. Consequently every completion hole between $\min \mathcal{K}_{P,q}$ *and* $\max \mathcal{K}_{P,q}$ *has a non-empty product cell. Hence*

$$S_\emptyset(h) = 0$$

and the hole-correction phase reduces to the four non-empty local packets

$$S_H(h) = S_{\mathrm{even}}(h) + S_{\mathrm{prime}}(h) + S_3(h) + S_5(h).$$

Proof. The function $m \mapsto qm/P$ is increasing. Since $q < P$, adjacent carrier integers satisfy

$$0 \le \left\lfloor \frac{q(m+1)}{P} \right\rfloor - \left\lfloor \frac{qm}{P} \right\rfloor \le 1.$$

Thus the integer-valued floor map can pause or move by one, but it cannot jump over an intermediate k. Its image on the integer interval $M_{P,q}$ is therefore a contiguous integer interval. The selected support $\mathcal{K}_{P,q}$ is a subset of that image, so every integer between $\min \mathcal{K}_{P,q}$ and $\max \mathcal{K}_{P,q}$ is also attained by some carrier $m \in M_{P,q}$. A completion hole is by definition a missing selected-support value inside this span; it is not a missing carrier value. Therefore the empty-cell class is identically zero. Combining this with the five-class partition gives the displayed four-class identity. □

Remark 8.165 (Finite nonempty ledger and remaining phase obstruction)**.** The audit up to $P \le 1009$ gives

6020 q-buckets, 299977 real k-support values, 1302951 completion holes.

The carrier floor map has no gap and the selected span has no empty product cell:

`carrier_gap_count_total` $= 0$, `selected_span_missing_count_total` $= 0$, `empty_hole_count_total` $=$

The four non-empty classes account for all holes:

373676 even singletons, 355919 prime candidates, 409713 LPF 3, 163643 LPF 5.

This closes the empty-cell packet only. It does not prove cancellation or absorption for the four non-empty packets.

The latest narrow mouth is

```
FourClassHolePhaseCancellationOrAbsorption
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-hole-nonempty-four-class-reduction-audit.md.
```

8.63 Blocking cofactors for the hole correction

Theorem 8.166 (Blocking-cofactor pushforward of the four-class correction)**.** *For each completion hole $k \in \mathcal{H}_{P,q}$, define its blocking cofactor $m_h = m_h(P,q,k)$ as follows. If the product cell $I_{P,k}(q)$ contains an odd integer, let m_h be that unique odd integer. Otherwise let m_h be the unique even singleton in the cell. Then m_h exists, is unique, satisfies*

$$k = \left\lfloor \frac{qm_h}{P} \right\rfloor,$$

and is never a residual rough LPF cofactor. Moreover

$$\mathrm{LPF}(m_h) \in \{2,3,5\} \quad \textit{or} \quad m_h \textit{ is prime.}$$

Consequently

$$S_H(h) = S_{30\mathrm{w}}(h) + S_{\text{prime blocker}}(h),$$

where $S_{30\mathrm{w}}$ is the finite 30-wheel blocker packet $\mathrm{LPF}(m_h) \in \{2,3,5\}$, and $S_{\text{prime blocker}}$ is the packet with m_h prime.

Proof. The preceding theorem removes the empty-cell alternative, so every completion hole has a non-empty product cell. Since $q > P/2$, the cell has length less than 2; hence it contains at most one odd integer. If that odd integer exists and were a composite number with least prime factor at least 7, it would be a residual rough LPF cofactor and would put k back into the true support $\mathcal{K}_{P,q}$, contradicting the definition of a hole. Thus an odd blocker is either prime or has least prime factor 3 or 5. If no odd integer exists, non-emptiness forces a unique even singleton, whose least prime factor is 2.

The equality $k = \lfloor qm_h/P \rfloor$ is the defining condition for the product cell. Therefore the additive character on the hole can be pushed forward to the blocker:

$$e(hPk/q) = e\Big(hP \left\lfloor \frac{qm_h}{P} \right\rfloor /q\Big).$$

Splitting the blockers according to $\mathrm{LPF}(m_h) \in \{2,3,5\}$ or m_h prime gives the displayed two-packet identity. □

Remark 8.167 (Finite blocker ledger and the new two-family mouth)**.** The audit up to $P \le 1009$ verifies the blocker pushforward for all 1302951 completion holes:

$$\texttt{blocking\_cofactor\_count\_total} = 1302951, \qquad \texttt{total\_bad\_pushforward\_count} = 0.$$

The two packets have sizes

$$947032 \text{ small-LPF blockers}, \qquad 355919 \text{ prime blockers}.$$

The small-LPF packet further splits as

$$373676 \ \mathrm{LPF} = 2, \qquad 409713 \ \mathrm{LPF} = 3, \qquad 163643 \ \mathrm{LPF} = 5.$$

The maximum numerical error in the phase pushforward identity is $9.334 \cdot 10^{-14}$. This is a normal-form theorem, not a phase-saving theorem.

The latest narrow mouth is

```
TwoFamilyBlockerPhaseCancellationOrAbsorption
AND PrimeBlockerDynamicSqrtSieveOrTraceEmbedding
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-hole-blocking-cofactor-pushforward-audit.md.
```

8.64 Primorial escalation after the blocker pushforward

Theorem 8.168 (Primorial escalation is inert until the prime-blocker oracle)**.** *Let m_h be a blocker from the preceding theorem, and let $y \geq 5$ be a primorial cutoff. The y-wheel deletes m_h exactly when*

$$\mathrm{LPF}(m_h) \leq y.$$

Since every blocker has least prime factor $2, 3, 5$ or is itself prime, the promotion

$$30,\ 210,\ 2310,\ 30030, \ldots$$

creates no new composite blocker shell after 30. Moreover the dynamic cutoff $y = \lfloor \sqrt{2P-1} \rfloor$ deletes no prime blocker. Therefore the $\sqrt{2P-1}$-primorial wheel has the same blocker action as the 30-wheel. All blockers are deleted only when the cutoff reaches every prime blocker itself, for example at $y = 2P-1$; this is a prime-blocker oracle, not an independent phase-saving theorem.

Proof. The first assertion is the definition of a primorial wheel cutoff on a cofactor: a cofactor is deleted precisely when it is divisible by a prime $\leq y$, equivalently when its least prime factor is at most y. The previous theorem says that no blocker has composite least prime factor shell $7, 11, 13, \ldots$. Hence, after the 30-wheel has removed the LPF $= 2, 3, 5$ blockers, all remaining blockers are prime blockers. Adding $7, 11, 13, \ldots$ can delete such a blocker only when the cutoff has reached that prime itself.

For $P \geq 11$, $\sqrt{2P-1} < P/2$. Every blocker lies in a product cell with $m_h \geq q > P/2$. Thus a prime blocker m_h is strictly larger than $\sqrt{2P-1}$, so the dynamic square-root cutoff cannot delete it. Finally, the cutoff $2P-1$ includes every possible blocker prime because $m_h \leq 2P-1$. This closes the wheel deletion ledger only by admitting the blocker primes themselves as wheel factors. $\square$

Remark 8.169 (Finite primorial escalation ledger)**.** The audit up to $P \leq 1009$ starts with

$$1302951 \text{ blockers} = 947032 \text{ small-LPF blockers} + 355919 \text{ prime blockers}.$$

The fixed wheels through 31 and the dynamic square-root wheel all have the same blocker count:

$$\text{deleted} = 947032, \qquad \text{surviving prime blockers} = 355919.$$

In particular

$$\texttt{fixed\_210\_2310\_and\_beyond\_new\_rough\_shell\_count\_total} = 0, \quad \texttt{sqrt\_cutoff\_extra\_killed\_over\_30} = 0.$$

The cutoff $y = P$ deletes 140983 prime blockers but leaves 214936. The cutoff $y = 2P - 1$ deletes all 355919 prime blockers, and is therefore recorded as

`full_2P_minus_1_cutoff_is_prime_oracle=true`.

Thus the primorial ladder is now fully diagnosed in the blocker normal form: continued finite-wheel refinement does not break the parity barrier. The remaining route must control the prime-blocker packet by a non-wheel phase saving, a trace/Kloosterman embedding, or a genuinely new theorem.

The latest narrow mouth is

```
PrimeBlockerNonWheelPhaseSavingOrTraceEmbedding
AND NonOracleControlOfPrimeBlockerDynamicSqrtSieve
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-hole-primorial-escalation-audit.md.
```

8.65 Dynamic square-root sieve for prime blockers

Theorem 8.170 (Prime blockers as dynamic square-root-sieve survivors). *Let $m_h = m_h(P, q, k)$ be a blocker in the preceding normal form and put*

$$W(m_h) = \prod_{\ell \le \sqrt{m_h}} \ell,$$

where the product is over primes. Then

$$m_h \text{ is a prime blocker} \iff (m_h, W(m_h)) = 1.$$

Equivalently m_h is a prime blocker if and only if

$$m_h \not\equiv 0 \pmod{\ell} \qquad (\ell \le \sqrt{m_h},\ \ell \text{ prime}).$$

Consequently the prime-blocker phase packet admits the exact rewriting

$$S_{\text{prime blocker}}(h) = S_{\sqrt{-\text{sieve}}\ \text{survivor}}(h).$$

All blockers rejected by this dynamic square-root sieve are exactly the LPF $= 2, 3, 5$ *blockers already belonging to the* 30*-wheel packet.*

Proof. The elementary square-root primality criterion gives, for any integer $m_h > 1$, that m_h is prime if and only if no prime $\ell \le \sqrt{m_h}$ divides it. This is the displayed $(m_h, W(m_h)) = 1$ condition. The blocker theorem has already shown that every blocker has least prime factor $2, 3, 5$, or else is prime. Therefore a rejected blocker has least prime factor $2, 3$, or 5; after the 30-wheel rejections, the remaining blockers are precisely the dynamic square-root-sieve survivors. The phase identity follows by summing the same additive character $e(hP\lfloor qm_h/P \rfloor/q)$ over this identical survivor set. $\square$

Remark 8.171 (Finite dynamic square-root-sieve ledger)**.** The audit up to $P \le 1009$ verifies the identity on all 1302951 blockers. The counts are

$$947032\,\text{small-LPF blockers}, \qquad 355919\,\text{prime blockers} = 355919\,\text{dynamic square-root-sieve survivors}.$$

There are no mismatches:

$$\texttt{total\_bad\_dynamic\_sqrt\_sieve\_count} = 0, \qquad \texttt{max\_prime\_packet\_phase\_identity\_error} = 0.$$

The first obstruction counts are

$$373676 \text{ at } 2, \qquad 409713 \text{ at } 3, \qquad 163643 \text{ at } 5, \qquad 355919 \text{ survivors}.$$

Thus the dynamic square-root sieve removes the prime-blocker oracle as a definition, but it does not yet give cancellation. The moving primorial $W(m_h)$ varies with the blocker; in the same finite audit the maximum number of square-root primes per blocker is 14, so a full Mobius expansion can have 16384 local terms per blocker, and the prime-blocker survivor packet accounts for 399176624 formal Mobius terms.

The external frontier is unchanged in substance: trace-function bilinear estimates, arbitrary-modulus Kloosterman estimates, composite Type-II amplification, unbalanced Kloosterman-fraction convolutions, and smooth/squarefree Kloosterman parameter estimates become relevant only after this moving survivor packet is embedded into a completed trace, Type-II, or convolution object. The known short-interval exponent 0.52 is still above the pointwise 1/2 scale needed for direct closure.

The latest narrow mouth is

```
PrimeBlockerSqrtSieveSurvivorPhaseSavingOrTraceEmbedding
AND MovingPrimorialMobiusExpansionCompression
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-prime-blocker-dynamic-sqrt-sieve-audit.md.
```

8.66 Euler–Mobius involution for prime blockers

Theorem 8.172 (Moving Mobius expansion reduces to involution and prime singletons)**.** *For a blocker m_h, let*

$$W(m_h) = \prod_{\ell \le \sqrt{m_h}} \ell$$

be the moving square-root primorial. Then the prime-blocker indicator has the exact Mobius form

$$1_{\text{prime blocker}}(m_h) = \sum_{\substack{d \mid m_h \\ d \mid W(m_h)}} \mu(d).$$

If m_h is rejected by the square-root sieve, let s be the least prime divisor of m_h with $s \le \sqrt{m_h}$. The terms in the displayed sum are paired by the involution

$$d \longleftrightarrow sd$$

on squarefree divisors not already containing s, and each pair has total Mobius weight zero. If m_h is a prime blocker, the only divisor contributing to the sum is $d = 1$. Hence the moving Mobius expansion itself does not create a new phase-saving layer: after pointwise cancellation it leaves exactly the prime-survivor singleton layer.

Proof. The identity is the finite Euler product

$$\prod_{\ell \mid W(m_h)} (1 - 1_{\ell \mid m_h}) = \sum_{\substack{d \mid m_h \\ d \mid W(m_h)}} \mu(d).$$

It equals 1 precisely when no prime $\ell \le \sqrt{m_h}$ divides m_h, which is the prime-blocker condition from the previous theorem; it equals 0 otherwise. In the rejected case, the least obstruction s divides m_h. Multiplication by s maps every squarefree divisor of m_h not containing s bijectively to one containing s, and changes the sign of the Mobius value. Thus the two terms cancel. In the prime case there is no nontrivial divisor $d \mid m_h$ with $d \le \sqrt{m_h}$, so the sum consists only of $d = 1$. □

Remark 8.173 (Finite Euler–Mobius involution ledger)**.** The audit up to $P \le 1009$ verifies the Mobius identity and the involution on all 1302951 blockers:

$$\texttt{total\_bad\_mobius\_involution\_count} = 0, \qquad \texttt{max\_mobius\_packet\_phase\_identity\_error} = 0.$$

The active squarefree divisor terms are

$$\texttt{active\_divisor\_terms\_total} = 4321483,$$

of which

$$\texttt{prime\_singleton\_terms\_total} = 355919$$

are the $d = 1$ prime-survivor terms, while

$$\texttt{composite\_cancelled\_terms\_total} = 3965564$$

are paired into

$$\texttt{composite\_involution\_pair\_count\_total} = 1982782$$

zero-sum Euler pairs. The largest active divisor basis has size 4, so the largest active pointwise expansion has 16 terms. This is far smaller than the formal moving-primorial expansion ledger, but it also shows why the Mobius expansion is not an independent proof: after the pointwise Euler involution, the only uncancelled object is again the prime-survivor singleton phase packet.

The latest narrow mouth is

```
PrimeSurvivorSingletonLayerPhaseSavingOrTraceEmbedding
AND NonPointwiseCompressionBeyondEulerMobiusInvolution
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-qsupport-
row-averaged-additive-k-prime-blocker-mobius-involution-audit.md.
```

8.67 Prime-survivor floor-span completion

Theorem 8.174 (Prime survivors fill their floor prime span up to the diagonal)**.** *Fix a row prime P and an active prime q-bucket in the row-averaged additive-k completion setting. Let*

$$M_{P,q}^{\mathrm{pr}}$$

be the set of prime blockers left by the Euler–Mobius singleton layer. If this set is nonempty, put

$$L_{P,q} = \min M_{P,q}^{\mathrm{pr}}, \qquad U_{P,q} = \max M_{P,q}^{\mathrm{pr}}.$$

Then

$$M_{P,q}^{\mathrm{pr}} = \{m : \ L_{P,q} \le m \le U_{P,q}, \ m \text{ prime}\} \setminus \{P\}.$$

Consequently the prime-survivor singleton phase packet has the exact floor-span form

$$S_{\mathrm{pr\ surv}}(h) = \sum_q \sum_{\substack{L_{P,q} \le m \le U_{P,q} \\ m \text{ prime}, \ m \ne P}} e\left(\frac{hP\lfloor qm/P \rfloor}{q}\right),$$

with the diagonal $m = P$ treated only as a completion ghost.

Proof. One inclusion is the definition of $L_{P,q}$ and $U_{P,q}$. Conversely let m be a prime in the displayed interval with $m \ne P$, and put

$$k = \lfloor qm/P \rfloor.$$

The map $m \mapsto \lfloor qm/P \rfloor$ is monotone, so k lies between the two k-values attached to the endpoint prime blockers, hence inside the same completion span. Since $m \ne P$, the defect

$$D = qm - kP$$

is positive. The corresponding product cell has length < 2 and contains the unique odd candidate m. If k were a genuine selected LPF-support index, that unique odd candidate would have to be a residual rough composite cofactor. But m is prime, so this is impossible. Thus k is a completion hole, and its blocker is exactly the prime m. The only prime in the span for which this argument fails is $m = P$, where $\lfloor qP/P \rfloor = q$ and $D = 0$; this is not a blocker and must be subtracted as the diagonal completion ghost. □

Remark 8.175 (Finite prime-survivor floor-span ledger)**.** The audit up to $P \le 1009$ verifies the floor-span identity on the entire prime-survivor singleton layer:

$$\texttt{prime\_survivor\_edge\_count\_total} = 355919, \qquad \texttt{q\_bucket\_count\_total} = 5848.$$

The raw completed prime spans contain

$$\texttt{span\_prime\_count\_total} = 361626$$

prime entries. The only raw missing entries are the diagonal ghosts:

$$\texttt{raw\_missing\_count\_total} = \texttt{diagonal\_ghost\_count\_total} = 5707, \qquad \texttt{raw\_extra\_count\_total} = 0.$$

After subtracting $m = P$, there are no span mismatches:

$$\texttt{total\_bad\_prime\_survivor\_floor\_span\_count} = 0, \qquad \texttt{max\_span\_completion\_phase\_error} = 1.421 \cdot 10^{-14}.$$

This does not turn the packet into a dense bilinear object. With the diagonal removed, the full prime-prime rectangle for the same audited rows has

$$\texttt{full\_prime\_prime\_rectangle\_edge\_count\_without\_diagonal\_total} = 849334$$

possible edges, while the survivor graph has density only

$$\texttt{prime\_survivor\_to\_full\_rectangle\_without\_diagonal\_density} = 0.41905658.$$

Thus 493415 rectangle edges are still absent. The result is a sharper normal form for the remaining prime-survivor layer, not a phase-saving theorem.

The latest narrow mouth is

```
PrimeSurvivorPrimeIntervalFloorSpanPhaseSavingOrTraceEmbedding
AND DiagonalPGhostSubtractionDiscipline
AND DenseRectangleCompletionOrBilinearTraceEmbeddingForPrimePrimeFloorGraph
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
floor-span-completion-audit.md.
```

8.68 Finite q-prefix cap structure of the prime-survivor graph

Reduction-closed Statement 8.176 (Finite q-prefix and unimodal-cap audit)**.** *In the finite audit range $P \le 1009$, transpose the prime-survivor floor-span graph by fixing P and a prime survivor coordinate m. For every audited nonempty (P, m) fibre, the q-neighbourhood is a prime prefix*

$$Q^{\mathrm{pr}}_{P,m} = \{q : \ q_0(P) \le q \le Q^*(P, m), \ q \text{ prime}\},$$

where $q_0(P)$ is the first eligible prime q with $P/2 < q < P$. Moreover, for every audited active row P, the cap $Q^*(P, m)$, as m runs through the ordered prime survivor coordinates, is unimodal: it is first nondecreasing and then nonincreasing.

Remark 8.177 (Finite q-prefix cap ledger)**.** The q-prefix audit is deliberately recorded as a finite structural certificate, not as a global theorem. It verifies the transposed support shape on the same 355919 prime-survivor edges:

$$\texttt{pm\_bucket\_count\_total} = 16328, \qquad \texttt{q\_prefix\_count\_total} = 355919.$$

There are no prefix defects:

$$\texttt{lower\_endpoint\_not\_row\_first\_total} = 0, \quad \texttt{q\_prefix\_missing\_count\_total} = 0, \quad \texttt{q\_prefix\_extra\_count\_}$$

The phase packet agrees exactly in the audit:

$$\texttt{bad\_q\_prefix\_identity\_count} = 0, \qquad \texttt{max\_q\_prefix\_phase\_error} = 0.$$

Among the 155 active audited rows, every cap is unimodal:

$$\texttt{cap\_unimodality\_bad\_row\_count} = 0, \qquad \texttt{cap\_turn\_count\_distribution} = \{0:1,\ 1:154\}.$$

Thus the remaining prime-survivor graph is no longer best viewed as a generic sparse point cloud. In the audited range it is a q-prefix cap graph, with a one-dimensional cap $Q^*(P,m)$. This is closer to a possible two-monotone Type-II or unbalanced convolution decomposition, but it still does not supply the required phase saving. A global proof of the prefix cap structure, or a replacement analytic estimate that does not need it, remains open.

The latest narrow mouth is

```
GlobalPrimeSurvivorQPrefixUnimodalCapProofOrReplacement
AND PrefixCapTraceOrTypeIIPhaseSaving
AND DiagonalPGhostSubtractionDiscipline
AND DenseRectangleCompletionOrBilinearTraceEmbeddingForPrimePrimeFloorGraph
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
qprefix-unimodal-audit.md.
```

8.69 Nested rough-envelope origin of the q-prefix cap

Theorem 8.178 (Residual rough envelope for the prime-survivor cap)**.** *Fix P, and let R_P be the set of residual rough composite cofactors in the row range $P/2 < m < 2P$. For an eligible prime q with $P/2 < q < P$, the selected residual support in the q-bucket is exactly*

$$R_{P,q} = \{m \in R_P :\ q \le m,\ qm < P^2\}.$$

If this set is nonempty, write

$$A_q = \min R_{P,q}, \qquad B_q = \max R_{P,q}.$$

Then the prime-survivor blockers in this q-bucket are precisely

$$\{m :\ A_q \le m \le B_q,\ m \text{ prime}\} \setminus \{P\}.$$

As q increases, the sets $R_{P,q}$ are nested decreasing; hence A_q is nondecreasing and B_q is nonincreasing. Consequently the transposed prime-survivor cap satisfies

$$Q^*(P,m) = \min\Big(\max\{q :\ A_q < m\},\ \max\{q :\ B_q > m\}\Big)$$

whenever the two threshold sets are nonempty. Thus the q-prefix and the unimodal cap shape come from the nested residual rough envelope.

Proof. For a residual rough cofactor m, the q-cell condition includes $m \geq q$. The additive row condition $1 \leq k < P$, with $k = \lfloor qm/P \rfloor$, is equivalent to $qm < P^2$. Since $q > P/2$, the product cell has length $P/q < 2$, so it contains at most one odd integer. Therefore a residual rough m is selected exactly when

$$m \in R_P, \qquad q \leq m, \qquad qm < P^2.$$

This proves the displayed formula for $R_{P,q}$.

The completion span of the q-bucket is the interval between the least and greatest selected residual k-values. Since the map

$$m \longmapsto \lfloor qm/P \rfloor$$

is nondecreasing, every prime m between A_q and B_q maps into that completion span. Its product cell has m as the unique odd candidate. It is not a residual rough composite, so it is a completion blocker; if $m = P$, then $k = q$ and $D = qm - kP = 0$, so it is only the diagonal completion ghost and must be removed. Conversely, the preceding floor-span theorem shows that every prime survivor in the q-bucket lies between the two residual endpoints. This proves the prime-survivor envelope.

If $q_1 < q_2$, then

$$[q_2, P^2/q_2) \subset [q_1, P^2/q_1),$$

so $R_{P,q_2} \subset R_{P,q_1}$. Hence A_q cannot decrease and B_q cannot increase along nonempty buckets. For fixed m, the conditions $A_q < m$ and $B_q > m$ are therefore both prefix conditions in the ordered list of eligible q's. Their intersection is a prefix whose last element is the displayed minimum of the two threshold maxima. As m increases, the first threshold is nondecreasing and the second is nonincreasing, so their minimum is unimodal. □

Remark 8.179 (Finite rough-envelope cap ledger)**.** The audit up to $P \leq 1009$ verifies the rough-envelope formula against the existing selected-k implementation:

$$\texttt{selected\_formula\_m\_count\_total} = \texttt{selected\_existing\_m\_count\_total} = 299977, \qquad \texttt{selected\_envelope\_f}$$

The prime-survivor edge set is recovered exactly:

$$\texttt{actual\_prime\_edge\_count\_total} = \texttt{predicted\_prime\_edge\_count\_total} = 355919,$$

with

$$\texttt{prime\_envelope\_missing\_count} = \texttt{prime\_envelope\_extra\_count} = 0.$$

The endpoint monotonicity has no audited exception:

$$\texttt{A\_monotonicity\_bad\_step\_count} = \texttt{B\_monotonicity\_bad\_step\_count} = 0.$$

The cap formula also matches all 16328 audited (P, m)-buckets:

$$\texttt{cap\_min\_threshold\_mismatch\_count} = 0, \qquad \texttt{predicted\_qprefix\_mismatch\_count} = 0.$$

This closes the support-shape origin of the q-prefix cap. It does not close the analytic phase problem: one still needs to turn the nested cap with the moving prime denominator q into a completed trace/Kloosterman, Type-II, dispersion, or convolution estimate.

The latest narrow mouth is

```
PrefixCapTraceOrTypeIIPhaseSavingFromNestedRoughEnvelope
AND CompletedTraceOrKloostermanVariableForMovingPrimeDenominator
AND DiagonalPGhostSubtractionDiscipline
AND FiniteThirtyWheelSmallLPFBlockerPacketControl
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND RoughBetaSiegelWalfiszUniformityOrReplacement
AND PointwisePKUniformTransferFromExternalAverageEstimate
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
rough-envelope-cap-audit.md.
```

8.70 Bulk rectangles inside the nested prime-survivor cap

Theorem 8.180 (Prime-prime bulk rectangle extracted from the cap)**.** *Fix P, and let E_P be the prime-survivor edge set in the (q,m)-plane after diagonal $m = P$ subtraction. For each eligible prime q, write its vertical fibre as the prime interval*

$$E_P(q) = \{m : \ (q,m) \in E_P\} = \{m \text{ prime} : \ L_q \le m \le U_q\} \setminus \{P\}.$$

For every consecutive block $Q = [q_i, q_j]$ of eligible prime q's, set

$$L(Q) = \max_{q\in Q} L_q, \qquad U(Q) = \min_{q\in Q} U_q.$$

Then

$$Q \times \{m \text{ prime} : \ L(Q) \le m \le U(Q), \ m \ne P\} \subset E_P.$$

Thus the nested cap contains genuine prime q by prime m product rectangles. The maximal such rectangle gives the natural first Type-II bulk candidate; its complement in E_P is the staircase boundary.

Proof. For every $q \in Q$, the definition of $L(Q)$ and $U(Q)$ gives

$$L_q \le L(Q) \le m \le U(Q) \le U_q.$$

Hence every prime $m \ne P$ in the displayed interval belongs to the vertical fibre $E_P(q)$. This proves the product inclusion. Maximising the product of the number of primes in Q and the number of primes in the common m-interval gives the largest bulk rectangle for that row. The remaining edges are exactly the boundary left after this rectangle is removed. □

Remark 8.181 (Finite bulk-rectangle Type-II ledger)**.** The audit up to $P \le 1009$ extracts the largest verified complete prime-prime rectangle in every active row. The total prime-survivor edge count remains

$$\texttt{actual\_prime\_edge\_count\_total} = 355919.$$

The extracted bulk and the leftover boundary have comparable sizes:

$$\texttt{bulk\_rectangle\_edge\_count\_total} = 178404, \qquad \texttt{boundary\_edge\_count\_total} = 177515,$$

so

$$\texttt{bulk\_fraction\_total} = 0.501248879661, \qquad \texttt{boundary\_fraction\_total} = 0.498751120339.$$

Every extracted bulk rectangle is actually complete:

$$\texttt{bulk\_missing\_count\_total} = 0, \qquad \texttt{total\_bad\_bulk\_rectangle\_typeii\_count} = 0.$$

Naively completing each row to its full $q \times m$ rectangle is too expensive:

$$\texttt{row\_full\_rectangle\_count\_total} = 795159, \qquad \texttt{row\_completion\_extra\_count\_total} = 439240.$$

Thus the support now has a genuine Type-II-shaped bulk entrance, but a bulk-only estimate cannot close the survivor layer: the staircase boundary is still rate-bearing and must be completed or estimated with cancellation.

The latest narrow mouth is

```
BoundaryPhaseSavingForNestedRoughEnvelopeStaircase
AND CompletedTraceFamilyForPrimePrimeBulkRectangle
AND StaircaseBoundaryCompletionWithoutComparableLoss
AND DiagonalPGhostSubtractionDiscipline
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
bulk-rectangle-typeii-audit.md.
```

8.71 Three monotone strips for the bulk boundary

Theorem 8.182 (Boundary strip decomposition)**.** *In the setting of the preceding theorem, suppose the vertical fibres*

$$E_P(q) = \{m \text{ prime}: \ L_q \le m \le U_q, \ m \ne P\}$$

are ordered by increasing eligible prime q. Since the rough-envelope endpoints satisfy L_q nondecreasing and U_q nonincreasing, a maximal product rectangle may be taken to start at the first eligible q. Write this bulk prefix as $q \le q_*$, and write its common m-interval as $[L_*, U_*]$. Then the complement of the bulk in E_P is the disjoint union of three strip packets:

$$B_{\text{lower}} = \{(q,m) : q \le q_*, \ L_q \le m < L_*, \ m \text{ prime}, \ m \ne P\},$$
$$B_{\text{upper}} = \{(q,m) : q \le q_*, \ U_* < m \le U_q, \ m \text{ prime}, \ m \ne P\},$$
$$B_{\text{right}} = \{(q,m) : q > q_*, \ L_q \le m \le U_q, \ m \text{ prime}, \ m \ne P\}.$$

There is no left-tail strip. Moreover, each active strip is supported on a contiguous block of q's, and its fibre lengths are nonincreasing along the q-order.

Proof. For any consecutive block of q's not beginning at the first eligible q, adding the preceding eligible q cannot shrink the common m-interval: its lower endpoint is no larger and its upper endpoint is no smaller. Thus the rectangle area weakly increases after adding it. A maximal rectangle therefore begins at the first eligible q.

Inside the bulk prefix $q \le q_*$, the bulk occupies the common interval $[L_*, U_*]$. The fibre interval $[L_q, U_q]$ splits disjointly into the lower part below L_*, the bulk part, and the upper part above U_*. For $q > q_*$, no point belongs to the chosen bulk, so the entire fibre is the right-tail packet. These three cases exhaust the boundary and are disjoint.

The monotonicity of L_q and U_q implies that $[L_q, L_*)$, $(U_*, U_q]$, and $[L_q, U_q]$ shrink as q increases within their respective active ranges. Hence the active q-sets are contiguous and the prime fibre lengths cannot increase. $\square$

Remark 8.183 (Finite boundary-strip ledger)**.** The audit up to $P \le 1009$ verifies that the boundary of size

$$\texttt{boundary\_edge\_count\_total} = 177515$$

is exactly recovered by the three strip packets:

$$\texttt{strip\_boundary\_count\_total} = 177515, \qquad \texttt{left\_tail\_count\_total} = 0.$$

The three nonzero strip counts are

$$\texttt{lower\_wing\_count\_total} = 29144, \qquad \texttt{upper\_wing\_count\_total} = 61620, \qquad \texttt{right\_tail\_count\_total} = 867$$

The bulk always starts at the first eligible q in the audited rows:

$$\texttt{q\_start\_not\_row\_first\_count} = 0.$$

The strip fibres have no completion mismatch and no monotonicity failure:

$$\texttt{strip\_prime\_interval\_mismatch\_count\_total} = 0, \qquad \texttt{noncontiguous\_strip\_count\_total} = 0, \qquad \texttt{strip\_l}$$

Thus the boundary is no longer an arbitrary leftover; it is a union of three monotone prime-interval endpoint packets. This still does not give cancellation. The remaining analytic task is to complete these endpoint packets, or to prove a summation-by-parts/trace estimate for them, without losing a boundary-sized main term.

The latest narrow mouth is

```
BoundaryPhaseSavingForThreeMonotonePrimeIntervalStrips
AND CompletedTraceFamilyForPrimePrimeBulkRectangle
AND StripEndpointSummationByPartsWithoutComparableLoss
AND DiagonalPGhostSubtractionDiscipline
AND UniformCancellationAcrossSparseKSupportRadialKernels
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-strip-decomposition-audit.md.
```

8.72 Layer-cake rectangles for the boundary strips

Theorem 8.184 (Boundary layer-cake product rectangles)**.** *Let B be one of the three boundary strips from the preceding theorem, and order its active fibres as*

$$I_1 \supseteq I_2 \supseteq \cdots \supseteq I_s$$

along increasing eligible prime q. Define $I_{s+1} = \emptyset$. Then

$$B = \bigsqcup_{j=1}^{s} \big(\{q_1, \dots, q_j\} \times (I_j \setminus I_{j+1})\big).$$

After splitting each prime shell $I_j \setminus I_{j+1}$ into consecutive prime intervals, the boundary is a disjoint union of strip-local q-prefix by prime-m-shell product rectangles.

Proof. Since the strip fibres are nested decreasing, a point $m \in I_j \setminus I_{j+1}$ belongs exactly to the first j fibres $I_1, \dots, I_j$ and to no later fibre. Hence its full contribution to the graph is precisely $\{q_1, \dots, q_j\} \times \{m\}$. Summing over the disjoint shell differences $I_j \setminus I_{j+1}$ proves the displayed disjoint union. Each shell is a union of consecutive prime intervals because the original fibres are prime intervals. Splitting the shell by its prime-interval components gives product rectangles. ☐

Remark 8.185 (Finite boundary layer-cake ledger)**.** The audit up to $P \le 1009$ verifies the exact rectangle layer expansion:

$$\texttt{boundary\_edge\_count\_total} = \texttt{layercake\_edge\_count\_total} = 177515.$$

The boundary is carried by

$$\texttt{layercake\_rectangle\_count\_total} = 6190$$

complete product rectangles, split by strip as

$$\texttt{lower\_wing\_rectangle\_count\_total} = 1032, \qquad \texttt{upper\_wing\_rectangle\_count\_total} = 1967, \qquad \texttt{right\_tail}$$

The edge counts carried by the same three families are

$$29144, \qquad 61620, \qquad 86751,$$

respectively. The layer expansion has no audited leak:

$$\texttt{nested\_bad\_steps\_total} = \texttt{missing\_count\_total} = \texttt{extra\_count\_total} = 0.$$

The largest audited layer contains 261 edges, and the maximum number of layers in one row is 81.

This removes the curved-boundary support obstruction, but it still does not close the analytic estimate. One must prove a uniform phase saving over all layer rectangles, or build a completed trace/Kloosterman family whose summation over the 6190 audited layer packets has no comparable loss.

The latest narrow mouth is

```
UniformPhaseSavingAcrossBoundaryLayerCakeRectangles
AND CompletedTraceFamilyForPrimePrimeBulkRectangle
AND CompletedKloostermanVariableForMovingPrimeDenominatorOnLayers
AND StripEndpointSummationByPartsWithoutComparableLoss
AND DiagonalPGhostSubtractionDiscipline
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still not an unconditional proof of the Phi–LPF target.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-layercake-rectangles-audit.md.
```

8.73 Phase interface of the boundary layer-cake packets

Theorem 8.186 (Short-shell obstruction inside the boundary layer-cake)**.** *The boundary layer-cake decomposition does not by itself put the boundary into a long two-variable Type-II range. In the audit range $P \le 1009$, the* 6190 *complete product layers split into*

phase class	*rectangles*	*edges*
genuine_qm_product_layer	3880	147181
q_prefix_line_layer	1955	29172
m_shell_line_layer	296	1103
point_layer	59	59.

The median q-prefix length is 11 *and the maximum is* 37*, but the median prime-m-shell length is only* 2*, and the maximum is* 12*. In particular no audited layer has both sides of length at least* 16*.*

Proof. For each layer rectangle R in the preceding layer-cake decomposition, write

$$Q(R) = \#\{q \text{ in the strip-local prefix}\}, \qquad M(R) = \#\{m \text{ in the prime shell}\}.$$

Classify R as a point layer if $Q(R) = M(R) = 1$, as a q-prefix line if $Q(R) > 1$ and $M(R) = 1$, as an m-shell line if $Q(R) = 1$ and $M(R) > 1$, and as a genuine q, m-product layer if both sides have length at least 2. The machine audit regenerates the same 6190 rectangles from the previous certificate and checks that their total edge count remains 177515. Direct enumeration of the pairs $(Q(R), M(R))$ gives the displayed table, the side-length statistics, and the zero count for $Q(R), M(R) \geq 16$. $\square$

Remark 8.187 (External theorem interface after the phase split)**.** This is a genuine narrowing of the obstruction. The issue is no longer that the boundary is a curved or arbitrary staircase: it is a finite family of product layers. The obstruction is that the product layers are predominantly q-long but m-short, with many q-prefix line layers. Thus a long-variable Type-II or trace theorem cannot be invoked layer by layer without an additional short prime-shell completion, a moving prime-q denominator completion, and a summation principle over all 6190 layer packets.

The finite ledger also records

$$\sum_R \sqrt{|R|} = 30147.122529825978, \qquad \sqrt{\sum_R |R|} = 421.325290007614,$$

so a naive per-layer square-root summation has an audited loss factor

$$71.553080825699.$$

This number is not an asymptotic theorem, but it prevents the manuscript from silently replacing uniform layer aggregation by a harmless constant.

The latest narrow mouth is now

```
UniformShortPrimeShellCompletionAcrossLayerCakeRectangles
AND MovingPrimeQDenominatorCompletedTraceFamily
AND NoLossLayerAggregationFor6190ShortShellPackets
AND EndpointSummationByPartsForQPrefixLineLayers
AND DiagonalPGhostSubtractionDiscipline
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

FKMS-type trace bilinear inputs, arbitrary-modulus Kloosterman inputs, composite Type-II inputs, unbalanced Kloosterman inputs, and $x^{0.52}$ short-interval prime theorems remain relevant interfaces only after these short-shell and moving-denominator conditions are supplied. This is still not an unconditional proof of the Phi–LPF target, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-layercake-phase-interface-audit.md.
```

8.74 Shell-step packets for the boundary layer-cake

Theorem 8.188 (Shell-step packet aggregation). *In the boundary layer-cake audit range $P \le 1009$, all layer rectangles with the same row, strip, q-prefix count, q-start, and q-end can be merged into a single q-prefix shell-step packet. This gives an edge-preserving aggregation*

$$6190 \text{ layer rectangles} \longrightarrow 5106 \text{ shell-step packets},$$

with total edge mass still equal to 177515*. Each packet has at most three prime-m blocks. The block distribution is*

number of m-blocks	*packets*	*edges*
1	4023	116221
2	1082	61258
3	1	36.

The median m-shell size rises to 3*, but the maximum remains* 12*; no audited packet has both sides of length at least* 16*.*

Proof. Start from the preceding layer-cake rectangles. Two rectangles are placed in the same packet exactly when they have the same tuple

$$(P,\ \text{strip},\ q\text{-prefix count},\ q_{\min},\ q_{\max}).$$

The q-side is therefore identical within a packet; the m-side is the disjoint union of the prime-m blocks contributed by the source rectangles. The machine audit recomputes all packets from the rectangle ledger and verifies

$$\sum_{\text{packets}} |Q_{\text{packet}}|\,|M_{\text{packet}}| = 177515 = \sum_{\text{rectangles}} |Q_{\text{rect}}|\,|M_{\text{rect}}|.$$

It also records the number of source m-blocks in each packet, giving the displayed block distribution and the bound of three blocks per packet. □

Remark 8.189 (What this aggregation does and does not prove). This is an accounting improvement for the endpoint summation problem. The naive finite square-root ledger improves from

$$71.553080825699 \quad \text{to} \quad 64.519277044879$$

when passing from individual layer rectangles to shell-step packets. Equivalently, the formal layer count drops by 1084 without changing the actual edge mass.

The remaining obstruction is still analytic. One must prove cancellation for q-prefix packets whose m-side has at most three prime blocks and at most 12 primes, complete the moving prime-q denominator as a trace or Kloosterman variable, and sum all 5106 packet estimates without a comparable endpoint or Cauchy loss.

The latest narrow mouth is

```
UniformShortShellPhaseSavingForAtMostThreeBlockPackets
AND MovingPrimeQDenominatorCompletedTraceFamilyOnShellStepPackets
AND NoLossAggregationAcross5106ShellStepPackets
AND EndpointSummationByPartsForQPrefixLinePackets
AND DiagonalPGhostSubtractionDiscipline
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

Thus the aggregation does not close the Phi–LPF parity barrier, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-shell-step-packet-audit.md.
```

8.75 Right-tail localization of multi-block shell gaps

Theorem 8.190 (All multi-block packets lie in the right tail). *In the shell-step packet audit range $P \le 1009$, the multi-block short-shell complexity is completely localized in the right-tail family. The packet ledger splits as*

strip	*m-blocks*	*packets*	*edges*
`lower_wing`	1	1032	29144
`upper_wing`	1	1967	61620
`right_tail`	1	1024	25457
`right_tail`	2	1082	61258
`right_tail`	3	1	36.

Thus the 1083 *multi-block packets, carrying* 61294 *edges, all lie in* `right_tail`*. The lower and upper wings contain no multi-block packet.*

Proof. For each shell-step packet, count the number of prime-m blocks in its m-support and record the strip label inherited from the boundary decomposition. The machine audit recomputes the full 5106-packet ledger, checks that the edge mass remains 177515, and tabulates the pair (strip, block count). The displayed table is the complete nonzero output of this tabulation. Since the only rows with block count larger than 1 have strip label `right_tail`, the asserted localization follows. □

Remark 8.191 (Gap statistics and the next analytic interface). The right-tail multi-block packets contain 1084 internal prime gaps. Their missing-prime gap sizes have

$$\min = 1, \qquad \text{median} = 26, \qquad \max = 79, \qquad \text{average} = 28.690959409594.$$

This proves that multi-block behavior is not a global feature of all short shell packets. It is a named right-tail endpoint phenomenon. The remaining single-block packet family consists of 4023 packets carrying 116221 edges; it includes all lower-wing packets, all upper-wing packets, and 1024 right-tail packets.

The latest narrow mouth is now

```
RightTailMultiBlockGapPacketPhaseSaving
AND SingleBlockEndpointPacketSummationByParts
AND MovingPrimeQDenominatorCompletedTraceFamilyOnRightTailAndSingleBlockPackets
AND NoLossAggregationAcross5106ShellStepPackets
AND DiagonalPGhostSubtractionDiscipline
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is a support localization and endpoint taxonomy theorem, not an unconditional phase-saving theorem. FKMS-type trace bilinear estimates, arbitrary-modulus Kloosterman estimates, composite Type-II estimates, and unbalanced Kloosterman estimates remain only candidate interfaces until the moving q-denominator and packet-level summation are supplied.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-right-tail-gap-localization-audit.md.
```

8.76 Diagonal-core decomposition of right-tail shell gaps

Theorem 8.192 (Right-tail gaps are successor cores plus diagonal ghosts). *In the right-tail multi-block shell packet audit range $P \le 1009$, every internal prime gap between adjacent m-blocks has* *the following form. If F_i is the current right-tail fibre, F_{i+1} is the successor fibre, and G is the set of missing primes in the internal gap, then*

$$G = (G \cap F_{i+1}) \sqcup E, \qquad E \in \{\varnothing, \{P\}\}.$$

Thus the only gap prime not carried by the successor fibre is the row-prime diagonal ghost P. The finite ledger gives

gap class	*gaps*	*gap-edge weight*
`diagonal_plus_carried_core`	1006	56349
`pure_diagonal_slit`	77	4945
`carried_core_without_diagonal`	1	36.

There are no unexplained gap primes.

Proof. For each of the 1083 right-tail multi-block packets, reconstruct the nested right-tail fibres from the boundary layer-cake ledger. The current packet shell is exactly $F_i \setminus F_{i+1}$. For each of the 1084 internal gaps, enumerate the missing primes between the adjacent shell blocks and split them into primes lying in F_{i+1}, the possible row prime P, and all remaining primes. The machine audit finds zero remaining primes. It also verifies that the inherited shell-step packet count remains 5106, the total edge mass remains 177515, and the right-tail multi-block edge mass remains 61294. Hence the displayed decomposition holds for every audited gap. □

Remark 8.193 (Latest right-tail phase interface)**.** The diagonal P-ghost is now an explicit support subtraction for the right-tail multi-block family. Across all internal gaps, the missing-prime mass 31101 splits into 30018 successor-fibre carried primes and 1083 diagonal ghosts. This does not prove cancellation. It reduces the right-tail multi-block analytic task to cancellation on the carried core:

```
RightTailSuccessorFibreCorePhaseSaving
AND SingleBlockEndpointPacketSummationByParts
AND MovingPrimeQDenominatorCompletedTraceFamilyOnSuccessorCoreAndSingleBlockPackets
AND NoLossAggregationAcross5106ShellStepPackets
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

FKMS-type trace bilinear estimates, arbitrary-modulus Kloosterman estimates, composite Type-II estimates, and unbalanced Kloosterman estimates remain candidate interfaces only after the successor-core family is completed with the moving q-denominator.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-right-tail-gap-diagonal-core-audit.md.
```

8.77 Punctured-interval completion of the right-tail fibres

Theorem 8.194 (Right-tail fibres are punctured prime intervals)**.** *In the audit range $P \leq 1009$, every right-tail fibre is a contiguous prime interval, possibly with the row prime P removed. The fibre ledger has*

fibre class	*fibres*	*prime mass*
`contiguous_prime_interval`	138	245
`p_punctured_prime_interval`	2873	86506.

There are no right-tail fibres with any other internal hole. Moreover, each of the 1083 right-tail multi-block shell packets completes to one prime interval after adding its successor-fibre core and the diagonal P-ghost. The completion carries all 61294 multi-block edges.

Proof. For every fixed row P, reconstruct the right-tail fibres from the boundary layer-cake ledger. For each nonempty fibre, compare it with the full prime interval between its least and greatest prime. The only possible missing prime is P, and the audit finds no other missing prime in any of the 3011 right-tail fibres.

For a multi-block packet, the shell is $F_i \setminus F_{i+1}$. Let $[a,b]_{\mathbb{P}}$ be the prime interval between the least and greatest prime of this shell. The audit checks that

$$(F_i \setminus F_{i+1}) \cup (F_{i+1} \cap [a,b]_{\mathbb{P}}) \cup \{P\} = [a,b]_{\mathbb{P}}$$

with the evident convention that P is added only when $P \in [a,b]$. This verification succeeds for all 1083 multi-block packets and has zero completion holes. □

Remark 8.195 (Updated analytic mouth)**.** The successor core is not an arbitrary sparse residue. It is the part of the next nested right-tail punctured interval lying inside the current completed interval. Thus the right-tail multi-block problem has been reduced from gap classification to phase saving for differences of nested P-punctured prime intervals:

```
RightTailPuncturedIntervalDifferencePhaseSaving
AND SingleBlockEndpointPacketSummationByParts
AND MovingPrimeQDenominatorCompletedTraceFamilyOnPuncturedIntervalsAndSingleBlockPackets
AND NoLossAggregationAcross5106ShellStepPackets
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still only a support-completion theorem. It does not prove the right-tail punctured-interval phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-right-tail-interval-completion-audit.md.
```

8.78 Endpoint-collar flux decomposition of the right-tail shells

Theorem 8.196 (Right-tail shell packets are endpoint-collar fluxes)**.** *In the audit range $P \leq 1009$, every right-tail shell-step packet is exactly one of the following endpoint flux objects, with the optional row prime P puncture removed:*

left collar, right collar, left collar ∪ right collar, terminal full interval.

The endpoint-collar ledger has

class	*packets*	*edge mass*
left_collar_only_no_P_puncture	314	8652
left_collar_only_with_P_puncture	19	776
right_collar_only_no_P_puncture	604	12879
right_collar_only_with_P_puncture	13	708
terminal_full_interval_no_P_puncture	75	2487
terminal_full_interval_with_P_puncture	75	4864
two_sided_left_and_right_collars_no_P_puncture	968	52957
two_sided_left_and_right_collars_with_P_puncture	39	3428.

Thus all 2107 *right-tail packets, carrying* 86751 *edges, satisfy the endpoint-collar identity. The audit finds zero bad endpoint-collar packets.*

Proof. For a fixed row P, let $F_i \supseteq F_{i+1}$ be adjacent nested right-tail fibres. By the preceding section each F_i is a prime interval with at most the row prime P removed. Complete F_i to its ambient prime interval I_i. If $F_{i+1} \neq \varnothing$, the shell $F_i \setminus F_{i+1}$ is exactly the union of the primes of I_i to the left of F_{i+1} and the primes of I_i to the right of F_{i+1}, again with P removed if present. If $F_{i+1} = \varnothing$, the same formula gives the terminal full interval. The machine audit reconstructs all right-tail packets from the shell-step ledger and verifies this equality packet by packet. □

Remark 8.197 (Narrowed analytic mouth)**.** The punctured-interval difference is no longer an arbitrary interval difference. It is an endpoint-collar flux problem. The latest remaining analytic mouth is

```
RightTailEndpointCollarFluxPhaseSaving
AND SingleBlockEndpointPacketSummationByParts
AND MovingPrimeQDenominatorCompletedTraceFamilyOnEndpointCollarsAndSingleBlockPackets
AND NoLossAggregationAcross5106ShellStepPackets
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still only a support decomposition. It does not prove endpoint collar phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-right-tail-endpoint-collar-audit.md.
```

8.79 Boundary endpoint-flux unification

Theorem 8.198 (All boundary shell-step packets are endpoint-flux packets)**.** *In the audit range $P \le 1009$, every boundary shell-step packet belongs to one unified endpoint-flux family. The lower and upper wings are single contiguous endpoint prime shells, while the right tail is the endpoint-collar family of the preceding section. The unified ledger has*

family	*packets*	*edge mass*
lower_wing	1032	29144
upper_wing	1967	61620
right_tail	2107	86751.

Thus all 5106 *boundary packets, carrying* 177515 *edges, are accounted for by endpoint flux. The audit finds zero bad endpoint-flux packets. The actual shell prime count has maximum* 12*; after restoring optional right-tail P-punctures, the completed endpoint-flux support has maximum* 13*.*

Proof. The shell-step packet ledger partitions the boundary into lower wing, upper wing, and right tail. The lower and upper wing packets all have one source rectangle, one contiguous prime shell, no internal prime gap, and no diagonal P-puncture. The preceding endpoint-collar theorem accounts for all right-tail packets. Combining these two verified cases gives a disjoint classification of every shell-step packet in the boundary ledger. □

Remark 8.199 (Unified endpoint-flux mouth)**.** The former separation between single-block endpoint packets and right-tail endpoint collars is now only a bookkeeping separation. The support problem has a single endpoint-flux interface:

```
BoundaryEndpointFluxPhaseSaving
AND MovingPrimeQDenominatorCompletedTraceFamilyOnBoundaryEndpointFluxPackets
AND NoLossAggregationAcross5106EndpointFluxPackets
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This does not prove endpoint-flux phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-unification-audit.md.
```

8.80 Q-prefix line atomization of endpoint flux

Theorem 8.200 (Endpoint flux splits into fixed-m prime-prefix line atoms)**.** *In the audit range $P \le 1009$, the 5106 boundary endpoint-flux packets split without overlap into 15439 fixed-m q-prefix line atoms of the form*

$$\{m\} \times [q_{\text{start}}, q_{\text{end}}]_{\mathbb{P}}.$$

The expanded atom edge set has size 177515, equal to the full boundary edge mass, and the audit finds no duplicate atom edge. The strip-level atom ledger is

strip	*atoms*	*edge mass*
`lower_wing`	2466	29144
`right_tail`	6962	86751
`upper_wing`	6011	61620.

Every q-prefix is a contiguous interval of primes. The packet support size has median 3 and maximum 12; the q-prefix length has median 11 and maximum 37.

Proof. For each endpoint-flux packet, expand the actual endpoint prime shell into its individual primes m. For each such m, pair it with the prime prefix from q_{start} to q_{end}. The audit verifies that the number of primes in this q-interval is exactly the packet q-prefix count. It then expands all atoms to triples (P, q, m) and checks that the resulting set has cardinality 177515, with no repeated triple. This equals the inherited endpoint-flux edge mass, so the atomization is lossless at the support level. □

Remark 8.201 (Fixed-m reciprocal-orbit mouth)**.** The endpoint-flux phase problem has been reduced to fixed-m prime-q prefix line atoms:

```
QPrefixLineAtomReciprocalOrbitPhaseSaving
AND MovingPrimeQDenominatorCompletedTraceFamilyOnFixedMAtoms
AND NoLossAggregationAcross15439QPrefixLineAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This remains a support atomization theorem only. It does not prove q-prefix reciprocal-orbit phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-atom-audit.md.
```

8.81 Phase normal form for q-prefix line atoms

Theorem 8.202 (Fixed-m q-prefix phase normal form)**.** *In the audit range $P \le 1009$, every edge (P, q, m) in the fixed-m q-prefix atom ledger has a unique division form*

$$qm = kP + D, \qquad 1 \le k < P, \qquad 1 \le D < P.$$

If $A(q)$ denotes the least positive residue of $-D$ modulo q, then

$$e\left(\frac{hkP}{q}\right) = e\left(-\frac{hD}{q}\right) = e\left(\frac{hA(q)}{q}\right).$$

The finite audit verifies this normal form for all 15439 *atoms and all* 177515 *atom edges:*

quantity	*value*
product_division_mismatch_count	0
phase_congruence_mismatch_count	0
k_out_of_strict_row_range_count	0
D_out_of_range_count	0
A_zero_count	0
k_nonincreasing_step_count	0.

Moreover every multi-q atom has strictly increasing $k(q) = \lfloor qm/P \rfloor$ along its prime-q prefix.

Proof. The atom ledger supplies P, a fixed prime m, and a contiguous prime-q interval. For every prime q in that interval, Euclidean division of qm by P gives $qm = kP + D$ with $0 \leq D < P$. The audit verifies that $D \neq 0$ and $1 \leq k < P$ for every edge. Since

$$\frac{kP}{q} = m - \frac{D}{q},$$

the displayed phase identity follows because $e(hm) = 1$. Finally, for fixed m, the function $q \mapsto \lfloor qm/P \rfloor$ is nondecreasing; the audit checks the stronger strict increase on every non-singleton prime-q prefix in the boundary atom ledger. □

Remark 8.203 (Moving numerator, not a completed Kloosterman input)**.** The phase normal form removes the previous ambiguity in the fixed-m reciprocal-orbit mouth, but it also identifies the next obstruction. The audit finds

orbit class	atoms	edge mass
`singleton_q`	1162	1162
`strict_beatty_k_multiq`	14277	176353
`constant_normalized_numerator`	1164	1166
`moving_beatty_numerator`	14275	176349.

Thus almost all atom mass is carried by a moving Beatty numerator $A(q)$, not by a fixed-numerator completed Kloosterman family. The latest remaining analytic mouth is

```
MovingBeattyNumeratorPrimeQPrefixReciprocalPhaseSaving
AND CompletionOfA(q)/qToExternalTraceOrKloostermanFamily
AND NoLossAggregationAcross15439QPrefixPhaseAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the deterministic phase normal form. It does not prove q-prefix phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-phase-normal-form-audit.md.
```

8.82 Successor carry dynamics on q-prefix phase atoms

Theorem 8.204 (Prime-q successor carry identity)**.** *In the audit range $P \leq 1009$, let $q < q'$ be adjacent primes inside the same fixed-m q-prefix phase atom. Write*

$$qm = kP + D, \qquad q'm = k'P + D',$$

with $1 \leq D, D' < P$, and put $g = q' - q$. Then every one of the 162076 adjacent transitions satisfies

$$D' = D + mg - Pc, \qquad c = k' - k = \left\lfloor \frac{D + mg}{P} \right\rfloor.$$

The audit verifies

quantity	*value*
successor_transition_count_total	162076
carry_formula_mismatch_count	0
D_successor_mismatch_count	0
A_successor_mismatch_count	0
q_gap_nonpositive_count	0
carry_nonpositive_count	0.

The observed prime gap range is $2 \leq g \leq 20$, with median 6, and the observed carry range is $1 \leq c \leq 33$, with median 6.

Proof. Subtract the two division identities:

$$q'm - qm = (k' - k)P + (D' - D).$$

Since $q'm = qm + mg$, this gives $D' = D + mg - P(k' - k)$. Because

$$q'm = kP + D + mg,$$

Euclidean division by P gives

$$k' - k = \left\lfloor \frac{D + mg}{P} \right\rfloor.$$

The audit checks these two formulas for every adjacent prime pair in every fixed-m q-prefix phase atom, and also recomputes the next normalized numerator $A(q')$ from D'. □

Remark 8.205 (Carry words are still not phase savings)**.** The successor identity removes another structural ambiguity, but it does not make the reciprocal orbit a constant-step rotation. Among the 15439 atoms, the audit finds

class	atoms
`variable_carry_word`	13317
`constant_carry_word`	960
`singleton_q_no_transition`	1162
`A_mixed_sawtooth`	12895
`A_constant`	2.

Thus the latest remaining analytic mouth is

```
PrimeGapDrivenCarryWordExponentialSumSaving
AND CompletionOfSuccessorCarryDynamicsToTraceOrKloostermanFamily
AND NoLossAggregationAcross15439QPrefixCarryAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only deterministic successor carry dynamics. It does not prove carry-word exponential-sum saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-dynamics-audit.md.
```

8.83 Finite carry letters and run fragmentation on q-prefix atoms

Theorem 8.206 (Finite carry-letter and run ledger)**.** *In the same audit range $P \le 1009$, every adjacent prime-q transition inside a fixed-m q-prefix carry atom is encoded by the raw carry letter*

$$L(q, q') = (q' - q,\ k' - k)$$

and by the signed carry letter

$$L^+(q, q') = (q' - q,\ k' - k,\ \mathrm{sgn}(A(q') - A(q))).$$

The letter/run audit verifies that the 162076 transitions are recovered without loss from their raw and signed run decompositions:

quantity	*value*
raw_letter_run_length_sum	162076
signed_letter_run_length_sum	162076
letter_run_decomposition_closed	*true*
raw_carry_letter_alphabet_count/capacity	117/297
signed_carry_letter_alphabet_count/capacity	256/891
raw_constant/variable_letter_atom_count	946/13331
signed_constant/variable_letter_atom_count	934/13343.

The raw and signed run lengths both have minimum 1, median 1, and maximum 3.

Proof. The previous successor carry identity supplies, for each adjacent pair $q < q'$,

$$q' - q = g, \qquad k' - k = c, \qquad A(q') - A(q).$$

Thus L and L^+ are defined for every transition. The audit scans each fixed-m atom in order, compresses consecutive identical letters into maximal runs, and checks that the sum of the run lengths is exactly the previous successor-transition count for both raw and signed alphabets. It also recomputes the alphabet capacities from the observed 9 prime-gap values, 33 carry values, and 3 signs. □

Remark 8.207 (Finite alphabet is not yet cancellation)**.** The finite alphabet removes another layer of structural ambiguity, but it does not itself provide cancellation. The audit finds raw and signed switch ratios

$$0.962097172511316, \qquad 0.9772664226415605,$$

and no raw or signed run longer than 3. Hence the latest remaining mouth is

```
FiniteCarryLetterWordExponentialSumSaving
AND PrimeGapCarrySwitchingLawOrTraceKloostermanCompletion
AND NoLossAggregationAcross15439QPrefixCarryLetterAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the finite carry-letter and run ledger. It does not prove finite-letter exponential-sum saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-letter-run-audit.md.
```

8.84 Directed switch graph of q-prefix carry letters

Theorem 8.208 (Finite carry-letter switch graph)**.** *In the same audit range $P \le 1009$, consider consecutive carry letters inside each fixed-m q-prefix carry atom:*

$$L_i = (g_i, c_i), \qquad L_{i+1} = (g_{i+1}, c_{i+1}),$$

and similarly the signed letters L_i^+. The switch-graph audit encodes all consecutive pairs as directed edges $L_i \to L_{i+1}$ and $L_i^+ \to L_{i+1}^+$. It verifies

quantity	*value*
adjacent_letter_pair_count_inside_atoms	147799
raw_switch_pair_count_total	147799
signed_switch_pair_count_total	147799
switch_graph_decomposition_closed	*true*
raw_graph_nodes/edges	117/1161
signed_graph_nodes/edges	256/4017
raw_outdegree_min/median/max	0/8/54
signed_outdegree_min/median/max	0/10/100.

The largest weak and strong components have sizes 117/115 *in the raw graph and* 255/250 *in the signed graph.*

Proof. The finite carry-letter ledger supplies the ordered word of raw and signed letters in each atom. The audit forms every adjacent pair in those words, counts it as a directed switch edge, and checks that the total number of switch pairs equals

$$162076 - 14277 = 147799,$$

namely the number of adjacent pairs among the 162076 transitions after one initial transition is removed from each of the 14277 multi-q atoms. The graph statistics are then computed from the observed directed edge set. □

Remark 8.209 (High branching remains the analytic obstruction)**.** The switch graph is finite, but it is not a low-branch deterministic automaton. The audit finds raw and signed maximum outdegrees 54 and 100, together with changed-letter and changed-sign ratios

$$0.962097172511316, \qquad 0.6169662852928639.$$

Thus the latest remaining mouth is

```
FiniteSwitchGraphPathExponentialSumSaving
AND TraceKloostermanCompletionOfHighBranchCarrySwitchGraph
AND NoLossAggregationAcross15439QPrefixSwitchAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the finite switch-graph ledger. It does not prove a deterministic switching law, switch-graph phase saving, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-switch-graph-audit.md.
```

8.85 Loop-erased cycle flow of q-prefix carry switch paths

Theorem 8.210 (Cycle-flow core and endpoint residual paths). *In the same audit range $P \leq 1009$, apply deterministic loop erasure to the raw and signed switch paths of every fixed-m q-prefix atom: whenever the active path revisits a carry letter, peel the resulting directed cycle; the remaining active stack is the endpoint residual path. The audit verifies*

quantity	*value*
adjacent_letter_pair_count_inside_atoms	147799
loop_erased_flow_decomposition_closed	*true*
raw_cycle_edge_mass	117733
raw_residual_edge_mass	30066
raw_cycle_edge_ratio	0.7965750783158209
signed_cycle_edge_mass	107677
signed_residual_edge_mass	40122
signed_cycle_edge_ratio	0.7285367289359197.

The raw residual path length has minimum/median/maximum $0/2/9$, *and the signed residual path length has minimum/median/maximum* $0/3/15$.

Proof. For a switch path $v_0, \ldots, v_n$, scan from left to right while keeping the current loop-erased stack. If the next vertex v_i has already appeared in the stack at position j, the segment from j to the top of the stack together with the new edge forms a directed cycle of length $\ell = \#\text{stack} - j$; remove the vertices after j. If v_i is new, append it to the stack. Every input edge is assigned exactly once, either to one peeled cycle or to the final residual path. The audit performs this decomposition for all raw and signed atom paths and checks that cycle mass plus residual mass is 147799 in both cases. □

Remark 8.211 (Cycle core is not yet cycle cancellation)**.** The loop-erased ledger shows that most switch-edge mass lies in a recurrent cycle core, but it does not estimate the phases carried by those cycles. The remaining analytic mouth is

```
CyclePacketPhaseSavingForLoopErasedCarrySwitchCore
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND TraceKloostermanCompletionOfCycleAndResidualPackets
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the loop-erased cycle/residual structure. It does not prove cycle-packet phase saving, residual endpoint summation, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-switch-flow-decomposition-audit.md.
```

8.86 Finite signatures of q-prefix carry cycles

Theorem 8.212 (Cycle core signature ledger)**.** *In the same audit range $P \le 1009$, take every directed cycle packet peeled from the loop-erased switch paths and quotient only by cyclic rotation of its starting point, not by reversal. Then attach the length, the gap/carry load, and, in the signed case, the A-step sign profile. The audit verifies*

quantity	*value*
cycle_signature_decomposition_closed	*true*
raw_cycle_packet_count	45178
raw_cycle_edge_mass	117733
raw_cycle_signature_count	3546
raw_cycle_shape_count	2718
raw_template_top20_edge_ratio	0.13926426745262585
raw_template_top100_edge_ratio	0.34888264123058105
signed_cycle_packet_count	36330
signed_cycle_edge_mass	107677
signed_cycle_signature_count	8691
signed_cycle_shape_count	5905
signed_A_sign_variable_cycle_count	26038
signed_template_top20_edge_ratio	0.06297538007188165
signed_template_top100_edge_ratio	0.18002916128792593.

Thus the recurrent cycle core is now a finite directed-signature ledger, but its mass is not concentrated in a few trivial templates.

Proof. For each peeled directed cycle $v_0, \dots, v_{\ell-1}, v_0$, serialize the raw letters $v_i = (q_{i+1}-q_i, k_{i+1}-k_i)$ and the signed letters $v_i^+ = (q_{i+1} - q_i, k_{i+1} - k_i, \operatorname{sgn}(A_{i+1} - A_i))$. Choose the lexicographically least cyclic rotation as the canonical directed template. Separately record

$$\ell, \quad \sum_i (q_{i+1} - q_i), \quad \sum_i (k_{i+1} - k_i),$$

the minimum and maximum of these two coordinates, and the signed A-profile in the signed ledger. The machine ledger replays the same loop-erasure decomposition as the previous theorem, checks that the raw and signed cycle edge masses remain 117733 and 107677, and verifies zero non-closed edge-delta errors. □

Remark 8.213 (Signature ledger is not weighted cancellation)**.** This step removes one more structural ambiguity: cycle packets are no longer unclassified loops in a high-branch graph. They are finite signature buckets. However, the largest raw template carries only about 1.61% of raw cycle edge mass, the raw top 20 templates carry about 13.93%, and the signed top 20 carry about 6.30%. Therefore the remaining mouth is not a single-template cancellation problem but

```
CycleSignatureWeightedPhaseSavingForLoopErasedCarrySwitchCore
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND TraceKloostermanCompletionOfCycleSignatureAndResidualPackets
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the finite cycle-signature ledger. It does not prove weighted phase saving, residual endpoint summation, the external-lemma version, or the internal self-contained version.

The machine certificate is

`docs/monograph/prime-matrix-phi-lpf-q-support-row-averaged-additive-k-prime-survivor-boundary-endpoint-flux-qprefix-carry-cycle-signature-audit.md`.

8.87 Weight carriers of q-prefix carry cycle signatures

Theorem 8.214 (Cycle-signature weight-carrier ledger)**.** *In the same audit range $P \leq 1009$, attach to every raw and signed cycle signature its P-support, strip-support, endpoint-flux class, and signed A-class. Also project each signed template to its raw-base template and record the signed-child fragmentation. The audit verifies*

quantity	*value*
cycle_signature_weight_carrier_ledger_closed	*true*
raw_cycle_signature_count	3546
raw_multi_P_edge_ratio	0.9341560989697026
raw_multi_strip_edge_ratio	0.04405731613056662
signed_cycle_signature_count	8691
signed_multi_P_edge_ratio	0.7385606954131337
signed_multi_strip_edge_ratio	0.014441338447393594
signed_P_support_width_min/median/max	1/1/95
raw_base_signed_refinement_count	5359
raw_base_with_multiple_signed_children_count	1823
raw_base_fragmented_signed_edge_ratio	0.7569583105027071
mixed_positive_negative_signed_edge_ratio	0.8184106169376932.

Thus the weighted cycle-signature problem is not a single undifferentiated bucket problem: it splits into signed carriers, raw-base to signed-child fragmentation, and thin P-support carriers.

Proof. Replay the previous cycle-signature ledger. For each raw template, count its cycle multiplicity and edge mass and store the set of primes P, the strip profile, and the endpoint-flux profile of the atoms in which it occurs. For each signed template, do the same, and additionally classify the A-step sign word as all-positive, all-negative, mixed positive/negative, or mixed with zero. Finally erase the sign coordinate from each signed template to obtain its raw-base template and record the edge mass carried by each signed child. The ledger checks that the raw and signed cycle counts and edge masses remain exactly those of the previous signature theorem. ☐

Remark 8.215 (Carrier ledger is not phase saving)**.** The signed cycle-edge mass is dominated by mixed positive/negative A-class carriers, about 81.84%. The raw-base to signed-child fragmentation carries about 75.70% of signed edge mass. At the same time the signed carrier P-support width has median 1, so a no-loss argument must also control thin P-support carriers. The remaining mouth is therefore

```
SignedCycleSignatureCarrierWeightedPhaseSaving
AND RawBaseToSignedChildWeightReconciliation
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND TraceKloostermanCompletionOfCycleSignatureAndResidualPackets
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the combinatorial weight-carrier ledger. It does not prove weighted phase saving, thin-support summation, residual endpoint summation, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-weight-carrier-audit.md.
```

8.88 Signed-child mirror obstruction inside cycle-signature carriers

Theorem 8.216 (Signed-child mirror reconciliation ledger)**.** *In the same audit range $P \leq 1009$, fix each raw-base cycle template and pair its signed children by the involution that exchanges A-step signs* `positive` $\leftrightarrow$ `negative` *while leaving* `zero` *fixed. The audit verifies*

quantity	*value*
`signed_child_mirror_reconciliation_ledger_closed`	`true`
`signed_cycle_edge_mass`	107677
`raw_base_signed_refinement_count`	5359
`raw_base_with_multiple_signed_children_count`	1823
`raw_base_multi_child_signed_edge_ratio`	0.7569583105027071
`mirror_balanced_edge_mass`	31334
`mirror_balanced_edge_ratio`	0.2909999349907594
`mirror_imbalance_edge_mass`	76343
`mirror_imbalance_edge_ratio`	0.7090000650092406
`missing_mirror_edge_ratio`	0.5848974247053689
`raw_base_exact_mirror_balance_count`	25
`raw_base_exact_mirror_balance_ratio`	0.004665049449524164‌5.

Thus sign-mirror pairing is a finite, auditable structure, but it is not a large enough exact pairing to reconcile the signed-child weights.

Proof. For a signed child template

$$(g_1, c_1, \epsilon_1), \ldots, (g_\ell, c_\ell, \epsilon_\ell),$$

replace each ϵ_i by its mirror sign and reduce again to the canonical directed rotation. Inside each raw-base template, compare the edge mass of a signed child with the edge mass of its mirror. The balanced contribution is twice the smaller mass in each nontrivial pair, while the absolute difference and any missing mirror mass are assigned to mirror imbalance. The machine ledger checks that balanced plus imbalanced mass is exactly the signed cycle-edge mass 107677. □

Remark 8.217 (Mirror pairing does not break parity)**.** Only about 29.10% of signed cycle-edge mass is balanced by the sign mirror; about 70.90% remains as mirror imbalance, and only 25 of 5359 raw-base templates have exact mirror balance. The remaining mouth is therefore

```
SignedChildMirrorImbalancePhaseSaving
AND NonMirrorSignedChildCarrierControl
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND TraceKloostermanCompletionOfMirrorImbalanceAndResidualPackets
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the mirror-pairing ledger. It does not prove mirror-imbalance phase saving, thin-support summation, residual endpoint summation, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-signed-child-mirror-
reconciliation-audit.md.
```

8.89 Mirror-imbalance support ledger

Theorem 8.218 (Mirror-imbalance support decomposition)**.** *In the same audit range $P \le 1009$, the mirror-imbalance mass left by the signed-child mirror ledger admits a deterministic support decomposition. The audit verifies*

quantity	*value*
mirror_imbalance_support_ledger_closed	*true*
signed_cycle_edge_mass	107677
mirror_balanced_edge_mass_imported	31334
mirror_imbalance_edge_mass	76343
missing_mirror_edge_mass	62980
missing_mirror_within_imbalance_ratio	0.824961031135795
unequal_mirror_pair_residual_edge_mass	13363
unequal_mirror_pair_within_imbalance_ratio	0.17503896886420497
residual_carrier_count	7795
multi_P_residual_edge_ratio	0.6484419003706954
P_support_width_min/median/max	1/1/94
multi_strip_residual_edge_ratio	0.006326709717983312.

Thus the residual signed-child mirror obstruction is no longer an anonymous imbalance: it is the union of missing-mirror carriers and unequal mirror-pair residual carriers.

Proof. For each non-self-mirror signed child, compare its edge mass with the edge mass of its A-sign mirror. If the mirror child is absent, the whole child mass is assigned to the missing-mirror class. If both children are present, cancel twice the smaller mass and assign the absolute difference to the unequal-pair residual class. The audit also records the P-support, strip support, endpoint-flux profile, A-class and raw-base parent of the dominant residual child. The ledger identity is

$$62980 + 13363 = 76343,$$

and $76343 + 31334 = 107677$, matching the imported signed cycle-edge mass. □

Remark 8.219 (New actual mouth)**.** The preceding theorem closes only the finite support ledger. It replaces the coarser obstruction

`SignedChildMirrorImbalancePhaseSaving`

by the more explicit pair

```
MissingMirrorCarrierPhaseSaving
AND UnequalMirrorPairResidualPhaseSaving.
```

The remaining mouth is

```
MissingMirrorCarrierPhaseSaving
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND TraceKloostermanCompletionOfMirrorImbalanceAndResidualPackets
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This does not prove phase saving for either class, does not complete the packets to a trace/Kloosterman family, and does not close the external-lemma or internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-mirror-imbalance-
support-audit.md.
```

8.90 Missing-mirror endpoint support ledger

Theorem 8.220 (Missing-mirror structure decomposition)**.** *In the audit range $P \leq 1009$, restrict the previous mirror-imbalance support ledger to the missing-mirror class. The audit verifies*

quantity	*value*
missing_mirror_structure_ledger_closed	*true*
missing_mirror_edge_mass	62980
missing_mirror_pair_count	7156
missing_raw_base_count	5114
single_P_missing_edge_ratio	0.4254842807240394
single_strip_missing_edge_ratio	0.9995554144172754
wing_single_shell_missing_edge_mass	30538
wing_single_shell_missing_edge_ratio	0.48488409018736106
right_tail_missing_edge_mass	32442
right_tail_missing_edge_ratio	0.5151159098126389
mixed_positive_negative A-class ratio	0.9735947919974595.

Thus the dominant missing-mirror obstruction is almost entirely a one-strip endpoint carrier family, split between wing single shells and right-tail collar/terminal shells.

Proof. For each signed child whose A-sign mirror has zero global edge mass, the audit records its strip support, endpoint-flux class, A-sign class, cycle length, P-support width and raw-base parent. Summing these records over all missing signed children recovers exactly the imported missing-mirror mass 62980. The strip, endpoint, sign, length and P-width row sums are each checked against the same total, giving the displayed structure ledger. □

Remark 8.221 (Endpoint support is not phase saving)**.** The preceding decomposition removes the anonymous missing-mirror black box but does not prove cancellation. The missing-mirror branch is now

```
MissingMirrorEndpointCarrierPhaseSaving
AND MissingMirrorTraceKloostermanCompletion.
```

The full remaining mouth is

```
MissingMirrorEndpointCarrierPhaseSaving
AND MissingMirrorTraceKloostermanCompletion
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

Known external trace/Kloosterman inputs can enter only after the endpoint carriers are completed to admissible trace families. This theorem does not close that completion, the external-lemma version, or the internal self-contained version.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-missing-mirror-
structure-audit.md.
```

8.91 Missing-mirror endpoint router

Theorem 8.222 (Pure and mixed endpoint routing). *In the audit range $P \le 1009$, the missing-mirror endpoint carriers split at the signed-child template level into pure endpoint templates and mixed endpoint templates. The audit verifies*

quantity	*value*
missing_mirror_endpoint_router_ledger_closed	*true*
missing_mirror_edge_mass	62980
missing_mirror_template_count	7156
pure_endpoint_template_edge_mass	48636
pure_endpoint_template_edge_ratio	0.7722451571927597
mixed_endpoint_template_edge_mass	14344
mixed_endpoint_template_edge_ratio	0.2277548428072404
mixed_right_tail_endpoint_edge_mass	14316
mixed_right_tail_endpoint_edge_ratio	0.22731025722451573
mixed_wing_tail_endpoint_edge_mass	28
single_strip_pure_endpoint_edge_mass	48636.

Thus most of the missing-mirror mass is already pure endpoint support, while the non-pure part is almost entirely a right-tail internal endpoint mixture.

Proof. For every missing-mirror signed child, the audit collects the endpoint groups that occur in its supporting cycle packets. A template with exactly one endpoint group is routed as pure endpoint support. A template with several right-tail endpoint groups is routed to the mixed right-tail packet; the audit separately records the exceptional wing/tail mixed mass. The displayed masses sum to $48636 + 14344 = 62980$, and the mixed part further splits as $14316 + 28 = 14344$. □

Remark 8.223 (Router closed, analytic gates open). The endpoint router turns the missing-mirror branch into

```
PureEndpointMissingMirrorCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND MissingMirrorTraceKloostermanCompletion.
```

The full remaining mouth is

```
PureEndpointMissingMirrorCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND MissingMirrorTraceKloostermanCompletion
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the finite endpoint-router ledger. It does not prove pure endpoint phase saving, does not split mixed right-tail templates without loss at the analytic level, and does not complete any carrier to an admissible trace/Kloosterman family.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-missing-mirror-
endpoint-router-audit.md.
```

8.92 Pure endpoint phase interface

Theorem 8.224 (Pure endpoint phase-interface ledger)**.** *In the audit range $P \leq 1009$, the pure endpoint part of the missing-mirror router splits into local single-P packets and multi-P trace candidates. The audit verifies*

quantity	*value*
pure_endpoint_phase_interface_ledger_closed	*true*
pure_endpoint_edge_mass	48636
pure_endpoint_template_count	6408
single_P_pure_endpoint_edge_mass	26753
single_P_pure_endpoint_edge_ratio	0.5500657948844477
multi_P_pure_endpoint_edge_mass	21883
multi_P_pure_endpoint_edge_ratio	0.44993420511555227
multi_P_wing_pure_endpoint_edge_mass	16982
multi_P_right_tail_pure_endpoint_edge_mass	4901
P_support_width_min/median/max	1/1.0/54.

Thus the pure endpoint mass is not yet a single analytic object: it has a local single-P part and a multi-P part which would still need explicit trace/Kloosterman completion.

Proof. For each pure endpoint signed child, the audit counts the distinct values of P supporting the template. Width 1 templates are routed to the local single-P packet class. Templates with wider P-support are routed by endpoint geometry into wing and right-tail trace-candidate classes. The edge mass identity

$$26753 + 21883 = 48636$$

is checked, and the multi-P mass further splits as $16982 + 4901 = 21883$. □

Remark 8.225 (External theorem entry point)**.** This ledger shows where external trace inputs may enter, and where they cannot. FKMS-type trace bilinear estimates, arbitrary-modulus Kloosterman estimates, or unbalanced Kloosterman-fraction estimates can only be invoked after the multi-P candidates have been completed to admissible variables. The withdrawn Dong–Robles–Zeindler Kloosterman-fraction near-miss is recorded as non-admissible external input. These inputs do not control the single-P local packets by themselves.

The active mouth is

```
SinglePLocalPureEndpointPacketBound
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
```

```
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This theorem closes only the finite phase-interface ledger and does not prove any of the displayed analytic gates.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-pure-endpoint-
phase-interface-audit.md.
```

8.93 Single-P local endpoint packets

Theorem 8.226 (Single-P local endpoint packet ledger)**.** *In the audit range $P \leq 1009$, the single-P branch inside the pure endpoint support has the following finite structure:*

quantity	*value*
single_P_local_endpoint_structure_ledger_closed	*true*
single_P_local_endpoint_edge_mass	26753
single_P_local_endpoint_template_count	4915
single_P_support_prime_count	128
single_P_support_P_min/max	97/1009
single_P_edge_mass_per_P_min/median/max	2/167.5/827
single_P_template_edge_mass_min/median/max	1/5/16
wing_single_P_local	13540
right_tail_single_P_local	13213.

The width-one endpoint mass is therefore not a completed trace family. It is a local template problem plus a P-slice summation problem.

Proof. The audit reruns the signed-child construction and imports the pure endpoint phase-interface certificate. It keeps exactly those pure endpoint templates whose P-support has width 1. The resulting edge mass is

$$26753,$$

matching the imported single-P mass. The same records are then sorted by endpoint route, A-class, P-slice, q-prefix load, shell-prime load, and local template mass. The route-superclass identity

$$13540 + 13213 = 26753$$

is checked by the generated ledger. □

Remark 8.227 (Local obstruction)**.** This narrows the first pure endpoint gate to

```
LocalTemplateMultiplicityUniformBound
AND SinglePSliceEndpointPacketSummationOrPDEC.
```

The observed local template edge mass is at most 16 in the finite audit, but this is not yet a global uniform bound. FKMS-type trace bilinear estimates, arbitrary-modulus Kloosterman estimates, unbalanced Kloosterman estimates, and the Xu–Zhang generalized Kloosterman bounds over arbitrary sets require completed long variables or finite-field set variables; they do not by themselves control width-one P-support local packets.

The active mouth is now

```
LocalTemplateMultiplicityUniformBound
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
endpoint-packet-audit.md.
```

8.94 Single-P local template multiplicity

Theorem 8.228 (Single-P local template multiplicity ledger). *In the audit range $P \leq 1009$, each single-P local endpoint template has edge mass equal to*

$$\texttt{cycle\_length} \cdot \texttt{occurrence\_count}.$$

The generated ledger verifies

quantity	*value*
single_P_local_template_multiplicity_factor_ledger_closed	*true*
single_P_local_endpoint_edge_mass	26753
single_P_local_endpoint_template_count	4915
single_occurrence_template_count	4818
single_occurrence_edge_mass	25881
repeated_template_count	97
repeated_template_edge_mass	872
template_edge_mass_min/median/max	1/5/16
cycle_length_min/median/max	1/5/14
occurrence_count_min/median/max	1/1/4.

Proof. The audit imports the single-P local endpoint packet certificate and reruns the same signed-child construction. For every width-one pure endpoint template it records the cycle length and the number of times that exact signed template occurs. The identity

$$\texttt{edge\_mass} = \texttt{cycle\_length} \cdot \texttt{occurrence\_count}$$

is checked template by template. Summing these identities gives the imported total edge mass 26753. The occurrence rows show that 4818 of the 4915 templates occur once, while the repeated part has edge mass 872. □

Remark 8.229 (Remaining multiplicity gates). This reduces the local-template gate to

```
LocalCycleLengthUniformBound
AND LocalOccurrenceMultiplicityUniformBound
AND CycleOccurrenceProductBoundOrPDEC.
```

The finite audit observes `cycle_length` ≤ 14, `occurrence_count` ≤ 4, and `edge_mass` ≤ 16. These are not yet global uniform bounds. Average trace/Kloosterman inputs, including the FKMS, Milićević–Qin–Wu, Wright, and Xu–Zhang type inputs, do not directly prove this local signed-cycle multiplicity statement without a completed averaging variable.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
template-multiplicity-audit.md.
```

8.95 Repeated single-P local template occurrences

Theorem 8.230 (Single-P local template occurrence class ledger)**.** *In the audit range $P \leq 1009$, the repeated part of the single-P local endpoint templates admits the following finite classification:*

quantity	*value*
single_P_local_template_occurrence_class_ledger_closed	*true*
repeated_template_count	97
repeated_template_edge_mass	872
repeated_template_edge_ratio	0.03259447538593802
single_packet_multi_m_repeated_template_count	78
single_packet_multi_m_repeated_edge_mass	720
multi_packet_repeated_template_count	17
multi_packet_repeated_edge_mass	136
single_packet_single_m_multi_cycle_template_count	2
single_packet_single_m_multi_cycle_edge_mass	16
unclassified_repeated_template_count	0.

Moreover every repeated template observed in the audit has strip support equal to 1*, packet support at most* 2*, q-prefix support at most* 2*, shell-prime support at most* 2*, and m-value support at most* 4*.*

Proof. The audit imports the single-P local template multiplicity ledger and reconstructs every occurrence of every repeated signed template. Each repeated template is then classified by the support of its occurrence records. If its packet support has size > 1, it is a multi-packet repeated template. If its packet support has size 1 but its m-value support has size > 1, it is a single-packet multi-m repeated template. The remaining repeated records are checked for multiple cycles at the same packet and the same m-value. The three classes have masses

$$720, \qquad 136, \qquad 16,$$

and their sum is 872, the repeated edge mass imported from the previous factor ledger. The unclassified counter is 0, and the support counters listed in the statement are verified by the generated JSON ledger. $\square$

Remark 8.231 (Occurrence collision gates)**.** This replaces the occurrence-multiplicity gate by the more concrete list

```
SinglePacketMultiMCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC.
```

Together with the previous cycle-length and product gates, the active mouth is now

```
LocalCycleLengthUniformBound
AND SinglePacketMultiMCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still a finite classification result. It does not prove a global same-packet multi-m collision bound, a multi-packet duplicate transport bound, a same-m multi-cycle suppression theorem, or a repeated-occurrence PDEC/SAE return. Average trace/Kloosterman inputs and short-interval prime existence theorems do not by themselves prove these width-one local occurrence collision statements.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
template-occurrence-class-audit.md.
```

8.96 Same-packet multi-m gap classes

Theorem 8.232 (Single-P same-packet multi-m gap ledger)**.** *In the audit range $P \leq 1009$, the same-packet multi-m repeated templates have the following finite gap classification:*

quantity	*value*
single_P_local_same_packet_multi_m_gap_ledger_closed	*true*
same_packet_multi_m_template_count	78
same_packet_multi_m_edge_mass	720
adjacent_prime_pair_collision_template_count	71
adjacent_prime_pair_collision_edge_mass	662
nonadjacent_prime_pair_collision_template_count	6
nonadjacent_prime_pair_collision_edge_mass	50
adjacent_prime_chain_collision_template_count	1
adjacent_prime_chain_collision_edge_mass	8.

The maximal observed prime-index gap between selected m-values is 3*, and the maximal observed integer m-gap is* 18*. The two dominant integer gap vectors have edge masses*

$$[4] \mapsto 316, \qquad [2] \mapsto 296,$$

while the dominant prime-index gap vector is

$$[1] \mapsto 662.$$

Proof. The audit imports the repeated-template occurrence class ledger and restricts to the 78 templates classified there as same-packet multi-m. For each template it sorts the selected m-values, records the integer gap vector, and also records the gap vector in the ambient prime sequence. A selected support of two consecutive primes is placed in the adjacent-prime-pair class; a selected support of two nonconsecutive primes is placed in the nonadjacent-prime-pair class; and a selected support of more than two consecutive primes is placed in the adjacent-prime-chain class. The three class masses are

$$662, \qquad 50, \qquad 8,$$

and these sum to the imported same-packet multi-m mass 720. The ledger also checks that all records remain in a single packet and that the factor identity `edge_mass` = `cycle_length` · `occurrence_count` is preserved. ☐

Remark 8.233 (Adjacent-pair collision gate)**.** The same-packet multi-m gate is now replaced by

```
AdjacentPrimePairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND AdjacentPrimePairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still only a finite m-gap classification. It does not prove a global adjacent-prime-pair signed collision bound, a nonadjacent-pair bound, an adjacent-chain suppression theorem, or a repeated-occurrence PDEC/SAE return. Trace/Kloosterman average estimates and ordinary prime-gap information do not by themselves control equality of signed local templates across adjacent m-values inside one endpoint packet.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
same-packet-multi-m-gap-audit.md.
```

8.97 Adjacent-prime-pair exact gap classes

Theorem 8.234 (Single-P adjacent-prime-pair exact-gap ledger)**.** *In the audit range $P \le 1009$, the* 71 *same-packet adjacent-prime-pair collision templates of the previous ledger split by exact integer* *prime gap as follows:*

quantity	*value*
single_P_local_adjacent_prime_pair_gap_class_ledger_closed	*true*
adjacent_prime_pair_template_count	71
adjacent_prime_pair_edge_mass	662
gap2_twin_adjacent_pair_template_count	33
gap2_twin_adjacent_pair_edge_mass	296
gap4_cousin_adjacent_pair_template_count	31
gap4_cousin_adjacent_pair_edge_mass	316
gap6_sexy_adjacent_pair_template_count	5
gap6_sexy_adjacent_pair_edge_mass	36
gap_ge8_adjacent_pair_template_count	2
gap_ge8_adjacent_pair_edge_mass	14
bad_adjacent_pair_template_count	0.

The observed integer gaps run from 2 *to* 10*. Gap* 2 *and gap* 4 *together carry edge mass* 612*, namely*

$$\frac{612}{662} = 0.9244712990936556\ldots.$$

Proof. The audit imports the same-packet multi-m gap ledger and restricts to its adjacent-prime-pair class. For each repeated template it sorts the two selected prime m-values and records their exact integer difference. Difference 2 is assigned to the twin-prime adjacent-pair class, difference 4 to the cousin-prime adjacent-pair class, difference 6 to the gap-6 class, and all remaining adjacent prime gaps to the ≥ 8 class. The four class edge masses are

$$296, \qquad 316, \qquad 36, \qquad 14,$$

which sum to the imported adjacent-pair mass 662. The ledger also checks that every imported record is genuinely a two-prime adjacent pair and that no bad adjacent-pair template remains. □

Remark 8.235 (Exact-gap collision gates)**.** The adjacent-prime-pair gate is now replaced by

```
Gap2TwinAdjacentPairCollisionBound
AND Gap4CousinAdjacentPairCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND Gap2TwinAdjacentPairCollisionBound
AND Gap4CousinAdjacentPairCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This remains a finite exact-gap ledger. It does not prove a signed collision bound for gap 2, gap 4, gap 6, or larger adjacent prime gaps. Trace/Kloosterman average estimates, small-gap prime theorems, and short-interval prime existence inputs do not by themselves prove equality control for signed Phi-LPF local templates inside a fixed endpoint packet.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
adjacent-prime-pair-gap-class-audit.md.
```

8.98 Gap 2/gap 4 route-superclass carriers

Theorem 8.236 (Gap 2/gap 4 route-superclass ledger)**.** *In the audit range $P \leq 1009$, the 64 gap 2 and gap 4 adjacent-prime-pair templates of the previous ledger split by route superclass as follows:*

quantity	*value*
single_P_local_gap2_gap4_route_superclass_ledger_closed	*true*
gap2_gap4_template_count	64
gap2_gap4_edge_mass	612
gap2_wing_template_count	32
gap2_wing_edge_mass	290
gap2_right_tail_template_count	1
gap2_right_tail_edge_mass	6
gap4_right_tail_template_count	28
gap4_right_tail_edge_mass	288
gap4_wing_template_count	3
gap4_wing_edge_mass	28.

Thus the dominant aligned carriers are gap 2 on the wing side and gap 4 on the right-tail side:

$$290 + 288 = 578, \qquad \frac{578}{612} = 0.9444444444444444\ldots.$$

Within gap 2, the wing carrier has mass ratio

$$\frac{290}{296} = 0.9797297297297297\ldots,$$

and within gap 4, the right-tail carrier has mass ratio

$$\frac{288}{316} = 0.9113924050632911\ldots.$$

Proof. The audit imports the exact-gap adjacent-pair ledger and restricts to its gap 2 and gap 4 classes. Each selected template already carries a local route label, either a wing single-shell route or a right-tail collar route. Passing from the route label to the coarser route superclass partitions the gap 2/gap 4 mass into four disjoint cells:

$$\text{gap2-wing}, \quad \text{gap2-right-tail}, \quad \text{gap4-right-tail}, \quad \text{gap4-wing}.$$

The four edge masses are

$$290, \qquad 6, \qquad 288, \qquad 28,$$

which sum to the imported gap 2/gap 4 mass 612. The ledger also checks the inherited product identity

$$\texttt{edge\_mass} = \texttt{cycle\_length} \cdot \texttt{occurrence\_count}$$

for every selected template. □

Remark 8.237 (Route-superclass collision gates)**.** The two dominant exact-gap gates are now replaced by

```
Gap2WingTwinAdjacentPairCollisionBound
AND Gap2RightTailTwinResidualCollisionBound
AND Gap4RightTailCousinAdjacentPairCollisionBound
AND Gap4WingCousinResidualCollisionBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND Gap2WingTwinAdjacentPairCollisionBound
AND Gap2RightTailTwinResidualCollisionBound
AND Gap4RightTailCousinAdjacentPairCollisionBound
AND Gap4WingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This remains a finite route-superclass ledger. Trace/Kloosterman average inputs, prime-gap theorems, and short-interval prime existence estimates do not by themselves prove signed collision control for one fixed gap2-wing or gap4-right-tail endpoint carrier.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-route-superclass-audit.md.
```

8.99 Gap 2/gap 4 exact route-class carriers

Theorem 8.238 (Gap 2/gap 4 exact route-class ledger). *In the audit range $P \le 1009$, the* 64 *gap* 2 *and gap* 4 *route-superclass templates of the previous ledger split into exact route classes as follows:*

quantity	*value*
single_P_local_gap2_gap4_exact_route_class_ledger_closed	*true*
gap2_gap4_template_count	64
gap2_gap4_edge_mass	612
gap4_right_tail_two_sided_edge_mass	258
gap2_upper_wing_edge_mass	212
gap2_lower_wing_edge_mass	78
gap4_right_tail_left_collar_edge_mass	30
gap4_upper_wing_edge_mass	22
gap2_right_tail_two_sided_edge_mass	6
gap4_lower_wing_edge_mass	6.

The two largest exact routes are the gap 4 *two-sided right-tail collar and the gap* 2 *upper wing:*

$$258 + 212 = 470, \qquad \frac{470}{612} = 0.7679738562091504\ldots.$$

The residual exact-route mass outside these two largest cells is 142.

Proof. The audit imports the gap 2/gap 4 route-superclass ledger and refines each local route label into its exact subclass: upper wing, lower wing, two-sided right-tail collar, or left-collar right-tail. The seven nonempty cells are

$$\text{gap4 two-sided right-tail}, \quad \text{gap2 upper wing}, \quad \text{gap2 lower wing},$$
$$\text{gap4 left-collar right-tail}, \quad \text{gap4 upper wing}, \quad \text{gap2 two-sided right-tail}, \quad \text{gap4 lower wing}.$$

Their edge masses are

$$258, \qquad 212, \qquad 78, \qquad 30, \qquad 22, \qquad 6, \qquad 6,$$

which sum to the imported gap 2/gap 4 mass 612. The ledger also checks that the inherited gap 2 wing/right-tail and gap 4 right-tail/wing superclass masses are exactly recovered after recombination. □

Remark 8.239 (Exact route-class collision gates). The route-superclass gates are now replaced by

```
Gap2UpperWingTwinCollisionBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
AND Gap4RightTailTwoSidedCousinCollisionBound
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND Gap2UpperWingTwinCollisionBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
AND Gap4RightTailTwoSidedCousinCollisionBound
```

```
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This remains a finite exact-route ledger. Trace/Kloosterman average inputs, prime-gap theorems, and short-interval prime existence estimates still do not prove signed collision control for one fixed exact route inside an endpoint packet.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-exact-route-class-audit.md.
```

8.100 Gap 2/gap 4 top-two sign-cycle carriers

Theorem 8.240 (Top-two exact-route sign-cycle ledger)**.** *In the audit range $P \le 1009$, restrict the previous gap 2/gap 4 exact-route ledger to its two largest cells:*

$$\textit{gap4 two-sided right-tail}, \qquad \textit{gap2 upper wing}.$$

Then their 48 templates and edge mass 470 satisfy

quantity	*value*
top_two_exact_route_sign_cycle_ledger_closed	*true*
top_two_exact_route_template_count	48
top_two_exact_route_edge_mass	470
gap4_right_tail_two_sided_edge_mass	258
gap2_upper_wing_edge_mass	212
occurrence_count_gt2_edge_mass	0
all_positive_edge_mass	0
cycle_length_min	2
cycle_length_max	8
sign_switch_count_max	6.

Moreover cycle lengths $4, 5, 6$ carry mass

$$362, \qquad \frac{362}{470} = 0.7702127659574468\ldots,$$

sign-switch count at most 3 *carries mass*

$$332, \qquad \frac{332}{470} = 0.7063829787234043\ldots,$$

and their intersection carries mass 270.

Proof. The audit imports the exact-route ledger and filters the two exact route keys

$$(\text{gap4 cousin}, \text{two-sided right-tail}), \qquad (\text{gap2 twin}, \text{upper wing}).$$

For each selected template it extracts the signed child word, counts adjacent sign switches, records the cycle length, and checks the occurrence multiplicity. All 48 selected templates have occurrence count 2, all are mixed-positive-negative, and none are all-positive. The route masses are

$$258, \qquad 212,$$

which sum to 470, the top-two mass imported from the exact-route ledger. The cycle and sign-switch subtables then give the stated masses. □

Remark 8.241 (Top-two sign-cycle collision gates)**.** The two largest exact-route gates are now refined to

```
Gap4RightTailTwoSidedCousinSignCycleCollisionBound
AND Gap2UpperWingTwinSignCycleCollisionBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND Gap4RightTailTwoSidedCousinSignCycleCollisionBound
AND Gap2UpperWingTwinSignCycleCollisionBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is a finite sign-cycle ledger, not a global cancellation theorem. Current trace/Kloosterman, prime-gap, and short-interval prime inputs still do not estimate a fixed endpoint packet sign-cycle equality.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-sign-cycle-audit.md.
```

8.101 Gap 2/gap 4 top-two core route-cycle-switch atoms

Theorem 8.242 (Top-two core route-cycle-switch ledger). *In the audit range $P \le 1009$, restrict the previous top-two sign-cycle ledger to the core*

$$\texttt{cycle\_length} \in \{4,5,6\}, \qquad \texttt{sign\_switch\_count} \le 3.$$

Then this core has 29 templates and edge mass 270, and it splits into 11 route × cycle × switch atoms. The principal audit values are

quantity	*value*
top_two_core_route_cycle_switch_ledger_closed	*true*
core_template_count	29
core_edge_mass	270
noncore_top_two_edge_mass	200
route_cycle_switch_atom_count	11
largest_route_cycle_switch_atom_edge_mass	70
gap2_upper_wing_core_edge_mass	140
gap4_right_tail_two_sided_core_edge_mass	130
cycle5_core_edge_mass	150
cycle4_core_edge_mass	96
cycle6_core_edge_mass	24
sign_switch3_core_edge_mass	146
sign_switch1_core_edge_mass	64
sign_switch2_core_edge_mass	60.

The largest atom is the gap 4 two-sided right-tail route with cycle length 5 and sign-switch count 3; its edge mass is 70.

Proof. The audit imports the top-two exact-route sign-cycle ledger and keeps exactly those rows with cycle length 4, 5, or 6, and sign-switch count at most 3. The imported ledger gives the selected mass as the previously recorded intersection mass 270. Each selected row is then labelled by its exact route, cycle length, and sign-switch count. Summing the edge masses over these labels gives the 11 atoms listed in the machine certificate; their masses sum to 270. The complementary top-two mass is therefore

$$470 - 270 = 200.$$

All selected templates inherit occurrence count 2 and mixed-positive-negative sign type from the previous ledger. □

Remark 8.243 (Core atom collision gates). The top-two sign-cycle gates are now refined to

```
Gap4RightTailTwoSidedCousinCoreRouteCycleSwitchAtomBound
AND Gap2UpperWingTwinCoreRouteCycleSwitchAtomBound
AND TopTwoNonCoreSignCycleResidualBound.
```

Consequently the current mouth is

```
LocalCycleLengthUniformBound
AND Gap4RightTailTwoSidedCousinCoreRouteCycleSwitchAtomBound
AND Gap2UpperWingTwinCoreRouteCycleSwitchAtomBound
AND TopTwoNonCoreSignCycleResidualBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
```

```
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is a finite atom ledger, not a global cancellation theorem. Current trace/Kloosterman average inputs are useful only after a completion from these local route-cycle-switch atoms to a bilinear or trace-family object; prime-gap and short-interval prime theorems do not control the fixed endpoint signed atom equality.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-route-cycle-switch-audit.md.
```

8.102 Gap 2/gap 4 largest core atom template witnesses

Theorem 8.244 (Largest core atom template-witness ledger)**.** *In the audit range $P \le 1009$, restrict the previous core atom ledger to its largest atom:*

$$\mathtt{route} = \mathtt{gap4\_right\_tail\_two\_sided}, \quad \mathtt{cycle\_length} = 5, \quad \mathtt{sign\_switch\_count} = 3.$$

This atom has edge mass 70. It splits into 7 template witnesses, each with edge mass 10. The principal audit values are

quantity	*value*
largest_atom_template_witness_ledger_closed	true
witness_template_count	7
witness_edge_mass	70
all_witness_edge_mass_equals_10	true
distinct_sign_word_count	5
dominant_sign_word	-+-+
dominant_sign_word_edge_mass	30
q_prefix_band_q_le_10_edge_mass	30
q_prefix_band_q_le_20_edge_mass	10
q_prefix_band_q_gt_20_edge_mass	30
m_shell_band_m_le_4_edge_mass	30
m_shell_band_m_le_8_edge_mass	40.

Thus the largest core atom no longer hides a larger internal carrier: its dominant sign word has mass 30, and the remaining four sign words have mass 10 each.

Proof. The audit imports the 11-atom top-two core ledger and filters the unique largest route-cycle-switch atom. The imported mass of this atom is 70. The filtered rows are then recorded by P, packet index, sign word, q-prefix count, m-shell prime count, and the adjacent m-pair. The seven rows all have edge mass 10, so their total is 70. Counting sign words gives the distribution

$$30, \quad 10, \quad 10, \quad 10, \quad 10,$$

with dominant word `-+-+`. The q-prefix and m-shell subtables are the stated finite refinements. □

Remark 8.245 (Template-witness family gates)**.** The largest core atom gate is now replaced by

```
DominantLargestAtomSignWordFamilyBound(-+-+)
AND OtherLargestAtomTemplateWitnessFamilyBounds.
```

Consequently the current mouth is

```
DominantLargestAtomSignWordFamilyBound(-+-+)
AND OtherLargestAtomTemplateWitnessFamilyBounds
AND OtherCoreRouteCycleSwitchAtomBounds
AND TopTwoNonCoreSignCycleResidualBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This remains a finite witness ledger. The current trace/Kloosterman and short-interval inputs still do not prove a uniform bound for the dominant sign word family or for the seven witness families.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-template-witness-audit.md.
```

8.103 Gap 2/gap 4 dominant sign-word path witnesses

Theorem 8.246 (Dominant largest-atom sign-word path ledger)**.** *In the audit range $P \leq 1009$, restrict the largest core atom template witness ledger to its dominant sign word* `-+-+`*. This family has edge mass* 30 *and splits into* 3 *raw-base/signed-child path witnesses. The principal audit values* *are*

quantity	*value*
dominant_sign_word_path_ledger_closed	*true*
path_template_count	3
path_edge_mass	30
all_path_edge_mass_equals_10	*true*
all_path_occurrence_count_equals_2	*true*
all_path_integer_gap_equals_4	*true*
all_path_m_shell_band_m_le_4	*true*
all_path_sign_balance_plus2_minus3	*true*
distinct_raw_base_template_count	3
distinct_signed_child_count	3
distinct_m_pair_count	2
dominant_m_pair	*[769, 773]*
dominant_m_pair_edge_mass	20
q_prefix_band_q_le_10_edge_mass	20
q_prefix_band_q_gt_20_edge_mass	10.

Thus the dominant sign-word block no longer hides an internal mass larger than one path witness of mass 10*.*

Proof. The audit imports the 7-witness largest-atom ledger and keeps only those rows with sign word `-+-+`. The imported mass of this sign word is 30. The filtered rows occur at three P-values and have three distinct raw-base templates and three distinct signed-child paths. Each row has edge mass 10, occurrence count 2, integer gap 4, m-shell band `m<=4`, and sign balance `plus=2,minus=3`. The three edge masses sum to 30, and the m-pair/q-prefix subtables give the stated refinement. □

Remark 8.247 (Dominant path-family gate)**.** The dominant sign-word family gate is now replaced by

```
DominantLargestAtomPathWitnessUniformFamilyBound(-+-+).
```

Consequently the current mouth is

```
DominantLargestAtomPathWitnessUniformFamilyBound(-+-+)
AND OtherLargestAtomTemplateWitnessFamilyBounds
AND OtherCoreRouteCycleSwitchAtomBounds
AND TopTwoNonCoreSignCycleResidualBound
AND Gap2LowerWingTwinCollisionBound
AND Gap2RightTailTwoSidedTwinResidualCollisionBound
AND Gap4RightTailLeftCollarCousinCollisionBound
AND Gap4UpperWingCousinResidualCollisionBound
AND Gap4LowerWingCousinResidualCollisionBound
AND Gap6SexyAdjacentPairCollisionBound
AND GapGe8AdjacentPairCollisionBound
AND NonAdjacentPrimePairCollisionBound
AND AdjacentPrimeChainCollisionBound
AND MultiPacketDuplicateTransportBound
AND SinglePacketSingleMMultiCycleSuppression
AND RepeatedOccurrenceAggregationOrPDEC
AND CycleOccurrenceProductBoundOrPDEC
AND SinglePSliceEndpointPacketSummationOrPDEC
AND MultiPPureEndpointTraceKloostermanCompletion
AND PureEndpointCarrierPhaseSaving
AND MixedRightTailEndpointRouterNoLoss
AND UnequalMirrorPairResidualPhaseSaving
AND ThinPSupportCarrierSummationWithoutLoss
AND ResidualEndpointPathSummationWithoutBoundaryLoss
AND NoLossAggregationAcross15439QPrefixFlowAtoms
AND PrimeQSupportSetReciprocalPhaseSavingBeyondParity.
```

This is still a finite path ledger. Current trace/Kloosterman inputs require a completion from these local paths to a nonlocal family, and short-interval prime inputs still do not estimate this fixed endpoint signed path equality.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-dominant-sign-word-path-audit.md.
```

8.104 Dominant m-pair endpoint coordinate witnesses

Theorem 8.248 (Dominant sign-word m-pair coordinate ledger)**.** *In the audit range $P \leq 1009$, restrict the dominant sign-word path ledger to its largest m-pair `[769, 773]`. This subledger has edge mass 20 and splits into two endpoint-coordinate path witnesses:*

quantity	*value*
dominant_m_pair_endpoint_coordinate_path_ledger_closed	*true*
target_sign_word	*-+-+*
target_m_pair	*[769, 773]*
endpoint_coordinate_witness_count	2
endpoint_coordinate_edge_mass	20
all_endpoint_edge_mass_equals_10	*true*
all_endpoint_integer_gap_equals_4	*true*
all_endpoint_occurrence_count_equals_2	*true*
all_endpoint_q_prefix_band_q_le_10	*true*
all_endpoint_m_shell_band_m_le_4	*true*
all_endpoint_cycle_length_equals_5	*true*
distinct_raw_base_template_count	2
distinct_signed_child_count	2.

The two witnesses are the $P = 607$ above-P path with offsets `[162,166]` and the $P = 953$ below-P path with offsets `[-184,-180]`.

Proof. The audit imports the dominant sign-word path ledger and keeps only rows whose m-pair is `[769, 773]`. The imported m-pair mass is 20. Exactly two rows remain. Each has edge mass 10, occurrence count 2, integer gap 4, q-prefix band `q<=10`, m-shell band `m<=4`, cycle length 5, sign word `-+-+`, sign switch count 3, and sign balance `plus=2,minus=3`. Their orientations relative to P are different, one above P and one below P, so the mass 20 no longer remains an unresolved coordinate block. □

Remark 8.249 (Coordinate-family gate)**.** The dominant path-family gate is now refined to

```
DominantMPairEndpointCoordinateUniformFamilyBound([769,773])
AND OtherDominantSignWordMPairBound([757,761]).
```

This is still a finite coordinate ledger, not a global cancellation theorem. Trace/Kloosterman inputs still require a completion from these fixed endpoint coordinate witnesses to a nonlocal family.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-dominant-sign-word-
dominant-m-pair-path-audit.md.
```

8.105 Dominant sign-word m-pair coordinate partition

Theorem 8.250 (Dominant sign-word coordinate partition ledger)**.** *In the audit range $P \leq 1009$, the whole dominant sign-word block* `-+-+` *of edge mass* 30 *splits into three m-pair/offset/orientation coordinate witnesses. The principal audit values are*

quantity	*value*
`dominant_sign_word_m_pair_coordinate_partition_ledger_closed`	`true`
`path_template_count`	3
`path_edge_mass`	30
`all_path_edge_mass_equals_10`	`true`
`all_path_integer_gap_equals_4`	`true`
`all_path_occurrence_count_equals_2`	`true`
`all_path_m_shell_band_m_le_4`	`true`
`distinct_m_pair_count`	2
`dominant_m_pair`	`[769, 773]`
`dominant_m_pair_edge_mass`	20
`residual_singleton_m_pair`	`[757, 761]`
`residual_singleton_m_pair_edge_mass`	10
`above_P_edge_mass`	20
`below_P_edge_mass`	10.

The coordinate witnesses are

P	*m-pair*	*orientation*	*offsets from P*
607	`[769,773]`	`above_P`	`[162,166]`
739	`[757,761]`	`above_P`	`[18,22]`
953	`[769,773]`	`below_P`	`[-184,-180]`.

Proof. The audit imports the three-path dominant sign-word ledger and the already closed `[769,773]` coordinate subledger. It then parses each m-pair, computes its offsets from the corresponding P, and classifies the orientation as above or below P. The residual m-pair `[757,761]` is a singleton at $P = 739$ with offsets `[18,22]`, q-prefix count 28, and edge mass 10. Together with the two `[769,773]` rows, this accounts for all mass 30 with three distinct coordinate fingerprints. □

Remark 8.251 (Full dominant sign-word coordinate gate)**.** The two gates

```
DominantMPairEndpointCoordinateUniformFamilyBound([769,773])
AND OtherDominantSignWordMPairBound([757,761])
```

are now replaced by the single finer gate

```
DominantSignWordEndpointCoordinateUniformFamilyBound(-+-+; m_pair partition).
```

This still closes only a finite coordinate partition. It does not give the uniform signed collision estimate, PDEC/SAE aggregation, or prime-q reciprocal phase saving needed for the global Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-dominant-sign-word-
m-pair-coordinate-partition-audit.md.
```

8.106 Dominant sign-word step-transition grammar

Theorem 8.252 (Dominant sign-word step-transition ledger). *In the audit range $P \leq 1009$, the three coordinate witnesses of the dominant sign word* `-+-+` *split into* 15 *signed step atoms and* 12 *adjacent transition atoms. The principal audit values are*

quantity	*value*
`dominant_sign_word_step_transition_ledger_closed`	`true`
`coordinate_witness_count`	3
`coordinate_witness_edge_mass`	30
`step_atom_count`	15
`step_atom_mass`	150
`transition_atom_count`	12
`transition_atom_mass`	120
`all_witness_length_equals_5`	`true`
`all_initial_steps_negative`	`true`
`all_terminal_steps_positive`	`true`
`sign_word_position_law_closed`	`true`
`transition_sign_law_closed`	`true`
`negative_step_mass`	90
`positive_step_mass`	60
`same_sign_transition_mass`	30
`sign_switch_transition_mass`	90.

The two repeated signed step atoms are

`g=2,c=2,A=negative`, `g=6,c=7,A=positive`,

each with step mass 20.

Proof. The audit imports the full m-pair coordinate partition and expands each length-5 signed coordinate path. Each of the three paths contributes five signed step atoms and four adjacent transition atoms, with inherited mass 10. Thus the total derived step mass is $3 \cdot 5 \cdot 10 = 150$, and the total derived transition mass is $3 \cdot 4 \cdot 10 = 120$. The sign at each position is fixed by `-+-+`: steps $1, 2, 4$ are negative and steps $3, 5$ are positive. The transition signs are consequently `-`, `-+`, `+-`, `-+` in every path, giving same-sign mass 30 and sign-switch mass 90. Direct aggregation of the (g, c, A) step labels gives exactly the two repeated signed step atoms listed above. □

Remark 8.253 (Step-transition gate). The coordinate-family gate is now refined to

```
DominantSignWordStepTransitionUniformFamilyBound(-+-+ grammar).
```

This is still a finite grammar ledger. It does not by itself give the uniform signed equality estimate, the required PDEC/SAE aggregation, or the prime-q reciprocal phase saving needed for the global Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-dominant-sign-word-
step-transition-audit.md.
```

8.107 Dominant sign-word repeated step occurrences

Theorem 8.254 (Dominant sign-word repeated-step occurrence ledger)**.** *In the audit range $P \leq 1009$, the two repeated signed step atoms inside the dominant sign word* `-+-+` *split into four exact occurrence positions and five adjacent transitions touching those repeated atoms. The principal audit values are*

quantity	*value*
`dominant_sign_word_repeated_step_occurrence_ledger_closed`	*true*
`repeated_signed_step_atom_count`	2
`repeated_step_occurrence_count`	4
`repeated_step_occurrence_mass`	40
`endpoint_role_law_closed`	*true*
`touching_transition_count`	5
`touching_transition_mass`	50
`touching_transition_repeated_endpoint_incidence_count`	6
`touching_transition_repeated_endpoint_incidence_mass`	60.

The two repeated atoms have endpoint-role profiles

atom	*mass*	*initial*	*internal*	*terminal*
`g=2,c=2,A=negative`	20	1	1	0
`g=6,c=7,A=positive`	20	0	1	1.

Proof. The audit imports the closed step-transition ledger and selects exactly the two signed step labels whose step mass is 20: `g=2,c=2,A=negative` and `g=6,c=7,A=positive`. Their four occurrences are

atom	P	step	m-pair
`g=2,c=2,A=negative`	607	1	$[769, 773]$
`g=6,c=7,A=positive`	607	3	$[769, 773]$
`g=2,c=2,A=negative`	739	4	$[757, 761]$
`g=6,c=7,A=positive`	739	5	$[757, 761]$.

All four have inherited mass 10, giving occurrence mass 40. Reading the adjacent transition ledger on the same paths gives five transition rows touching at least one repeated atom. One of these transitions, `g=2,c=2,A=negative` $\to$ `g=6,c=7,A=positive`, touches both repeated atoms, so the transition count is 5 but the repeated-endpoint incidence count is 6. The endpoint-role profile follows immediately from the step positions $1, 3, 4, 5$. ☐

Remark 8.255 (Repeated-step gate)**.** The step-transition gate is now refined by isolating the finite repeated-step carrier

```
RepeatedStepUniformFamilyBound(g=2,c=2,A=negative and g=6,c=7,A=positive).
```

This is still a finite occurrence ledger. It does not prove the uniform signed collision estimate, PDEC/SAE aggregation, or prime-q reciprocal phase saving required for the global Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-q-
support-row-averaged-additive-k-prime-survivor-
boundary-endpoint-flux-qprefix-carry-cycle-signature-single-p-local-
gap2-gap4-top-two-core-largest-atom-dominant-sign-word-
repeated-step-occurrence-audit.md.
```

8.108 Repeated-step directed incidence graph

Theorem 8.256 (Repeated-step directed-incidence graph ledger)**.** *In the audit range $P \leq 1009$, the five adjacent transitions touching the two repeated signed step atoms form a finite directed incidence graph with five nodes and five directed edges. The principal audit values are*

quantity	*value*
repeated_step_directed_incidence_graph_ledger_closed	*true*
node_count	5
repeated_node_count	2
neighbor_node_count	3
directed_edge_count	5
directed_edge_mass	50
repeated_endpoint_incidence_count	6
weak_component_count	1
directed_acyclic	*true*
cross_repeated_bridge_edge_count	1.

The edge-class masses are

edge class	*mass*
neighbor_to_repeated	20
repeated_to_neighbor	20
repeated_to_repeated	10.

The unique source is `g=6,c=6,A=positive`*, the unique sink is* `g=8,c=10,A=negative`*, and the longest directed path has length* 4.

Proof. The audit imports the repeated-step occurrence ledger and uses each touching transition as a directed edge from its left signed step atom to its right signed step atom. The resulting edges are

edge	class	mass
g=2,c=2,A=negative → g=4,c=5,A=negative	repeated_to_neighbor	10
g=4,c=5,A=negative → g=6,c=7,A=positive	neighbor_to_repeated	10
g=6,c=7,A=positive → g=8,c=10,A=negative	repeated_to_neighbor	10
g=6,c=6,A=positive → g=2,c=2,A=negative	neighbor_to_repeated	10
g=2,c=2,A=negative → g=6,c=7,A=positive	repeated_to_repeated	10.

Thus there are two repeated nodes and three neighbouring nodes, and the total edge mass is 50. Reading directions gives the topological chain

g=6,c=6,A=positive → g=2,c=2,A=negative → g=4,c=5,A=negative → g=6,c=7,A=positive → g=8,c=10,A=n

which has length 4. The same ordering contains every directed edge, so the carrier is acyclic and has one weak component. □

Remark 8.257 (Directed-incidence gate)**.** The repeated-step gate is now refined to

```
RepeatedStepDirectedIncidenceGraphUniformBound(acyclic two-repeated-node carrier).
```

This is still a finite graph ledger. It does not prove the corresponding uniform family estimate, PDEC/SAE aggregation, or prime-q reciprocal phase saving required for the global Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-directed-incidence-graph-audit.md.
```

8.109 Repeated-step source-sink path cover

Theorem 8.258 (Repeated-step source-sink path-cover ledger). *In the audit range $P \leq 1009$, the directed repeated-step graph has exactly two source-sink graph paths. The source is `g=6,c=6,A=positive` and the sink is `g=8,c=10,A=negative`. The principal path-cover values are*

quantity	*value*
repeated_step_source_sink_path_cover_ledger_closed	*true*
source_sink_path_count	2
source_sink_path_edge_incidence_count	7
source_sink_path_edge_incidence_mass	70
edge_cover_complete	*true*
shared_edge_count	2
shared_edge_mass	20
shared_edge_path_incidence_mass	40
single_witness_source_sink_path_count	0
mixed_witness_source_sink_path_count	2
all_source_sink_paths_mixed_P	*true*
all_source_sink_paths_have_one_P_switch	*true*
diamond_decomposition_closed	*true.*

The single-witness orbit interpretation is therefore invalid for these graph paths.

Proof. Enumerating the source-sink paths in the five-edge DAG gives

$$\texttt{g=6,c=6,A=positive} \to \texttt{g=2,c=2,A=negative} \to \texttt{g=4,c=5,A=negative} \\ \to \texttt{g=6,c=7,A=positive} \to \texttt{g=8,c=10,A=negative},$$

and

$$\texttt{g=6,c=6,A=positive} \to \texttt{g=2,c=2,A=negative} \to \texttt{g=6,c=7,A=positive} \to \texttt{g=8,c=10,A=negative}.$$

The two common edges are the source edge into `g=2,c=2,A=negative` and the terminal edge out of `g=6,c=7,A=positive`; their ordinary mass is 20, while their path-incidence mass is 40. The branch edge through `g=4,c=5,A=negative` and the direct repeated-to-repeated bridge close the branch-join diamond. Both source-sink paths use data from $P = 607$ and $P = 739$, with one P-switch. Hence the path cover is exact as a finite signature graph ledger, but it is not a single-P witness orbit. □

Remark 8.259 (Mixed-P path-cover gate). The repeated-step gate is now refined to

```
RepeatedStepMixedPSourceSinkPathCoverUniformBound(two mixed-P graph paths).
```

This still does not prove the required uniform family estimate. The available trace, bilinear, Kloosterman, small-gap, and short-interval prime inputs do not by themselves turn this mixed-P signature stitching into a global Phi–LPF bound.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-source-sink-path-cover-audit.md.
```

8.110 Repeated-step repeated-node P-switch cuts

Theorem 8.260 (Repeated-step P-switch cut ledger)**.** *In the audit range $P \leq 1009$, the two mixed-P source-sink paths have exactly two P-switch cuts. Both are monotone*

$$739 \longrightarrow 607$$

switches, and they occur exactly at the two repeated nodes. The principal audit values are

quantity	*value*
repeated_step_p_switch_cut_ledger_closed	*true*
path_switch_cut_count	2
switch_node_count	2
repeated_switch_node_cover_complete	*true*
neighbor_switch_node_count	0
all_switches_at_repeated_nodes	*true*
all_switches_high_to_low	*true*
all_switches_739_to_607	*true*
switch_pair_edge_mass_total	40
switch_edge_incidence_count	4
switch_edge_incidence_mass	40.

Proof. The long corridor path has P-sequence $(739, 607, 607, 607)$, so its only switch is at

`g=2,c=2,A=negative`.

The bridge shortcut path has P-sequence $(739, 739, 607)$, so its only switch is at

`g=6,c=7,A=positive`.

These two nodes are exactly the repeated nodes in the directed-incidence graph. The corresponding local cuts are

`g=6,c=6,A=positive` $\to$ `g=2,c=2,A=negative` $\|$ `g=2,c=2,A=negative` $\to$ `g=4,c=5,A=negative`,

`g=2,c=2,A=negative` $\to$ `g=6,c=7,A=positive` $\|$ `g=6,c=7,A=positive` $\to$ `g=8,c=10,A=negative`.

Each cut has adjacent edge mass 20, giving total switch-pair edge mass 40. Four distinct edges participate in a P-switch; the only non-switch edge is

`g=4,c=5,A=negative` $\to$ `g=6,c=7,A=positive`.

□

Remark 8.261 (Repeated-node P-switch gate)**.** The repeated-step gate is now refined to

```
RepeatedStepRepeatedNodePSwitchCutUniformBound(two 739->607 cuts).
```

This is a sharper finite localization of the mixed-P obstruction, not a proof of the uniform family estimate. The switch-cut object is still a local signed grammar event rather than a completed trace, bilinear, Kloosterman, or short-interval prime family.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-p-switch-cut-audit.md.
```

8.111 Repeated-step occurrence splices

Theorem 8.262 (Repeated-step occurrence-splice ledger). *In the audit range $P \leq 1009$, the two repeated-node P-switch cuts are exactly two same-signed-atom cross-witness occurrence splices. The principal audit values are*

quantity	*value*
repeated_step_occurrence_splice_ledger_closed	*true*
occurrence_splice_count	2
occurrence_splice_mass_total	40
occurrence_splice_endpoint_count	4
occurrence_splice_endpoint_cover_complete	*true*
all_splices_same_signed_atom	*true*
all_splices_same_gap_carry	*true*
all_splices_same_orientation	*true*
all_splices_739_to_607	*true*
all_splices_m_pair_delta_12_12	*true*
all_splices_q_delta_minus_21	*true*
all_splices_step_rewind	*true*
step_rewind_total	5.

Proof. Resolving the left and right endpoints of the two P-switch cuts gives

signed atom	left occurrence	right occurrence
`g=2,c=2,A=negative`	`P739:step4:[757,761]:above_P`	`P607:step1:[769,773]:above_P`
`g=6,c=7,A=positive`	`P739:step5:[757,761]:above_P`	`P607:step3:[769,773]:above_P`.

Both rows preserve the signed atom, the gap/carry coordinate, the sign, and the endpoint orientation. Both change P by $607 - 739 = -132$, change the underlying q-slot from 28 to 7, and translate the m-pair by $[12, 12]$. The step slots rewind from 4 to 1, and from 5 to 3, so the total rewind is $3 + 2 = 5$. The four repeated-step occurrences are all used exactly once as splice endpoints, giving endpoint mass 40. ☐

Remark 8.263 (Occurrence-splice gate). The repeated-step gate is now refined to

```
RepeatedStepSameAtomOccurrenceSpliceUniformBound(two P739-to-P607 splices).
```

This removes another finite ambiguity in the mixed-P obstruction, but it does not prove a uniform family bound. The object is still a local cross-witness splice, not a trace/Kloosterman family or a short-interval prime statement.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-occurrence-splice-audit.md.
```

8.112 Repeated-step affine skeleton

Theorem 8.264 (Repeated-step shared witness-pair affine skeleton). *In the audit range $P \leq 1009$, the two same-atom $P739$-to-$P607$ occurrence splices use one common left/right witness pair. The* *principal audit values are*

quantity	value
shared_witness_pair_affine_skeleton_ledger_closed	*true*
splice_count	2
splice_mass_total	40
distinct_witness_pair_count	1
all_same_affine_deltas	*true*
affine_identity_offset_delta_equals_m_delta_minus_P_delta	*true*
coordinate_atom_recheck_closed	*true.*

Proof. Both splices share the witness-pair skeleton

$$\texttt{P739\_packet2842\_q28\_mpair\_757\_761} \longrightarrow \texttt{P607\_packet1887\_q7\_mpair\_769\_773}.$$

For each splice the affine deltas are

$$\Delta P = -132, \qquad \Delta q = -21, \qquad \Delta\text{packet} = -955, \qquad \Delta m = [12, 12],$$

and the offsets from P move by

$$[162, 166] - [18, 22] = [144, 144] = [12, 12] - (-132).$$

The m-width $773 - 769 = 761 - 757 = 4$ and the endpoint orientation are preserved. Rechecking the signed step coordinates gives the two atoms `g=2,c=2,A=negative` and `g=6,c=7,A=positive`, with the same gap/carry/sign on the left and right occurrences. Hence the two local splices are not independent witness pairs; they are two signed-atom projections of one affine witness-pair skeleton. □

Remark 8.265 (Affine-skeleton gate)**.** The repeated-step gate is now refined to

```
RepeatedStepSharedWitnessPairAffineSkeletonUniformBound
(P739/q28/[757,761]/packet2842 -> P607/q7/[769,773]/packet1887).
```

This is a genuine finite compression of the obstruction, but it still does not create a summable trace/Kloosterman family and it does not prove a uniform family bound. The external short-interval and prime-gap inputs also do not estimate this fixed witness-pair affine skeleton.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-occurrence-splice-affine-skeleton-audit.md.
```

8.113 Repeated-step packet enclosure

Theorem 8.266 (Repeated-step affine-skeleton packet enclosure)**.** *In the audit range $P \le 1009$, the shared affine skeleton is enclosed in two concrete right-tail shell-step packets. The principal audit values are*

quantity	value
packet_enclosure_ledger_closed	*true*
unique_left_packet_indices	*[2842]*
unique_right_packet_indices	*[1887]*
left_q_prefix_count	28
right_q_prefix_count	7
all_q_windows_disjoint	*true*
all_m_shells_disjoint	*true*
all_selected_pairs_terminal	*true.*

Proof. The left packet is `packet2842` with $P = 739$, right-tail q-window $[541, 709]$, 28 prime q's, and m-shell

$$\{719, 751, 757, 761\}.$$

The right packet is `packet1887` with $P = 607$, right-tail q-window $[439, 467]$, 7 prime q's, and m-shell

$$\{479, 769, 773\}.$$

The selected affine skeleton uses the terminal pairs $[757, 761]$ and $[769, 773]$. The two q-windows are disjoint, with the $P607$ window strictly to the left of the $P739$ window; the two m-shell supports are also disjoint. Thus the shared witness-pair skeleton is not already a completed bilinear or Kloosterman family: it is a fixed asymmetric right-tail packet enclosure. □

Remark 8.267 (Packet-enclosure gate)**.** The repeated-step gate is now refined to

```
RepeatedStepAffineSkeletonPacketEnclosureUniformBound
(packet2842:[q=541..709,m={719,751,757,761}] ->
packet1887:[q=439..467,m={479,769,773}]).
```

This closes another finite localization question, but it still does not prove a uniform family bound, create a summable trace/Kloosterman family, or break the Phi–LPF parity barrier globally.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-affine-skeleton-packet-enclosure-audit.md.
```

8.114 Repeated-step terminal line atoms

Theorem 8.268 (Repeated-step packet-enclosure terminal line atoms)**.** *In the audit range $P \le 1009$, the repeated-step packet enclosure splits into seven unique fixed-m q-prefix line atoms. The principal audit values are*

quantity	*value*
terminal_line_atom_ledger_closed	*true*
support_packet_edge_mass	133
unique_line_atom_count_total	7
selected_terminal_line_atom_count	4
selected_terminal_line_atom_edge_mass	70
extra_line_atom_count	3
extra_line_atom_edge_mass	63
all_selected_line_atoms_terminal	*true.*

Proof. The left packet `packet2842` contributes two selected terminal line atoms $m = 757, 761$, each with 28 q-prefix edges, and two extra line atoms $m = 719, 751$. The right packet `packet1887` contributes two selected terminal line atoms $m = 769, 773$, each with 7 q-prefix edges, and one extra line atom $m = 479$. Hence the selected terminal edge mass is $2 \cdot 28 + 2 \cdot 7 = 70$, while the extra shell edge mass is $2 \cdot 28 + 1 \cdot 7 = 63$. The selected and extra expanded edge sets are disjoint and together recover the packet support mass 133. □

Remark 8.269 (Terminal line-atom gate)**.** The repeated-step gate is now refined to

```
RepeatedStepPacketEnclosureTerminalLineAtomUniformBound
(packet2842:selected m={757,761}, q=541..709;
packet1887:selected m={769,773}, q=439..467).
```

The two splices share the same packet support, so the splice-incidence mass is only a repeated count of this unique line-atom support. This closes the finite terminal line-atom ledger, but it still does not prove a uniform bound, absorb the extra line atoms, create a summable trace/Kloosterman family, or break the Phi–LPF parity barrier globally.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-packet-enclosure-terminal-line-atom-audit.md.
```

8.115 Repeated-step terminal phase normal form

Theorem 8.270 (Repeated-step terminal phase normal form)**.** *In the audit range $P \leq 1009$, all seven terminal/extra line atoms in the repeated-step packet enclosure satisfy the local phase normal form*

$$qm = kP + D, \qquad 0 < D < P, \qquad A(q) \equiv -D \pmod q,$$

and therefore

$$e(hkP/q) = e(-hD/q) = e(hA(q)/q).$$

The principal audit values are

quantity	*value*
terminal_phase_normal_form_closed	*true*
terminal_phase_normal_form_edge_count_total	133
selected_terminal_phase_atom_count	4
selected_terminal_phase_edge_count	70
extra_phase_atom_count	3
extra_phase_edge_count	63
selected_terminal_all_moving_numerator	*true*
selected_terminal_fixed_numerator_atom_count	0.

Proof. The audit applies the deterministic division identity $qm = kP + D$ to every edge in the seven line atoms and checks that all division and phase congruence mismatches are zero. On the four selected terminal atoms the normalized numerator $A(q)$ is moving and full-distinct along the prime q-prefix: for $m = 757, 761$ in `packet2842` the prefix length is 28, and for $m = 769, 773$ in `packet1887` the prefix length is 7. Thus no selected terminal atom is a fixed-numerator Kloosterman input. □

Remark 8.271 (Moving-numerator terminal gate)**.** The repeated-step terminal gate is now

```
SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
(packet2842:m=757,761; packet1887:m=769,773).
```

This is a narrower obstruction than terminal support: the local phase is known exactly, but the numerator $A(q)$ moves with q. Therefore this step does not prove moving-numerator phase saving, absorb the extra phase atoms, create a summable trace/Kloosterman family, or break the Phi–LPF parity barrier globally.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-packet-enclosure-terminal-phase-normal-form-audit.md.
```

8.116 Repeated-step terminal phase carry orbits

Theorem 8.272 (Repeated-step terminal phase carry recurrence)**.** *In the audit range $P \leq 1009$, every adjacent prime-q transition in the seven terminal/extra phase atoms satisfies the exact carry recurrence*

$$q' = q + g, \qquad qm = kP + D, \qquad c = \left\lfloor \frac{D + gm}{P} \right\rfloor,$$

and hence

$$k' = k + c, \qquad D' = D + gm - cP, \qquad A(q') \equiv -D' \pmod{q'}.$$

The principal audit values are

quantity	*value*
terminal_phase_carry_orbit_closed	*true*
terminal_carry_atom_count_total	7
terminal_carry_transition_count_total	126
selected_terminal_transition_count	66
extra_transition_count	60
selected_terminal_A_wrap_count	49
selected_terminal_D_wrap_count	11
selected_terminal_A_wrap_fraction	*49/66*
selected_terminal_D_wrap_fraction	*11/66*
carry_identity_mismatch_count	0.

Proof. The previous normal-form layer gives, on every phase edge,

$$qm = kP + D, \qquad 0 < D < P, \qquad A(q) \equiv -D \pmod{q}.$$

For adjacent prime denominators $q < q'$, write $g = q' - q$. Then

$$q'm = qm + gm = kP + D + gm.$$

Dividing $D + gm$ by P gives the unique integer

$$c = \left\lfloor \frac{D + gm}{P} \right\rfloor$$

and residue $D' = D + gm - cP$ with $0 < D' < P$. Therefore

$$q'm = (k + c)P + D',$$

so $k' = k + c$ and $A(q') \equiv -D' \pmod{q'}$. The certificate checks this identity for all 126 adjacent transitions. The selected terminal part has 66 transitions: $27 + 27$ in `packet2842` for $m = 757, 761$, and $6 + 6$ in `packet1887` for $m = 769, 773$. Among those selected transitions 49 wrap in A and 11 wrap in D, with carry spectrum $2, \ldots, 13$. The extra-shell part contributes 60 further verified transitions. No carry identity mismatch occurs. □

Remark 8.273 (Carry-word gate)**.** The repeated-step terminal gate is now refined to

```
SelectedTerminalPrimeGapCarryWordPhaseSaving
(66 transitions, 49 A-wraps, carry spectrum 2..13).
```

This closes the finite deterministic carry-orbit ledger, but it still does not prove cancellation along the prime-gap/carry word, absorb the 60 extra carry transitions, create a summable trace/Kloosterman family, or break the Phi–LPF parity barrier globally. Current external inputs such as trace/bilinear estimates, arbitrary-modulus Kloosterman bounds, composite Type-II estimates, unbalanced Kloosterman fractions, short-interval prime theorems, and bounded prime-gap theorems do not by themselves estimate this finite signed carry word.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-packet-enclosure-terminal-phase-carry-orbit-audit.md.
```

8.117 Repeated-step terminal phase-turn word

Theorem 8.274 (Repeated-step terminal phase-turn word). *In the audit range $P \le 1009$, the terminal carry-orbit phase has a two-lift representation*

$$A(q) = \lambda(q)q - D(q), \qquad \lambda(q) \in \{1,2\},$$

on every terminal/extra phase edge. Moreover, for every adjacent prime transition $q < q'$, the sign of

$$\frac{A(q')}{q'} - \frac{A(q)}{q}$$

agrees with the A-wrap word: the phase decreases exactly on $A(q') < A(q)$, and increases exactly on no-wrap transitions. The principal audit values are

quantity	*value*
terminal_phase_turn_word_closed	*true*
terminal_phase_turn_transition_count_total	126
terminal_phase_turn_run_count_total	59
selected_terminal_transition_count	66
selected_terminal_positive_transition_count	17
selected_terminal_negative_transition_count	49
selected_terminal_phase_run_count	35
selected_terminal_phase_run_max_length	6
extra_transition_count	60
extra_positive_transition_count	27
extra_negative_transition_count	33
extra_phase_run_count	24
phase_direction_Awrap_mismatch_count	0
lift_identity_mismatch_count	0
zero_phase_delta_count	0.

Proof. The carry-orbit layer gives $0 < D(q) < P$ and $A(q) \equiv -D(q) \pmod q$. Since all q-prefix denominators in these terminal atoms lie above $P/2$, we have $D(q) < P < 2q$. Hence the lift

$$\lambda(q) = \left\lceil \frac{D(q)}{q} \right\rceil$$

belongs to $\{1,2\}$, and the residue representative is exactly

$$A(q) = \lambda(q)q - D(q).$$

The audit checks this identity on every edge. It then computes the exact rational difference $A(q')/q' - A(q)/q$ for each of the 126 adjacent prime transitions. There are no zero differences. In all 126 cases the phase decreases precisely when $A(q') < A(q)$, and increases precisely when there is no A-wrap. The selected terminal word therefore consists of 49 negative A-wrap turns and 17 positive no-wrap turns, split into 35 monotone runs of maximum length 6. The extra-shell word has 33 negative turns and 27 positive turns, split into 24 runs. □

Remark 8.275 (A-wrap phase-turn gate)**.** The repeated-step terminal gate is now refined to

```
SelectedTerminalAwrapPhaseTurnWordSaving
(66 transitions: 49 negative/A-wrap, 17 positive/no-wrap;
35 runs; max run 6).
```

This is a sharper finite phase ledger than the carry recurrence alone. It does not prove signed cancellation for the A-wrap word, does not absorb the extra phase-turn runs, does not create a completed trace/Kloosterman family, and does not close the row/column Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-packet-enclosure-terminal-phase-turn-word-audit.md.
```

8.118 Repeated-step terminal phase variation budget

Theorem 8.276 (Repeated-step terminal signed variation budget)**.** *In the audit range $P \leq 1009$, the terminal phase-turn word admits an exact signed variation ledger. For every atom and for each role aggregate,*

$$\mathrm{Var}_{\mathrm{tot}} = \mathrm{Var}_{+} + \mathrm{Var}_{-}, \qquad \Delta_{\mathrm{net}} = \mathrm{Var}_{+} - \mathrm{Var}_{-}\,.$$

The principal audit values are

quantity	*value*
terminal_phase_variation_budget_closed	*true*
terminal_phase_variation_transition_count_total	126
terminal_phase_variation_run_count_total	59
selected_terminal_transition_count	66
selected_terminal_phase_run_count	35
selected_terminal_phase_run_max_length	6
selected_terminal_positive_variation	8.261301579660
selected_terminal_negative_variation	9.717867552237
selected_terminal_total_variation	17.979169131897
selected_terminal_net_phase_displacement	−1.456565972578
extra_phase_run_count	24
extra_total_variation	14.109301881162
extra_net_phase_displacement	−0.907719323182
bad_atom_variation_identity_count	0
bad_role_variation_identity_count	0.

Proof. For each adjacent transition the previous phase-turn certificate gives the exact rational increment

$$\delta_i = \frac{A(q_{i+1})}{q_{i+1}} - \frac{A(q_i)}{q_i}.$$

Define

$$\mathrm{Var}_+ = \sum_{\delta_i>0} \delta_i, \qquad \mathrm{Var}_- = \sum_{\delta_i<0} (-\delta_i), \qquad \mathrm{Var}_{\mathrm{tot}} = \sum_i |\delta_i|.$$

The net phase displacement is $\sum_i \delta_i$, which telescopes to the endpoint phase difference. The audit checks these identities for each of the seven atoms, for the selected-terminal aggregate, and for the extra-shell aggregate. It also verifies that the run-level variation sums equal the atom-level variation sums. On the selected terminal aggregate the negative variation is larger than the positive variation by

$$1.456565972578\ldots,$$

and the total variation is approximately 17.979169131897. Thus the selected phase-turn word is not naturally balanced at the signed-variation level. ☐

Remark 8.277 (Variation-budget gate)**.** The repeated-step terminal gate is now refined to

```
SelectedTerminalNegativeVariationExcessPhaseSaving
(total variation 17.979169131897; net -1.456565972578;
35 runs; max run 6).
```

This closes the finite variation ledger and rules out an unproved “automatic balance” interpretation of the A-wrap word. It does not prove a monotone-run phase-saving cap, absorb the extra variation budget, create a completed trace/Kloosterman family, or close the row/column Phi–LPF theorem.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-dominant-sign-word-
repeated-step-packet-enclosure-terminal-phase-variation-budget-audit.md.
```

8.119 Terminal signed payload measure and absorption frontier

Theorem 8.278 (Terminal signed payload measure schema)**.** *The seven terminal and extra fixed-m line atoms in the repeated-step packet enclosure can be written as a single finite signed payload measure*

$$\mu = \mu(q, m, \texttt{packet})$$

on 126 *adjacent phase transitions. Each transition carries the fields*

$$\texttt{packet}, P, m, q, q', A, A', D, D', \texttt{lift}, \texttt{lift}', \texttt{carry}, \texttt{phase\_direction}, \delta,$$

where

$$\delta = \frac{A(q')}{q'} - \frac{A(q)}{q}.$$

The measure schema is closed as a finite ledger.

Proof. The phase-turn certificate supplies, for each terminal or extra atom, the adjacent prime-q transitions together with A, D, the lift class, carry, wrap flags, and the exact rational phase increment. Collecting these transition records over the four selected-terminal atoms and the three extra-shell atoms gives 126 records. The audit checks that every record has the required fields and that no required field is missing. ☐

Theorem 8.279 (Net absorption succeeds but total-variation absorption fails)**.** *For the finite measure $\mu(q, m, \texttt{packet})$, the selected-terminal negative net excess is*

$$1.456565972578\ldots,$$

whereas the extra-shell negative net excess is

$$0.907719323182\ldots.$$

Thus the net-excess margin is positive:

$$0.548846649396\ldots.$$

However the extra-shell total variation is

$$14.109301881162\ldots,$$

so the strong total-variation margin is negative:

$$-12.652735908584\ldots.$$

Consequently this ledger proves a net-level surplus, but not the strong absorption needed for phase saving.

Proof. The variation-budget ledger gives exact rational values for the selected and extra role aggregates. Subtracting the extra negative net excess from the selected negative net excess gives the positive margin above. Subtracting the extra total variation from the selected negative net excess gives the negative strong margin above. Hence there is no contradiction in the finite ledger: net surplus and total-variation failure coexist. □

Computational-certificate Statement 8.280 (Terminal signed measure absorption audit)**.** *The machine certificate records*

```
terminal_signed_payload_measure_schema_closed=true
mu_transition_count=126
selected_net_excess_beats_extra_net_excess=true
selected_net_excess_beats_extra_total_variation=false
monotone_run_total_to_net_compression_proved=false
extra_total_variation_absorption_proved=false
admissible_averaged_trace_family_created=false
trace_or_kloosterman_completion_ready=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-signed-
payload-measure-absorption-frontier-router.md.
```

Remark 8.281 (Next non-cyclic absorption gate)**.** The new frontier is

```
MonotoneRunTotalToNetCompressionOrPDEC
AND ExtraTotalVariationAbsorptionOrLocalSurvivor
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

External trace/Kloosterman or Type-II inputs can enter only after this finite measure is promoted to a genuine averaged family, or after a named return absorbs the total variation loss.

8.120 Terminal monotone-run total-to-net compression frontier

Theorem 8.282 (Finite monotone-run decomposition). *For the terminal signed payload measure above, the phase-variation ledger decomposes into* 59 *monotone runs:* 35 *selected-terminal runs and* 24 *extra-shell runs. On every monotone run,*

$$\text{variation} = |\delta_{\text{run}}|.$$

Thus the run-local total-to-net compression ratio is identically 1*; there is no cancellation inside a single monotone run.*

Proof. The machine audit reads every `run_variation_row` from the terminal phase-variation certificate and matches it with the corresponding phase-turn run. For each row it compares the recorded total variation with the absolute value of the recorded signed displacement. All 59 identities match. □

Proposition 8.283 (Adjacent-run cancellation explains the finite surplus). *Within each terminal atom, the ordered monotone runs admit an exact finite decomposition into adjacent opposite-run cancelled chunks plus an atom-local survivor. In the extra-shell role,*

$$V_{\text{extra}} = 14.109301881162\ldots$$

compresses, under this finite cancellation decomposition, to the atom-local survivor

$$S_{\text{extra}} = 0.907719323182\ldots.$$

The selected-terminal negative excess is

$$E_{\text{sel}} = 1.456565972578\ldots,$$

so the finite post-cancellation margin would be

$$E_{\text{sel}} - S_{\text{extra}} = 0.548846649396\ldots > 0.$$

Proof. For each atom, order the monotone runs by their transition index and greedily cancel adjacent positive and negative run masses. The audit checks the exact identity

$$V_{\text{atom}} - 2C_{\text{atom}} = S_{\text{atom}}, \qquad \Delta_{\text{atom}} = S^{\text{signed}}_{\text{atom}}.$$

Summing the three extra-shell atom survivors gives the displayed S_{extra}. Subtracting this from the selected-terminal negative excess gives the positive margin above. □

Computational-certificate Statement 8.284 (Terminal monotone-run compression audit). *The machine certificate records*

```
terminal_run_count_total=59
selected_terminal_run_count=35
extra_shell_run_count=24
strict_run_local_compression_count=0
atom_adjacent_cancellation_decomposition_closed=true
finite_absorption_would_close_after_uniform_cancellation_law=true
monotone_run_total_to_net_compression_proved=false
uniform_run_cancellation_family_created=false
row_column_unconditional_closed=false.
```

It is recorded at

docs/monograph/prime-matrix-phi-lpf-terminal-monotone-run-total-to-net-compression-frontier-router.md.

Remark 8.285 (What remains)**.** The finite cancellation ledger shows exactly what a proof would need to promote: a uniform adjacent-run cancellation family, an admissible averaged trace/Kloosterman or Type-II family, or a named PDEC/SAE/LocalSurvivor return. It does not by itself break the Phi–LPF parity barrier, because every monotone run has local compression ratio 1.

8.121 Terminal adjacent-run Jordan cancellation and source obstruction

Theorem 8.286 (Formal adjacent-run cancellation law)**.** *For the seven terminal phase paths $A(q)/q$, the sign of every adjacent transition is determined by the wrap flag:*

$$\delta(q,q') < 0 \iff A(q') < A(q), \qquad \delta(q,q') > 0 \iff A(q') > A(q).$$

The maximal same-sign blocks are exactly the 59 monotone runs recorded in the run ledger. Moreover each atom satisfies the signed telescoping identity

$$\sum_{q\to q'} \delta(q,q') = \frac{A(q_{\mathrm{end}})}{q_{\mathrm{end}}} - \frac{A(q_{\mathrm{start}})}{q_{\mathrm{start}}}.$$

Consequently the formal adjacent-run cancellation law is just the Jordan decomposition of the one-dimensional phase path $A(q)/q$.

Proof. The audit rebuilds the maximal monotone runs directly from the phase-turn transition list and compares them with the terminal variation runs. All signs match the A-wrap flag, all maximal runs match, and the exact rational sum of signed phase increments equals the endpoint phase difference for each atom. The Jordan identity follows by subtracting the cancelled opposite-sign chunks from the total variation. □

Proposition 8.287 (Source-preserving cancellation is still open)**.** *The finite cancellation ledger contains 51 cancellation events. None is a complete whole-run involution:*

$$\texttt{complete\_whole\_run\_pair\_event\_count} = 0.$$

All 51 events require synthetic splitting of run masses. In addition, the selected atom `right:1887:selected_termin` *has an internal negative survivor fragment on the run $q \in [449, 457]$ of mass*

$$0.017995938314\ldots.$$

Thus the tail-only survivor law and the source-preserving adjacent-run pairing are not proved by the finite Jordan identity.

Proof. The audit runs the same ordered opposite-sign stack cancellation used in the finite run-compression certificate, but records each cancellation chunk. In every event, one side is a proper residual chunk rather than a whole run matched to a whole run. After all cancellations, one survivor fragment is not on the last run of its atom, giving the internal survivor stated above. □

Computational-certificate Statement 8.288 (Adjacent-run Jordan cancellation audit)**.** *The machine certificate records*

```
sign_matches_Awrap_all_transitions=true
maximal_run_reconstruction_closed=true
signed_telescoping_identity_closed=true
formal_jordan_cancellation_law_closed=true
cancellation_event_count=51
complete_whole_run_pair_event_count=0
synthetic_split_cancellation_event_count=51
internal_survivor_fragment_count=1
source_preserving_adjacent_run_pairing_constructed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-adjacent-run-
jordan-cancellation-source-obstruction-router.md.
```

Remark 8.289 (New narrow gate)**.** The next gate is no longer to rediscover telescoping. It is to construct a source-preserving adjacent-run pairing, prove an orientation local-factor law before pushforward, turn the moving Beatty numerator into an admissible averaged trace/Kloosterman or Type-II family, or route the internal survivor to a named PDEC/LocalSurvivor certificate.

8.122 Terminal prefix-record reflection and source-key obstruction

Theorem 8.290 (Prefix-record reflection ledger)**.** *The formal adjacent-run Jordan cancellation law admits an equivalent prefix-record/reflection ledger on the seven terminal atoms. The certificate checks*

$$\texttt{terminal\_atom\_count} = 7, \qquad \texttt{terminal\_run\_count\_total} = 59,$$

and records

$$\texttt{prefix\_record\_reflection\_schema\_closed} = \texttt{true}.$$

Thus the one-dimensional path accounting has been fully compressed to prefix minima, prefix maxima, cancelled opposite-sign chunks, and survivor fragments.

Proof. For each terminal atom, the audit orders the monotone runs and forms the exact rational cumulative sum of the signed run displacements. It records the prefix extrema and then repeats the opposite-sign stack cancellation. The total number of cancellation events is 51, matching the previous Jordan audit, and the total survivor count is 8. Hence the reflection ledger is the same finite signed path, written in prefix-record coordinates. □

Proposition 8.291 (Source-key lift remains open)**.** *The prefix-record ledger does not yet give an actual source-preserving pairing. It has*

$$\begin{aligned} \texttt{whole\_run\_pair\_event\_count} &= 0, & \texttt{synthetic\_split\_event\_count} &= 51,\\ \texttt{q\_boundary\_pair\_event\_count} &= 47, & \texttt{non\_q\_boundary\_pair\_event\_count} &= 4,\\ \texttt{survivor\_fragment\_count\_total} &= 8, & \texttt{internal\_survivor\_fragment\_count} &= 1. \end{aligned}$$

Consequently the source-key lift, the tail-only survivor law, and the global Phi–LPF parity-breaking conclusion all remain unproved.

Proof. The audit labels each cancellation chunk by its old run, new run, boundary status, and consumed mass. No event consumes both full runs, so every cancellation requires synthetic mass splitting. Four cancellations pair non-adjacent q-boundaries, and one survivor fragment remains internal rather than terminal. These are source-level obstructions rather than rounding artefacts of the prefix representation. □

Computational-certificate Statement 8.292 (Prefix-record source-key obstruction audit). *The machine certificate records*

```
previous_formal_jordan_cancellation_law_closed=true
prefix_record_reflection_schema_closed=true
cancellation_event_count=51
whole_run_pair_event_count=0
synthetic_split_event_count=51
q_boundary_pair_event_count=47
non_q_boundary_pair_event_count=4
survivor_fragment_count_total=8
internal_survivor_fragment_count=1
prefix_record_source_key_lift_constructed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-prefix-record-
source-key-obstruction-router.md.
```

Remark 8.293 (Current non-cyclic gate). The current narrow gate is

```
PrefixRecordSourceKeyLiftOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

The prefix-record ledger is useful because it removes another formal escape: the missing object is now specifically a source-key lift, a local orientation law before pushforward, or a genuine averaged family to which external trace/Kloosterman/Type-II tools can apply.

8.123 Terminal source-key obstruction partition

Theorem 8.294 (Three-way source-key obstruction partition). *The prefix-record source-key obstruction has an exact finite partition into three actual gates:*

$$\mathtt{q\_boundary\_synthetic\_split\_event\_count} = 47,$$
$$\mathtt{nonboundary\_record\_jump\_event\_count} = 4,$$
$$\mathtt{internal\_survivor\_fragment\_count} = 1.$$

Together with the seven tail survivors, these rows account for all 51 *cancellation events and all* 8 *survivor fragments in the prefix-record ledger.*

Proof. The audit recomputes the opposite-sign prefix-record cancellation events and labels each event by whether its old q-endpoint equals its new q-start. The 47 matching-boundary events form the q-boundary synthetic-split class, while the remaining 4 events are non-boundary record jumps. It also compares every survivor run with the last run of its atom; exactly one survivor is internal, and the other seven are tail survivors. The counts add to the previous prefix-record certificate. □

Proposition 8.295 (Finite mass is not the final obstruction). *The four non-boundary record jumps and the internal survivor have total mass*

$$0.215539338772\ldots.$$

Subtracting this finite mass from the selected-terminal post-cancellation margin leaves

$$0.333307310624\ldots > 0.$$

This scalar check does not prove the row/column claim: the q-boundary split ratio law, the non-boundary jump source-key lift, and the internal survivor return are still unproved.

Proof. The audit sums the exact rational chunk masses of the four non-boundary cancellation events and adds the unique internal survivor mass. It then subtracts this total from the previously certified finite margin `selected_negative_excess_minus_extra_atom_survivor`. The resulting rational number is positive. The computation is only a finite budget check; it does not construct a uniform source-key law. ☐

Computational-certificate Statement 8.296 (Source-key obstruction partition audit)**.** *The machine certificate records*

```
source_key_obstruction_partition_closed=true
cancellation_event_count=51
whole_run_pair_event_count=0
synthetic_split_event_count=51
q_boundary_synthetic_split_event_count=47
nonboundary_record_jump_event_count=4
nonboundary_record_jump_atom_count=2
tail_survivor_fragment_count=7
internal_survivor_fragment_count=1
finite_margin_after_nonboundary_internal_payment_positive=true
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-source-key-
obstruction-partition-router.md.
```

Remark 8.297 (Updated source-key gate)**.** The next gate is

```
BoundarySyntheticSplitRatioSourceKeyLawOrPDEC
AND NonBoundaryPrefixRecordJumpSourceKeyLiftOrPDEC
AND InternalPrefixRecordSurvivorPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

The largest remaining structural task is to turn the 47 q-boundary synthetic splits into a source-key ratio law before pushforward. The four non-boundary jumps and one internal survivor are now separate named returns, not hidden parity reservoirs.

8.124 Terminal boundary split ratio obstruction

Theorem 8.298 (Boundary split ratio spectrum)**.** *The* 47 *q-boundary synthetic splits are all genuinely adjacent boundary events:*

$$\texttt{boundary\_adjacency\_closed} = \texttt{true}, \qquad \texttt{q\_boundary\_synthetic\_split\_event\_count} = 47.$$

None is an equal whole-run involution:

$$\texttt{whole\_equal\_pair\_event\_count} = 0.$$

The split patterns are

$$42 \text{ old-consumed/new-residual events}, \qquad 5 \text{ old-residual/new-consumed events}.$$

Proof. The audit filters the prefix-record cancellation events to those satisfying `old_q_end` = `new_q_start`. For each such event it records the two pre-split run masses, their minimum chunk, their residual gap, and the smaller-to-larger ratio. All filtered rows have run distance one. No row has equal old and new run mass, and the consumed-side counts are as displayed. ☐

Proposition 8.299 (Ratio-law obstruction). *The boundary ratio spectrum has*

$$\min \frac{\min(M_{\mathrm{old}}, M_{\mathrm{new}})}{\max(M_{\mathrm{old}}, M_{\mathrm{new}})} = 0.013003592969\dots$$

and

$$\max |M_{\mathrm{old}} - M_{\mathrm{new}}| = 0.861355534983\dots.$$

The total finite boundary residual gap mass is

$$13.480078809654\dots.$$

Thus boundary adjacency alone cannot provide a source-key pairing; one needs an adjacent-run mass ratio law or a source-key residual-flow conservation law.

Proof. For each q-boundary event, the audit computes the exact rational ratio $\min(M_{\mathrm{old}}, M_{\mathrm{new}})/\max(M_{\mathrm{old}}, M_{\mathrm{new}})$ and the exact residual gap $|M_{\mathrm{old}} - M_{\mathrm{new}}|$. Taking the minimum ratio, maximum gap, and total gap over all 47 events gives the displayed values. Since the equal-pair count is zero, these data cannot be upgraded to whole-run involutions without an additional mass-ratio/source-key law. □

Computational-certificate Statement 8.300 (Boundary split ratio obstruction audit). *The machine certificate records*

```
boundary_ratio_spectrum_closed=true
q_boundary_synthetic_split_event_count=47
boundary_adjacency_closed=true
whole_equal_pair_event_count=0
old_consumed_new_residual_event_count=42
old_residual_new_consumed_event_count=5
negative_to_positive_event_count=24
positive_to_negative_event_count=23
boundary_synthetic_split_ratio_source_key_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
split-ratio-obstruction-router.md.
```

Remark 8.301 (Updated boundary gate). The current boundary gate is

```
BoundaryAdjacentRunMassRatioLawOrPDEC
AND BoundaryResidualFlowSourceKeyConservationOrPDEC
AND NonBoundaryPrefixRecordJumpSourceKeyLiftOrPDEC
AND InternalPrefixRecordSurvivorPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

External trace/Kloosterman or Type-II estimates still cannot enter directly: the missing object is a pre-pushforward source-key law explaining the adjacent run mass ratios and residual flows.

8.125 Terminal boundary residual-flow obstruction

Theorem 8.302 (Boundary residual side ledger). *For the same* 47 *q-boundary synthetic splits, the residual side ledger is finite and exact:*

$$\mathtt{residual\_flow\_side\_decomposition\_closed} = \mathtt{true}.$$

The side counts are

$$42 \text{ new-side residual events}, \qquad 5 \text{ old-side residual events}.$$

Their exact total masses are

$$R_{\text{new}} = 13.264539470882\ldots, \qquad R_{\text{old}} = 0.215539338772\ldots.$$

Hence the finite boundary ledger has

$$R_{\text{new}} - R_{\text{old}} = 13.049000132111\ldots.$$

Proof. The audit reuses the exact q-boundary ratio rows. For each row it compares the old and new pre-split masses, records which side retains the residual $|M_{\text{old}} - M_{\text{new}}|$, and sums these rational residuals by side. The displayed values are the exact rational totals rendered in decimal form. □

Proposition 8.303 (No local opposite-side boundary conservation)**.** *The q-boundary residuals do not form a local opposite-side conservation law inside the boundary ledger:*

$$\texttt{finite\_boundary\_local\_opposite\_side\_cancellation\_refuted} = \texttt{true}.$$

Even after atomwise matching of old-side and new-side residual mass, the unmatched residual is

$$13.049000132111\ldots.$$

The direction counts are nearly balanced, 24 *negative-to-positive events against* 23 *positive-to-negative events, but the residual mass by direction is not balanced.*

Proof. The atomwise matching amount is

$$\sum_A \min(R_{\text{new}}(A), R_{\text{old}}(A)).$$

The audit computes this amount as $0.215539338772\ldots$, equal to the total old-side residual mass. Thus all old-side residual mass can be matched, but a new-side residual source of $13.049000132111\ldots$ remains. This proves that the boundary ledger cannot be closed by an internal opposite-side pairing. The separate direction tally only counts event orientation; it does not control the residual masses. □

Computational-certificate Statement 8.304 (Boundary residual-flow obstruction audit)**.** *The machine certificate records*

```
residual_flow_side_decomposition_closed=true
new_residual_side_event_count=42
old_residual_side_event_count=5
new_residual_mass_total=13.264539470882
old_residual_mass_total=0.215539338772
net_new_minus_old_residual_mass=13.049000132111
atomwise_unmatched_residual_mass=13.049000132111
boundary_residual_flow_source_key_conservation_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
residual-flow-obstruction-router.md.
```

Remark 8.305 (Updated residual-flow gate)**.** This removes one cyclic route: the next proof cannot be another search for equal adjacent whole-run pairs or boundary-internal residual conservation. The narrowed gate is

```
BoundaryDominantNewResidualSourceLawOrPDEC
AND BoundaryResidualTransportToNonBoundaryInternalReturnsOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND BoundaryResidualFlowSourceKeyConservationOrPDEC
AND NonBoundaryPrefixRecordJumpSourceKeyLiftOrPDEC
AND InternalPrefixRecordSurvivorPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

External Kloosterman, Type-II, and sieve theorems become relevant only after this source-key residual family has been constructed or returned as a named PDEC obstruction.

8.126 Terminal boundary old-residual return alignment

Theorem 8.306 (Old residual return alignment)**.** *The five old-side q-boundary residuals are exactly accounted for by the source-key partition outside the q-boundary split branch:*

$$\mathit{old\_residual\_return\_alignment\_closed} = \mathit{true}.$$

More precisely,

$$\mathit{old\_residual\_event\_count} = 5, \qquad \mathit{nonboundary\_record\_jump\_event\_count} = 4, \qquad \mathit{internal\_survivor\_re}$$

The total old residual mass equals the nonboundary plus internal obstruction mass:

$$R_{\mathrm{old}} = 0.215539338772\ldots = \mathit{nonboundary\_plus\_internal\_obstruction\_mass}.$$

Proof. For each old-side boundary residual the audit forms the key

$$(\texttt{atom\_key}, \texttt{old\_run\_id}, \texttt{boundary\_q}).$$

For each nonboundary record jump it forms

$$(\texttt{atom\_key}, \texttt{old\_run\_id}, \texttt{old\_q\_end}),$$

and for each internal survivor it forms

$$(\texttt{atom\_key}, \texttt{run\_id}, \texttt{q\_end}).$$

All five old residual keys have matching return keys, and in each case the exact rational mass is identical. Four matches are nonboundary record jumps; one match is the internal survivor. ☐

Proposition 8.307 (Remaining new-side source)**.** *The old-side transport subgate is closed, but this does not close the row/column theorem. The unmatched old-side mass is zero:*

$$\mathit{unmatched\_old\_residual\_return\_mass} = 0.$$

The remaining boundary mass is a new-side source of size

$$\mathit{new\_residual\_mass\_total\_still\_open} = 13.264539470882\ldots.$$

Thus the next non-cyclic target is a dominant new-residual source law, a new-residual return alignment, or a named PDEC.

Computational-certificate Statement 8.308 (Old residual return alignment audit). *The machine certificate records*

```
old_residual_return_alignment_closed=true
old_residual_event_count=5
nonboundary_record_jump_event_count=4
internal_survivor_return_count=1
old_residual_equals_nonboundary_plus_internal_obstruction=true
matched_old_residual_return_mass=0.215539338772
unmatched_old_residual_return_mass=0
new_residual_return_alignment_closed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
old-residual-return-alignment-router.md.
```

Remark 8.309 (Updated old-return gate). The current shortest Prime Matrix gate is now

```
BoundaryDominantNewResidualSourceLawOrPDEC
AND BoundaryNewResidualReturnOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

The nonboundary record jumps and the internal survivor are no longer an unpaid old-side reservoir in the finite ledger; the unpaid object is the dominant new-side residual source.

8.127 Terminal boundary new-residual tail alignment

Theorem 8.310 (New residual tail-return partial alignment). *After the old-side q-boundary residuals have been returned to the nonboundary/internal source-key partition, the new-side residuals admit an exact terminal-tail subalignment:*

$$\texttt{new\_residual\_tail\_alignment\_partial\_closed} = \texttt{true}.$$

Among the 42 *new-side residual events, exactly* 6 *are matched by terminal tail survivors. Their total matched mass is*

$$1.580044997219\ldots.$$

Proof. The audit compares each new-side residual key with the terminal tail survivor key using the common atom, run, boundary prime q, and exact rational mass. Six keys agree exactly, including the two left extra-shell returns, two left selected-terminal returns, one right extra-shell return, and one right selected-terminal return. In each case the rational mass on the residual side is identical to the rational mass on the tail-survivor side. □

Proposition 8.311 (Bulk new-residual source and tail overhang remain). *The tail-return subalignment is not a row/column proof. The remaining new-side bulk residual is*

$$\texttt{new\_residual\_unmatched\_after\_tail\_event\_count} = 36, \qquad \texttt{new\_residual\_unmatched\_after\_tail\_mass} = 1$$

There is also one unmatched right selected-terminal tail survivor of mass

$$0.831750175347\ldots.$$

Consequently

$$\texttt{all\_new\_residual\_return\_alignment\_closed} = \texttt{false}, \qquad \texttt{row\_column\_unconditional\_closed} = \texttt{false}.$$

Computational-certificate Statement 8.312 (New residual tail alignment audit)**.** *The machine certificate records*

```
previous_residual_flow_side_decomposition_closed=true
previous_old_residual_return_alignment_closed=true
new_residual_event_count=42
tail_survivor_count=7
new_residual_tail_matched_event_count=6
new_residual_tail_alignment_partial_closed=true
new_residual_tail_matched_mass=1.580044997219
new_residual_unmatched_after_tail_event_count=36
new_residual_unmatched_after_tail_mass=11.684494473663
tail_survivor_unmatched_count=1
tail_survivor_unmatched_mass=0.831750175347
all_new_residual_return_alignment_closed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
new-residual-tail-alignment-router.md.
```

Remark 8.313 (External and group-theoretic entry after tail alignment)**.** The current shortest Prime Matrix gate is now

```
BoundaryBulkNewResidualSourceLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Trace/Kloosterman, Type-II, affine-sieve, or thin-group expansion inputs can enter only after the 36 bulk new-residual events and the one overhang have been organized into an admissible averaged signed family, a finite group orbit family with expansion, or a named PDEC obstruction. The present finite tail-return ledger supplies that sharper target; it does not itself break the Phi–LPF parity barrier.

8.128 Terminal boundary bulk carry-chain normal form

Theorem 8.314 (Bulk new-residual carry-chain normal form)**.** *The* 42 *new-side boundary residual events admit an atomwise carry-chain normal form. They split into* 11 *carry segments on* 7 *atoms, with*

$$\texttt{carry\_transition\_count} = 31, \qquad \texttt{carry\_break\_count} = 4.$$

For every carry transition inside a segment,

$$\texttt{old\_mass(next)} = \texttt{residual\_mass(previous)}$$

as exact rational numbers. Hence the 36 *unmatched bulk events are not independent scatter; they are roots or interiors of this finite carry normal form.*

Proof. For each atom the audit sorts the new-side residual rows by new run id. A transition from one row to the next is declared a carry transition exactly when the next old run id equals the previous new run id and the next old mass equals the previous residual mass. This gives 31 exact transitions. The remaining noninitial discontinuities are the four recorded carry breaks. The segment partition covers all 42 new-side residual rows, and the six previously matched tail returns occur at segment terminal rows. ☐

Proposition 8.315 (Source law still open after carry compression). *The carry normal form is a finite structural compression, not a row/column proof:*

$$\mathtt{bulk\_carry\_chain\_normal\_form\_closed} = \mathtt{true}, \qquad \mathtt{row\_column\_unconditional\_closed} = \mathtt{false}.$$

The open objects are now the 11 *segment roots, the* 4 *carry breaks, and the single right selected-terminal tail overhang. A proof must supply a uniform segment-root source law, a named PDEC, or an admissible averaged trace/Type-II/group-orbit family.*

Computational-certificate Statement 8.316 (Bulk carry-chain normal form audit). *The machine certificate records*

```
new_residual_event_count=42
bulk_unmatched_new_residual_event_count=36
atom_count=7
carry_segment_count=11
carry_transition_count=31
carry_break_count=4
tail_closed_segment_count=6
open_segment_count=5
all_carry_transitions_exact=true
all_tail_matched_events_are_segment_terminal=true
bulk_carry_chain_normal_form_closed=true
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bulk-carry-chain-normal-form-router.md.
```

Remark 8.317 (Updated carry-chain gate). The current shortest Prime Matrix gate is now

```
BoundaryBulkCarrySegmentRootSourceLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

The spectral and group-theoretic frontier is therefore sharply located: one must first turn the carry-segment roots into a completed signed family or a finite group orbit with expansion. Current trace/Kloosterman, Type-II, and thin-group tools do not act directly on the finite ledger alone.

Theorem 8.318 (Carry-break source-packet reduction). *In the terminal boundary carry-chain normal form, the four carry breaks are not four independent source obligations. They split into two bridge/unit packets:*

$$4 \text{ breaks} = 2 \text{ unit old-return echoes} + 2 \text{ bridge-root debts.}$$

The two unit breaks have debt exactly equal to already aligned old-side nonboundary-record-jump return masses. The two bridge roots occur at the `new_q_start` *endpoint of those old returns. Thus the carry-break layer is reduced to the two bridge-root debt source laws, together with the previous right-tail overhang.*

Proof. The machine certificate reads the four breaks from the bulk carry-chain normal form and the five old-return rows from the old-residual return alignment. For each break it compares

$$\text{debt} = \text{previous residual mass} - \text{root old mass}.$$

Two breaks have `run_gap`=1, and their debts are exactly

$$\frac{20701}{308911} \quad \text{and} \quad \frac{15500}{325141},$$

the old-return masses at $q = 569$ and $q = 599$. The corresponding bridge roots are at $q = 577$ and $q = 607$, precisely the recorded `new_q_start` endpoints of those returns. The remaining two breaks have `run_gap`=3 and carry the open bridge-root debt mass

$$0.209830550963\ldots.$$

Therefore the finite source-packet reduction is closed, but the bridge-root source law is not. □

Computational-certificate Statement 8.319 (Carry-break source-packet audit)**.** *The machine certificate records*

```
carry_break_count=4
unit_old_return_echo_break_count=2
bridge_root_debt_break_count=2
paired_bridge_unit_packet_count=2
unmatched_unit_break_count=0
all_unit_echo_breaks_source_aligned_to_old_returns=true
all_bridge_roots_packetized_with_following_unit_echo=true
carry_break_source_packet_reduction_closed=true
bridge_root_source_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
carry-break-source-packet-router.md.
```

Remark 8.320 (Updated bridge-root gate)**.** The current shortest Prime Matrix gate is now

```
BoundaryBridgeRootDebtSourceLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

This is a non-cyclic narrowing of the source problem. It still does not construct the averaged signed trace, Type-II, or finite group orbit family needed for external spectral or group-theoretic inputs.

Theorem 8.321 (Bridge-root q-spine microtemplate)**.** *The two bridge-root debts in the carry-break source-packet reduction lie on a single selected-terminal q-spine. Both occur in the left $P = 739$, packet 2842, selected-terminal pair $m = 757, 761$. Both bridge roots are preceded by the same local run template:*

$$\textit{negative } A1/D0 \textit{ carry } 2 \textit{ singleton} \quad \textit{then} \quad \textit{positive } A0/D1 \textit{ carry } 7 \textit{ singleton.}$$

Moreover the first unit root $q = 607$ is the second bridge root, so the spine is

$$577 \longrightarrow 607 \longrightarrow 631.$$

Proof. The audit reads the two bridge/unit packets from the carry-break source certificate and the run templates from the terminal phase-variation ledger. For $m = 757$, the bridge microtemplate is

$$r2:\ q569 \to 571,\ A1/D0,\ c = 2 \quad \text{then} \quad r3:\ q571 \to 577,\ A0/D1,\ c = 7.$$

For $m = 761$, it is

$$r6:\ q599 \to 601,\ A1/D0,\ c = 2 \quad \text{then} \quad r7:\ q601 \to 607,\ A0/D1,\ c = 7.$$

Thus both bridge roots share the same AD-singleton microtemplate. The first packet has bridge/unit roots $577, 607$, while the second has $607, 631$. This proves the finite q-spine microtemplate ledger. The source law for the spine remains open. ☐

Computational-certificate Statement 8.322 (Bridge-root q-spine audit)**.** *The machine certificate records*

```
bridge_root_packet_count=2
all_bridge_roots_share_ad_singleton_template=true
unit_to_bridge_pivot_alignment_closed=true
shared_pivot_q=607
m_gap_between_bridge_packets=4
root_micro_q_shift=30
bridge_to_unit_q_gaps=[30,24]
bridge_root_qspine_microtemplate_closed=true
bridge_root_qspine_source_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-qspine-microtemplate-router.md.
```

Remark 8.323 (Updated q-spine gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootADSingletonQSpineSourceLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

The external trace/Kloosterman, Type-II, and thin-group tools still require an admissible averaged signed family or a genuine finite group orbit built from this spine; the finite microtemplate itself is not enough.

Theorem 8.324 (Bridge-root q-spine Beatty margin)**.** *Let $m = P + r$, $q' = q + g$,*

$$a = \left\lfloor \frac{qr}{P} \right\rfloor, \qquad D = qr - aP, \qquad s = \left\lfloor \frac{D + gr}{P} \right\rfloor.$$

For each of the four AD-singleton micro-transitions in the bridge-root q-spine, the phase numerator satisfies

$$\Delta_{\text{num}} = (\lambda' - \lambda)qq' + P(sq - ag),$$

where $\lambda, \lambda' \in \{1, 2\}$ are the two-lift numerators $A(q) = \lambda q - D(q)$. In the finite spine the four integer rows are

m	$q \to q'$	r	a	s	$sq - ag$	Δ_{num}
757	$569 \to 571$	18	13	0	-26	-19214
757	$571 \to 577$	18	13	1	493	34860
761	$599 \to 601$	22	17	0	-34	-25126
761	$601 \to 607$	22	17	1	499	3954.

Thus the two pre-bridge A-singletons are pure negative P-multiples, while the two bridge-old D-singletons have positive Beatty margins.

Proof. Write $q' = q + g$, $m = P + r$, and $qm = kP + D$. Since $qm = qP + qr$, we have $k = q + \lfloor qr/P \rfloor = q + a$ and $D = qr - aP$. The carry recurrence gives

$$D' = D + gm - cP, \qquad c = g + s.$$

With $A(q) = \lambda q - D$ and $A(q') = \lambda' q' - D'$,

$$\begin{aligned} A(q')q - A(q)q' &= (\lambda' - \lambda)qq' - D'q + Dq' \\ &= (\lambda' - \lambda)qq' + P(cq - kg) \\ &= (\lambda' - \lambda)qq' + P(sq - ag). \end{aligned}$$

Substitution of the four audited AD-singleton transitions gives the table. The second D-singleton margin is only 3954, so this is a finite factorization of the source problem, not a uniform source law. □

Computational-certificate Statement 8.325 (Bridge-root q-spine Beatty margin audit)**.** *The machine certificate records*

```
micro_transition_count=4
beatty_numerator_identity_closed=true
A_singleton_negative_pure_P_multiple_closed=true
D_singleton_positive_margin_closed=true
bridge_root_qspine_beatty_margin_closed=true
bridge_root_uniform_beatty_margin_source_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-qspine-beatty-margin-router.md.
```

Remark 8.326 (Updated Beatty-margin gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootADSingletonBeattyMarginSourceLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

The external trace/Kloosterman, Type-II, and thin-group tools still require a completed averaged family or a genuine finite group orbit. The Beatty-margin identity is a pointwise source factorization, not by itself a parity-breaking estimate.

Theorem 8.327 (Bridge-root endpoint slack reduction)**.** *For the AD-singleton rows in the bridge-root q-spine, the A-singleton microsteps satisfy*

$$\Delta_{\text{num}} = -Pag < 0,$$

while every D-singleton microstep satisfies the endpoint slack identity

$$\Delta_{\text{num}} = q\big(P - q - g(1 + r)\big) + gD.$$

In the audited spine the two D-singleton endpoint slacks are 54 *and* 0, *and the zero-slack row is*

$$m = 761, \qquad q = 601, \qquad q' = 607, \qquad D = 659,$$

so its positive margin is exactly $6 \cdot 659 = 3954$.

Proof. The A-singleton case has $s = 0$ and $\lambda' = \lambda$ in the Beatty-margin identity, hence

$$\Delta_{\text{num}} = P(0 \cdot q - ag) = -Pag < 0.$$

For the D-singleton case, $s = 1$ and $\lambda' - \lambda = -1$. Since $D = qr - aP$, the previous identity gives

$$\begin{aligned}\Delta_{\text{num}} &= -q(q+g) + P(q - ag) \\ &= q\big(P - q - g(1+r)\big) + g(qr - aP) \\ &= q\big(P - q - g(1+r)\big) + gD.\end{aligned}$$

The finite audit records endpoint slacks 54 and 0 for the two D-singleton rows, with residual $D > 0$ in both cases. Thus positivity in the audited rows follows from endpoint slack nonnegativity plus residual nonvanishing. The uniform endpoint slack law remains open. □

Computational-certificate Statement 8.328 (Bridge-root endpoint slack audit)**.** *The machine certificate records*

```
bridge_root_endpoint_slack_reduction_closed=true
D_singleton_min_endpoint_slack=0
D_singleton_min_positive_margin=3954
D_singleton_zero_slack_rows=['m761 q601->607']
bridge_root_uniform_endpoint_slack_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-endpoint-slack-router.md.
```

Remark 8.329 (Updated endpoint-slack gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootEndpointSlackNonnegativeLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

This replaces a generic Beatty-margin source law by the smaller requirement that the relevant endpoint slack $P - q - g(1 + r)$ be nonnegative, or else that a negative-slack row be named as PDEC.

Theorem 8.330 (Bridge-root moving endpoint barrier reduction)**.** *In every audited D-singleton bridge-root microstep, put*

$$q_{\text{br}} = q' = q + g, \qquad q_{\text{bar}} = P - gr.$$

Then the endpoint slack is exactly the moving endpoint barrier distance

$$P - q - g(1 + r) = q_{\text{bar}} - q_{\text{br}}.$$

For the two audited bridge-root rows the data are

transition	q_{br}	q_{bar}	$q_{\text{bar}} - q_{\text{br}}$
$m757,\ q571 \to 577$	577	631	54
$m761,\ q601 \to 607$	607	607	0.

Both bridge roots and both moving barriers lie on the same q-spine

$$577 \longrightarrow 607 \longrightarrow 631,$$

and the second row is exact barrier contact.

Proof. The D-singleton bridge root is the next q-node, so $q_{\mathrm{br}} = q + g$. Therefore

$$P - q - g(1 + r) = P - gr - (q + g) = q_{\mathrm{bar}} - q_{\mathrm{br}}.$$

The finite certificate checks that both values q_{br} and q_{bar} belong to the audited q-spine, that $q_{\mathrm{br}} \leq q_{\mathrm{bar}}$ in the two bridge rows, and that the endpoint slack distances are 54 and 0. Thus the audited endpoint-slack nonnegativity is reduced to a finite moving-barrier order ledger. The uniform moving-barrier order law remains open. □

Computational-certificate Statement 8.331 (Bridge-root moving endpoint barrier audit)**.** *The machine certificate records*

```
bridge_root_moving_endpoint_barrier_reduction_closed=true
q_spine_nodes=[577, 607, 631]
endpoint_slack_equals_barrier_distance_closed=true
all_barriers_lie_on_qspine=true
all_bridge_roots_lie_on_qspine=true
finite_bridge_root_barrier_order_closed=true
zero_barrier_contact_count=1
bridge_root_uniform_barrier_order_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-moving-endpoint-barrier-router.md.
```

Remark 8.332 (Updated moving-barrier gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootMovingEndpointBarrierOrderLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

This replaces endpoint slack nonnegativity by the sharper order problem $q_{\mathrm{br}} \leq P - gr$, or else by a named moving-barrier PDEC.

Theorem 8.333 (Bridge-root q-spine index-gap reduction)**.** *For the audited bridge-root rows, the moving endpoint barrier order is equivalent to a nonnegative q-spine index gap. With*

$$Q = (577, 607, 631), \qquad \Delta Q = (30, 24),$$

the endpoint slack is the adjacent-gap sum from the bridge-root index to the moving-barrier index:

$$\text{slack} = \sum_{i=i_{\mathrm{br}}}^{i_{\mathrm{bar}}-1} (Q_{i+1} - Q_i).$$

The two audited rows have index gaps

transition	i_{br}	i_{bar}	$i_{\mathrm{bar}} - i_{\mathrm{br}}$	slack
$m757,\ q571 \to 577$	0	2	2	$30 + 24 = 54$
$m761,\ q601 \to 607$	1	1	0	0.

Thus the thin second row is exact q-spine index contact.

Proof. The previous moving-barrier certificate shows that bridge roots and barriers belong to the same ordered q-spine Q. Therefore

$$q_{\text{bar}} - q_{\text{br}} = Q_{i_{\text{bar}}} - Q_{i_{\text{br}}} = \sum_{i=i_{\text{br}}}^{i_{\text{bar}}-1} (Q_{i+1} - Q_i)$$

whenever $i_{\text{br}} \le i_{\text{bar}}$. In the finite audit the first row uses the two adjacent gaps $607 - 577 = 30$ and $631 - 607 = 24$, while the second row has $i_{\text{br}} = i_{\text{bar}} = 1$. Hence endpoint slack nonnegativity has been reduced to the finite index-order ledger. The uniform q-spine index-order law remains open. □

Computational-certificate Statement 8.334 (Bridge-root q-spine index-gap audit)**.** *The machine certificate records*

```
bridge_root_qspine_index_gap_reduction_closed=true
q_spine_gap_vector=[30, 24]
qspine_index_gaps=[2, 0]
endpoint_slack_equals_qspine_gap_sum_closed=true
finite_qspine_index_order_closed=true
zero_index_contact_count=1
bridge_root_uniform_qspine_index_order_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-qspine-index-gap-router.md.
```

Remark 8.335 (Updated q-spine index gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootQSpineIndexBarrierOrderLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

This replaces moving-barrier distance positivity by the discrete assertion $i_{\text{br}} \le i_{\text{bar}}$ on the named q-spine, or else by a named index-barrier PDEC.

Theorem 8.336 (Bridge-root q-spine pivot-enclosure reduction)**.** *The audited q-spine index order is certified by the shared pivot*

$$q_* = 607, \qquad i_* = 1.$$

For each audited bridge-root row,

$$i_{\text{br}} \le i_* \le i_{\text{bar}},$$

and the endpoint slack splits as

$$\text{slack} = \sum_{i=i_{\text{br}}}^{i_*-1} (Q_{i+1} - Q_i) + \sum_{i=i_*}^{i_{\text{bar}}-1} (Q_{i+1} - Q_i).$$

Concretely,

transition	q_{br}	q_*	q_{bar}	slack
$m757,\ q571 \to 577$	577	607	631	$30 + 24 = 54$
$m761,\ q601 \to 607$	607	607	607	$0 + 0 = 0.$

Thus the thin second row is exact pivot contact.

Proof. The previous index-gap certificate imports the q-spine

$$Q = (577, 607, 631).$$

The earlier microtemplate certificate identifies 607 as the shared pivot: the first packet unit root is the second packet bridge root. Therefore the finite index-order inequality is equivalent in the audited rows to enclosing this shared pivot. The first row has indices $0 \le 1 \le 2$, and its slack splits into the left gap $607 - 577 = 30$ and the right gap $631 - 607 = 24$. The second row has indices $1 = 1 = 1$, hence zero left and right gap. This proves the finite pivot-enclosure reduction. The uniform pivot-enclosure law remains open. □

Computational-certificate Statement 8.337 (Bridge-root q-spine pivot-enclosure audit)**.** *The machine certificate records*

```
bridge_root_qspine_pivot_enclosure_reduction_closed=true
shared_pivot_q=607
shared_pivot_index=1
finite_pivot_enclosure_closed=true
endpoint_slack_equals_pivot_gap_sum_closed=true
exact_pivot_contact_count=1
bridge_root_uniform_qspine_pivot_enclosure_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-boundary-
bridge-root-qspine-pivot-enclosure-router.md.
```

Remark 8.338 (Updated q-spine pivot gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

This replaces the index-order assertion by the sharper statement that the shared pivot remains enclosed between the bridge root and the moving barrier, or else by a named pivot-enclosure PDEC.

Theorem 8.339 (Right-tail overhang excess decomposition)**.** *The single right selected-terminal tail overhang is not an independent residual pool. For the audited atom*

$$\text{atom} = (\text{right} : 1887 : \text{selected\_ terminal} : m773),$$

the selected-terminal negative variation excess decomposes exactly as

$$\frac{174209}{205013} = \frac{3642}{202379} + \frac{179065}{215287}.$$

The first summand is the internal survivor already paid by old-residual return alignment. The second summand is precisely the final negative run

$$q = 461 \to 467, \qquad \text{run} = 4,$$

whose signed delta is $-179065/215287$.

Proof. The source-key partition certificate identifies two $m773$ survivor fragments: the internal survivor $q = 449 \to 457$, of mass 3642/202379, and the tail survivor $q = 461 \to 467$, of mass 179065/215287. The old-side return-alignment certificate pays the internal survivor exactly. The terminal variation certificate gives the $m773$ selected-terminal negative excess as 174209/205013. Direct rational comparison gives

$$\frac{174209}{205013} - \frac{3642}{202379} = \frac{179065}{215287},$$

and the terminal run ledger identifies the right-hand side with the final negative run $q = 461 \to 467$. This proves the finite decomposition. The uniform final-tail payment law remains open. □

Computational-certificate Statement 8.340 (Right-tail overhang excess audit)**.** *The machine certificate records*

```
negative_excess_equals_internal_plus_tail=true
tail_overhang_equals_excess_after_internal_return=true
internal_survivor_old_return_paid=true
final_run_tail_matches=true
right_tail_overhang_excess_decomposition_closed=true
right_tail_final_negative_run_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-right-tail-overhang-
excess-decomposition-router.md.
```

Remark 8.341 (Updated final-tail gate)**.** The current shortest Prime Matrix gate becomes

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND RightSelectedTerminalFinalNegativeRunExcessPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

The old right-tail overhang label is therefore replaced by a sharper final negative-tail payment law or named PDEC.

Theorem 8.342 (Final negative run endpoint-collar reduction)**.** *The final negative run in the preceding theorem is a two-edge terminal endpoint-collar wrap debt. More precisely, for*

$$\text{atom} = (\text{right} : 1887 : \text{selected\_terminal} : m773),$$

the final path is

$$q : 461 \to 463 \to 467, \qquad A : 417 \to 87 \to 34, \qquad D : 44 \to 376 \to 433.$$

Both edges have turn word

```
negative_Awrap1_Dwrap0_L1to1
```

with gaps $2, 4$ *and carries* $2, 5$*. The middle phase* $87/463$ *telescopes away, and the exact remaining mass is*

$$\frac{179065}{215287} = \frac{417}{461} - \frac{34}{467} = 1 - \frac{44}{461} - \frac{34}{467}.$$

Proof. The phase-turn certificate gives the two final transitions

$$\frac{87}{463} - \frac{417}{461} = -\frac{152964}{213443}, \qquad \frac{34}{467} - \frac{87}{463} = -\frac{24887}{216221}.$$

Adding them cancels the middle term 87/463 and gives

$$\frac{34}{467} - \frac{417}{461} = -\frac{179065}{215287}.$$

Since $417 = 461 - 44$, the positive tail mass is equivalently

$$\frac{417}{461} - \frac{34}{467} = 1 - \frac{44}{461} - \frac{34}{467}.$$

This is exactly the final-tail mass recorded in the variation and right-tail overhang certificates. The proof is finite and does not supply a uniform payment law for all terminal double-A-wrap endpoint collars. □

Computational-certificate Statement 8.343 (Final negative-run endpoint-collar audit). *The machine certificate records*

```
q_path=[461,463,467]
A_path=[417,87,34]
D_path=[44,376,433]
two_edge_sum_matches_variation_and_tail=true
endpoint_telescoping_closed=true
middle_phase_cancels=true
endpoint_collar_debt_formula_closed=true
double_awrap_same_lift_word_closed=true
right_tail_final_negative_run_endpoint_collar_reduction_closed=true
terminal_double_awrap_endpoint_collar_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-right-tail-final-negative-run-
endpoint-collar-router.md.
```

Remark 8.344 (Updated terminal-collar gate). The shortest Prime Matrix gate is sharpened to

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalDoubleAwrapEndpointCollarPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the former final-tail payment problem is now a terminal double-A-wrap endpoint-collar payment or exclusion problem.

Theorem 8.345 (Terminal double-A-wrap sibling q-spine kernel). *The terminal double-A-wrap endpoint-collar debt of $m773$ is aligned with the already-paid sibling $m769$. The two selected-terminal atoms*

$$\text{right} : 1887 : \text{selected\_terminal} : m769, \qquad \text{right} : 1887 : \text{selected\_terminal} : m773$$

share the same terminal path $461 \to 463 \to 467$, the same gaps $2, 4$, the same carries $2, 5$, and the same two turn words

```
negative_Awrap1_Dwrap0_L1to1.
```

Moreover the $m773$ final tail has the exact sibling-kernel decomposition

$$\frac{179065}{215287} = \frac{123221}{205013} + \frac{36420}{202379} + \frac{10926}{215287},$$

or, after factoring the row prime $P = 607$,

$$\frac{179065}{215287} = 607\left(\frac{203}{439 \cdot 467} + \frac{60}{439 \cdot 461} + \frac{18}{461 \cdot 467}\right).$$

Proof. The tail-alignment ledger pays the $m769$ sibling tail by the new-residual return

$$\frac{123221}{205013}.$$

The variation ledger gives the $m769$ final collar mass as $168139/215287$ and the $m773$ final collar mass as $179065/215287$. Therefore

$$\frac{168139}{215287} - \frac{123221}{205013} = \frac{36420}{202379} = 10 \cdot \frac{3642}{202379},$$

where $3642/202379$ is the $m773$ internal survivor unit already isolated in the source-key partition. Also

$$\frac{179065}{215287} - \frac{168139}{215287} = \frac{10926}{215287} = 3 \cdot \frac{3642}{215287}.$$

Adding the paid sibling tail, the ten internal units, and the three endpoint units gives the claimed identity. Finally,

$$123221 = 203 \cdot 607, \quad 36420 = 60 \cdot 607, \quad 10926 = 18 \cdot 607, \quad 179065 = 295 \cdot 607,$$

which gives the displayed P-scaled q-spine kernel form. □

Computational-certificate Statement 8.346 (Sibling q-spine kernel audit)**.** *The machine certificate records*

```
same_q_path=true
same_turn_words=true
same_gap_carry=true
paid_sibling_tail_closed=true
sibling_gap_is_ten_internal_units=true
endpoint_offset_is_three_endpoint_units=true
target_kernel_identity_closed=true
p_scaled_kernel_identity_closed=true
terminal_double_awrap_sibling_qspine_kernel_closed=true
terminal_double_awrap_sibling_qspine_kernel_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-double-awrap-
sibling-qspine-kernel-router.md.
```

Remark 8.347 (Updated sibling-kernel gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalDoubleAwrapSiblingQSpineKernelPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the remaining terminal collar problem is a P-scaled q-spine kernel payment or exclusion problem, not an isolated endpoint fluctuation.

Theorem 8.348 (Terminal sibling q-spine integer balance)**.** *The P-scaled sibling q-spine kernel in the preceding theorem is equivalent to a single integer balance on the right-side q-spine*

$$439, \quad 461, \quad 467.$$

Precisely,

$$\frac{295}{461 \cdot 467} = \frac{203}{439 \cdot 467} + \frac{60}{439 \cdot 461} + \frac{18}{461 \cdot 467},$$

or, after clearing denominators,

$$295 \cdot 439 = 203 \cdot 461 + 60 \cdot 467 + 18 \cdot 439 = 129505.$$

Moreover the endpoint-collar coefficient for both $m769$ and $m773$ is given by the same formula

$$c = \frac{461 \cdot 467 - D_{\text{start}} \cdot 467 - A_{\text{end}} \cdot 461}{607}.$$

For $m769$ this gives

$$\frac{461 \cdot 467 - 21 \cdot 467 - 81 \cdot 461}{607} = 277,$$

and for $m773$ it gives

$$\frac{461 \cdot 467 - 44 \cdot 467 - 34 \cdot 461}{607} = 295.$$

The offset coefficient is the corresponding endpoint difference

$$\frac{(21 - 44) \cdot 467 + (81 - 34) \cdot 461}{607} = 18.$$

Proof. The previous certificate gives the normalized sibling-kernel identity

$$\frac{179065}{215287} = 607 \left(\frac{203}{439 \cdot 467} + \frac{60}{439 \cdot 461} + \frac{18}{461 \cdot 467} \right)$$

and $179065 = 295 \cdot 607$, while

$$215287 = 461 \cdot 467, \qquad 205013 = 439 \cdot 467, \qquad 202379 = 439 \cdot 461.$$

Dividing by 607 gives the displayed three-denominator identity. Clearing the common denominator $439 \cdot 461 \cdot 467$ gives the integer balance

$$295 \cdot 439 = 203 \cdot 461 + 60 \cdot 467 + 18 \cdot 439.$$

For the endpoint coefficients, the terminal signatures recorded by the ledger are

$$(D_{\text{start}}, A_{\text{end}}) = (21, 81) \quad \text{for } m769$$

and

$$(D_{\text{start}}, A_{\text{end}}) = (44, 34) \quad \text{for } m773.$$

Substitution gives the two coefficients 277 and 295. Subtracting the two endpoint formulas gives the offset identity, whose quotient by 607 is 18. $\square$

Computational-certificate Statement 8.349 (Sibling q-spine integer-balance audit). *The machine certificate records*

```
q_spine_for_balance=[439,461,467]
q_path_from_double_awrap=[461,463,467]
denominators_closed=true
normalized_identity_closed=true
integer_balance_closed=true
endpoint_coefficients_closed=true
endpoint_offset_formula_closed=true
terminal_sibling_qspine_integer_balance_closed=true
terminal_sibling_qspine_integer_balance_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-sibling-
qspine-integer-balance-router.md.
```

Remark 8.350 (Updated integer-balance gate). The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalSiblingQSpineIntegerBalancePaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the remaining terminal payment problem is no longer a generic rational three-denominator kernel; it is an integer-balance payment or exclusion problem on the local right-side q-spine.

Theorem 8.351 (Terminal sibling q-spine primitive gap drift). *The terminal sibling q-spine integer balance is equivalent to a primitive gap-drift identity. Writing*

$$461 = 439 + 22, \qquad 467 = 439 + 28,$$

the balance

$$295 \cdot 439 = 203 \cdot 461 + 60 \cdot 467 + 18 \cdot 439$$

is equivalent to

$$(295 - 203 - 60 - 18) \cdot 439 = 22 \cdot 203 + 28 \cdot 60,$$

that is,

$$14 \cdot 439 = 22 \cdot 203 + 28 \cdot 60.$$

After division by the common q-spine gap factor 2*, this becomes*

$$7 \cdot 439 = 11 \cdot 203 + 14 \cdot 60.$$

Finally, since $203 = 7 \cdot 29$ *and* $14 = 7 \cdot 2$*, the same finite identity has the* 7*-peeled form*

$$439 = 11 \cdot 29 + 2 \cdot 60.$$

Proof. Substitute $461 = 439 + 22$ and $467 = 439 + 28$ into the integer-balance identity:

$$295 \cdot 439 = 203(439 + 22) + 60(439 + 28) + 18 \cdot 439.$$

Collecting the 439-terms on the left gives

$$(295 - 203 - 60 - 18) \cdot 439 = 22 \cdot 203 + 28 \cdot 60.$$

The coefficient defect is $295 - 203 - 60 - 18 = 14$. Both q-spine gaps have common factor 2, so division by 2 gives

$$7 \cdot 439 = 11 \cdot 203 + 14 \cdot 60.$$

The left defect is the LPF-threshold value 7; moreover $203 = 7 \cdot 29$ and $14 = 7 \cdot 2$. Dividing by this common 7-factor yields the final peeled identity $439 = 11 \cdot 29 + 2 \cdot 60$. □

Computational-certificate Statement 8.352 (Sibling q-spine gap-drift audit)**.** *The machine certificate records*

```
q_spine_for_balance=[439,461,467]
q_gaps_from_prefix=[22,28]
primitive_gap_vector=[11,14]
offset_cancels_from_gap_drift=true
gap_drift_identity_closed=true
primitive_gap_drift_identity_closed=true
lpf_factor_peeling_closed=true
terminal_sibling_qspine_gap_drift_closed=true
terminal_sibling_qspine_gap_drift_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-sibling-
qspine-gap-drift-router.md.
```

Remark 8.353 (Updated primitive gap-drift gate)**.** The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalSiblingQSpinePrimitiveGapDriftPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the remaining terminal payment problem is no longer just an integer balance; it is a primitive q-spine gap-drift payment or exclusion problem.

Theorem 8.354 (Terminal sibling q-spine wheel-residue carrier)**.** *The 7-peeled primitive gap-drift identity is equivalent to a 30-wheel residue-carrier identity. Namely,*

$$439 = 11 \cdot 29 + 2 \cdot 60$$

can be written as

$$439 = 11 \cdot 29 + 4 \cdot 30.$$

The non-wheel residue is carried by $11 \cdot 29$, since

$$439 \equiv 11 \cdot 29 \equiv 19 \pmod{30}.$$

The remaining lift is an integral wheel lift,

$$4 = 2 \cdot (60/30).$$

Proof. The previous theorem gives $439 = 11 \cdot 29 + 2 \cdot 60$. Since $60 = 2 \cdot 30$, the second summand is $4 \cdot 30$, giving the displayed wheel-residue identity. Furthermore $439 = 14 \cdot 30 + 19$, while $11 \cdot 29 = 319 = 10 \cdot 30 + 19$. Thus the two terms have the same residue 19 modulo 30, and their quotient difference is $14 - 10 = 4$. This equals $2 \cdot (60/30)$, which is the right primitive gap after the 7-peel multiplied by the number of wheel units in the middle kernel. □

Computational-certificate Statement 8.355 (Sibling q-spine wheel-residue audit). *The machine certificate records*

```
q_prefix=439
wheel_modulus=30
residue_carrier=319
wheel_neutral_mass=120
residue_match_closed=true
wheel_lift_closed=true
carrier_identity_closed=true
terminal_sibling_qspine_wheel_residue_closed=true
terminal_sibling_qspine_wheel_residue_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-sibling-
qspine-wheel-residue-router.md.
```

Remark 8.356 (Updated wheel-residue gate). The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalSiblingQSpineWheelResidueCarrierPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the remaining terminal payment problem is no longer just a primitive gap-drift identity; it is a 30-wheel residue-carrier payment or exclusion problem.

Theorem 8.357 (Terminal sibling q-spine wheel-gap lock). *The terminal* 30*-wheel residue carrier is locked to the common double-A-wrap q-gap path. The terminal sibling atoms have*

$$\text{q\_gap\_path} = [2, 4], \qquad \text{carry\_path} = [2, 5].$$

The first gap satisfies

$$2 = 60/30 = \text{right gap after the 7-peel},$$

and the second gap satisfies

$$4 = \text{wheel lift height} = 2^2.$$

Moreover the right-side q-spine is generated by the first gap:

$$461 - 439 = 11 \cdot 2, \qquad 467 - 439 = 14 \cdot 2 = 7 \cdot 2 \cdot 2,$$

and the carry path is locked by

$$[2, 5] = [2, 4 + 1].$$

Proof. The wheel-residue certificate gives middle wheel units $60/30 = 2$, right gap after the 7-peel equal to 2, and wheel lift height equal to 4. The terminal sibling-kernel certificate gives the common double-A-wrap q-gap path $[2, 4]$ and carry path $[2, 5]$ for $m769$ and $m773$. Hence the first terminal gap is exactly the wheel-unit/right-gap value 2, and the second terminal gap is the wheel lift height $4 = 2^2$. The right-side q-spine identities follow by direct subtraction:

$$461 - 439 = 22 = 11 \cdot 2, \qquad 467 - 439 = 28 = 14 \cdot 2 = 7 \cdot 2 \cdot 2.$$

Finally, the carry path is $[2, 5]$, which is $[2, 4 + 1]$. $\square$

Computational-certificate Statement 8.358 (Sibling q-spine wheel-gap-lock audit). *The machine certificate records*

```
right_side_q_spine=[439,461,467]
terminal_q_gap_path=[2,4]
terminal_carry_path=[2,5]
first_gap_lock_closed=true
second_gap_lock_closed=true
q_spine_generation_closed=true
carry_gap_lock_closed=true
terminal_sibling_qspine_wheel_gap_lock_closed=true
terminal_sibling_qspine_wheel_gap_lock_payment_law_proved=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-sibling-
qspine-wheel-gap-lock-router.md.
```

Remark 8.359 (Updated wheel-gap-lock gate). The current shortest Prime Matrix gate is now

```
BridgeRootQSpinePivotEnclosureLawOrPDEC
AND TerminalSiblingQSpineWheelGapLockPaymentOrPDEC
AND BoundaryAdjacentRunMassRatioLawOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily
AND AdmissibleFiniteGroupOrbitExpansionOrThinGroupSieveFamily.
```

Thus the remaining terminal payment problem is no longer just a 30-wheel residue carrier; it is a terminal wheel-gap-lock payment or exclusion problem.

Theorem 8.360 (Affine odd Euler normalization does not supply a half-error). *Let $m = 2n + 1$ and let p be an odd prime. Then*

$$m \equiv 0 \pmod p \iff n \equiv (p-1)/2 \pmod p.$$

Consequently the odd-prime zero class on the m-axis is pulled back exactly to one forbidden affine class on the n-axis. For every finite odd-prime cutoff Y, the Euler product main term on the full interval of length $2X$, with the $p = 2$ factor included, is

$$2X\left(1-\frac{1}{2}\right)\prod_{\substack{3\le p\le Y\\ p\ \text{prime}}}\left(1-\frac{1}{p}\right) = X\prod_{\substack{3\le p\le Y\\ p\ \text{prime}}}\left(1-\frac{1}{p}\right).$$

Thus the factor two is a parity normalization, not evidence that the finite Euler product truncation error is one half of the main term.

Proof. The congruence $2n + 1 \equiv 0 \pmod p$ is equivalent to $2n \equiv -1 \pmod p$. Since p is odd, $2^{-1} \equiv (p + 1)/2 \pmod p$, so $n \equiv -(p + 1)/2 \equiv (p - 1)/2 \pmod p$. This proves the exact pullback. The displayed Euler-product identity is then only the $p = 2$ factor normalizing the full interval of length $2X$ to the odd affine axis of length X. □

Computational-certificate Statement 8.361 (Affine odd Euler normalization audit). *The machine certificate records*

```
affine_forbidden_class_bijection_verified=true
finite_euler_product_full_with_p2_equals_affine_odd_main=true
finite_euler_product_half_error_claim_supported=false
p2_normalization_explains_factor_two=true
lpf_bucket_identity_verified=true
phi_lpf_iteration_route_closes_parity_barrier=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-affine-odd-
euler-normalization-router.md.
```

Remark 8.362 (LPF synchronization and the remaining signed gate)**.** For a prime P and $n = kP + (P-1)/2$,

$$2n + 1 = P(2k + 1).$$

If $k \geq 1$ and $2k+1$ has no prime factor below P, then P is the least prime factor of $2n+1$. The endpoint $k = 0$ gives the prime $2n+1 = P$, so it is an endpoint exception to the composite wording but not to the least-prime-factor identity. This synchronization is useful for aligning Phi and LPF ledgers; it still does not create the signed cofactor saving needed to cross the parity barrier. The active gate is

```
AffineOddLiftOnlyNormalizesParityNoSignedSaving
AND TerminalSiblingQSpineWheelGapLockPaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

Theorem 8.363 (Affine endpoint first-hit LPF partition)**.** *Let $M = 2X + 1$, and let m range over odd integers $3 \leq m \leq M$. If $p = \mathrm{LPF}(m)$, then exactly one of the following alternatives holds:*

$$m = p$$

or

$$m = p(2k+1), \qquad k \geq 1, \qquad \mathrm{LPF}(2k+1) \geq p.$$

In the second case, with $n = (m-1)/2$,

$$n = kp + (p-1)/2.$$

Thus the increasing LPF peel partitions the affine odd axis into endpoint prime leaks and composite LPF tails.

Proof. If m is prime then $p = m$, giving the endpoint case. Otherwise write $m = pc$, where $c > 1$ is odd. Since p is the least prime factor of m, no prime below p divides c; hence $\mathrm{LPF}(c) \geq p$. Writing $c = 2k+1$ gives $k \geq 1$. Finally

$$n = (m-1)/2 = (p(2k+1)-1)/2 = kp + (p-1)/2.$$

The two cases are disjoint and exhaustive. □

Computational-certificate Statement 8.364 (Affine endpoint LPF first-hit audit)**.** *The machine certificate records*

```
endpoint_prime_leak_separated=true
lpf_tail_composite_partition_closed=true
zero_class_duplicate_overcount_positive=true
cofactor_parity_mixture_present_in_tail=true
prime_extraction_from_lpf_tail_proved=false
signed_payload_or_von_mangoldt_weight_constructed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-affine-endpoint-
lpf-first-hit-router.md.
```

Remark 8.365 (Why this still does not break parity)**.** The first-hit partition removes the $k = 0$ ambiguity: $m = p$ is a prime emission, while $m = p(2k + 1)$ with $k \geq 1$ is a composite LPF tail. But the tail still contains both semiprime and higher-composite cofactor layers. Moreover, raw zero-class counts overcount composites unless they are assigned by their least prime factor. Hence the next non-cyclic object remains a signed payload, a von-Mangoldt-like cofactor weight, a completed Type-II/trace family, or a named PDEC:

```
EndpointPrimeLeakSeparatedFromLPFTailButPrimeExtractionStillParityBlocked
AND TerminalSiblingQSpineWheelGapLockPaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

Proposition 8.366 (Affine LPF first-hit von Mangoldt lift)**.** *For every odd integer $m \geq 3$,*

$$\Lambda(m) = \sum_{d|m} \mu(d) \log(m/d).$$

Equivalently, $\Lambda(m) = \log p$ if $m = p^a$ is a prime power and $\Lambda(m) = 0$ otherwise. Therefore the affine first-hit partition can be lifted exactly to a von-Mangoldt payload, but the lift is a global Mobius divisor signed payload. It is not an unsigned local count on the LPF(m) *bucket.*

Proof. The displayed identity is the standard Mobius inversion of $\log n = \sum_{d|n} \Lambda(d)$. If $m = p^a$, the nonzero terms in the Mobius sum occur only for $d = 1, p$, giving $a \log p - (a-1) \log p = \log p$. If m has at least two distinct prime factors, the squarefree divisor sum cancels the logarithmic contribution in each prime coordinate, hence the value is zero. Combining this with the first-hit decomposition above separates endpoint primes from composite tails, but the cancellation on non-prime-power tails uses all Mobius divisor signs. □

Computational-certificate Statement 8.367 (Affine LPF first-hit von Mangoldt lift audit)**.** *The machine certificate records*

```
mobius_von_mangoldt_identity_closed=true
lpf_first_hit_identity_imported_and_verified=true
tail_prime_power_leak_present=true
nonprimepower_tail_cancelled_only_by_mobius_divisor_sum=true
lpf_local_unsigned_count_sufficient_for_prime_extraction=false
global_divisor_signed_payload_required=true
admissible_typeii_or_trace_family_constructed=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-affine-lpf-
first-hit-von-mangoldt-lift-router.md.
```

Remark 8.368 (Updated non-cyclic gate after the von-Mangoldt lift)**.** The lift answers the natural follow-up to the endpoint LPF partition. A prime extraction mechanism must now build a signed divisor family, a Type-II/trace family to which external Kloosterman technology applies, or a pointwise θ-in-progressions substitute at the P^2 scale. Raw LPF bucket counts and finite Euler-product normalization still do not break the parity barrier. The active gate is

```
VonMangoldtLiftRequiresGlobalDivisorSignedPayloadNotLPFLocalCount
AND PointwiseThetaAPPositivityAtP2OrAdmissibleSignedDivisorPayloadTypeIIFamily
AND TerminalSiblingQSpineWheelGapLockPaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving.
```

Proposition 8.369 (Exact LPF bucket count is a Legendre-Phi count)**.** *Let $N \geq 4$ and let $p \leq \sqrt{N}$ be prime. Put*

$$C_p(N) = \#\{n \leq N : n \text{ composite, } \mathrm{LPF}(n) = p\}.$$

Then

$$C_p(N) = \Phi(\lfloor N/p \rfloor;\ q < p) - 1,$$

where $\Phi(x; q < p)$ counts positive integers $m \leq x$ divisible by no prime $q < p$. Consequently the continuous main term is

$$(N/p - p) \prod_{q<p} (1 - 1/q) = \frac{N - p^2}{p} \prod_{q<p} (1 - 1/q),$$

not $(N - p^2) \prod_{q \leq p} q^{-1}$ except for the accidental low-prime cases $p = 2, 3$.

Proof. If $\mathrm{LPF}(n) = p$, write $n = pm$. Since n is composite and p is least, $m \leq \lfloor N/p \rfloor$, m is divisible by no prime below p, and $m \neq 1$. Conversely, every $m \leq \lfloor N/p \rfloor$ with no prime factor below p and $m \neq 1$ gives a composite $pm \leq N$ whose least prime factor is p. This proves the exact formula. The displayed main term is the interval-length approximation to the same rough-number count. For each $q < p$, the sieve keeps all nonzero residue classes modulo q, of density $1 - 1/q$; multiplying by $1/q$ would keep only one residue class. □

Computational-certificate Statement 8.370 (LPF bucket count formula audit)**.** *The machine certificate records*

```
exact_lpf_bucket_identity_closed=true
user_reciprocal_density_formula_supported=false
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-lpf-bucket-
count-formula-audit.md.
```

Remark 8.371 (Why the corrected count still does not break parity)**.** The correction is useful because it closes the arithmetic meaning of the $\mathrm{LPF} = p$ bucket: it is an exact Legendre-Phi rough-number bucket, not a full reciprocal-density Euler product. But it is still an unsigned sieve count. It does not distinguish prime emission from semiprime and higher-composite tails after the first-hit assignment. The active gate is

```
ExactLPFBucketCountIsLegendrePhiNotReciprocalDensity
AND UnsignedLPFBucketCountStillParityBlind
AND VonMangoldtLiftRequiresGlobalDivisorSignedPayloadNotLPFLocalCount
AND PointwiseThetaAPPositivityAtP2OrAdmissibleSignedDivisorPayloadTypeIIFamily.
```

Proposition 8.372 (LPF owner class is divisibility, not survival)**.** *For the composite bucket $C_p(N)$, the small-prime conditions split into two different local roles. For every prime $q < p$ one must avoid the class*

$$n \equiv 0 \pmod{q},$$

which contributes the survival density $1 - 1/q$. For the owner prime p, however, one must hit the class

$$n \equiv 0 \pmod{p},$$

which contributes the divisibility density $1/p$. Hence the n-axis continuous main term is

$$(N-p^2)\frac{1}{p}\prod_{q<p}(1-1/q),$$

whereas

$$(N-p^2)\prod_{q\le p}(1-1/q)$$

is larger by the factor $p-1$.

Proof. The condition $\mathrm{LPF}(n)=p$ means exactly that $p\mid n$ and no prime below p divides n. The first condition selects one residue class modulo p, while each second condition deletes one residue class modulo q. Since these congruence conditions are independent modulo the squarefree product of the primes involved, the local density is

$$\frac{1}{p}\prod_{q<p}(1-1/q).$$

Replacing the p-factor by $1-1/p$ changes the required zero class modulo p into the forbidden nonzero-class survival condition and multiplies the correct density by $p-1$. □

Computational-certificate Statement 8.373 (LPF inclusive survival formula audit)**.** *The machine certificate records*

```
exact_lpf_bucket_identity_closed=true
inclusive_survival_formula_supported=false
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-lpf-bucket-
inclusive-survival-formula-audit.md.
```

Remark 8.374 (Updated LPF density grammar)**.** The corrected local grammar is: survival classes for primes $q<p$, and an owner divisibility class for p. This fixes the density bookkeeping but does not create a signed prime-extraction mechanism. The active gate is

```
PrimeDivisibilityClassIsOneOverPNotOneMinusOneOverP
AND ExactLPFBucketCountIsLegendrePhiNotInclusiveSurvivalProduct
AND UnsignedLPFBucketCountStillParityBlind
AND VonMangoldtLiftRequiresGlobalDivisorSignedPayloadNotLPFLocalCount.
```

Proposition 8.375 (Legendre-Phi truncation error is periodic boundary error)**.** *Let p be prime and put*

$$W_{<p}=\prod_{q<p}q.$$

For every $x\ge 0$,

$$\Phi(x;q<p)=\left\lfloor\frac{x}{W_{<p}}\right\rfloor\varphi(W_{<p})+R_p(x \bmod W_{<p}),$$

where $R_p(r)$ counts the reduced residues $1\le a\le r$ modulo $W_{<p}$. Hence

$$C_p(N)=\Phi(\lfloor N/p\rfloor;q<p)-\Phi(p-1;q<p)$$

equals

$$(\lfloor N/p \rfloor - p + 1)\frac{\varphi(W_{<p})}{W_{<p}} + B_p(\lfloor N/p \rfloor) - B_p(p-1),$$

with

$$|B_p(t)| \le \varphi(W_{<p}).$$

Thus the finite Legendre-Phi truncation error is a primorial-periodic boundary term, not a universal half-main term.

Proof. The condition counted by $\Phi(x; q < p)$ depends only on the residue class of the integer modulo $W_{<p}$. Each complete block of length $W_{<p}$ therefore contributes exactly $\varphi(W_{<p})$ admissible classes. The remaining incomplete block contributes $R_p(x \bmod W_{<p})$. Defining

$$B_p(t) = \Phi(t; q < p) - t\frac{\varphi(W_{<p})}{W_{<p}}$$

gives the displayed formula after subtracting the endpoint $\Phi(p-1; q < p)$. Since the incomplete block contains at most $\varphi(W_{<p})$ admissible residue classes and its continuous main is also between 0 and $\varphi(W_{<p})$, the stated bound follows. □

Computational-certificate Statement 8.376 (Legendre-Phi periodic truncation error audit)**.** *The machine certificate records*

```
legendre_phi_periodic_truncation_error_closed=true
exact_lpf_bucket_identity_closed=true
half_main_truncation_error_claim_supported=false
truncation_error_is_periodic_residue_boundary=true
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-legendre-phi-
periodic-truncation-error-audit.md.
```

Remark 8.377 (External frontier after the periodic error correction)**.** The periodic correction closes the local truncation grammar but supplies no signed prime-extraction payload. Milićević–Qin–Wu bilinear Kloosterman estimates, Zheng simultaneous arithmetic-progressions input, Runbo Li large modulus arithmetic progressions, Wright trilinear Kloosterman fractions, and Becker–Breuillard spectral-gap/anti-concentration tools remain candidate inputs only after an admissible signed trace, Type-II, or finite-group orbit family has been constructed. The active gate is

```
LegendrePhiTruncationErrorIsPeriodicBoundaryNotHalfMain
AND UnsignedLPFBucketCountStillParityBlind
AND VonMangoldtLiftRequiresGlobalDivisorSignedPayloadNotLPFLocalCount
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

Proposition 8.378 (LPF endpoint singleton fixes the exact bucket formula)**.** *Let $p \le \sqrt{N}$ be prime and put $X = \lfloor N/p \rfloor$. Then the exact composite LPF bucket is*

$$C_p(N) = \Phi(X; q < p) - 1.$$

Equivalently,

$$C_p(N) = \Phi(X; q < p) - \Phi(p-1; q < p),$$

because

$$\Phi(p-1; q<p) = 1.$$

Consequently

$$(N-p^2)\frac{1}{p}\prod_{q<p}(1-1/q)$$

is only the continuous Euler-product main term. It is not the exact bucket count unless the floor, the endpoint singleton, and the periodic boundary term are also supplied.

Proof. The formula $C_p(N) = \Phi(X; q < p) - 1$ follows from writing $n = pm$: the cofactor $m \le X$ must be indivisible by every prime $q < p$, and the single cofactor $m = 1$ gives the prime $n = p$, not a composite bucket element. For the endpoint form, every integer $2 \le m \le p-1$ has a prime factor below p, so the only p-rough integer in $[1, p-1]$ is 1. Hence $\Phi(p-1; q < p) = 1$, proving the equivalence. The final displayed expression replaces X by N/p and replaces the exact periodic count by its continuous density; it is therefore a main term, not an exact identity. $\square$

Computational-certificate Statement 8.379 (LPF exact bucket endpoint equivalence audit)**.** *The machine certificate records*

```
exact_endpoint_singleton_fixed=true
minus_one_and_endpoint_forms_equivalent=true
exact_lpf_bucket_identity_closed=true
continuous_euler_main_is_exact_count=false
floor_endpoint_and_periodic_boundary_required=true
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-exact-bucket-
endpoint-equivalence-audit.md.
```

Remark 8.380 (Updated LPF exact-count gate)**.** The corrected exact formula layer is now fixed:

$$\Phi(\lfloor N/p \rfloor; q<p) - 1 = \Phi(\lfloor N/p \rfloor; q<p) - \Phi(p-1; q<p).$$

This removes the endpoint ambiguity but does not construct a signed payload. The active gate is

```
ExactLPFBucketEndpointSingletonFixed
AND FloorEndpointAndPeriodicBoundaryRetained
AND ContinuousEulerMainNotExactCount
AND UnsignedLPFBucketCountStillParityBlind
AND VonMangoldtLiftRequiresGlobalDivisorSignedPayloadNotLPFLocalCount.
```

Proposition 8.381 (LPF pure-power compression of the von Mangoldt lift)**.** *Let $m \ge 2$, put $p = \mathrm{LPF}(m)$, and write*

$$m = p^a r, \qquad a \ge 1, \qquad p \nmid r.$$

Then

$$\Lambda(m) = \begin{cases} \log p, & r = 1, \\ 0, & r > 1. \end{cases}$$

Equivalently, the pointwise von Mangoldt lift may be computed by stripping all powers of the LPF owner and testing whether the residual cofactor is 1.

Proof. The von Mangoldt function is $\log \ell$ exactly on prime powers ℓ^b, $b \geq 1$, and is 0 otherwise. Since p is the least prime factor of m, the factorization $m = p^a r$ with $p \nmid r$ has $r = 1$ if and only if m is a pure power of its LPF owner. This gives $\Lambda(m) = \log p$ in the pure-power case and 0 in the mixed-prime case. The identity is pointwise equivalent to the standard Mobius divisor formula

$$\Lambda(m) = \sum_{d|m} \mu(d) \log(m/d),$$

but it is not a linear divisor family suitable for immediate averaging. □

Computational-certificate Statement 8.382 (LPF von Mangoldt pure-power compression audit)**.** *The machine certificate records*

```
lpf_pure_power_compression_closed=true
lambda_mass_reconstructed_from_endpoint_plus_prime_power_tail=true
mixed_composites_cancel_to_zero_pointwise=true
prime_power_tail_separated=true
pure_power_selector_supplies_additive_signed_distribution_family=false
pointwise_ap_theta_lower_bound_proved=false
admissible_typeii_or_trace_family_constructed=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-von-mangoldt-
pure-power-compression-audit.md.
```

Remark 8.383 (Updated von Mangoldt lift gate)**.** This closes a pointwise correction: the lift from LPF ownership to Λ does not have to remain written as the full Mobius divisor cube. However, the pure-power selector is a nonlinear factorization predicate. It is not an additive signed Type-II family, a trace/Kloosterman family, a simultaneous arithmetic-progression family, or a finite-group orbit. Thus the external frontier inputs of Milićević–Qin–Wu, Zheng, Runbo Li, Wright, Pascadi, and Becker–Breuillard still require a separately constructed admissible signed family before they can be applied. The active gate is

```
LPFPurePowerVonMangoldtCompressionClosed
AND PrimePowerTailSeparated
AND PurePowerSelectorNotAnAdditiveSignedDistributionFamily
AND PointwiseAPThetaLowerBoundOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.384 (Strict-row prime-power tail absorption)**.** *Let P be prime and let $1 \leq k < P$. Put*

$$I_{P,k} = (kP, (k+1)P) \cap \mathbb{Z}.$$

Then

$$\psi(I_{P,k}) = \theta(I_{P,k}) + \sum_{\substack{p^a \in I_{P,k} \\ a \geq 2}} \log p.$$

Consequently,

$$\theta(I_{P,k}) > 0 \quad \Longleftrightarrow \quad \psi(I_{P,k}) > \sum_{\substack{p^a \in I_{P,k} \\ a \geq 2}} \log p.$$

Moreover the prime-power tail obeys the deterministic bound

$$\sum_{\substack{p^a \in I_{P,k} \\ a \geq 2}} \log p \leq \log P \sum_{a \geq 2} \left(\left\lfloor ((k+1)P - 1)^{1/a} \right\rfloor - \left\lfloor (kP)^{1/a} \right\rfloor \right).$$

Proof. The first identity is the defining decomposition of ψ into prime and higher prime-power contributions on the row. The equivalence follows because $\theta(I_{P,k}) = \psi(I_{P,k})-$ the displayed prime-power tail and θ is nonnegative. For the bound, if $p^a \in I_{P,k}$ with $a \geq 2$, then $p < P$ because the row lies below P^2. For each fixed exponent a, the number of possible prime bases is at most the number of integer bases b satisfying

$$kP < b^a < (k+1)P,$$

which is the corresponding difference of integer roots. Multiplying by $\log P$ and summing over a gives the claim. □

Computational-certificate Statement 8.385 (LPF prime-power tail absorption audit)**.** *The machine certificate records*

```
psi_theta_tail_identity_all_samples=true
prime_power_tail_bound_all_samples=true
psi_tail_absorption_equivalent_to_prime_presence_all_samples=true
prime_power_tail_absorption_threshold_closed=true
pointwise_psi_row_lower_bound_beyond_tail_proved=false
pointwise_theta_ap_lower_bound_proved=false
admissible_signed_typeii_or_trace_family_constructed=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-prime-power-
tail-absorption-audit.md.
```

Remark 8.386 (Updated prime-extraction load)**.** This reduces the final prime-extraction load from direct rowwise $\theta(I_{P,k}) > 0$ to a rowwise lower bound for $\psi(I_{P,k})$ that exceeds the explicit prime-power tail. It does not prove such a pointwise ψ bound. Average theorems for Λ, Kloosterman/Type-II tools, or simultaneous arithmetic-progressions inputs still need a zero-exception rowwise upgrade or an admissible signed trace family. The active gate is

```
LPFPurePowerVonMangoldtCompressionClosed
AND PrimePowerTailAbsorptionThresholdClosed
AND NeedPointwisePsiRowLowerBoundBeyondPrimePowerTail
AND PointwiseAPThetaLowerBoundOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.387 (Strict-row prime-power tail is sublinear)**.** *Let $P \geq 3$ be prime, $1 \leq k < P$, and*

$$I_{P,k} = (kP, (k+1)P) \cap \mathbb{Z}.$$

Then the composite prime-power tail satisfies

$$\sum_{\substack{p^a \in I_{P,k} \\ a \geq 2}} \log p \leq \log P \left((\sqrt{2}-1)\sqrt{P} + 1 + (\lfloor \log_2(P^2-1) \rfloor - 2)\big((2^{1/3}-1)P^{1/3}+1\big) \right).$$

In particular this tail is

$$O\big(\sqrt{P}\log P + P^{1/3}(\log P)^2\big) = o(P).$$

Consequently, any pointwise lower bound

$$\psi(I_{P,k}) \geq \eta P$$

with fixed $\eta > 0$, uniformly in all strict rows and all sufficiently large P, would imply $\theta(I_{P,k}) > 0$ for all those rows.

Proof. For every prime power $p^a \in I_{P,k}$ with $a \geq 2$, the base satisfies $p < P$, so $\log p \leq \log P$. It remains to bound the possible integer bases. For $a = 2$,

$$\lfloor ((k+1)P - 1)^{1/2} \rfloor - \lfloor (kP)^{1/2} \rfloor \leq (\sqrt{2} - 1)\sqrt{P} + 1,$$

because the square-root increment $\sqrt{k+1} - \sqrt{k}$ is largest at $k = 1$. For each $a \geq 3$, concavity gives

$$\lfloor ((k+1)P - 1)^{1/a} \rfloor - \lfloor (kP)^{1/a} \rfloor \leq (2^{1/a} - 1)P^{1/a} + 1 \leq (2^{1/3} - 1)P^{1/3} + 1.$$

There are at most $\lfloor \log_2(P^2 - 1) \rfloor - 2$ exponents $a \geq 3$. Multiplying the resulting base-count bound by $\log P$ proves the displayed inequality and the $o(P)$ estimate. Combining this with the preceding prime-power tail absorption equivalence gives the final implication. □

Computational-certificate Statement 8.388 (LPF prime-power tail sublinear threshold audit). *The machine certificate records*

```
actual_tail_bound_all_samples=true
sublinear_tail_bound_all_samples=true
prime_power_tail_sublinear_threshold_closed=true
positive_proportion_psi_would_close_rows_eventually=true
known_short_interval_input_reaches_sqrt_window=false
pointwise_psi_row_positive_proportion_proved=false
admissible_signed_typeii_or_trace_family_constructed=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-prime-power-
tail-sublinear-threshold-audit.md.
```

Remark 8.389 (Updated square-root-scale psi gate). This closes a real load estimate: the prime-power correction is not a main-term obstruction after the LPF counting formula has been corrected. It is uniformly $o(P)$. The remaining obstacle is sharper: one needs a pointwise positive-proportion lower bound for ψ on every interval of length $P = x^{1/2}$ near $x = P^2$, or an admissible signed Type-II/trace family that returns such rowwise load. Current short-interval and averaged AP/Kloosterman inputs remain insufficient without that additional construction. The active gate is

```
LPFPurePowerVonMangoldtCompressionClosed
AND PrimePowerTailAbsorptionThresholdClosed
AND PrimePowerTailSublinearThresholdClosed
AND NeedPointwisePsiRowPositiveProportionAtSqrtScale
AND PointwiseAPThetaLowerBoundOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.390 (Runbo Li short intervals close the low-row band). *Assume the external theorem of Runbo Li that, for all sufficiently large X, the interval $[X - X^{13/25}, X]$ contains a prime. Let P be prime and $1 \leq k < P$, with $X = (k+1)P$ sufficiently large. If*

$$((k+1)P)^{13/25} < P,$$

equivalently

$$(k+1)^{13} < P^{12},$$

then the strict row

$$(kP, (k+1)P)$$

contains a prime. Thus the external short-interval theorem closes the low-row band $k + 1 < P^{12/13}$, up to the integer floor and the external threshold.

Proof. Put $X = (k+1)P$. The displayed inequality gives

$$X - X^{13/25} > X - P = kP.$$

By Li's theorem, $[X - X^{13/25}, X]$ contains a prime. Since $X = (k+1)P$ is composite for $1 \le k < P$, that prime is strictly below X. Hence it lies in $(kP, (k+1)P)$. The equivalence of the two conditions follows by raising to the 25-th power and cancelling P^{13}. □

Computational-certificate Statement 8.391 (Runbo Li low-row band bridge audit). *The machine certificate records*

```
runbo_li_low_row_band_closed=true
top_band_sqrt_scale_gap_remains=true
closes_all_strict_rows=false
phi_lpf_parity_barrier_globally_broken=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-runbo-li-
low-row-band-bridge-audit.md.
```

Remark 8.392 (Updated top-band localization). This is a genuine unconditional external bridge, but it is not a full closure. It closes about $P^{12/13}$ low rows, whose density among the $P-1$ strict rows is $P^{-1/13}$. The dominant top band

$$k + 1 \ge P^{12/13}$$

still requires a square-root-scale pointwise ψ/θ lower bound or an admissible signed Type-II/trace family. The active gate is

```
RunboLiLowRowBandClosedForKPlusOneLessThanPTo12Over13
AND TopBandKPlusOneAtLeastPTo12Over13StillRequiresSqrtScalePointwisePsi
AND PrimePowerTailSublinearThresholdClosed
AND PointwiseAPThetaLowerBoundOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.393 (Fixed $\theta > 1/2$ short intervals close only zero-density low rows). *Assume a pointwise short-interval input which, for all sufficiently large X, guarantees a prime in $[X - X^\theta, X]$, where $\theta > 1/2$ is fixed. Embed it into the strict row*

$$I_{P,k} = (kP, (k+1)P)$$

by taking $X = (k+1)P$. For the guaranteed interval to lie inside $I_{P,k}$, it is necessary and sufficient up to endpoint constants that

$$X^\theta < P,$$

and hence

$$k + 1 < P^{(1-\theta)/\theta}.$$

Since $(1-\theta)/\theta < 1$, every fixed $\theta > 1/2$ input closes only a zero-density family of low rows among the $P-1$ strict rows. The density-one top band remains outside the reach of ordinary fixed-exponent short-interval theorems.

Proof. With $X = (k+1)P$, the interval $[X - X^\theta, X]$ is contained in the strict row if and only if $X - X^\theta > kP$, which is $X^\theta < P$. Substituting $X = (k+1)P$ gives

$$(k+1)^\theta P^\theta < P,$$

or $k + 1 < P^{(1-\theta)/\theta}$. The exponent $(1-\theta)/\theta$ is strictly less than 1 when $\theta > 1/2$, so the closed low-row count is $O(P^{(1-\theta)/\theta}) = o(P)$. This includes the Runbo Li 13/25 bridge as the special case $P^{12/13}$, but also shows that any fixed exponent still above 1/2 leaves a density-one top band. □

Computational-certificate Statement 8.394 (Fixed theta short-interval zero-density band router). *The machine certificate records*

```
all_fixed_theta_gt_half_close_only_zero_density_low_rows=true
theta_half_identified_as_short_interval_lane_threshold=true
density_one_top_band_remains_for_all_fixed_theta_gt_half=true
row_column_unconditional_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-theta-short-
interval-zero-density-band-router.md.
```

Remark 8.395 (Updated short-interval boundary). This rules out a full closure by merely improving the ordinary short-interval exponent while it remains fixed above 1/2. Baker–Harman–Pintz $\theta = 21/40$, Runbo Li $\theta = 13/25$, and any hypothetical fixed $\theta = 0.5001$ input all close only P^α low rows with $\alpha < 1$. Therefore the active gate is

```
AllFixedThetaGreaterThanHalfShortIntervalInputsCloseOnlyZeroDensityLowRows
AND DensityOneTopBandStillRequiresThetaHalfPointwisePsiOrStructuralParityBreak
AND PrimePowerTailSublinearThresholdClosed
AND PointwiseAPThetaLowerBoundOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.396 (Sharp constant gate on the square-root short-interval lane). *Assume a pointwise square-root short-interval input with constant $C > 0$. For a left-endpoint version*

$$(x, x + C\sqrt{x}] \text{ contains a prime,}$$

applied at $x = kP$, the guaranteed interval is contained in $(kP, (k+1)P)$ whenever

$$C^2 k \le P.$$

For a right-endpoint version

$$[X - C\sqrt{X}, X] \text{ contains a prime,}$$

applied at $X = (k+1)P$, the guaranteed interval is contained in the closed row whenever

$$C^2(k+1) \le P.$$

Endpoint equality is harmless, since kP and $(k+1)P$ are composite row endpoints. Consequently, a pointwise input with $C \le 1$ would close every sufficiently large strict row, while any fixed $C > 1$ closes only approximately P/C^2 low rows and leaves a top band of density

$$1 - \frac{1}{C^2}.$$

Proof. For the left-endpoint input, containment is $C\sqrt{kP} \le P$, equivalent to $C^2 k \le P$. For the right-endpoint input, containment is $C\sqrt{(k+1)P} \le P$, equivalent to $C^2(k+1) \le P$. If $C \le 1$, these inequalities hold for all $1 \le k < P$, up to the equality case at the top right endpoint when $C = 1$. Since both row endpoints are multiples of P, any prime supplied by the closed interval must lie in the open row. If $C > 1$, the inequalities restrict k to $k \le P/C^2$, or $k + 1 \le P/C^2$, so the asymptotic closed density is $1/C^2$ and the uncovered top-band density is $1 - 1/C^2$. $\square$

Computational-certificate Statement 8.397 (Sqrt constant threshold router). *The machine certificate records*

```
sqrt_constant_one_pointwise_input_would_close_all_strict_rows=true
fixed_constant_greater_than_one_leaves_positive_density_top_band=true
known_unconditional_C_at_most_one_pointwise_input_available=false
row_column_unconditional_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-sqrt-constant-
threshold-router.md.
```

Remark 8.398 (Updated square-root constant boundary)**.** This is the sharp constant form of the previous $\theta > 1/2$ obstruction. The pure short-interval route does not merely need an exponent equal to $1/2$; it needs the aligned constant $C \leq 1$. A fixed $C > 1$ theorem, even at exact square-root scale, still leaves a positive-density top band. The active gate is

```
SqrtScaleConstantAtMostOnePointwiseInputWouldCloseStrictRows
AND AnyFixedSqrtConstantGreaterThanOneLeavesPositiveDensityTopBand
AND PrimePowerTailSublinearThresholdClosed
AND PointwisePsiAtSharpSqrtScaleOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.399 (Sqrt $C = 1$ equals the prime-indexed Oppermann top row)**.** *For the top strict row $k = P - 1$,*

$$I_{\text{top}} = (P^2 - P, P^2).$$

At the square endpoint $X = P^2$, the right-endpoint square-root interval with constant $C = 1$ is exactly

$$[X - \sqrt{X}, X] = [P^2 - P, P^2].$$

Since both endpoints $P^2 - P = P(P - 1)$ and P^2 are composite, a prime in this closed interval is equivalent to a prime in the open top row. Thus the top-row obstruction is precisely the prime-indexed left Oppermann half

$$\pi(P^2 - 1) - \pi(P^2 - P) \geq 1 \qquad (P \text{ prime}).$$

By contrast, the Legendre square interval decomposes as

$$((P - 1)^2, P^2) = ((P - 1)^2, P^2 - P] \ \cup\ (P^2 - P, P^2),$$

and the two integer halves have the same size $P - 1$. Therefore Legendre's wider interval, or any fixed $C > 1$ square-root input, does not isolate the top row.

Proof. The identity $I_{\text{top}} = (P^2 - P, P^2)$ is the specialization $k = P - 1$ of $(kP, (k + 1)P)$. Since $\sqrt{P^2} = P$, the $C = 1$ right-endpoint interval at $X = P^2$ is $[P^2 - P, P^2]$. The endpoint compositeness removes any closed-endpoint ambiguity. Finally,

$$(P - 1)^2 = P^2 - 2P + 1,$$

so the integer set in $((P - 1)^2, P^2 - P]$ has cardinality $P - 1$, and the integer set in $(P^2 - P, P^2)$ also has cardinality $P - 1$. A Legendre prime may lie entirely in the lower half, so it gives no implication for top-row positivity without an additional localization input. □

Computational-certificate Statement 8.400 (Sqrt-Oppermann top-row alignment router)**.** *The machine certificate records*

```
top_row_equals_prime_indexed_oppermann_left_half=true
sqrt_C_one_right_endpoint_equivalent_to_top_row=true
legendre_interval_splits_into_equal_lower_leak_and_target_halves=true
legendre_wide_square_interval_implies_top_row=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-sqrt-oppermann-
toprow-alignment-router.md.
```

Remark 8.401 (Updated top-row external boundary)**.** This removes a common shortcut. A Legendre-scale result is a $C \approx 2$ right-endpoint square-root input at P^2, and it contains a lower leakage half of the same integer size as the target top row. The remaining top-row external input is not Legendre but prime-indexed Oppermann left, equivalently the sharp $C = 1$ right-endpoint square-root statement at $X = P^2$. The active gate is

```
PrimeIndexedOppermannLeftTopRowOrSharpCOneSqrtInputStillOpen
AND LegendreWideSquareIntervalDoesNotImplyTopRow
AND AnyFixedSqrtConstantGreaterThanOneHasLowerLeakStrip
AND PointwisePsiAtSharpSqrtScaleOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.402 (Oppermann top-row subcore is necessary but not sufficient)**.** *Let P be prime and write $p_+(x)$ for the least prime strictly larger than x. Put*

$$h(x) = p_+(x) - x.$$

Full strict-row positivity for the Prime Matrix row/column target is equivalent to

$$h(kP) < P \qquad (1 \le k < P),$$

or, equivalently,

$$\pi((k+1)P - 1) - \pi(kP) \ge 1 \qquad (1 \le k < P).$$

The prime-indexed Oppermann-left top row is only the single specialization

$$k = P - 1, \qquad h(P^2 - P) < P,$$

that is,

$$\pi(P^2 - 1) - \pi(P^2 - P) \ge 1.$$

Therefore full strict-row positivity implies the top-row Oppermann-left statement, but a top-row input alone is a necessary subcore rather than a full closure of all strict rows.

Proof. The row $I_{P,k} = (kP, (k+1)P)$ contains a prime if and only if the least prime after its left endpoint satisfies $p_+(kP) < (k+1)P$, which is exactly $h(kP) < P$. Requiring this for all $1 \le k < P$ gives the full strict-row positivity statement. Taking $k = P - 1$ gives

$$I_{P,P-1} = (P^2 - P, P^2),$$

which is the prime-indexed Oppermann-left half already identified above. Thus the full statement contains the top row as one of its conjuncts. Conversely, the top-row interval is disjoint from every row $I_{P,k}$ with $k < P - 1$; as a local short-interval or Oppermann-left input it supplies a prime only in the $k = P - 1$ row and gives no containment information for the remaining $P - 2$ strict rows. $\square$

Computational-certificate Statement 8.403 (Oppermann subcore not-full-closure router)**.** *The machine certificate records*

```
row_column_strict_positivity_implies_toprow=true
top_row_input_alone_closes_all_strict_rows=false
top_row_input_is_necessary_not_sufficient=true
all_rows_equivalent_to_prime_gap_bound_h_kP_less_than_P=true
row_column_unconditional_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-oppermann-
subcore-not-full-closure-router.md.
```

Remark 8.404 (Updated necessary-subcore boundary)**.** This prevents a final shortcut. Even if the top-row prime-indexed Oppermann-left statement were supplied externally, the complete row/column claim would still require $h(kP) < P$ for every strict row $1 \leq k < P$, or an internal signed Type-II/trace family that returns this positivity without passing through ordinary unsigned LPF/Phi counts. The active gate is

```
TopRowOppermannLeftIsNecessarySubcoreNotFullClosure
AND FullRowsRequireGapBoundHkPLessThanPForEveryK
AND LPFPhiExactCountsRemainUnsignedParityBlind
AND PointwisePsiAtSharpSqrtScaleOrAdmissibleSignedTypeIIFamilyStillOpen.
```

Proposition 8.405 (Corrected LPF counts route to signed trace, not unsigned closure)**.** *The corrected least-prime-factor bucket identity*

$$C_p(N) = \Phi(\lfloor N/p \rfloor; q < p) - 1 = \Phi(\lfloor N/p \rfloor; q < p) - \Phi(p-1; q < p),$$

with $\Phi(p-1; q < p) = 1$, supplies an exact owner and capacity ledger. Together with the primorial-periodic boundary formula for finite Euler truncation, it does not supply a signed prime-emission theorem. Consequently, after the Oppermann top-row subcore has also been separated from full row closure, the fastest remaining non-cyclic route among the three manuscript claims is not a pure LPF/Phi count, a wheel refinement, or a top-row short-interval input, but the Prime Matrix terminal signed monotone-run payload. Its next mathematical object must be an admissible signed Type-II/trace family, a uniform adjacent-run cancellation theorem, or a named PDEC/SAE/LocalSurvivor return.

Proof. The LPF formula is exact because $n = pm$ has least prime factor p exactly when $m \leq \lfloor N/p \rfloor$ is indivisible by every prime below p, and the single cofactor $m = 1$ gives the prime p, not a composite bucket element. The equality with the endpoint form follows from $\Phi(p-1; q < p) = 1$. The finite Euler-product approximation is periodic modulo the primorial $W_{<p}$, so its error is an endpoint boundary term, not a universal positive half-main saving. Therefore this layer counts owners and rough cofactors, but it does not distinguish primes from semiprime and higher composite tails.

The von Mangoldt lift gives such a distinction only through a global Mobius divisor signed sum. Similarly, the top-row Oppermann-left input controls only the single row $k = P - 1$, while full row closure requires all rows $1 \leq k < P$. Existing external candidates, including short-interval, Kloosterman, Type-II, trilinear, and finite-group spectral inputs, require an admissible internal signed family before they can apply. Among the current project artifacts, the terminal monotone-run ledger is the one that already has signed mass, finite adjacent-run decomposition, and a recorded surplus; its missing step is precisely the uniform signed family or a named return. □

Computational-certificate Statement 8.406 (Corrected-LPF signed-trace breakthrough router)**.** *The machine certificate records*

```
lpf_exact_count_fixed=true
finite_euler_truncation_error_type=primorial_periodic_boundary
von_mangoldt_lift_requires_global_signed_payload=true
top_row_oppermann_necessary_not_sufficient=true
all_external_inputs_require_internal_admissible_family=true
fastest_first_break_candidate=PrimeMatrixTerminalSignedMonotoneRunPayload
row_column_unconditional_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-corrected-lpf-
signed-trace-breakthrough-router.md.
```

Remark 8.407 (Updated signed-trace boundary)**.** Runbo Li type short intervals, large-modulus AP inputs, Kloosterman bilinear estimates, composite-modulus Type-II estimates, trilinear Kloosterman fractions, and spectral finite-group tools remain valuable, but none attaches directly to an unsigned LPF bucket. The active gate is

```
CorrectedLPFExactCountsAndPeriodicErrorsDoNotGivePrimeEmission
AND PureShortIntervalOrTopRowInputsDoNotCloseAllRows
AND FastestPrimeMatrixRouteRequiresSignedTraceOrTypeIIFamily
AND UniformAdjacentRunCancellationOrNamedPDECSAEStillOpen.
```

Proposition 8.408 (Terminal run trace-admissibility contract)**.** *The terminal monotone-run ledger gives a genuine finite signed object, but it is not yet an admissible trace or Type-II family. The ledger records*

$$59 \text{ runs} = 35 \text{ selected} + 24 \text{ extra}$$

with no strict compression inside a monotone run, and its finite adjacent-run decomposition gives

$$1.456565972578\ldots - 0.907719323182\ldots = 0.548846649396\ldots > 0.$$

Thus the finite surplus would pay the extra atom-local survivor after a uniform adjacent-run cancellation theorem. However, direct use of trace function, Kloosterman, composite Type-II, trilinear, AP/Harman, or spectral inputs still requires a forward kernel or coefficient family rather than this finite ledger alone.

Proof. The terminal certificate imports the closed monotone-run ledger and the corrected LPF signed-trace router. It then tests external admissibility against six requirements: a terminal kernel formula $K_P(q, m, \alpha)$, one pre-Cauchy trace key shared by source, orientation, local factor, ExactUV pair and signed value, a uniform family in P, well-factorable Type-II coefficients, conductor or modulus control, and uniform adjacent-run cancellation. Each requirement is presently false. Hence the finite ledger is valid evidence for the next target, but it cannot be promoted to an unconditional row/column theorem or used as the input object of the cited external estimates. □

Computational-certificate Statement 8.409 (Terminal-run trace-admissibility contract router)**.** *The machine certificate records*

```
terminal_finite_signed_run_ledger_closed=true
finite_adjacent_cancellation_decomposition_closed=true
selected_negative_excess_beats_extra_atom_survivor=true
trace_or_typeii_family_admissible_now=false
external_theorems_directly_attach_now=false
all_trace_contracts_proved=false
row_column_unconditional_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-run-
trace-admissibility-contract-router.md.
```

Remark 8.410 (Pinned admissibility gate)**.** The active non-cyclic gate is now

```
TerminalRunTraceAdmissibilityContractPinned
AND FiniteSignedRunLedgerIsNotYetUniformTraceFamily
AND NeedTraceKernelOrTypeIICoefficientFormulaOrNamedPDECSAE
AND UniformAdjacentRunCancellationStillOpen.
```

Proposition 8.411 (Terminal trace kernel reduces to source-key lift)**.** *The terminal trace-kernel interface is not an empty black box. The finite terminal ledger already supplies a formal phase kernel through the signed telescoping of $A(q)/q$ and the prefix-record Jordan reflection. What is still missing is the promotion of that post-pushforward ledger to a pre-Cauchy source-key kernel. Quantitatively, the source-key obstruction is split into* 47 *adjacent q-boundary synthetic splits,* 4 *non-boundary record jumps, one internal survivor fragment, and the unpaid sibling q-spine kernel for the m*773 *terminal packet.*

Proof. The adjacent-run Jordan certificate proves the formal telescoping identity and the maximal-run reconstruction for 126 transitions arranged into 59 runs. The prefix-record certificate then rewrites all 51 cancellation events as deterministic reflection events, but none is a whole-run involution. The source-key partition certificate further separates them into 47 adjacent boundary splits and 4 non-boundary record jumps; it also records one internal survivor. The boundary ratio certificate closes the finite ratio spectrum, with no equal whole-run pairing. Finally, the sibling q-spine certificate closes a finite P-scaled kernel identity but not its payment law. Therefore the first two trace-admissibility contracts reduce to these actual source-key gates, rather than to a directly applicable external trace theorem. □

Computational-certificate Statement 8.412 (Terminal trace-kernel source-key lift router)**.** *The machine certificate records*

```
formal_jordan_phase_kernel_closed=true
prefix_record_reflection_schema_closed=true
source_key_obstruction_partition_closed=true
boundary_ratio_spectrum_closed=true
sibling_qspine_finite_kernel_closed=true
terminal_run_kernel_formula_reduced_to_actual_source_key_gates=true
terminal_run_kernel_formula_proved=false
same_trace_key_source_consistency_proved=false
trace_or_typeii_family_admissible_now=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-terminal-
trace-kernel-source-key-lift-router.md.
```

Remark 8.413 (Next non-cyclic atom)**.** The next primary attack target is now the boundary ratio source-key law:

```
TraceKernelSourceKeyLiftReductionClosed
AND BoundaryRatioSourceKeyLawOrPDEC
AND NonBoundaryRecordJumpSourceKeyLiftOrPDEC
AND InternalSurvivorPDEC
AND TerminalSiblingQSpinePaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND UniformAdjacentRunCancellationStillOpen.
```

Proposition 8.414 (Boundary ratio reduces to q-spine pivot enclosure). *The boundary ratio source-key gate has a further finite reduction. The old-side residuals are exactly returned by the non-boundary jumps and the internal survivor. On the new side, six residuals are terminal-tail returns, and the remaining* 36 *bulk residuals form an atomwise carry-chain normal form with* 11 *carry segments,* 31 *carry transitions, and* 4 *carry breaks. Those breaks split into two unit old-return echoes and two bridge-root debts. The bridge-root debts are then enclosed by the finite q-spine*

$$577, \quad 607, \quad 631,$$

with shared pivot 607*.*

Proof. The boundary residual-flow certificate separates the 47 adjacent boundary splits into 42 new-side and 5 old-side residuals, and refutes local opposite-side cancellation. The old-side certificate matches all 5 old residuals to four non-boundary returns and one internal survivor. The new-side certificate matches 6 residuals to terminal tails and leaves 36 bulk events. The bulk normal-form certificate organizes these events into carry segments; the carry-break certificate reduces the four breaks to two already-accounted unit echoes and two bridge-root debts. Finally, the q-spine pivot certificate proves the finite enclosure of those bridge roots between the q-spine endpoints and the shared pivot. Thus the actual gate is no longer a generic boundary ratio law; it is the uniform q-spine pivot enclosure law, plus the right-tail overhang and sibling q-spine payment. ☐

Computational-certificate Statement 8.415 (Boundary ratio q-spine pivot reduction router). *The machine certificate records*

```
boundary_ratio_source_key_law_reduced_to_qspine_pivot=true
old_residual_side_closed=true
new_residual_tail_alignment_partial_closed=true
bulk_carry_chain_normal_form_closed=true
carry_break_source_packet_reduction_closed=true
bridge_root_qspine_pivot_enclosure_reduction_closed=true
bridge_root_uniform_qspine_pivot_enclosure_law_proved=false
trace_or_typeii_family_admissible_now=false
row_column_unconditional_closed=false.
```

It is recorded at

```
docs/monograph/prime-matrix-phi-lpf-boundary-ratio-
qspine-pivot-reduction-router.md.
```

Remark 8.416 (Updated boundary source-key gate). The active gate is now

```
BoundaryRatioQSpinePivotReductionClosed
AND BridgeRootQSpinePivotEnclosureLawOrPDEC
AND RightSelectedTerminalTailOverhangPDEC
AND TerminalSiblingQSpinePaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

8.129 Three-claim breakthrough-route synthesis

Proposition 8.417 (Non-cyclic selection among the three claims). *As of the present audit layer, none of the three manuscript claims is promoted to an unconditional final theorem. The Prime Matrix row/column line remains the shortest line for genuine local progress, but the next step is no* *longer a wheel refinement or another finite terminal ledger. It must be one of the following three outcomes:*

$$\texttt{ActualSignedPayloadTraceConstructorBeforeAssignmentOrReturn},$$
$$\texttt{MonotoneRunPhaseSavingWithExtraBudgetAbsorption},$$
$$\texttt{PDEC/SAE/LocalSurvivorNamedReturn}.$$

The two-point sieve line remains the second candidate: its external DI/BFI/Kloosterman direction is useful only after the numerator estimate and the denominator floor are proved in the same singular-series convention. The RH contradiction-field line remains a controlled-exit verification package rather than the fastest first-break target.

Status synthesis. The previous terminal phase-variation audit closes the finite positive and negative variation ledger, but it also shows that the selected terminal word is not automatically balanced. Its total variation is approximately

$$17.979169131897,$$

and its net displacement is approximately

$$-1.456565972578,$$

with 35 selected runs and 24 extra-shell runs. Thus a proof still needs either a monotone-run phase-saving cap, an absorption of the extra variation budget, or a return to a named PDEC/SAE/LocalSurvivor defect.

The direct rank-one route is also not closed: the rank-one obstruction is equivalent to sharp pointwise positivity of $\theta(P^2; P, a)$ in every non-zero residue class. Standard GRH-shape errors, Bombieri–Vinogradov averages, complete CRT uniformity, and standard Linnik bounds do not prove this at the P^2 threshold. Therefore the non-cyclic Prime Matrix route must first build an admissible averaged signed-payload family before invoking trace, Kloosterman, or Type-II external inputs.

The synthesis certificate is

```
docs/monograph/three-claims-breakthrough-route-synthesis-20260525.md.
```

It records

```
row_column_unconditional_closed=false
external_lemma_version_unconditional_closed=false
internal_self_contained_closed=false
phi_lpf_parity_barrier_globally_broken=false.
```

□

Remark 8.418 (External theorem boundary)**.** The recent trace, Kloosterman, and composite Type-II inputs of Fouvry–Kowalski–Michel–Sawin, Milicevic–Qin–Wu, Pascadi, and Wright are kept as candidate tools only after an admissible averaged family has been constructed. They do not directly estimate the present finite terminal run ledger or the real reciprocal phase with prime-q Phi–LPF weights. Short-interval prime theorems and bounded-gap theorems likewise do not close the row-by-row or residue-by-residue positivity required here.

8.130 Affine $2n + 1$ Euler–LPF parity diagnostic

Lemma 8.419 (Shifted-residue sieve identity)**.** *Let P be an odd prime and write*

$$n = kP + \frac{P-1}{2}.$$

Then

$$2n + 1 = (2k + 1)P.$$

Thus P is a factor of $2n + 1$, not of n. If $k \geq 1$ and $2k + 1$ has no prime factor below P, then $P = \mathrm{LPF}(2n + 1)$. Moreover, for every odd prime p,

$$p \mid (2n+1) \iff n \equiv \frac{p-1}{2} \pmod{p}.$$

Proof. The first identity follows by direct substitution. The divisibility criterion is the congruence $2n+1 \equiv 0 \pmod p$, and $2^{-1}(-1) \equiv (p-1)/2 \pmod p$ for odd p. The least-prime-factor statement follows because every prime factor of $2k + 1$ is then at least P. □

Theorem 8.420 (Affine Euler-product normalization audit)**.** *In the finite audit for $P = 11, 31, 101, 251, 1009$, with $x = P^2$, the map*

$$m = 2n + 1, \qquad 0 \leq n \leq x,$$

gives an exact bijection between

$$\{m \leq 2x + 1 : \ (m, \prod_{p \leq P} p) = 1, \ m \text{ odd}\}$$

and

$$\{n \leq x : \ n \not\equiv (p-1)/2 \pmod p \text{ for every odd } p \leq P\}.$$

The audit records

```
affine_sieve_bijection_verified_all_samples=true
apparent_half_main_gap_explained_by_missing_p2_all_samples=true
euler_product_half_main_error_proved=false
phi_lpf_parity_barrier_globally_broken=false.
```

Audit interpretation. If the m-side interval of length approximately $2x$ is multiplied only by the odd-prime Euler product, its naive main term is approximately twice the correct odd-space main term. The exact count then appears to miss the naive main term by about one half. This is not a theorem saying that the finite Euler-product truncation error has half-main size. It is the missing $p = 2$, or equivalently the failure to restrict first to the odd sample space. After the 2-adic normalization is included, the m-side main term and the shifted-residue n-side main term agree.

The machine certificate is

```
docs/monograph/prime-matrix-affine-2n-plus-1-euler-lpf-parity-audit.md.
```

□

Remark 8.421 (Remaining gate)**.** The affine identity is useful as a shifted-residue Phi–LPF ledger, but it still counts rough survivors. It does not separate prime $2n + 1$ values from rough composites and therefore does not break the parity barrier. The new open gate is

```
AffineShiftedResidueSieveSignedPayloadConstructorOrReturn
AND PrimeExtractionFrom2nPlus1RoughSurvivorsBeyondParity.
```

8.131 Power-two affine Phi–LPF iteration

Theorem 8.422 (Power-two affine iteration is shifted-residue conjugacy)**.** *For $t \geq 1$, put*

$$m_t = 2^t n + (2^t - 1).$$

For every odd prime p,

$$p \mid m_t \quad \Longleftrightarrow \quad n \equiv -(2^t - 1)(2^t)^{-1} \pmod p.$$

Thus each power-two affine iterate again deletes exactly one residue class modulo each odd prime. It does not create a new sieve dimension or a new Phi–LPF cancellation identity.

Proof. Modulo an odd prime p, 2^t is invertible. Solving

$$2^t n + (2^t - 1) \equiv 0 \pmod p$$

gives the displayed residue class. This is a conjugate of the ordinary zero-residue sieve under an affine change of variables. □

Theorem 8.423 (Power-two affine iteration audit)**.** *In the finite audit for $P = 31, 101, 251, 1009$ and $t = 1, 2, 3, 4$, the shifted residue formula above is verified for every tested P, t. The observed naive Euler-product gap follows the missing 2-adic density $1 - 2^{-t}$. The audit records*

```
all_shifted_residue_formula_verified=true
naive_gap_tracks_two_adic_density=true
iteration_creates_new_phi_lpf_information=false
euler_product_half_main_error_proved=false
phi_lpf_parity_barrier_globally_broken=false.
```

Audit interpretation. For $t = 1$, forgetting the 2-adic restriction gives the apparent half-main gap. For $t = 2, 3, 4$, the same mistake gives the apparent gaps

$$1 - 2^{-t} = 3/4,\ 7/8,\ 15/16.$$

Hence the phenomenon is not a finite Euler-product truncation error. It is the cost of using a full interval main term for a single 2^t-adic residue class. After the correct residue-space normalization, the remaining task is unchanged: one must separate prime values of m_t from affine rough survivors.

The machine certificate is

```
docs/monograph/prime-matrix-affine-power-two-
phi-lpf-iteration-no-gain-audit.md.
```

□

Remark 8.424 (Iterated affine open gate)**.** The iterated affine route is therefore useful only if it produces a genuine signed payload family or returns to a named defect. The current open gate is

```
PowerTwoAffineShiftedResidueSignedPayloadConstructorOrNamedReturn
AND PrimeExtractionFromAffineRoughSurvivorsBeyondParity.
```

8.132 Small-to-large Phi–LPF factor peeling

Theorem 8.425 (Small-to-large factor peeling is an unsigned state ledger)**.** *Fix N. For every composite $n \leq N$, let $p = \mathrm{LPF}(n)$ and write $n = pm$. Then m is p-rough and has a unique nondecreasing prime factor word*

$$m = q_1 q_2 \cdots q_r, \qquad p \leq q_1 \leq \cdots \leq q_r.$$

Consequently the squarefree flag, the Mobius value of m, the Liouville value of m, and the depth parity $r \bmod 2$ are all functions of the closed unsigned factor word. They are not, by themselves, pre-Cauchy signed payload data.

Proof. The existence and uniqueness of the ordered word is the fundamental theorem of arithmetic, with the factors sorted increasingly. Since p is the least prime factor of n, every prime factor of m is at least p. The listed parity states are then read directly from the multiplicities in this word. None of these operations selects a pre-Cauchy source key, an orientation, a local factor, or an ExactUV trace. □

Theorem 8.426 (Small-to-large factor-peeling audit)**.** *The finite audit for $N = 100, 997, 5003, 10000, 30030$ verifies the ordered factor-peeling ledger for all tested composite LPF buckets. For $N = 30030$ it records* 26781 *composite support keys,* 40 *owner buckets,* 69651 *small-to-large factor steps, and maximum depth* 13*. The same audit records*

```
small_to_large_factor_peeling_verified_all_samples=true
mobius_liouville_state_computable_from_factor_word_all_samples=true
peeling_generates_new_precauchy_signed_payload=false
orientation_local_factor_law_proved=false
built_in_signed_pairing_proved=false
row_column_unconditional_closed=false.
```

Audit interpretation. At $N = 30030$, the tail Mobius split is 8367/11306/7108 for positive, negative, and zero, while the Liouville split is 11810/14971. These numbers confirm that the natural parity states are fully visible after factorization. This visibility is not a signed-emission law: it is still a post-factorization label attached to the LPF bucket. Thus small-to-large peeling closes an unsigned boundary but does not break the parity barrier.

The machine certificate is

```
docs/monograph/prime-matrix-phi-lpf-small-to-large-
factor-peeling-signed-state-boundary-router.md.
```

□

Remark 8.427 (Remaining signed gate)**.** The LPF-adjacent remaining gate is

```
PrimitiveOrientationLocalFactorProductLawBeforePushforward
OR BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows.
```

The global strict basis still also requires source entropy, ExactUV multiplicity, model-gap, rate-preservation, and D-structure/Rankin gates.

8.133 Factor-word parity shadows do not give orientation

Theorem 8.428 (Factor-word parity shadow no-go)**.** *The Mobius value, Liouville value, depth parity, and squarefree flag obtained from the LPF factor word are functions only of the unsigned post-factorization word. They do not determine a primitive orientation/local-factor product law before pushforward, nor do they determine a built-in signed coefficient pairing for atomic joint rows.*

Proof. The preceding small-to-large peeling theorem gives a unique unsigned factor word. Every listed parity shadow is a deterministic function of this word and therefore remains unchanged under any formal change of the missing orientation slot. The signed coefficient required by the row-level contract is orientation/local-factor sensitive before Cauchy, Phi, or payment pushforward. Thus a function that reads only the factor word cannot supply the pre-Cauchy source key, branch trace, ExactUV payload, or named return required by the signed law.

The associated audit imports the largest peeling sample $N = 30030$, with 26781 composite support keys, and registers 26781 abstract orientation-twin collisions: the same unsigned word gives the same parity shadow on both formal orientation slots. This is a contract-level obstruction to using the shadow as an orientation-sensitive signed coefficient formula. □

Computational-certificate Statement 8.429 (Factor-word parity shadow audit)**.** *The machine certificate records*

```
factor_word_mobius_shadow_closed=true
factor_word_liouville_shadow_closed=true
depth_parity_shadow_closed=true
shadow_depends_only_on_unsigned_factor_word=true
shadow_lacks_precauchy_source_key=true
shadow_lacks_orientation_branch_trace=true
shadow_lacks_exactuv_payload=true
factor_word_shadow_proves_orientation_local_factor_law=false
factor_word_shadow_proves_builtin_pairing=false
row_column_unconditional_closed=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-factor-word-
parity-shadow-orientation-nogo-router.md.
```

Remark 8.430 (Non-cyclic next target)**.** This removes one natural Mobius/Liouville shortcut. The remaining non-cyclic target is still

```
PrimitiveOrientationLocalFactorProductLawBeforePushforward
OR BuiltInSignedCoefficientPairingClosedFormForAtomicJointRows,
```

or else a newly constructed admissible averaged signed trace/Kloosterman/Type-II family with named returns.

8.134 Bridge-root shared-pivot hinge contract

Theorem 8.431 (Bridge-root shared-pivot hinge reduction)**.** *The finite bridge-root q-spine pivot enclosure in the boundary-ratio source-key route reduces to a shared-pivot hinge contract. In the audited ledger the q-spine is*

$$577,\ 607,\ 631,$$

with shared pivot 607*. The two bridge-root packets satisfy*

$$\text{packet } 1 : 577 \to 607 \to 631, \qquad \text{slack} = 54,$$
$$\text{packet } 2 : 607 = 607 = 607, \qquad \text{slack} = 0.$$

For each packet the endpoint slack is the actual moving-barrier distance

$$\text{slack} = q_{\text{barrier}} - q_{\text{bridge}} = P - q - g(1 + r).$$

Thus the finite pivot enclosure is equivalent, on this ledger, to the endpoint algebra, the shared-pivot role assignment, and the q-spine gap payment identity.

Proof. For the first bridge packet the D-singleton transition is $(P, m, r, q, g) = (739, 757, 18, 571, 6)$. Hence

$$q_{\text{bridge}} = q + g = 577, \qquad q_{\text{barrier}} = P - gr = 631,$$

and the slack is $631 - 577 = 54 = P - q - g(1 + r)$. The shared pivot 607 lies between these two nodes, with gap split $30 + 24 = 54$.

For the second bridge packet the D-singleton transition is $(P, m, r, q, g) = (739, 761, 22, 601, 6)$. Hence

$$q_{\text{bridge}} = 607, \qquad q_{\text{barrier}} = 607,$$

and the slack is 0. The same 607 is now both bridge root and moving barrier. The carry-packet ledger also gives

$$\text{packet 1 unit} = 607, \quad \text{packet 2 bridge} = 607, \quad \text{packet 2 unit} = 631,$$

so the first packet's barrier is the second packet's unit root. These equalities give the stated shared-pivot hinge. They are finite identities; they do not by themselves prove a uniform pivot law. □

Computational-certificate Statement 8.432 (Shared-pivot hinge contract audit)**.** *The machine certificate records*

```
q_spine_nodes=[577,607,631]
shared_pivot_q=607
finite_endpoint_algebra_closed=true
finite_gap_payment_closed=true
bridge_root_qspine_pivot_to_shared_hinge_contract_closed=true
bridge_root_shared_pivot_hinge_law_proved=false
bridge_root_endpoint_slack_uniform_nonnegative_law_proved=false
row_column_unconditional_closed=false.
```

It also imports the corrected LPF bucket boundary

$$C_p(N) = \Phi(\lfloor N/p \rfloor;\ q < p) - 1,$$

and records that this is still an unsigned count:

```
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
euler_product_half_main_error_proved=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-bridge-root-
shared-pivot-hinge-contract-router.md.
```

Remark 8.433 (Remaining hinge gates)**.** The latest non-cyclic boundary-ratio gate is

```
BoundaryRatioQSpinePivotReductionClosed
AND BridgeRootSharedPivotHingeLawOrPDEC
AND BridgeRootEndpointSlackNonnegativeLawOrPDEC
AND BridgeRootQSpineGapPaymentLawOrPDEC
AND TerminalDoubleAwrapSiblingQSpineKernelPaymentOrPDEC
AND PrimitiveOrientationLocalFactorProductLawBeforePushforward
AND SelectedTerminalMovingBeattyNumeratorPrimeQPrefixPhaseSaving
AND AdmissibleAveragedSignedTraceKloostermanOrTypeIIFamily.
```

Thus the hinge contract is a strict reduction of the previous pivot hard point, not an unconditional closure of the row/column theorem.

8.135 Parity barrier and the prime-distribution contract

Theorem 8.434 (Phi–LPF parity-barrier contract). *The Phi–LPF identities, after correcting their exact endpoint and truncation terms, are exact unsigned rough-number identities. They do not by themselves separate primes from integers with two or more large prime factors. The correct composite LPF bucket formula is*

$$C_p(N) = \Phi(\lfloor N/p \rfloor;\ q < p) - 1,$$

equivalently

$$C_p(N) = \Phi(\lfloor N/p \rfloor;\ q < p) - \Phi(p-1;\ q < p),$$

since $\Phi(p-1;\ q<p) = 1$. The finite Euler-product error is the Legendre-Φ periodic boundary term, not a universal half-main saving.

Proof. If $n = pm$ has least prime factor p, then $m \le N/p$ and m has no prime factor below p. The case $m = 1$ gives the prime p, not a composite bucket member, so it must be removed. This gives the displayed formula. Writing $\Phi(x;\ q < p)$ over a period of $W_{<p}$ gives a main term plus a residue-class boundary term depending on $x \bmod W_{<p}$. Thus refining wheels or LPF buckets controls support and boundary bookkeeping; it does not create a signed prime selector. □

Theorem 8.435 (What would actually break parity). *For the Prime Matrix row problem at the P^2 scale, a genuine parity breakthrough must supply at least one of the following inputs:*

$$\theta((kP, (k+1)P)) > 0$$

for the relevant rows, or the equivalent von-Mangoldt tail absorption

$$\psi((kP, (k+1)P)) > \mathrm{PrimePowerTail}((kP, (k+1)P)),$$

or a source-key consistent signed divisor/trace/Kloosterman/Type–II family whose cancellation distinguishes primes from P_2/P_3 rough composites. Alternatively, the failure of such a uniform law must return as a named PDEC/SAE/local-survivor contradiction.

Proof. The prime-power tail identity writes

$$\psi(I) = \theta(I) + \sum_{p^a \in I,\ a \ge 2} \log p.$$

Hence $\psi(I)$ exceeding the pure prime-power tail forces $\theta(I) > 0$. This is a signed prime-distribution assertion. By contrast, LPF ownership, Phi recursion, affine shifted-residue conjugacy, and small-to-large factor peeling only partition or relabel the same unsigned support. Such data are invariant under the usual sieve parity obstruction: rough primes and rough products of two large primes remain together unless a signed divisor payload, a trace-family cancellation, or a pointwise prime lower bound is added. □

Computational-certificate Statement 8.436 (Parity-barrier prime-distribution contract). *The machine certificate records*

```
parity_barrier_diagnosis_closed=true
lpf_correction_closed=true
legendre_periodic_boundary_not_half_main=true
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
pure_power_selector_supplies_additive_signed_distribution_family=false
pointwise_psi_row_lower_bound_beyond_tail_proved=false
trace_or_typeii_family_admissible_now=false
row_column_unconditional_closed=false.
```

The certificate is

`docs/monograph/prime-matrix-phi-lpf-parity-barrier-prime-distribution-contract-router.md.`

Remark 8.437 (External frontier boundary)**.** FKMS trace functions, Milićević–Qin–Wu Kloosterman bilinear forms, Wright trilinear Kloosterman fractions, Runbo Li large-modulus AP/Harman-sieve inputs, Pascadi Type–II inputs, and thin-group/expander methods are all potentially relevant only after the Prime Matrix payload is promoted to an admissible signed family or a genuine group orbit. In the current ledger they do not directly close any of the three target claims.

8.136 Minimal parity-breaker route forcing

Theorem 8.438 (Minimal parity-breaker alternatives)**.** *After the Phi–LPF endpoint correction and the parity-barrier diagnosis, a Prime Matrix closure cannot be obtained by further unsigned wheel or LPF refinement alone. A closure must supply one of the following actual objects:*

$$\theta((kP,(k+1)P)) > 0 \quad (1 \le k < P),$$

equivalently $h(kP) < P$ for every strict row; or

$$\psi(I_{P,k}) > \mathrm{PrimePowerTail}(I_{P,k});$$

or a pushforward-before signed coefficient transport law

$$a_p(qm) \quad \textit{from the source key, local factor, orientation, and branch data};$$

or a completed source-keyed trace/Type–II family, or a named PDEC/SAE return. The first internal non-cyclic target is therefore

$$\mathrm{PhiLPFRoughCofactorMultiplicationSignedTransportLawBeforePushforward},$$

with the parallel equivalent target

$$\mathrm{PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward}.$$

Proof. The preceding theorem shows that Φ- and LPF-based counts determine only the support and endpoint boundary. The bucket signed-transport ledger adds the formal identity

$$\sum_{m \text{ rough}} a_p(m) = \sum_{m \text{ next-rough}} a_p(m) + \sum_{m'} a_p(qm'),$$

but the second term is still unknown unless a forward law for $a_p(qm')$ is supplied. Thus the first recursive signed field is exactly the cofactor multiplication transport law: it must state how the source key, orientation, local factor, alpha/delta side, and branch transition move when a small prime factor is peeled in or out. External spectral or Type–II theorems require such factorable signed coefficients before they can attach; ordinary short-interval inputs instead require the unavailable pointwise $C \le 1$ square-root-scale theorem. Hence additional unsigned refinement is not a first-break route. □

Computational-certificate Statement 8.439 (Minimal parity-breaker route-forcing certificate)**.** *The machine certificate records*

```
minimal_route_forcing_closed=true
more_wheel_or_lpf_refinement_rejected_as_first_break=true
chosen_next_primary_attack_target=PhiLPFRoughCofactorMultiplicationSignedTransportLawBeforePushforward
chosen_parallel_attack_target=PointwiseSignedCoefficientValueTableOnPhiLPFBucketSupportBeforePushforward
row_column_unconditional_closed=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-minimal-parity-breaker-
route-forcing-router.md.
```

8.137 Parity barrier atom-cut frontier

Theorem 8.440 (Phi-LPF parity barrier atom-cut frontier)**.** *The corrected least-prime-factor bucket identities are*

$$A_p(N) = \Phi(\lfloor N/p \rfloor; \ell < p), \qquad C_p(N) = \Phi(\lfloor N/p \rfloor; \ell < p) - 1,$$

where $A_p(N)$ includes the prime value p, and $C_p(N)$ counts only composite values with least prime factor p. Equivalently

$$\Phi(y; \ell < p) = \sum_{d \mid Q_{<p}} \mu(d) \lfloor y/d \rfloor = y \prod_{\ell < p} (1 - 1/\ell) + E_p(y),$$

with a primorial-periodic floor boundary E_p. Thus the expression

$$(N - p^2) \prod_{\ell \le p} (1 - 1/\ell)$$

is not an exact LPF bucket formula: it uses the wrong counting variable, removes multiples of p from the cofactor, and replaces the floor boundary by a continuous main term.

Consequently the current non-cyclic Prime Matrix route cannot make further progress by more unsigned LPF, Phi, or wheel refinement alone. The signed transport hard point has been synchronized to the following atom-cut frontier:

PhiLPFOffDiagonalPureSemiprimePairSignedSeedAtomBeforePushforward,
PhiLPFEdgeLocalTwoPrimeSignedAtomFieldsOrNamedReturnTagBeforePushforward,
PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward.

These must still be accompanied by orientation, ExactUV return, source-packet three-atom data, row-mass normalization, and signed row survival; otherwise a pointwise signed table, a completed branch/atomic trace, a pointwise θ/ψ row theorem, or a named PDEC/SAE return must be supplied.

Proof. Writing $m = pa$, the condition that p is the least prime factor of m is exactly that a has no prime factor below p. The cofactor a may itself be divisible by p, so the finite Euler product must run over $\ell < p$, not $\ell \le p$. The value $a = 1$ is the prime p, and subtracting this singleton gives the composite bucket formula for $C_p(N)$. The Legendre expansion with floors is exact; replacing floors by y/d only creates a main term plus a periodic boundary, not a parity-breaking signed distribution.

The earlier minimal route-forcing theorem therefore identifies the first possible internal break as a signed cofactor transport law. The subsequent offdiagonal pure-pair and constructor edge-local field-cut certificates refine that law further: the seed side has only the pure pair pq as the first atom, while nonunit tails are internal continuation; the constructor side has only the unsigned edge label until signed atom fields or a named return are emitted. Neither refinement emits a sign, orientation, ExactUV payload, or source-row coefficient. Hence the row/column theorem remains open, and the first non-cyclic work item is the stated signed atom frontier or one of the listed genuine bypasses. □

Computational-certificate Statement 8.441 (Parity-barrier atom-cut frontier certificate). *The machine certificate records*

```
atom_cut_frontier_synced=true
corrected_lpf_formula=C_p(N)=Phi(floor(N/p); primes<p)-1
legendre_periodic_boundary_not_half_main=true
unsigned_lpf_bucket_count_sufficient_for_prime_extraction=false
next_seed_side_attack_target=PhiLPFOffDiagonalPureSemiprimePairSignedSeedAtomBeforePushforward
next_constructor_side_attack_target=PhiLPFEdgeLocalTwoPrimeSignedAtomFieldsOrNamedReturnTagBeforePushforward
row_column_unconditional_closed=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-parity-barrier-
atom-cut-frontier-router.md.
```

8.138 Parity barrier noncircular kernel sync

Theorem 8.442 (Phi-LPF parity barrier noncircular kernel sync). *The atom-cut frontier does not by itself close the row/column theorem. On the constructor side, the edge-local signed fields synchronize to a same-trace-key and named-return matrix. The only productive output of that matrix is a new primitive atomic signed payload or trace artifact. Such a payload, if it is not merely a signed-lane rename, must carry source-rank/no-collapse data and therefore first reaches the source-domain entropy gate. Expanding that gate inside the current constructor corpus returns to the joint declaration line, while the row-level generation table returns to itself through the signed-source spine.*

Consequently the current internal non-cyclic target is not another LPF main term or another atom-field name. After the already proved Phi-LPF support stripping removes the unsigned row-search and capacity parts of the kernel, the fastest remaining internal target is the bucket signed law

$$\text{PhiLPFBucketSignedCoefficientLawBeforePushforward},$$

with

$$\text{NonzeroSignedRowSurvivalOnPhiLPFSupportBeforePushforward},$$
$$\text{SameFormalUnitPrimitiveRowMassNormalizationAndNoHeavyRowLedger}.$$

The non-circular pre-Cauchy kernel remains the discipline condition forbidding a return to the row-level table, the source-entropy payload loop, or downstream Phi/payment data. In parallel, the source-rank convergence route remains the three-atom pointwise kernel package

$$\text{AlphaRowAnchorPhaseEmissionFormulaLedger},$$
$$\text{IndependentNoncanonicalPreCauchyArithmeticIdentityStatementLedger},$$
$$\text{SameUnitExactUVRankMultiplicityCertificateForPrimitiveKernelRows}.$$

The remaining genuine bypasses are a pointwise Phi-LPF signed value table, a pointwise θ/ψ row theorem at the $x^{1/2}$ scale, a same-set PDEC/SAE return, or an externally admissible source-keyed trace/Type–II family.

Proof. The preceding atom-cut theorem reduces the unsigned LPF/Phi layer to two signed interfaces: a pure semiprime seed atom and constructor edge-local signed fields. Existing signed-atom trace synchronization proves that the constructor fields must belong to one pre-Cauchy trace key, with every missing field, conflicting field, zero local factor, or cross-key read sent to a named return. Thus the only productive non-return object is a new primitive payload or trace.

The new-payload alignment theorem then forces that object to carry the actual source-rank/no-collapse package: source entropy, complete primitive emitter key partition, and fixed-key ExactUV local multiplicity. The constructor source-entropy payload-loop certificate shows that expanding this source entropy through the present constructor corpus returns to the joint declaration used upstream to create the same payload. The row-level fixed-point certificate shows the same obstruction on the row table: the signed-source spine returns to the row-level table itself. Neither loop is a proof.

Deleting these loops first leaves a non-circular pre-Cauchy signed emission kernel. The Phi-LPF bucket support-stripping certificate then removes the unsigned search and capacity portion of that kernel, so the remaining first positive object is the bucket signed coefficient law together with signed survival and row-mass control. Independently, the trace-exit/source-rank convergence certificate keeps the alpha-row/source-rank three-atom package as a parallel pointwise kernel route. External short-interval, short-interval arithmetic-progression, or trace/Type–II theorems can attach only after one of these signed or pointwise interfaces is supplied. Therefore the row/column theorem remains open, but the next non-cyclic hard point is now strictly smaller and explicitly named. □

Computational-certificate Statement 8.443 (Parity-barrier noncircular-kernel sync certificate). *The machine certificate records*

```
noncircular_kernel_sync_closed=true
lpf_phi_unsigned_scope_exhausted=true
pointwise_phi_lpf_bucket_signed_value_table_proved=false
noncircular_signed_emission_kernel_proved=false
phi_lpf_bucket_signed_coefficient_law_proved=false
alpha_row_anchor_phase_emission_formula_proved=false
external_trace_typeii_family_directly_attaches_now=false
row_column_unconditional_closed=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-parity-barrier-
noncircular-kernel-sync-router.md.
```

8.139 Parity barrier transport-edge sync

Theorem 8.444 (Phi-LPF parity barrier transport-edge sync). *The bucket signed law isolated in the preceding theorem is not treated as an indivisible terminal name. If it is not supplied directly by a pointwise Phi-LPF signed value table, it must pass through the rough-cofactor signed transport stack. The existing transport certificates have already removed the unsigned ordered-factorization component, the unit/square-base private escape, and the common source-packet self-proof loop. Hence the recursive branch reaches the edge multiplier table*

$$\text{PhiLPFRoughCofactorStepSignedMultiplierTableBeforePushforward.}$$

The first-edge slab certificate first splits this table into

$$\text{PhiLPFSemiprimeFirstEdgeSignedSeedTableBeforePushforward,}$$
$$\text{PhiLPFInternalPrimeAdjoinSignedTransitionLawBeforePushforward.}$$

The semiprime seed diagonal/offdiagonal certificate then sends the diagonal $p = q$ lane back to the common source packet and leaves only the offdiagonal seed lane. The offdiagonal tuple-field and pure-pair atom certificates remove the unsigned owner/first-prime/tail fields and show that the true tail-one atom is

$$\text{PhiLPFOffDiagonalPureSemiprimePairSignedSeedAtomBeforePushforward,}$$

with orientation, ExactUV return, and internal transition still attached. Moreover, the exact Phi fiber formula

$$\mathrm{mass}(p,q) = \Phi\left(\left\lfloor \frac{N}{pq} \right\rfloor, q\right)$$

supplies only support and occurrence mass. It does not supply the first-edge signed seed value, orientation parity, ExactUV return, the internal prime-adjoin local factor, a branch trace, or an ExactUV payload.

Proof. The bucket transport certificate states that the bucket signed law is either a new pointwise signed table or a rough-cofactor transport law. Along the transport branch, ordered factorization coherence is an LPF path fact and the unit/square-base entry cannot create a private signed source. The branch therefore reaches the edge multiplier table together with the already known source-rank package. The first-edge slab certificate gives a unique partition of every ordered path into the initial semiprime edge and the remaining prime-adjoin transitions. The diagonal/offdiagonal split removes the diagonal private lane, and the offdiagonal tuple/pure-pair certificates show that tail > 1 is continuation lift, not another first seed. These are structural partitions, not sign formulas: the Phi fibers count continuations, but they are insensitive to the missing signed coefficient. Thus the next non-cyclic target is the offdiagonal pure-pair signed atom together with orientation, ExactUV return, and the internal transition law, unless one instead supplies a pointwise signed table, a complete branch/atomic trace, a controlled PDEC/SAE return, or a genuine pointwise θ/ψ or source-keyed trace/Type–II input. □

Computational-certificate Statement 8.445 (Parity-barrier transport-edge sync certificate). *The machine certificate records*

```
transport_edge_sync_closed=true
bucket_signed_law_imported=true
edge_multiplier_table_reached=true
edge_table_split_into_first_seed_and_internal_transition=true
first_edge_phi_fiber_formula_proved=true
first_edge_phi_fiber_supplies_signed_seed=false
diagonal_offdiagonal_support_split_closed=true
offdiagonal_source_tuple_bijection_synced=true
pure_pair_atom_bijection_synced=true
tail_lift_no_new_first_seed_closed=true
pure_semiprime_pair_signed_seed_atom_proved=false
offdiagonal_orientation_parity_law_proved=false
offdiagonal_exactuv_fixed_pair_return_ledger_proved=false
internal_prime_adjoin_signed_transition_law_proved=false
row_column_unconditional_closed=false.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-parity-barrier-
transport-edge-sync-router.md.
```

8.140 Row Delta-Phi cover contract

Theorem 8.446 (Phi-LPF row Delta-Phi cover identity and boundary). *For an integer interval $I = (A, B]$, $1 \le A < B$, the LPF bucket prefix is*

$$C_p(N) = \begin{cases} 0, & N < p^2, \\ \Phi(\lfloor N/p \rfloor;\ primes\ \ < p) - 1, & N \ge p^2. \end{cases}$$

Consequently the row prime count is exactly

$$\pi(A, B] = B - A - \sum_{p \le \sqrt{B}} \bigl(C_p(B) - C_p(A)\bigr).$$

Thus row positivity is equivalent to the strict Delta-Phi cover defect

$$\sum_{p \le \sqrt{B}} \bigl(C_p(B) - C_p(A)\bigr) \le B - A - 1.$$

This equivalence is an exact reduction, not by itself a proof of the strict defect. A proof must either produce a uniform Delta-Phi cover saving, return a named LPF-owner residue PDEC/SAE when equality occurs, or attach a pointwise θ/ψ, signed divisor, trace, or Type–II input.

Proof. The prefix formula follows from the bijection $n = pm$. The condition $\mathrm{LPF}(n) = p$ is exactly that m is not divisible by any prime less than p; the composite bucket begins at p^2, and the subtraction of 1 removes the single prefix survivor $m = 1$, which corresponds to the prime endpoint $n = p$. Taking the difference at B and A counts the composites in the row with LPF p. Summing over $p \le \sqrt{B}$ gives every composite in $(A, B]$ exactly once. Therefore the displayed formula for $\pi(A, B]$ is exact. The strict inequality is just the statement that at least one integer in the row is not covered by any composite LPF bucket. It is therefore equivalent to the existence of a prime in the row and cannot be deduced from the equality alone without an independent saving or a controlled return. $\square$

Computational-certificate Statement 8.447 (Row Delta-Phi cover contract certificate). *The machine certificate records*

```
row_delta_phi_identity_closed=true
row_delta_phi_prefix_difference_is_exact=true
strict_cover_inequality_proved_uniformly=false
row_delta_phi_positive_lower_bound_proved=false
row_column_unconditional_closed=false.
```

It also records the prime-free check $(90, 96]$, *where the Delta-Phi composite cover is* 6, *the row length is* 6, *and the prime count is* 0. *The certificate is*

```
docs/monograph/prime-matrix-phi-lpf-row-delta-phi-
cover-contract-router.md.
```

8.141 Row Delta-Phi inequality breakthrough frontier

Theorem 8.448 (Row inequality breakthrough formula frontier). *Assume the row Delta-Phi identity of the preceding section. To close a target Prime Matrix row $(A, B]$ unconditionally, it is not enough to restate the prefix identity. One must supply at least one of the following independent inputs:*

$$\sum_{p \le \sqrt{B}} \bigl(C_p(B) - C_p(A)\bigr) \le B - A - 1,$$

or an equality-case LPF-owner residue PDEC/SAE, or a pointwise square-root row input such as

$$\vartheta(B) - \vartheta(A) > 0 \quad \textit{or} \quad \psi(B) - \psi(A) > \textit{prime-power tail}(A, B),$$

or a source-keyed signed divisor, trace, Kloosterman, or Type–II family whose error is smaller than the row main term. Unsigned LPF/Phi survivor mass, fixed-wheel refinement, Euler product main terms, and support-only P_2 counts do not by themselves break the parity barrier.

Proof. The preceding identity gives

$$\pi(A,B] = B - A - \sum_{p\le\sqrt{B}} \big(C_p(B) - C_p(A)\big).$$

Thus row positivity is exactly the displayed strict cover defect. Any proof which uses only the same unsigned LPF cover has no information distinguishing a prime survivor from a rough semiprime or deeper rough composite survivor; this is the parity obstruction in this formulation. A valid proof must therefore add either a genuine strict saving in the cover, a structured contradiction when equality occurs, or signed oscillation/pointwise prime distribution information before the LPF pushforward. Published and preprint short-interval inputs with exponent $\theta > 1/2$ give intervals of length x^θ. At $x = P^2$ these occupy $P^{2\theta-1}$ rows beyond one row, so exponents such as $17/30$ or $13/25$ still prove only a thickened-row statement here, not the fixed-row inequality. □

Computational-certificate Statement 8.449 (Row inequality breakthrough frontier certificate)**.** *The machine certificate records*

```
breakthrough_formula_frontier_synced=true
strict_cover_inequality_proved_uniformly=false
row_column_unconditional_closed=false
chosen_primary_attack_target=UniformDeltaPhiCoverDefectOrNamedLPFOwnerResiduePDEC
chosen_parallel_signed_target=SourceKeyedMobiusVonMangoldtTraceTypeIIFamily
chosen_parallel_distribution_target=PointwiseThetaPsiCOneInputAtSqrtRowScale.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-row-inequality-
breakthrough-frontier-router.md.
```

8.142 Target-row residue-cover normal form

Theorem 8.450 (Target punctured-row residue-cover frontier)**.** *The row Delta-Phi strict cover inequality cannot be promoted to a theorem for arbitrary integer short intervals. There are ordinary intervals, for example* $(90, 96]$*, on which the Delta-Phi composite cover equals the whole interval and* $\pi(90, 96] = 0$*. Therefore the current target must remain the Prime Matrix punctured row*

$$R_{P,k} = \{kP + r : 1 \le r \le P-1\}, \qquad 1 \le k \le P-1.$$

For every prime $p < P$*, define*

$$D_p(P,k) = \{1 \le r \le P-1 : r \equiv -kP \pmod{p}\}.$$

The row inequality fails exactly when these small-prime residue fibers cover all offsets:

$$\bigcup_{p\le\sqrt{(k+1)P-1}} D_p(P,k) = [1, P-1].$$

Thus the next non-circular target is a full-cover owner-residue PDEC, or a source-keyed Mobius/trace/spectral cancellation formula attached to these same fibers.

Proof. The ordinary interval counterexample shows that a universal short-interval Delta-Phi defect is false. On the punctured row $R_{P,k}$, all entries are less than P^2. Hence any composite entry has a prime factor $\le \sqrt{(k+1)P-1} < P$. Since $p < P$, the residue of P is invertible modulo p, and divisibility of $kP+r$ by p is the single congruence $r \equiv -kP \pmod p$. Therefore an offset not lying in any displayed $D_p(P,k)$ has no prime factor up to the square-root bound and is prime. The only failure mode for the strict cover inequality is complete coverage of all offsets by these residue fibers. Proving that complete coverage is impossible without merely restating prime existence is exactly the remaining owner-residue PDEC or signed-trace problem. □

Computational-certificate Statement 8.451 (Target residue-cover certificate). *The machine certificate records*

```
target_row_residue_cover_standard_form_synced=true
generic_interval_uniform_defect_false=true
target_punctured_row_full_cover_found_in_scan=false
strict_cover_inequality_proved_uniformly=false
row_column_unconditional_closed=false
selected_primary_next_gate=FullCoverOwnerResiduePDEC
selected_signed_parallel_gate=MobiusResidueCoverSignedTrace
selected_spectral_parallel_gate=SpectralKloostermanResidueLift.
```

The finite scan covers $P = 31, 101, 251, 499, 1009$ *only and is not used as a global proof. The certificate is*

```
docs/monograph/prime-matrix-phi-lpf-row-inequality-
target-residue-cover-router.md.
```

8.143 Full-cover owner PDEC stress test

Theorem 8.452 (Owner-only full-cover PDEC is insufficient). *A contradiction extracted from a full-cover row cannot depend only on the facts that LPF owner fibers are disjoint, that they cover the row, or that each prime defines one residue class. Ordinary zero-prime intervals such as* $(90, 96]$*,* $(114, 126]$*, and* $(200, 210]$ *already satisfy these owner-support properties and have full Delta-Phi cover. Hence any non-circular full-cover PDEC for the Prime Matrix row must use the target-affine data*

$$A = kP, \qquad |R_{P,k}| = P-1, \qquad P \text{ prime}, \qquad 1 \le k \le P-1,$$

together with global residue coupling $a_p = -kP \bmod p$*, owner minimality, and a signed or phase payload before pushforward.*

Proof. The listed ordinary intervals contain no primes, so their LPF owner buckets partition the whole interval. For each prime p, divisibility in any ordinary interval $(A, B]$ is still one residue class in the offset variable. Thus disjoint owner support, full cover, and one-residue-per-prime structure are not contradictions. These are tautological consequences of LPF ownership and local congruence, not target-specific constraints. The Prime Matrix row adds the global affine anchor $A = kP$ with P prime and $k < P$. Any surviving PDEC must therefore use this global coupling across all primes, or else import signed Mobius/Von Mangoldt, trace, or spectral cancellation. The finite scan through $P = 5003$ found no target-row full cover, but this is evidence only. □

Computational-certificate Statement 8.453 (Full-cover owner PDEC stress certificate). *The machine certificate records*

```
full_cover_owner_pdec_stress_synced=true
owner_only_pdec_rejected=true
target_affine_owner_pdec_proved=false
target_scan_no_full_cover=true
row_column_unconditional_closed=false
selected_next_primary_gate=TargetAffineFullCoverOwnerResiduePDEC
selected_parallel_signed_gate=MobiusResidueCoverSignedTraceWithTargetAffineAnchor
selected_parallel_spectral_gate=SpectralKloostermanResidueLiftWithSourceKeys.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-full-cover-owner-
pdec-stress-router.md.
```

8.144 Target-affine gap equivalence

Theorem 8.454 (Target-affine anchor reduces to a row-sized prime gap). *The target-affine data*

$$A = kP, \qquad |R_{P,k}| = P - 1, \qquad P \text{ prime}, \qquad 1 \le k \le P - 1$$

do not by themselves close the full-cover PDEC. They identify the remaining failure state with a prime gap crossing the whole target row:

$$\bigcup_{p \le \sqrt{(k+1)P-1}} D_p(P,k) = [1, P-1] \iff \pi(kP,(k+1)P) = 0.$$

Near $k = P$, this is a pointwise short-interval problem at $x \asymp P^2$ with $H = P \asymp \sqrt{x}$. The top row $k = P - 1$ is the prime-indexed Oppermann-left half-window $(P^2 - P, P^2)$. Hence the target-affine anchor must be supplemented by a signed phase payload, a genuine $C = 1$ square-root-scale prime input, or a source-keyed spectral lift.

Proof. For an offset $1 \le r \le P - 1$, the row element $kP + r$ is composite with a least prime owner $p \le \sqrt{(k+1)P-1}$ exactly when

$$r \equiv -kP \pmod p$$

after deleting smaller owner classes. Therefore full owner coverage of all offsets is precisely the assertion that every integer in the target row is composite. This is equivalent to $\pi(kP,(k+1)P) = 0$, or to an adjacent prime gap covering the row. The affine formula supplies the location of the row, but not a contradiction to such a gap. Existing unconditional short-interval exponents $21/40$, $17/30$, and $13/25$ still give intervals thicker than one P-row at $x = P^2$, so they cannot directly close this gate. □

Computational-certificate Statement 8.455 (Target-affine gap equivalence certificate). *The machine certificate records*

```
target_affine_gap_equivalence_synced=true
owner_only_pdec_rejected_imported=true
target_affine_anchor_alone_closes=false
target_affine_owner_pdec_proved=false
row_column_unconditional_closed=false
selected_next_primary_gate=TargetAffineSignedPhasePayload
selected_parallel_distribution_gate=PointwiseSqrtPrimeInputCOne
selected_parallel_spectral_gate=SpectralKloostermanResidueLiftWithSourceKeys.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-target-affine-
gap-equivalence-router.md.
```

8.145 Target-affine signed phase contract

Theorem 8.456 (Row Fourier phase is an exact defect identity, not a closure)**.** *For every target row $R_{P,k} = \{kP+r : 1 \le r \le P-1\}$ and every frequency h mod P, the offset Fourier decomposition satisfies*

$$\sum_{r=1}^{P-1} e(hr/P) = \sum_{p} \sum_{r \in O_p(P,k)} e(hr/P) + \sum_{\substack{1 \le r \le P-1 \\ kP+r \text{ prime}}} e(hr/P),$$

where $O_p(P,k)$ is the LPF owner fiber in the row. Hence the row Fourier defect is exactly the Fourier transform of the prime-survivor set. This phase identity detects survivors, but a nonzero lower bound for the defect is still equivalent to proving the row contains a prime.

Proof. The LPF owner fibers and the prime survivors form a disjoint partition of $\{1, \dots, P-1\}$. Summing the additive character $e(hr/P)$ over this partition gives the displayed identity. The zero frequency gives precisely the survivor count. For the nonzero frequencies, Parseval gives

$$\sum_{h=1}^{P-1} \left| \sum_{\substack{1 \le r \le P-1 \\ kP+r \text{ prime}}} e(hr/P) \right|^2 = Ps - s^2,$$

where s is the number of prime survivors. Thus the phase energy is positive if and only if $s > 0$. Consequently, offset Fourier phase alone is only a rephrasing of the same prime-gap exclusion problem. A non-circular proof must produce a source-keyed signed owner phase before pushforward, complete the owner fibers into an admissible trace/Kloosterman family, or import a genuine $C = 1$ square-root-scale pointwise prime theorem. ☐

Computational-certificate Statement 8.457 (Target-affine signed phase contract certificate)**.** *The machine certificate records*

```
target_affine_signed_phase_contract_synced=true
row_fourier_defect_identity_closed=true
row_fourier_positive_lower_bound_proved=false
source_keyed_owner_phase_emission_formula_proved=false
completed_trace_kloosterman_family_from_owner_fibers_proved=false
row_column_unconditional_closed=false
selected_next_primary_gate=SourceKeyedOwnerPhaseEmissionFormula
selected_parallel_distribution_gate=PointwiseSqrtPrimeInputCOne
selected_parallel_trace_gate=CompletedTraceKloostermanFamilyFromOwnerFibers.
```

The certificate is

```
docs/monograph/prime-matrix-phi-lpf-target-affine-
signed-phase-contract-router.md.
```

Chapter 9

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.